CGP

GCSE AQA

Biology

Complete

Revision & Practice

with new ***CGP RevisionHub***

New ***CGP RevisionHub*** includes:

- Quick Quizzes
- Summary Tests
- Videos
- And much more...

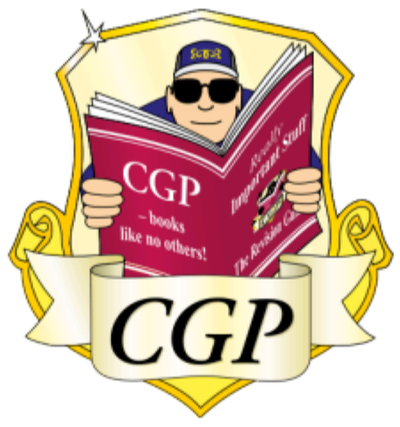
Ta-dah...

CGP RevisionHub

...is here!

- ✓ New online revision
- ✓ Quizzes, tests, videos and more
- ✓ All matched to your book

CGP RevisionHub included with this book



GCSE AQA

Biology

This fantastic CGP book explains all the important facts, theory and practical skills for GCSE Biology with the clearest study notes around!

We've got exam-style practice questions for every topic — plus a set of mock exam papers to *really* test you. And we've included fully-worked answers, so it's easy to mark your work.

There's also plenty of online revision in CGP RevisionHub. You'll find quick quizzes, summary tests, videos and more to use alongside your book. We know, we're too good to you.

Complete

Revision & Practice

with new ***CGP RevisionHub***

Contents

How to Revise with CGP RevisionHub 2
What to Expect in the Exams 3

Paper One

Topic 1 — Cell Biology

Topic 1a — Cells and Microscopy

Cells 4
Microscopy 5
More on Microscopy 6
Warm-Up & Exam Questions 7
Cell Specialisation and Differentiation 8
Stem Cells 9
Chromosomes and Mitosis 10
Binary Fission 11
Warm-Up & Exam Questions 12
Culturing Microorganisms 13
Warm-Up & Exam Questions 15
Revision Summary Test for Topic 1a 16

Topic 1b — Exchanging Substances

Diffusion 17
Osmosis 18
Active Transport 19
Exchange Surfaces 20
Exchanging Substances 21
More on Exchanging Substances 22
Warm-Up & Exam Questions 23
Exam Questions 24
Revision Summary Test for Topic 1b 25

Topic 2 — Organisation

Topic 2a — Organisation, Enzymes and Circulation

Cell Organisation 26
Enzymes 27
Investigating Enzymatic Reactions 28
Enzymes and Digestion 29
More on Enzymes and Digestion 30
Food Tests 31
Warm-Up & Exam Questions 32
Exam Questions 33
The Lungs 34
Circulatory System — The Heart 35
Circulatory System — Blood Vessels 36
Circulatory System — Blood 37
Warm-Up & Exam Questions 38
Exam Questions 39
Revision Summary Test for Topic 2a 40

Topic 2b — Disease and Plant Organisation

Cardiovascular Disease 41
More on Cardiovascular Disease 42
Warm-Up & Exam Questions 43
Health and Disease 44
Non-Communicable Disease Risk Factors 45
Cancer 46
Warm-Up & Exam Questions 47
Plant Cell Organisation 48
Transpiration and Translocation 49
Transpiration and Stomata 50
Warm-Up & Exam Questions 51
Exam Questions 52
Revision Summary Test for Topic 2b 53

Topic 3 — Infection and Response

Communicable Disease 54
The Spread of Disease 55
Viral and Fungal Diseases 56
Bacterial and Protist Diseases 57
Warm-Up & Exam Questions 58
Fighting Disease 59
Fighting Disease — Vaccination 60
Fighting Disease — Drugs 61
Developing Drugs 62
Warm-Up & Exam Questions 63
Exam Questions 64
Producing Monoclonal Antibodies 65
Uses of Monoclonal Antibodies 66
More Uses of Monoclonal Antibodies 67
Plant Diseases and Defences 68
Warm-Up & Exam Questions 69
Revision Summary Test for Topic 3 70

Topic 4 — Bioenergetics

Photosynthesis and Limiting Factors.....71
The Rate of Photosynthesis 72
More on the Rate of Photosynthesis.....73
Investigating Photosynthesis 74
Light Intensity and Farming.....75
Warm-Up & Exam Questions 76
Exam Questions 77
Respiration and Metabolism 78
Aerobic and Anaerobic Respiration 79
Exercise.....80
Warm-Up & Exam Questions 81
Revision Summary Test for Topic 4 82

Paper Two

Topic 5 — Homeostasis and Response

Topic 5a — The Nervous System

Homeostasis.....83
The Nervous System 84
Synapses and Reflexes.....85
Investigating Reaction Time 86
Warm-Up & Exam Questions 87
The Brain.....88
The Eye.....89
Vision Defects 90
Vision Defects and the Iris Reflex 91
Controlling Body Temperature 92
Warm-Up & Exam Questions 93
Exam Questions.....94
Revision Summary Test for Topic 5a.....95

Topic 5b — Hormones

The Endocrine System.....96
Controlling Blood Glucose 97
Warm-Up & Exam Questions 98
The Kidneys.....99
Kidney Failure 100
Warm-Up & Exam Questions 101
Puberty and the Menstrual Cycle.....102
Controlling Fertility.....103
More on Controlling Fertility.....104
Adrenaline and Thyroxine.....105
Warm-Up & Exam Questions 106
Plant Hormones.....107
Investigating Plant Growth.....108
Commercial Uses of Plant Hormones 109
Warm-Up & Exam Questions 110
Revision Summary Test for Topic 5b 111

Topic 6 — Inheritance, Variation and Evolution

Topic 6a — DNA and Reproduction

DNA.....112
The Structure of DNA and Protein Synthesis.....113
Mutations 114
Warm-Up & Exam Questions 115
Reproduction.....116
Meiosis.....117
More on Reproduction 118
X and Y Chromosomes 119
Warm-Up & Exam Questions 120
Genetic Diagrams.....121
More Genetic Diagrams 122
Inherited Disorders.....123
The Work of Mendel.....124
Warm-Up & Exam Questions 125
Exam Questions.....126
Revision Summary Test for Topic 6a.....127

Topic 6b — Variation, Evolution and Classification

Variation.....128
Evolution 129
More About Evolution.....130
Warm-Up & Exam Questions 131
Selective Breeding 132
Genetic Engineering 133
Cloning 134
Warm-Up & Exam Questions 135
Fossils.....136
Speciation 137
Antibiotic-Resistant Bacteria 138
Classification 139
Warm-Up & Exam Questions 140
Revision Summary Test for Topic 6b 141

Topic 7 — Ecology

Topic 7a — Ecology and Cycles

Competition 142
Abiotic and Biotic Factors.....143
Adaptations 144
Food Chains 145
Warm-Up & Exam Questions 146
Using Quadrats 147
Using Transects.....148
Environmental Change & The Water Cycle...149
The Carbon Cycle.....150
Warm-Up & Exam Questions 151
Exam Questions.....152
Decay.....153
Investigating Decay 154
Warm-Up & Exam Questions 155
Revision Summary Test for Topic 7a 156

Topic 7b — Ecosystems, Biodiversity & Food Production

Biodiversity and Waste Management 157
Global Warming.....158
Deforestation and Land Use 159
Maintaining Ecosystems and Biodiversity.....160
Warm-Up & Exam Questions 161
Trophic Levels 162
Pyramids of Biomass.....163
Biomass Transfer.....164
Food Security and Farming 165
Biotechnology 166
Warm-Up & Exam Questions 167
Revision Summary Test for Topic 7b 168

Working Scientifically

The Scientific Method.....169
Communication & Issues Created by Science 170
Risk 171
Designing Investigations 172
Collecting Data 173
Processing Data.....174
Presenting Data 175
Interpreting Graphs.....176
Units 177
Maths Skills 178
Drawing Conclusions 179
Uncertainties and Evaluations.....180

Practical Skills

Measuring Substances 181
Heating, Safety and Ethics 182
Potometers.....183
Sampling 184
Comparing Results 185

Practice Exams

Practice Paper 1 186
Practice Paper 2 208

Answers 231
Glossary 252
Index 257

Throughout this book you'll see grade stamps like these:

These grade stamps help to show how difficult the questions are.
Remember — to get a top grade you need to be able to answer **all** the questions, not just the hardest ones.

In the real exams, some questions test how well you can write (as well as your scientific knowledge).
In this book, we've marked these questions with an asterisk (*).

Published by CGP

From original material by Richard Parsons.

Editors: Katherine Faudemer, Rachel Hickman, Hannah Lawson, Sarah Pattison and Camilla Sheridan.

Contributor: Paddy Gannon.

BAS47 ~ 0325 - 30593

With thanks to Jade Sim for the copyright research.

With thanks to EYE OF SCIENCE/SCIENCE PHOTO LIBRARY for permission to reproduce the image on page 196.
With thanks to SCIENCE PHOTO LIBRARY for permission to reproduce the image of a contraceptive implant on page 227.

Clipart from Corel®
Illustrations by: Sandy Gardner Artist, email sandy@sandygardner.co.uk

Text, design, layout and original illustrations © Coordination Group Publications Ltd. (CGP) 2025
All rights reserved.

0800 1712 712 • www.cgpbooks.co.uk

How to Revise with *CGP RevisionHub*

*There's more to **revision** than just reading this book — you need to **understand** and **remember** it too. Read on for how to get the most out of your **book** and **RevisionHub**...*

Here's a handy guide to using RevisionHub.

Video

1) You Need to **Learn Your Stuff**

This book covers everything you need to know for GCSE AQA Biology.

1) Read it — but not all at once... break it into manageable chunks.
2) Once you've read over a chunk, try making your own mind maps and flash cards — writing down and thinking about the content will help it to really stick.
3) We've also included some videos to help with the trickier bits — watch out for the QR codes.

2) Check That You Can **Remember** What You've Learnt

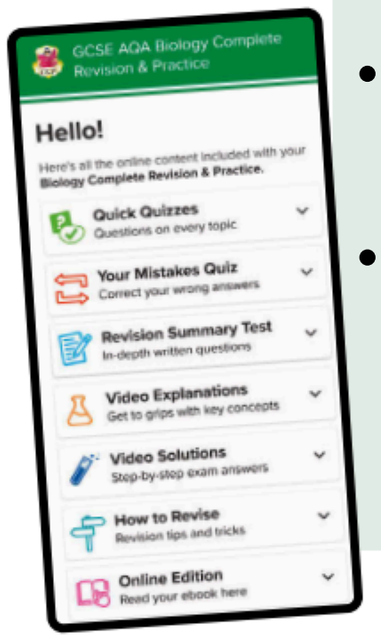
Ta-dah!

Revision Hub

If you don't test yourself, you won't know what you don't know. So try covering up a page and seeing if you can write it all out. Then try quizzing yourself using RevisionHub:

Quick Quizzes

- The QR code at the top of each page takes you to a quick quiz of the stuff on that page.
- These quizzes are a quick check of the key facts and your maths skills.
- Revisit questions that you get wrong by doing the 'Your Mistakes' quiz, and there are Paper 1 and Paper 2 quizzes too.



Revision Summary Tests

- At the end of each topic in the book is a fiendish memory test on as much stuff as we could cram in.
- You can do these tests on paper or you can complete them on the RevisionHub — follow the QR code.
- Online, you can also find sample answers, assess your progress, and look back at your results for previous topics to see the areas you need to work on.

3) Make Sure That You're **Exam Ready**

To see if you're exam ready, try some exam-style questions. Luckily for you, there are loads in this book.

- You'll find one or more exam-style questions at the bottom of some pages.
- Every few pages, you'll find some warm-up and exam questions.
- There are also two practice papers at the back of the book.

Use these resources to check that you can apply what you've learnt to exams.

There are answers in the back of the book, and many questions at the bottom of the main pages also have a video solution for you to watch — just use the QR code next to the question.

You probably won't stick strictly to this order of steps — for example, you might want to try some exam-style questions as you go, and you'll need to look back at bits of this book if you get a question wrong or are finding a topic hard.

Revision — there's no better path to exam success...

Yep, revision can seem quite daunting. But it's a key part of preparing for your exams. If it's making you feel like you don't know where to start, then fear not — follow this QR code to see what we recommend.

Start Revising

What to Expect in the Exams

*Before you get cracking with your **revision** and **exam practice**, here's a **handy guide** to what you'll have to face in the exams. You're welcome.*

Topics are Covered in Different Papers

You're expected to know the basic concepts of biology in both papers.

For GCSE AQA Biology, you'll sit two exam papers at the end of your course.

Paper	Time	No. of Marks	Topics Assessed	Maths Skills	Practical Skills
1	1 hr 45 mins	100	1, 2, 3 and 4	Around 10% of total marks	At least 15% of total marks
2	1 hr 45 mins	100	5, 6 and 7		

There are Different Question Types

In each exam, you'll be expected to answer a mixture of multiple choice questions, structured questions, questions that have short, closed answers, as well as open response questions.

For some open response questions, you'll be marked on the overall quality of your answer, not just its scientific content. So...

Always make sure:

- You answer the question fully.
- You include detailed, relevant information.
- Your answer is clear and has a logical structure.

You'll be Tested on Your Maths...

At least 10% of the total marks for GCSE Biology will come from questions that test your maths skills. For these questions, always remember to:

We've indicated maths skills throughout this book with a handy stamp that looks like this:

MATHS SKILLS

- Show your working — you could get marks for this, even if your final answer's wrong.
- Check that the units of your answer are the same as the ones they asked for in the question.
- Make sure your answer is given to an appropriate number of significant figures.

...and on Your Practical Skills

PRACTICAL

Required practical activities are marked up with a stamp like this.

GCSE Biology contains 10 required practical activities that you'll do during the course. You can be asked about these, and the practical skills they involve, in the exams.

E.g. you might have to comment on experimental design (the apparatus and method), make predictions, analyse or interpret results... Pretty much anything to do with planning and carrying out investigations.

At least 15% of the total marks will be for questions that test your understanding of the practical activities and practical skills.

You'll Need to Know About Working Scientifically

Working Scientifically is all about how science is applied in the outside world by real scientists.

For example, you might be asked about ways that scientists communicate an idea to get their point across without being biased, or about the limitations of a scientific model.

You need to think about the situation that you've been given and use all your scientific savvy. Always read the question and any data you've been given really carefully before you start writing.

Quick Quiz

Cells

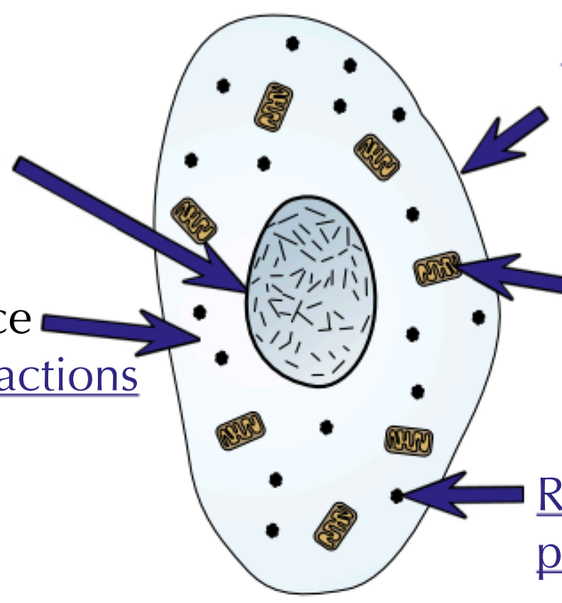
*When someone first peered down a microscope at a slice of cork and drew the **boxes** they saw, little did they know that they'd seen the **building blocks** of **every organism** on the planet...*

Organisms can be **Prokaryotes** or **Eukaryotes**

1) All living things are made of cells.
2) Cells can be either prokaryotic or eukaryotic. Eukaryotic cells are complex and include all animal and plant cells. Prokaryotic cells are smaller and simpler, e.g. bacteria (see below).
3) Eukaryotes are organisms that are made up of eukaryotic cells.
4) A prokaryote is a prokaryotic cell (it's a single-celled organism).

The **Different Parts** of a Cell are called **Subcellular Structures**

Most animal cells have these subcellular structures:



Nucleus — contains genetic material that controls the activities of the cell.

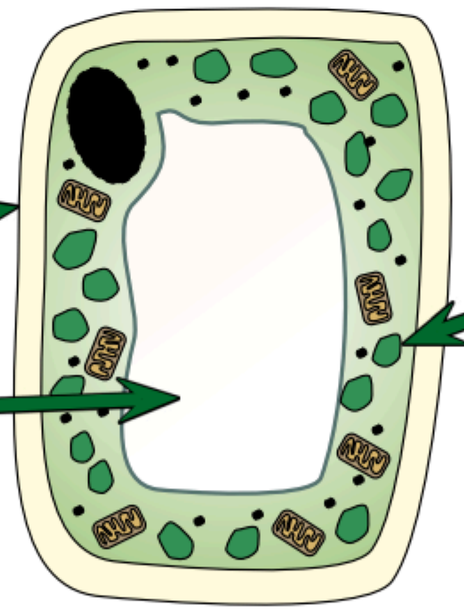
Cytoplasm — gel-like substance where most of the chemical reactions happen. It contains enzymes (see page 27) that control these chemical reactions.

Cell membrane — holds the cell together and controls what goes in and out.

Mitochondria — these are where most of the reactions for aerobic respiration take place (see page 79). Respiration transfers energy that the cell needs to work.

Ribosomes — these are where proteins are made in the cell.

Plant cells usually have all the bits that animal cells have, plus a few extra things:



Rigid cell wall — made of cellulose. It supports the cell and strengthens it.

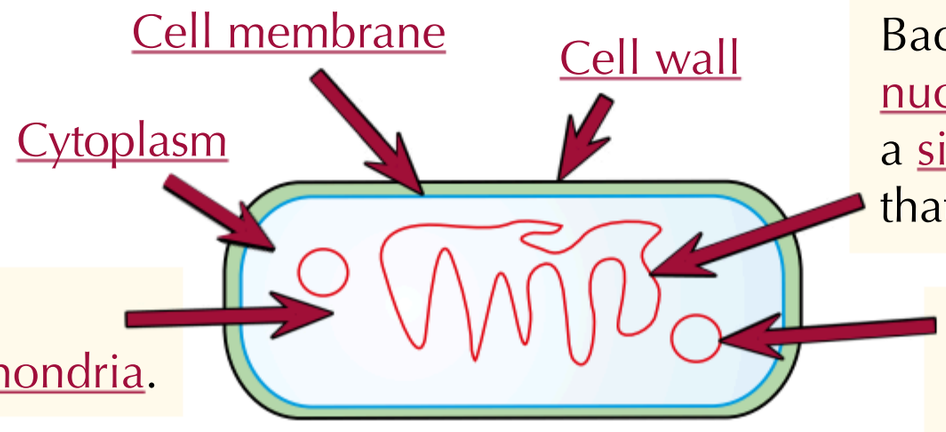
Permanent vacuole — contains cell sap, a weak solution of sugar and salts.

Chloroplasts — where photosynthesis occurs, which makes food for the plant (see page 71). They contain a green substance called chlorophyll, which absorbs the light needed for photosynthesis.

The cells of algae (e.g. seaweed) also have a rigid cell wall and chloroplasts.

Bacterial Cells Are Much **Smaller**

Bacteria are prokaryotes. Here's what a bacterial cell might look like:



You might see the sizes of cells written in standard form (see the next page).

Bacterial cells don't have a 'true' nucleus — instead they have a single circular strand of DNA that floats freely in the cytoplasm.

Bacteria don't have chloroplasts or mitochondria.

They may also contain one or more small rings of DNA called plasmids.

There's quite a bit to learn in biology — but that's life, I guess...

Make sure you're familiar with the structures of a typical animal cell, plant cell and bacterial cell. Check that you know what all the bits and pieces are by copying the diagrams and seeing if you can remember all the labels. No cheating.

Microscopy

Quick Quiz

Microscopes *are pretty important for biology. They let us see things that we* ***can't see*** *with the* ***naked eye****.*

Cells are **Studied** Using **Microscopes**

1) Microscopy techniques have developed as technology and knowledge have improved.
2) Light microscopes use light and lenses to form an image of a specimen and magnify it (make it look bigger). They let us see individual cells and large subcellular structures, like nuclei.
3) Electron microscopes use electrons to form an image. They have a much higher magnification than light microscopes. They also have a higher resolution.
4) They let us see much smaller things in more detail, like the internal structure of mitochondria and chloroplasts. They even let us see tinier things like ribosomes and plasmids.

Resolution is how well a microscope can distinguish between two very close points. A higher resolution gives a sharper image.

You Need to be Able to Use the **Formula** for **Magnification**

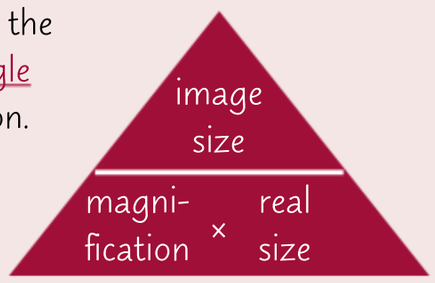
You can calculate the magnification of an image:

$$\text{magnification} = \frac{\text{image size}}{\text{real size}}$$

Image size and real size should have the same units. If they don't, you'll need to convert them first (see page 177).

If you want to work out the image size or the real size of the object, this formula triangle can be useful for rearranging the equation.

Cover up the thing you're trying to find. The parts you can still see are the formula you need to use.



EXAMPLE

A specimen is 50 µm wide. Calculate the width of the image of the specimen under a magnification of × 100. Give your answer in mm.

MATHS SKILLS

1) Rearrange the formula for image size by multiplying both sides by real size.
2) Fill in the values you know.
3) Remember the units in your answer.
4) Convert the units from µm to mm.

magnification = image size ÷ real size
image size = magnification × real size
image size = 100 × 50
= 5000 µm
= 5000 µm ÷ 1000 = **5 mm**

To convert from micrometres (µm) to millimetres, divide by 1000 (see p.177).

Standard Form Can Be Useful for Writing **Tiny Measurements**

1) This is where you change very big or small numbers with lots of zeros into something more manageable, e.g. 0.017 can be written 1.7×10^{-2}.
2) To do this you just need to move the decimal point left or right.
3) The number of places the decimal point moves is then represented by a power of 10 — this is positive if the decimal point's moved to the left, and negative if it's moved to the right.

EXAMPLE

A mitochondrion is 0.0025 mm long. Write this figure in standard form.

MATHS SKILLS

1) The first number needs to be between 1 and 10 so the decimal point needs to move after the '2'.
2) Count how many places the decimal point has moved — this is the power of 10. The decimal point has moved right so you need to add a minus sign.

0.0025

2.5×10^{-3}

Check the units used in that equation are both the same

That's the units for 'image size' and 'real size' — if they're not the same, you'll need to convert one of them.

Q1 A cheek cell is viewed under a microscope with × 40 magnification. The image of the cell is 2.4 mm wide. Calculate the real width of the cheek cell. Give your answer in µm. [2 marks]

Q1 Video Solution

More on Microscopy

PRACTICAL

Quick Quiz

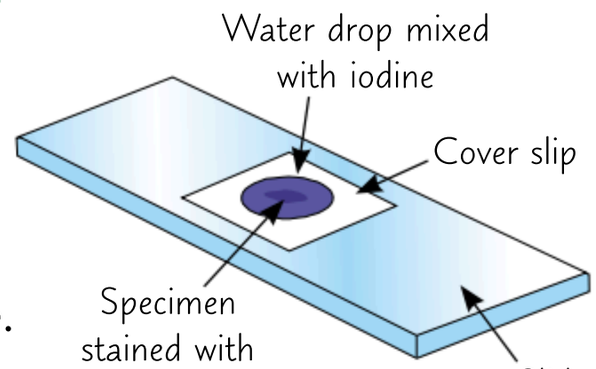
*It's all very well knowing what light microscopes **do** — you also have to know how to actually **use** one.*

You Need to Put Your **Specimen** On a **Slide**

A slide is a strip of clear glass or plastic onto which the specimen is mounted.

Here's how you could prepare a slide to view onion cells:

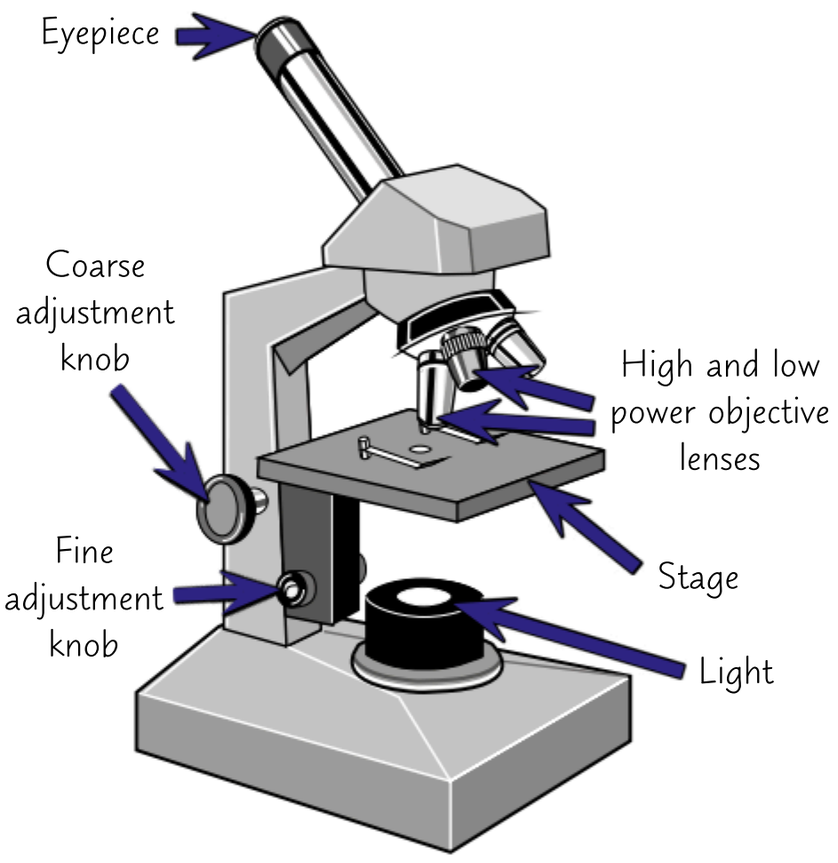
1) Add a drop of water to the middle of a clean slide.
2) Cut up an onion and separate it out into layers. Use tweezers to peel off some epidermal tissue from the bottom of one of the layers.
3) Using the tweezers, place the epidermal tissue into the water on the slide.
4) Add a drop of iodine solution. Iodine solution is a stain. Stains are used to highlight objects in a cell by adding colour to them.
5) Place a cover slip (a square of thin, transparent plastic or glass) on top. To do this, stand the cover slip upright on the slide, next to the water droplet. Then carefully tilt and lower it so it covers the specimen. Try not to get any air bubbles under there — they'll obstruct your view of the specimen.



Your specimen could be plant cells or animal cells.

Use a **Light Microscope** to Look at Your **Slide**

1) Clip the slide you've prepared onto the stage.
2) Select the lowest-powered objective lens (the one that produces the lowest magnification).
3) Use the coarse adjustment knob to move the stage up to just below the objective lens.
4) Look down the eyepiece. Use the coarse adjustment knob to move the stage down until the image is roughly in focus.
5) Adjust the focus with the fine adjustment knob, until you get a clear image of what's on the slide.
6) If you need to see the slide with greater magnification, swap to a higher-powered objective lens and refocus.

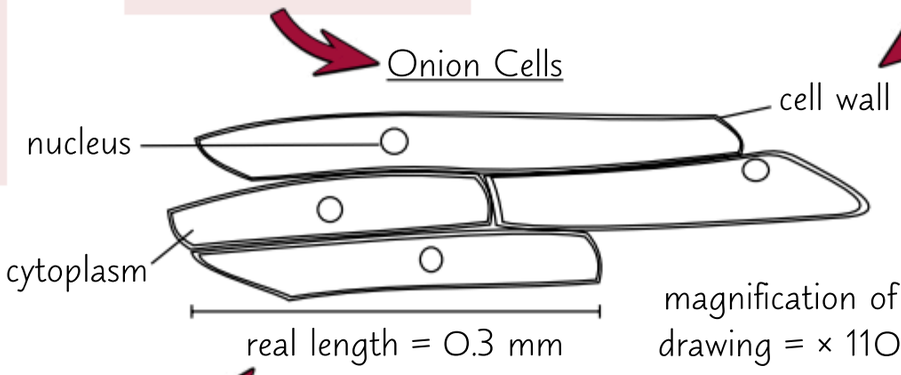


Draw Your Observations **Neatly** using a **Pencil** With a **Sharp Point**

Make sure your drawing isn't too small and that it's drawn with clear, unbroken lines.

Don't include any colouring or shading.

Include a title.



Label the important features (e.g. nucleus, cell wall), using straight, uncrossed lines.

Any subcellular structures should be drawn in proportion.

magnification of drawing = × 110

Write down the magnification of your drawing.

You can find the real size of a cell by counting the number of cells in 1 mm of the sample (see p.183).

magnification = length of drawing of cell ÷ real length of cell
So here, magnification = 33 mm ÷ 0.3 mm = x 110

Your microscope might look a bit different

The appearance of light microscopes can vary but they should all have the same basic features shown on this page.

Warm-Up & Exam Questions

So, hopefully you've read the last three pages. But could you cope if a question on cells or microscopes came up in the exam? With amazing new technology we can simulate that very situation...

Warm-Up Questions

1) Name the subcellular structures where aerobic respiration takes place.
2) Give three ways in which animal cells are different from plant cells.
3) Give two differences in structure between prokaryotic and eukaryotic cells.
4) What type of microscope should be used to look at the internal structure of chloroplasts?
5) Write the number 0.00045 μm in standard form.

Exam Questions

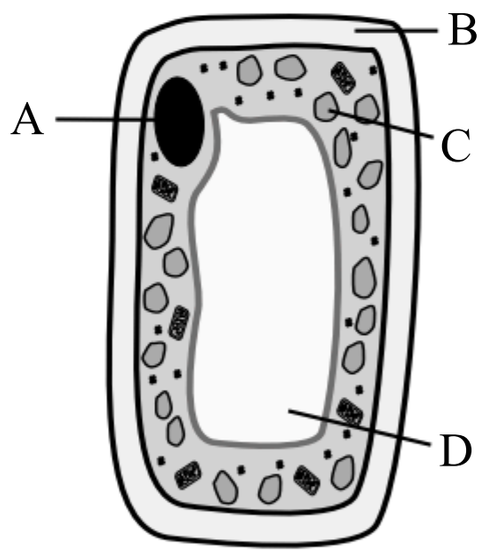
1 Which of the following subcellular structures would you not expect to find in a prokaryotic cell? Tick **one** box. GRADE 4-6 GRADE

☐ plasmid ☐ nucleus ☐ cell wall ☐ cell membrane

[1 mark]

2 **Figure 1** shows a typical plant cell. GRADE 4-6 GRADE

Figure 1



2.1 Which label points to a chloroplast? Tick **one** box.

☐ A ☐ B ☐ C ☐ D

[1 mark]

2.2 What is the function of a chloroplast?

[1 mark]

2.3 **Figure 1** also shows ribosomes.
What is the function of a ribosome?

[1 mark]

PRACTICAL

3 A light microscope can be used to observe a layer of onion cells on a slide. GRADE 6-7 GRADE

3.1* Describe how you would use a light microscope to view onion cells.
Include how you would prepare the slide.

[6 marks]

3.2 When the onion cell is viewed through the light microscope, the image of the cell is 7.5 mm wide.
The real width of the onion cell is 75 μm.
Calculate the magnification used to view the cell using the formula:

$$\text{magnification} = \frac{\text{image size}}{\text{real size}}$$

[2 marks]

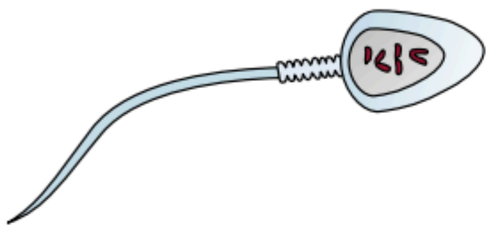
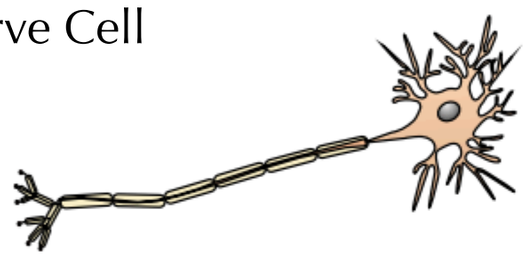
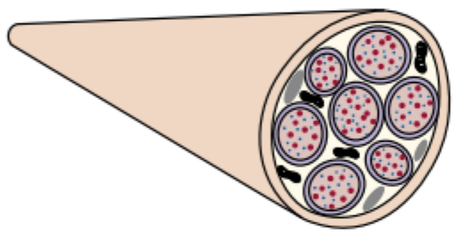
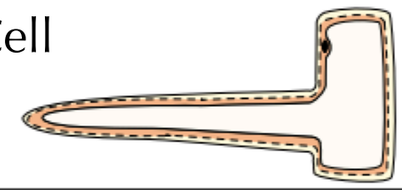
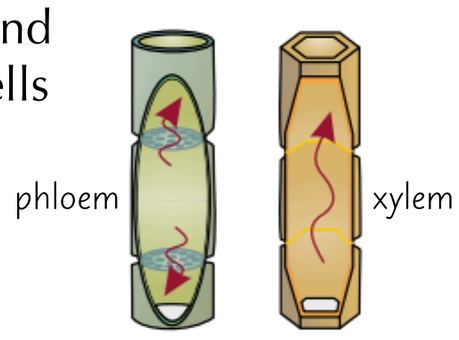
Quick Quiz

Cell Specialisation and Differentiation

*Cells **don't** all look the **same**. They have **different structures** to suit their **different functions**.*

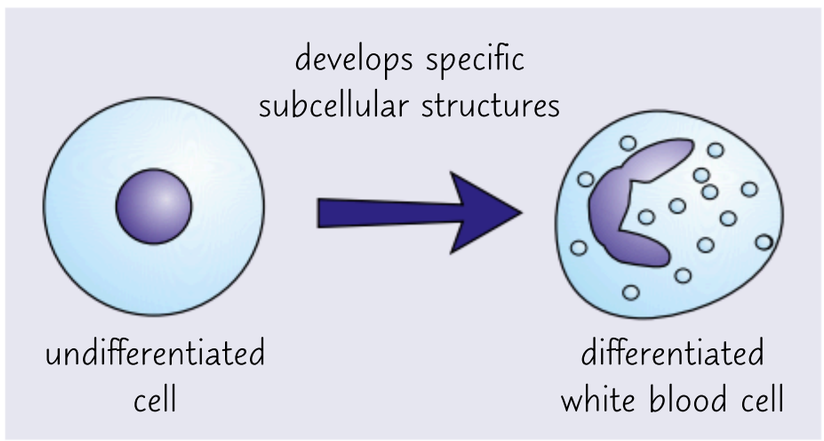
Specialised Cells Have Particular Jobs

Plant and animal cells don't all look like the ones on page 4. There are lots of different types of eukaryotic cells, all with particular shapes and subcellular structures that help them do specific jobs.

Specialised Cell	Function	Adaptations
Sperm Cell 	To get the male DNA to the female DNA.	A long tail and streamlined head to help it swim to the egg. Lots of mitochondria to provide the energy needed. Enzymes in its head to digest through the egg cell membrane.
Nerve Cell 	To carry electrical signals from one part of the body to another.	Long to cover more distance. Have branched connections at their ends to connect to other nerve cells and form a network throughout the body.
Muscle Cell 	To contract quickly.	Long so that they have space to contract. Contain lots of mitochondria to generate energy for contraction.
Root Hair Cell 	To absorb water and mineral ions from the soil.	They have long "projections" which give the roots a big surface area for absorption.
Phloem and Xylem Cells 	Form phloem and xylem tubes, which transport substances such as food and water around plants. There's more about phloem and xylem on p.49.	The cells are long and joined end to end. Xylem cells are hollow in the centre and phloem cells have very few subcellular structures, so that stuff can flow through them.

Cells Differentiate to Become Specialised

1) Differentiation is the process by which a cell changes to become specialised for carrying out its function.
2) Most differentiation occurs as an organism develops. In most animal cells, the ability to differentiate is then lost at an early stage, after they become specialised. However, lots of plant cells don't ever lose this ability.
3) The cells that differentiate in mature animals are mainly used for repairing and replacing cells, such as skin or blood cells.
4) Undifferentiated cells are called stem cells. There's more about them on the next page.



Cells have the same basic bits but are specialised for their function

Not all cells contain all of the subcellular structures that you learnt about on page 4. This is because some specialised cells don't have a use for certain subcellular structures — it depends on their function. For example, root hair cells grow underground in the soil, so they don't need chloroplasts because they don't photosynthesise.

Stem Cells

Quick Quiz

*Stem cell research has exciting **possibilities**, but it's also pretty **controversial**.*

Embryonic Stem Cells Can Turn into **ANY** Type of Cell

1) Undifferentiated cells, called stem cells, can divide to produce lots more undifferentiated cells. They can differentiate into different types of cell, depending on what instructions they're given.
2) Stem cells are found in early human embryos. They have the potential to turn into any kind of cell at all — all the different types of cell found in a human being have to come from those few cells in the early embryo. These stem cells are exciting to doctors and medical researchers (see below).
3) Adults also have stem cells, but only in certain places, like bone marrow. They can't turn into any cell type, only certain ones, such as blood cells.
4) Stem cells from embryos and bone marrow can be grown in a lab to produce clones (genetically identical cells) and made to differentiate into specialised cells to use in medicine or research.

Stem Cells are **Powerful** but **Controversial**

1) Medicine already uses adult stem cells to cure some diseases. E.g. stem cells transferred from the bone marrow of a healthy person can replace faulty blood cells in the patient who receives them.
2) Embryonic stem cells could also be used to replace faulty cells in sick people — you could make insulin-producing cells for people with diabetes, nerve cells for people paralysed by spinal injuries, etc.
3) In a type of cloning called therapeutic cloning, an embryo can be made with the same genetic information as the patient. Stem cells produced from this embryo would also share the same genes, so wouldn't be rejected by the patient's body if used to replace faulty cells.
4) However, because embryonic stem cells come from embryos, research is a tricky ethical issue:

There are risks in using stem cells in medicine, e.g. stem cells grown in the lab may become contaminated with a virus, which could be passed on to the patient.

Arguments against embryonic stem cell research	Arguments for embryonic stem cell research
Some people feel that human embryos shouldn't be used for experiments since each one is a human life or has the potential to be a human life.	Some people think that curing patients who are suffering is more important than the rights of embryos.
Some people feel that scientists should concentrate more on finding and developing other sources of stem cells, so people could be helped without having to use embryos.	Some people argue that the embryos used in the research are usually unwanted ones from fertility clinics which would otherwise probably just be destroyed.

Stem Cells Can Produce **Identical Plants**

In plants, stem cells are found in the meristems (parts of the plant where growth occurs — see p.48). Cells in the meristem tissues can differentiate into any type of plant cell through the plant's entire life. These stem cells can be used to produce clones (identical copies) of whole plants quickly and cheaply. This can be to:

- grow more plants of rare species (to prevent them being wiped out).
- grow crops of identical plants that have desired features for farmers, for example, disease resistance.

Stem cells have a lot of uses...

...so before turning the page, make sure that you know about these uses in case there's a question on them in the exam.

Chromosomes and Mitosis

Quick Quiz

*In order to survive and grow, our cells have got to be able to **divide**. And that means our DNA as well...*

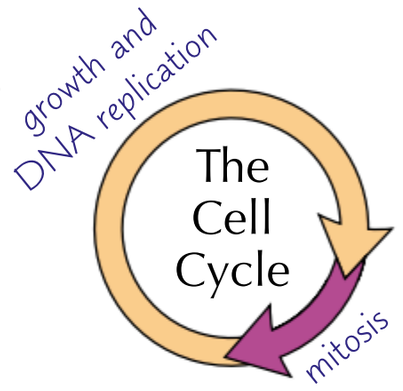
Chromosomes Contain Genetic Information

1) Most cells in your body have a nucleus. The nucleus contains your genetic material in the form of chromosomes. Chromosomes are coiled up lengths of DNA molecules.
2) Each chromosome carries a large number of genes. Different genes control the development of different characteristics, e.g. hair colour.
3) Body cells normally have two copies of each chromosome — one from the organism's 'mother', and one from its 'father'. So, humans have two copies of chromosome 1, two copies of chromosome 2, etc.

The Cell Cycle Makes New Cells

Video

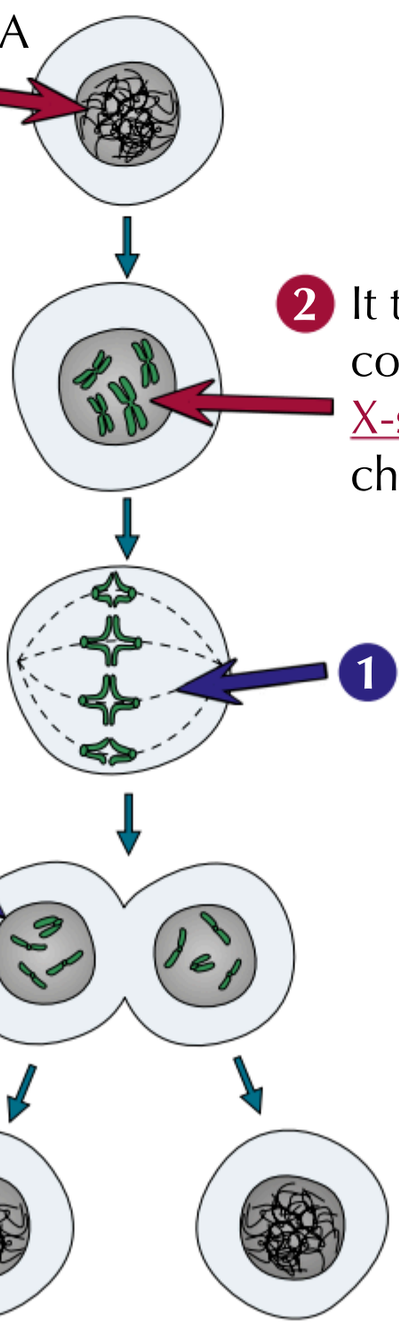
1) Body cells in multicellular organisms divide to produce new cells in a series of stages called the cell cycle.
2) The stage of the cell cycle when the cell divides is called mitosis.
3) Multicellular organisms use mitosis to grow or replace cells that have been damaged.
4) The end of the cell cycle results in two new cells identical to the original cell, with the same number of chromosomes.
5) You need to know about these two main stages of the cell cycle:



Growth & DNA Replication

1. In a cell that isn't dividing, the DNA is all spread out in long strings.
   Before it divides, the cell has to grow and increase the amount of subcellular structures such as mitochondria and ribosomes.
2. It then duplicates its DNA — so there's one copy for each new cell. The DNA forms X-shaped chromosomes, where each 'arm' of the chromosome is an exact duplicate of the other.

The left arm has the same DNA as the right arm of the chromosome.



Once its contents and DNA have been copied, the cell is ready for...

Mitosis

1. The chromosomes line up at the centre of the cell and cell fibres pull them apart. The two arms of each chromosome go to opposite ends of the cell.
2. Membranes form around each of the sets of chromosomes. These become the nuclei of the two new cells — the nucleus has divided.
3. Lastly, the cytoplasm and cell membrane divide.

The cell has now produced two new daughter cells. The daughter cells contain exactly the same DNA — they're identical. Their DNA is also identical to the parent cell.

Mitosis produces two identical daughter cells

Q1 Video Solution

Mitosis can seem tricky at first. But don't worry — just go through it slowly, one step at a time.

Q1 A student looks at cells in the tip of a plant root under a microscope. She counts 11 cells that are undergoing mitosis and 62 cells that are not.

a) Calculate the percentage of cells that are undergoing mitosis. [1 mark]

b) Suggest how the student can tell whether a cell is undergoing mitosis or not. [1 mark]

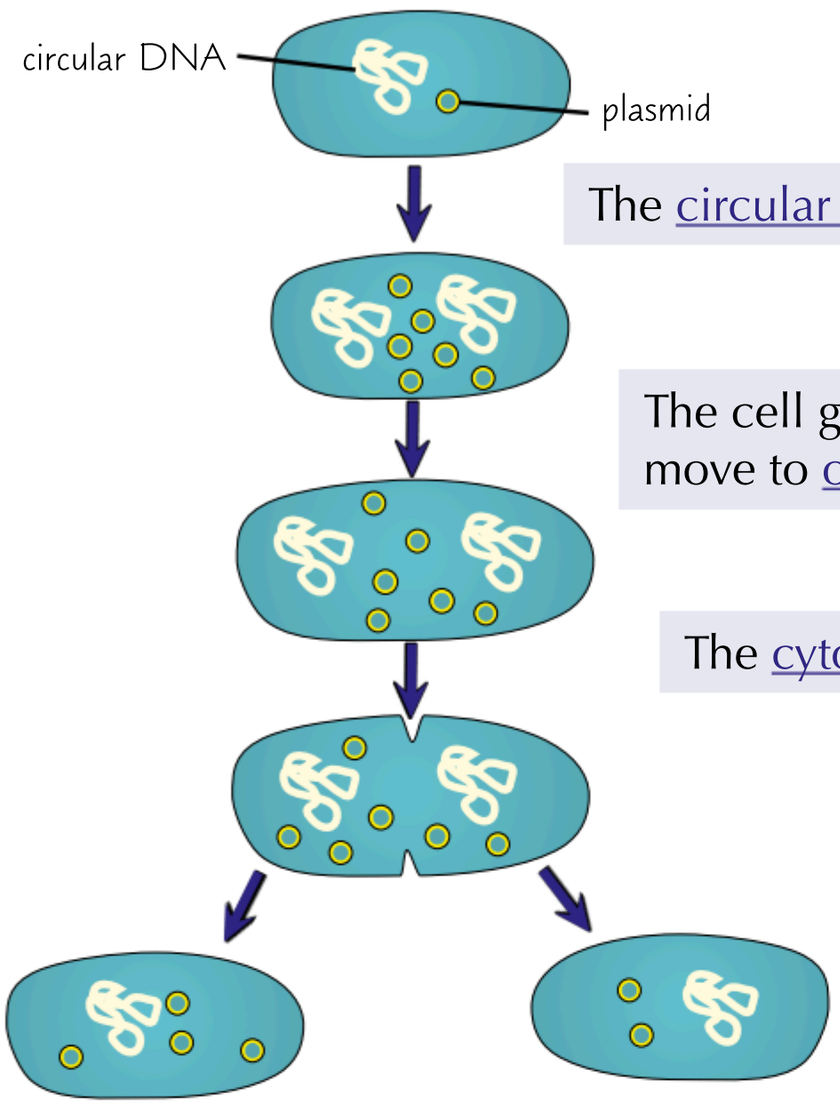
Binary Fission

Quick Quiz

Prokaryotic cells *can reproduce using a type of simple cell division called* ***binary fission***.

Prokaryotic Cells Replicate by **Binary Fission**

In binary fission, the cell... splits into two. There's a bit more to it than that though:



The circular DNA and plasmid(s) replicate.

The cell gets bigger and the circular DNA strands move to opposite 'poles' (ends) of the cell.

The cytoplasm begins to divide and new cell walls begin to form.

The cytoplasm divides and two daughter cells are produced. Each daughter cell has one copy of the circular DNA, but can have a variable number of copies of the plasmid(s).

Bacteria can divide very quickly if given the right conditions (e.g. a warm environment and lots of nutrients). Some bacteria, such as *E. coli*, can take as little as 20 minutes to replicate in the right environment. However, if conditions become unfavourable, the cells will stop dividing and eventually begin to die.

Use **Mean Division Time** to Find the **Number of Bacteria**

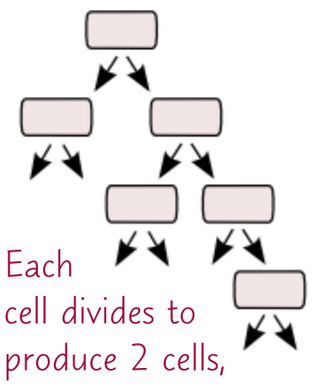
The mean division time is just the average amount of time it takes for one bacterial cell to divide into two. If you know the mean division time of a cell, you can work out how many times it has divided in a certain amount of time, and so the number of cells it has produced in that time.

EXAMPLE

A bacterial cell has a mean division time of 30 minutes. How many cells will it have produced after 2.5 hours?

MATHS SKILLS

1) Make sure both times are in the same units. 2.5 hours × 60 = 150 minutes.
2) Divide the total time that the bacteria are producing cells by the mean division time. This gives you the number of divisions.
   150 minutes ÷ 30 minutes = 5 divisions
3) Multiply 2 by itself for the number of divisions to find the number of cells.
   $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$ cells



Each cell divides to produce 2 cells, so the number of cells increases by a power of 2 for each division.

Binary = two parts, fission = splitting

Binary fission is the sort of subject where standard form could show up. Head back to p.5 if you're unsure.

Q2 Video Solution

Q1 Give two things that help to maximise the rate of binary fission. [2 marks]

Q2 *E. coli* is a type of bacteria. The mean division time of an *E. coli* cell is 20 minutes. How many cells will a single *E. coli* cell produce in 2 hours and 40 minutes? [2 marks]

Warm-Up & Exam Questions

There's only one way to do well in the exam — learn the facts and then practise lots of exam questions to see what it'll be like on the big day. We couldn't have made it easier for you — so do it.

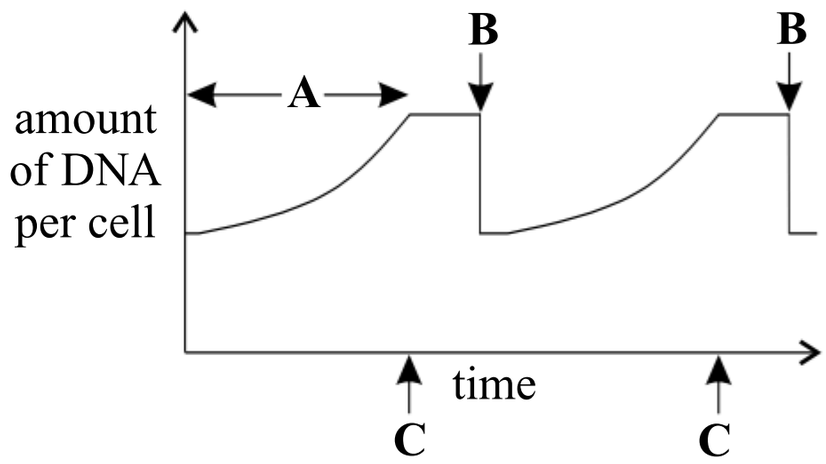
Warm-Up Questions

1) Describe how a root hair cell is specialised for its function.
2) What does cell 'differentiation' mean?
3) How can stem cells be used to preserve rare plant species?
4) Where in the cell are chromosomes found?
5) What type of cell replicates by binary fission — prokaryotic or eukaryotic?

Exam Questions

1 **Figure 1** shows how the amount of DNA per cell changes as a cell undergoes two cell divisions by mitosis. Point **C** is the time when the chromosomes first become visible in the cells.

Figure 1



GRADE 6-7

1.1 Describe what is happening to the DNA during stage **A**. Suggest why this needs to happen.

[2 marks]

1.2 Suggest what happens at time **B**.

[1 mark]

1.3 How many cells are there after the first cell division?

[1 mark]

2 Adult stem cells have already been used to cure disorders, and it is thought that embryonic stem cells have the potential to treat many more disorders.

GRADE 6-7

2.1 Describe how stem cells could be used to treat disorders.

[1 mark]

2.2 Explain why embryonic stem cells have the potential to treat more disorders than adult stem cells.

[1 mark]

2.3 Give **one** place where adult stem cells are found in the body.

[1 mark]

3 A bacterial cell has a mean division time of 30 minutes. How many cells will it have produced in 3 hours?

GRADE 7-9

[2 marks]

PRACTICAL

Culturing Microorganisms

Quick Quiz

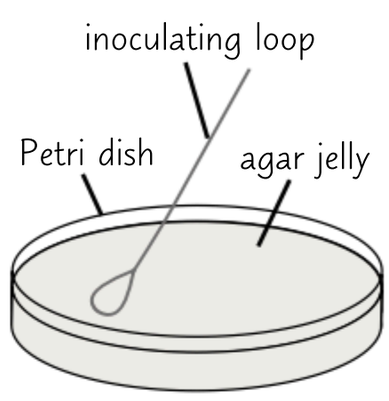
*And now for some hands-on stuff. You can **grow your own** microorganisms and test how **effective** different **antibiotics**, **antiseptics** or **disinfectants** are at killing them...*

You Can **Grow Bacteria** in the Lab

1) Microorganisms such as bacteria are grown (cultured) in a "culture medium", containing the proteins, carbohydrates, minerals and vitamins they need. It can be a nutrient broth solution or solid agar jelly.

2) Bacteria grown on agar 'plates' will form visible colonies on the surface of the jelly, or will spread out to give an even covering of bacteria. To make an agar plate:

- Hot agar jelly is poured into shallow round plastic dishes called Petri dishes.
- When the jelly's cooled and set, inoculating loops (wire loops) can be used to transfer microorganisms to the culture medium. Alternatively, a sterile dropping pipette and spreader can be used to get an even covering of bacteria.
- The microorganisms then multiply.



3) In the lab at school, cultures of microorganisms are not kept above 25 °C, because harmful pathogens are more likely to grow above this temperature.

Pathogens are microorganisms that cause disease (see p.54).

4) In industrial conditions, cultures are incubated at higher temperatures so that they can grow a lot faster.

You Can Investigate the **Effect** of **Antibiotics** on **Bacterial Growth**

Here's a way of testing the action of antibiotics (or antiseptics) on cultures of bacteria:

Method

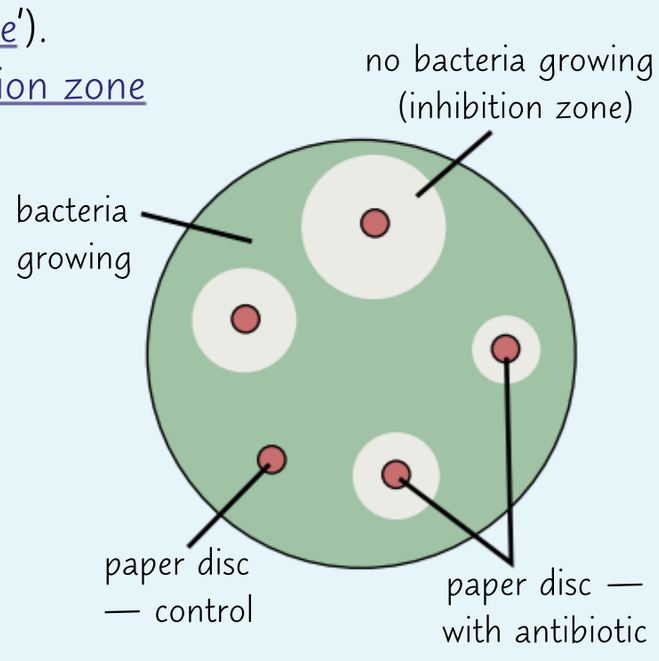
1) Place paper discs soaked in different types (or different concentrations) of antibiotics on an agar plate that has an even covering of bacteria. Leave some space between the discs. The antibiotic should diffuse into the agar jelly.

2) Leave the plate for 48 hours at 25 °C.

3) Anywhere the bacteria can't grow is called an inhibition zone (or a 'clear zone'). The more effective the antibiotic is against the bacteria, the larger the inhibition zone around the paper disc will be — see the next page.

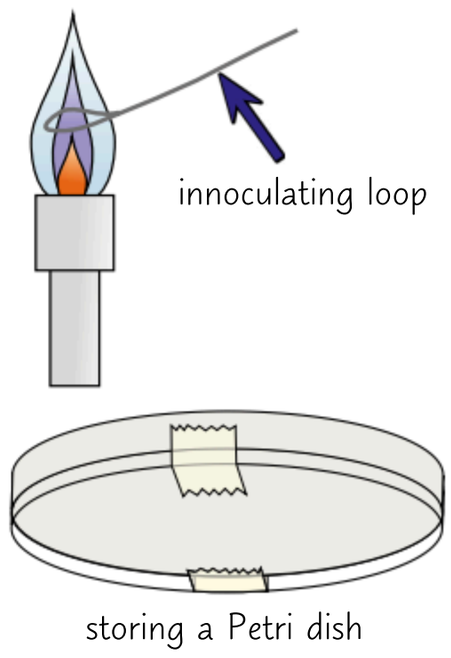
4) Antibiotic-resistant bacteria (bacteria that aren't affected by the antibiotic — see p.61) will continue to grow on the agar around the paper discs.

5) Make sure you use a control. This is a paper disc that has not been soaked in an antibiotic. Instead, soak it in sterile water. You can then be sure that any difference between the growth of the bacteria around the control disc and around the antibiotic discs is due to the effect of the antibiotic alone (and not something weird in the paper, for example).



Contamination by unwanted microorganisms will affect your results and can potentially result in the growth of pathogens. To avoid this:

- The Petri dishes and culture medium must be sterilised before use (e.g. by heating to a high temperature), to kill any unwanted microorganisms that may be lurking on them.
- If an inoculating loop is used to transfer the bacteria to the culture medium, it should be sterilised first by passing it through a hot flame.
- After transferring the bacteria, the lid of the Petri dish should be lightly taped on — to stop microorganisms from the air getting in.
- The Petri dish should be stored upside down — to stop drops of condensation falling onto the agar surface.



storing a Petri dish

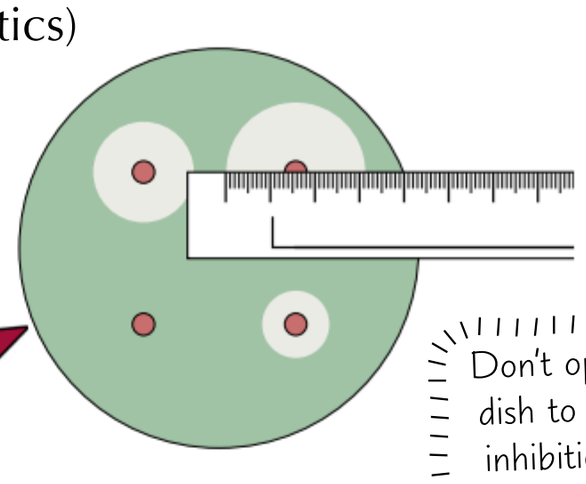
Culturing Microorganisms

PRACTICAL

Calculate the Sizes of the Inhibition Zones to Compare Results

You can compare the effectiveness of different antibiotics (or antiseptics) on bacteria by looking at the relative sizes of the inhibition zones. The larger the inhibition zone around a disc, the more effective the antibiotic is against the bacteria.

You can do this by eye if there are large differences in size. To get more accurate results though, calculate the area of the inhibition zones using their diameter (the distance across).



Don't open the Petri dish to measure the inhibition zones — they should be visible through the bottom of the dish.

To calculate the area of an inhibition zone, you need to use this equation:

This is the equation for the area of a circle. You're likely to use the units cm^2 or mm^2.

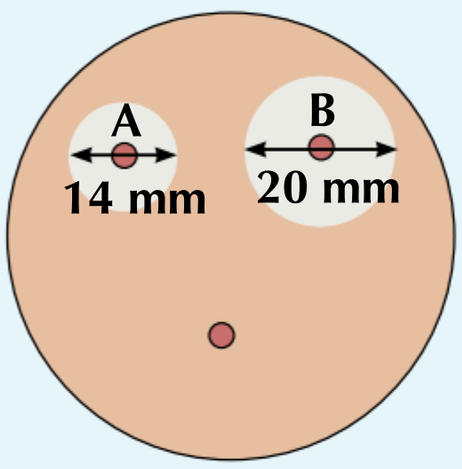
$$\text{Area} = \pi r^2$$

r is the radius of the inhibition zone — it's equal to half the diameter.

π is just a number. You should have a button for it on your calculator. If not, just use the value 3.14.

EXAMPLE

The diagram below shows the inhibition zones produced by antibiotics A and B. Use the areas of the inhibition zones to compare the effectiveness of the antibiotics.

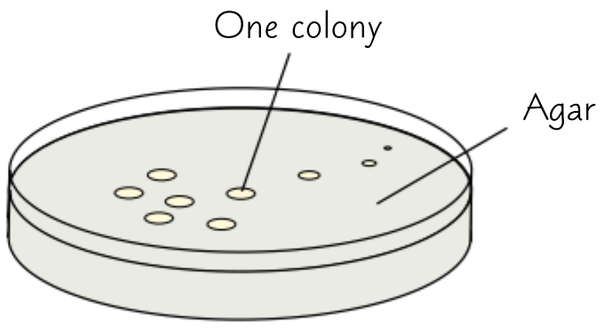


1) Divide the diameter of zone A by two to find the radius.
   Radius of A = 14 ÷ 2 = 7 mm
2) Stick the radius value into the equation area = πr^2.
   Area of A = π × 7^2 = 153.938...
   = 154 mm^2
3) Repeat steps 1 and 2 for zone B.
   Radius of B = 20 ÷ 2 = 10 mm
   Area of B = π × 10^2 = 314.159...
   = 314 mm^2
4) Compare the sizes of the areas. 314 mm^2 is just over twice 154 mm^2, so you could say that:
   The inhibition zone of antibiotic B is roughly twice the size of the inhibition zone of antibiotic A.

MATHS SKILLS

You Can Also Find the Area of a Colony

The equation above can also be used to calculate the area of a bacterial colony. You just need to measure the diameter of the colony you are interested in first.



The greater the area, the more effective the antibiotic

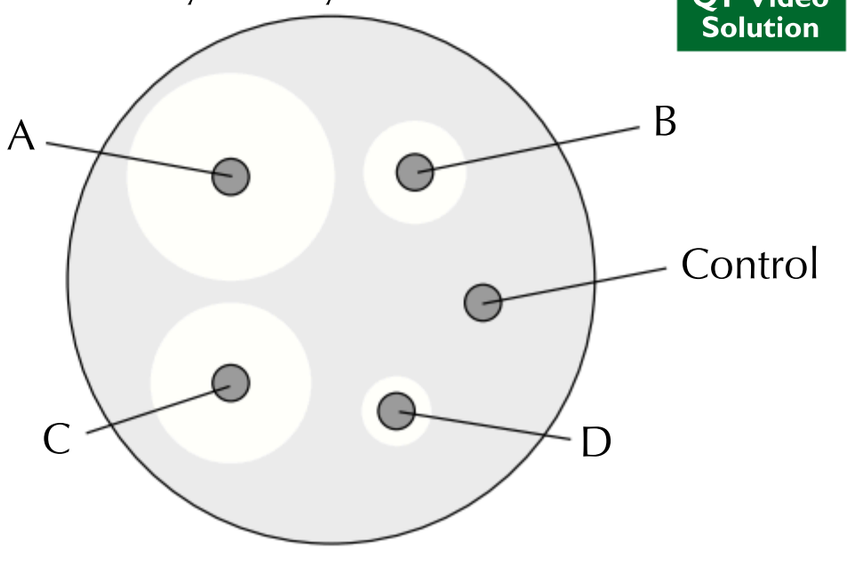
One of the main things to think about when setting up this practical is how to keep everything completely sterile. If you don't, your agar plate will get contaminated and your results won't tell you very much.

Q1 Helena investigated the effect of four different concentrations (A-D) of an antiseptic on the growth of bacteria. The diagram on the right shows the results.

a) Suggest which concentration of antiseptic was the strongest. [1 mark]

b) Calculate the size of the inhibition zone for Concentration C. Give your answer to the nearest mm^2. [2 marks]

c) Suggest what Helena used as a control in this investigation. [1 mark]



Q1 Video Solution

PRACTICAL

Warm-Up & Exam Questions

Just like bacteria, this topic can grow on you. Here are some questions to enjoy now, you lucky thing...

Warm-Up Questions

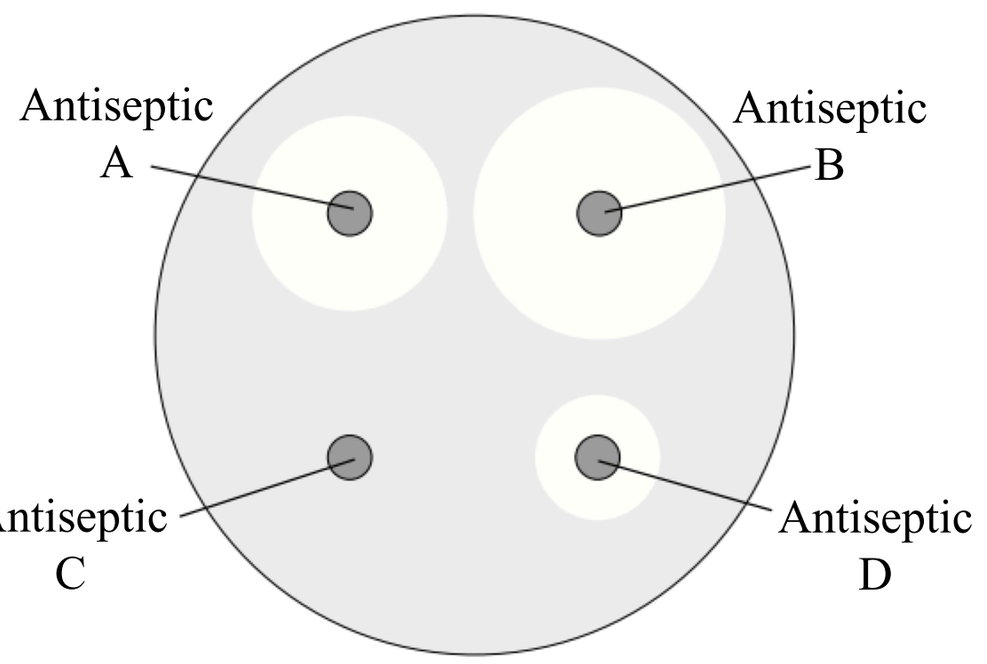
1) What lab equipment can be used to transfer microorganisms to an agar jelly culture medium?
2) Why is it important to sterilise Petri dishes and culture medium before use?

Exam Questions

1 A researcher was investigating the effect of different antiseptics on the growth of bacteria. He soaked four paper discs in different antiseptics. **Figure 1** shows the results.

GRADE (6-7) GRADE

Figure 1



1.1 Which antiseptic was most effective against the bacteria?

[1 mark]

1.2 Explain why the researcher should have used a control.

[1 mark]

2 A student was investigating the effectiveness of different antibiotics on a strain of bacteria. The student placed paper discs soaked in the different antibiotics on an agar plate that had an even covering of the bacteria. The results are shown in **Table 1**.

GRADE (6-7) GRADE

Table 1

Antibiotic	X	Y	Z
Diameter of inhibition zone (mm)	15	9	0

2.1 Calculate the area of the inhibition zone for antibiotic X. Give your answer in mm^2.
Use the equation: area = πr^2

[2 marks]

2.2 No inhibition zone was formed around the disc soaked in antibiotic Z.
Suggest a reason why.

[1 mark]

Revision Summary Test for Topic 1a

I hope you've absorbed all of that lovely knowledge. To make sure it sticks, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Types of Cell and Microscopy (p.4-6) ☑

1) How does the size of a bacterial cell compare to an animal cell? ☑
2) Give five subcellular structures found in both animal and plant cells. What are their functions? ☑
3) Give three subcellular structures found in plant cells but not in animal cells. What are their functions? ☑
4) Describe two subcellular structures that are present in prokaryotic cells but not eukaryotic cells. ☑
5) How have electron microscopes been able to increase our understanding of subcellular structures? ☑
6) What formula is used to calculate the magnification of an image? ☑
7) Outline how you would prepare a sample of onion cells to view on a slide. ☑
8) Outline how you would use a light microscope to view a slide. ☑
9) What rules should you follow when you're drawing observations of cells through a microscope? ☑

Differentiation and Stem Cells (p.8-9) ☑

10) Give three ways that a sperm cell is adapted for getting the male DNA to the female DNA. ☑
11) What does a nerve cell do? And how is it adapted for this? ☑
12) Describe three more examples of cells that are specialised to carry out a particular function. How are they adapted to do these jobs? ☑
13) Name the process by which cells become specialised. When does this mostly occur in animal cells? ☑
14) What is a stem cell? ☑
15) What is the difference between embryonic stem cells and adult stem cells in animals? ☑
16) Give two examples of ways that embryonic stem cells could be used to treat medical conditions. ☑
17) Give one risk of using stem cells in medicine. ☑
18) What is meant by therapeutic cloning? Why is it useful for medical treatment with stem cells? ☑
19) For embryonic stem cell research, outline: a) two arguments for, and b) two arguments against. ☑
20) Where are stem cells found in plants? How are they different to human stem cells? ☑
21) Give two uses of stem cells from plants. ☑

Cell Division (p.10-11) ☑

22) What are chromosomes? ☑
23) What is mitosis used for in multicellular organisms? ☑
24) Briefly describe the two main stages involved in the cell cycle. Include the end result. ☑
25) What is the name of the process by which bacteria divide? What conditions are best for this? ☑

Culturing Microorganisms (p.13-14) ☑

26) What is the maximum temperature that microorganisms should be grown at in a school lab? Why shouldn't a temperature above this be used? ☑
27) Outline a method to test the action of different antibiotics on bacteria. What would you observe? ☑
28) How can you ensure an experiment on the effect of antibiotics on bacteria doesn't get contaminated? ☑
29) What's the formula for calculating the area of an inhibition zone? ☑

Diffusion

Quick Quiz

***Diffusion** is **really important** in living organisms — it's how a lot of **substances** get **in** and **out** of cells.*

Diffusion is About the **Movement** of **Particles**

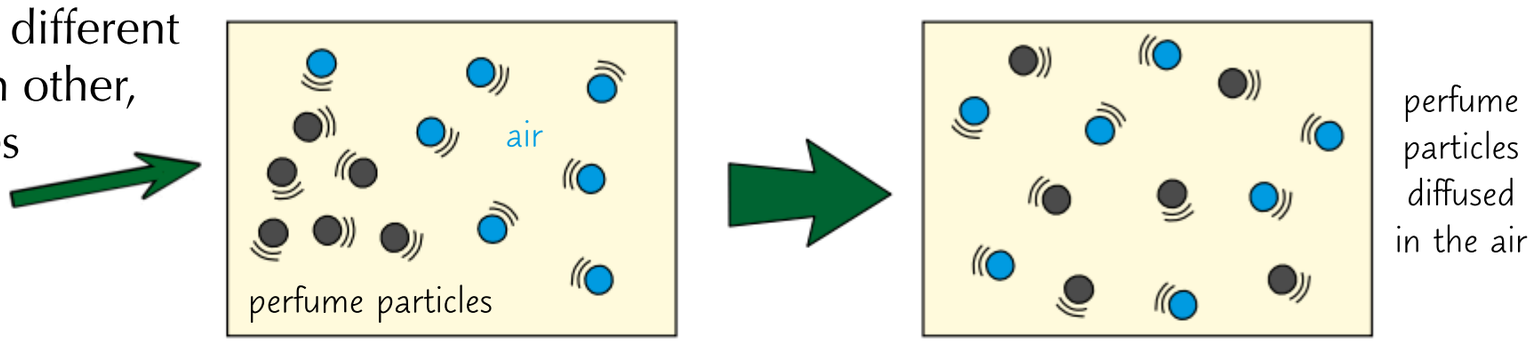
1) Diffusion is the gradual movement of particles from places where there are lots of them to places where there are fewer of them — it's just the natural tendency for stuff to spread out.

2) You also have to learn the fancy way of saying the same thing:

> Diffusion is the spreading out of particles from an area of higher concentration to an area of lower concentration.

3) Diffusion happens in both solutions and gases because the particles in these substances are free to move about randomly.

4) The simplest type is when different gases diffuse through each other, e.g. when perfume diffuses through the air in a room:



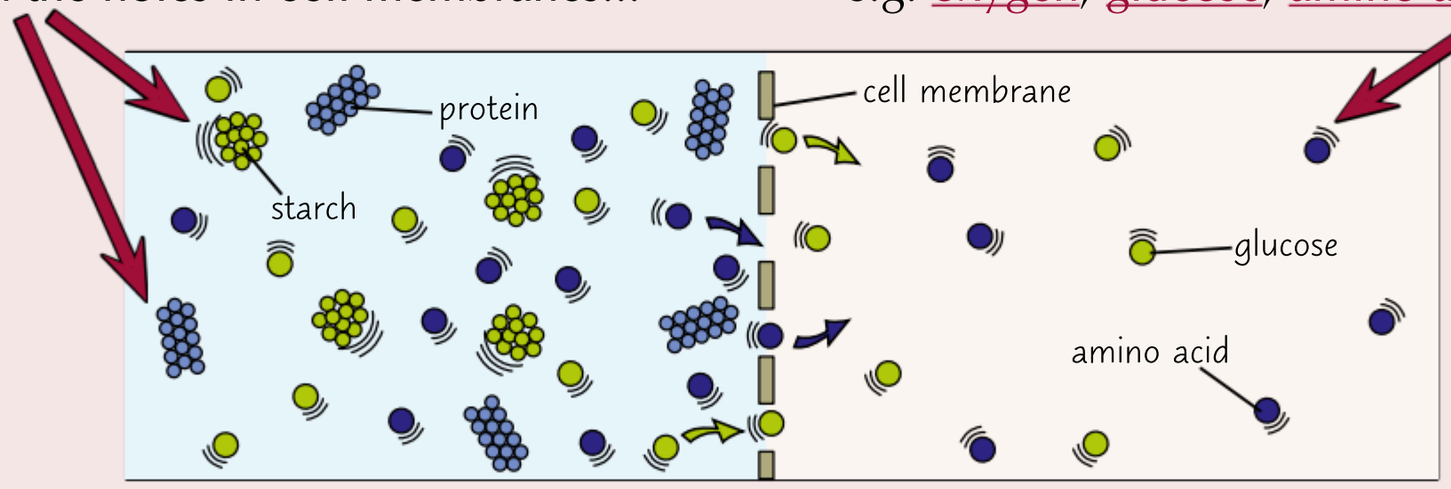
- The bigger the concentration gradient (the difference in concentration), the faster the diffusion rate.
- A higher temperature will also give a faster diffusion rate because the particles have more energy, so move around faster.

Cell Membranes Hold the Cell **Together**...

Video

...BUT they let stuff in and out as well. This is because they are partially permeable membranes — they have very small holes. Dissolved substances can move in and out of cells by diffusion:

1) Big molecules like starch and proteins can't fit through the holes in cell membranes... ...but very small molecules can diffuse through — e.g. oxygen, glucose, amino acids and water.



2) Particles flow from where there's a higher concentration (a lot of them)... ...to where there's a lower concentration (not such a lot of them).

3) The particles are only moving about randomly, so they go both ways — but if there are a lot more particles on one side of the membrane, there's a net (overall) movement from that side.

> The larger the surface area of the membrane, the faster the diffusion rate, because more particles can pass through at once — see page 20 for more on why this is important.

Diffusion is just particles spreading out

Q1 Video Solution

Remember, really tiny particles can go through cell membranes to even up the concentration on either side.

Q1 A student adds a drop of ink to a glass of cold water.

a) What will the student observe to happen to the drop of ink. Explain your answer. [2 marks]

b) How might the observation differ if the ink was added to a glass of warm water? [1 mark]

Osmosis

Quick Quiz

*If you've got the hang of **diffusion**, osmosis will be a **breeze**. If not, have another read of the previous page.*

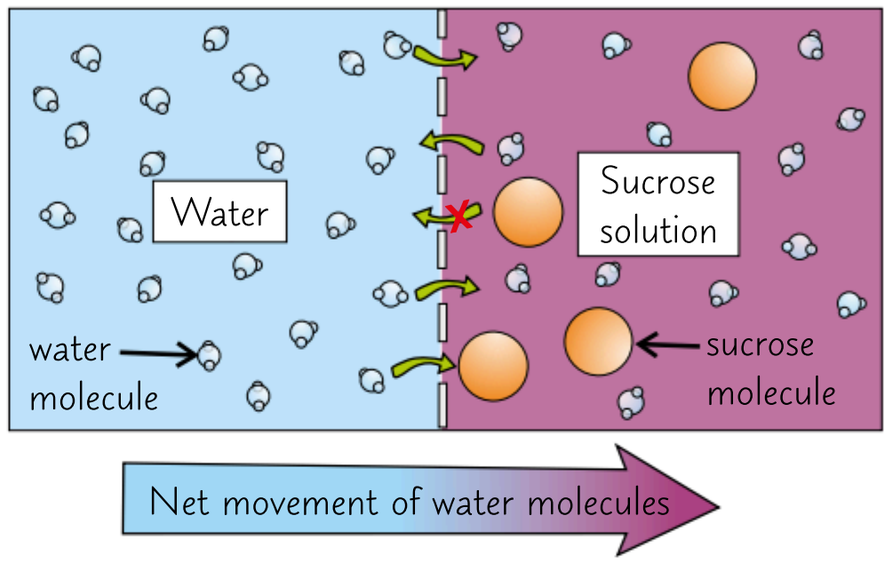
Osmosis is a Special Case of Diffusion, That's All

Video

Osmosis is the movement of water molecules across a partially permeable membrane from a more dilute solution to a more concentrated solution.

More dilute just means less concentrated.

1) Water can pass through the holes in a partially permeable membrane but bigger molecules (e.g. sucrose) can't.
2) The water molecules actually pass both ways through the membrane during osmosis. This happens because water molecules move about randomly all the time.
3) But overall, there is a steady net flow of water from the less concentrated solution (where there are lots of water molecules) to the more concentrated solution (where there are fewer water molecules).
4) This means the more concentrated solution gets more dilute. The water acts like it's trying to "even up" the concentration either side of the membrane.



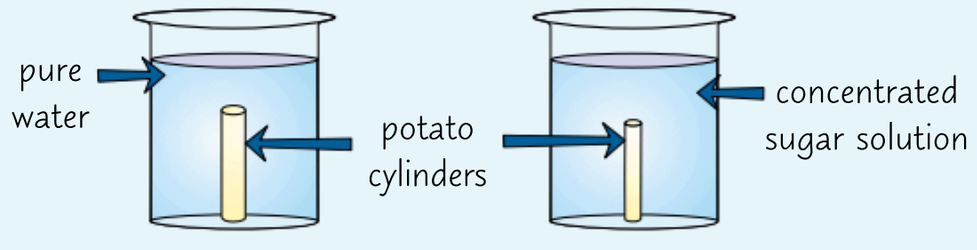
You can Observe How Sugar Solutions Affect Plant Tissue

PRACTICAL

This is just one way of showing osmosis at work. Don't worry if you see salt being used rather than sugar.

Method

1) Cut up a potato into identical cylinders, and get some beakers with different sugar solutions in them. One should be pure water and another should be a very concentrated sugar solution (e.g. 1 mol/dm^3). Then you can have a few others with concentrations in between (e.g. 0.2 mol/dm^3, 0.4 mol/dm^3, 0.6 mol/dm^3, etc.).
2) Measure the mass of the cylinders, then leave one cylinder in each beaker for twenty-four hours.
3) After that time, take them out, dry them with a paper towel and measure their masses again.



4) If the cylinders have drawn in water by osmosis, they'll have increased in mass. If water has been drawn out, they'll have decreased in mass. You can calculate the percentage change in mass, then plot a few graphs.

By calculating the percentage change (see p.185), you can compare the effect of sugar concentration on cylinders that didn't have the same initial mass. An increase in mass will give a positive percentage change and a decrease will give a negative percentage change.

- The dependent variable is the cylinder mass and the independent variable is the concentration of the sugar solution. All other variables (volume of solution, temperature, time, etc.) must be kept the same or the experiment won't be a fair test. See page 172 for more on variables.
- Like any experiment, you need to be aware of errors (see p.173). For example, if some potato cylinders were not fully dried, the excess water would give a higher mass, or if water evaporated from the beakers, the concentrations of the sugar solutions would change. You can reduce the effect of these errors by repeating the experiment and calculating a mean percentage change at each concentration.

Osmosis = diffusion of water molecules

Just remember, osmosis is just a fancy word for the diffusion of water molecules across a partially permeable membrane.

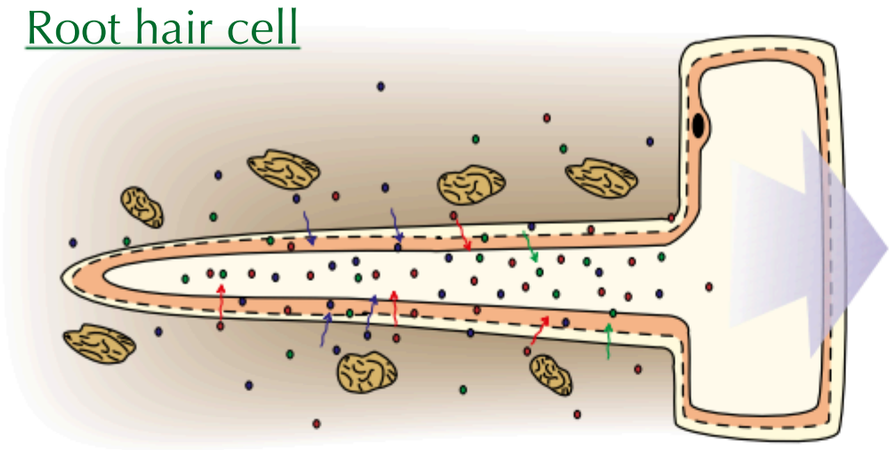
Active Transport

Quick Quiz

Sometimes substances need to be absorbed against a concentration gradient, i.e. from a lower to a higher concentration. This process is referred to as ***active transport****.*

Root Hairs Take In **Minerals** and **Water**

Root hair cell



1) As you saw on page 8, the cells on plant roots have projections which stick out into the soil.
2) Each branch of a root will be covered in millions of these microscopic hairs.
3) This gives the plant a large surface area for absorbing water and mineral ions from the soil.
4) Plants need these mineral ions for healthy growth.

The concentration of minerals is usually higher in the root hair cells than in the soil around them. So the root hair cells can't use diffusion to take up minerals from the soil...

- Minerals should move out of the root hairs if they followed the rules of diffusion. The cells must use another method to draw them in.
- That method is a conveniently mysterious process called "active transport".
- Active transport allows the plant to absorb minerals from a very dilute solution, against a concentration gradient. This is essential for its growth.
- But active transport needs ENERGY from respiration (see p.78) to make it work.

Water is taken into root hair cells by osmosis (see page 18).

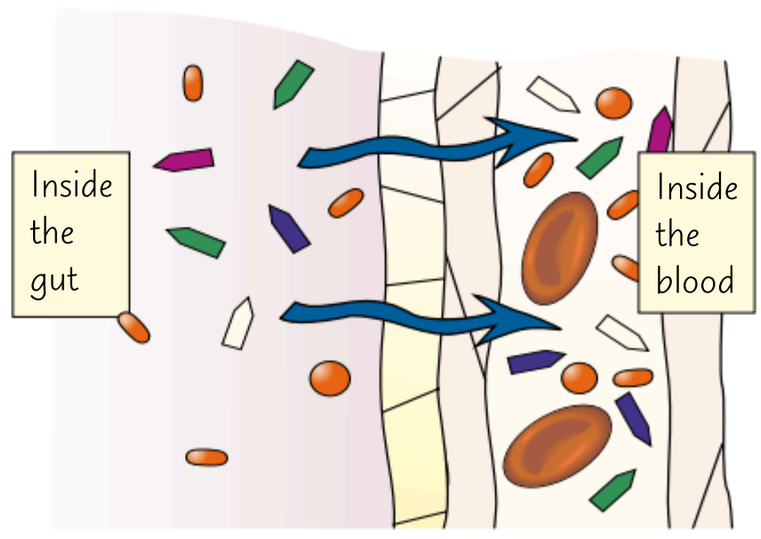
We Need **Active Transport** to Stop Us Starving

Video

Active transport also happens in humans, for example, in the gut:

1) When there's a higher concentration of glucose and amino acids in the gut than in the blood, they diffuse naturally into the blood.
2) BUT — sometimes there's a lower concentration of nutrients in the gut than there is in the blood. This means that the concentration gradient is the wrong way.
3) Active transport is used to allow nutrients to be taken into the blood, despite the fact that the concentration gradient is the wrong way.

For example, glucose can be taken into the bloodstream when its concentration in the blood is already higher than in the gut. It can then be transported to cells, where it's used for respiration.



Active transport uses energy

An important difference between active transport and diffusion is that active transport uses energy. Another key difference between these processes is that active transport allows for absorption against a concentration gradient.

Exchange Surfaces

Quick Quiz

*How easily stuff **moves** between an **organism** and its **environment** depends on its **surface area to volume ratio**.*

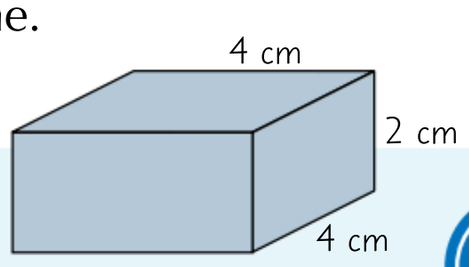
Organisms **Exchange Substances** with their **Environment**

Cells can use diffusion to take in substances they need and get rid of waste products. For example:

- Oxygen and carbon dioxide are transferred between cells and the environment during gas exchange.
- In humans, urea (a waste product produced from the breakdown of proteins, see p.99) diffuses from cells into the blood plasma for removal from the body by the kidneys.

How easy it is for an organism to exchange substances with its environment depends on the organism's surface area to volume ratio (SA : V).

A ratio shows how big one value is compared to another.
The larger an organism is, the smaller its surface area is compared to its volume.
You can show this by calculating surface area to volume ratios:



EXAMPLE

A hippo can be represented by a 2 cm × 4 cm × 4 cm block.

1) The area of a surface is found by the equation: LENGTH × WIDTH
Use that to calculate total surface area.
Total surface area:
(4 × 4) × 2 (top and bottom surfaces of block)
+ (4 × 2) × 4 (four sides of the block) = 64 cm^2

2) The volume of a block is found by the equation: LENGTH × WIDTH × HEIGHT
Volume: 4 × 4 × 2 = 32 cm^3

3) Use the surface area and volume to write a ratio.
Surface area to volume ratio: 64 : 32

4) Simplify the ratio by dividing both sides of the ratio by the volume.
64 ÷ 32 = 2. 32 ÷ 32 = 1.
So the surface area to volume ratio of the hippo is 2 : 1

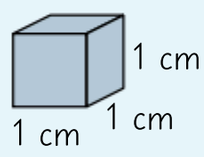
MATHS SKILLS

A mouse can be represented by a 1 cm × 1 cm × 1 cm block.

1) Calculate the surface area.
Total surface area: (1 × 1) × 6 = 6 cm^2

2) Calculate the volume.
Volume: 1 × 1 × 1 = 1 cm^3

3) Write the surface area to volume ratio.
Surface area to volume ratio: 6 : 1



The cube mouse's surface area is six times its volume, but the cube hippo's surface area is only twice its volume. So the mouse has a larger surface area compared to its volume.

Multicellular Organisms Need **Exchange Surfaces**

1) In single-celled organisms, gases and dissolved substances can diffuse directly in and out of the cell across the cell membrane. They have a large surface area compared to their volume, so enough substances can be exchanged across the membrane to supply the volume of the cell.
2) Multicellular organisms have a smaller surface area compared to their volume — not enough substances can diffuse from their outside surface to supply their entire volume. They need some sort of exchange surface for efficient diffusion (see pages 21-22 for some examples).
3) Exchange surfaces are adapted to maximise effectiveness:
   - They have a thin membrane, so substances only have a short distance to diffuse.
   - They have a large surface area so lots of a substance can diffuse at once.
   - Exchange surfaces in animals have lots of blood vessels, to get stuff in and out of the blood quickly.
   - Gas exchange surfaces in animals (e.g. alveoli) are often ventilated too — air moves in and out.

Surface area to volume ratios crop up a lot in biology

Remember, generally speaking, a smaller object has a larger surface area to volume ratio than a bigger object.

Q1 A bacterial cell can be represented by a 2 μm × 2 μm × 1 μm block.
Calculate the cell's surface area to volume ratio. [3 marks]

Q1 Video Solution

Exchanging Substances

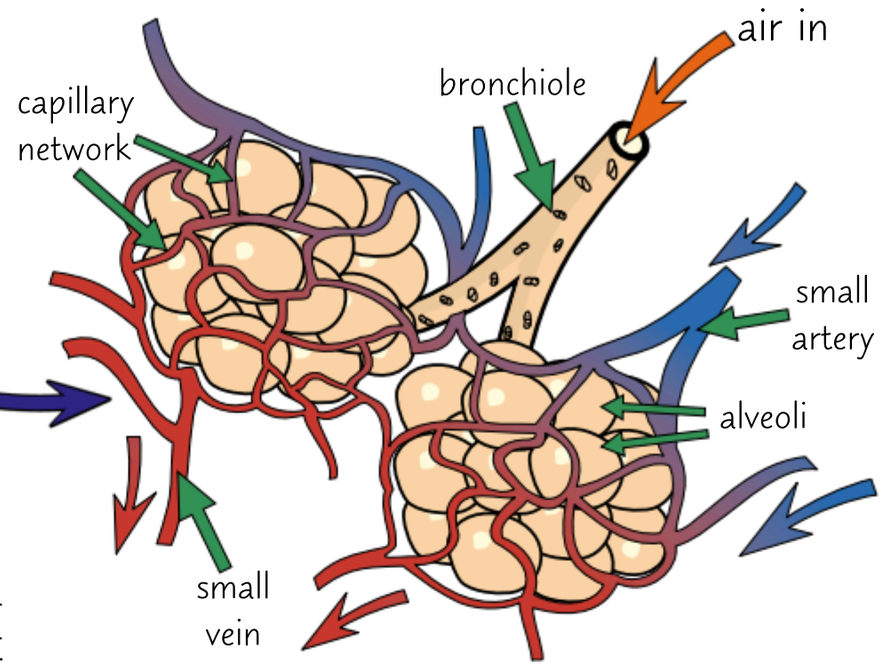
Quick Quiz

*Here's how **gases** in the lungs get **into** and **out** of the **blood** and how **food** gets from the **gut to the blood**.*

Gas Exchange Happens in the **Lungs**

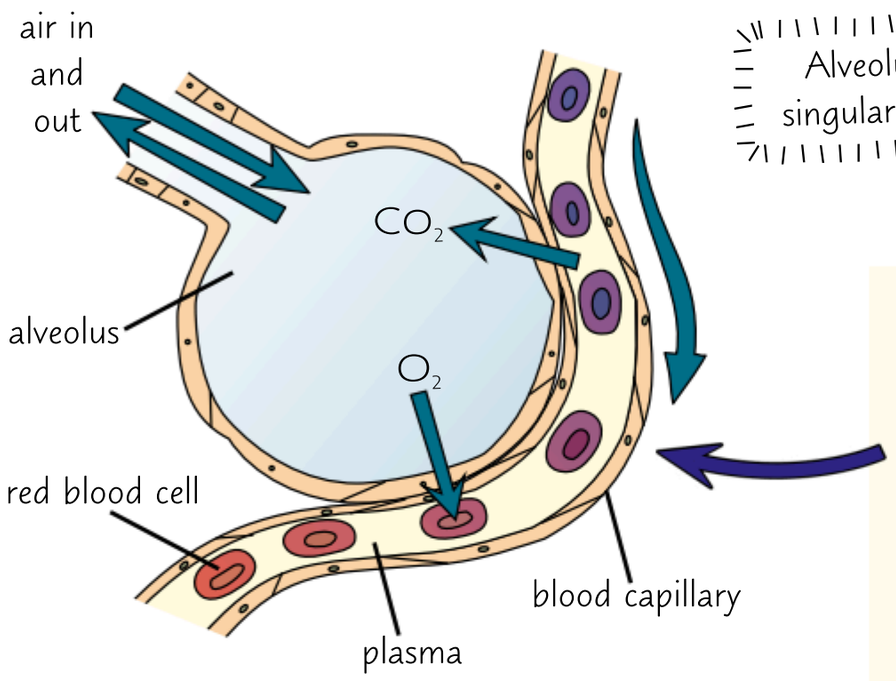
The job of the lungs is to transfer oxygen to the blood and to remove waste carbon dioxide from it.

To do this, the lungs contain millions of little air sacs called alveoli where gas exchange takes place.



Blue = blood with carbon dioxide.

Red = blood with oxygen.



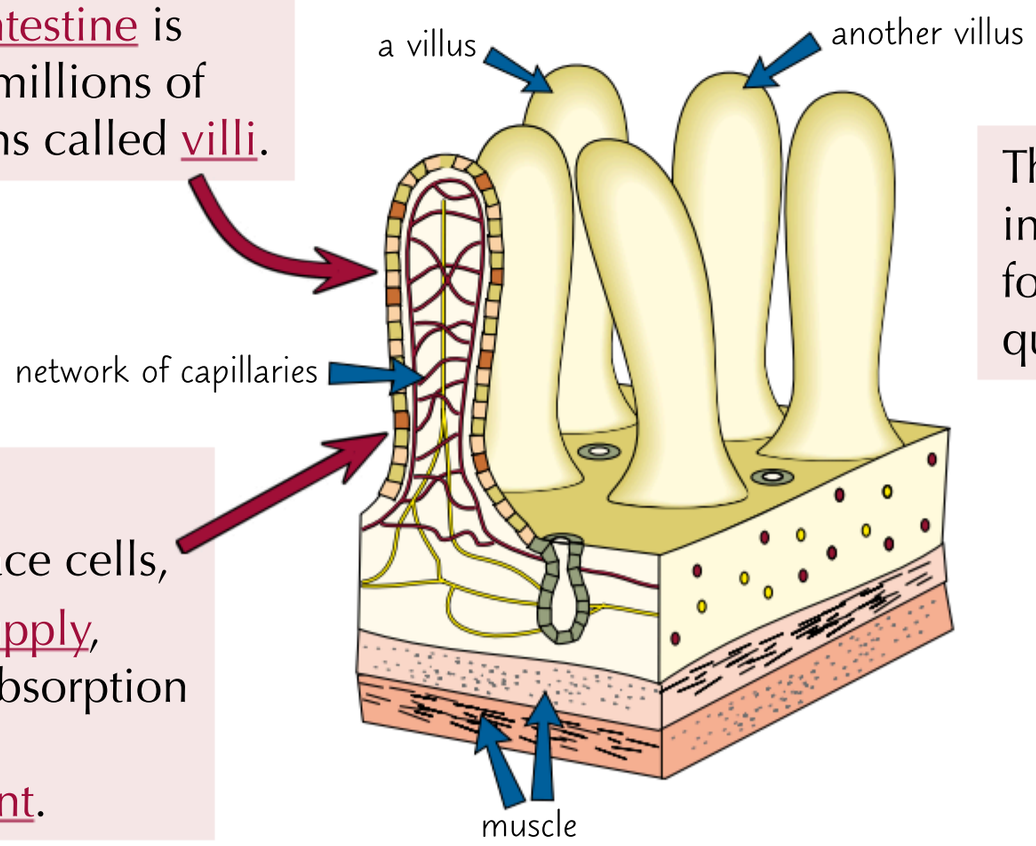
Alveolus is the singular of alveoli.

The alveoli are specialised to maximise the diffusion of O_2 and CO_2. They have:

- An enormous surface area (about 75 m^2 in humans).
- A moist lining for dissolving gases.
- Very thin walls.
- A good blood supply which maintains the concentration gradient.

The **Villi** Provide a **Really Really Big Surface Area**

The inside of the small intestine is covered in millions and millions of these tiny little projections called villi.



They have:

- a single layer of surface cells,
- a very good blood supply, which assists quick absorption by maintaining the concentration gradient.

They increase the surface area in a big way so that digested food is absorbed much more quickly into the blood.

The digested food moves into the blood by diffusion and by active transport (see page 19).

Big surface areas mean substances can diffuse through quickly

Q2 Video Solution

You may well get asked to explain how the adaptations of the alveoli help with gas exchange or how the adaptations of villi help with substance exchange. So make sure that you're happy with this stuff.

Q1 Give one way in which alveoli are adapted for gas exchange. [1 mark]

Q2 Coeliac disease causes inflammation of the small intestine, which can damage the villi. Suggest why a person with coeliac disease might have low levels of iron in their blood. [2 marks]

Quick Quiz

More on Exchanging Substances

*Here are some more adaptations for diffusion — this time, it's **plants** and **fish**.*

The **Structure of Leaves** Lets Gases **Diffuse** In and Out of Cells

Carbon dioxide diffuses into the air spaces within the leaf, then it diffuses into the cells where photosynthesis happens. The leaf's structure is adapted so that this can happen easily:

The underneath of a leaf is an exchange surface.

The walls of the cells inside the leaf form another exchange surface. The air spaces inside the leaf increase the area of this surface so there's more chance for carbon dioxide to get into the cells.

Carbon dioxide diffuses through little holes that cover the underside of a leaf called stomata. Guard cells (p.50) close the stomata if the plant is losing water faster than is being replaced by the roots. Without these guard cells the plant would soon wilt.

Oxygen (produced in photosynthesis) and water vapour also diffuse out through the stomata. (Water vapour is actually lost from all over the leaf surface, but most of it is lost through the stomata.)

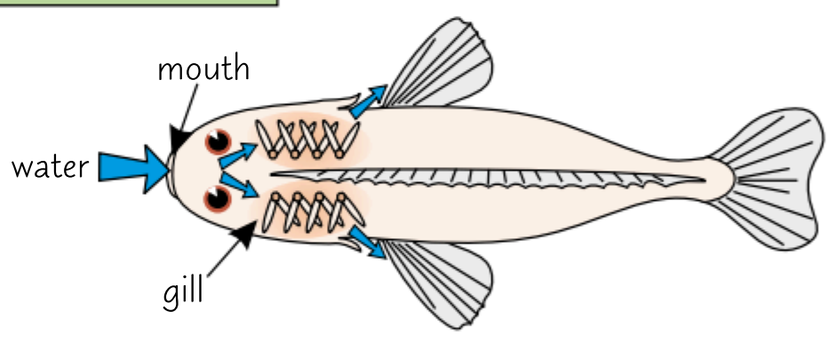
The water vapour evaporates from the cells inside the leaf. Then it escapes by diffusion because there's a lot of it inside the leaf and less of it in the air outside.

The flattened shape of the leaf increases the area of the exchange surface so that it's more effective.

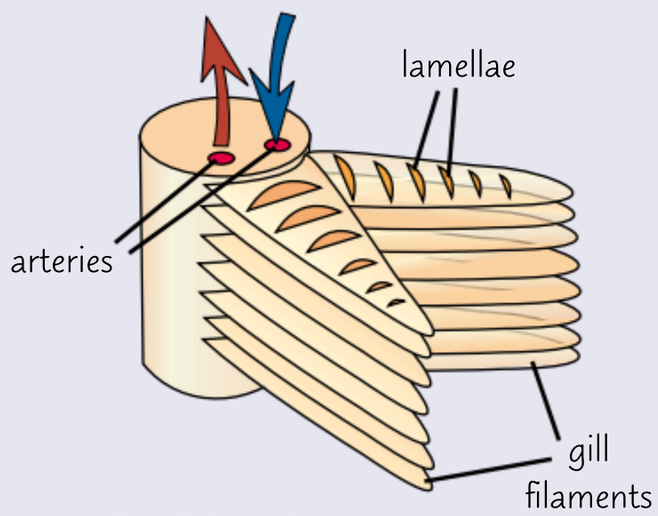
Gills Have a **Large Surface Area** for **Gas Exchange**

The gills are the gas exchange surface in fish.

Water (containing oxygen) enters the fish through its mouth and passes out through the gills. As this happens, oxygen diffuses from the water into the blood in the gills and carbon dioxide diffuses from the blood into the water.



1) Each gill is made of lots of thin plates called gill filaments, which give a big surface area for exchange of gases.
2) The gill filaments are covered in lots of tiny structures called lamellae, which increase the surface area even more.
3) The lamellae have lots of blood capillaries to speed up diffusion.
4) They also have a thin surface layer of cells to minimise the distance that the gases have to diffuse.
5) Blood flows through the lamellae in one direction and water flows over in the opposite direction. This maintains a large concentration gradient between the water and the blood.
6) The concentration of oxygen in the water is always higher than that in the blood, so as much oxygen as possible diffuses from the water into the blood.



Exchange surfaces are specialised for efficient diffusion

Multicellular organisms are incredibly well adapted for getting the substances they need to their cells. It makes sense — if they couldn't do this well, they'd die out. A large surface area is a key way that organisms' exchange surfaces are made more effective — molecules can only diffuse through a membrane when they're right next to it, and a large surface area means that a lot more of the molecules that are exchanged are close to the membrane.

Warm-Up & Exam Questions

Question time again — Warm-Up first, then Exam (or the other way round if you want to be different).

Warm-Up Questions

1) Explain why temperature affects the rate of diffusion.
2) Other than diffusion, by what two processes do substances move across exchange surfaces?
3) Give four ways in which exchange surfaces in animals are adapted to maximise their effectiveness.
4) Name two exchange surfaces in humans.
5) Give two substances that diffuse out of a plant's leaves through the stomata.
6) Explain why lamellae have a good blood supply.

Exam Questions

1 In a fume cupboard, a student injects a sample of ammonia gas into one end of a glass tube. A piece of damp litmus paper in the opposite end of the tube turns blue when the ammonia gas reaches it. It takes 35 seconds for the litmus paper to turn completely blue.

GRADE 6-7 GRADE

Suggest what will happen to the length of time taken for the litmus paper to turn completely blue if the student injects a greater amount of the ammonia gas. Explain your answer.

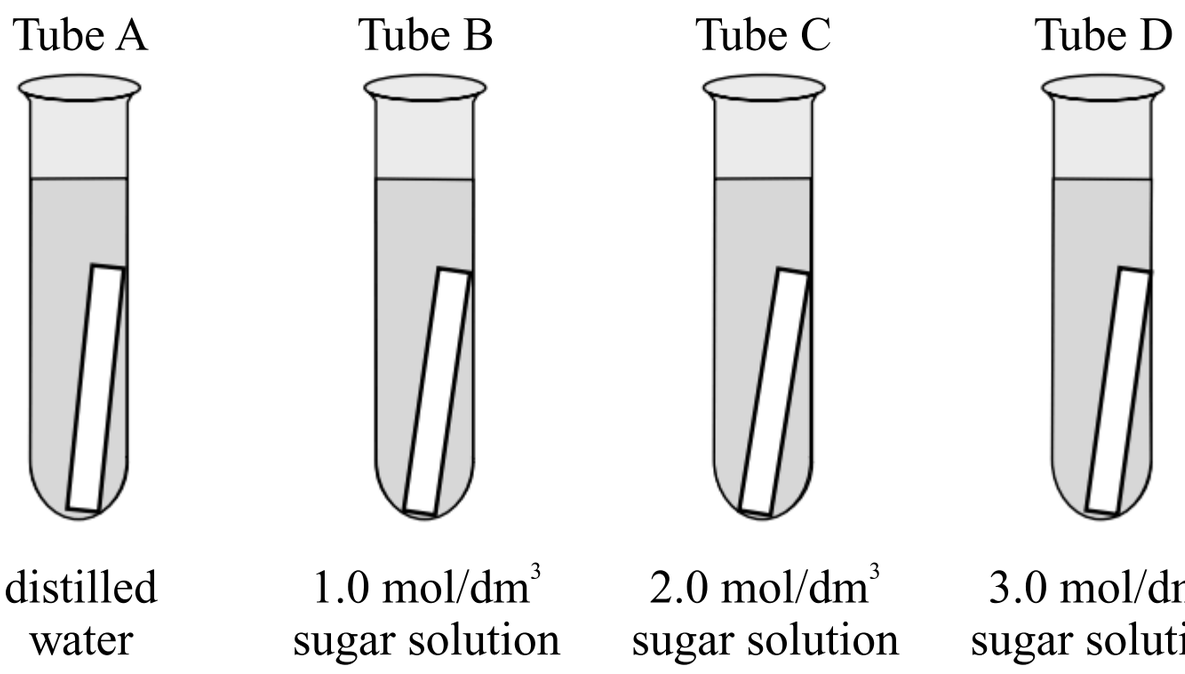
[3 marks]

PRACTICAL

2 In an experiment, four 5 cm long cylinders were cut from a fresh potato. The cylinders were then placed in different sugar solutions, as shown in **Figure 1**. After 24 hours the potato cylinders were removed and measured.

GRADE 6-7 GRADE

Figure 1



2.1 Which potato cylinder would you expect to be shortest after 24 hours? Explain your answer.

[2 marks]

2.2 The potato cylinder in tube A increased in length during the 24 hours. Explain why this happened.

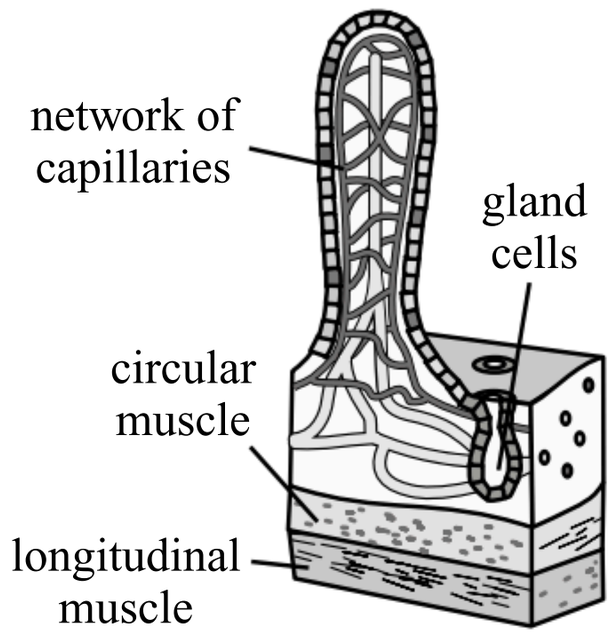
[2 marks]

Exam Questions

3 **Figure 2** shows a villus from the small intestine.
Glucose and other products of digestion are absorbed into the blood through the villi.

GRADE 6-7 GRADE

Figure 2



3.1 Suggest how the products of digestion move into the blood.

[2 marks]

3.2 Explain how the structure of villi is related to their function.

[3 marks]

4 A student made up some gelatine with cresol red solution and ammonium hydroxide.
Cresol red solution is a pH indicator that is red in alkaline solutions and yellow in acidic solutions. The gelatine was cut into cubes of different sizes, and placed in a beaker of dilute hydrochloric acid. The student measured the time taken for the cubes to change from red to yellow as the acid moved into the gelatine and neutralised the ammonium hydroxide.
The results are shown in **Table 1**.

GRADE 7-9 GRADE

Table 1

Size of cube (mm)	Time taken for cube to become yellow (s)			
	Trial 1	Trial 2	Trial 3	Trial 4
5 × 5 × 5	174	167	177	182
7 × 7 × 7	274	290	284	292
10 × 10 × 10	835	825	842	838

4.1 Name the process by which hydrochloric acid moves into the gelatine cubes in this experiment.

[1 mark]

4.2 Calculate the mean time for a 10 × 10 × 10 mm gelatine cube to become yellow in this experiment.

[2 marks]

4.3 Calculate the surface area to volume ratio of a 10 × 10 × 10 mm cube.
Give the ratio in its simplest form.

[3 marks]

4.4 Describe and explain the relationship between the size of the gelatine cube and the time taken for the cube to become yellow.

[3 marks]

Revision Summary Test for Topic 1b

I hope all that information has now diffused across into your brain cells. To make sure it has:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Diffusion and Osmosis (p.17-18) ☑

1) What's the definition of diffusion?
2) Why is diffusion able to happen in gases and in solutions?
3) Outline three factors which affect the rate of diffusion.
4) What is a partially permeable membrane?
5) Explain, giving examples, why only certain substances can move across cell membranes by diffusion.
6) What is osmosis?
7) Sugar solutions A and B are separated by a partially permeable membrane.
   There is a net movement of water molecules from solution B into solution A.
   Which solution initially had the higher water concentration?
8) a) Describe an experiment you could do to investigate the effect of the concentration of sugar solutions on the mass of plant tissue.
   b) Give the dependent variable and the independent variable in this experiment.
   Then give three variables that you'd need to control.
9) Explain why calculating a percentage change in mass, rather than just calculating the difference in mass, is useful for the experiment you've just described.

Active Transport (p.19) ☑

10) Give two substances that a plant takes in via its root hair cells.
    How are plant roots adapted to maximise uptake?
11) What is usually taken into a plant via active transport? Why is active transport needed?
    How does this benefit the plant?
12) Explain how active transport is different from diffusion.
13) Give an example of active transport in humans, and explain why it's needed.

Exchange Surfaces and Exchanging Substances (p.20-22) ☑

14) Name three substances that organisms exchange with their environment.
    Where are they transferred from and to?
15) Explain why single-celled organisms don't require specialised exchange surfaces to exchange substances, but multicellular organisms do.
16) Explain four adaptations of exchange surfaces in animals that increase the efficiency of diffusion.
17) Give an example of a specialised exchange surface found in humans and explain how it's adapted to maximise the exchange of substances.
18) Now give another example of a human specialised exchange surface and explain how it's adapted to maximise the exchange of substances.
19) Explain how leaves are adapted to maximise the amount of carbon dioxide that gets to their cells.
20) Explain how the gills in fish are adapted to maximise the exchange of substances between water and the blood.

Quick Quiz

Cell Organisation

*Some organisms contain loads of **cells**, but how, you might wonder, do all these cells end up making a working human or squirrel... the answer's **organisation**.*

Large Multicellular Organisms are Made Up of Organ Systems

1) Cells are the basic building blocks that make up all living organisms.
2) As you know from p.8, specialised cells carry out a particular function.
3) These specialised cells form tissues, which form organs, which form organ systems (see below).
4) Large multicellular organisms (e.g. humans) have different systems inside them for exchanging and transporting materials.

Remember, the process by which cells become specialised for a particular job is called differentiation. It occurs during the development of a multicellular organism.

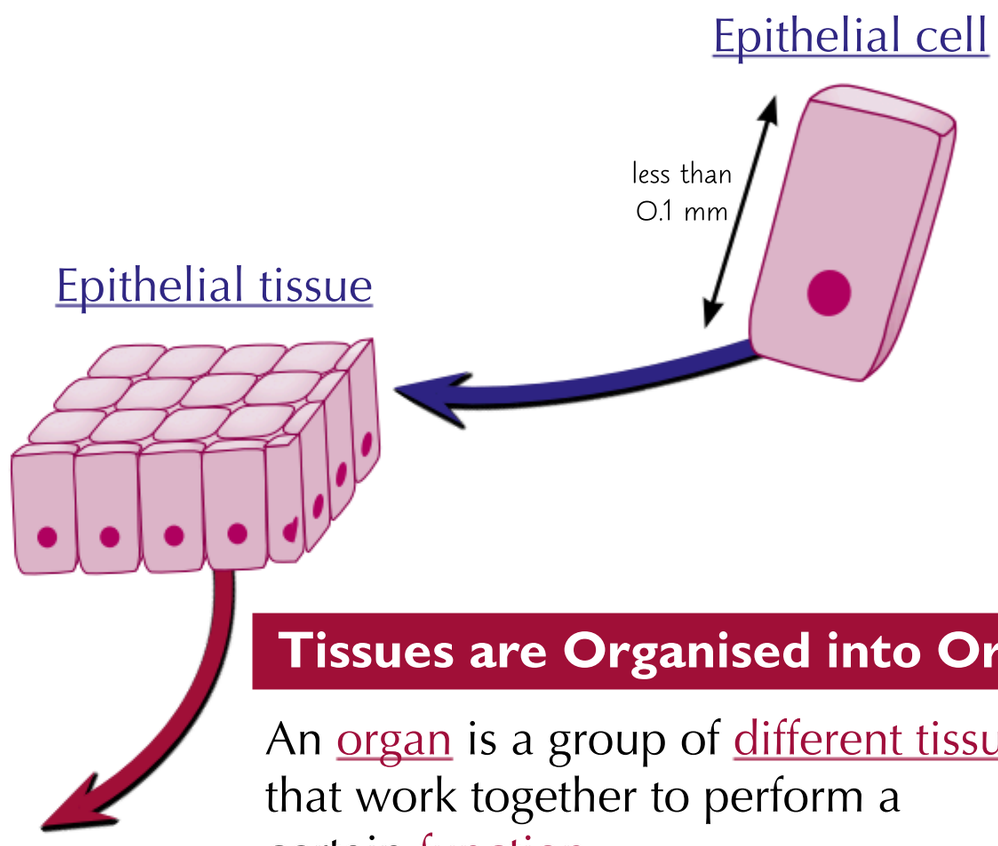
Everything Starts With Cells

Similar Cells are Organised into Tissues

A tissue is a group of similar cells that work together to carry out a particular function. It can include more than one type of cell.

In mammals (like humans), tissues include:

- Muscular tissue, which contracts (shortens) to move whatever it's attached to.
- Glandular tissue, which makes and secretes chemicals like enzymes and hormones.
- Epithelial tissue, which covers some parts of the body, e.g. the inside of the gut.

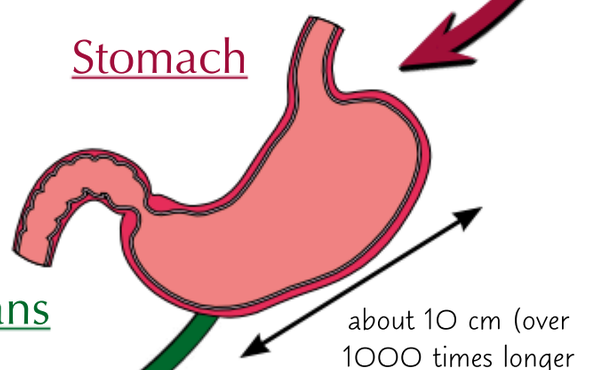


Tissues are Organised into Organs

An organ is a group of different tissues that work together to perform a certain function.

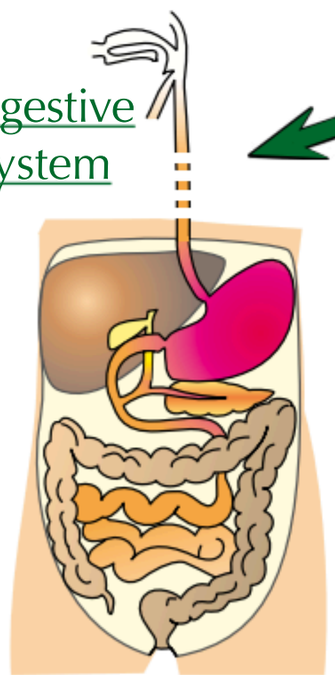
E.g. the stomach is an organ made of:

- Muscular tissue, which moves the stomach wall to churn up the food.
- Glandular tissue, which makes digestive juices to digest food.
- Epithelial tissue, which covers the outside and inside of the stomach.



Organs are Organised into Organ Systems

An organ system is a group of organs working together to perform a particular function.



E.g. the digestive system breaks down and absorbs food.
It's made up of these organs:

- Glands (e.g. the pancreas, salivary glands) which produce digestive juices.
- The stomach and small intestine, which digest food.
- The liver, which produces bile.
- The small intestine, which absorbs soluble food molecules.
- The large intestine, which absorbs water from undigested food, leaving faeces.

Organ systems work together to make entire organisms.

Remember — cells, tissues, organs, organ systems

OK, so from this page, you now know what cell organisation is all about. Basically, cells are organised into tissues, the tissues into organs, the organs into organ systems and the organ systems into a whole organism.

Enzymes

Quick Quiz

__Chemical reactions__ are what make you work. And __enzymes__ are what make them work.

__Enzymes__ are __Catalysts__ Produced by __Living Things__

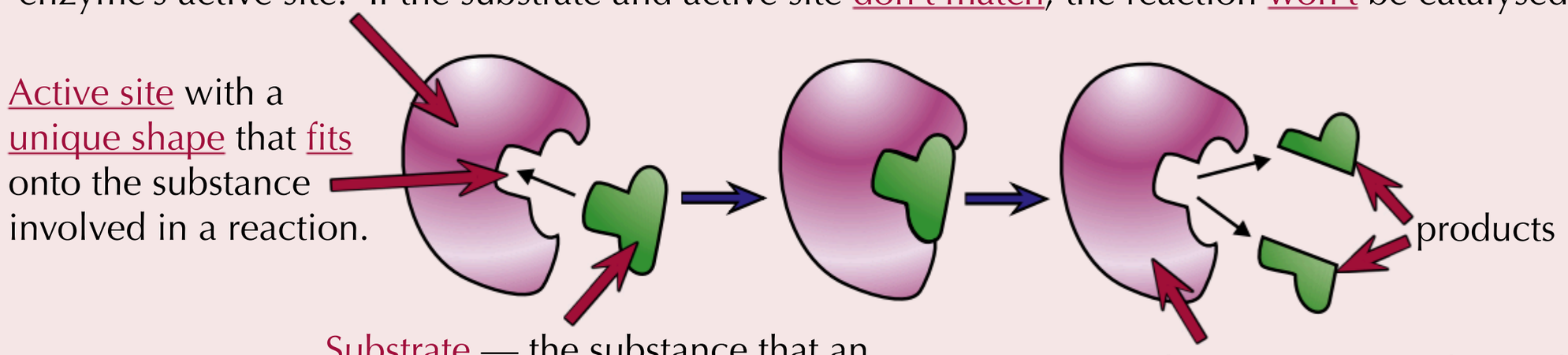
1) Living things have thousands of different chemical reactions going on inside them all the time. These reactions need to be carefully controlled — to get the right amounts of substances.
2) You can usually make a reaction happen more quickly by raising the temperature. This would speed up the useful reactions but also the unwanted ones too... not good. There's also a limit to how far you can raise the temperature inside a living creature before its cells start getting damaged.
3) So... living things produce enzymes that act as biological catalysts. Enzymes reduce the need for high temperatures and we only have enzymes to speed up the useful chemical reactions in the body.

A CATALYST is a substance which INCREASES the speed of a reaction, without being CHANGED or USED UP in the reaction.

Video

4) Enzymes are all large proteins and all proteins are made up of chains of amino acids. These chains are folded into unique shapes, which enzymes need to do their jobs. The 'lock and key' model of enzyme action shows why these unique shapes are important:

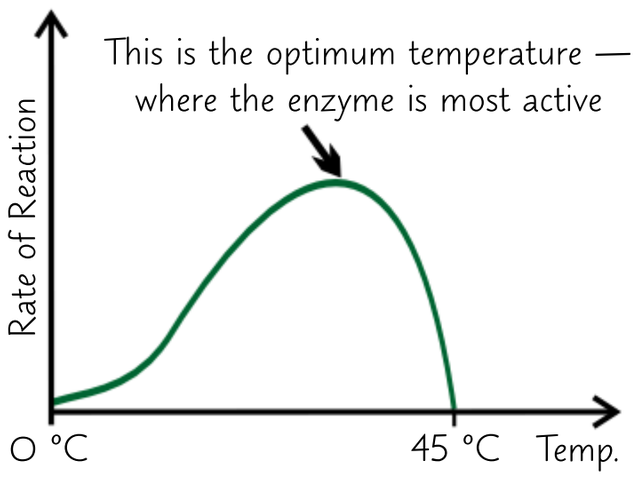
Enzyme — usually only catalyses one specific reaction because the substrate has to fit into the enzyme's active site. If the substrate and active site don't match, the reaction won't be catalysed.



Substrate — the substance that an enzyme acts on. In reactions, it's usually split apart or joined together.

The enzyme is unchanged after the reaction.

__Temperature__ and __pH__ Affect the __Rate__ of __Enzyme-Catalysed Reactions__

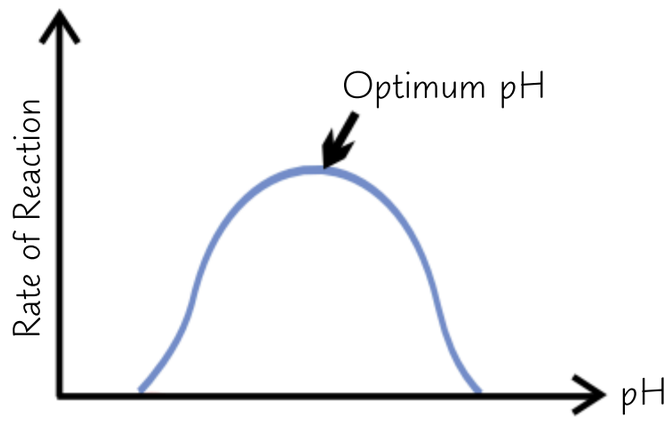


A higher temperature increases the rate of reaction at first. But if it gets too hot, some of the bonds holding the enzyme together break. This changes the shape of the enzyme's active site, so the substrate won't fit any more. The enzyme is said to be denatured.

All enzymes have an optimum temperature that they work best at.

If pH is too high or too low, it damages the bonds in the enzyme. This changes the shape of the active site and denatures the enzyme.

All enzymes have an optimum pH that they work best at. It's often neutral pH 7, but not always — e.g. pepsin works in the stomach where there are acidic conditions, and works best at pH 2.



Enzymes speed up chemical reactions

Just like you've got to have the correct key for a lock, you've got to have the right substance for an enzyme. As you can see in the diagram at the top, if the substance doesn't fit, the enzyme won't catalyse the reaction...

Quick Quiz

Investigating Enzymatic Reactions

PRACTICAL

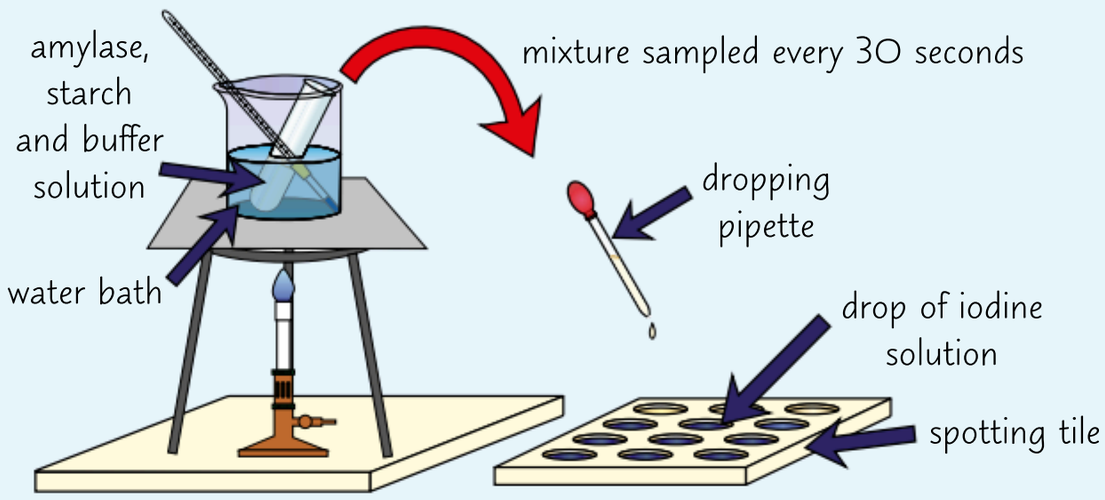
*Enzymes are clearly very clever, but they're **not** very versatile. They need just the right **conditions** if they're going to work properly. Here's a **practical** you can do to investigate this.*

Here's a Way of **Investigating** the Effect of **pH** on **Enzyme Activity**

The enzyme amylase breaks down starch to maltose. You can tell if this reaction is complete using iodine solution — if starch is still present, the iodine solution will change from browny-orange to blue-black.

Method

1) Put a drop of iodine solution into every well of a spotting tile.
2) Set up a water bath and heat it to 35 °C. Try to keep the temperature of the water constant throughout the experiment.
3) Use a syringe to add 1 cm^3 of amylase solution and 1 cm^3 of a pH 5 buffer solution to a boiling tube. Using test tube holders, put the tube into the water bath and wait for five minutes.
4) Next, use a different syringe to add 5 cm^3 of a starch solution to the boiling tube.
5) Immediately mix the contents of the boiling tube and start a stop clock.



6) Use continuous sampling to record how long it takes for the amylase to break down all of the starch. To do this, use a dropping pipette to take a fresh sample from the boiling tube every 30 seconds and put a drop into a well. When the iodine solution remains browny-orange, starch is no longer present.
7) Repeat the whole experiment with buffer solutions of different pH values to see how pH affects the time taken for the starch to be broken down.
8) Remember to control any variables each time (e.g. concentration and volume of amylase solution) to make it a fair test.

There's more about water baths on page 182.

You could use a pH meter to accurately measure the pH of your solutions.

Here's How to **Calculate** the **Rate of Reaction**

1) Rate is a measure of how much something changes over time.
2) For the method above, you can calculate the rate of reaction using this formula:

$$\text{Rate} = \frac{1000}{\text{time}}$$

E.g. At pH 6, the time taken for amylase to break down the starch in a solution was 90 seconds. So the rate of the reaction = $1000 \div 90 = 11\ s^{-1}$ (2 s.f.)

Rate is given 'per unit time', so s^{-1} means 'per second'.

3) If an experiment measures how much something changes over time, you can calculate rate by dividing the amount that it has changed by the time taken.

$$\text{Rate} = \frac{\text{change}}{\text{time taken}}$$

EXAMPLE

During an investigation into the activity of the enzyme catalase, 24 cm^3 of oxygen was released in 50 seconds. Calculate the rate of the reaction in cm^3/s.

MATHS SKILLS

1) Write down how much product was formed (the change), and how much time it took.
   Product formed = 24 cm^3, time = 50 seconds
2) Substitute these numbers into the formula to calculate rate.
   Rate = change ÷ time taken
   = 24 cm^3 ÷ 50 s = 0.48 cm^3/s

You can investigate other factors too...

You could adapt this experiment to investigate how factors other than pH affect the rate of amylase activity. E.g. you could use a water bath set to different temperatures to investigate the effect of temperature.

Q1 An enzyme-controlled reaction was carried out at pH 4. After 2 minutes, 36 cm^3 of product had been released. Calculate the rate of reaction in cm^3/s. [1 mark]

Q1 Video Solution

Enzymes and Digestion

Quick Quiz

*The **enzymes** used in **digestion** are produced by **cells** and then released into the **gut** to mix with food.*

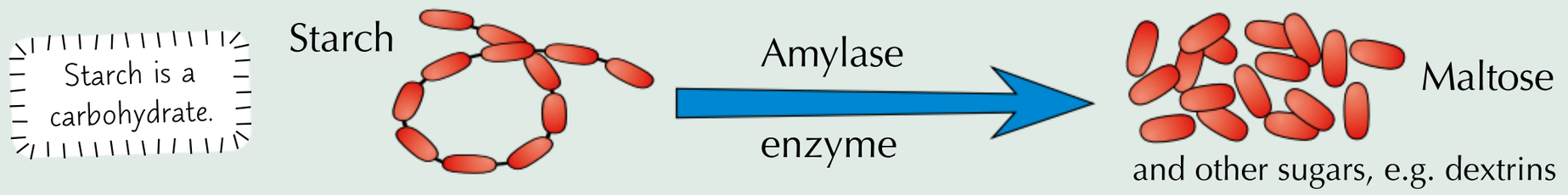
Digestive Enzymes Break Down **Big Molecules**

Starch, proteins and fats are big molecules. They're too big to pass through the walls of the digestive system, so digestive enzymes break these big molecules down into smaller ones like sugars (e.g. glucose and maltose), amino acids, glycerol and fatty acids. These smaller, soluble molecules can pass easily through the walls of the digestive system, allowing them to be absorbed into the bloodstream.

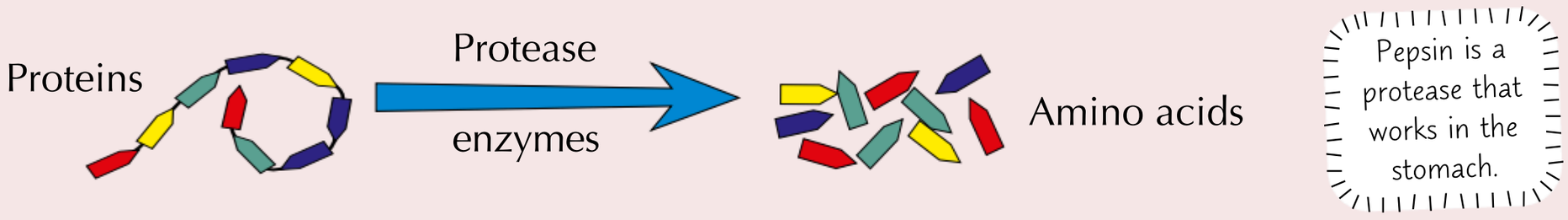
Different enzymes catalyse the breakdown of different food molecules:

Carbohydrases Convert Carbohydrates into Simple Sugars

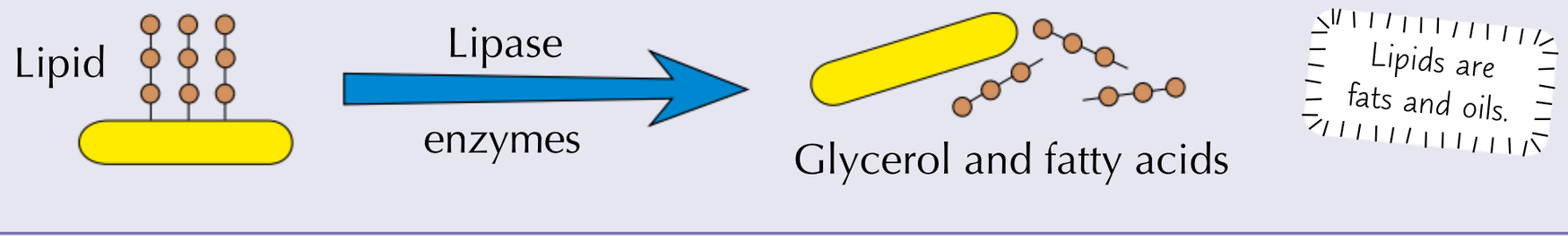
Amylase is an example of a carbohydrase. It breaks down starch.



Proteases Convert Proteins into Amino Acids



Lipases Convert Lipids into Glycerol and Fatty Acids



The products of digestion can be used to make new carbohydrates, proteins and lipids. Some of the glucose (a carbohydrate) that's made is used in respiration (see p.78).

Bile Neutralises the Stomach Acid and **Emulsifies** Fats

Bile is produced in the liver. It's stored in the gall bladder before it's released into the small intestine.

1) The hydrochloric acid in the stomach makes the pH too acidic for enzymes in the small intestine to work properly. Bile is alkaline — it neutralises the acid and makes conditions alkaline. The enzymes in the small intestine work best in these alkaline conditions.
2) Bile also emulsifies fats (breaks the fat into tiny droplets). This gives a much bigger surface area of fat for the enzyme lipase to work on — which makes its digestion faster.

Big molecules get broken into smaller molecules that are absorbed

Now you know a few examples of how enzymes are really useful. Make sure you know the examples of amylase, protease and lipase, and the reactions that they catalyse. This stuff could get you a few quick marks in your exam.

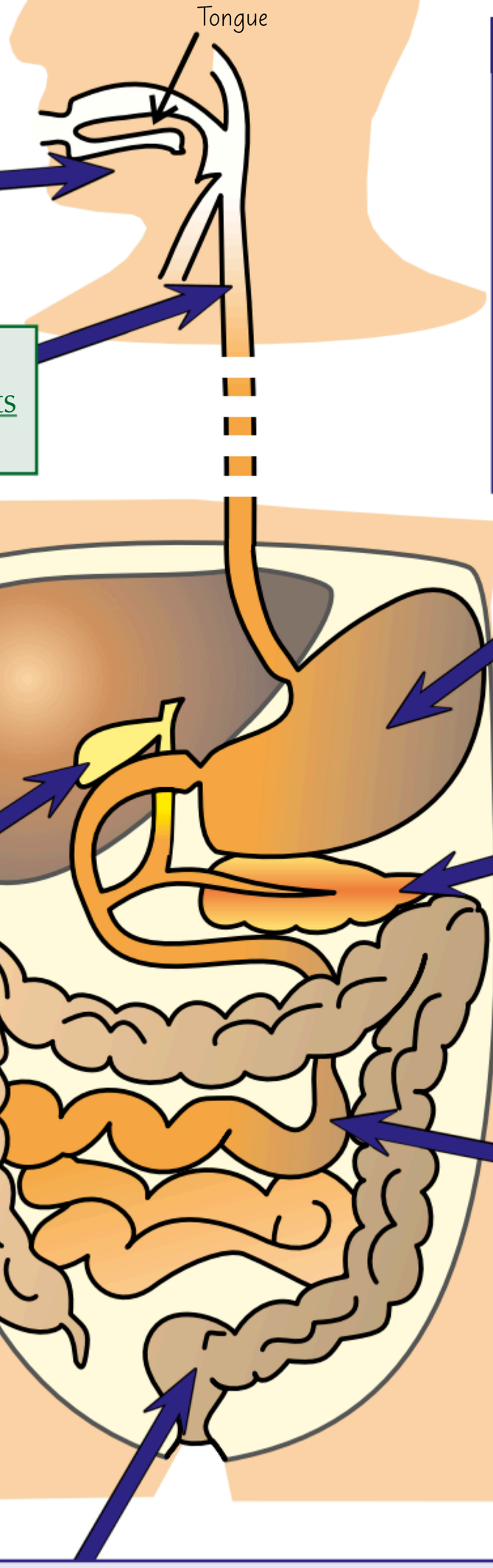
More on Enzymes and Digestion

Quick Quiz

*So now you know what the enzymes do, here's a nice **big picture** of the **whole** of the digestive system.*

Digestive Enzymes Mix With Food in the Gut

Enzymes used in the digestive system are produced by specialised cells in glands and in the gut lining. These are released into the gut where they mix with and help to break down food.



Salivary glands

These produce amylase enzyme in the saliva.

Gullet

Also known as the oesophagus, and connects the mouth and stomach.

Liver

Where bile is produced. Bile neutralises stomach acid and emulsifies fats.

Gall bladder

Where bile is stored, before it's released into the small intestine.

Large intestine

Where excess water is absorbed from the food.

Stomach

1) It pummels the food with its muscular walls.
2) It produces the protease enzyme, pepsin.
3) It produces hydrochloric acid to:
   a) kill bacteria
   b) give the right pH for the protease enzyme to work (pH 2 — acidic).

Pancreas

Produces protease, amylase and lipase enzymes. It releases these into the small intestine.

Small intestine

1) Produces protease, amylase and lipase enzymes to complete digestion.
2) This is also where the digested food is absorbed out of the digestive system into the blood.

Rectum

Where the faeces (made up mainly of indigestible food) are stored before they bid you a fond farewell through the anus.

That's nine different bits of the digestive system with different jobs

Did you know that the whole of your digestive system is actually a big hole that goes right through your body? It just gets loads of food, digestive juices and enzymes piled into it... Make sure you know what bit does what.

PRACTICAL

Food Tests

Quick Quiz

*There are some clever little **tests** you can do to tell what a food **contains**. Read on...*

You Might Need to Prepare a **Food Sample**

There are some nifty ways to identify what type of food molecule a sample contains. For each of the tests, you need to prepare a food sample. It's the same each time though — here's what you'd do:

1) Break up a piece of food using a pestle and mortar.
2) Transfer the ground-up food to a beaker and add some distilled water.
3) Give the mixture a good stir with a glass rod to dissolve some of the food.
4) Filter the solution using a funnel lined with filter paper to get rid of the solid bits of food.

Once You've Got Your **Sample** You Can **Test It**

You need to know how to test for sugars, proteins, starch and lipids.

Sugars — the Benedict's Test

Sugars are found in foods such as biscuits, cereal and bread. Here's how to test for sugars:

1) Prepare a food sample and transfer 5 cm^3 to a test tube.
2) Prepare a water bath so that it's set to 75 °C.
3) Add about 10 drops of Benedict's solution to the test tube using a pipette.
4) Place the test tube in the water bath using a test tube holder and leave it in there for 5 minutes. Make sure the tube is pointing away from you.

RESULTS:

If the food sample contains a sugar, the solution in the test tube will change from the normal blue colour to green, yellow or brick-red — it depends on how much sugar is in the food.

Proteins — the Biuret Test

Meat and cheese are protein-rich and good foods to use in this test. Here's how you can do it:

1) Prepare a sample of your food and transfer 2 cm^3 of your sample to a test tube.
2) Add 2 cm^3 of biuret reagent to the sample and mix the contents of the tube by gently shaking it.

RESULTS:

If the food sample contains protein, the solution will change from blue to purple. If no protein is present, the solution will stay blue.

Starch — test with Iodine Solution

Foods like pasta, rice and potatoes contain a lot of starch. Here's how to do the test:

1) Make a food sample and transfer 5 cm^3 of your sample to a test tube.
2) Add a few drops of iodine solution and gently shake the tube to mix the contents.

RESULTS:

If the sample contains starch, the colour of the solution will change from browny-orange to black or blue-black.

Lipids — the Sudan III Test

Lipids are found in foods like olive oil, margarine and milk. Here's a way to test for them:

1) Prepare a sample of the food you're testing (but you don't need to filter it). Transfer about 5 cm^3 into a test tube.
2) Use a pipette to add 3 drops of Sudan III stain solution to the test tube and gently shake the tube.

RESULTS:

Sudan III stain solution stains lipids.
If the sample contains lipids, the mixture will separate out into two layers.
The top layer will be bright red.

If no lipids are present, no separate red layer will form at the top of the liquid.

You can also use ethanol to test for lipids — if they're present, you'll get a cloudy layer on your sample.

Make sure you think about all of the hazards...

Iodine is an irritant to the eyes, and the chemicals in the biuret reagent are dangerous, so wear safety goggles. If you spill any chemicals on your skin, wash it off straight away. Also, be careful around the water bath in the Benedict's test. And if that's not enough to be cautious of, Sudan III stain solution is flammable, so keep it away from flames.

Warm-Up & Exam Questions

Doing well in exams isn't just about remembering all the facts, although that's important. You have to get used to the way the exams are phrased and make sure you always read the question carefully.

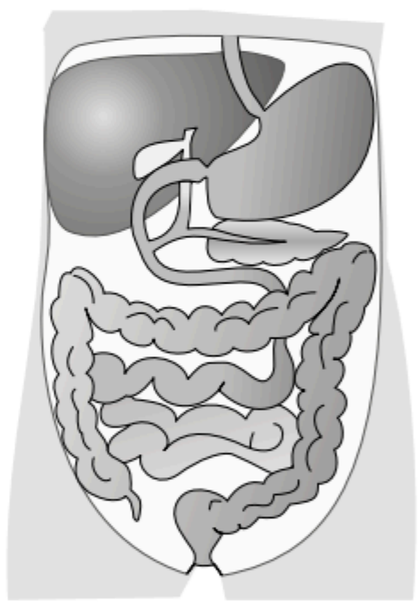
Warm-Up Questions

1) What is an organ system?
2) What is meant by the optimum pH of an enzyme?
3) Which enzyme digests: (a) starch (b) protein (c) lipids?
4) What are the products of the digestion of: (a) starch (b) protein (c) lipids?
5) Name the three parts of the digestive system that produce protease enzymes.
6) Describe how you would prepare a food sample before testing it for the presence of different food molecules.

Exam Questions

1 **Figure 1** shows the human digestive system. GRADE 4-6 GRADE

Figure 1



1.1 Label the place where bile is produced.

[1 mark]

1.2 Describe why bile needs to have an alkaline pH.

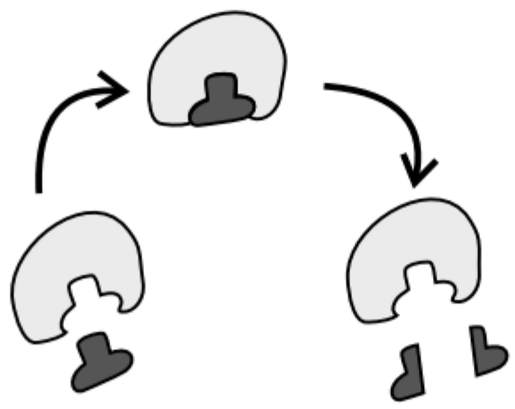
[1 mark]

1.3 Outline how bile helps with the digestion of fats.

[2 marks]

2 **Figure 2** represents the action of an enzyme in catalysing a biological reaction. GRADE 4-6 GRADE

Figure 2



2.1 In terms of the enzyme's shape, explain why an enzyme only catalyses one reaction.

[1 mark]

2.2 The optimum pH of the enzyme is pH 7.
Explain what effect a very low pH would have on the activity of the enzyme.

[2 marks]

Exam Questions

3 **Figure 3** shows the effect of temperature on the action of two different enzymes.

GRADE 6-7 GRADE

3.1 What is the optimum temperature for enzyme **A**?

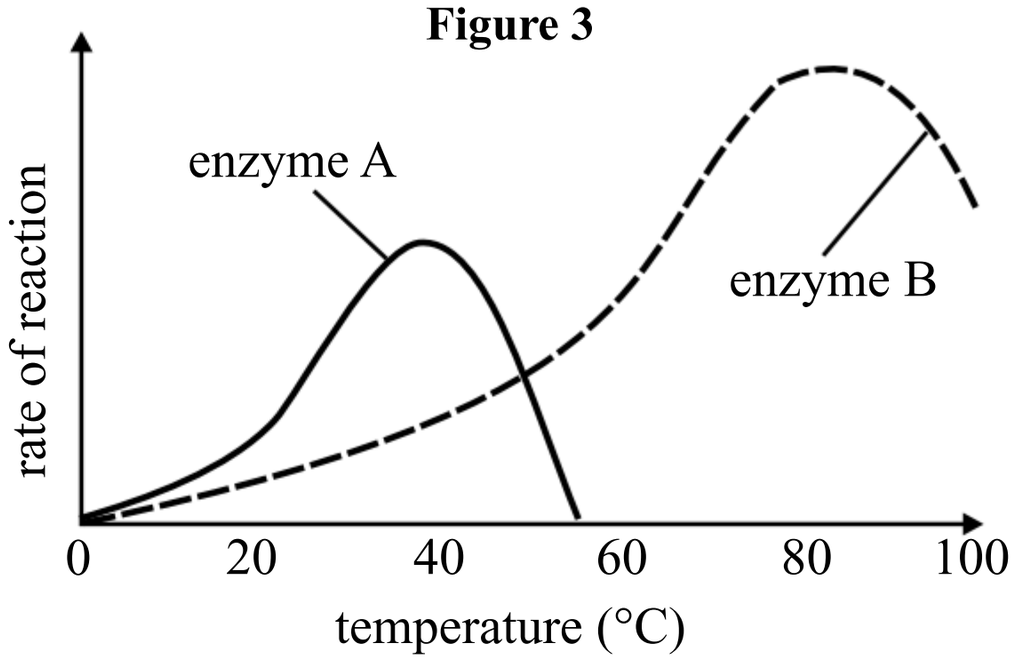
[1 mark]

3.2 One of these enzymes was extracted from human liver cells. The other was extracted from bacteria living in hot underwater vents.

Suggest which enzyme came from the bacteria. Give a reason for your answer.

[1 mark]

Figure 3



PRACTICAL

4 A student wanted to know which substances were present in a food sample. A solution containing the food sample was prepared, and then some of the solution was added to each of three test tubes. The student then added iodine solution to test tube **A**, biuret solution to test tube **B** and Sudan III stain solution to test tube **C**. The results are shown in **Table 1**.

GRADE 6-7 GRADE

Table 1

Test tube	Description of solution in test tube
A	Blue-black
B	Blue
C	In two layers. Top layer red.

Describe what the student's results show about which substances are present in the food sample.

[3 marks]

PRACTICAL

5 A student was investigating the effect of pH on the rate of amylase activity. A syringe was used to put amylase solution and a buffer solution with a pH of 6 into a test tube. A different syringe was then used to add a starch solution to the boiling tube. The student mixed the contents and then started a stop clock. Every 30 seconds a sample was taken from the boiling tube and tested for the presence of starch. When there was no starch present the stop clock was stopped. The student repeated the experiment three times.

GRADE 6-7 GRADE

5.1 Suggest why the student used two different syringes when adding substances to the boiling tube.

[1 mark]

5.2 The experiment showed that the average time taken for the starch in the boiling tube to be broken down was 60 seconds. Calculate the rate of the reaction. Give your answer in s^{-1} to 2 significant figures. Use the formula: $\textbf{rate} = \frac{\textbf{1000}}{\textbf{time}}$

[1 mark]

5.3 Describe what the student needs to do next in this investigation to determine the effect of pH on the rate of amylase activity.

[1 mark]

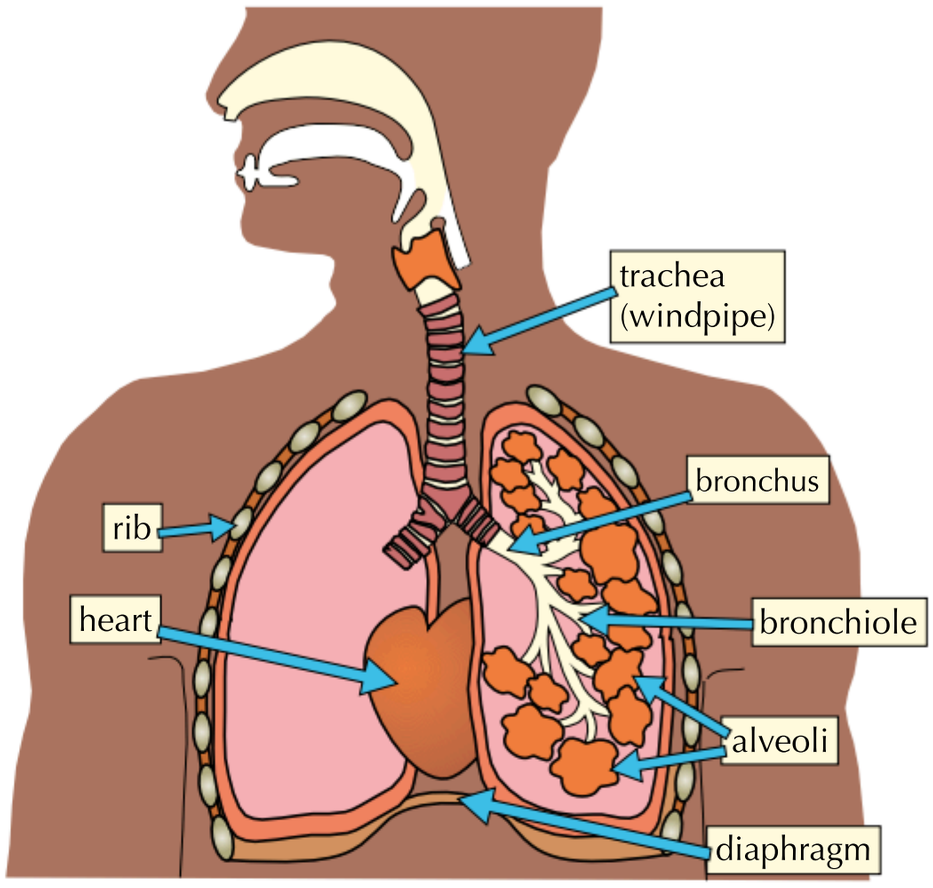
The Lungs

Quick Quiz

*When it comes to **breathing**, much more than just a pair of lungs is needed. There are some other vital parts that help to get the **oxygen** you need in, and the waste **carbon dioxide** out...*

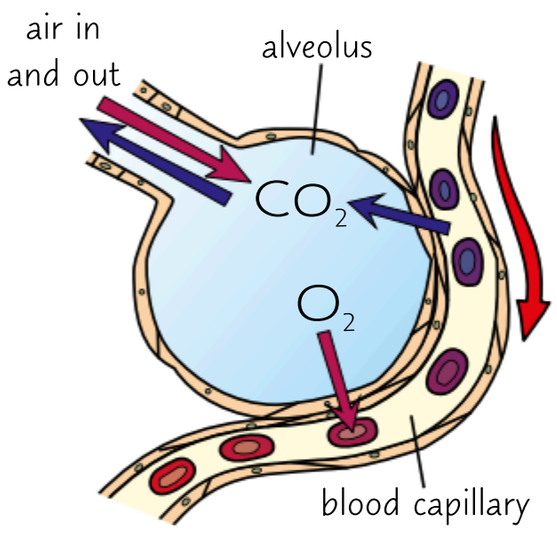
The **Lungs** are in the **Thorax**

1) The thorax is the top part of your body. It's separated from the lower part of the body by the diaphragm.
2) The lungs are like big pink sponges and are protected by the ribcage.
3) The air that you breathe in goes through the trachea. This splits into two tubes called bronchi (each one is a bronchus), one going to each lung.
4) The bronchi split into progressively smaller tubes called bronchioles.
5) The bronchioles end at small air sacs called alveoli.



Gas Exchange Happens in the **Alveoli**

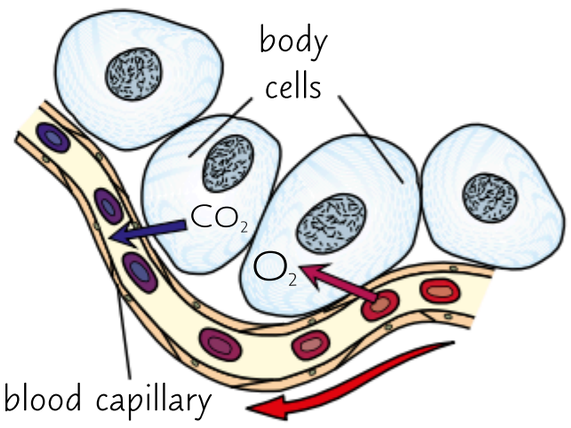
The lungs have millions of alveoli, surrounded by a network of blood capillaries.



- The blood passing next to the alveoli has just returned to the lungs from the rest of the body, so it contains lots of carbon dioxide and very little oxygen.
- Oxygen diffuses out of the alveolus (high concentration) into the blood (low concentration). Carbon dioxide diffuses out of the blood (high concentration) into the alveolus (low concentration) to be breathed out.

Video

- When the blood reaches body cells, oxygen is released from the red blood cells (high concentration) and diffuses into the body cells (low concentration).
- At the same time, carbon dioxide diffuses out of the body cells (high concentration) into the blood (low concentration). It's then carried back to the lungs.



You Can **Calculate** the **Breathing Rate** in **Breaths Per Minute**

You need to know how to do rate calculations for the exam — and breathing rate is an example of this:

EXAMPLE

Kerry takes 91 breaths in 7 minutes.
Calculate her average breathing rate in breaths per minute.

MATHS SKILLS

1) To calculate the rate, divide the number of breaths by the amount of time.
2) Substitute in the numbers and work out the answer.

breaths per minute = number of breaths ÷ number of minutes
= 91 ÷ 7
= 13 breaths per minute

Alveoli are exchange surfaces

Back on page 20, you might remember that you read about how exchange surfaces can be adapted to maximise their effectiveness. Alveoli have all of these adaptations to speed up the process of getting oxygen into the blood and getting the waste carbon dioxide out of the body. If you feel like you need a reminder, flick back to page 21.

Circulatory System — The Heart

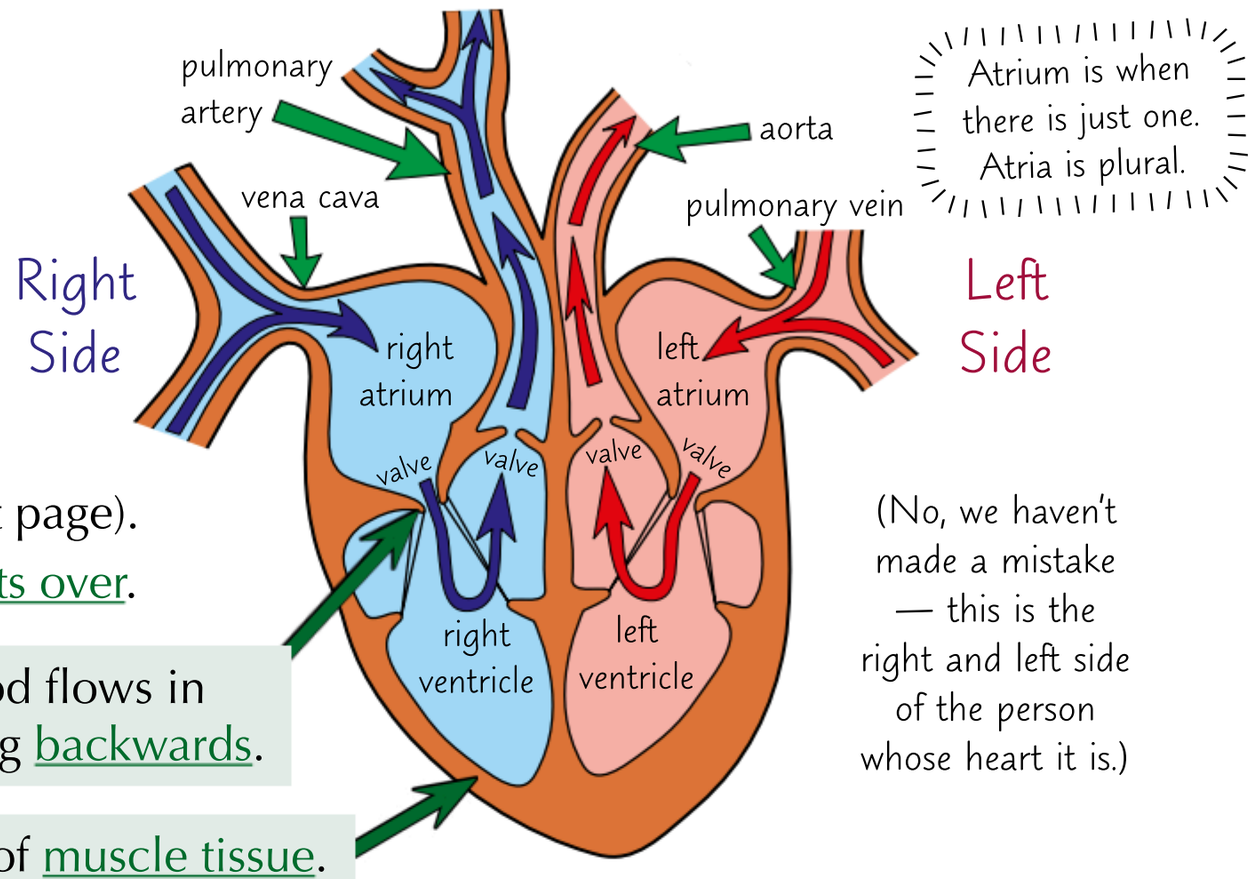
Quick Quiz

*The circulatory system carries **food** and **oxygen** to every cell in the body. As well as being a delivery service, it's also a waste collection service — it carries **waste products** to where they can be removed from the body.*

The **Heart Contracts** to **Pump Blood** Around the Body

The circulatory system is made up of the heart, blood vessels and blood.
The heart uses its four chambers (left and right atrium, left and right ventricle) to pump blood around:

1) Blood flows into the two atria from the vena cava and the pulmonary vein.
2) The atria contract, pushing the blood into the ventricles.
3) The ventricles contract, forcing the blood into the pulmonary artery and the aorta, and out of the heart.
4) The blood then flows to the organs through arteries, and returns through veins (see next page).
5) The atria fill again and the whole cycle starts over.



Atrium is when there is just one. Atria is plural.

(No, we haven't made a mistake — this is the right and left side of the person whose heart it is.)

The heart has valves to make sure that blood flows in the right direction — they prevent it flowing backwards.

The walls of the heart are mostly made of muscle tissue.

The heart also needs its own supply of oxygenated blood (see below). Arteries called coronary arteries branch off the aorta and surround the heart, making sure that it gets all the oxygenated blood it needs.

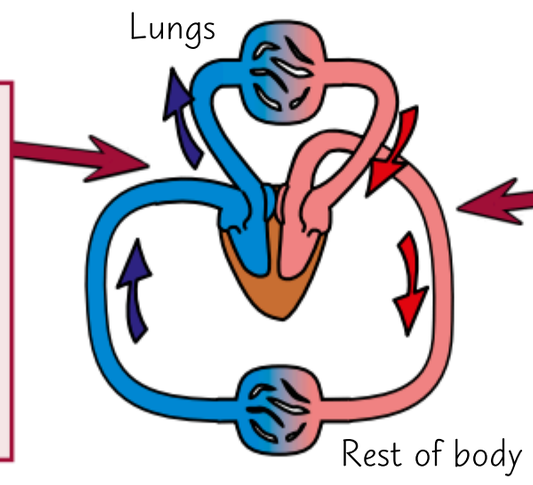
The **DOUBLE Circulatory System**

Video

Humans have a double circulatory system — two circuits joined together:

1st Circuit

The right ventricle pumps deoxygenated blood (blood without oxygen) to the lungs to take in oxygen. The oxygenated blood then returns to the heart.



2nd Circuit

The left ventricle pumps oxygenated blood around all the other organs of the body. The blood gives up its oxygen to body cells and the deoxygenated blood returns to the heart to be pumped out to the lungs again.

The **Heart** Has a **Pacemaker**

1) Your resting heart rate is controlled by a group of cells in the right atrium wall that act as a pacemaker.
2) They produce an electric impulse which spreads to the surrounding muscle cells, making them contract.
3) An artificial pacemaker is often used to control the heartbeat if the natural pacemaker cells don't work properly (e.g. if the patient has an irregular heartbeat). It's a little device that's implanted under the skin and has a wire going to the heart. It produces an electric current to keep the heart beating regularly.

Electric impulses from pacemaker cells cause the heart to beat

When doctors or medics use a stethoscope to listen to your heart, they're not hearing your heart contract. What they are actually listening to is your heart valves snapping shut. Lub-dub, lub-dub, lub-dub...

Circulatory System — Blood Vessels

Quick Quiz

Blood *needs a good set of 'tubes' to carry it round the body. Here's a page on the different types:*

Blood Vessels are Adapted for Their Function

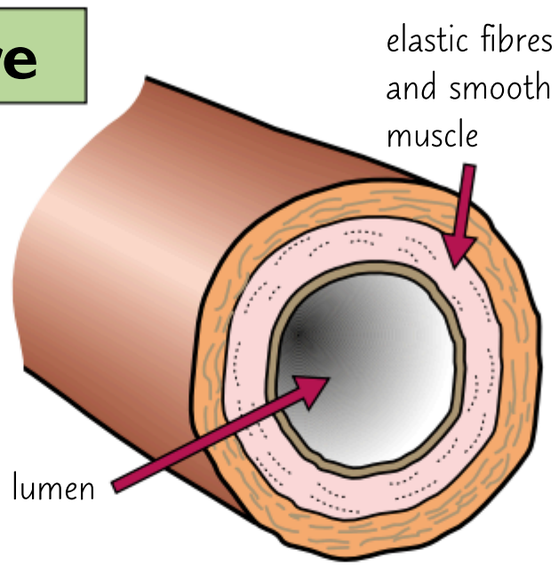
There are three different types of blood vessel:

1) ARTERIES — carry the blood away from the heart.
2) CAPILLARIES — involved in the exchange of materials at the tissues.
3) VEINS — carry the blood to the heart.

Arteries Carry Blood Under Pressure

The heart pumps the blood out at high pressure so the artery walls are strong and elastic.

The walls are thick compared to the size of the hole down the middle (the "lumen").



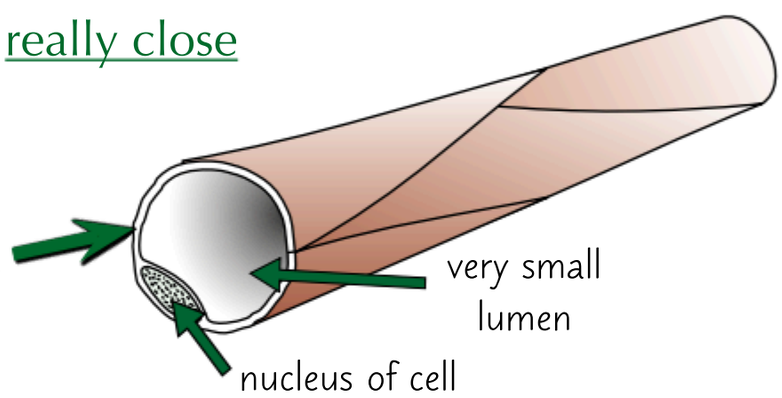
They contain thick layers of muscle to make them strong, and elastic fibres to allow them to stretch and spring back.

Capillaries are Really Small

Arteries branch into capillaries, which are tiny. They carry the blood really close to every cell in the body to exchange substances with them. They supply food and oxygen, and take away waste like CO_2.

They have permeable walls, so substances can diffuse in and out.

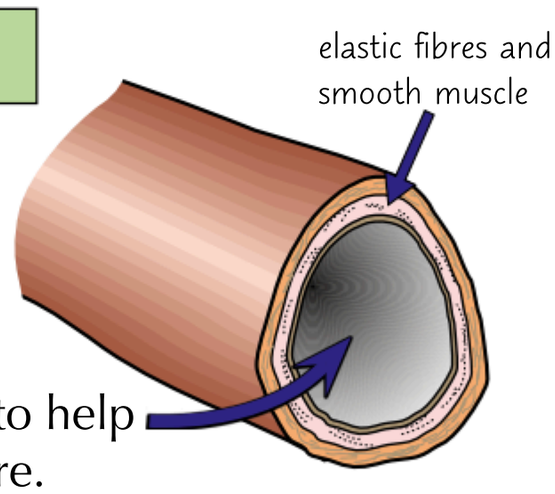
These walls are usually only one cell thick. This increases the rate of diffusion by decreasing the distance over which it occurs.

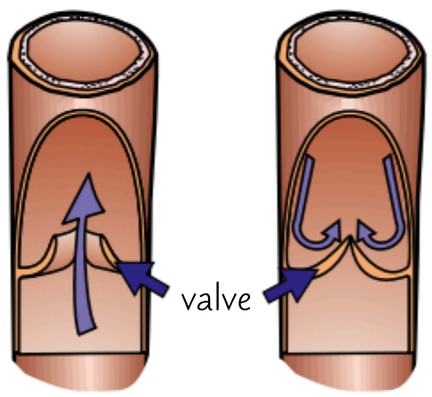


Veins Take Blood Back to the Heart

Capillaries eventually join up to form veins. The blood is at lower pressure in the veins so the walls don't need to be as thick as artery walls.

They have a bigger lumen than arteries to help the blood flow despite the lower pressure.





Valves help to keep the blood flowing in the right direction.

You Can Calculate the Rate of Blood Flow

EXAMPLE

1464 ml of blood passed through an artery in 4.5 minutes. Calculate the rate of blood flow through the artery in ml/min.

MATHS SKILLS

1) To calculate the rate, divide the volume of blood by the amount of time.
2) Substitute in the numbers to work out the rate.

rate of blood flow = volume of blood ÷ number of minutes
= 1464 ÷ 4.5 = 325.333...
= 325 ml/min (3 s.f.)

Veins have valves — arteries carry blood away from the heart

Here's an interesting fact for you — your body contains about 60 000 miles of blood vessels.

Q1 2.175 litres of blood passed through a vein in 8.7 minutes. Calculate the rate of blood flow through the vein in ml/min. [2 marks]

Q1 Video Solution

Circulatory System — Blood

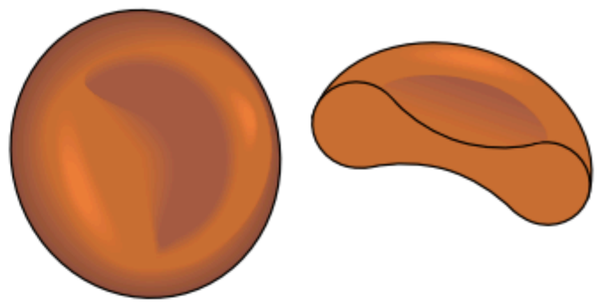
Quick Quiz

*Blood is a **tissue**. It's part of a huge **transport system**. There are four main things in blood...*

Red Blood Cells Carry **Oxygen**

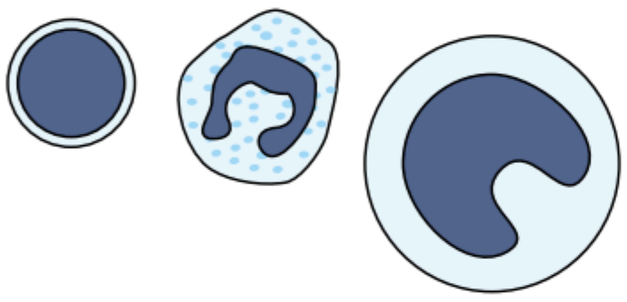
Red blood cells carry oxygen from the lungs to all the cells in the body.

- Their shape is a biconcave disc (like a doughnut) — this gives a large surface area for absorbing oxygen.
- They have no nucleus — this gives more room to carry oxygen.
- They contain a red pigment called haemoglobin. In the lungs, haemoglobin binds to oxygen to become oxyhaemoglobin. In body tissues, oxyhaemoglobin splits up into haemoglobin and oxygen, to release oxygen to the cells.



More red blood cells means more oxygen can get to your cells.

White Blood Cells **Defend Against Infection**



1) Some can change shape to gobble up unwelcome microorganisms, in a process called phagocytosis.
2) Others produce antibodies to fight microorganisms, as well as antitoxins to neutralise any toxins produced by the microorganisms.
3) Unlike red blood cells, they do have a nucleus.

Platelets Help Blood **Clot**

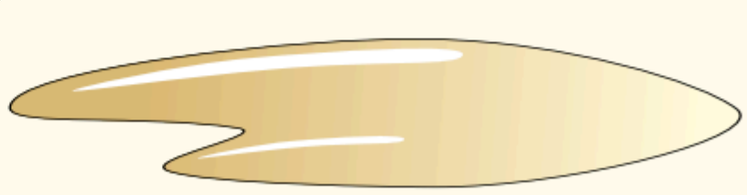
1) These are small fragments of cells. They have no nucleus.
2) They help the blood to clot at a wound — to stop all your blood pouring out and to stop microorganisms getting in.

Lack of platelets can cause excessive bleeding and bruising.

Plasma is the Liquid That **Carries Everything** in Blood

This is a pale straw-coloured liquid which carries just about everything:

- Red and white blood cells and platelets.
- Nutrients like glucose and amino acids. These are the soluble products of digestion which are absorbed from the gut and taken to the cells of the body.
- Carbon dioxide from the organs to the lungs.
- Urea from the liver to the kidneys.
- Hormones
- Proteins
- Antibodies and antitoxins produced by the white blood cells.

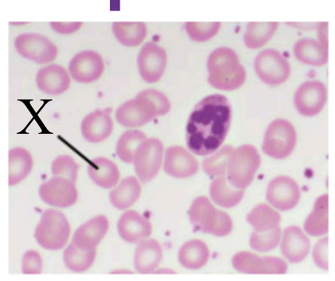


Blood — red blood cells, white blood cells, platelets and plasma

Blood tests can be used to diagnose loads of things — not just disorders of the blood. This is because the blood transports so many chemicals produced by so many organs...

Q1 Describe the purpose of platelets in blood. [1 mark]

Q2 State the function of the cell labelled X in the image on the right. [1 mark]



Q2 Video Solution

Warm-Up & Exam Questions

There are some nice diagrams to learn on the previous few pages. If you don't bother, you'll feel pretty silly if you turn over the exam paper and the first question asks you to label a diagram of the heart. Just saying... Anyway, let's see if these questions get your blood pumping...

Warm-Up Questions

1) What is the name of the tubes that the trachea splits into?
2) Which chamber of the heart does the pulmonary artery come from?
3) What is an artificial pacemaker?
4) What do veins do?
5) Which component of blood is made from small fragments of cells?

Exam Questions

1 A student ran for 12 minutes. GRADE 4-6

1.1 During this 12 minute run, the student took 495 breaths.
Calculate the student's average breathing rate in breaths per minute.

[1 mark]

1.2 The student's heart rate was measured before and during the run.
Before the run, the student's heart rate was at its natural resting rate.
Outline how natural resting heart rate is controlled.

[1 mark]

2 The cell shown in **Figure 1** transports oxygen around the body. GRADE 6-7

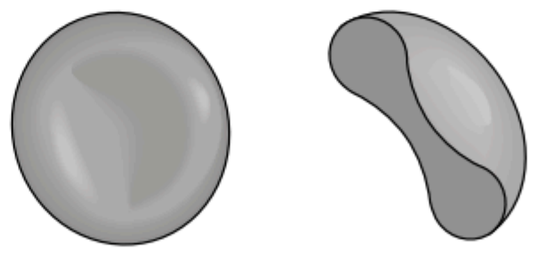
2.1 Explain how this cell's shape is adapted for transporting oxygen.

[1 mark]

2.2 Describe and explain **one** other way in which this cell is adapted for carrying oxygen.

[2 marks]

Figure 1



View from above

Cut through view

2.3 Name the other main type of blood cell, and state its function.

[2 marks]

Blood cells are carried in the bloodstream inside blood vessels.

2.4 Capillaries are one type of blood vessel.
Outline how the structure of a capillary enables it to carry out its function.

[4 marks]

2.5 Blood flows through different types of blood vessels at different rates.
The volume of blood that passed through an artery in 150 seconds was 1155 ml.
Calculate the rate of blood flow through the artery in ml/min.

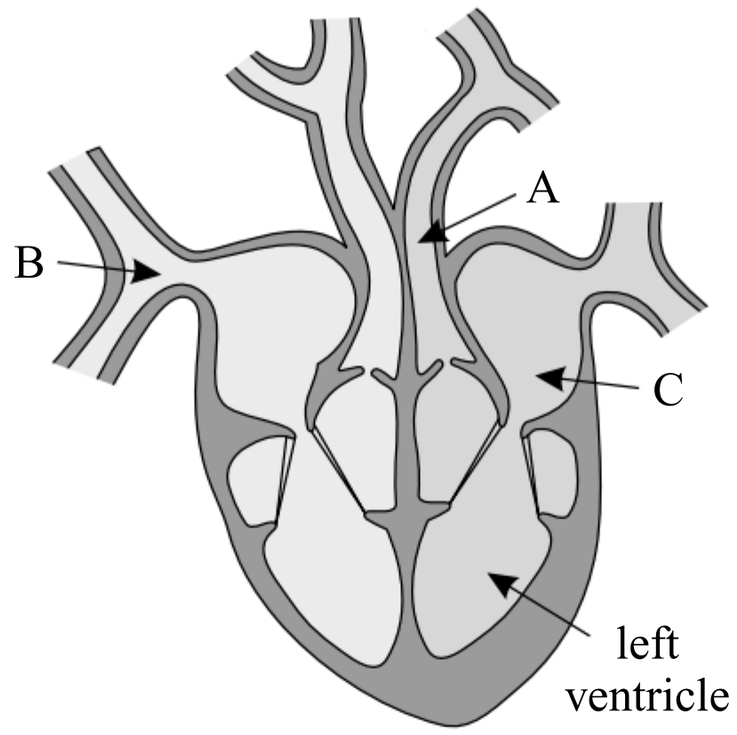
[2 marks]

Exam Questions

3 **Figure 2** shows the human heart and four blood vessels, as seen from the front. The left ventricle has been labelled.

GRADE 6-7 GRADE

Figure 2



3.1 Name the parts labelled **A**, **B** and **C**.

[1 mark]

3.2 Describe the function of the left ventricle.

[1 mark]

3.3 What is the function of the valves in the heart?

[1 mark]

3.4* Describe how deoxygenated blood from the body passes through the heart to reach the lungs.

[4 marks]

4 A student did an experiment to compare the elasticity of arteries and veins. An artery and a vein were dissected from a piece of fresh meat. A 5 cm length of each vessel was taken and different masses were hung on it. The student measured how much it stretched. The results are shown in **Table 1**.

GRADE 7-9 GRADE

Table 1

Mass added (g)	Length of blood vessel (mm)	
	Artery	Vein
0	50	50
5	51	53
10	53	56
15	55	59
20	56	-

4.1 Suggest **one** way in which the student could tell which was the artery and which was the vein when dissecting the meat.

[1 mark]

4.2 Which vessel stretched more easily?
Suggest why this was.

[1 mark]

Revision Summary Test for Topic 2a

Now that you've digested that little lot, there are things you can do to check you've taken it all in:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Cell Organisation (p.26) ☑

1) Name the basic building blocks that make up all living organisms.
2) What is a tissue? Give three examples of tissues in the human body.
3) What is an organ? Give three examples of human organs.
4) What is meant by the term 'organ system'? Give an example of an organ system.

The Role of Enzymes and Food Tests (p.27-31) ☑

5) What are enzymes? Why are enzymes vital for cells?
6) Draw a diagram to show the 'lock and key' model of enzyme action.
7) Why do enzymes usually only catalyse one reaction?
8) What does it mean when an enzyme has been denatured?
9) Describe and explain how each of the following factors affects the rate of an enzyme-controlled reaction: a) temperature, b) pH.
10) Describe how you could investigate the effect of pH on the rate of amylase activity.
11) Give two formulas that you could use for calculating the rate of an enzyme-controlled reaction.
12) Name the smaller molecules that make up: a) carbohydrates, b) proteins, c) lipids.
13) Name three types of digestive enzyme. For each one, say what it breaks down.
14) What are the products of digestion used for?
15) Where is bile produced and stored? What does it do?
16) Where are these enzymes made and where do they act? a) amylase b) protease c) lipase
17) Describe how to produce a food sample to use in a food test.
18) Describe tests for: a) sugars, b) starch, c) proteins, d) lipids.

The Lungs and Circulatory System (p.34-37) ☑

19) Describe the route that air takes from when it is breathed in to when it reaches the alveoli.
20) Explain the role that alveoli play in gas exchange.
21) Sketch a diagram of the heart. Label the chambers and major blood vessels, then draw arrows to show the direction that blood flows into and out of each chamber.
22) Why does the heart have valves?
23) Which blood vessel do the coronary arteries branch off, and what is their role?
24) Explain why the circulatory system in humans is described as a 'double circulatory system'.
25) How is the resting heart rate controlled in a healthy heart?
26) Name the three main types of blood vessel. Describe how each one is adapted to its function.
27) How would you calculate the rate of blood flow in a vessel?
28) What is the role of red blood cells? How are they adapted to this function?
29) What is the role of white blood cells? How are they adapted to this function?
30) What are platelets, and what do they do?
31) What is the role of plasma?

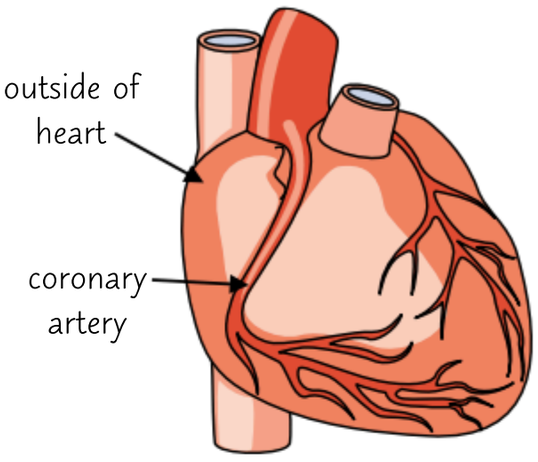
Cardiovascular Disease

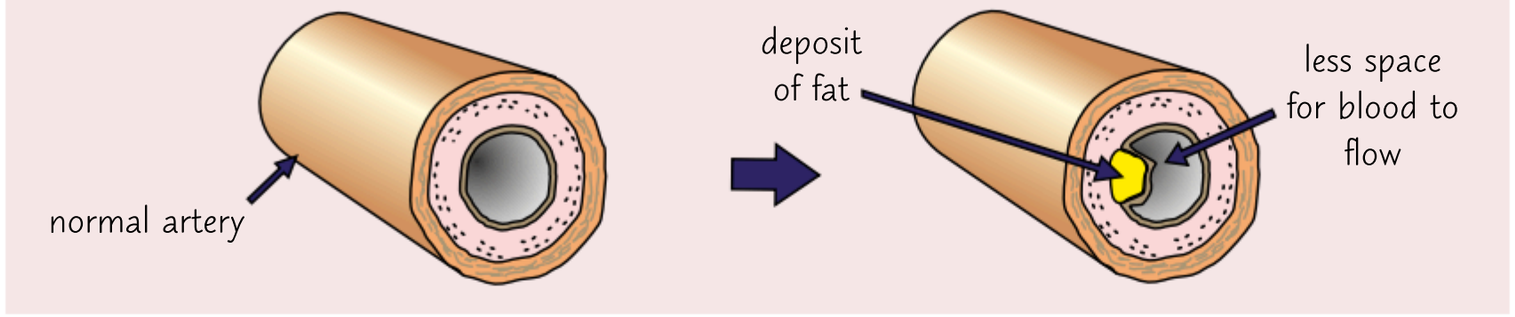
Quick Quiz

Cardiovascular disease *is a term used to describe diseases of the heart or blood vessels, for example coronary heart disease. This page tells you all about how* ***stents*** *and* ***statins*** *are used to combat coronary heart disease.*

Coronary Heart Disease is when Coronary Arteries Get Blocked

1) The coronary arteries supply blood to the muscle of the heart.
2) Layers of fatty material can build up in the coronary arteries. This makes them narrow, which restricts blood flow.



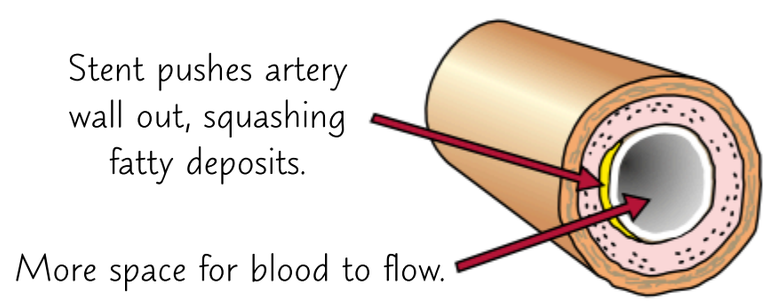


3) Restricted blood flow reduces the amount of oxygen that reaches the heart muscle, which can cause a heart attack.

Stents are Tubes Inserted into Arteries

Stents are tubes that can be inserted to keep coronary arteries open, making sure blood can get to the heart muscle. This lowers the risk of a heart attack in people with coronary heart disease.

Stents have pros and cons:



Advantages of stents	Disadvantages of stents
• Effective for a long time. • Recovery time from the surgery is relatively quick.	• Risk of complications (e.g. heart attack) during operation, or infection following surgery. • Risk of a blood clot developing near the stent.

This is called a thrombosis.

Statins are Drugs That Reduce 'Bad' Cholesterol

1) Too much of a certain type of cholesterol (known as 'bad' or LDL cholesterol) in the bloodstream can cause fatty deposits to form in arteries.
2) Statins reduce the amount of this 'bad' cholesterol, slowing down the formation of fatty deposits.

Cholesterol is an essential lipid that your body produces and needs to function properly.

Advantages of statins	Disadvantages of statins
• As well as reducing the amount of 'bad' cholesterol, statins can increase the amount of beneficial cholesterol (known as 'good' or HDL cholesterol). This type can help remove 'bad' cholesterol from the blood. • Some studies suggest that statins may also help prevent some other diseases.	• Statins are a long-term drug that must be taken regularly. There's a risk of forgetting to take them. • Statins can cause negative side effects, such as headaches. Some side effects can be serious, e.g. kidney failure, liver damage and memory loss. • The effect of statins isn't instant. It takes time for their effect to kick in.

Coronary heart disease is a type of cardiovascular disease

Coronary heart disease is caused by the arteries being blocked by fatty material — if the heart muscle can't get enough oxygen, then it can't work properly. And if the heart can't work properly, well, unfortunately, you're in trouble.

Quick Quiz

More on Cardiovascular Disease

*You've read about stents and statins, so now here's a page about some **other treatments** for cardiovascular disease — transplanting **donor hearts**, using **artificial hearts** and replacing **heart valves**.*

Donor Organs Can be Transplanted into a Patient with Heart Failure

Heart transplants use donor hearts from people who have recently died.

PROS

- Can allow someone with heart failure to live longer.
- The patient will often enjoy a better quality of life and have more energy.

If the lungs are also damaged a heart and lung transplant may be needed.

CONS

- Only possible if a suitable donor heart is available.
- Transplant surgery can lead to bleeding and infection.
- The immune system may recognise a donor heart as 'foreign' and reject it.
- Patients must take drugs for the rest of their lives to prevent rejection — these drugs can cause other health problems.

An Artificial Heart Can Pump Blood for a Person with Heart Failure

Artificial hearts are mechanical devices that can keep a person alive.

PRO

Artificial hearts are less likely to be rejected than donor hearts.

This is because they're made from metals or plastics, rather than from living tissue.

CONS

- Surgery to fit an artificial heart can lead to bleeding and infection.
- Parts of an artificial heart can wear out or the electrical motor could fail.
- Blood doesn't flow through artificial hearts as smoothly as it does through natural hearts — this can cause blood clots and lead to strokes. To avoid this, blood-thinning drugs must be taken — however, these can cause problems with bleeding if the person is hurt in an accident.

Artificial hearts are usually used temporarily, until a donor heart is found, or to allow someone's heart to rest and heal. However, they're sometimes a permanent solution.

Faulty Heart Valves Can Be Replaced

1) The valves in the heart can be damaged or weakened by heart attacks, infection or old age. E.g.:
   - Valve tissue may stiffen so a valve can't open properly.
   - A valve may become leaky, allowing blood to flow in both directions rather than just forward.
2) Faulty valves mean blood doesn't circulate effectively.
3) There are two types of replacement valve that can be used:

BIOLOGICAL VALVES — from humans or other mammals (e.g. cows or pigs)

MECHANICAL VALVES — these are man-made

4) Replacing a valve is a much less drastic procedure than a whole heart transplant. But it is still major surgery and there can still be problems with blood clots.

Don't lose heart...

You could be asked to evaluate treatments for cardiovascular disease. Don't panic — just use any information you're given and your own knowledge to weigh up the advantages and disadvantages. Make sure your answer doesn't just focus on one side — e.g. don't just talk about the advantages — and don't forget to include a justified conclusion.

Warm-Up & Exam Questions

Hopefully I've persuaded you by now that it's a good idea to try these questions. Believe me, when you're sitting with your real exam paper in front of you, you'll feel so much better knowing that you've already been through loads of practice questions. So off you go...

Warm-Up Questions

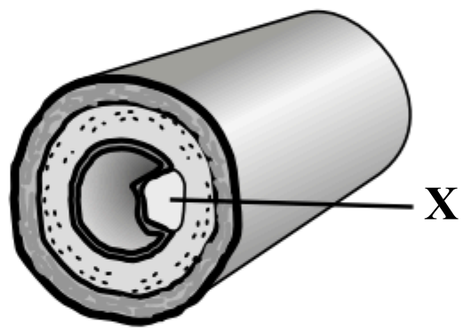
1) Which vessels are affected in coronary heart disease?
2) a) What device can be used to keep an artery open?
   b) Give three risks associated with getting one of these devices fitted.
3) What is an artificial heart?

Exam Questions

1 Coronary heart disease is the main cause of death worldwide. **Figure 1** shows a cross-section of a blood vessel in someone with coronary heart disease.

GRADE 4-6 GRADE

Figure 1



1.1 Name the substance labelled **X** on **Figure 1**.

[1 mark]

1.2 Explain how the presence of this substance can affect oxygen delivery to the heart muscle.

[2 marks]

2 A patient is taken into hospital.
They are diagnosed as having a leaky heart valve.

GRADE 6-7 GRADE

2.1 Explain why a leaky valve could cause a health problem.

[2 marks]

The doctor decides that a surgeon should replace the valve.

2.2 Name and describe the **two** types of valve that the surgeon might use.

[4 marks]

2.3 Suggest **one** disadvantage of replacing the faulty valve.

[1 mark]

3 Some people with cardiovascular disease take statins.
These are drugs that need to be taken every day.

GRADE 6-7 GRADE

3.1 Outline why statins are sometimes used in the treatment of coronary heart disease.

[2 marks]

3.2 Suggest **one** disadvantage to a patient of taking statins.

[1 mark]

Health and Disease

Quick Quiz

Try as we might, it's pretty unlikely that we'll be in tip-top condition for all of our lives — unfortunately, ***disease*** *tends to get us all at some point. There are lots of* ***different types*** *of diseases we could get. Read on...*

Diseases are a Major Cause of Ill Health

1) Health is the state of physical and mental wellbeing. Diseases are often responsible for ill health.
2) Diseases can be communicable or non-communicable:

Communicable diseases

- These spread from person to person or between animals and people. They're caused by things like bacteria, viruses, parasites and fungi.
- Examples include measles and malaria (p.56-57).

Communicable diseases are often described as contagious or infectious.

Non-communicable diseases

- These can't spread between people or between animals and people.
- They generally last a long time and get worse slowly.
- Examples include asthma, cancer and coronary heart disease (see page 41).

Different Types of Disease Sometimes Interact

Diseases can cause other health issues that aren't obviously related to them. Here are a few examples:

Immune system problems (these affect the body's ability to defend itself against pathogens — see p.59). —increase the risk of→ Communicable diseases such as influenza (flu).

Pathogen is just the fancy term for a microorganism that can cause a disease when it infects its host.

Immune system reactions caused by infection by a pathogen. —can trigger→ Allergic reactions such as skin rashes, or worsened symptoms of asthma.

Severe physical health problems, particularly those that impact a person's ability to carry out everyday activities or affect their life expectancy. —can trigger→ Mental health issues such as depression.

Certain viruses —can trigger→ Some types of cancer

- Some types of hepatitis virus can cause long-term liver infections, where the virus lives in the cells. This can increase the risk of developing liver cancer.
- Infection with HPV (human papilloma virus) can cause cervical cancer in women.

Factors Other than Disease can Affect Your Health

1) Diet A good, balanced diet provides your body with everything it needs in the right amounts. A poor diet can negatively affect your physical and mental health.

2) Stress Being constantly under lots of stress can lead to health issues.

3) Life Situation E.g. whether you have access to:

- medicines to treat illness
- things that can prevent you from getting ill, such as healthy food, or condoms.

Condoms prevent the transmission of some sexually transmitted diseases.

Communicable diseases can spread...

...but non-communicable diseases can't. Remember that — you don't want to lose any marks by mixing them up.

Non-Communicable Disease Risk Factors

Quick Quiz

*You've probably heard the term '**risk factor**' before. This page has all you need to know about them.*

Risk Factors **Don't Guarantee** that Someone Will Get a **Disease**

RISK FACTORS are things that are linked to an INCREASE in the LIKELIHOOD that a person will develop a certain disease during their lifetime.

There are different types of risk factor, including:

1. aspects of a person's lifestyle (e.g. how much exercise they do).
2. substances in the environment (e.g. air pollution).
3. substances in a person's body (e.g. asbestos fibres).

Asbestos was a material used in buildings, but it is no longer used as the fibres can build up in airways and cause diseases such as cancer later in life.

Many non-communicable diseases are caused by several risk factors interacting rather than one factor alone.

The Incidence of **Non-communicable** Diseases Varies at **Global**, **National** and **Local** Levels. E.g.

- In developed countries, non-communicable diseases are more common as people generally have a higher income and can buy high-fat food.
- Nationally, people from deprived areas are more likely to smoke, have a poor diet and not exercise. This means a higher incidence of cardiovascular disease, obesity and Type 2 diabetes in those areas.
- Your individual choices affect the local incidence of disease.

Some Risk Factors Can **Cause** a Disease **Directly**...

Smoking causes cardiovascular disease, lung disease and lung cancer. It damages the walls of arteries and the cells in the lining of the lungs.

Obesity is thought to cause Type 2 diabetes. It makes the body less sensitive, or more resistant, to insulin — so it struggles to control the glucose concentration in the blood.

Drinking too much alcohol causes liver disease.

It can affect brain function too — it damages nerve cells in the brain, causing the brain to lose volume.

Exposure to carcinogens can cause cancer. Carcinogens are substances or radiation that cause cancer, e.g. ionising radiation from X-rays.

Smoking or drinking alcohol when pregnant can cause lots of health problems for the unborn baby.

...But Other Risk Factors **Can't Directly** Cause Disease

See p.179.

Scientists identify risk factors by looking for correlations in data, but correlation may not equal cause.

A lack of exercise and a high fat diet are linked to an increased chance of cardiovascular disease, but they can't cause it directly. It's the resulting high blood pressure and high 'bad' cholesterol levels (see p.41) that can actually cause it.

Non-Communicable Diseases Can Be **Costly**

This not only affects those with the diseases, but their loved ones too.

Human Cost	Tens of millions of people die from non-communicable diseases each year. People with these diseases may also have a lower quality of life.
Financial Cost	• The family of someone with a disease may have to move or adapt their home. Also, the family's income will be reduced if the person has to give up work or dies. • On a national level, the cost of researching and treating diseases is huge. A reduction in the number of people able to work also affects a country's economy.

Lots of things affect health, and ill health can be costly

You might be asked to interpret data about risk factors. See page 179 for a few tips on what you can and can't say.

Quick Quiz

Cancer

*The more we understand about **cancer**, the better our chances of **avoiding** and **beating** it. Hoorah.*

Cancer Starts with **Changes to Cells**

Changes occur to cells. → Abnormal cells form. → Uncontrolled cell growth and division. → A tumour may form (a mass of cells).

There are two types of tumour:

Benign tumours are NOT cancerous	Malignant tumours ARE cancerous
• A benign tumour grows until there's no more room.	• A malignant tumour grows and spreads to neighbouring healthy tissues.
• It stays in one place (usually within a membrane) rather than invading other tissues.	• Malignant cells can break off and spread around the body through the bloodstream. These cells then invade other healthy tissues and form secondary tumours.
• This type isn't normally dangerous.	• Malignant tumours can be fatal — they are cancers.

Risk Factors Can **Increase** the **Chance** of **Some Cancers**

Anyone can develop cancer. Having risk factors doesn't mean that you'll definitely get cancer. It just means that you're at an increased risk of developing the disease.

Risk Factors Can Be Associated With **Lifestyle**

Smoking	It's well-known that smoking is linked to lung cancer, but research has also linked it to other types of cancer too, including mouth, bowel, stomach and cervical cancer.
Obesity	Obesity has been linked to many cancers, including bowel, liver and kidney cancer.
UV exposure	Too much exposure to UV radiation from the Sun (or from sun beds) leads to an increased chance of developing skin cancer.
Infection with certain viruses	This has been shown to increase the chances of developing certain types of cancer. For example, infection with hepatitis B and hepatitis C viruses increases the risk of developing liver cancer. The likelihood of becoming infected with these viruses may depend on lifestyle — e.g. they can be spread between people through unprotected sex or sharing needles.

Risk Factors Can Also Be Associated With **Genetics**

1) Sometimes you can inherit faulty genes that make you more susceptible to cancer.
2) For example, mutations (changes) in the BRCA genes have been linked to an increased likelihood of developing breast and ovarian cancer.

Cancer survival rates have increased over the years. This is due to medical advances leading to improved treatment, earlier diagnosis and increased screening.

It's hard to avoid all the risk factors associated with cancer...

...but remember, risk factors that cause the disease don't mean you'll definitely get it — they just increase the chance.

Warm-Up & Exam Questions

It's time for some more questions — don't just assume that you've remembered everything you just read on the past few pages. Give these a go, and then go back over anything that you struggled with.

Warm-Up Questions

1) What is meant by 'health'?
2) What does it mean if a disease is 'communicable'?
3) Give an example of a type of risk factor other than an aspect of a person's lifestyle.
4) Give an example of a national financial cost associated with non-communicable diseases.

Exam Questions

1 Many diseases have risk factors. GRADE 4-6 GRADE

1.1 What is meant by the term risk factor?

[1 mark]

1.2 Give **one** risk factor for lung disease.

[1 mark]

1.3 Name **one** carcinogen that is a risk factor for cancer.

[1 mark]

2 Cancer results from a tumour. GRADE 4-6 GRADE

2.1 What causes tumours to form? Tick **one** box.

- ☐ Uncontrolled cell division.
- ☐ Slow cell division.
- ☐ Cell division stopping.
- ☐ Cells dying.

[1 mark]

2.2 Name the type of tumour that causes cancer.

[1 mark]

2.3 Explain how secondary tumours form in the body.

[2 marks]

3 Diseases can be communicable or non-communicable. GRADE 6-7 GRADE

3.1 What is meant by a non-communicable disease?

[1 mark]

3.2 Influenza is a communicable disease.
Why might someone who has a problem with their immune system have an increased chance of suffering from influenza?

[1 mark]

Quick Quiz

Plant Cell Organisation

*You saw on page 26 how animals keep their **specialised cells** neat and tidy — plants are in on the act too.*

Plant Cells are Organised Into **Tissues**, **Organs** and **Organ Systems**

Plants CELLS are arranged to form TISSUES — see below.

↓

Tissues are organised to form ORGANS, like stems, roots and leaves.

↓

Plant organs work together to make ORGAN SYSTEMS. These perform the tasks that a plant needs to carry out to survive and grow — for example, transporting substances around the plant.

The **Leaf** is an **Organ** Made Up of **Several Types of Tissue**

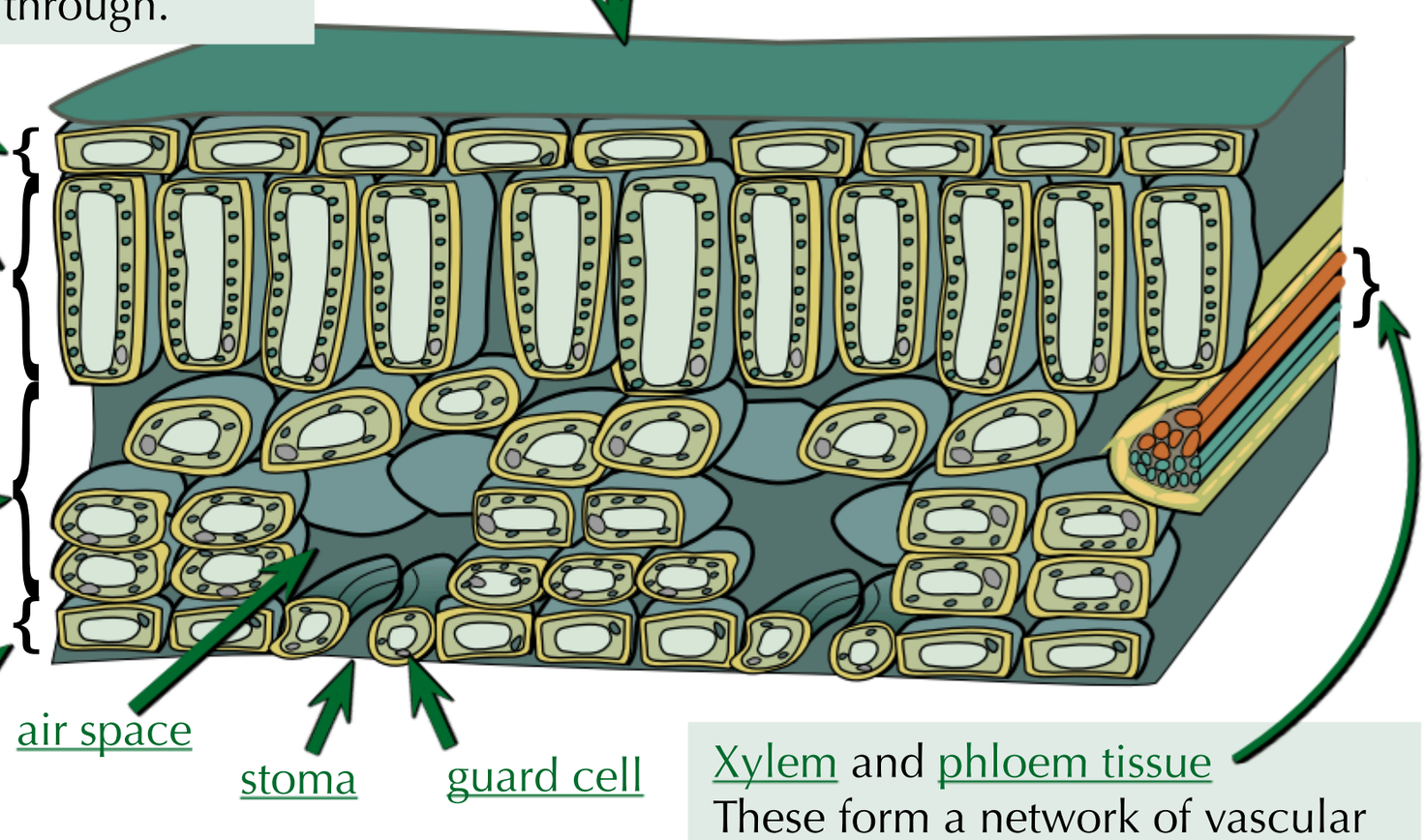
You need to know how the structures of the tissues that make up the leaf are related to their function:

Epidermal tissue
This covers the whole plant. The upper epidermis is transparent to allow light to pass through.

Palisade mesophyll tissue
This contains lots of chloroplasts in which photosynthesis takes place. It's near the top of the leaf where it can get most light.

Spongy mesophyll tissue
This contains big air spaces to allow gases to diffuse in and out of cells.

Epidermal tissue
The lower epidermis has little holes called stomata, which let CO_2 diffuse into the leaf. Guard cells control the opening and closing of stomata in response to environmental conditions.



The big air spaces and the presence of stomata/guard cells are examples of adaptations for efficient gas exchange.

Meristem Tissue Isn't Found in the Leaves

Meristem tissue is found at the growing tips of shoots and roots.
It's able to differentiate (change) into lots of different types of plant cell, allowing the plant to grow.

Make sure you know the structure and function of the tissues in a leaf

To help you to remember this stuff, try drawing your own leaf diagram and include a description of each type of tissue.

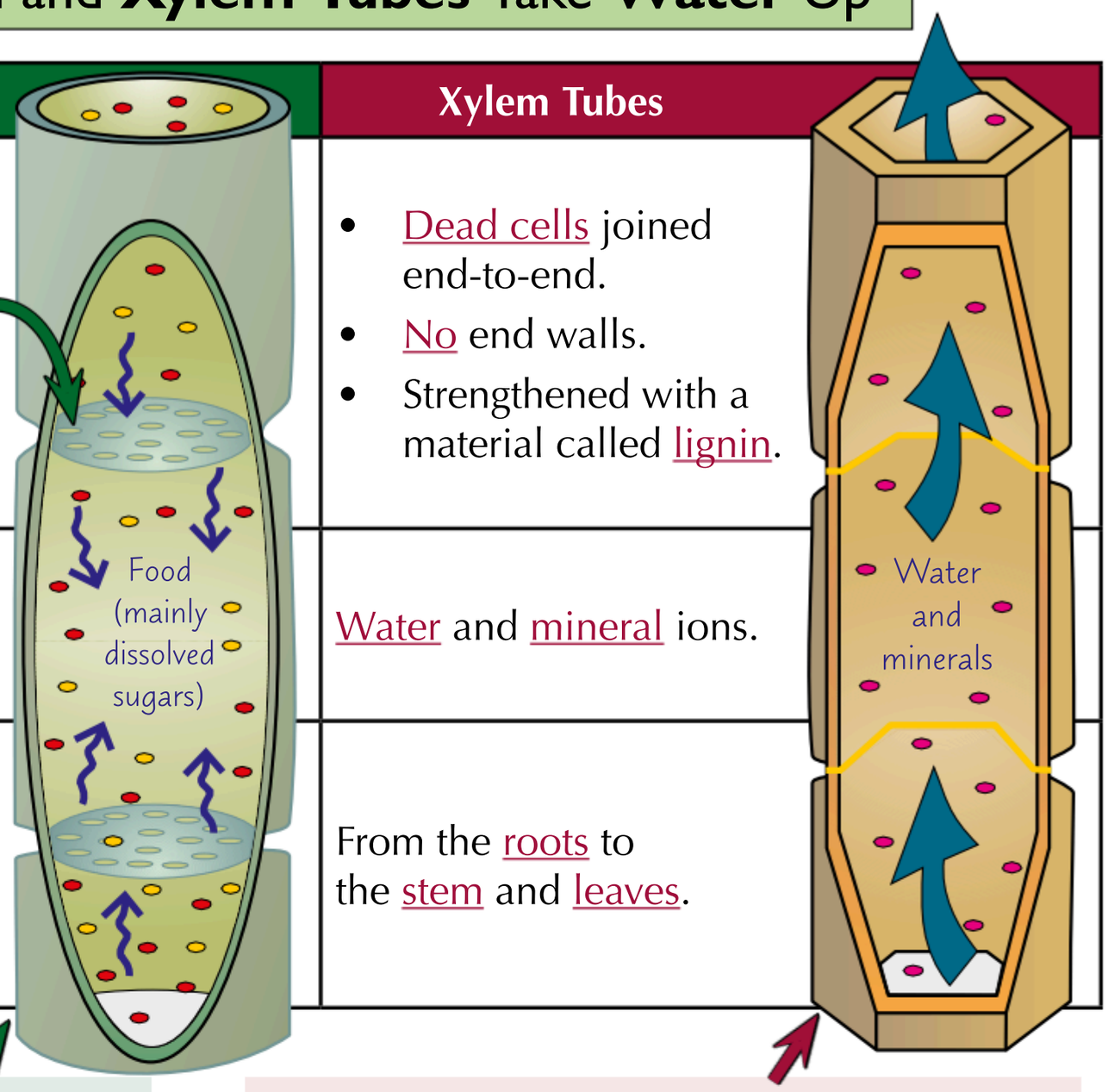
Transpiration and Translocation

Quick Quiz

*Flowering plants have **two** separate types of vessel — **xylem** and **phloem** — for transporting stuff. **Both** types of vessel go to **every part** of the plant, but they're totally **separate**. Xylem is part of the **transpiration stream**.*

Phloem Tubes Transport Food and Xylem Tubes Take Water Up

	Phloem Tubes	Xylem Tubes
Structure	• Columns of elongated living cells. • Small pores in end walls to allow cell sap to flow. Cell sap is made up of the substances being transported and water.	• Dead cells joined end-to-end. • No end walls. • Strengthened with a material called lignin.
Substances transported	Food substances (mainly dissolved sugars made in the leaves).	Water and mineral ions.
Where from and to?	From the leaves to the rest of the plant (either for immediate use, e.g. in growing regions, or for storage). Goes in both directions.	From the roots to the stem and leaves.



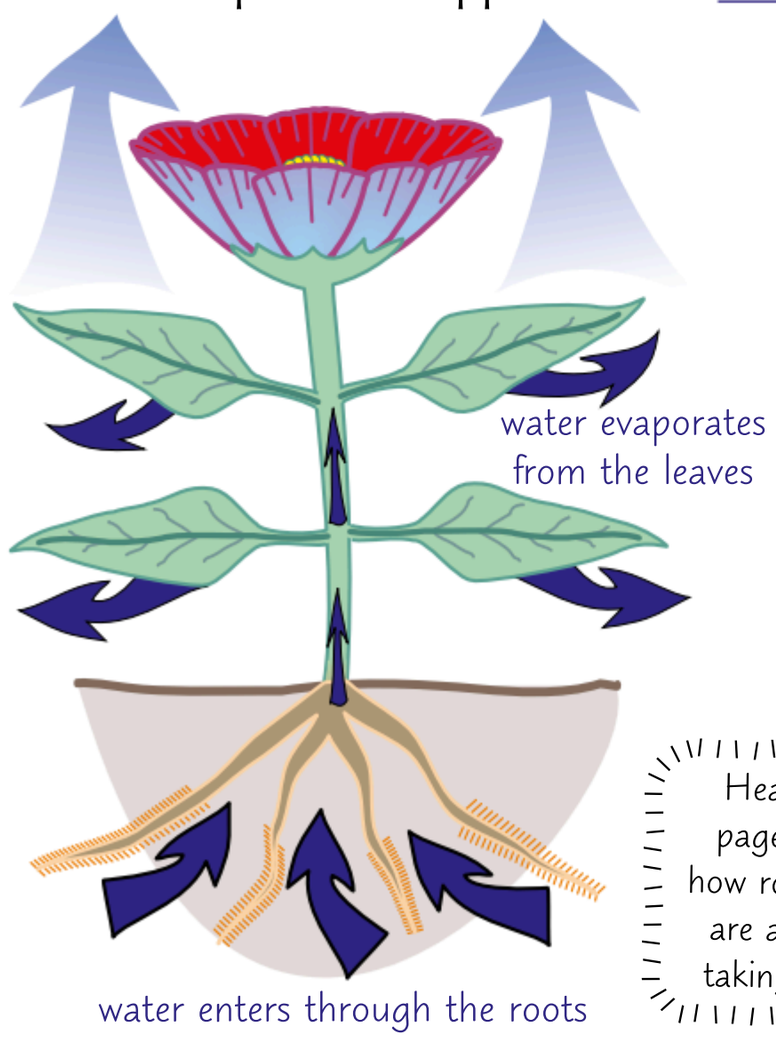
The transport of dissolved sugars through the phloem is called translocation.

The movement of water through the xylem is part of the transpiration stream — see below.

Transpiration is the Loss of Water from the Plant

Video

Most transpiration happens at the leaves. Here's the process:



1. Water evaporates and diffuses (page 17) out through stomata.
2. There's now a slight shortage of water in the leaf, so more water is drawn up from the rest of the plant through the xylem to replace it.
3. This in turn means more water is drawn up from the roots.

This results in a...

constant TRANSPIRATION STREAM through the plant.

Head back to page 19 to see how root hair cells are adapted for taking up water.

Transpiration is a side-effect of the way leaves are adapted for photosynthesis. The stomata allow gases to be exchanged easily (see p.22), but because there's more water inside the plant than in the air outside, the water diffuses out of the stomata and is lost.

Xylem vessels carry water, phloem vessels carry sugars

Make sure you don't get your phloem mixed up with your xylem. Remember that phloem transports substances in both directions, but xylem only transports things upwards — xylem to the sky. It might just bag you a mark or two...

Quick Quiz

Transpiration and Stomata

*Sorry, more on **transpiration**. But then it's a quick dash through **stomata** and out of the other end of the topic.*

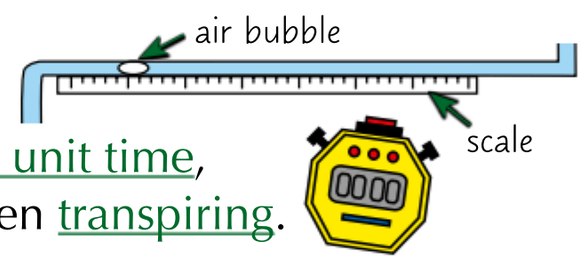
Transpiration Rate is Affected by **Four Main Things**

Factor	Explanation
LIGHT INTENSITY Brighter = faster transpiration	Stomata begin to close as it gets darker. (Photosynthesis can't happen in the dark, so stomata don't need to be open to let CO_2 in.) When the stomata are closed, very little water escapes. This slows down transpiration.
TEMPERATURE Warmer = faster transpiration	When it's warm the water particles have more energy to evaporate and diffuse out of the stomata.
HUMIDITY Drier air = faster transpiration	Humid air has a lot of water in it already, so there's not much of a difference between the inside and outside of the leaf. Diffusion happens fastest if there's a really high concentration in one place, and a really low concentration in the other.
AIR MOVEMENT Good air movement = faster transpiration	Poor air movement (e.g. no wind) means water vapour remains around a leaf. A high concentration of water particles outside the leaf as well as inside it means slow diffusion out of the leaf. Good air movement blows the water vapour away, maintaining a low concentration of water in the air outside the leaf compared to inside. Diffusion then happens quickly.

Estimate **Transpiration Rate** using a **Potometer**

A potometer allows you to measure the water uptake of a plant per unit time, e.g. per hour. You can assume this is the same as the rate of transpiration.

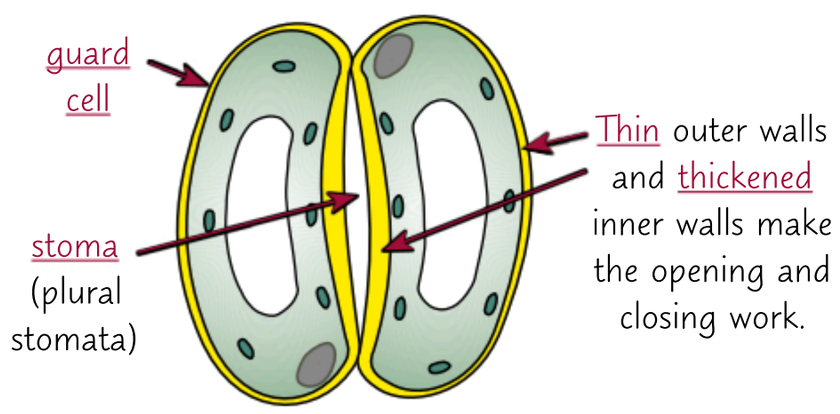
Set up the potometer as in the diagram on page 183, and record the starting position of the air bubble. Record the distance the bubble moves along the scale per unit time, e.g. per hour. This indicates how quickly the plant has been transpiring.



Keep the conditions constant throughout the experiment, e.g. the temperature.

Guard Cells Are Adapted for **Gas Exchange** and **Controlling Water Loss**

1) Guard cells have a kidney shape which opens and closes the stomata (page 22).



- When the plant has lots of water, guard cells fill with it and go plump and turgid. This opens the stomata so gases can be exchanged for photosynthesis.
- When the plant is short of water, the guard cells lose water and become flaccid, closing the stomata. This helps stop too much water vapour escaping.

2) Guard cells are also sensitive to light and close stomata at night.

This saves water without losing out on photosynthesis.

3) You usually find more stomata on the undersides of leaves than on the top. The lower surface is shaded and cooler — so less water is lost through the stomata than would be if they were on the upper surface.

The opening and closing of stomata allows plants to survive

Q1 Video Solution

Different leaves will have different distributions of stomata. You can peel the epidermal tissue off some leaves and mount them on microscope slides (see page 6) to compare them.

Q1 Aloe vera plants grow in hot, dry areas. Primroses grow in cool, wet areas. Predict which plant will have fewer stomata per cm^2 on the underside of its leaves. Explain your answer. [2 marks]

Warm-Up & Exam Questions

Just a few simple Warm-Up Questions and a few slightly harder Exam Questions stand between you and mastering cell organisation and transport in plants...

Warm-Up Questions

1) Which layer of plant tissue contains lots of chloroplasts?
2) Describe the characteristics of meristem tissue.
3) True or false? Substances pass in both directions through xylem vessels.

Exam Questions

1 Leaves contain many types of tissue. GRADE 4-6 GRADE

Which type of plant tissue contains air spaces for the diffusion of gases?
Tick **one** box.

- ☐ epidermal tissue
- ☐ palisade mesophyll tissue
- ☐ spongy mesophyll tissue
- ☐ meristem tissue

[1 mark]

2 Plants absorb water and mineral ions through their root hair cells. GRADE 4-6 GRADE

2.1 Name the vessels that transport water and mineral ions from the roots of a plant to the leaves.

[1 mark]

2.2 Describe the structure of the vessels you named in **2.1**.

[3 marks]

2.3 Name the process of water transport through a plant.

[1 mark]

3 Aphids are insects which feed on plant cell sap. Cell sap is the name given to the liquids carried around the plant in transport vessels. GRADE 6-7 GRADE

3.1 The cell sap the aphids feed on contains sugars.
What type of transport vessel does the cell sap come from?

[1 mark]

3.2 Explain how the structure of these transport vessels is adapted to their function.

[2 marks]

3.3 Name the movement of cell sap through the plant's transport vessels.

[1 mark]

Exam Questions

4 A student used a rose plant to carry out some investigations. GRADE 7-9 GRADE

4.1 The student cut several stems from the plant and put them in a glass vase on a kitchen windowsill.
A layer of oil was added to the surface of the water to prevent evaporation.
Over the next few days, the student noticed that the level of water in the vase gradually decreased.
How would the reduction of water in the vase differ if the kitchen was not as warm?
Explain your answer.

[3 marks]

4.2* The student investigated the appearance of guard cells on the underside of a rose leaf at different times on a humid day. A leaf was taken from the rose plant at 10 am, 4 pm and 11 pm and immediately examined under a microscope. The student drew diagrams to show the appearance of the guard cells at each time, as shown in **Figure 1**.

Figure 1

A **B** **C**

Suggest which diagram (A-C), shows the appearance of the guard cells observed at 11 pm.
Explain your answer.

[4 marks]

5 A student was investigating transpiration in basil plants under different conditions. GRADE 7-9 GRADE
Twelve plants were used, with three plants in each of the four different conditions.
The plants were weighed before and after the experiment. The student calculated the % loss in the mass per day and recorded the results in **Table 1**.

Table 1

plant	in a room (% loss in mass)	next to a fan (% loss in mass)	by a lamp (% loss in mass)	next to a fan and by a lamp (% loss in mass)
1	5	8	10	13
2	5	9	11	15
3	4	11	9	13
mean	4.7	9.3		13.6

5.1 Calculate the mean % loss in plant mass for the three plants by a lamp.

[2 marks]

5.2 Explain why the plants located next to a fan lost more mass than those in a still room.

[3 marks]

5.3 The student then covered the undersides of the leaves with petroleum jelly.
Suggest how this would affect the rate of transpiration from the leaves.

[2 marks]

5.4 Suggest how you could alter the student's experiment to investigate the effects of humidity on the rate of transpiration in basil plants.

[2 marks]

Revision Summary Test for Topic 2b

Summary Test

Well, that was a fact-packed section. To see what you remember, you can:

- Tackle the revision summary test below, or scan the QR code to do it online. The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Cardiovascular Disease (p.41-42) ☑

1) What is meant by 'cardiovascular disease'?
2) Explain what coronary heart disease is. How can it lead to a heart attack?
3) Describe what stents are. How do they treat coronary heart disease? Give two advantages and two disadvantages of them.
4) What do statins do? Give two advantages and two disadvantages of them.
5) Give the advantages and disadvantages of a heart transplant for treating someone with heart failure.
6) Give an advantage of an artificial heart. Then give three disadvantages of one.
7) Give two types of damage that can be seen in heart valves.
8) What problem does a damaged valve cause?
9) Give two types of replacement heart valve.

Health and Disease (p.44-46) ☑

10) Define 'health'.
11) Describe what communicable and non-communicable diseases are. Give an example of each type.
12) Some types of disease can cause other health problems. Describe four examples of this.
13) Other than disease, describe three other factors that can affect your health.
14) What is meant by risk factors for a disease? Give three examples of risk factors.
15) Suggest how the incidence of non-communicable disease varies at global, national and local levels.
16) Some risk factors cause disease or health issues directly. Describe two examples of this.
17) 'Some risk factors can't cause disease directly.' Explain this statement using an example.
18) Explain two types of cost associated with non-communicable diseases.
19) Describe how a tumour forms. Name the two types of tumour and say how they differ.
20) Describe four lifestyle factors that increase the risk of someone developing certain types of cancer.
21) Explain a risk factor for cancer that isn't related to lifestyle.

Plant Cell Organisation and Transport (p.48-50) ☑

22) Describe the relationship between plant cells, organs, tissues and organ systems.
23) Name four tissues found in leaves. How does the structure of each relate to its function?
24) Give two adaptations of leaves for efficient gas exchange.
25) What's so special about meristem tissue? Where is it found?
26) Describe the differences between the structures and functions of phloem and xylem tubes.
27) What's the difference between translocation and the transpiration stream?
28) What causes the transpiration stream through a plant?
29) List the four factors that affect transpiration rate. Explain why each of them affects the rate.
30) Outline how you could estimate the rate of transpiration, including the name of the apparatus.
31) Name the type of cell that helps open and close stomata. Explain how it does this.
32) Which part of the leaf has most stomata? Why is this?

Quick Quiz

Communicable Disease

*If you're hoping I'll ease you gently into this new topic... no such luck. Straight on to the **baddies** of biology.*

There Are **Several Types** of **Pathogen**

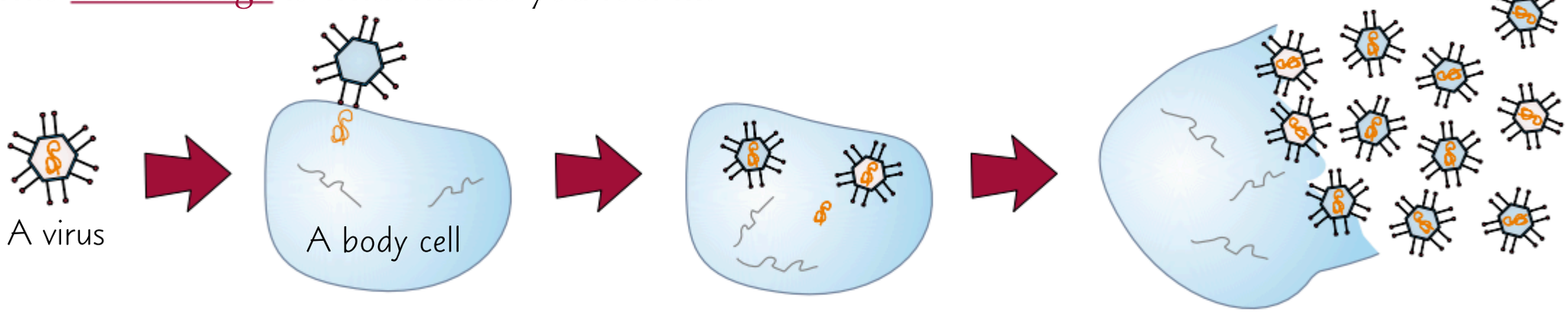
1) Pathogens are microorganisms that enter the body and cause disease.
2) They cause communicable (infectious) diseases — diseases that can easily spread (see p.44).
3) Both plants and animals can be infected by pathogens.

1. **Bacteria** Are Very Small **Living Cells**

- Bacteria are very small cells (about 1/100th the size of your body cells), which can reproduce rapidly inside your body.
- They can make you feel ill by producing toxins (poisons) that damage your cells and tissues.

2. **Viruses** Are **Not** Cells — They're Much Smaller

- Viruses are not cells. They're tiny, about 1/100th the size of a bacterium.
- Like bacteria, they can reproduce rapidly inside your body.
- They live inside your cells and replicate themselves using the cells' machinery to produce many copies of themselves. The cell will usually then burst, releasing all the new viruses.
- This cell damage is what makes you feel ill.



3. **Protists** are **Single-Celled Eukaryotes**

- There are lots of different types of protists. But they're all eukaryotes (see p.4) and most of them are single-celled.
- Some protists are parasites. Parasites live on or inside other organisms and can cause them damage. They are often transferred to the organism by a vector, which doesn't get the disease itself — e.g. an insect that carries the protist.

4. **Fungi** Come in Different Shapes

- Some fungi are single-celled. Others have a body which is made up of hyphae (thread-like structures).
- These hyphae can grow and penetrate human skin and the surface of plants, causing diseases.
- The hyphae can produce spores, which can be spread to other plants and animals.

Watch yourself, there are a lot of nasties out there...

Plants need to be worried too, as you'll find out in the next few pages. It's strange to think such small things can have such massive effects on your body, but bacteria and viruses in particular can multiply extremely quickly.

The Spread of Disease

Quick Quiz

*The bad news is that diseases can **spread** from person to person, and they can do that in lots of different ways. The good news is that there are ways of **reducing** the spread of disease and **protecting** ourselves from infection.*

Pathogens Can Be **Spread** in **Different Ways**

Here are a few that you need to know about:

Water

Some pathogens can be picked up by drinking or bathing in dirty water. E.g. cholera is a bacterial infection that's spread by drinking water contaminated with the diarrhoea of other sufferers.

Air

Pathogens can be carried in the air and can then be breathed in.
Some airborne pathogens are carried in the air in droplets produced when you cough or sneeze — e.g. the influenza virus that causes flu is spread this way.

Direct Contact

Some pathogens can be picked up by touching contaminated surfaces, including the skin. E.g. athlete's foot is a fungus which makes skin itch and flake off. It's most commonly spread by touching the same things as an infected person, e.g. shower floors and towels.

The **Spread** of Disease Can Be **Reduced** or **Prevented**

There are things that we can do to reduce, and even prevent, the spread of disease.

For example:

1) Being hygienic — Using simple hygiene measures can prevent the spread of disease. For example, doing things like washing your hands thoroughly before preparing food or after you've sneezed can stop you infecting another person.

2) Destroying vectors — By getting rid of the organisms that spread disease, you can prevent the disease from being passed on. Vectors that are insects can be killed using insecticides or by destroying their habitat so that they can no longer breed.

3) Isolating infected individuals — If you isolate someone who has a communicable disease, it prevents them from passing it on to anyone else.

4) Vaccination — Vaccinating people and animals against communicable diseases means that they are less likely to develop the infection and then pass it on to someone else.

There's more about how vaccination works on page 60.

There you go, basic hygiene can be a real life saver...

Make sure you know how pathogens can be spread and how that can be reduced. There are a few specific diseases on the next couple of pages that you need to know about for your exam, so when you're happy with this page, read on.

Quick Quiz

Viral and Fungal Diseases

*There are heaps of diseases caused by **viruses** or **fungi** — here are four of them...*

You Need to Know About **Three Viral Diseases**...

Tobacco Mosaic Virus (TMV)

- Tobacco mosaic virus (TMV) is a virus that affects many species of plants, e.g. tomatoes.
- It causes a mosaic pattern on the leaves of the plants — parts of the leaves become discoloured.
- The discolouration means the plant can't carry out photosynthesis as well, so the virus affects growth.

Measles

- Measles is a viral disease spread by droplets from an infected person's sneeze or cough.
- People with measles develop a red skin rash, and may have a fever (a high temperature).
- Measles can be very serious, or even fatal, if there are complications. For example, measles can sometimes lead to pneumonia (a lung infection) or inflammation of the brain (encephalitis).
- Most people are vaccinated against measles when young.

HIV

- HIV is a virus spread by sexual contact, or by exchanging bodily fluids such as blood. This can happen when people share needles, e.g. when taking drugs.
- HIV initially causes flu-like symptoms for a few weeks. Usually, the person doesn't then experience any symptoms for several years. During this time, HIV can be controlled with antiretroviral drugs. These stop the virus replicating in the body.
- The virus attacks the immune cells (see page 59).
- If the body's immune system is badly damaged, it can't cope with other infections or cancers. At this stage, the virus is known as late stage HIV infection, or AIDS.

...and a **Fungal Disease**

1) Rose black spot is a fungus that causes purple or black spots to develop on the leaves of rose plants. The leaves can then turn yellow and drop off.

2) This means that less photosynthesis can happen, so the plant doesn't grow very well.

Photosynthesis is important for plant growth because it produces glucose — see page 71.

3) It is spread through the environment in water or by the wind.

4) Gardeners can treat the disease using fungicides and by stripping the plant of its affected leaves. These leaves then need to be destroyed so that the fungus can't spread to other rose plants.

Plants, animals and people are all affected by pathogens...

...but they're not all affected by the same ones. For example, plants can't be infected by the measles virus and humans and other animals can't get TMV. In fact, there aren't any known plant disease that humans can catch. Phew...

Bacterial and Protist Diseases

Quick Quiz

Sorry — I'm afraid there are some more diseases to learn about here. This time, they're diseases caused by ***bacteria*** *and* ***protists****. I don't know about you, but I'm starting to feel a bit itchy all over...*

You Need to Know About **Two Bacterial Diseases**...

Salmonella

- *Salmonella* is a type of bacteria that causes food poisoning.
- Infected people can suffer from fever, stomach cramps, vomiting and diarrhoea.
- These symptoms are caused by the toxins that the bacteria produce (see page 54).
- You can get *Salmonella* food poisoning by eating food that's been contaminated with *Salmonella* bacteria, e.g. eating chicken that caught the disease whilst it was alive, or eating food that has been contaminated by being prepared in unhygienic conditions.
- In the UK, most poultry (e.g. chickens and turkeys) is given a vaccination against *Salmonella*. This is to control the spread of the disease.

Gonorrhoea

- Gonorrhoea is a bacterial sexually transmitted disease (STD).
- STDs are passed on by sexual contact, e.g. having unprotected sex.
- A person with gonorrhoea will get pain when they urinate.
  Another symptom is a thick yellow or green discharge from the vagina or the penis.
- Gonorrhoea was originally treated with an antibiotic called penicillin, but this has become trickier now because strains of the bacteria have become resistant to it (see page 61).
- To prevent the spread of gonorrhoea, people can be treated with antibiotics and should use barrier methods of contraception (see page 103), such as condoms.

...and a **Disease** Caused by a **Protist**

Malaria causes repeating episodes of fever and can be fatal.
It is caused by a protist (see page 54):

1) Part of the malarial protist's life cycle takes place inside the mosquito. The mosquitoes are vectors (see page 54) — they pick up the malarial protist when they feed on the blood of an infected animal.
2) Every time the mosquito bites another animal, it infects it by transferring the protist through its saliva into the animal's blood.

The spread of malaria can be reduced by stopping the mosquitoes from breeding.
People can be protected from mosquitoes using insecticides and mosquito nets.

Hang in there, this stuff is pretty gross, but it's important to know

Try drawing out a table with columns for 'disease', 'type of organism it's caused by', 'symptoms' and 'how it's spread', then fill it in for all the diseases covered on both this page and on the previous one. See how much you can write down without looking back at the page — the more details you can recall for each disease, the better.

Warm-Up & Exam Questions

Have a go at these questions to test whether you know about each of the diseases covered on the previous pages — their symptoms, how they are transmitted and how their spread can be limited.

Warm-Up Questions

1) What is meant by a communicable disease?
2) How can vaccinations help to limit the spread of a disease?
3) List three viral diseases.
4) What symptom of measles is shown on the skin?
5) What effect does rose black spot disease have on plants?
6) How is gonorrhoea transmitted between individuals?

Exam Questions

1 Diseases are often recognised by their symptoms. GRADE 4-6 GRADE

1.1 Describe the initial symptoms of HIV infection.

[1 mark]

1.2 Give **two** symptoms of gonorrhoea.

[2 marks]

1.3 A person has food poisoning caused by *Salmonella*. Give **two** symptoms that they may have.

[2 marks]

2 Viruses are a type of pathogen. They can infect every type of living organism. GRADE 6-7 GRADE

2.1 What is meant by the term 'pathogen'?

[1 mark]

2.2 The leaves of a tobacco plant can become discoloured if it is infected by a particular virus. Name the virus that affects tobacco plants in this way and describe what effect the discolouration of the leaves can have on a plant.

[2 marks]

2.3 HIV is a virus that can infect humans. Outline how HIV can be spread.

[2 marks]

3 The methods used to prevent the spread of a disease depend on how the disease is transmitted. GRADE 6-7 GRADE

3.1 It is important for chefs to wash their hands thoroughly before cooking. Suggest why.

[1 mark]

3.2 Explain why hand washing may not be helpful in limiting the spread of malaria.

[2 marks]

3.3 Suggest and explain **one** reason why efforts to limit the spread of malaria often focus on the mosquito.

[1 mark]

Fighting Disease

Quick Quiz

The human body has some pretty neat features when it comes to ***fighting disease****.*

Your Body Has a Sophisticated **Defence System**

The first line of defence stops a lot of nasties getting inside in the first place:

Body part	How it defends the body
Skin	Acts as a barrier to pathogens. Secretes antimicrobial substances to kill pathogens.
Nose	Hairs and mucus trap particles that could contain pathogens.
Trachea and bronchi	Secrete mucus to trap pathogens. Lined with cilia (hair-like structures), which waft the mucus up to the back of the throat where it can be swallowed.
Stomach	Produces hydrochloric acid which kills pathogens that make it to the stomach.

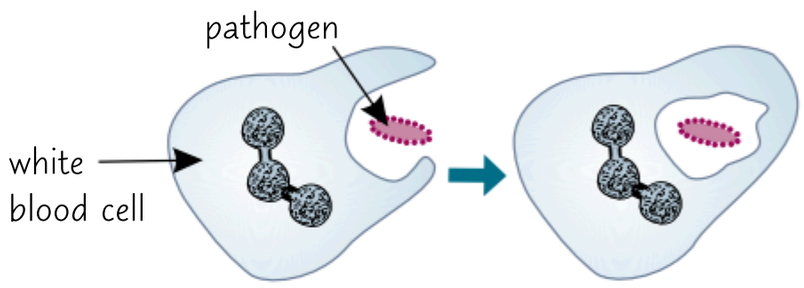
The **Immune System Attacks Pathogens** That Make It **Into the Body**

The most important part of your immune system is the white blood cells. They travel around in your blood and crawl into every part of you, constantly patrolling for microbes.

When they come across an invading microbe they have three lines of attack:

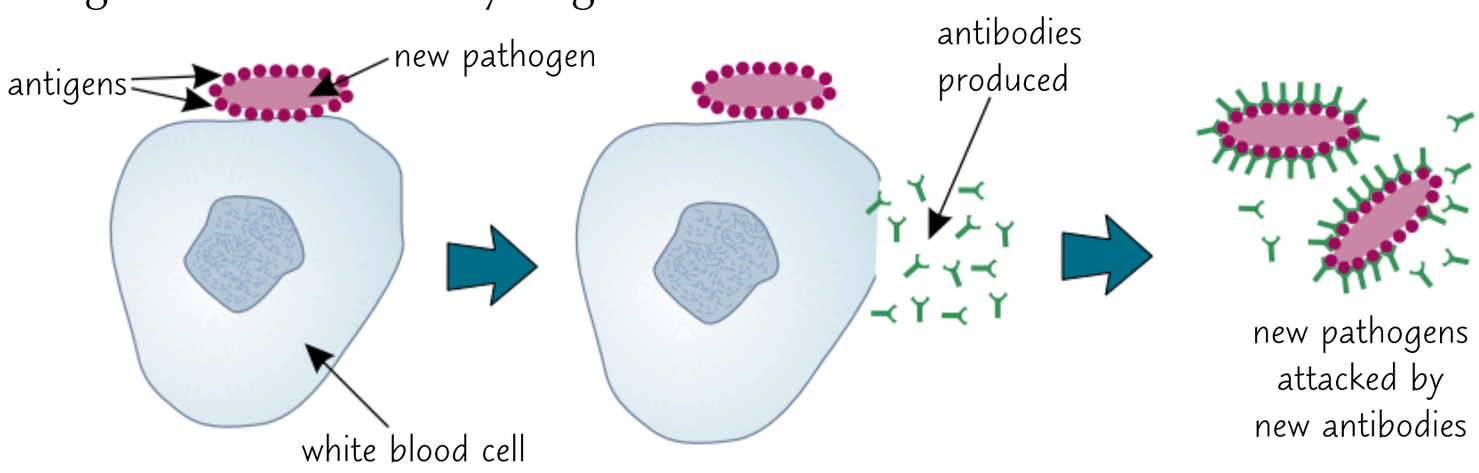
1. Consuming Them

Some types of white blood cells can engulf foreign cells and digest them. This is called phagocytosis.



2. Producing Antibodies

1) Every invading pathogen has unique molecules called antigens on its surface.
2) When some types of white blood cell find a foreign antigen (one they don't recognise), they start to rapidly produce proteins called antibodies. These lock onto the invading cells so that they can be found and destroyed by other white blood cells.
3) These antibodies are carried around the body in the bloodstream to find all similar bacteria or viruses.
4) If the person is infected with the same pathogen again, the white blood cells can more rapidly produce the antibodies to kill it — the person is naturally immune to that pathogen and is less likely to get ill.



Antibodies are specific to one type of antigen — they won't lock on to any others.

White blood cells that produce antibodies are called B-lymphocytes.

3. Producing Antitoxins

These counteract toxins produced by the invading bacteria.

Invading pathogens are destroyed by white blood cells

If you have a low level of white blood cells, you'll be more susceptible to infections. HIV (see page 56) attacks white blood cells and weakens the immune system, making it easier for other pathogens to invade.

Fighting Disease — Vaccination

Quick Quiz

*Vaccinations have changed the way we fight disease. We don't always have to deal with the problem once it's happened — we can **prevent** it happening in the first place.*

Vaccination can Protect from Future Infections

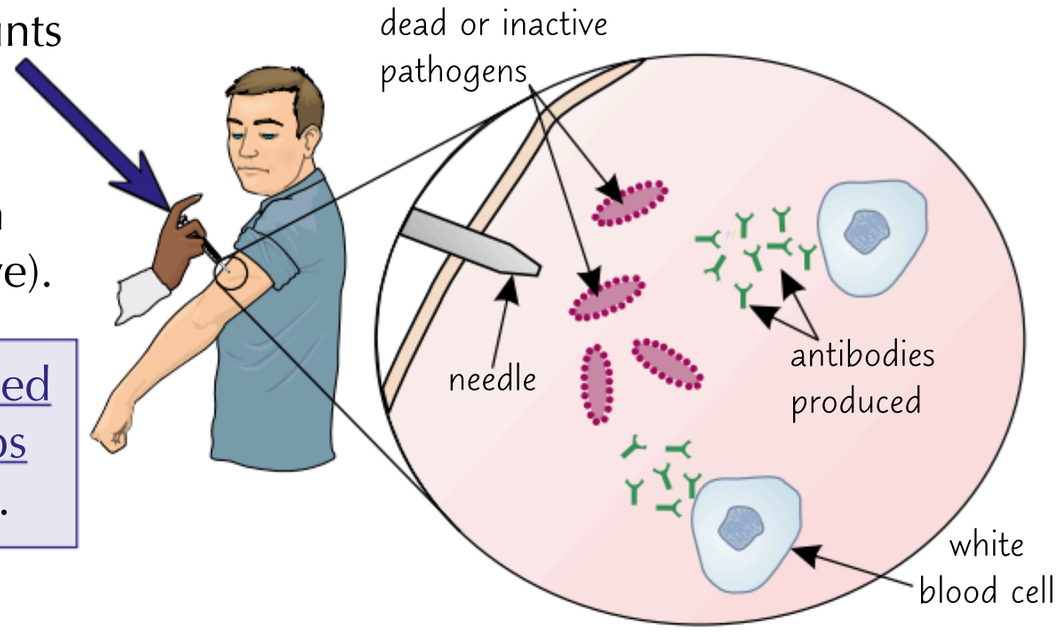
Video

When you're infected with a new pathogen, it takes your white blood cells a few days to learn how to deal with it. But by that time, you can be pretty ill.

1) Vaccinations usually involve injecting small amounts of dead or inactive pathogens.

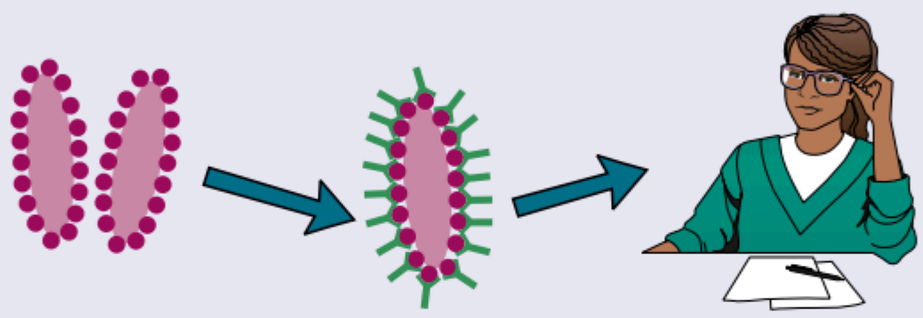
2) These carry antigens, which cause your body to produce antibodies to attack them — even though the pathogen is harmless (since it's dead or inactive).

For example, the MMR vaccine contains weakened versions of the viruses that cause measles, mumps and rubella (German measles) all in one vaccine.



3) After vaccination...

...if live pathogens of the same type try to attack...



...so you are less likely to get ill.

...they are quickly recognised, and white blood cells rapidly mass-produce antibodies to kill off the pathogen...

There are Pros and Cons of Vaccination

PROS	CONS
Vaccines have helped control communicable diseases that were once common in the UK (e.g. polio, measles, whooping cough, rubella, mumps, tetanus...). Worldwide, smallpox no longer occurs at all, and polio infections have fallen by 99%.	Vaccines don't always work — sometimes they don't give you immunity.
Big outbreaks of disease (epidemics) can be prevented if a large percentage of the population is vaccinated. That way, even the people who aren't vaccinated are unlikely to catch the disease because there are fewer people able to pass it on.	Rarely, you can have a bad reaction to a vaccine (e.g. swelling, or maybe something more serious like a fever or seizures).

If a significant number of people aren't vaccinated, the disease can spread quickly through them and lots of people will be ill at the same time.

Prevention is better than cure...

Deciding whether to have a vaccination means balancing risks — the risk of catching the disease if you don't have a vaccine against the risk of having a bad reaction if you do. There's more about risk on p.171.

Q1 Basia is vaccinated against flu and Cassian isn't. They are both exposed to a flu virus. Cassian falls ill whereas Basia doesn't. Explain why. [2 marks]

Q1 Video Solution

Fighting Disease — Drugs

Quick Quiz

*You've probably had to take some sort of **medicine** if you've been ill, e.g. cough remedies, painkillers. And you're about to find out why it's important to only take **antibiotics** when you **really need them**.*

Some Drugs **Relieve Symptoms** — Others **Cure** the Problem

Drugs like pain killers and "cold remedies" can help to reduce the symptoms of a disease (e.g. pain), but they don't kill pathogens. So they don't actually tackle the underlying cause of the disease.

- Antibiotics (e.g. penicillin) actually kill (or prevent the growth of) the bacteria causing the problem without killing body cells.
- Different antibiotics kill different types of bacteria, so you need to be treated with the right one.
- Antibiotics have greatly reduced the number of deaths from communicable bacterial diseases.

Antibiotics don't destroy viruses (e.g. flu or cold viruses). Viruses reproduce using your body cells, which makes it very difficult to develop drugs that destroy just the virus without killing the body's cells.

Bacteria Can Become **Resistant** to **Antibiotics**

Bacteria can mutate — sometimes the mutations cause them to be resistant to (not killed by) an antibiotic.

1) If you have an infection, some of the bacteria might be resistant to antibiotics. This means that when you treat the infection, only the non-resistant strains of bacteria will be killed.
2) The individual resistant bacteria will survive and reproduce, and the population of the resistant strain will increase. This is an example of natural selection (see page 129).
3) This resistant strain could cause a serious infection that can't be treated by antibiotics. E.g. MRSA causes serious wound infections and is resistant to the powerful antibiotic meticillin.

To slow down the rate of development of resistant strains, doctors avoid over-prescribing antibiotics. So you won't get them for a sore throat, only for something more serious.

It's also important to finish the whole course of antibiotics and don't just stop once you feel better.

Many **Drugs** Originally Came From **Plants**

Plants produce a variety of chemicals to defend themselves against pests and pathogens (see page 68).

Some of these chemicals can be used as drugs to treat human diseases or relieve symptoms.
A lot of our current medicines were discovered by studying plants used in traditional cures. For example:

- Aspirin is used as a painkiller. It was developed from a chemical found in willow.
- Digitalis is used to treat heart conditions. It was developed from a chemical found in foxgloves.

Some drugs were extracted from microorganisms. For example:

- Alexander Fleming was clearing out some Petri dishes containing bacteria. He noticed that one of the dishes of bacteria also had mould on it and the area around the mould was free of the bacteria.
- He found that the mould (called *Penicillium notatum*) on the Petri dish was producing a substance that killed the bacteria — this substance was penicillin.

These days, drugs are made on a large scale in the pharmaceutical industry — they're synthesised by chemists in labs. However, the process still might start with a chemical extracted from a plant.

Good old mould — saving lives since 1928...

Drug development is a very big industry. A lot of money, time and effort goes into discovering, developing and improving drugs, but it can all start with something as humble as a plant or as small as a microorganism.

Developing Drugs

Quick Quiz

*New drugs are constantly being developed. But before they can be given to the general public, they have to go through a thorough **testing** procedure. This is what usually happens...*

There are **Three Main Stages** in Drug Testing

1 In preclinical testing, drugs are tested on samples of human cells and tissues in the lab. However, you can't use human cells and tissues to test drugs that affect whole or multiple body systems, e.g. testing a drug for blood pressure must be done on a whole animal because it has an intact circulatory system.

2 The next step in preclinical testing is to test the drug on live animals. This is to:

- test efficacy — whether the drug works and produces the effect you're looking for.
- find out about its toxicity — how harmful it is.
- find the best dosage — the concentration that should be given, and how often it should be given.

The law in Britain states that any new drug must be tested on two different live mammals.

Some people think it's cruel to test on animals, or that animals are so different from humans that testing on animals is pointless. Others believe this is the safest way to make sure a drug isn't dangerous before it's given to humans.

3 If the drug passes the tests on animals then it's tested on human volunteers in a clinical trial:

1) First, the drug is tested on healthy volunteers. This is to make sure that it doesn't have any harmful side effects when the body is working normally. At the start of the trial, a very low dose of the drug is given and this is gradually increased.
2) If the results of the tests on healthy volunteers are good, the drugs can be tested on people suffering from the illness.

In clinical trials, the optimum dose is found — this is the dose of drug that is the most effective and has few side effects.

1) To test how well the drug works, patients are randomly put into two groups. One is given the new drug, the other is given a placebo (a substance that's like the drug being tested but doesn't do anything). This is so the doctor can see the actual difference the drug makes — it controls for any possible placebo effect (when the patient expects the treatment to work and so feels better, even though the treatment isn't doing anything).
2) Clinical trials are blind — the patient in the study doesn't know whether they're getting the drug or the placebo. In fact, they're often double-blind — neither the patient nor the doctor knows until all the results have been gathered. This is so the doctors monitoring the patients and analysing the results aren't unconsciously influenced by their knowledge.
3) The results of drug testing and drug trials aren't published until they've been through peer review. This helps to prevent false claims.

Peer review is when other scientists check that the work is valid and has been carried out rigorously — see p.169.

The placebo effect doesn't work with revision...

... unfortunately, you can't just expect to get a good mark and then magically get it. I know, I know, there's a lot of information to take in on this page, but just read it through nice and slowly. There's nothing too tricky here — it's just a case of going over it again and again and again until you've got it all firmly lodged in your memory.

Warm-Up & Exam Questions

It's easy to think you've learnt everything in the section until you try the Warm-Up Questions.
Don't panic if there's a bit you've forgotten, just go back over that bit until it's firmly fixed in your brain.

Warm-Up Questions

1) How is the skin adapted to defend against the entry of pathogens?
2) What is the role of the immune system?
3) From what organism was the drug digitalis sourced originally?
4) What is meant by the efficacy of a drug?
5) What is a placebo?
6) Why are placebos used in drug trials?

Exam Questions

1 There are many different lines of defence in the human body that help to prevent pathogens from entering the blood. GRADE 4-6 GRADE

1.1 What is the role of the hairs and mucus in the nose?

[1 mark]

1.2 How do the cilia in the trachea and bronchi help to defend the body?

[1 mark]

1.3 What does the stomach produce to kill pathogens?

[1 mark]

2 A scientist is carrying out a clinical trial. GRADE 6-7 GRADE

2.1 What is a drug tested on in a clinical trial? Tick **one** box.

- ☐ human cells
- ☐ human volunteers
- ☐ live animals
- ☐ human tissue

[1 mark]

2.2 The clinical trial is double blind. Explain what this means.

[2 marks]

2.3 Apart from the toxicity of the drug, give **two** other factors that scientists research during drug testing.

[2 marks]

2.4 The results from drug testing are assessed by peer review. Explain what this means and why it is done.

[2 marks]

Exam Questions

3 White blood cells play an important role in defence against pathogens, including the production of antibodies and antitoxins.

GRADE 6-7 GRADE

3.1 Apart from producing antibodies and antitoxins, give **one** other method that white blood cells use to defend the body against pathogens.

[1 mark]

3.2 Explain why the production of antibodies is specific to a certain disease.

[1 mark]

4 The development of antibiotics has helped to save many lives.
However, the overuse of antibiotics may be a threat to global health in the future.
Penicillin is an example of an antibiotic.

GRADE 6-7 GRADE

4.1 Name the organism that penicillin originates from.

[1 mark]

4.2 Explain why the overuse of antibiotics may be a threat to global health in the future.

[2 marks]

4.3 Infection with rhinovirus causes the common cold.
Explain why antibiotics such as penicillin are not used to treat a cold.

[1 mark]

5 Rubella is a communicable viral disease.

GRADE 7-9 GRADE

The rubella virus is spread in droplets through the air when an infected person coughs, sneezes or talks. The virus causes several symptoms including fever and painful joints. Fortunately, the spread of the disease can be reduced by vaccination.

5.1 Suggest a drug that patients with rubella may be given to relieve their symptoms.

[1 mark]

5.2 Explain why it might be difficult to develop a drug which destroys rubella in the body.

[2 marks]

5.3* Explain how being vaccinated against rubella can help to prevent a person from catching the disease and suggest why vaccinating a large proportion of the population reduces the risk of someone who hasn't been vaccinated from catching rubella.

[6 marks]

5.4 Suggest **one** reason why some individuals may choose not to receive a vaccination against a disease.

[1 mark]

Producing Monoclonal Antibodies

Quick Quiz

__Antibodies__ aren't only used by the immune system — scientists have engineered them for lots of new uses.

Monoclonal Antibodies are Identical Antibodies

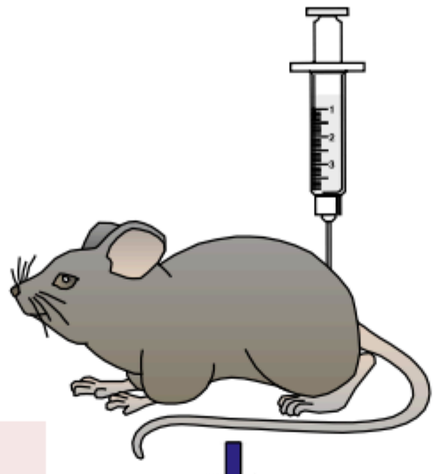
Video

Antibodies are produced by B-lymphocytes — a type of white blood cell (see page 59).

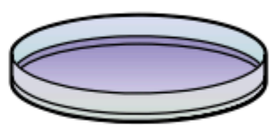
Monoclonal antibodies are produced from lots of clones of a single white blood cell. This means all the antibodies are identical and will only target one specific protein antigen.

Here's how monoclonal antibodies are made:

1) A mouse is injected with a certain antigen so that it produces the specific antibodies needed.

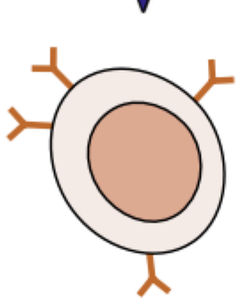


2) Fast-dividing tumour cells are grown in a lab.

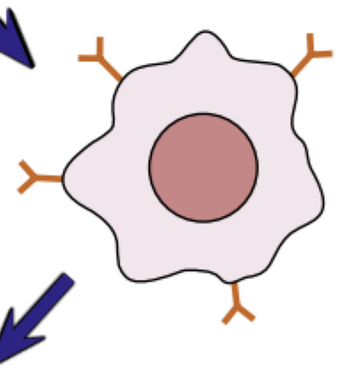


You can't just grab the lymphocyte that made the antibody and grow more — lymphocytes don't divide very easily. That's why tumour cells are used in the process — they don't produce antibodies but they do divide lots — so they can be grown really easily.

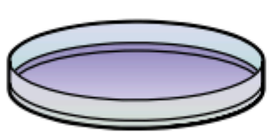
3) B-lymphocytes are taken from the mouse. These produce the antibodies specific to the antigen that the mouse was injected with.

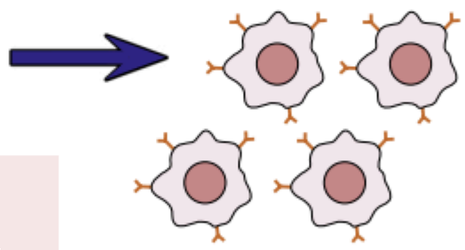


4) A B-lymphocyte is fused with a tumour cell...



... which makes a hybridoma.





5) The hybridoma cell can then be cloned to get lots of identical cells. These cells all produce the same antibodies (monoclonal antibodies).

6) Finally, the antibodies are collected and purified.

You can make monoclonal antibodies that bind to anything you want, e.g. an antigen that's only found on the surface of one type of cell.

Monoclonal antibodies are really useful because they will only bind to this molecule — this means you can use them to target a specific cell or chemical in the body.

A monoclonal antibody will only bind to one specific antigen

There's more on monoclonal antibodies coming up next, but don't move on until you totally understand this page.
You could be asked in the exam to describe how monoclonal antibodies are made — best make sure you know it...
Cover up the page and see if you can write it all out without looking. If you can't, go back over the page and try again.

Uses of Monoclonal Antibodies

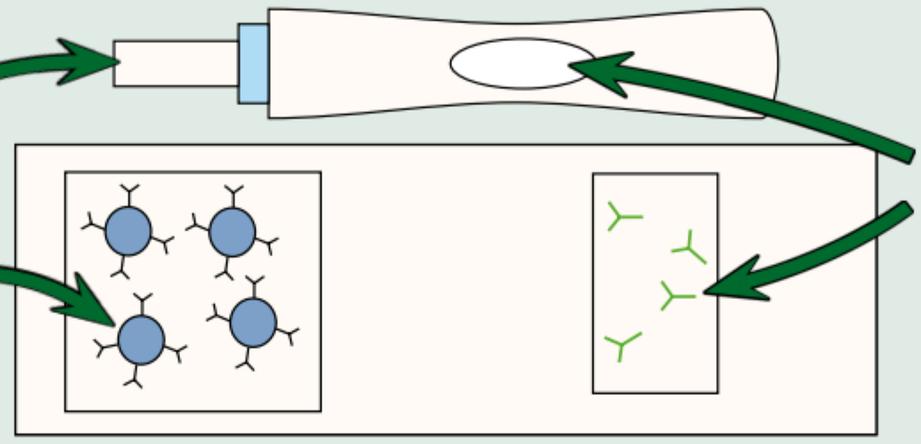
Quick Quiz

*Now that you know how **antibodies** are **made**, it's time to learn about how they can be **used**.*
*First up is **pregnancy tests**...*

Monoclonal Antibodies Are Used In **Pregnancy Tests**...

A hormone called HCG is found in the urine of women only when they are pregnant.
Pregnancy testing sticks detect this hormone. Here's how they work:

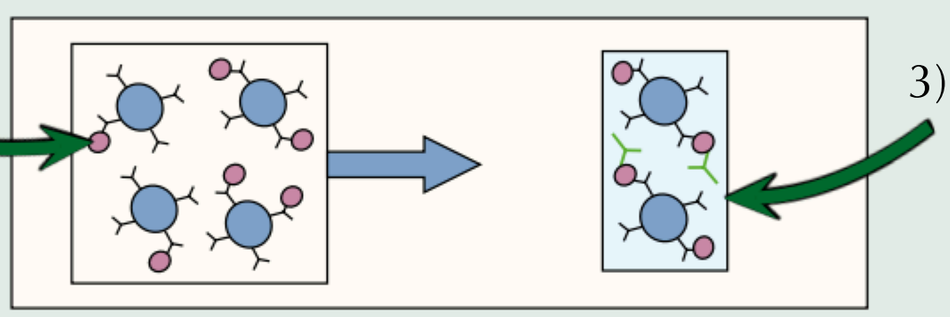
The bit of the stick you wee on has some antibodies to the hormone, with blue beads attached.



The test strip (the bit that turns blue if you're pregnant) has some more antibodies to the hormone stuck onto it (so that they can't move).

If you're PREGNANT:

1) The hormone in your wee binds to the antibodies on the blue beads.

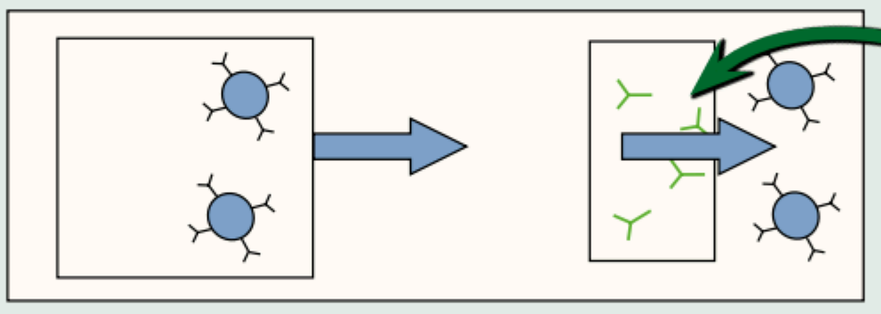


2) The urine moves along the stick, carrying the hormone and the beads.

3) The beads and hormone bind to the antibodies on the strip. So the blue beads get stuck on the strip, turning it blue.

If you're NOT PREGNANT:

1) The urine still moves along the stick, carrying the blue beads.



2) But there's nothing to stick the blue beads onto the test strip, so it doesn't go blue.

...and in **Laboratories** and **Research** to **Find Specific Substances**

Monoclonal antibodies can be used to:

- Bind to hormones and other chemicals in blood to measure their levels.
- Test blood samples in laboratories for certain pathogens.
- Locate specific molecules on a cell or in a tissue:

Monoclonal antibodies are made that will bind to the specific molecules you're looking for. The antibodies are then bound to a fluorescent dye. If the molecules are present, the monoclonal antibodies will attach to them, and they can be detected using the dye.

Handy little things, these monoclonal antibodies...

Monoclonal antibodies have lots more uses in medicine — for example, a test similar to a pregnancy test can be used to detect the antigens of pathogens in a blood sample. There's more about their uses coming up on the next page.

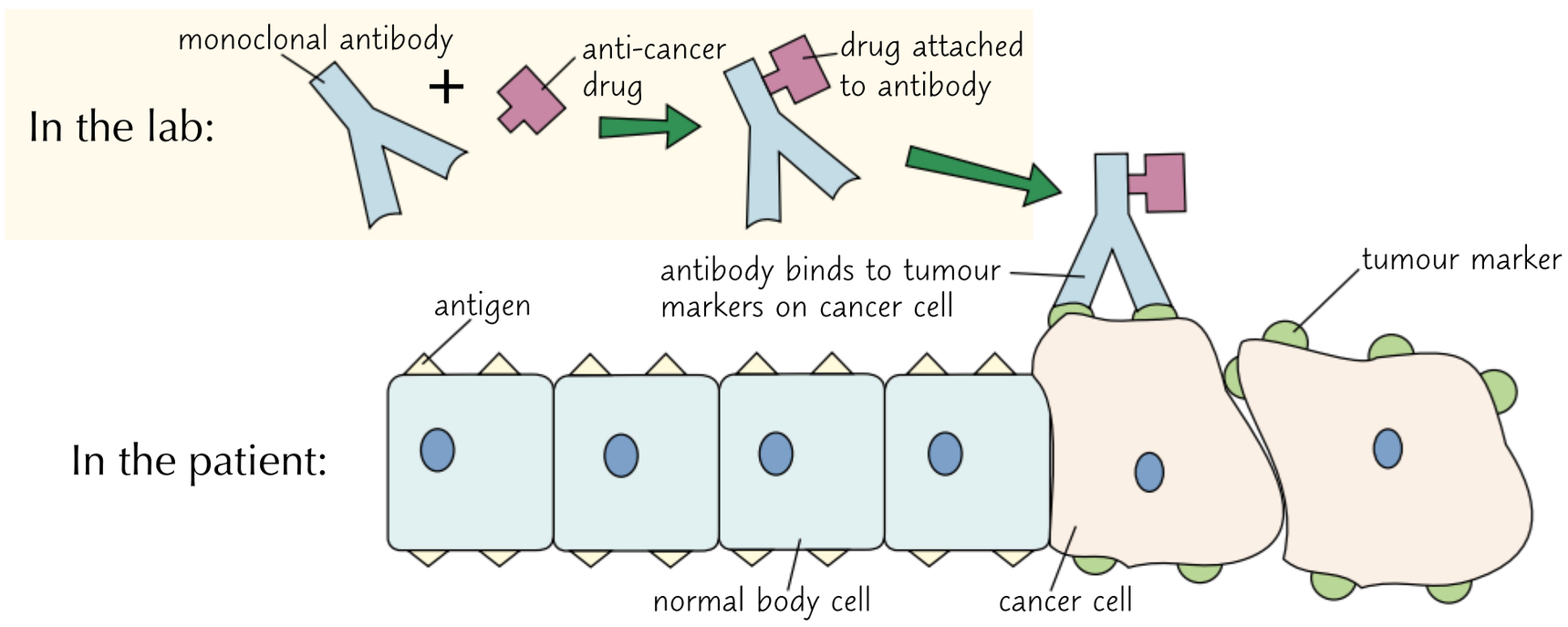
More Uses of Monoclonal Antibodies

Quick Quiz

*Because monoclonal antibodies can be produced to **target** a **specific** chemical or cell, they have loads of uses.*

Monoclonal Antibodies Can be Used to Treat Diseases

1) Different cells in the body have different antigens on their cell surface. So you can make monoclonal antibodies that will bind to specific cells in the body (e.g. just liver cells).
2) Cancer cells have antigens on their cell membranes that aren't found on normal body cells. They're called tumour markers.
3) In the lab, you can make monoclonal antibodies that will bind to these tumour markers.
4) An anti-cancer drug can be attached to these monoclonal antibodies. This might be a radioactive substance, a toxic drug or a chemical which stops cancer cells growing and dividing.
5) The antibodies are given to the patient through a drip.
6) The antibodies target specific cells (the cancer cells) because they only bind to the tumour markers.
7) The drug kills the cancer cells but doesn't kill any normal body cells near the tumour.



Monoclonal Antibodies Do Have Some **Side Effects** Though

Monoclonal antibodies target specific cells...

- Other cancer treatments (like standard chemotherapy and radiotherapy) can affect normal body cells as well as killing cancer cells, whereas monoclonal antibodies target specific cells.
- This means the side effects of an antibody-based drug are lower than for standard chemotherapy or radiotherapy.

...but do cause side effects

- However, monoclonal antibodies do cause more side effects than were originally expected, e.g. they can cause fever, vomiting and low blood pressure.
- When they were first developed, scientists thought that because they targeted a very specific cell or molecule, they wouldn't create a lot of side effects.
- This means that they are not as widely used as treatments as scientists had originally thought they might be.

Monoclonal antibodies have many uses in research and medicine

Q1 A scientist has a sample of tissue from a patient with a suspected bacterial infection. He sticks it to a microscope slide. Then he adds a solution of monoclonal antibodies against the bacteria that have been joined to a fluorescent dye. Finally, he washes off any unbound antibodies and looks at the slide under a microscope. Explain how the test works to confirm that the bacteria are present. [2 marks]

Q1 Video Solution

Plant Diseases and Defences

Quick Quiz

*It's not just us humans that can get diseases — **plants** get their fair share too.*

Plants Need **Mineral Ions**

Plants need mineral ions from the soil. If there aren't enough, plants suffer deficiency symptoms.

- Nitrates are needed to make proteins and therefore for growth. A lack of nitrates causes stunted growth.
- Magnesium ions are needed for making chlorophyll, which is needed for photosynthesis. Plants without enough magnesium suffer from chlorosis and have yellow leaves.

Plants Can Get **Diseases**

1) Plants can be infected by viral, bacterial and fungal pathogens — see page 56 for some examples. It's usually pretty clear that a plant has a disease. The common signs are:

- Stunted growth
- Abnormal growths, e.g. lumps
- Spots on the leaves
- Malformed stems or leaves
- Patches of decay (rot)
- Discolouration

2) Different plant diseases have different signs. They can be identified by:

- Looking up the signs in a gardening manual or on a gardening website.
- Taking the infected plant to a laboratory, where scientists can identify the pathogen.
- Using testing kits that identify the pathogen using monoclonal antibodies (see page 65).

3) Plants can also be infested and damaged by insects. E.g. aphids can cause huge damage to plants. Infestations of pests are easy to spot — you should be able to see them on the plants.

Plants Have Different Types of **Defence**...

Physical defences

- Most plant leaves and stems have a waxy cuticle, which provides a barrier to stop pathogens entering.
- Plant cells are surrounded by cell walls made from cellulose. These form a barrier against pathogens that make it past the waxy cuticle.
- Plants have layers of dead cells around their stems, for example, the outer part of the bark on trees. These act as a barrier to stop pathogens entering.

Chemical defences

- Some plants produce antibacterial chemicals which kill bacteria — e.g. the mint plant and witch hazel.
- Others produce poisons to deter herbivores — e.g. tobacco plants, foxgloves and deadly nightshade.

Mechanical defences

- Some plants have thorns and hairs. These stop animals touching and eating them.
- Other plants have leaves that droop or curl when something touches them. They can prevent themselves from being eaten by knocking insects off themselves and moving away from things.
- Some plants can mimic other organisms. E.g. the passion flower has bright yellow spots on its leaves which look like butterfly eggs. This stops other butterflies laying eggs there. There are also some species of plant that look like stones and pebbles. This tricks other organisms into not eating them.

The way a plant looks can tell you a lot about its health

Plants need the right mineral ions to grow — if they aren't getting the right stuff, they don't grow properly. They can also end up looking rather sickly if a pathogen gets past their defences and the symptoms of the disease affects their growth.

Warm-Up & Exam Questions

It's that time again — time to test just how much of the previous pages you really remember...

Warm-Up Questions

1) What type of white blood cell produces antibodies?
2) What is a hybridoma cell made from?
3) Give a symptom of nitrate deficiency in plants.
4) How does bark prevent a tree from being infected with pathogens?

Exam Questions

1 Which of the following is a term used to describe yellow leaves on a plant? Tick **one** box. GRADE 4-6 GRADE

☐ chlorophyll ☐ chlorosis ☐ cellulose ☐ cuticle

[1 mark]

2 Plant diseases can be caused by deficiencies in nutrients or by pathogens. Plants have many methods to protect themselves against diseases. GRADE 6-7 GRADE

2.1 Give **three** methods of physical defence used by plants to stop pathogens entering.

[3 marks]

2.2 Explain why plants with a magnesium deficiency are less able to photosynthesise.

[1 mark]

2.3 Apart from discoloured leaves, state **two** signs which may indicate that a plant is diseased.

[2 marks]

3 Monoclonal antibodies have many uses in medicine. GRADE 6-7 GRADE

3.1 List **three** things that can be bound to a monoclonal antibody to treat cancer.

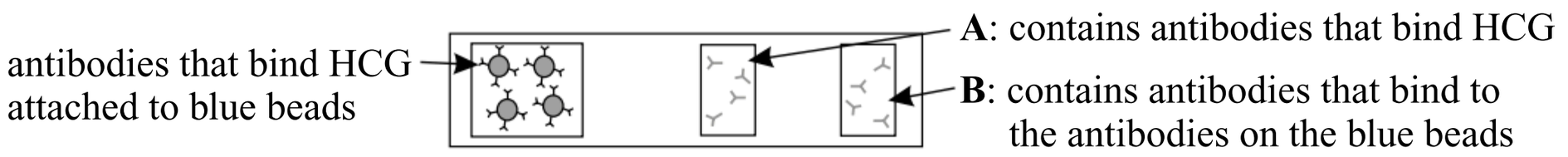
[3 marks]

3.2 Explain **one** advantage of using monoclonal antibodies to treat cancer instead of using normal treatment methods.

[2 marks]

4 **Figure 1** shows a pregnancy testing stick, which uses monoclonal antibodies to test a urine sample for the pregnancy hormone HCG. When urine is applied to the end containing the beads, it travels up the stick towards strips **A** and **B**. If the woman is pregnant, both **A** and **B** will turn blue. GRADE 7-9 GRADE

Figure 1



Strip **B** in **Figure 1** is designed to always turn blue, even if the woman is not pregnant. Suggest why strip **B** is included on the testing stick. Explain your answer.

[3 marks]

Revision Summary Test for Topic 3

Another topic down and plenty to fill your brain with. To check what you remember, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Types of Disease (p.54-57) ☑

1) How can bacteria make us feel ill? And what about viruses? ☑
2) Give one way that fungi can cause disease. ☑
3) Outline three ways that pathogens can be spread. For each one, give a disease spread this way. ☑
4) Give four ways of reducing or preventing the spread of disease. ☑
5) What are the signs of tobacco mosaic virus? How does it affect a plant's growth? ☑
6) How is measles spread, and what are the symptoms? ☑
7) How is HIV spread? Outline the progression of the disease. ☑
8) What type of disease is rose black spot? What effect does it have on plants and how is it treated? ☑
9) What type of disease is *Salmonella*? How is it spread, and how is the spread controlled? ☑
10) How is gonorrhoea spread? What are the symptoms, and how is it treated? ☑
11) How are mosquitoes involved in the spread of malaria? How is the spread of malaria controlled? ☑

Fighting Disease (p.59-62) ☑

12) Give four body parts that act as a first line of defence. How do they defend the body? ☑
13) Describe three ways that white blood cells can defend against pathogens. ☑
14) How do vaccines work? Why is it beneficial to vaccinate a large percentage of the population? ☑
15) Why don't drugs like painkillers tackle the cause of disease? What do they do instead? ☑
16) What are antibiotics used for? Give an example of an antibiotic. ☑
17) Why is it difficult to develop drugs that kill viruses without also damaging body tissues? ☑
18) Describe one concern linked to the use of antibiotics. ☑
19) Give two examples of drugs developed from chemicals found in plants. What are they used for? ☑
20) Give an example of a drug extracted from microorganisms. Who discovered it? ☑
21) Outline the three main stages in drug testing. ☑
22) What happens in a double-blind trial? Why are double-blind trials used? ☑

Antibodies (p.65-67) ☑

23) Describe how monoclonal antibodies are made. ☑
24) Why does a pregnancy test strip turn blue when you wee on it if you're pregnant? ☑
25) Why might a fluorescent dye be added to a monoclonal antibody? ☑
26) How can monoclonal antibodies be used in the treatment of cancer? ☑
27) Why aren't monoclonal antibodies as widely used as originally thought? ☑

Plant Diseases and Defences (p.68) ☑

28) What do plants use nitrates for? And what about magnesium ions? ☑
29) Give six signs that a plant has a disease. Then give three ways of identifying a plant disease. ☑
30) Give three physical defences that a plant may have to defend itself. ☑
31) Give two types of chemical substance that a plant may produce to defend itself. ☑
32) Give three mechanical defences that a plant may have to defend itself. ☑

Photosynthesis and Limiting Factors

Quick Quiz

*First, **photosynthesis equations**. Then onto how plants use **glucose**...*

Photosynthesis Produces **Glucose** Using **Light**

Photosynthesis uses energy to change carbon dioxide and water into glucose and oxygen.

Word equation:

$$\text{carbon dioxide + water} \xrightarrow{\text{light}} \text{glucose + oxygen}$$

Symbol equation:

$$6CO_2 + 6H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6O_2$$

Photosynthesis takes place in chloroplasts in plant cells — they contain chlorophyll that absorbs light. It's an endothermic reaction, because energy is transferred from the environment to the chloroplasts. This energy is transferred by light.

Plants Use Glucose in **Five** Main Ways...

1. For respiration — This transfers energy from glucose (see p.78) which enables the plants to convert the rest of the glucose into various other useful substances.
2. Making cellulose — Glucose is converted into cellulose to make strong plant cell walls (see p.4).
3. Making amino acids — Glucose is combined with nitrate ions (absorbed from the soil) to make amino acids, which are then made into proteins.
4. Stored as oils or fats — Glucose is turned into lipids (fats and oils) for storing in seeds.
5. Stored as starch — Glucose is turned into starch and stored in roots, stems and leaves, ready for use when photosynthesis isn't happening, like in the winter.

Starch is insoluble, which makes it better for storing than glucose — a cell with lots of glucose would draw in loads of water by osmosis and swell up.

Limiting Factors Affect the **Rate of Photosynthesis**

The rate of photosynthesis is affected by intensity of light, concentration of CO_2 and temperature.

Any of these factors can become the limiting factor — this means that it's stopping photosynthesis from happening any faster. These factors have a combined effect on the rate of photosynthesis, but which factor is limiting at a particular time depends on the environmental conditions:

- at night it's pretty obvious that light is the limiting factor,
- in winter it's often the temperature,
- if it's warm enough and bright enough, the amount of CO_2 is usually limiting.

Chlorophyll can also be a limiting factor of photosynthesis.

The amount of chlorophyll in a plant can be affected by disease (e.g. tobacco mosaic virus) or environmental stress (e.g. a lack of nutrients). These factors can cause chloroplasts to become damaged or to not make enough chlorophyll. This means the rate of photosynthesis is reduced because they can't absorb as much light.

'Photo' means light and 'synthesis' means putting together...

...so photosynthesis means 'putting together using light'. And of course the thing being put together is glucose. Well, I guess that's one way of remembering it... (Maybe it'd be easier to just learn the word equation instead.)

The Rate of Photosynthesis

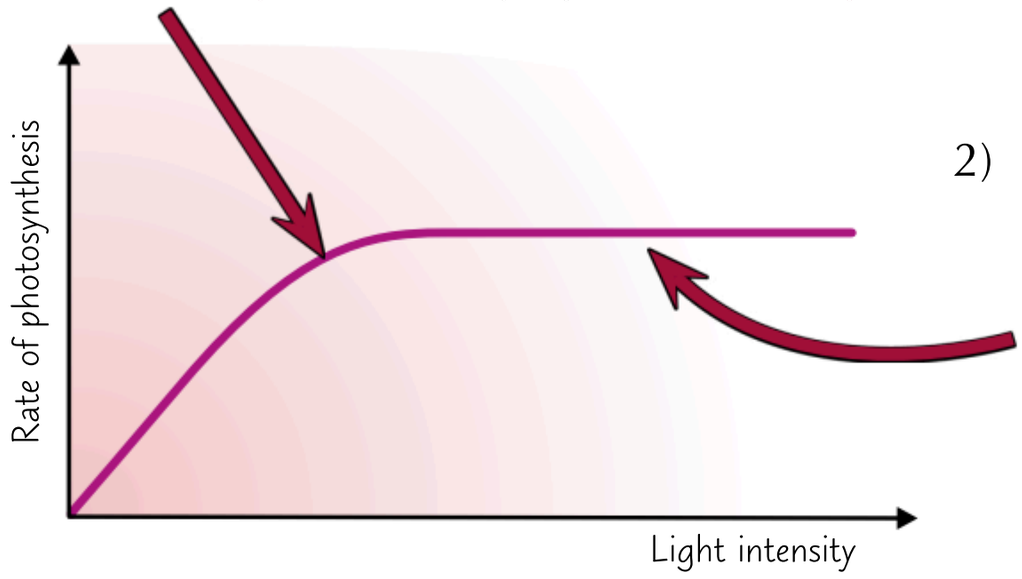
Quick Quiz

*The rate of photosynthesis is affected by the intensity of **light**, the concentration of **CO_2**, and the **temperature**. Plants also need **water** for photosynthesis, but when a plant is so short of water that it becomes the **limiting factor** in photosynthesis, it's already in such **trouble** that this is the least of its worries.*

1) Not Enough **Light** Slows Down the Rate of Photosynthesis

Light provides the energy needed for photosynthesis.

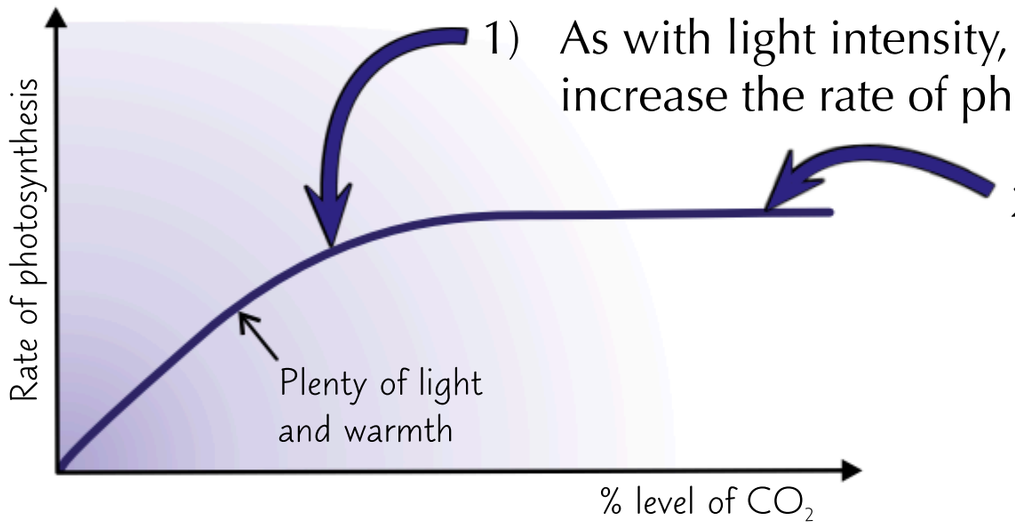
1) As the light level is raised, the rate of photosynthesis increases steadily — but only up to a certain point.



2) Beyond that, it won't make any difference — as light intensity increases, the rate will no longer increase. It'll be either the temperature or the CO_2 level which is now the limiting factor, not light.

2) Too Little **Carbon Dioxide** Also Slows it Down

CO_2 is one of the raw materials needed for photosynthesis.

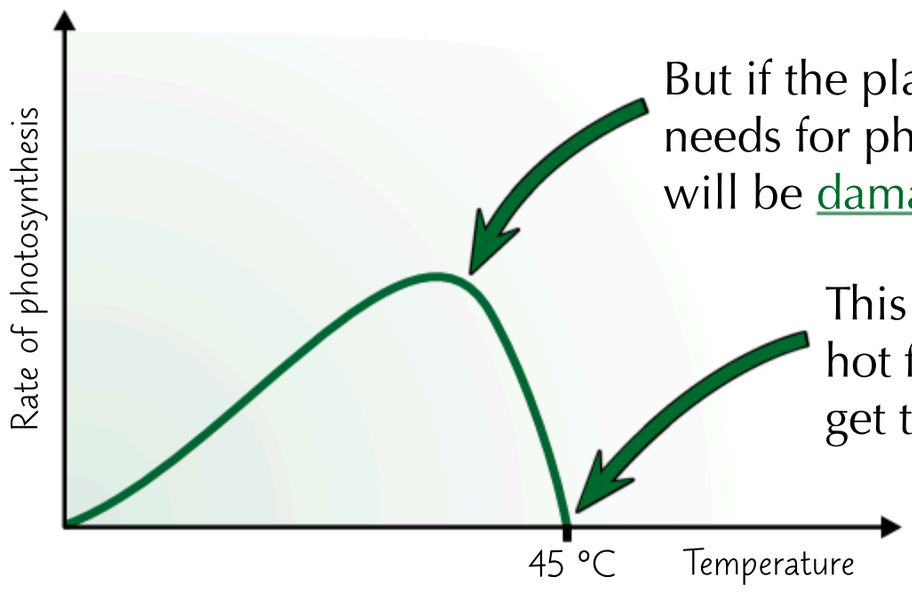


1) As with light intensity, the amount of CO_2 will only increase the rate of photosynthesis up to a point.

2) After this the graph flattens out — as the amount of CO_2 increases, the rate no longer increases. CO_2 is no longer the limiting factor.
As long as light and CO_2 are in plentiful supply then the limiting factor must be temperature.

3) The **Temperature** has to be Just Right

Usually, if the temperature is the limiting factor it's because it's too low — the enzymes needed for photosynthesis work more slowly at low temperatures.



But if the plant gets too hot, the enzymes it needs for photosynthesis and its other reactions will be damaged.

This happens at about 45 °C (which is pretty hot for outdoors, although greenhouses can get that hot if you're not careful).

Graphs, graphs and more graphs...

It's really vital that you've got your head around limiting factors, particularly the graphs. I say this because on the next page you're going to have to tackle graphs that have more than one limiting factor — so there's no point moving on until you're completely happy with dealing with just one on its own first. Give it another read over if you've got doubts...

More on the Rate of Photosynthesis

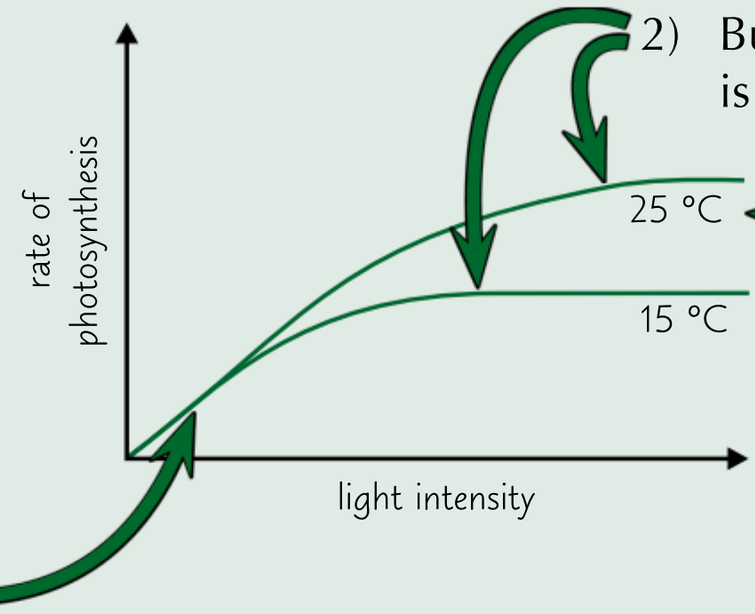
Quick Quiz

*Now that you know all about **limiting factors**, it's time to take it to the next level...*

One Graph May Show the Effect of Many Limiting Factors

You could get a graph that shows more than one limiting factor on the rate of photosynthesis, for example:

This graph shows how the rate of photosynthesis is affected by light intensity and temperature.



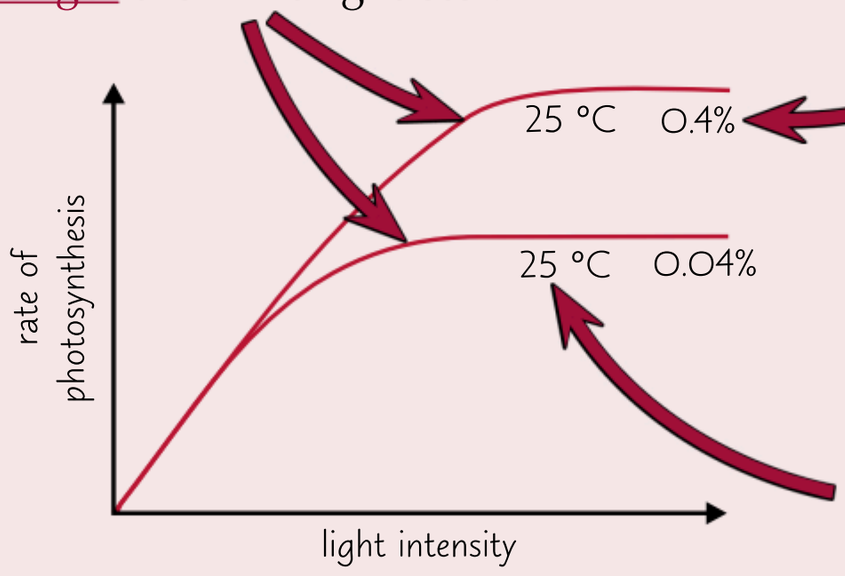
1) At the start, both of the lines show that as the light intensity increases, the rate of photosynthesis increases steadily.

2) But the lines level off when light is no longer the limiting factor.

3) The line at 25 °C levels off at a higher point than the one at 15 °C, showing that temperature must have been a limiting factor at 15 °C.

This graph shows how the rate of photosynthesis is affected by light intensity and CO_2 concentration.

1) Again, both the lines level off when light is no longer the limiting factor.



2) The line at the higher CO_2 concentration of 0.4% levels off at a higher point than the one at 0.04%. This means CO_2 concentration must have been a limiting factor at 0.04% CO_2.

3) The limiting factor here isn't temperature because it's the same for both lines (25 °C).

In the lab you can change the light intensity by moving a lamp closer to or further away from your plant (see the next page for this experiment). But if you just plot the rate of photosynthesis against "distance of lamp from the plant", you get a weird-shaped graph. To get a graph like the one above you either need to measure the light intensity at the plant using a light meter or do a bit of nifty maths with your results.

Make sure you know how to read from graphs

In the exam, you might be asked to describe what's going on in a graph involving several limiting factors. If that happens, don't panic — just look at the graph carefully, making sure you pay close attention to the axes, as well as any additional labels, to see how the factors shown are affecting the rate of photosynthesis.

Quick Quiz

Investigating Photosynthesis

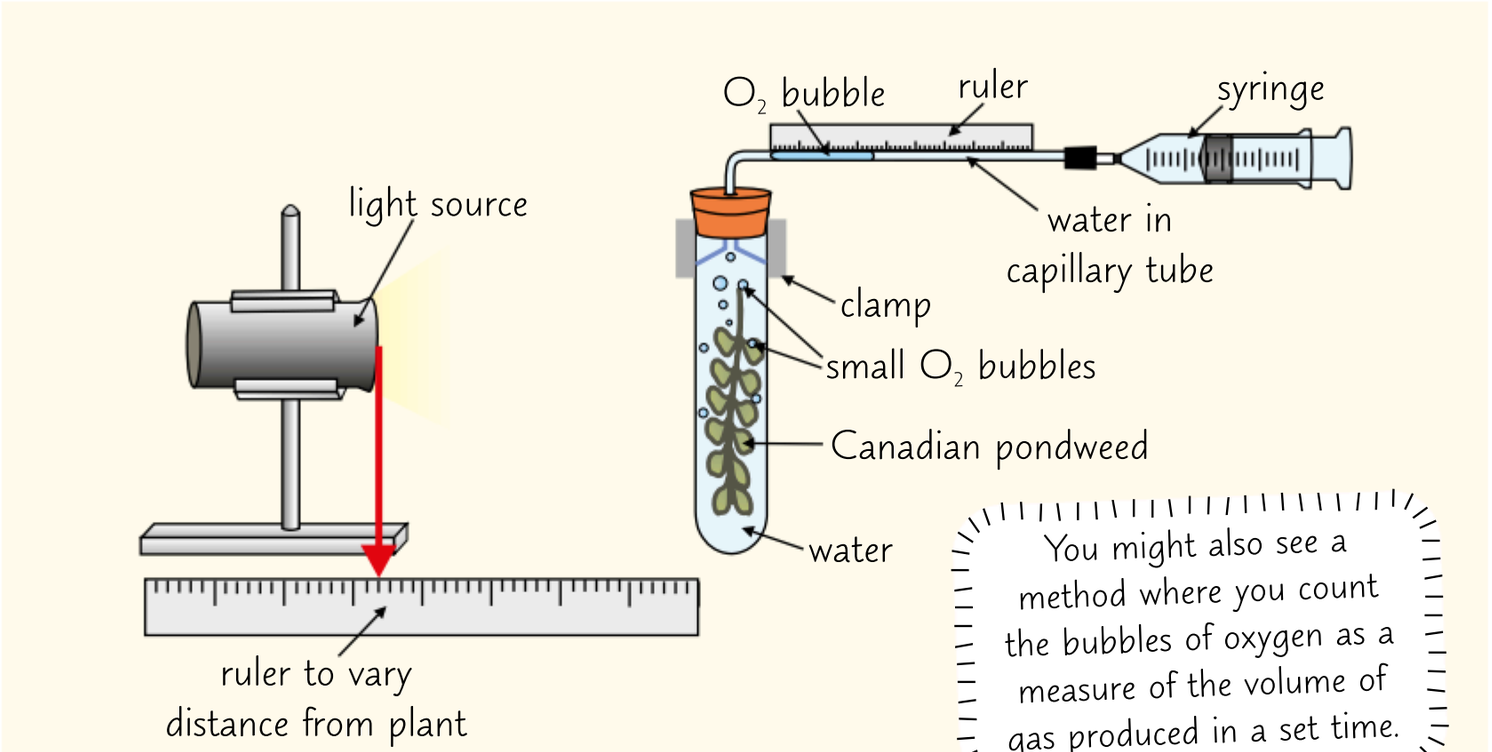
PRACTICAL

*It's practical time again. This one lets you see how changing **light intensity** affects the **rate** of **photosynthesis**.*

Oxygen Production Shows the **Rate** of Photosynthesis

Canadian pondweed can be used to measure the effect of light intensity on the rate of photosynthesis. The rate at which the pondweed produces oxygen corresponds to the rate at which it's photosynthesising — the faster the rate of oxygen production, the faster the rate of photosynthesis.

Here's how you could set up the apparatus:



Method

1) A source of white light is placed at a specific distance from the pondweed.
2) The pondweed is left to photosynthesise for a set amount of time. As it photosynthesises, the oxygen released will collect in the capillary tube.
3) At the end of the experiment, the syringe is used to draw the gas bubble in the tube up alongside a ruler and the length of the gas bubble is measured. This is proportional to the volume of O_2 produced.
4) The experiment is repeated twice with the light source at the same distance and the mean volume of O_2 produced is calculated.
5) Then the whole experiment is repeated with the light source at different distances from the pondweed.

You can compare the results at different light intensities by giving the rate as the length of the bubble per unit time, e.g. cm/min.

For this experiment, any variables that could affect the results should be controlled, e.g. the temperature and time the pondweed is left to photosynthesise.

The apparatus above can be altered to measure the effect of temperature or CO_2 on photosynthesis. E.g. the test tube of pondweed can be put into a water bath at a set temperature, or a measured amount of sodium hydrogencarbonate can be dissolved in the water (which gives off CO_2). The experiment can then be repeated with different temperatures of water / concentrations of sodium hydrogencarbonate.

The more gas produced, the higher the rate of photosynthesis

Rate is all about how much of something is happening over a set period of time. If a lot is happening in a short amount of time then the rate is high — and if not a lot is happening, then the rate is low. For example, in the investigation shown above, if the rate of photosynthesis is high then you'll see lots of oxygen bubbles being produced quickly.

Light Intensity and Farming

Quick Quiz

*The **inverse square law** can be tricky to get your head around, so focus...*

The **Inverse Square Law** Links **Light Intensity** and **Distance**

In the experiment on the previous page, when the lamp is moved away from the pondweed, the amount of light that reaches the pondweed decreases. So, as the distance increases, the light intensity decreases. In other words, distance and light intensity are inversely proportional to each other.

However, it's not quite as simple as that. Light intensity actually decreases in proportion to the square of the distance. This is called the inverse square law and is written out like this:

This is the 'proportional to' symbol.

$$\text{light intensity} \propto \frac{1}{\text{distance (d)}^2}$$

Putting one over the distance shows the inverse.

The distance is squared.

If you halve the distance, the light intensity will be four times greater and if you third the distance, the light intensity will be nine times greater. Likewise, if you double the distance, the light intensity will be four times smaller and if you treble the distance, the light intensity will be nine times smaller.

You can use $1/d^2$ as a measure of light intensity:

EXAMPLE

Calculate the light intensity when the lamp is 10 cm from the pondweed.

MATHS SKILLS

1) Use the formula $\frac{1}{d^2}$.
2) Fill in the values you know — you're given the distance, so put that in.
3) Calculate the answer.

$$\text{light intensity} = \frac{1}{d^2}$$

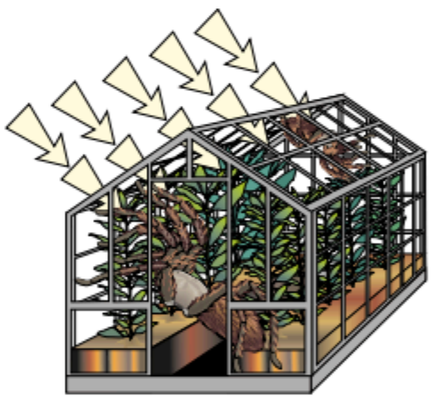
$$\text{light intensity} = \frac{1}{10^2}$$

$$= 0.01 \text{ a.u.}$$

'a.u.' stands for 'arbitrary units'.

Greenhouses can **Artificially Create** the **Ideal Conditions** for **Plants**

Greenhouses help to trap the Sun's heat, and make sure that temperature doesn't become the limiting factor. In winter, a farmer or gardener might use a heater as well to keep the temperature at the ideal level. In summer it could get too hot, so shades and ventilation can cool things down.



Light is always needed for photosynthesis, so commercial farmers often supply artificial light after the Sun sets to give their plants more quality photosynthesis time.

Farmers and gardeners can also increase the level of carbon dioxide in the greenhouse, e.g. by using a paraffin heater to heat the greenhouse. As the paraffin burns, it makes carbon dioxide as a by-product.

Keeping plants enclosed in a greenhouse also makes it easier to keep them free from pests and diseases. The farmer can add fertilisers to the soil as well, to provide all the minerals needed for healthy growth.

Sorting all this out costs money — but if the farmer can keep the conditions just right for photosynthesis, the plants will grow much faster and a decent crop can be harvested much more often, which can then be sold. It's important that a farmer supplies just the right amount of heat, light, etc. — enough to make the plants grow well, but not more than the plants need, as this would just be wasting money.

As one thing goes up, the other goes down...

That's what inverse proportion is all about. So here, as distance decreases, light intensity increases.

Q1 Briefly explain why it can be beneficial for a farmer to grow crops in a greenhouse. [2 marks]

Q2 A plant is moved from 15 cm away from its light source to 5 cm away from its light source. Using the inverse square law, show that the light intensity becomes nine times greater. [3 marks]

Q2 Video Solution

Warm-Up & Exam Questions

Time for a break in the topic and some questions. Do them now, whilst all that learning is fresh in your mind. Using that knowledge will help you to remember it all, and that's what this game is all about.

Warm-Up Questions

1) What does the chemical symbol $C_6H_{12}O_6$ represent?
2) What is meant by a limiting factor for the rate of photosynthesis?
3) Explain why the rate of photosynthesis decreases if the temperature is too high.
4) What could you measure to show the rate of photosynthesis?
5) Write down the inverse square law for light intensity.
6) Why might a farmer want to increase the rate of photosynthesis in her greenhouse of tomatoes?

Exam Questions

1 Photosynthesis produces glucose using light. GRADE 4-6 GRADE

1.1 Complete the word equation for photosynthesis.

carbon dioxide + ... $\xrightarrow{\text{light}}$ glucose + ...

[2 marks]

1.2 Plants use some of the glucose they produce to make a substance which strengthens their cell walls.
Which of the following strengthens cells walls?
Tick **one** box.

☐ cellulose ☐ oils ☐ starch ☐ fats

[1 mark]

1.3 Give **two** others way that plants use the glucose they produce in photosynthesis.

[2 marks]

PRACTICAL

2 A student did an experiment to see how the rate of photosynthesis depends on light intensity. GRADE 4-6 GRADE
Figure 1 shows some of her apparatus.

2.1 How can the student measure the rate of photosynthesis?

[1 mark]

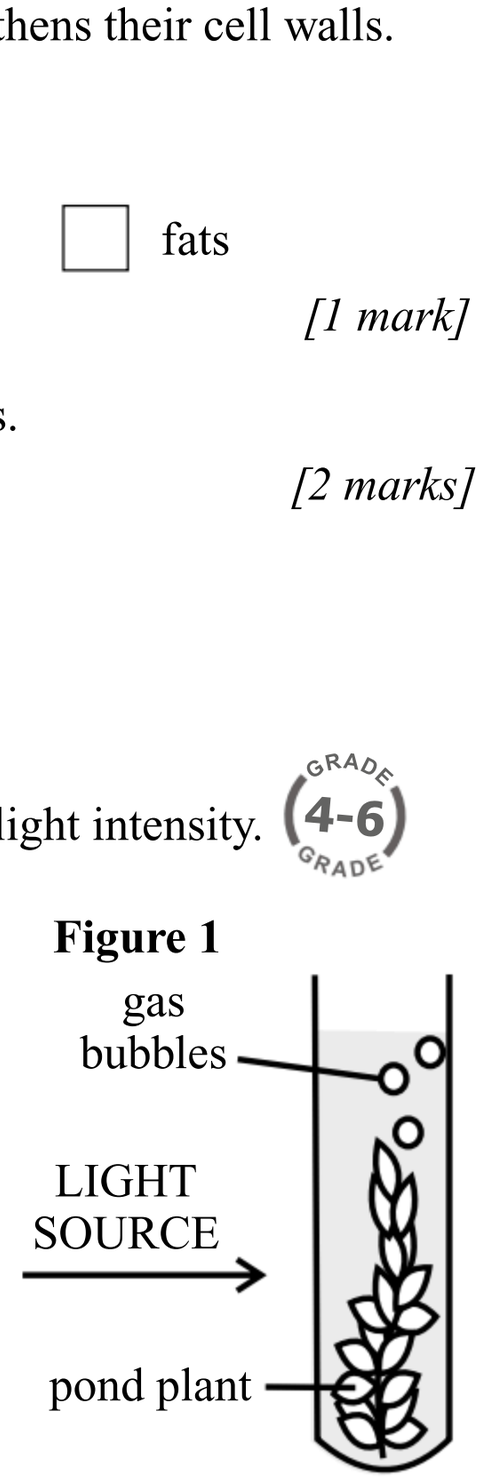
2.2 State the dependent variable and the independent variable in this experiment.

[2 marks]

2.3 State **one** factor that should be kept constant during this experiment.

[1 mark]

Figure 1

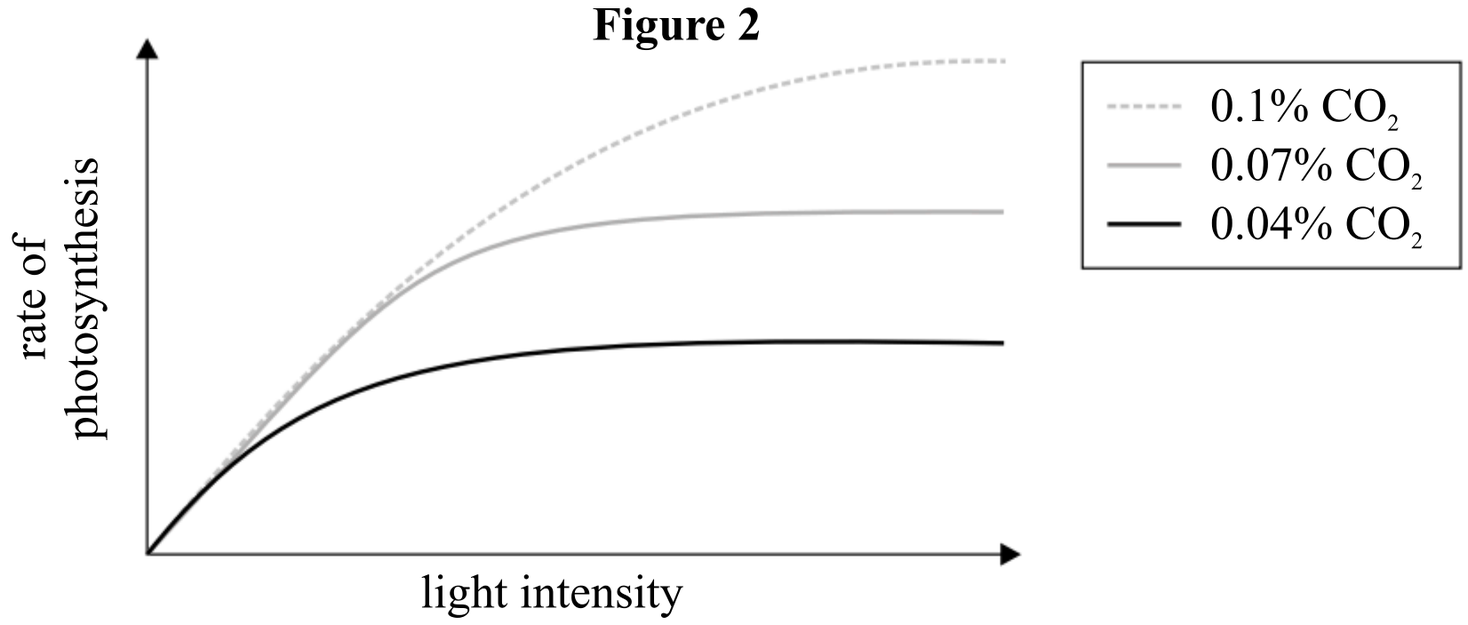


Exam Questions

3 A student investigated the effect of different concentrations of carbon dioxide on the rate of photosynthesis of his Swiss cheese plant. The results are shown in **Figure 2**.

GRADE 6-7

Figure 2



3.1 Describe the effect that increasing the concentration of CO_2 has on the rate of photosynthesis as light intensity increases.

[2 marks]

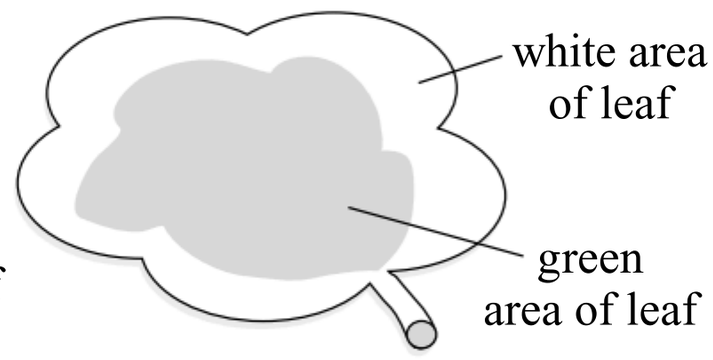
3.2 Explain why all the lines on the graph level off eventually.

[1 mark]

4 **Figure 3** shows a variegated leaf. It is partly green and partly white. Chlorophyll is present in the green parts of the leaf but not the white parts.

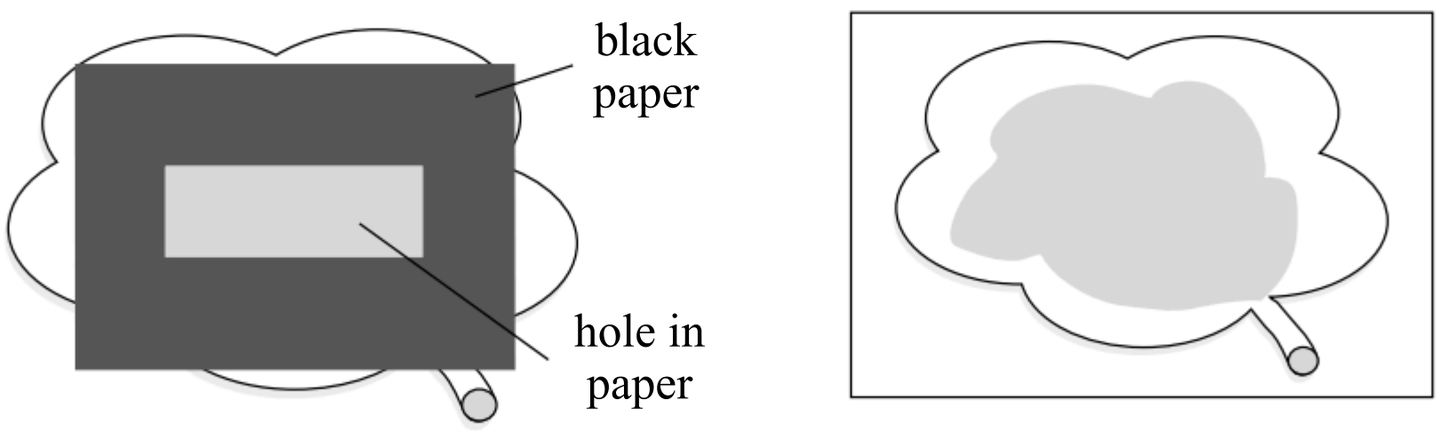
GRADE 6-7

Figure 3



A student did an experiment in which part of the leaf was covered with black paper, as shown in **Figure 4**. The leaf was then exposed to light for four hours and was then tested for starch.

Figure 4



4.1 Complete **Figure 4** by shading in the part(s) of the leaf that you would expect to contain **starch**.

[1 mark]

4.2 Explain your answer to **4.1**.

[2 marks]

5 A student is investigating the effect of light intensity on the rate of photosynthesis by placing a lamp at various distances from a plant and measuring the rate of photosynthesis.

GRADE 7-9

Use the inverse square law to calculate the light intensity when the lamp is 7.5 cm from the plant.
Give your answer in arbitrary units (a.u.) to 2 significant figures.

[2 marks]

Quick Quiz

Respiration and Metabolism

*You need **energy** to keep your body going. Energy comes from **food**, and it's **transferred** by **respiration**.*

Respiration is **NOT** "Breathing In and Out"

1) Respiration involves many reactions. These are really important reactions, as respiration transfers the energy that cells need to do just about everything — this energy is used for all living processes.
2) The energy that's transferred in respiration comes from the breakdown of glucose (sugar).
3) Respiration goes on in every cell in your body continuously. It happens in all animals, and in plants too. It's how living things transfer energy from their food to their cells.

RESPIRATION is the process of TRANSFERRING ENERGY FROM GLUCOSE, which goes on IN EVERY CELL.

4) Respiration is exothermic — it transfers energy to the environment.

Respiration **Transfers Energy** for All Kinds of Things

Here are three examples of how organisms use the energy transferred by respiration:

- To build up larger molecules from smaller ones (like proteins from amino acids — see below).
- In animals it's used to allow the muscles to contract (so they can move about).
- In mammals and birds the energy is used to keep their body temperature steady in colder surroundings.

Unlike other animals, mammals and birds keep their bodies constantly warm.

Metabolism is **ALL** the **Chemical Reactions** in an **Organism**

In a cell there are lots of chemical reactions happening all the time, which are controlled by enzymes.
Many of these reactions are linked together to form bigger reactions:

Enzymes are biological catalysts — see p.27.

In some of these reactions, larger molecules are made from smaller ones. For example:

- Lots of small glucose molecules are joined together in reactions to form starch (a storage molecule in plant cells), glycogen (a storage molecule in animal cells) and cellulose (a component of plant cell walls).
- Lipid molecules are each made from one molecule of glycerol and three fatty acids.
- Glucose is combined with nitrate ions to make amino acids, which are then made into proteins.

In other reactions, larger molecules are broken down into smaller ones. For example:

- Glucose is broken down in respiration.
- Excess protein is broken down in a reaction to produce urea. Urea is then excreted in urine.

The sum (total) of all of the reactions that happen in a cell or the body is called its metabolism.

The energy released in respiration is used in metabolism

So... respiration is a pretty important thing. Enzymes are key to metabolism, so if you need a reminder about them, now is a good time to head back to page 27. Even if you are happy with enzymes, don't move on just yet — check you've taken in everything on this page before you flip over to continue on through the world of respiration. It's important stuff.

Aerobic and Anaerobic Respiration

Quick Quiz

*There are **two types** of **respiration** — aerobic and anaerobic. You need to know about them both, so read on.*

Aerobic Respiration Needs Plenty of **Oxygen**

Aerobic respiration is respiration using oxygen. It's the most efficient way to transfer energy from glucose.
Here are the word and symbol equations for aerobic respiration:

glucose + oxygen → carbon dioxide + water

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

Aerobic respiration goes on all the time in plants and animals.
Most of the reactions in aerobic respiration happen inside mitochondria (see page 4).

Anaerobic Respiration is Used if There's **Not Enough Oxygen**

When you do vigorous exercise and your body can't supply enough oxygen to your muscles, they start doing anaerobic respiration as well as aerobic respiration.

1) "Anaerobic" just means "without oxygen". It's the incomplete breakdown of glucose, making lactic acid.

2) Here's the word equation for anaerobic respiration in muscle cells:

glucose → lactic acid

3) Anaerobic respiration does not transfer nearly as much energy as aerobic respiration. This is because glucose isn't fully oxidised (because it doesn't combine with oxygen).

Anaerobic respiration is only useful in emergencies, e.g. during exercise when it allows you to keep on using your muscles for a while longer.

Anaerobic Respiration in **Plants** and **Yeast** is Slightly **Different**

Plants and yeast cells can respire without oxygen too, but they produce ethanol (alcohol) and carbon dioxide instead of lactic acid.

Here is the word equation for anaerobic respiration in plants and yeast cells:

glucose → ethanol + carbon dioxide

Yeast are single-celled organisms.

Anaerobic respiration in yeast cells is called fermentation.

In the food and drinks industry, fermentation by yeast is of great value because it's used to make bread and alcoholic drinks, e.g. beer and wine.

In bread-making, it's the carbon dioxide from fermentation that makes bread rise.

In beer and wine-making, it's the fermentation process that produces alcohol.

Yeast fermentation is used to make a load of different things

Fermentation is a really important process because of its use in making alcoholic drinks and bread.
We drink and eat so much of these that making them is big bucks. And it's all down to tiny yeast cells.

Quick Quiz

Exercise

*When you **exercise**, your body responds in a number of helpful ways...*

When You **Exercise** You **Respire More**

1) Muscles need energy from respiration to contract. When you exercise, some of your muscles contract more frequently than normal so you need more energy. This energy comes from increased respiration.
2) The increase in respiration in your cells means you need to get more oxygen into them. To do this...

- breathing rate and breath volume increase → to get more oxygen into the blood
- heart rate increases → to get oxygenated blood around the body faster and remove CO_2 quickly

3) When you do really vigorous exercise (like sprinting), your body can't supply oxygen to your muscles quickly enough, so they start respiring anaerobically (see the previous page).
4) This is NOT the best way to transfer energy from glucose because lactic acid builds up in the muscles, which gets painful.
5) Long periods of exercise also cause muscle fatigue — the muscles get tired and then stop contracting efficiently.

Remember, lactic acid is formed from the incomplete oxidation of glucose.

Anaerobic Respiration Leads to an **Oxygen Debt**

After resorting to anaerobic respiration, you'll have an "oxygen debt" when you stop exercising. This is the amount of extra oxygen your body needs to remove the build up of lactic acid from the cells. Oxygen reacts with the lactic acid to form harmless CO_2 and water.

In other words you have to "repay" the oxygen that you didn't get to your muscles in time, because your lungs, heart and blood couldn't keep up with the demand earlier on. This means:

- you have to keep breathing hard for a while after you stop, to get more oxygen into your blood, which is transported to the muscle cells.
- your pulse and breathing rate stay high whilst there are high levels of lactic acid and CO_2.

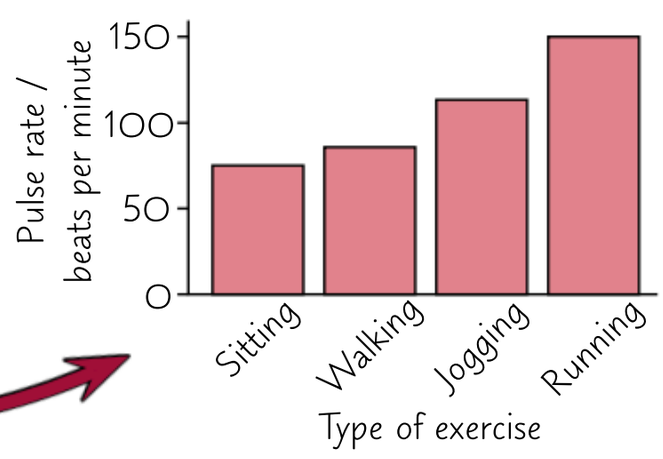
Your body also has another way of coping with the high level of lactic acid — the blood that enters your muscles transports the lactic acid to the liver. In the liver, the lactic acid is converted back to glucose.

You Can Investigate The **Effect of Exercise** on **The Body**

1) You can measure heart rate by taking the pulse. E.g. you could take your pulse after:

- sitting down for 5 minutes,
- then after 5 minutes of gentle walking,
- then again after 5 minutes of slow jogging,
- then again after running for 5 minutes,

and plot your results on a bar chart.



To take your pulse, put two fingers on your inner wrist or neck and count the number of pulses in 1 minute.

You could also do the same with breathing rate by counting breaths.

2) To reduce the effect of any random errors (see p.173) on your results, do it as a group and plot the average pulse rate for each exercise.

Your pulse rate will increase the more intense the exercise is, as your body needs to get more oxygen to the muscles and take more carbon dioxide away from the muscles.

Very vigorous exercise = anaerobic respiration = oxygen debt

After a race, you see athletes breathing hard — and now you know that this is to get rid of the lactic acid.

Q1 Look at the graph above. Predict which type of exercise would lead to the highest concentration of lactic acid in the blood after 10 minutes. Explain your answer. [4 marks]

Q1 Video Solution

Warm-Up & Exam Questions

You know the drill by now — work your way through the Warm-Up questions, then the Exam Questions.

Warm-Up Questions

1) True or false? Respiration is exothermic.
2) What are the reactants of aerobic respiration?
3) State three changes that take place in the body during vigorous exercise.

Exam Questions

1 Respiration is a process carried out by all living cells.
It can take place aerobically or anaerobically.

GRADE 4-6 GRADE

1.1 State the purpose of respiration.

[1 mark]

1.2 Give **two** differences between aerobic and anaerobic respiration.

[2 marks]

1.3 Write the word equation for aerobic respiration.

[2 marks]

2 In the human body, respiration may be aerobic or anaerobic at different times.

GRADE 6-7 GRADE

2.1 Write down the word equation for anaerobic respiration in humans.

[1 mark]

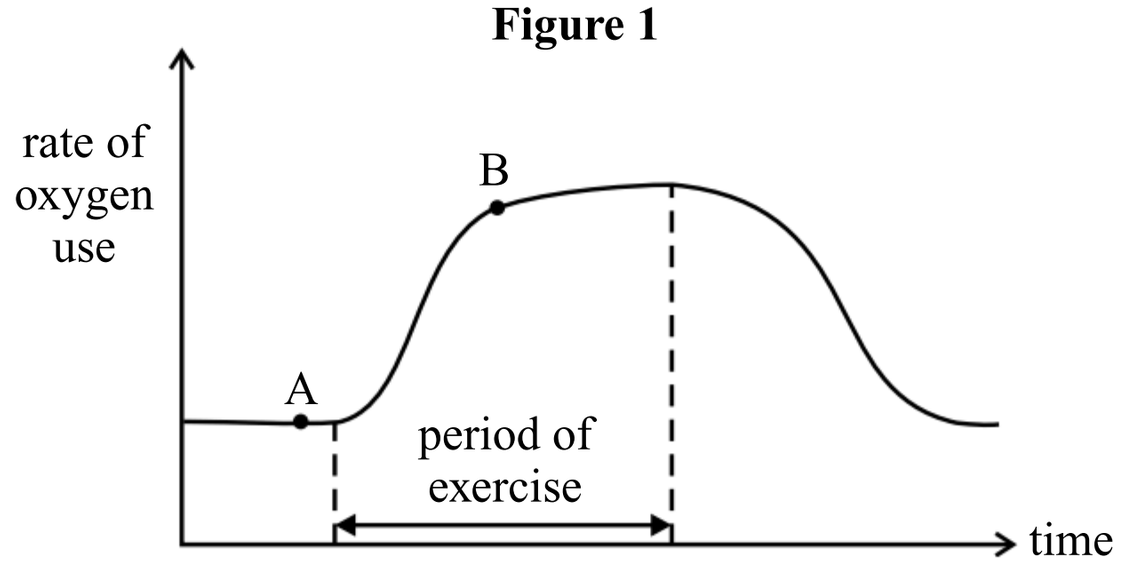
2.2 Explain why the body uses anaerobic respiration during vigorous exercise.

[2 marks]

3 **Figure 1** shows the rate of oxygen use by a person before, during and after a period of exercise.

GRADE 6-7 GRADE

Figure 1



3.1 Explain why the rate of oxygen consumption is higher at **B** than at **A**.

[2 marks]

3.2 Suggest why oxygen use remains high, even after the period of exercise ends.

[1 mark]

Revision Summary Test for Topic 4

Another busy old section. To see if you've got any limiting factors, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Photosynthesis (p.71-75) ☑

1) Describe the process of photosynthesis, including the word and symbol equations. ☑
2) Where in a plant cell does photosynthesis take place? ☑
3) Is photosynthesis exothermic or endothermic? How can you tell? ☑
4) Give five ways that plants use glucose. ☑
5) What is meant by a 'limiting factor' of photosynthesis? ☑
6) Why might the amount of chlorophyll in a plant be reduced?
   Why can this make chlorophyll a limiting factor of photosynthesis? ☑
7) Describe how each of the following factors affects the rate of photosynthesis:
   a) light intensity, b) concentration of CO_2, c) temperature. ☑
8) Describe how you could investigate the effect of light intensity on the rate of photosynthesis. ☑
9) The distance of a lamp from a subject and the light intensity are inversely proportional to each other.
   What does this mean? ☑
10) What is the inverse square law? ☑
11) How do greenhouses help to ensure that temperature doesn't become a limiting factor? ☑
12) Give one way that the temperature can be decreased in a greenhouse. ☑
13) Give one way that the level of carbon dioxide can be increased in a greenhouse. ☑
14) Explain why creating the ideal conditions for plants in a greenhouse needs to be done carefully. ☑

Respiration and Metabolism (p.78-80) ☑

15) What is respiration? Is it exothermic or endothermic? ☑
16) Give three examples of how organisms use the energy transferred by respiration. ☑
17) Give five examples of metabolic reactions. ☑
18) What is meant by metabolism? ☑
19) Name the type of respiration that requires oxygen.
    Give the word and symbol equation for this reaction. ☑
20) How do the products of anaerobic respiration differ between animals and plants? ☑
21) Which type of respiration transfers less energy?
    Why is less energy transferred in this type of respiration? ☑
22) What's the name for anaerobic respiration in yeast cells?
    Give two examples of where this is a useful reaction. ☑
23) Why does respiration need to increase during exercise? ☑
24) Give three things that increase to supply the muscles with more oxygenated blood during exercise. ☑
25) What builds up in muscles during anaerobic respiration? ☑
26) What happens to muscles when they become fatigued? ☑
27) Which type of respiration leads to an 'oxygen debt'? What does this mean, and how is it repaid? ☑
28) In which organ is lactic acid converted back to glucose? ☑
29) Describe an experiment to investigate how exercise affects heart rate.
    What would you expect to see? ☑

Homeostasis

Quick Quiz

Homeostasis *— a word that strikes fear into the heart of many a GCSE student. But it's really not that bad at all. This page is a brief* ***introduction*** *to the topic, so you need to* ***nail all of this*** *before you can move on.*

Homeostasis — Maintaining a Stable Internal Environment

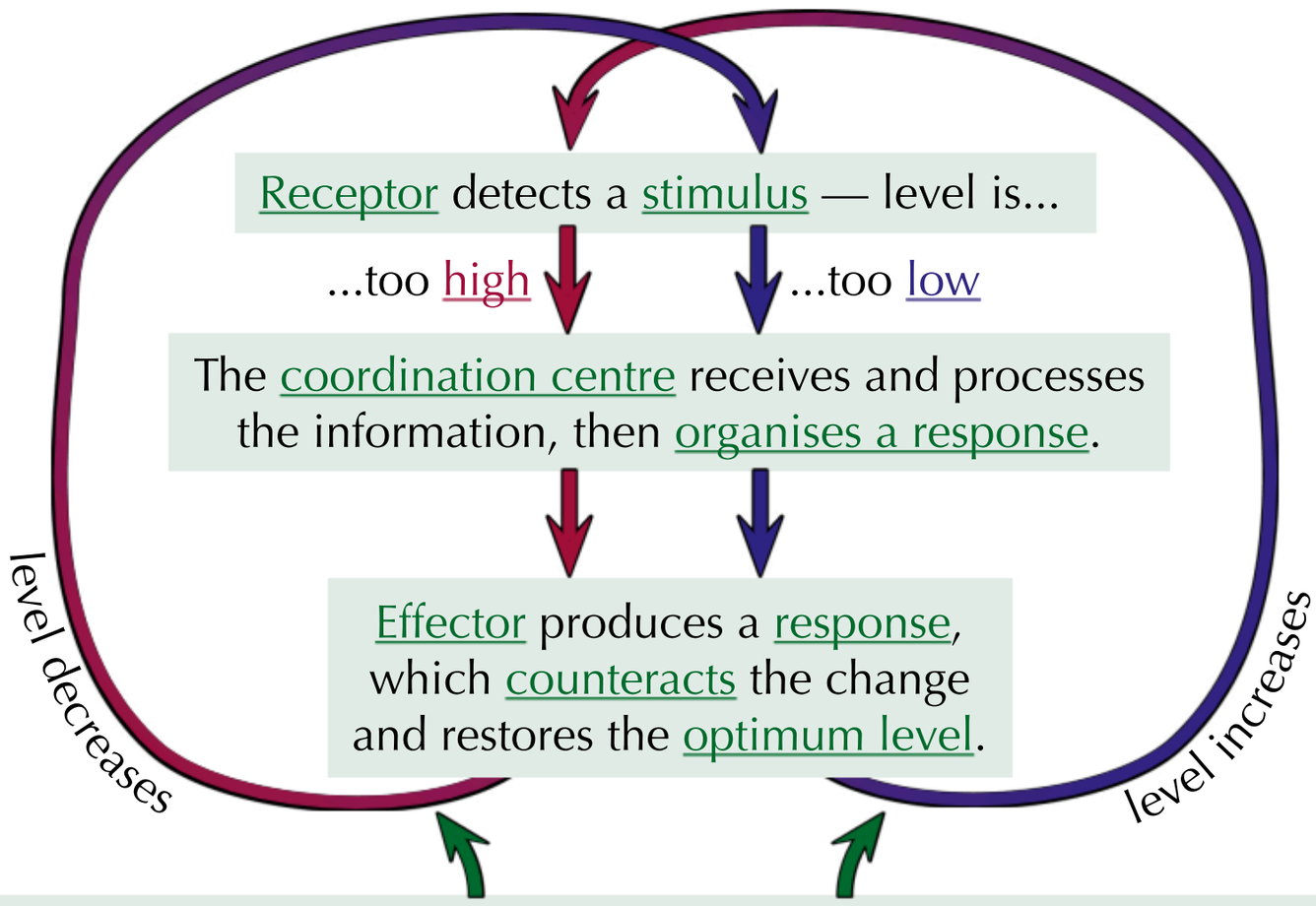
- Conditions in your body need to be kept steady, so that your cells have the right conditions to function properly, including the right conditions for maximum enzyme activity (see p.27).

HOMEOSTASIS is the REGULATION of the conditions inside your body (and cells) to maintain a STABLE INTERNAL ENVIRONMENT, in response to CHANGES in both internal and external conditions.

- Your body has loads of automatic control systems that regulate your internal environment — these include both nervous and hormonal communication systems. For example, there are control systems that maintain your body temperature (see p.92), your blood glucose level (see p.97) and your blood water content (see p.99).
- These systems are made up of three main components which work together to maintain a steady condition — cells called receptors, coordination centres (including the brain, spinal cord and pancreas) and effectors (see next page).

Negative Feedback Counteracts Changes

Your automatic control systems keep your internal environment stable using a mechanism called negative feedback. When the level of something (e.g. water or temperature) gets too high or too low, negative feedback brings it back to normal. This is all automatic — it happens without you thinking about it.



A stimulus is a change in the environment.

The effectors will just carry on producing the responses for as long as they're stimulated by the coordination centre. This might cause the opposite problem — making the level change too much (away from the ideal). Luckily, the receptor detects if the level becomes too different and negative feedback starts again.

If you do enough revision, you can avoid negative feedback...

Negative feedback is a fancy-sounding name for a not-very-complicated idea. Basically, when something increases or decreases from the normal level, negative feedback kicks in to bring it back to normal.

Quick Quiz

The Nervous System

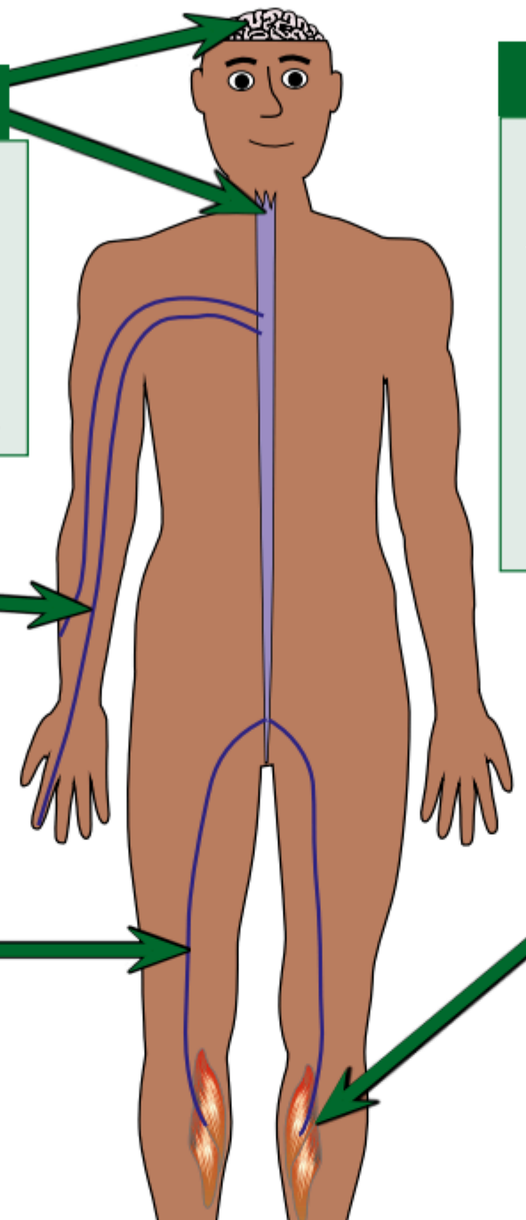
*The cells of **multicellular** organisms need to **communicate** with each other in order to respond to stimuli. So as multicellular organisms evolved, they developed **nervous** and **hormonal communication systems**.*

The **Nervous System** is Made up of **Different Parts**

The nervous system allows humans to react to their surroundings and coordinate their behaviour. It's made up of all the neurones (nerve cells) in your body.

Central Nervous System (CNS)

In vertebrates (animals with backbones) this consists of the brain and spinal cord only. In mammals, the CNS is connected to the body by sensory neurones and motor neurones.



Sensory Neurones

These carry information as electrical impulses from the receptors to the CNS.

Motor Neurones

These carry electrical impulses from the CNS to effectors.

Receptors

1) These are the cells that detect stimuli.
2) There are many different types, such as taste receptors on the tongue and sound receptors in the ears.
3) Receptors can form part of larger, complex organs, e.g. the retina of the eye is covered in light receptor cells.

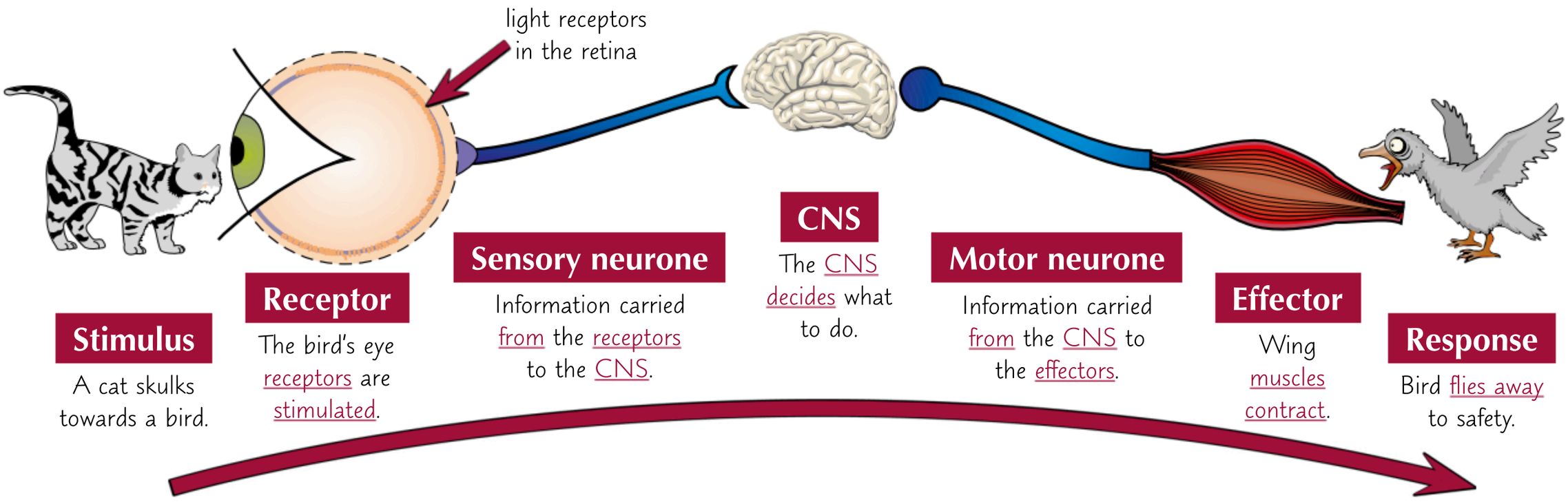
Effectors

1) All your muscles and glands, which respond to nervous impulses and bring about change.
2) Effectors respond in different ways. Muscles contract in response to a nervous impulse, whereas glands secrete hormones.

Effectors can form part of complex organs too — e.g. the pancreas.

The **Central Nervous System** (CNS) **Coordinates** the **Response**

The CNS is a coordination centre — it receives information from the receptors and then coordinates a response (decides what to do about it). The response is carried out by effectors.



The exam isn't just a test of what you know...

...it's also a test of how well you can apply what you know. E.g. you might have to take what you know about a human's nervous system and apply it to a cat. The key is not to panic — think carefully about the information that you're given.

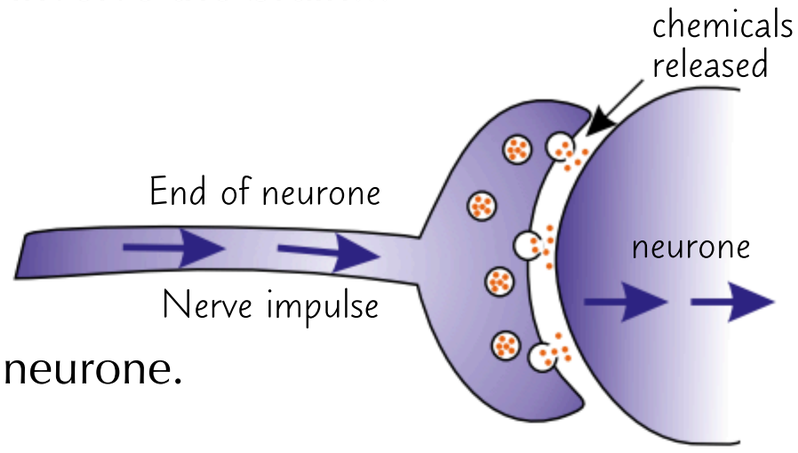
Synapses and Reflexes

Quick Quiz

__Neurones__ transmit information __very quickly__ to and from the brain, and your brain __quickly decides__ how to respond to a stimulus. __Reflexes__ are rapid responses that don't involve the brain...

__Synapses__ Connect Neurones

- The connection between two neurones is called a synapse.
- The nerve signal is transferred by chemicals, which diffuse (move) across the gap.
- These chemicals then set off a new electrical signal in the next neurone.



__Reflexes__ Help __Prevent Injury__

- Reflexes are rapid, automatic responses to certain stimuli that don't involve the conscious part of the brain — they can reduce the chances of being injured.
- For example, if someone shines a bright light in your eyes, your pupils automatically get smaller so that less light gets into the eye — this stops it getting damaged.
- Or if you get a shock, your body releases the hormone adrenaline automatically — it doesn't wait for you to decide that you're shocked.
- The passage of information in a reflex (from receptor to effector) is called a reflex arc:

Relay neurones connect sensory neurones to motor neurones.

Video

1 Stimulus (e.g. painful bee sting) is detected by receptors (e.g. a pain receptor).

2 Impulses are sent along a sensory neurone.

3 Impulses reach a synapse between the sensory and relay neurones. This triggers chemicals to be released (see above) causing the impulses to be passed along a relay neurone in the CNS.

4 Impulses reach a synapse between the relay and motor neurones. Chemicals are released causing the impulses to be sent along the motor neurone.

5 Impulses travel to the effector, which responds (e.g. a muscle contracts and moves your hand away from the bee).

The neurones in reflex arcs go through the spinal cord or through an unconscious part of the brain. Since you don't have to think about the response (which takes time) it's quicker than normal responses.

Don't get all twitchy — just learn it...

Q2 Video Solution

Reflexes bypass the conscious parts of your brain completely when a quick response is essential — your body just gets on with things. If you had to stop and think first, you'd end up a lot more sore (or worse).

Q1 What is a reflex action? [1 mark]

Q2 A chef touches a hot pan. A reflex reaction causes him to immediately move his hand away.
a) State the effector in this reflex reaction. [1 mark]
b) Describe the pathway of the reflex from stimulus to effector. [4 marks]

Quick Quiz

Investigating Reaction Time

PRACTICAL

*The time it takes you to **respond** to a stimulus is your **reaction time** — here's one way you can investigate it...*

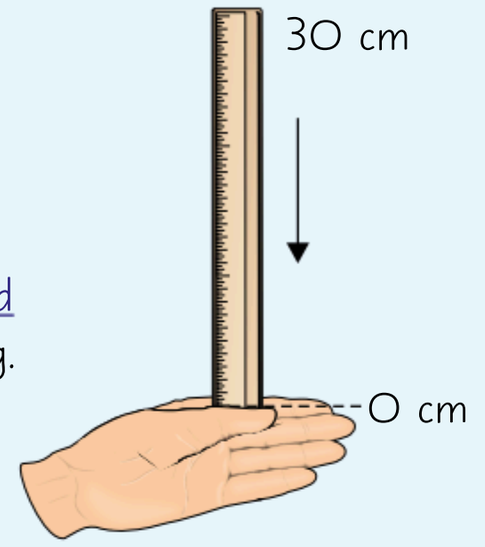
Reaction Time is How Quickly You Respond

Reaction time is the time it takes to respond to a stimulus — it's often less than a second.
It can be affected by factors such as age, gender or drugs.

Method

Caffeine is a drug that can speed up a person's reaction time.
This is one way the effect of caffeine on reaction time can be measured...

1) The person being tested should sit with their arm resting on the edge of a table (this should stop them moving their arm up or down during the test).
2) Hold a ruler vertically between their thumb and forefinger. Make sure that the zero end of the ruler is level with their thumb and finger. Then let go without giving any warning.
3) The person being tested should try to catch the ruler as quickly as they can.
4) Reaction time is measured by the number on the ruler where it's caught. The number should be read from the top of the thumb. The further down the ruler it's caught (i.e. the higher the number), the slower their reaction time.
5) Repeat the test several times then calculate the mean distance that the ruler fell.
6) The person being tested should then have a caffeinated drink (e.g. 300 ml of cola). After ten minutes, repeat steps 1 to 5.
7) You need to control any variables to make sure that this is a fair test, e.g.
   - The same person should catch the ruler each time, and they should use the same hand to catch the ruler.
   - The ruler should always be dropped from the same height.
   - The person being tested should not have had any caffeine (or anything else that may affect their reaction time) before the start of the experiment.
8) Too much caffeine can cause unpleasant side-effects, so the person being tested should avoid drinking any more caffeine for the rest of the day after the experiment is completed.



With a little bit of maths, it's possible to work out the reaction time in seconds using the mean distance.

Reaction Time Can Also Be Measured Using Simple **Computer Tests**

- For example, the person being tested has to click the mouse (or press a key) as soon as they see a stimulus on the screen, e.g. a box change colour.
- Computers can give a more precise reaction time because they remove the possibility of human error from the measurement.
- As the computer can record the reaction time in milliseconds, it can also give a more accurate measurement.
- Computer tests can also remove the possibility that the person can predict when to respond — using the ruler test, the catcher may learn to anticipate the drop by reading the tester's body language.

See page 173 for more about precise and accurate measurements.

As with any practical, you need to control the variables

Q1 Video Solution

Remember, the independent variable is the one you change and the dependent variable is the one you measure.

Q1 Some students investigated the effect of an energy drink on reaction time. They measured their reaction times using a computer test. They had to click the mouse when the screen changed from red to green. Each student repeated the test five times before having an energy drink, and five times afterwards.

a) The results for one of the students before having the energy drink were as follows: 242 ms, 256 ms, 253 ms, 249 ms, 235 ms. Calculate the mean reaction time. [2 marks]

b) Suggest two variables that the students needed to control during their investigation. [2 marks]

Warm-Up & Exam Questions

Welcome to some questions. There are quite a few of them, but that's because they're pretty important...

Warm-Up Questions

1) What is homeostasis?
2) State three things that are controlled by homeostasis in the human body.
3) What name is given to the connection between two neurones?
4) Name the three types of neurone in a reflex arc.
5) State one factor that affects human reaction time.

Exam Questions

PRACTICAL

1 A student is taking part in an experiment to test reaction times. Every time a red triangle appears on the computer screen in front of her, she has to click the mouse.

GRADE 4-6 GRADE

1.1 Suggest what the stimulus, receptors and effectors are in this experiment.

[3 marks]

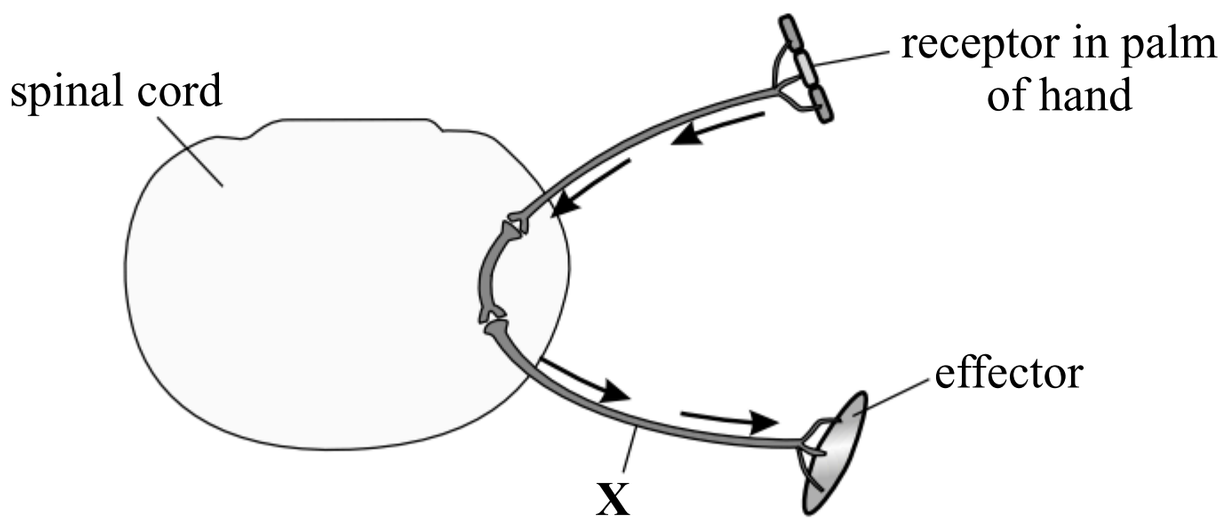
1.2 The student took the test three times. Her reaction time in test 1 was 328 ms. Her reaction time in test 2 was 346 ms. Her mean reaction time was 343 ms. Calculate her reaction time for test 3.

[2 marks]

2 Young babies have several reflexes not usually present in adults. For example, if an object is placed in the palm of a newborn baby's hand, the baby will move their fingers to grasp the object. The reflex arc for this reflex is shown in **Figure 1**.

GRADE 6-7 GRADE

Figure 1



2.1 Name the structure labelled **X** on **Figure 1**.

[1 mark]

2.2 State the type of effector in this response and describe its action.

[2 marks]

2.3 Explain how an electrical impulse in one neurone is able to pass to the next neurone.

[2 marks]

2.4 If an object is placed in the palm of a baby over 6 months old, it can choose whether it wants to grasp hold of the object. Describe **one** way in which the pathway of nervous impulses involved in grasping an object differs between a newborn baby and a baby older than 6 months.

[1 mark]

Quick Quiz

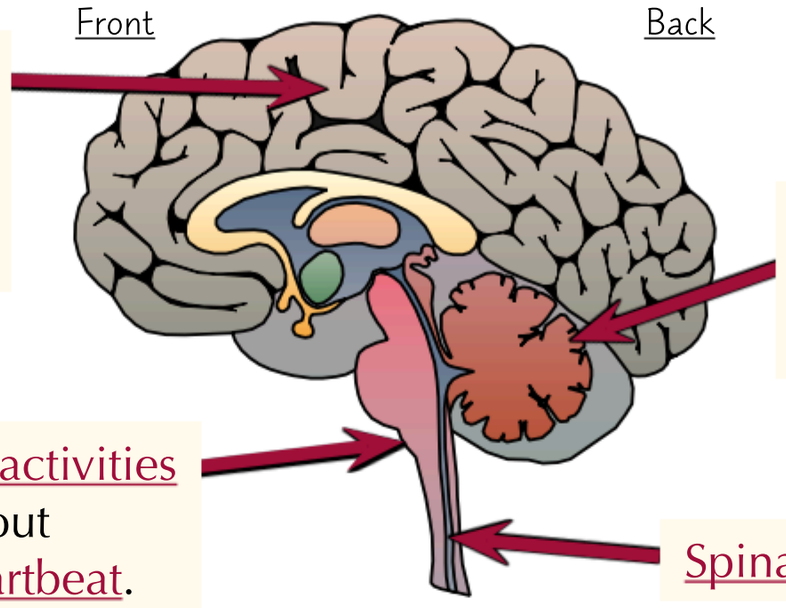
The Brain

*Scientists know a bit about the **brain** but **not as much** as they'd like.*

The **Brain** is Responsible for **Complex Behaviours**

1) The brain is made up of billions of interconnected neurones (neurones that are connected together).
2) It controls and coordinates everything you do — running, breathing, sleeping, remembering your gym kit...
3) We know that different regions of the brain carry out different functions:

Cerebral cortex — This is the outer wrinkly bit. It's responsible for things like consciousness, intelligence, memory and language.



Cerebellum — Responsible for muscle coordination.

Medulla — Controls unconscious activities (things you don't have to think about doing) like breathing and your heartbeat.

Spinal cord

Scientists Use a Range of **Methods** to **Study** the **Brain**

Scientists that study the brain are called neuroscientists.
They use different techniques to map out which bits of the brain do what:

Studying patients with brain damage	If a small part of the brain has been damaged, the effect this has on the patient can tell you a lot about what the damaged part of the brain does. E.g. if an area at the back of the brain was damaged by a stroke and the patient went blind, you know that that area has something to do with vision.
Electrical stimulation	Scientists can observe the effect of stimulating different parts of the brain electrically with a tiny electrode to get an idea of what the parts do. E.g. when a certain part of the brain, known as the motor area, is stimulated, it causes muscle contraction and movement.
MRI scans	An MRI scanner is a big fancy tube-like machine that can produce a very detailed picture of the brain's structures. Scientists use it to find out what areas of the brain are active when people are doing things like listening to music or trying to recall a memory.

Messing With the **Brain** Can Have **Consequences**

1) Knowledge of how the brain works has led to the development of treatments for disorders of the nervous system. For example, electrical stimulation of the brain can help reduce muscle tremors caused by nervous system disorders such as Parkinson's disease.
2) However, the brain is incredibly complex and delicate — the investigation of brain function and any treatment of brain damage or disease is difficult. It also carries risks, such as physical damage to the brain or increased problems with brain function (e.g. difficulties with speech).

Strange that the brain is the only organ that named itself...

Anyway, in your exam you might be asked to identify regions of the brain on a diagram or describe what each part does. So now would be a great time to activate your cerebral cortex and get this page firmly lodged in there.

The Eye

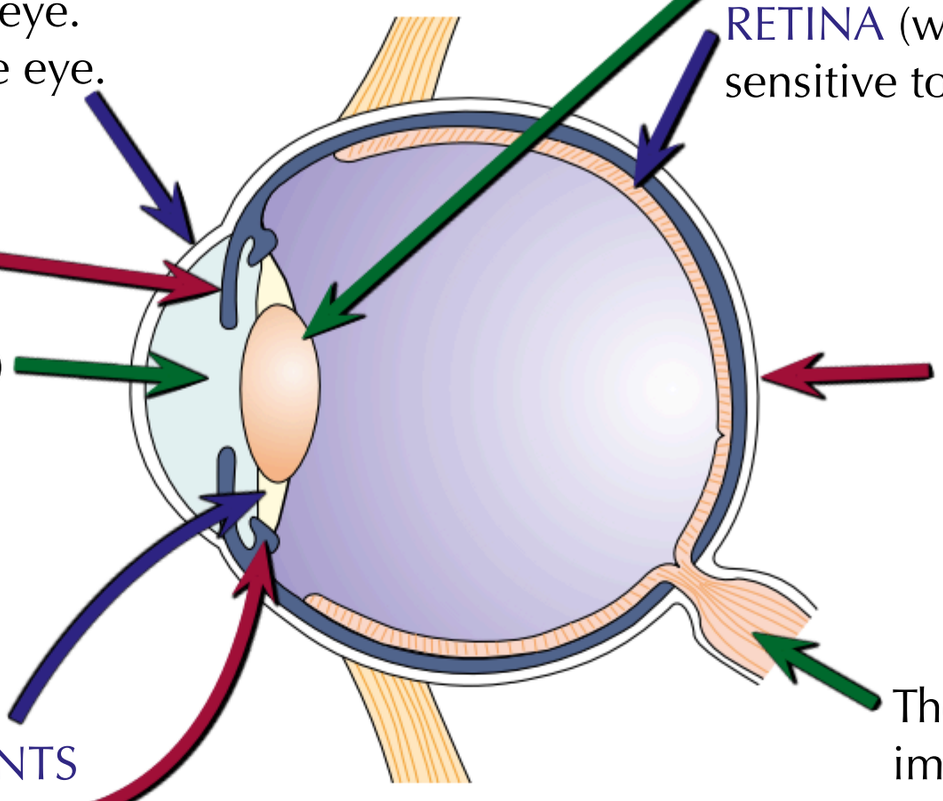
Quick Quiz

*The **eye** is a **sense organ**. There's a lot to get your head around, so get **focused**.*

Learn the Eye with All Its Labels

The CORNEA is the transparent outer layer found at the front of the eye. It refracts (bends) light into the eye.

The LENS focuses the light onto the RETINA (which contains receptor cells sensitive to light intensity and colour).



The IRIS contains muscles that allow it to control the diameter of the PUPIL (the hole in the middle) and therefore how much light enters the eye.

The SCLERA is the tough, supporting wall of the eye.

The shape of the lens is controlled by the SUSPENSORY LIGAMENTS and CILIARY MUSCLE.

The OPTIC NERVE carries impulses from the receptors on the retina to the brain.

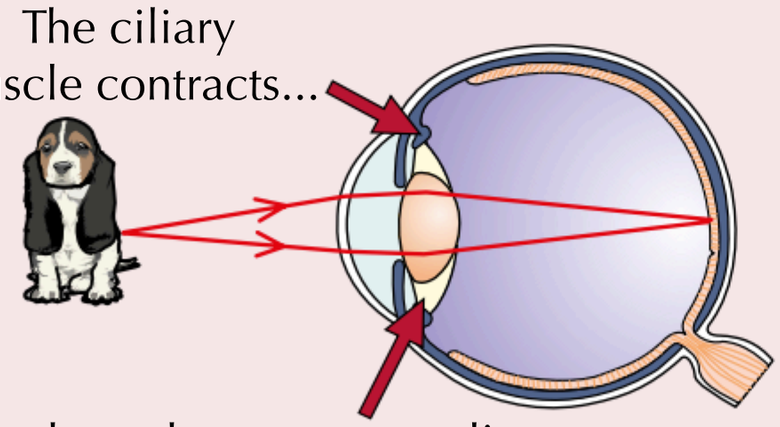
Focusing on Near and Distant Objects — Accommodation Reflex

Video

The eye focuses light on the retina by changing the shape of the lens — this is known as accommodation. When an object is nearer to the eye, the lens needs to refract (bend) the light more to ensure it hits the retina. When an object is further away, the amount by which the lens needs to refract the light decreases.

To look at near objects:

The ciliary muscle contracts...

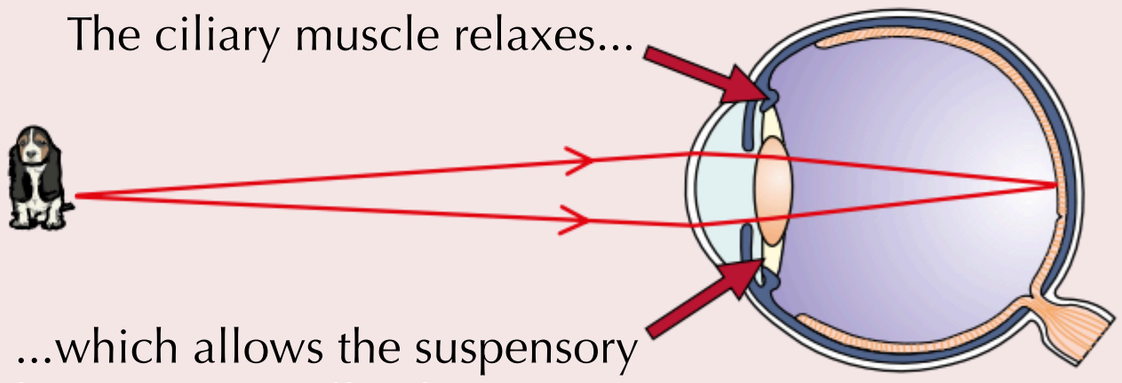


...which slackens the suspensory ligaments.

This makes the lens fatter (more curved), which means it refracts (bends) light by a larger amount, so it focuses on the retina.

To look at distant objects:

The ciliary muscle relaxes...



...which allows the suspensory ligaments to pull tight.

This makes the lens thinner (less curved), which means it refracts light by a smaller amount, so it focuses on the retina.

Learn that diagram of the eye...

Yep, another diagram you've got to know... It'll help if you sketch the diagram out roughly, close this book and then try to label your sketch. If your artistic skills aren't up to much, you could even trace it, then label the traced version.

Vision Defects

Quick Quiz

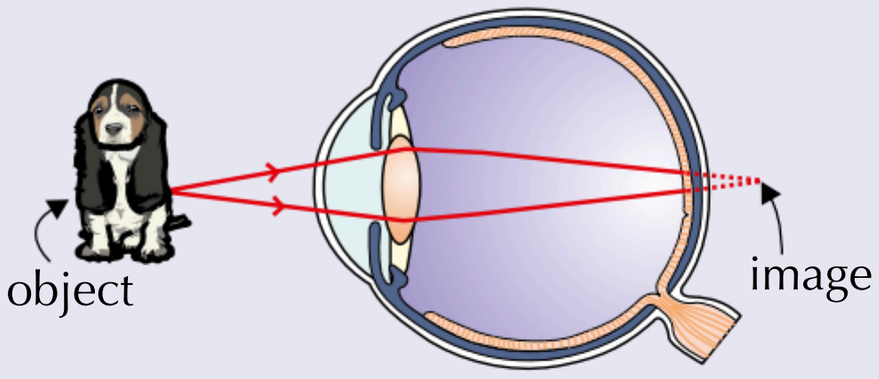
*As you saw on the previous page, the way the eye works is quite **complex**. It's not surprising really that sometimes it **doesn't work so well**. Thankfully, we have ways of **correcting** it...*

Video

Long-sighted People **Can't Focus** on **Near** Objects

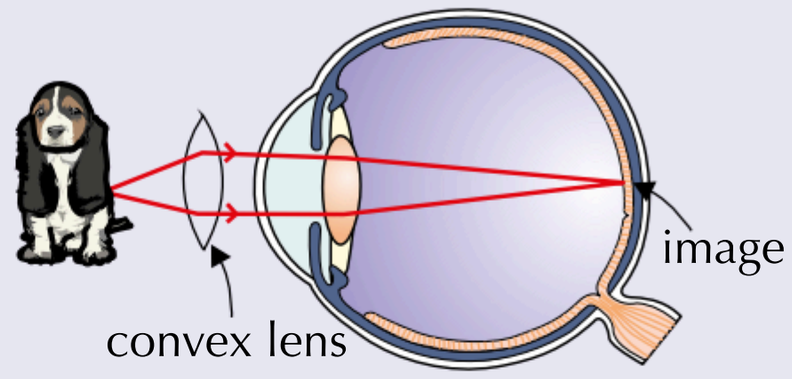
Uncorrected

1) Long-sightedness occurs when the lens is the wrong shape and doesn't refract (bend) the light enough or the eyeball is too short.
2) The images of near objects are brought into focus behind the retina.



Corrected

You can use glasses with a convex lens (a lens which curves outwards) to correct it. The lens refracts the light rays so they focus on the retina.



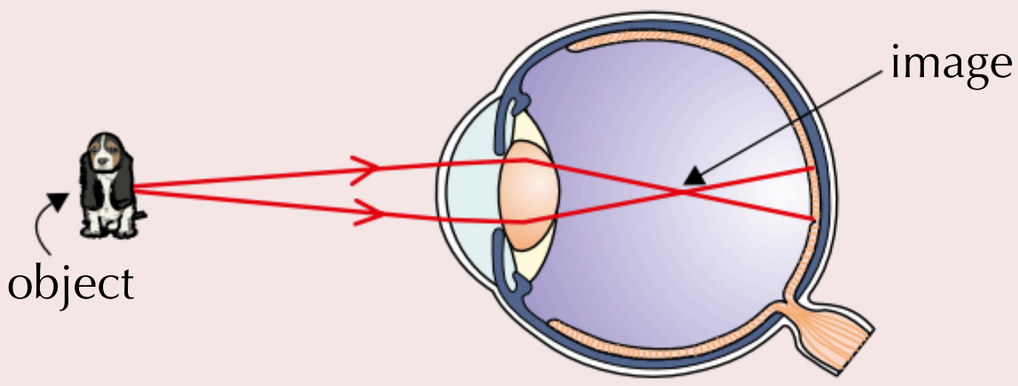
The medical term for long-sightedness is hyperopia.

If an image isn't brought into focus on the retina, it'll appear blurry.

Short-sighted People **Can't Focus** on **Distant** Objects

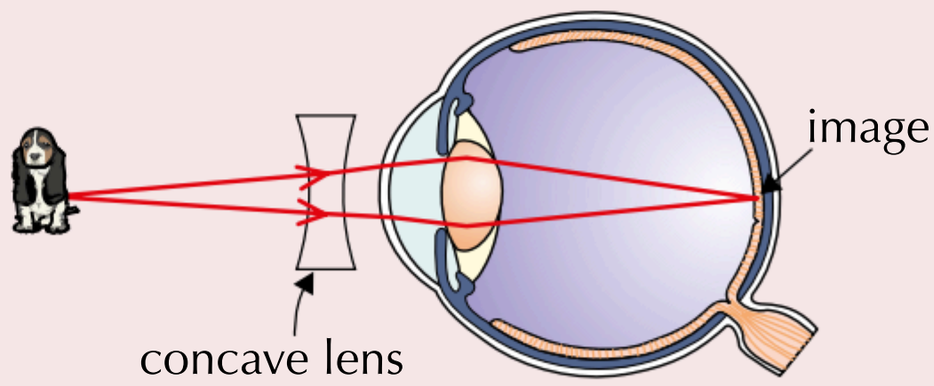
Uncorrected

1) This occurs when the lens is the wrong shape and refracts the light too much or the eyeball is too long.
2) The images of distant objects are brought into focus in front of the retina.



Corrected

You can use glasses with a concave lens (a lens which curves inwards) to correct it, so that the light rays focus on the retina.



The medical term for short-sightedness is myopia.

A convex lens increases refraction...

...and a concave lens reduces it. So if the lenses in your eye aren't refracting the light enough, you need a convex lens to correct it. Simple. Well, it will be once you've read through this page a few times.

Vision Defects and the Iris Reflex

Quick Quiz

*Thanks to new technologies, there are ways other than glasses to help **correct** vision defects...*

There are Several **Treatments** for **Vision Defects**

Wearing glasses isn't for everyone. You need to know about these alternatives:

Contact lenses

Contact lenses are thin, almost invisible lenses that sit on the surface of the eye and are shaped to compensate for the fault in focusing. The two main types are hard lenses and soft lenses.

Laser eye surgery

- Bad eyesight can sometimes be corrected with laser eye surgery.
- A laser can be used to vaporise tissue, changing the shape of the cornea (and so changing how strongly it refracts light into the eye).
- Slimming it down makes it less powerful and can improve short sight. Changing the shape so that it's more powerful will improve long sight.
- The surgeon can precisely control how much tissue the laser takes off, completely correcting the vision.

Replacement lens surgery

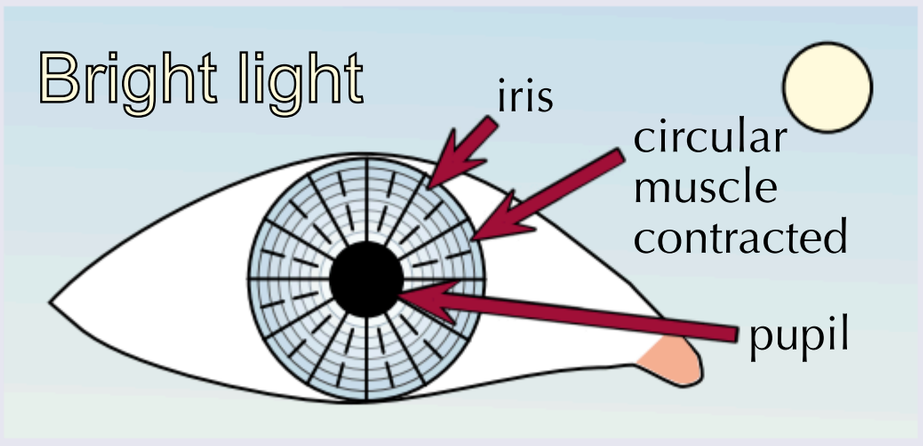
Sometimes long-sightedness may be more effectively treated by replacing the natural lens of the eye with an artificial lens made of clear plastic (rather than altering the shape of the cornea with laser eye surgery).

The **Iris Reflex** is an Adaptation to **Bright Light**

Very bright light can damage the retina — so you have a reflex to protect it.

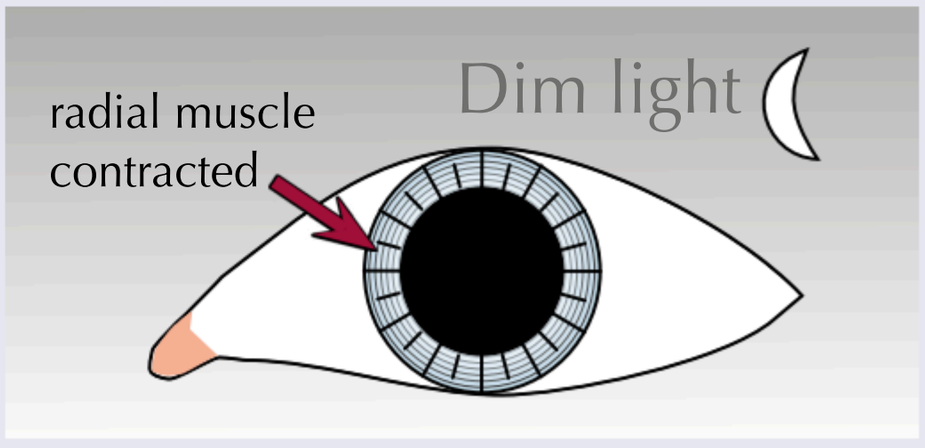
Bright light

When light receptors in the eye detect very bright light, a reflex is triggered that makes the pupil smaller. The circular muscles in the iris contract and the radial muscles relax. This reduces the amount of light that can enter the eye.



Dim light

The opposite process happens in dim light. This time, the radial muscles contract and the circular muscles relax, which makes the pupil wider.



The iris changes size depending on how bright the light is

In different light conditions, the circular muscles and radial muscles in the iris either relax or contract, which causes the pupil to change size — remember to include the details of what's doing what if you're asked about it in the exam.

Controlling Body Temperature

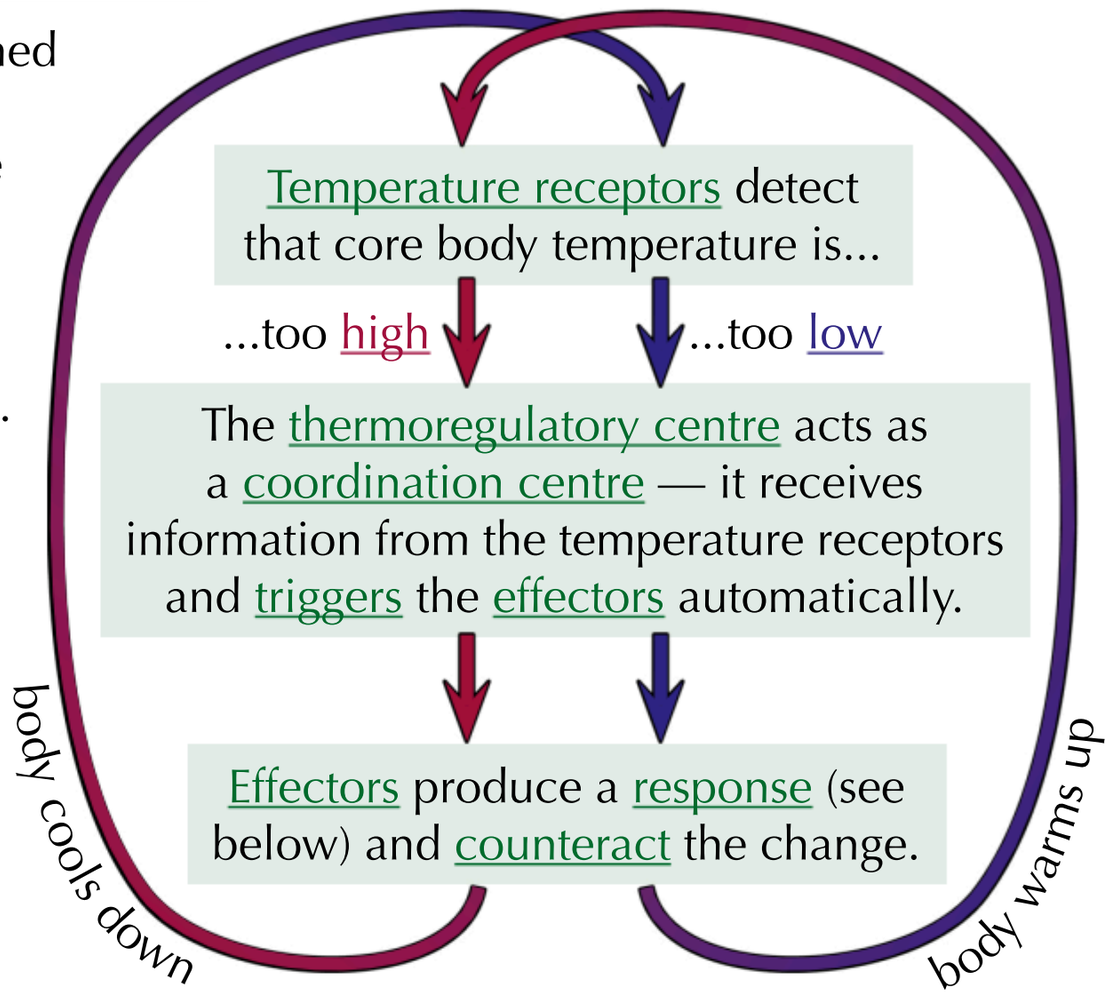
Quick Quiz

*The body has to keep its insides at around **37 °C** — the **optimum temperature** for **enzymes** in the body.*

Body Temperature Must be Kept Constant

1) The body has to balance the amount of energy gained (e.g. through respiration) and lost to keep the core body temperature constant. This is the temperature inside your body, where your internal organs are.
2) There is a thermoregulatory centre in the brain, which contains receptors that are sensitive to the temperature of the blood flowing through the brain.
3) It also receives impulses from temperature receptors in the skin, giving information about skin temperature.

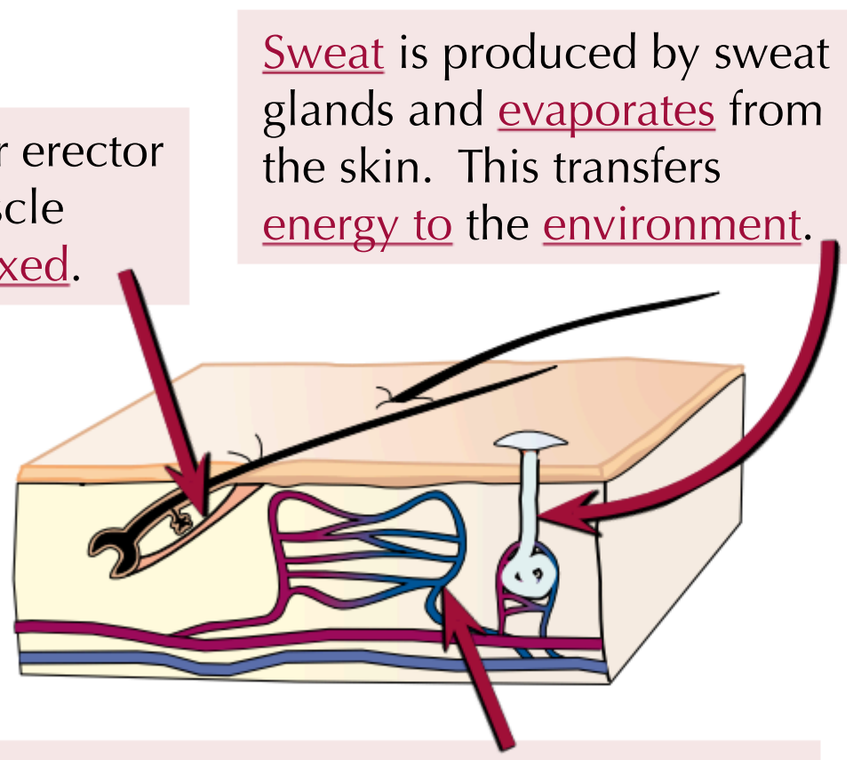
Some effectors work antagonistically, e.g. one effector heats and another cools — they'll work at the same time to achieve a very precise temperature. This mechanism allows a more sensitive response.



The Body has Some Nifty Tricks for Altering its Temperature

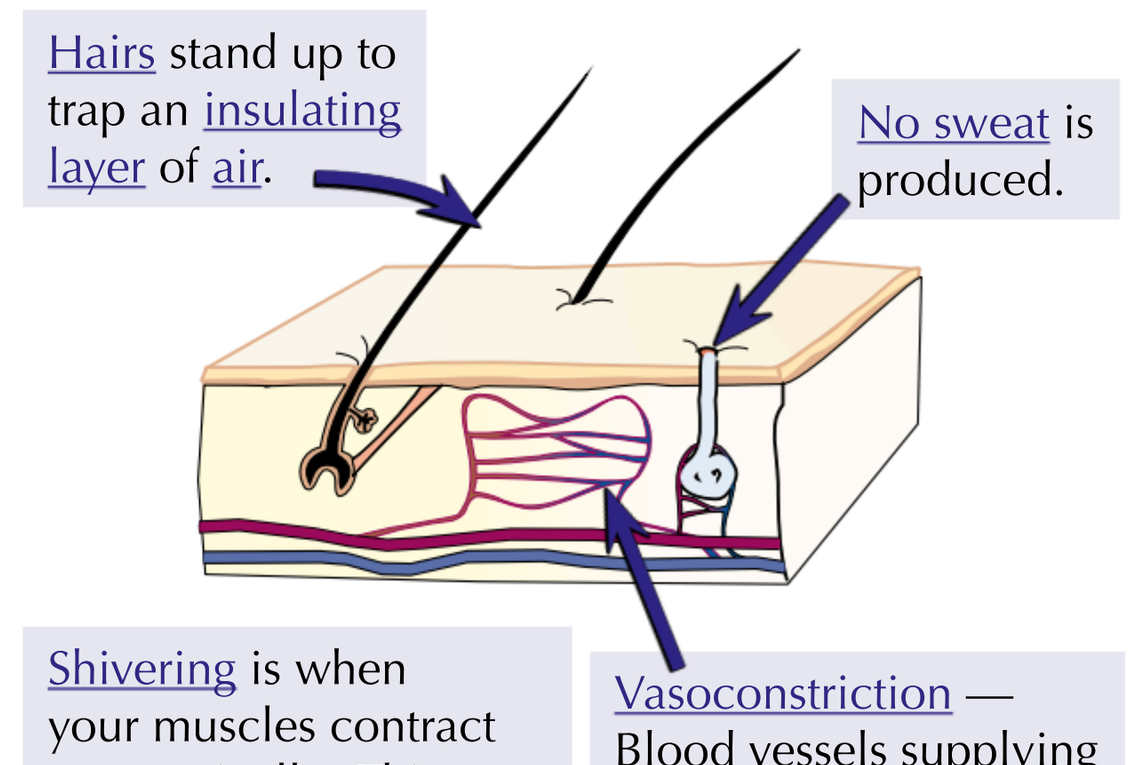
Different responses are produced by effectors to counteract an increase or decrease in body temperature.

When You're Too Hot



Vasodilation — Blood vessels supplying the skin dilate so more blood flows close to the skin's surface. This transfers energy from the skin to the environment.

When You're Too Cold



Shivering is when your muscles contract automatically. This needs respiration, which transfers some energy to warm the body.

Vasoconstriction — Blood vessels supplying skin capillaries constrict to close off the skin's blood supply.

Goose bumps are the result of hairs standing up when you're cold

People exposed to extreme cold for a long time without protection can get frostbite — the blood supply to the fingers and toes is cut off to reduce the amount of energy lost (this kills the cells, and they go black).

Warm-Up & Exam Questions

Without a good warm-up you're likely to strain a brain cell or two. So take the time to run through these simple questions and get the basic facts straight before plunging into the Exam Questions.

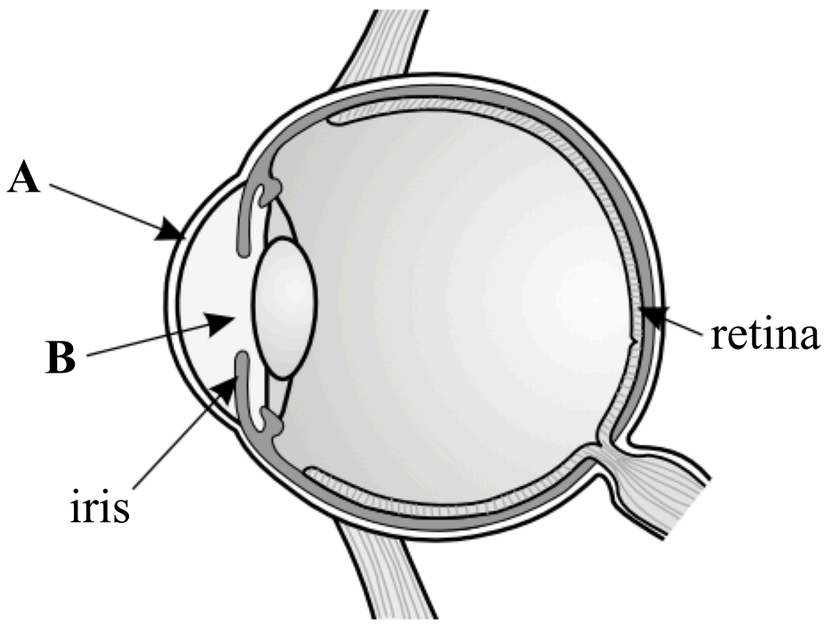
Warm-Up Questions

1) Name the region of the brain that controls coordinated movement.
2) What name is given to the transparent layer at the front on the eye?
3) How can glasses help a long-sighted person focus on something nearby?
4) How can laser eye surgery help correct a vision defect?
5) Name the area of the brain that controls body temperature.

Exam Questions

1 **Figure 1** shows a cross section through the eye. GRADE 4-6 GRADE

Figure 1



1.1 Name the parts labelled **A** and **B**.

[2 marks]

1.2 Describe the function of the iris.

[1 mark]

1.3 Describe how information about light entering the eye is passed to the brain.

[2 marks]

2 A patient with a brain tumour is experiencing difficulties with memory and language. GRADE 6-7 GRADE

2.1 Which of the following areas of the brain is most likely to be affected by the tumour?
Tick **one** box.

☐ medulla ☐ cerebral cortex ☐ cerebellum ☐ hypothalamus

[1 mark]

2.2 The patient's doctor has concerns about operating to remove the tumour.
Suggest **two** possible concerns the doctor may have about possible effects on the brain.

[2 marks]

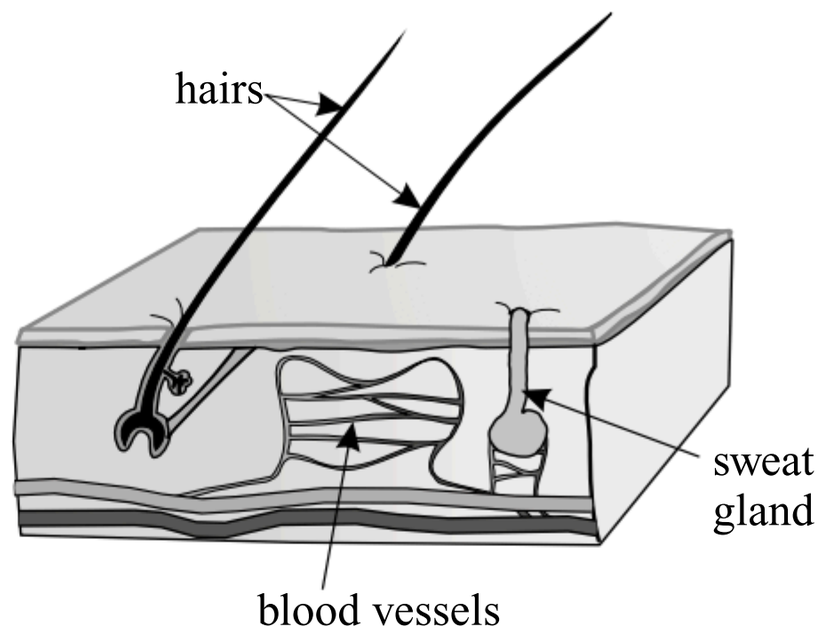
Exam Questions

3 Changes in the skin are an important part of temperature regulation.

GRADE 6-7

Figure 2 shows a cross section through the skin of a person who is cold.

Figure 2



3.1 In **Figure 2** the blood vessels close to the surface of the skin have narrowed. Give the scientific name for this process and explain why it happens.

[3 marks]

3.2 Describe and explain the response of a sweat gland when a person is cold.

[2 marks]

Temperatures in a sauna can reach up to 100 °C.

3.3 Explain why a person using a sauna may be advised to drink water regularly.

[2 marks]

3.4 Suggest how the blood vessels close to the surface of the skin would respond to being in a sauna. Explain your answer.

[3 marks]

4 The eye adjusts to different light levels.
Figure 3 shows the eye in two different light conditions.

GRADE 6-7

Figure 3

4.1 Which eye, **A** or **B**, is in dim light? Explain your answer.

[2 marks]

4.2 Which muscles in the eye contract in order to bring about the eye's response to dim light?

[1 mark]

4.3 The response of the eye to bright light is a reflex.
Suggest why it is an advantage to have this type of response controlling the action of the eye.

[2 marks]

5 The eye adjusts its shape slightly to focus light.

GRADE 7-9

5.1 Describe how the eye adjusts to focus on distant objects.

[3 marks]

5.2 Presbyopia is a condition in which the lens of the eye becomes less elastic and isn't easily able to form a rounded shape. Suggest how having presbyopia will affect a person's vision.

[2 marks]

Revision Summary Test for Topic 5a

Your eyes and brain have been working hard reading that section. To make sure you've got it down:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Homeostasis (p.83)

1) What is homeostasis and why is it important?
2) Which three components make up the automatic control systems in the body?
3) What is a stimulus?
4) Explain how negative feedback helps to maintain a stable internal environment.

The Nervous System and Investigating Reaction Times (p.84-86)

5) What makes up the central nervous system in vertebrates and what does it do?
6) Describe the role of sensory neurones in the nervous system. How about motor neurones?
7) Name the type of cell that detects stimuli.
8) Give two examples of effectors and the responses they may have.
9) Explain how the nervous system's structure allows it to produce a coordinated response to a stimulus.
10) What is a synapse?
11) What is a reflex action? How is it different to a normal response?
12) Describe the pathway of a reflex arc from stimulus to response.
13) What is a reaction time? Give three examples of factors that can affect it.
14) Briefly outline a method to measure your reaction time using a ruler.
15) Give three benefits of using a computer test to measure reaction time over the ruler drop method.

The Brain (p.88)

16) Give four things the cerebral cortex is responsible for.
17) What are the functions of the cerebellum and the medulla?
18) Describe three methods used by scientists to study the brain.

The Eye and Vision Defects (p.89-91)

19) Explain the role of the following parts of the eye:
    a) cornea b) iris c) lens d) retina e) optic nerve
20) How does the eye focus on near objects? What about distant objects?
21) What causes somebody to be long-sighted? How can this be corrected?
22) What causes somebody to be short-sighted? How can this be corrected?
23) Give three ways, other than glasses, that help to correct vision defects.
24) Describe the iris reflex. Why is this needed?

Controlling Body Temperature (p.92)

25) How does the thermoregulatory centre in the brain ensure body temperature stays constant?
26) What can the body do to respond if it gets too hot?
    How about when it's too cold?

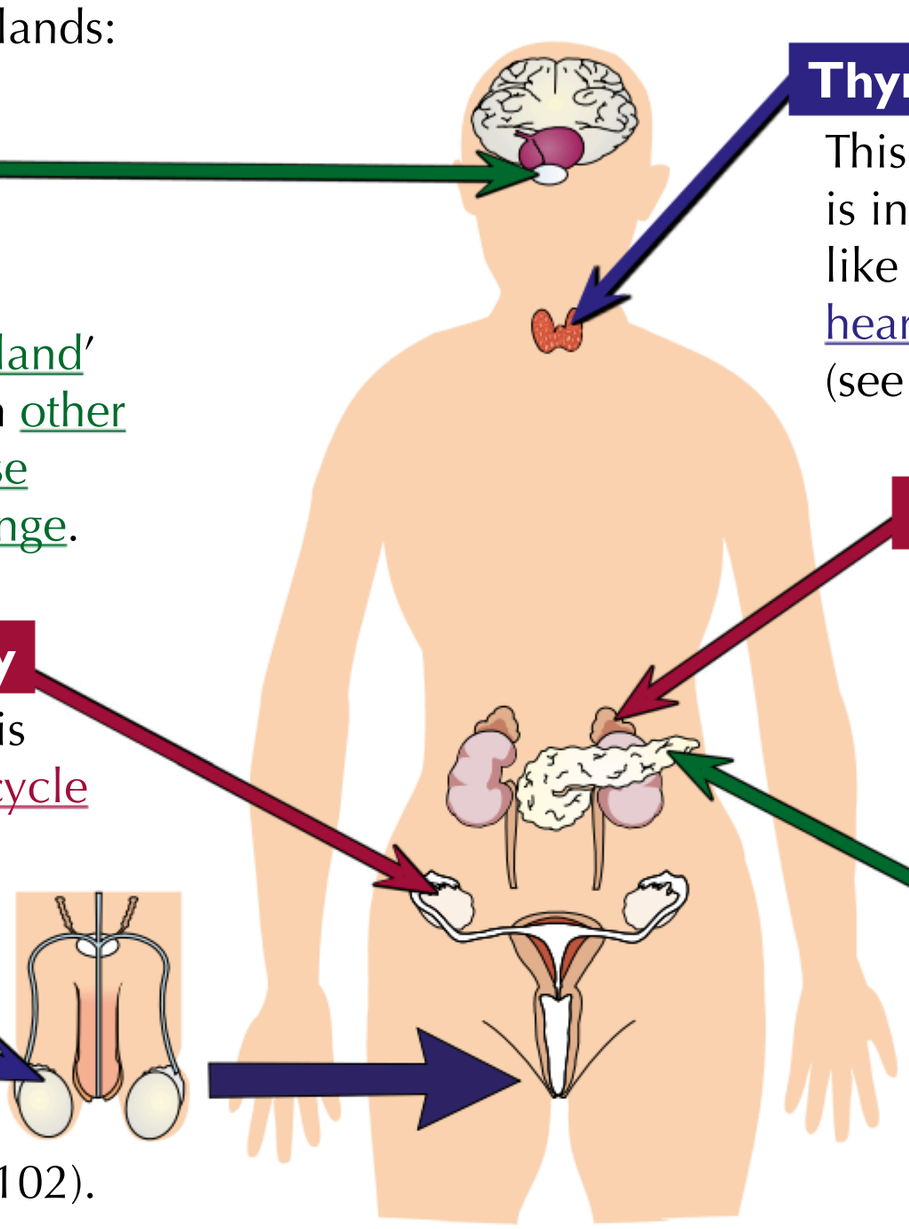
Quick Quiz

The Endocrine System

*The other way to send information around the body (apart from along nerves) is by using **hormones**.*

Hormones Are Chemical Molecules Sent in the Blood

1) Hormones are released directly into the blood and carried to all parts of the body, but they only affect particular cells in particular organs (called target organs).
2) Hormones control things in organs and cells that need constant adjustment.
3) Hormones are produced in (and secreted by) various glands, called endocrine glands. These glands make up your endocrine system.
4) Here are some examples of glands:



The Pituitary Gland

Produces many hormones that regulate body conditions. It is sometimes called the 'master gland' because these hormones act on other glands, directing them to release hormones that bring about change.

Thyroid

This produces thyroxine, which is involved in regulating things like the rate of metabolism, heart rate and temperature (see page 105).

Adrenal Glands

These produce adrenaline, which is used to prepare the body for a 'fight or flight' response (see page 105).

Ovaries — females only

Produce oestrogen, which is involved in the menstrual cycle (see page 102).

The Pancreas

This produces insulin, which is used to regulate the blood glucose level (see next page).

Testes — males only

Produce testosterone, which controls puberty and sperm production in males (see page 102).

Hormones and Nerves Have Differences

	Nerves:	Hormones:
	Very fast action.	Slower action.
	Act for a very short time.	Act for a long time.
	Act on a very precise area.	Act in a more general way.

So if you're not sure whether a response is nervous or hormonal, have a think...

1) If the response is really quick, it's probably nervous. Some information needs to be passed to effectors really quickly (e.g. pain signals, or information from your eyes telling you about the lion heading your way), so it's no good using hormones to carry the message — they're too slow.
2) But if a response lasts for a long time, it's probably hormonal. For example, when you get a shock, a hormone called adrenaline is released into the body (causing the fight or flight response, where your body is hyped up ready for action). You can tell it's a hormonal response (even though it kicks in pretty quickly) because you feel a bit wobbly for a while afterwards.

Nerves, hormones — no wonder revision makes me tense...

Hormones control various organs and cells in the body, though they tend to control things that aren't immediately life-threatening (so things like sexual development, blood sugar level, water content, etc.).

Controlling Blood Glucose

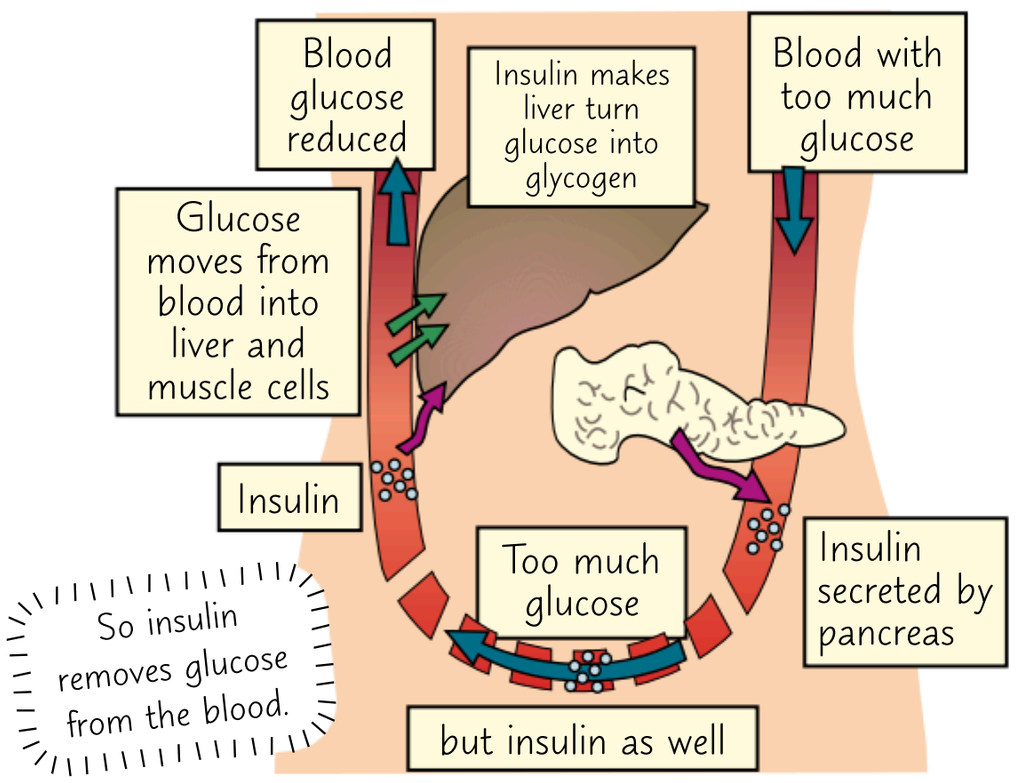
Quick Quiz

*On page 92, you saw that **body temperature** is controlled as part of **homeostasis** — well, so is blood glucose level. **Insulin** and **glucagon** are **hormones** that control how much **glucose** there is in your **blood**.*

Insulin and **Glucagon** Control **Blood Glucose** Level

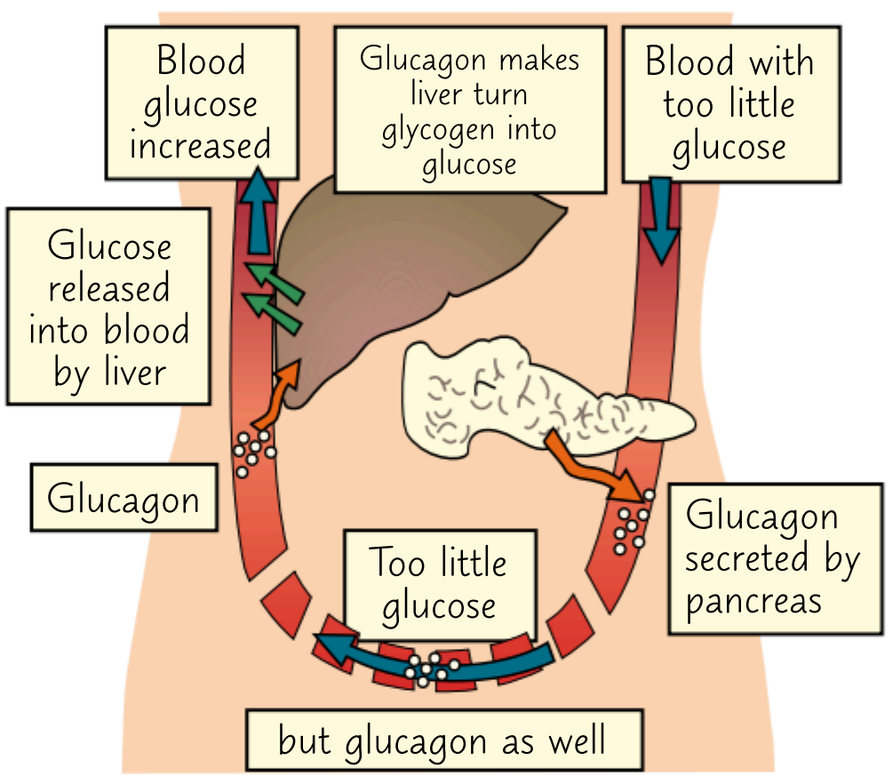
1) Eating foods containing carbohydrate puts glucose (a type of sugar) into the blood from the gut.
2) The normal metabolism of cells removes glucose from the blood.
3) Vigorous exercise removes much more glucose from the blood.
4) Excess glucose can be stored as glycogen in the liver and in the muscles.
5) The level of glucose in the blood must be kept steady. Changes are monitored and controlled by the pancreas, using the hormones insulin and glucagon, in a negative feedback cycle:

Blood glucose level too HIGH — INSULIN is added



Blood glucose level too LOW — GLUCAGON is added

Video



With **Diabetes**, You **Can't Control** Your **Blood Sugar Level**

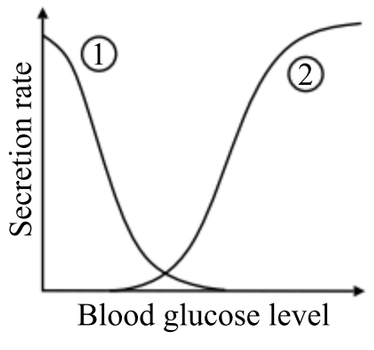
Diabetes is a condition that affects your ability to control your blood sugar level. There are two types:

Type 1	Type 2
• This is where the pancreas doesn't produce enough, or any, insulin. So a person's blood glucose can rise to a level that can kill them. • Insulin therapy is a very effective treatment usually involving several injections of insulin throughout the day, most likely at mealtimes. This removes glucose from the blood quickly once the food has been digested, stopping the level getting too high. • The amount of insulin that needs to be injected depends on the person's diet and how active they are.	• This is where a person becomes resistant to their own insulin (they still produce insulin, but their body's cells don't respond properly to the hormone). So a person's blood glucose can rise to a dangerous level. • Obesity is a major risk factor in the development of Type 2 diabetes. • Type 2 diabetes can be controlled by eating a carbohydrate-controlled diet and getting regular exercise.

Insulin brings blood glucose down — glucagon brings it up

This stuff can be confusing, but if you learn the diagrams above, it'll be a bit easier.

Q1 The graph shows the relative secretion rates of insulin and glucagon as the blood glucose level increases. Which curve represents glucagon? Explain your answer. [2 marks]



Q1 Video Solution

Warm-Up & Exam Questions

Right then, another lot of pages down. Now there's just the small matter of answering some questions...

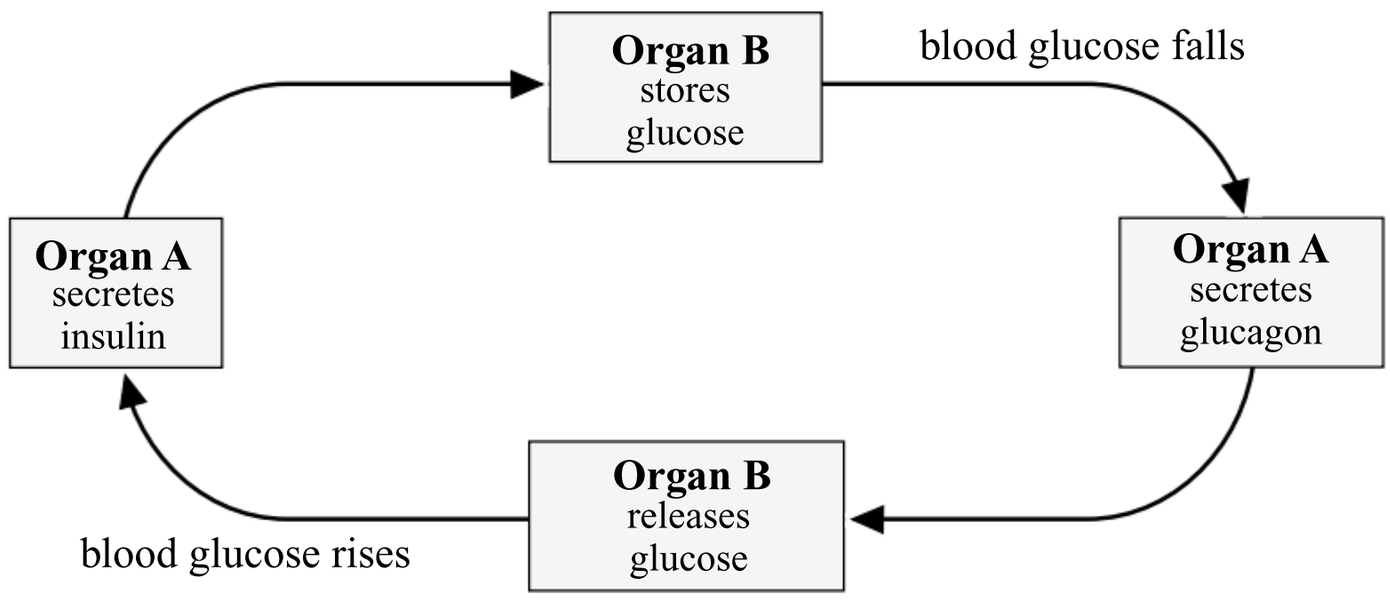
Warm-Up Questions

1) How do hormones travel to their target organs?
2) Name five endocrine glands found in the male human body.
3) Which has longer lasting effects — a nervous response or a hormonal response?
4) What does insulin do?
5) What is Type 2 diabetes?

Exam Questions

1 **Figure 1** shows how the blood glucose level is regulated in humans. GRADE 6-7

Figure 1



1.1 Identify organs A and B in **Figure 1**.

[2 marks]

1.2 Apart from organ B releasing glucose, suggest **one** reason why the blood glucose level could rise.

[1 mark]

1.3 With reference to **Figure 1**, explain what goes wrong with the regulation of blood glucose level in people with Type 1 diabetes.

[3 marks]

1.4 Describe what the hormone glucagon does.

[1 mark]

2 Hypopituitarism is a condition in which the pituitary gland stops secreting one or more of the pituitary hormones, or doesn't secrete enough of these hormones. People with hypopituitarism may experience tiredness and weight gain. They may also feel the cold more. These symptoms are all linked to low thyroid hormone levels. GRADE 7-9

Suggest and explain why someone with hypopituitarism may experience these symptoms.

[3 marks]

The Kidneys

Quick Quiz

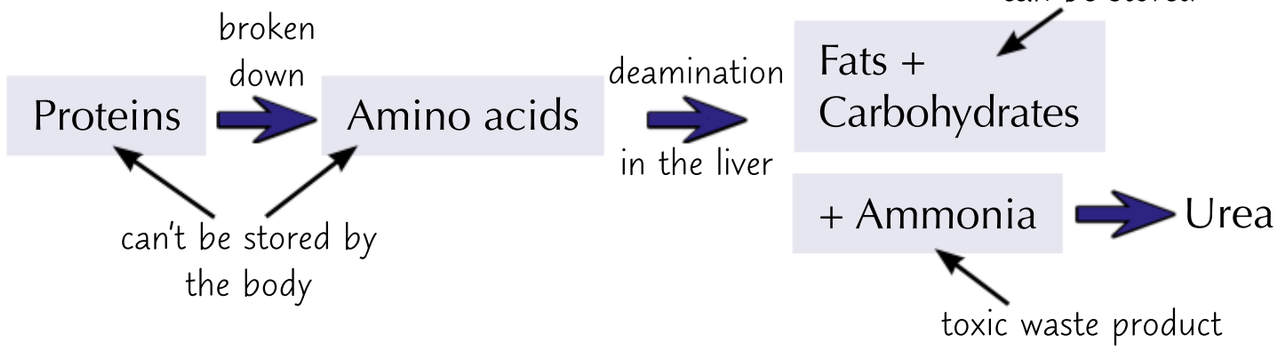
*The **kidneys** are really important in this whole **homeostasis** thing.*

Kidneys Act as Filters to "Clean the Blood"

Substances are filtered out of the blood as it passes through the kidneys. This process is called filtration. Useful substances like glucose, some ions and the right amount of water are then absorbed back into the blood. This process is called selective reabsorption. Waste products (and other unwanted substances) are removed from the body in urine. They include:

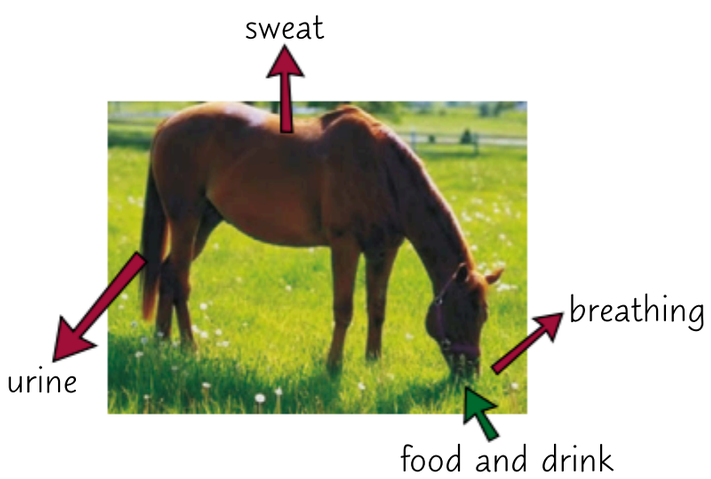
1 Urea

Urea is a waste product produced when the body deals with excess amino acids in a process called deamination. It's transported to the kidneys, where it's filtered out of the blood and safely excreted from the body in urine.



2 Water

- Having the wrong amount of water can damage cells or mean they don't work as well as normal.
- We lose water (as well as urea and ions) from the skin in sweat, and water from the lungs when breathing out. But the amount lost in these ways is unregulated.
- The amount of water we consume and the amount removed by the kidneys in urine helps to balance what is lost in other ways.

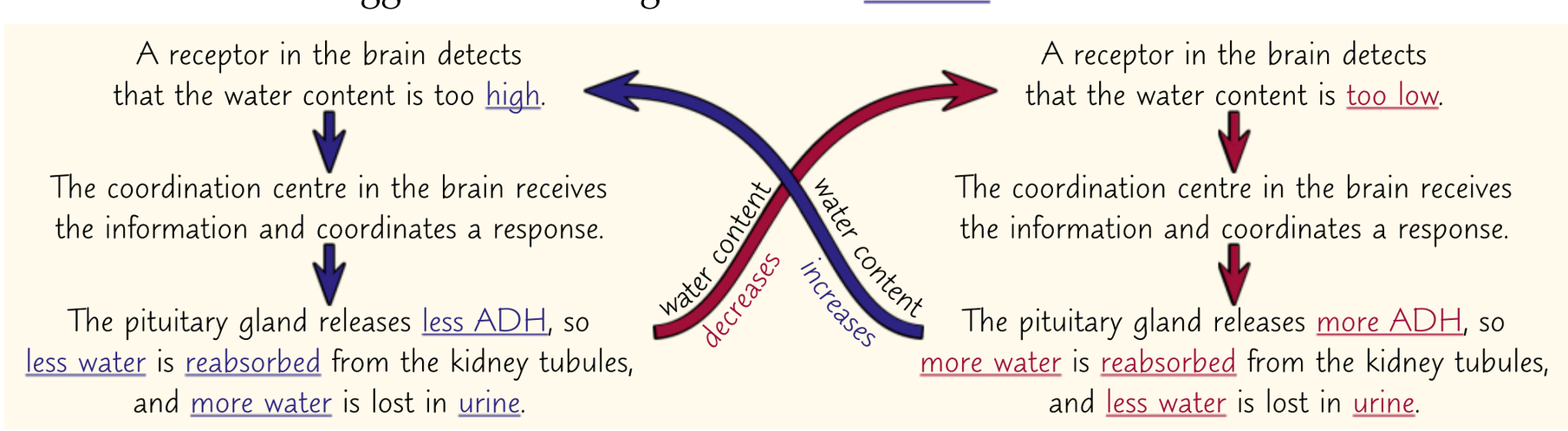


3 Ions

- Ions such as sodium are taken into the body in food, and then absorbed into the blood.
- If the ion (or water) content of the body is wrong, this could upset the balance between ions and water, meaning too much or too little water is drawn into cells by osmosis (see page 18).
- The right amount of ions are reabsorbed into the blood after filtration and the rest is removed from the body in urine.

The Concentration of Urine is Controlled by a Hormone

1) The brain monitors the water content of the blood and instructs the pituitary gland to release anti-diuretic hormone (ADH) into the blood according to how much is needed.
2) ADH affects the permeability of the kidney tubules.
3) The whole process of water content regulation is controlled by negative feedback (see page 83). This means that if the water content gets too high or too low a mechanism will be triggered that brings it back to normal.



More ADH = less urine produced...

When blood water content drops (e.g. when you sweat lots), the pituitary gland releases more ADH. More water gets reabsorbed from the kidney tubules and a smaller volume of more concentrated urine is produced.

Q1 Alcohol suppresses the production of ADH. Suggest how this can lead to dehydration. [2 marks]

Q1 Video Solution

Quick Quiz

Kidney Failure

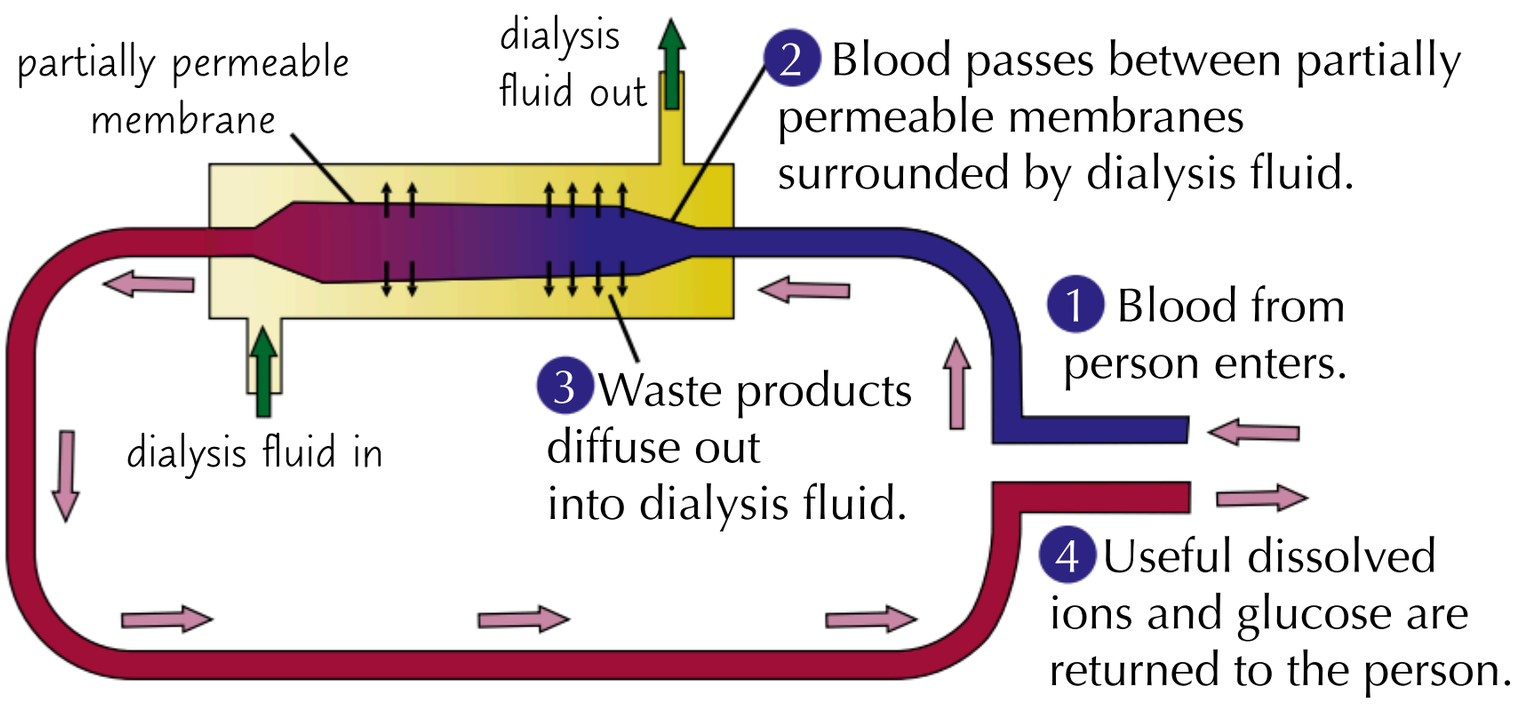
*If someone's kidneys stop working, there are basically two treatments — **regular dialysis** or a **transplant**.*

The Kidneys Remove **Waste Substances** from the Blood

1) If the kidneys don't work properly, waste substances build up in the blood and you lose your ability to control the levels of ions and water in your body. Eventually, this results in death.
2) People with kidney failure can be kept alive by having dialysis treatment — where machines do the job of the kidneys. Or they can have a kidney transplant.

Dialysis Machines **Filter** the Blood

Dialysis has to be done regularly to keep the concentrations of dissolved substances in the blood at normal levels, and to remove waste substances, such as urea. Here's how a dialysis machine works:



- The membrane is permeable to ions and waste substances, but not big molecules like proteins (like in the kidney), so waste substances, excess water and ions can diffuse into the dialysis fluid.
- The dialysis fluid has the same concentration of dissolved ions and glucose as healthy blood, so these useful substances aren't lost.

Pros	Cons
Dialysis can buy a patient with kidney failure valuable time until a donor organ is found.	Many patients need dialysis sessions three times a week. Each session takes 3-4 hours — not much fun.
	Dialysis may cause blood clots or infections.
	Being on a dialysis machine is not a pleasant experience.
	It is expensive for the NHS to run.

Kidney Transplants are a **Cure**, but can be **Rejected**

1) At the moment, the only cure for kidney failure is to have a kidney transplant.
2) Healthy kidneys are usually transplanted from people who have died suddenly.
3) Kidneys can also be transplanted from people who are still alive (as we all have two of them) but there's a small risk to the person donating the kidney.
4) There is also a risk that the donor kidney can be rejected by the patient's immune system. The patient is treated with drugs to prevent this but it can still happen.
5) Transplants are cheaper (in the long run) than dialysis and they can put an end to the hours patients have to spend on dialysis, but there are long waiting lists for kidneys.

Dialysis or transplant? Both have their downsides...

Donor kidneys are ideally matched by blood type (and a few other things) to the recipient, which make them less likely to be rejected. However, it means a potentially long waiting time for a suitable kidney. Not much fun.

Warm-Up & Exam Questions

Just a few simple Warm-Up Questions and a few slightly harder Exam Questions stand between you and the menstrual cycle...

Warm-Up Questions

1) Describe how urea is produced from excess amino acids.
2) Aside from urea, give two substances removed in the urine.
3) Which hormone controls the reabsorption of water in the kidneys?
4) Name the part of the brain where this hormone is released.
5) What is the purpose of the partially permeable membrane in a dialysis machine?

Exam Questions

1 Substances filtered out of the blood in the kidneys form a fluid called a filtrate. GRADE 4-6 GRADE

1.1 **Table 1** shows some substances that are present either in the blood or in the filtrate just after filtration has taken place. Complete **Table 1** by filling in the blanks in each column.

[2 marks]

Table 1

Substances	Blood	Filtrate
glucose	✗	✓
urea		
	✗	✓

1.2 Glucose is present in the blood that leaves the kidneys. By which of the following processes does glucose move from the filtrate into the blood? Tick **one** box.

☐ filtration ☐ osmosis ☐ selective reabsorption ☐ deamination

[1 mark]

2 A hospital patient has kidney failure. He has dialysis three times a week. GRADE 6-7 GRADE

2.1 Explain how the dialysis machine removes urea from the patient's blood.

[2 marks]

2.2 Explain why the patient does not lose glucose from his blood during dialysis.

[2 marks]

The patient is on the waiting list to receive a kidney transplant.

2.3 Suggest **one** reason why this form of treatment may be preferable to dialysis.

[1 mark]

2.4 Give **one** disadvantage of kidney transplants.

[1 mark]

3* A runner went for a 10 mile run on a warm day. When she got home she noticed that her urine was darker in colour than normal. Explain why she produced darker coloured urine. GRADE 7-9 GRADE

[6 marks]

Puberty and the Menstrual Cycle

Quick Quiz

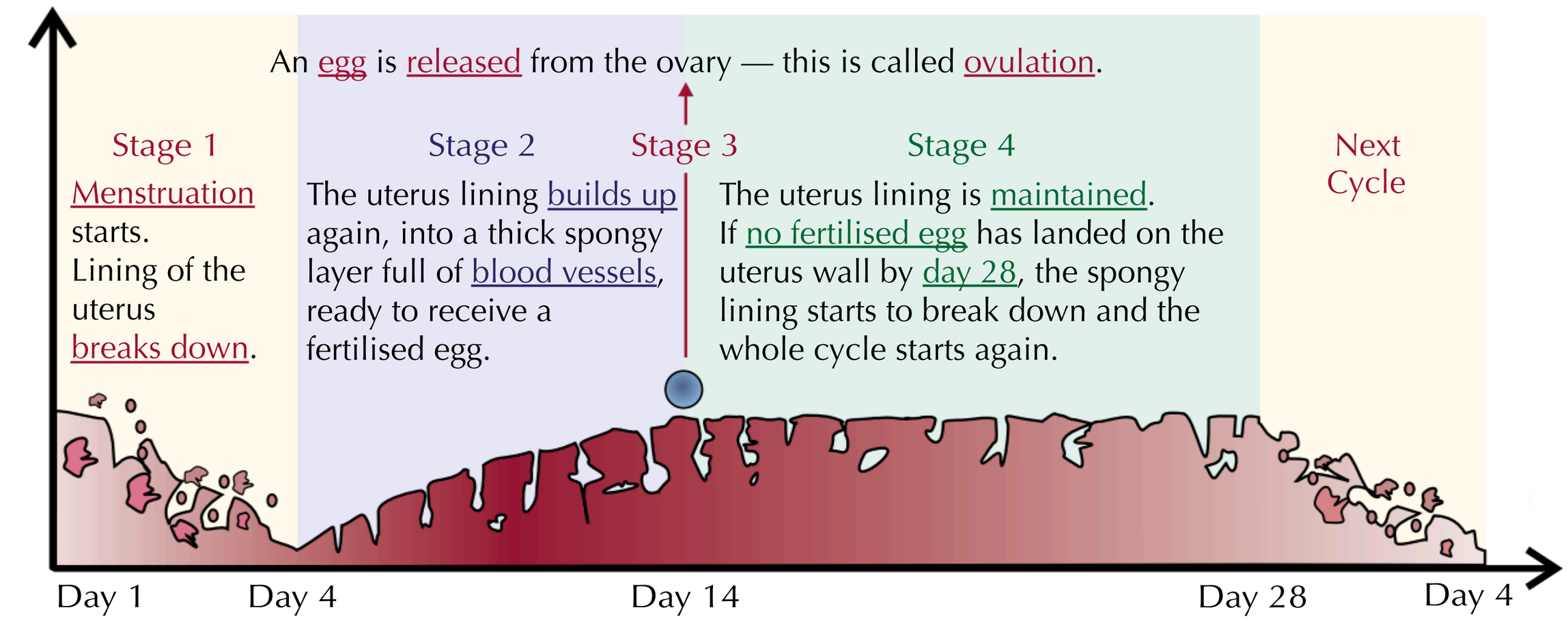
*The monthly **release of an egg** from a woman's ovaries is part of the **menstrual cycle**.*

Hormones Promote Sexual Characteristics at Puberty

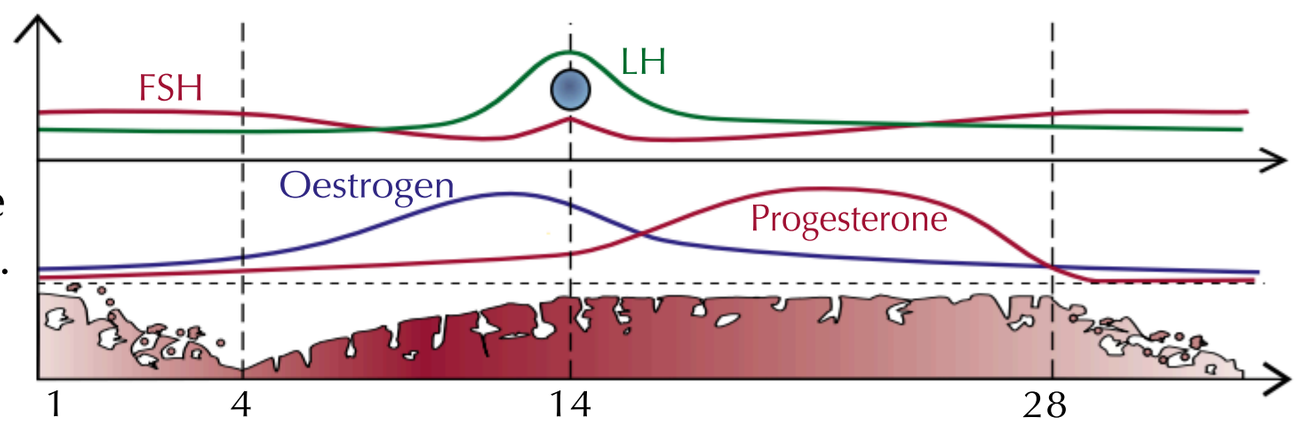
At puberty, your body starts releasing sex hormones that trigger off secondary sexual characteristics (such as the development of facial hair in men and breasts in women) and cause eggs to mature in women.

- In men, the main reproductive hormone is testosterone.
  It's produced by the testes and stimulates sperm production.
- In women, the main reproductive hormone is oestrogen. It's produced by the ovaries.
  As well as bringing about physical changes, oestrogen is also involved in the menstrual cycle.

The Menstrual Cycle Has Four Stages



It's Controlled by Four Hormones



1. FSH (Follicle-Stimulating Hormone)
   1) Produced in the pituitary gland.
   2) Causes an egg to mature in one of the ovaries, in a structure called a follicle.
   3) Stimulates the ovaries to produce oestrogen.

2. Oestrogen
   1) Causes the lining of the uterus to grow.
   2) Stimulates the release of LH (which causes the release of an egg) and inhibits release of FSH.

3. LH (Luteinising Hormone)
   1) Produced by the pituitary gland.
   2) Stimulates the release of an egg at day 14 (ovulation).

4. Progesterone
   1) Produced in the ovaries by the remains of the follicle after ovulation.
   2) Maintains the lining of the uterus during the second half of the cycle. When the level of progesterone falls, the lining breaks down.
   3) Inhibits the release of LH and FSH.

Examiners love a good menstrual cycle graph...

Learn this page until you know which hormone does what, and you understand the hormone graph above.

Controlling Fertility

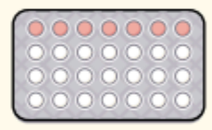
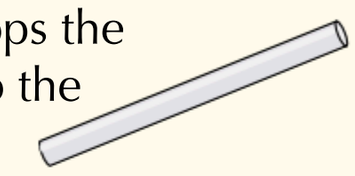
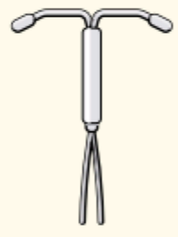
Quick Quiz

Pregnancy *can happen if sperm reaches the ovulated egg.* ***Contraception*** *tries to* ***stop*** *this happening.*

Hormonal Methods Can Be Used to **Reduce Fertility**

Oestrogen can be used to prevent the release of an egg. Taking oestrogen every day keeps the level of it permanently high. This inhibits the production of FSH, and after a while egg development and production stop.

Progesterone reduces fertility, e.g. by stimulating the production of thick mucus which prevents any sperm getting through and reaching an egg.

The pill (combined oral contraceptive pill) contains oestrogen and progesterone	• Over 99% effective at preventing pregnancy. • Can cause side effects like headaches and nausea. • Doesn't protect against sexually transmitted diseases.
Progesterone-only pill	• Over 99% effective at preventing pregnancy. • Fewer side effects than the combined oral contraceptive pill. 
Contraceptive implant	• A flexible rod inserted under the skin of the arm that can last for three years. • Releases a continuous amount of progesterone, which stops the ovaries releasing eggs, makes it hard for sperm to swim to the egg, and stops a fertilised egg implanting in the uterus. 
Contraceptive patch	• A small (5×5 cm) patch containing oestrogen and progesterone. • Each patch is stuck to the skin and lasts one week.
Contraceptive injection	• Contains progesterone. Each dose lasts 2 to 3 months.
Intrauterine device	• A T-shaped device that is inserted into the uterus. Prevents implantation of a fertilised egg and releases copper to prevent sperm surviving. • There are also intrauterine systems that release progesterone. 

Barrier Methods are **Non-hormonal** Forms of Contraception

Condoms Worn over the penis during intercourse to prevent the sperm entering the vagina. There are also female condoms that are worn inside the vagina. Condoms are the only form of contraception that will protect against sexually transmitted diseases.

Diaphragm A shallow plastic cup that fits over the cervix (the entrance to the uterus) to form a barrier. It has to be used with spermicide (a substance that disables or kills the sperm).

There are Other Ways to **Avoid Pregnancy**

- Spermicide — can be used alone as a form of contraception, but it is not as effective (only 70-80%).
- Sterilisation — cutting or tying the fallopian tubes (which connect the ovaries to the uterus) in a female, or the sperm ducts (the tube between the testes and penis) in a male. This is a permanent procedure. However, there is a very small chance that the tubes can rejoin.
- 'Natural' methods — finding out when in the menstrual cycle the woman is most fertile and avoiding sexual intercourse on those days. It's popular with people who think that hormonal and barrier methods are unnatural, but it's not very effective.
- Abstinence — the only way to be totally sure that sperm and egg don't meet is not to have intercourse.

The winner of best contraceptive ever — just not doing it...

Make sure you know the pros and cons of each method above — you could be asked to evaluate them in your exam.

Quick Quiz

More on Controlling Fertility

*Scientific advances in **understanding fertility** have led to many **infertile** couples being helped to **have babies**.*

Hormones Can Be Used to **Increase Fertility**

1) Some women have levels of FSH that are too low to cause their eggs to mature. This means that no eggs are released and the women can't get pregnant.
2) The hormones FSH and LH can be given to women in a fertility drug to stimulate ovulation.

PRO

It helps a lot of women to get pregnant when previously they couldn't.

CONS

It doesn't always work — some women may have to do it many times, which can be expensive.

Too many eggs could be stimulated, resulting in unexpected multiple pregnancies (twins, triplets, etc.).

IVF Can Help if Medication **Doesn't Work**

IVF stands for "*in vitro* fertilisation".

1) FSH and LH are given to stimulate several eggs to mature (so more than one egg can be collected).
2) The eggs are collected from the woman's ovaries and fertilised in a lab using the man's sperm.
3) The fertilised eggs are then grown into embryos in a laboratory incubator.
4) Once the embryos are tiny balls of cells, one or two of them are transferred to the woman's uterus (womb) to improve the chance of pregnancy.

PRO

Fertility treatment can give an infertile couple a child — a pretty obvious benefit.

CONS

- Multiple births can happen if more than one embryo grows into a baby — these are risky for the mother and babies (there's a higher risk of miscarriage, stillbirth...).
- The success rate of IVF is low. This makes the process incredibly stressful and often upsetting, especially if it ends in multiple failures.
- The process is also physically stressful for the woman — some have a strong reaction to the hormones, e.g. abdominal pain, vomiting, dehydration.

See p.123 for more.

Advances in microscope techniques have helped to improve the success rate of IVF.

- Specialised micro-tools have been developed to use on the eggs and sperm under the microscope. They're also used to remove single cells from the embryo for genetic testing (to check it's healthy).
- The development of time-lapse imaging (using a microscope and camera built into the incubator) means that the growth of the embryos can be continuously monitored to help identify those that are more likely to result in a successful pregnancy.

- The process of IVF often results in unused embryos that are eventually destroyed. Because of this, some people think it is unethical because each embryo is a human life or has the potential to be a human life.
- The genetic testing of embryos before implantation also raises ethical issues as some people think it could lead to the selection of preferred characteristics, such as biological sex or eye colour.

Understanding how hormones work helps put them to use artificially

Knowing the effects that LH and FSH have on the menstrual cycle allows us to use them when treating infertility.

Adrenaline and Thyroxine

Quick Quiz

*You've met a lot of human hormones so far, but **two more** won't hurt.*
Then that's it, I promise — before it's time to go onto plant hormones...

Adrenaline Prepares You for **"Fight or Flight"**

Adrenaline is a hormone released by the adrenal glands, which are just above the kidneys (see p.96).

1) It's released in response to stressful or scary situations — your brain detects fear or stress and sends nervous impulses to the adrenal glands, which respond by secreting adrenaline.

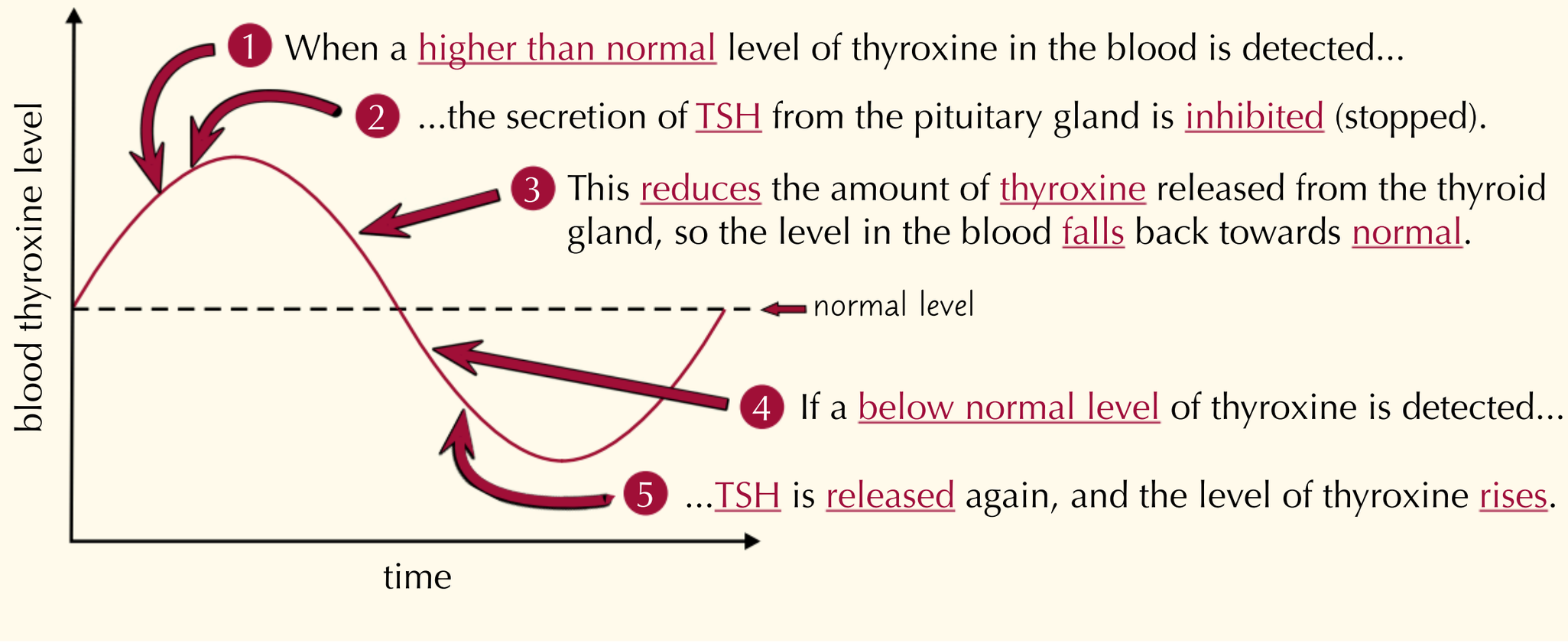
2) It gets the body ready for 'fight or flight' by triggering mechanisms that increase the supply of oxygen and glucose to cells in the brain and muscles. For example, adrenaline increases heart rate.

Thyroxine Regulates **Metabolism** Using **Negative Feedback**

Your body can control the levels of hormones (and other substances) in the blood using negative feedback systems (see p.83). When the body detects that the level of a substance has gone above or below the normal level, it triggers a response to bring the level back to normal again.

- Thyroxine is a hormone released by the thyroid gland, which is in the neck (see p.96).
- It plays an important role in regulating the basal metabolic rate — the speed at which chemical reactions in the body occur while the body is at rest. Thyroxine is also important for loads of processes in the body, e.g. stimulating protein synthesis for growth and development.
- It is released in response to thyroid stimulating hormone (TSH), which is released from the pituitary gland.

A negative feedback system keeps the amount of thyroxine in the blood at the right level:



Hormone release is often controlled by negative feedback systems

Negative feedback systems make sure the levels of some hormones (e.g. thyroxine) don't peak or dip too much. You can think about negative feedback working like a thermostat — if the temperature gets too low, the thermostat will turn the heating on, then if the temperature gets too high, it'll turn the heating off again. Fantastic.

Warm-Up & Exam Questions

If these questions don't get your adrenaline pumping, I don't know what will. Better get started...

Warm-Up Questions

1) Name the hormone that stimulates an egg to mature in the ovary.
2) Name three forms of contraception that reduce fertility using progesterone, not oestrogen.
3) Give one drawback to using hormones to increase fertility.
4) What is the role of FSH and LH during IVF?
5) Name the gland that releases thyroxine.

Exam Questions

1 The menstrual cycle is controlled by several different hormones. GRADE 4-6 GRADE

1.1 What effect does oestrogen have on the release of FSH?

[1 mark]

1.2 Which hormone is responsible for maintaining the uterus lining?

[1 mark]

1.3 On what day of the menstrual cycle is the egg released?

[1 mark]

2 The combined oral contraceptive pill contains oestrogen and progesterone. GRADE 6-7 GRADE

2.1 Explain how taking oestrogen can prevent pregnancy.

[2 marks]

2.2 Progesterone can prevent pregnancy by preventing ovulation.
Explain **one** other way in which taking progesterone can prevent pregnancy.

[2 marks]

The combined pill is taken once a day in a '21 day pill, 7 day no pill' cycle. One of the problems with the combined pill is that women may forget to take it on the days they are supposed to.

2.3 Suggest an alternative method of hormonal contraception that may be more suitable for a woman who is worried about remembering to take the combined pill. Explain your choice.

[2 marks]

2.4 A couple in their late 30s who have already had three children are looking for a more permanent method of contraception. Suggest **one** method of contraception that may be suitable for them.

[1 mark]

3 A dog suddenly runs towards a cat across the street, which frightens the cat. GRADE 7-9 GRADE

Explain what would happen to the cat's heart rate when it sees the dog and explain why this response is beneficial.

[5 marks]

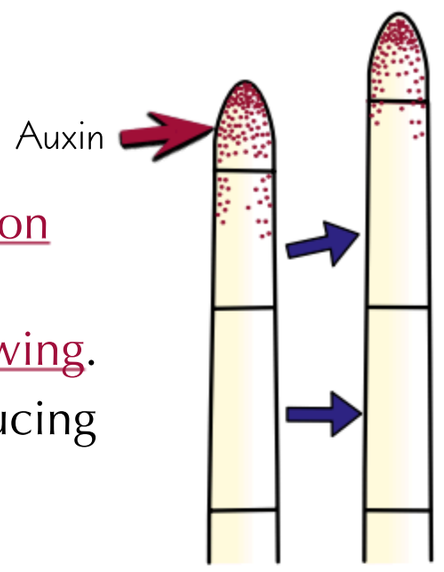
Plant Hormones

Quick Quiz

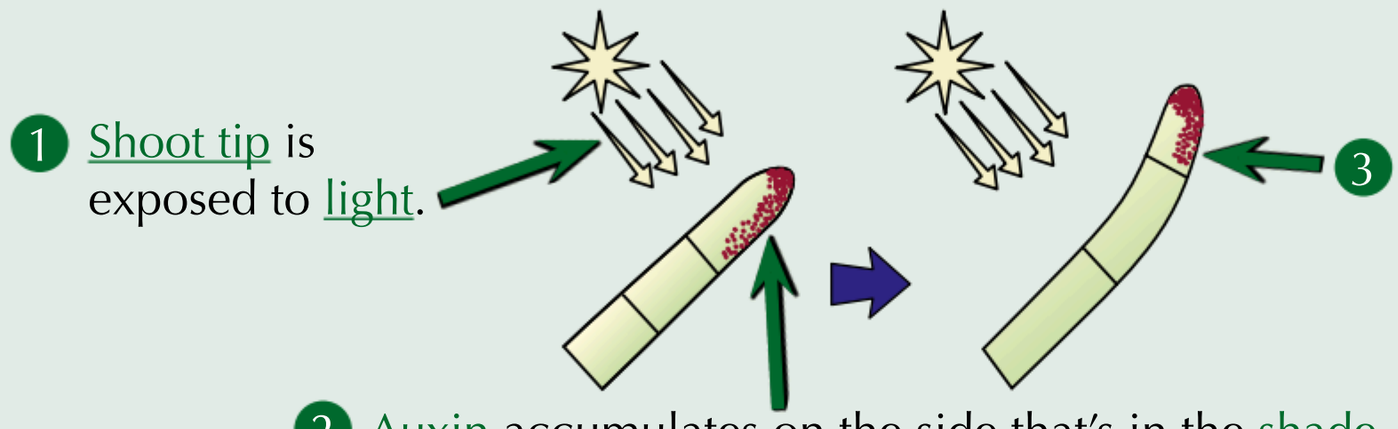
*Plants **don't** just grow randomly. Plant hormones make sure they grow in the **right direction**.*

Auxin is a Plant **Growth Hormone**

- Auxin controls growth near the tips of shoots and roots in response to light (phototropism) and gravity (gravitropism or geotropism).
- Auxin is produced in the tips and moves backwards to stimulate the cell elongation (enlargement) process which occurs in the cells just behind the tips.
- If the tip of a shoot is removed, no auxin is available and the shoot may stop growing.
- Extra auxin promotes growth in the shoot but inhibits growth in the root — producing the desired result...

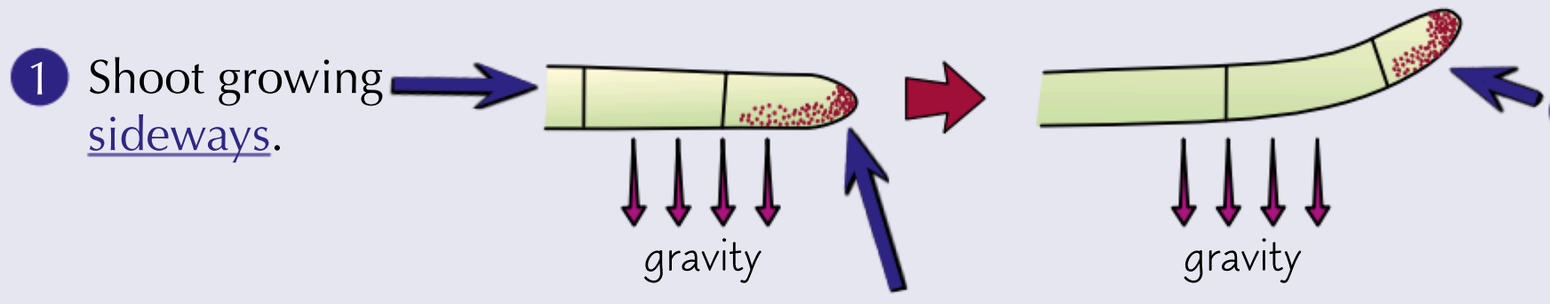


Shoots Grow **Towards Light**

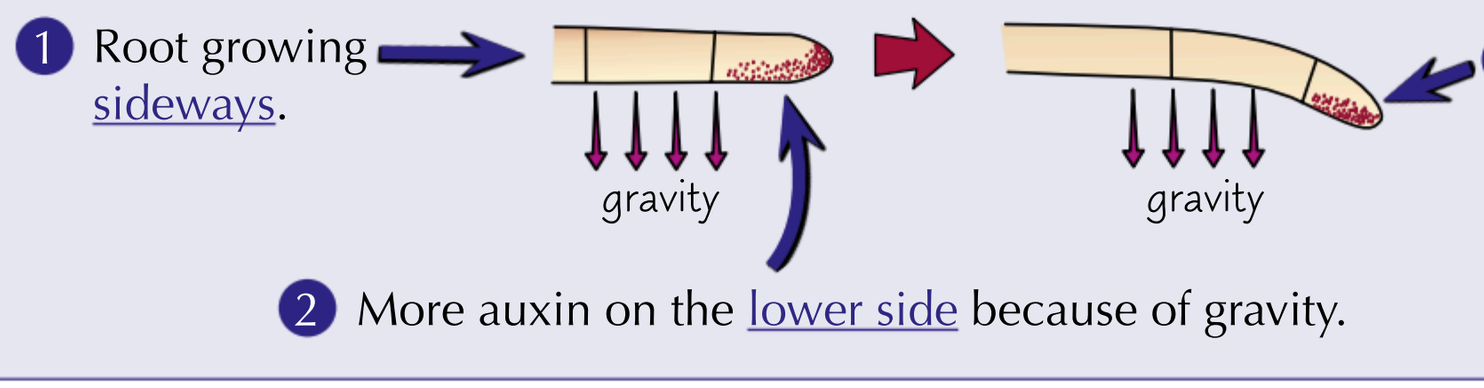


Gravity Affects Shoots and Roots **Differently**

Shoots grow away from gravity



Roots grow towards gravity



One hormone, three different responses

By responding to stimuli in their environment, plants increase their chances of survival. For example, by growing towards the light, plants increase the amount of light they receive for photosynthesis. Clever things, these plants...

Quick Quiz

Investigating Plant Growth

PRACTICAL

*Want to see auxin in action? You can **investigate** plant growth responses with a few **cress seeds**...*

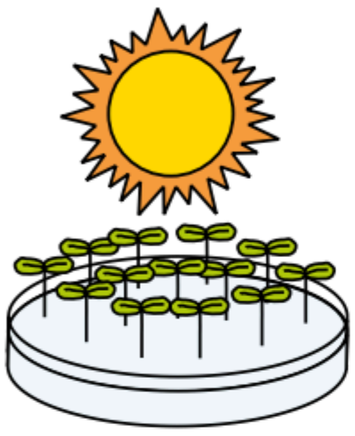
You Can **Investigate** the Effect of **Light** on **Plant Growth**

Here's one way you might see this being done:

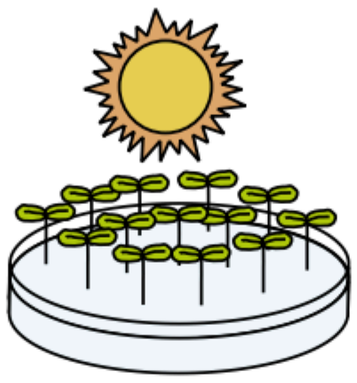
1) Put 10 cress seeds into three different Petri dishes, each lined with moist filter paper. Allow the seeds to germinate (start growing) in a warm room.

2) Measure the height of each seedling in mm.

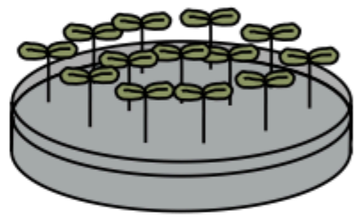
3) Put one dish on a windowsill in direct sunlight...



...put the second dish in partial light...



...and put the third in the dark, e.g. in a cupboard.



Remember to label your Petri dishes.

4) Measure the height of each seedling every day for five days. Calculate the mean height for each Petri dish each day, then compare the difference in growth. The seedlings will grow taller at lower light levels — this is because there is more auxin in the shoots of plants grown in the dark.

5) You know that the growth of the cress seeds is due to light only, if you control all other variables, e.g.

Variable	How to control it
Number of seeds	Use the same number of seeds in each dish.
Type of seed	Use seeds that all come from the same packet.
Temperature	Keep your Petri dishes in a place where the temperature is stable (i.e. away from heat sources and draughts).
Water	Use a measuring cylinder to add the same amount of water.

You might see other experiments about the effects of light on plant growth in your exams. Just remember how auxin works and you'll be able to figure out what's going on.

You Can Also **Investigate** the Effect of **Gravity** on **Plant Growth**

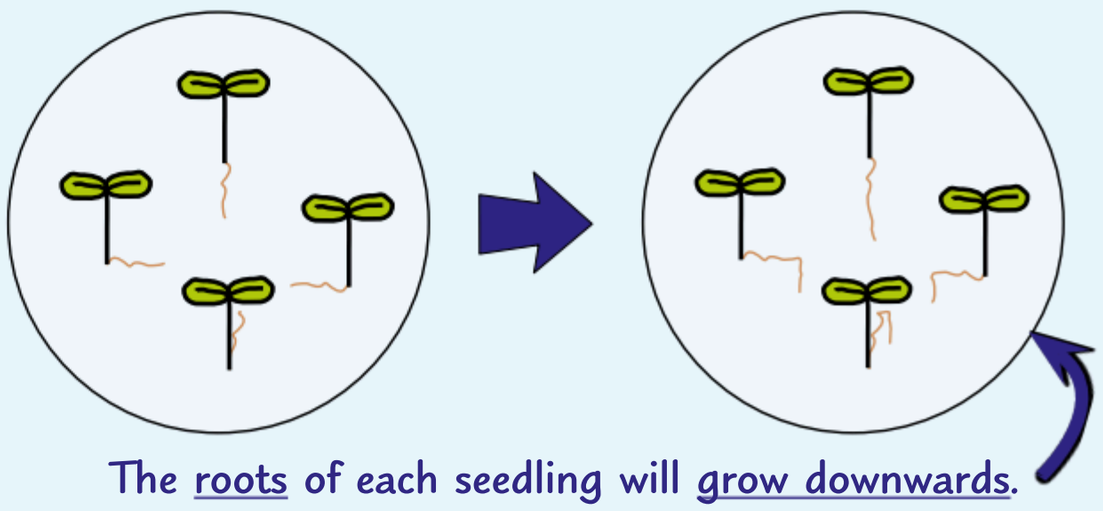
Here's one way you can investigate the effect of gravity using some more cress seedlings:

Method

1) Place four seedlings on damp cotton wool in a Petri dish, each with their roots pointing in a different direction.

2) Store the Petri dish vertically for a few days in the dark (this ensures that light does not affect their growth).

Don't forget to control all the other variables — they'd be the same as in the table above.



The roots of each seedling will grow downwards.

Record your results using scientific drawings

Labelled diagrams are a really good way to show the results of experiments like this. There's more about scientific drawings on page 6. Make sure your drawings are neat and useful — remember, don't do any sketching or shading (not even if you're really good at it...) and label your drawing using straight, uncrossed lines.

Commercial Uses of Plant Hormones

Quick Quiz

*Plant hormones can be **extracted**, or **artificial copies** can be made. This comes in pretty handy...*

Auxins are Used in Different Ways to **Control Plant Growth**

1 Killing Weeds

Most weeds growing in fields of crops or in a lawn are broad-leaved...

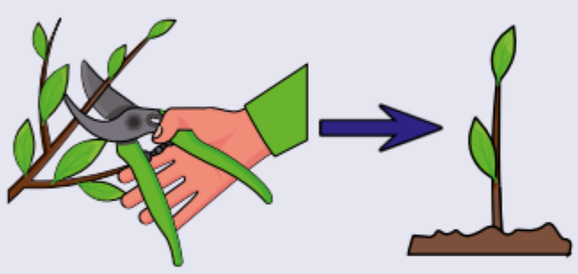
...while grasses and cereals have narrow leaves.

Apply selective weedkillers containing auxins.

The weedkillers disrupt the growth patterns of broad-leaved plants, which kills them, whilst leaving the grass and crops untouched.

2 Growing from Cuttings with Rooting Powder

A cutting is part of a plant that has been cut off it, like the end of a branch with a few leaves on it.



This enables growers to produce lots of clones (exact copies) of a really good plant very quickly.

If you stick cuttings in the soil and add rooting powder, which contains auxins, they will produce roots rapidly and start growing as new plants.

3 Growing Cells in Tissue Culture

Take a few cells from a plant.

See p.134 for more on tissue culture.

Add auxins to the growth medium (along with nutrients)...

... to stimulate the cells to divide to form both roots and shoots.

Gibberellin Controls **Flowering**, **Fruit Size** and **Germination**

- Inducing flowering — Some plants require certain conditions to flower, such as longer days or low temperatures. If these plants are treated with gibberellin, they will flower without any change in their environment. Gibberellin can also be used to grow bigger flowers.
- Growing larger fruit — Seedless varieties of fruit (e.g. seedless grapes) often do not grow as large as seeded fruit. However, if gibberellin is added to these fruits, they will grow larger to match the normal types.
- Controlling dormancy — Lots of seeds won't germinate until they've been through certain conditions (e.g. a period of cold or of dryness). This is called dormancy. Seeds can be treated with gibberellin to alter dormancy and make them germinate at times of year that they wouldn't normally. It also helps to make sure all the seeds in a batch germinate at the same time.

Seed germination is when a seed starts to grow into a plant.

Ethene Stimulates **Ripening of Fruit**

1) Ethene is a gas produced by plants that stimulates enzymes that cause fruit to ripen.
2) Commercially, it can be used to speed up the ripening of fruits — either while they are still on the plant, or during transport to the shops.
3) This means that fruit can be picked while it's still unripe (and therefore firmer and less easily damaged). The gas is then added to the fruit on the way to the supermarket so that it will be perfect just as it reaches the shelves.
4) Ripening can also be delayed while the fruit is in storage by adding chemicals that block ethene's effect on the fruit or reduce the amount of ethene that the fruit can produce. Alternatively, some chemicals can be used that react with ethene to remove it from the air.

Ethene also controls cell division in plants.

There you go — three different hormones, many different uses

You may be asked to describe what any of these hormones do, so make sure you know your auxins from your ethene.

Warm-Up & Exam Questions

You could skim through this page in a few minutes, but there's no point unless you check over any bits you don't know and make sure you understand everything. It's not quick but it's the only way.

Warm-Up Questions

1) Name the plant hormone responsible for both phototropism and gravitropism.
2) Give one way that gibberellin is used commercially.
3) Give one way that ethene is used commercially.

Exam Questions

1 A gardener frequently uses products containing plant hormones in her garden.

GRADE 4-6 GRADE

1.1 The gardener has a problem with weeds in her lawn.
She uses a weedkiller that targets the weeds without affecting the growth of the lawn.
Name the hormone likely to be present in the weedkiller.

[1 mark]

1.2 The gardener has a cutting of a plant she would like to grow more of.
Explain how a plant hormone could help the gardener to grow a new plant from the cutting.

[2 marks]

1.3 The gardener is entering some flowering plants into a competition. It is important that the plants all flower in time for the competition. What plant hormone could the gardener use to achieve this?

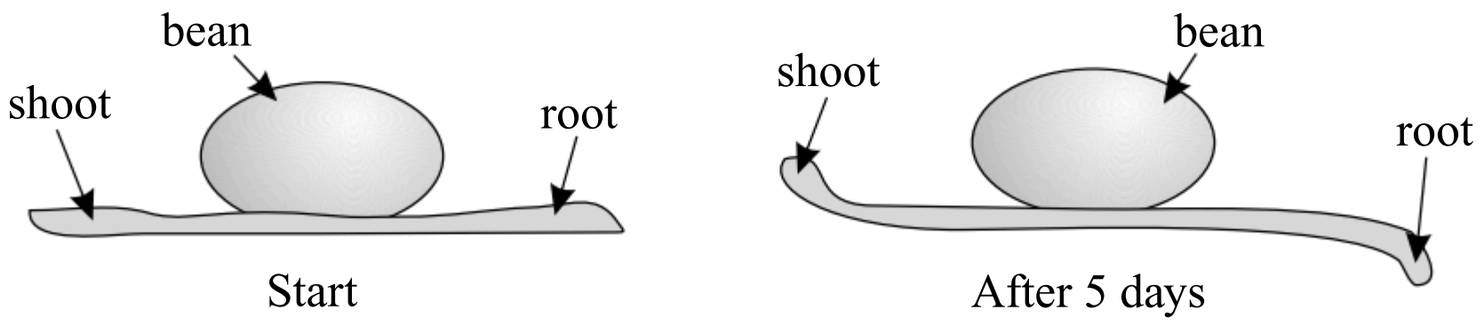
[1 mark]

PRACTICAL

2 A student placed some germinating beans on the surface of some damp soil and left them in the dark for five days. The appearance of the beans before and after the five day period is shown in **Figure 1**.

GRADE 6-7 GRADE

Figure 1



Both the shoot and the root have undergone a gravitropic response after 5 days.

2.1 Describe the gravitropic response of the shoot.

[1 mark]

2.2 Explain the mechanism behind this response.

[2 marks]

2.3 The student wants to compare his findings with beans grown in the light.
He puts some new germinating beans on damp soil and shines a light on them.
Give **two** variables that the student should control in this second experiment.

[2 marks]

Revision Summary Test for Topic 5b

Riding high on the adrenaline of a good revision session? It's a perfect time to:

- Tackle the revision summary test below, or scan the QR code to do it online. The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

The Endocrine System and Controlling Blood Glucose (p.96-97) ☑

1) How do hormones travel around the body? Give one hormone that each of these glands produces: a) thyroid, b) ovaries, c) adrenal glands, d) testes, e) pancreas. ☑
2) List three differences between the action of nerves and hormones. ☑
3) How does the body respond if glucose levels are too high? Is this positive or negative feedback? ☑
4) How does the body increase the blood glucose concentration? ☑
5) What is Type 1 diabetes? How is it treated? ☑
6) What happens in Type 2 diabetes? Give one risk factor for developing Type 2 diabetes. ☑

The Kidneys and Kidney Failure (p.99-100) ☑

7) What are filtration and selective reabsorption in the kidneys? ☑
8) Name three substances that are removed from the body in urine. Bonus points for saying why. ☑
9) How else are these waste substances lost from the body? Why do the kidneys need to get involved? ☑
10) Which hormone controls the concentration of urine? From which gland is it released? ☑
11) How does the body respond if it detects that the water content of the blood is too high? ☑
12) How does a dialysis machine help treat kidney failure? What are the downsides of being on dialysis? ☑
13) What are the advantages and disadvantages of having a kidney transplant to cure kidney failure? ☑

Puberty, Fertility and Other Hormonal Responses (p.102-105) ☑

14) What are the main male and female reproductive hormones? What effects do they have on the body? ☑
15) What is happening in the body during menstruation? ☑
16) What happens on day 14 of the menstrual cycle? Which hormone is primarily responsible for this? ☑
17) Which hormone causes the uterus lining to grow? And which hormone maintains it? ☑
18) Describe the effects that the four hormones involved in the menstrual cycle have on each other. ☑
19) Which two hormones are used in hormonal methods of contraception? How do they act? ☑
20) List six methods of hormonal contraception. ☑
21) Describe two barrier methods of contraception. What other contraceptive options are there? ☑
22) How can hormones be used to increase fertility? ☑
23) Briefly outline the process of IVF. What are its pros and cons? ☑
24) What is the effect of adrenaline? ☑
25) What effects does thyroxine have on the body? How is it regulated? ☑

Plant Hormones (p.107-109) ☑

26) How do shoots grow towards light? ☑
27) How do roots grow downwards? ☑
28) Briefly describe how you could investigate the growth of plants in response to light and gravity. ☑
29) Give three ways auxins are used commercially to control plant growth. ☑
30) Give three benefits of treating plants or seeds with gibberellin. ☑
31) Explain how hormones can be used to make sure fruit sold in shops is undamaged and perfectly ripe. ☑

Topic 6a — DNA and Reproduction

Quick Quiz

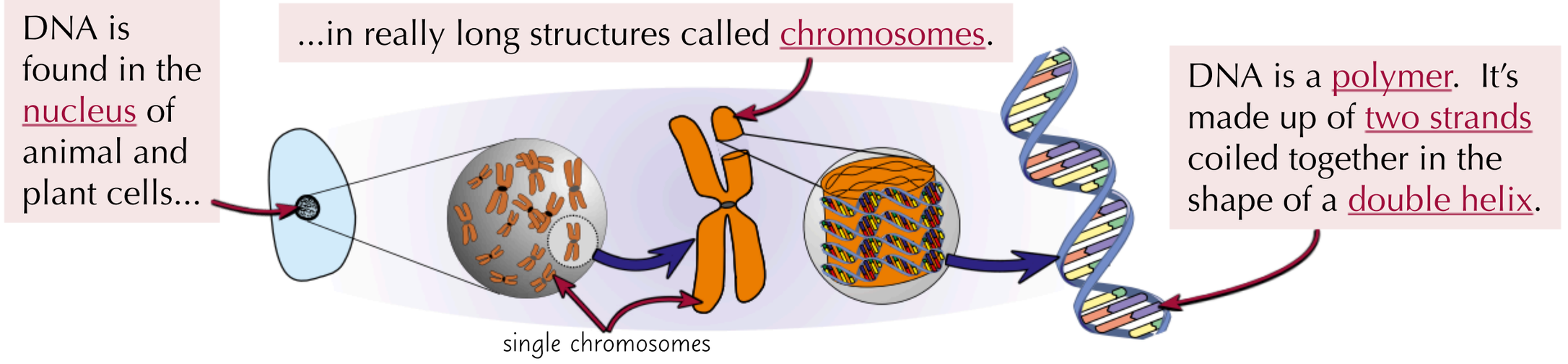
DNA

*The first step in understanding **genetics** is getting to grips with **DNA** and **genes**. The key to life...and this topic.*

Chromosomes Are Really Long Molecules of **DNA**

1) DNA stands for deoxyribonucleic acid. It's the chemical that all of the genetic material in a cell is made up from.
2) It contains coded information — basically all the instructions to put an organism together and make it work.
3) So it's what's in your DNA that determines what inherited characteristics you have.

Head to the next page for more on the structure of DNA.



A **Gene** Codes for a **Specific Protein**

- A gene is a small section of DNA found on a chromosome.
- Each gene codes for (tells the cells to make) a particular sequence of amino acids which are put together to make a specific protein.
- Genes simply tell cells in what order to put the amino acids together (more on this on the next page).
- DNA also determines what proteins the cell produces, e.g. haemoglobin, keratin. That in turn determines what type of cell it is, e.g. red blood cell, skin cell.

Only 20 amino acids are used, but they make up thousands of different proteins.

Every Organism Has a **Genome**

A genome is the entire set of genetic material in an organism.

1) Scientists have worked out the complete human genome.
2) Understanding the human genome is a really important tool for science and medicine for many reasons.

- Diseases — scientists can identify genes in the genome that are linked to different types of disease.
- Inherited disorders — knowing which genes are linked to inherited diseases could help us to understand them better and could help us to develop effective treatments for them.
- Migration patterns — scientists can look at genomes to trace the migration of certain populations of people around the world. All modern humans are descended from a common ancestor who lived in Africa, but humans can now be found all over the planet. The human genome is mostly identical in all individuals, but as different populations of people migrated away from Africa, they gradually developed tiny differences in their genomes. By investigating these differences, scientists can work out when new populations split off in a different direction and what route they took.

Your genes make you different from everyone else

Remember, DNA contains all the instructions to 'build' an organism. The instructions are different for each type of organism on Earth (otherwise all living things would be the same). Now, there's a lot more about DNA and chromosomes coming up in this topic so it's really important that you understand this page before you move on.

The Structure of DNA and Protein Synthesis

Quick Quiz

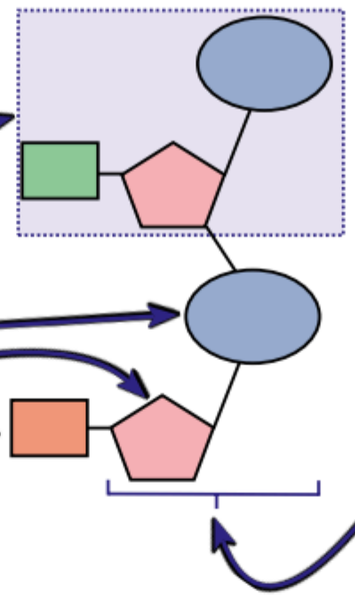
*So here's how life works — DNA molecules contain a **genetic code** that determines which **proteins** are built.*

DNA is Made Up of Nucleotides

DNA strands are polymers made up of lots of repeating units called nucleotides.

Each nucleotide consists of:
a phosphate group,
a sugar,
one of four different bases
— A, T, C or G.

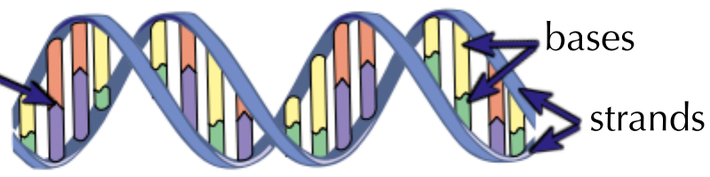
Part of a DNA strand



The sugar and phosphate groups in the nucleotides form a 'backbone' to the DNA strands. The sugar and phosphate groups alternate. A base is attached to each sugar.

Each base links to a base on the opposite strand in the helix.
A always pairs up with T, and C always pairs up with G.
This is called complementary base pairing.

Part of a DNA molecule



mRNA Carries The Code to The Ribosomes Where Proteins are Made

1) Each amino acid in a protein is coded for by a sequence of three bases in the gene.
So, the order of bases in a gene decides the order of amino acids in a protein.
2) Proteins are made in the cell cytoplasm on tiny structures called ribosomes.
3) DNA is found in the cell nucleus and can't move out of it because it's really big.
So the cell needs to get the code from the DNA to the ribosome.
4) This is done using a molecule called mRNA — which is made by copying the code from the DNA. The mRNA acts as a messenger between the DNA and the ribosome — it carries the code between the two.
5) The correct amino acids are brought to the ribosomes in the correct order by carrier molecules.

There are parts of DNA that don't code for proteins. Some of these non-coding parts switch genes on and off, so they control whether or not a gene is expressed (used to make a protein).

Proteins Have Many Different Functions

When a chain of amino acids has been assembled, it folds into a unique shape which allows the protein to perform the task it's meant to do. Here are a few examples of types of protein:

- ENZYMES — act as biological catalysts to speed up chemical reactions in the body (see p.27).
- HORMONES — used to carry messages around the body. E.g. insulin is a hormone released into the blood by the pancreas to regulate the blood sugar level.
- STRUCTURAL PROTEINS — are physically strong. E.g. collagen is a structural protein that strengthens connective tissues (like ligaments and cartilage).

Apple Turnover and Gloopy Custard...

Don't forget that in a DNA molecule, A always pairs with T and C always pairs with G.

Q1 The diagram on the right shows a short sequence of DNA bases.

C T C C T A C G T G A T

a) The sequence is from a coding region of DNA.
How many amino acids does it code for? [1 mark]

b) Give the sequence of bases for the complementary strand. [1 mark]

Q1 Video Solution

Quick Quiz

Mutations

*Sometimes the **sequence** of **DNA bases** can be changed. These changes are called **mutations**. Read on...*

Mutations are Random **Changes** to the **Genetic Code**

1) Mutations occur continuously. They can occur spontaneously, e.g. when a chromosome isn't quite replicated properly. However, the chance of mutation is increased by exposure to certain substances or some types of radiation. Mutations can sometimes be inherited.
2) Mutations change the sequence of the DNA bases in a gene, which produces a genetic variant (a different form of the gene).
3) As the sequence of DNA bases codes for the sequence of amino acids that make up a protein (see page 112), mutations to a gene sometimes lead to changes in the protein that it codes for.
   - Most mutations have very little or no effect on the protein.
   - Some will change it to such a small extent that its function or appearance is unaffected.
   - However, a few mutations can seriously affect a protein — the mutation may code for an altered protein with a change in its shape. This could affect its ability to perform its function. E.g.
     - If the shape of an enzyme's active site is changed, its substrate may no longer be able to bind to it.
     - Structural proteins like collagen could lose their strength if their shape is changed, making them pretty useless at providing structure and support.

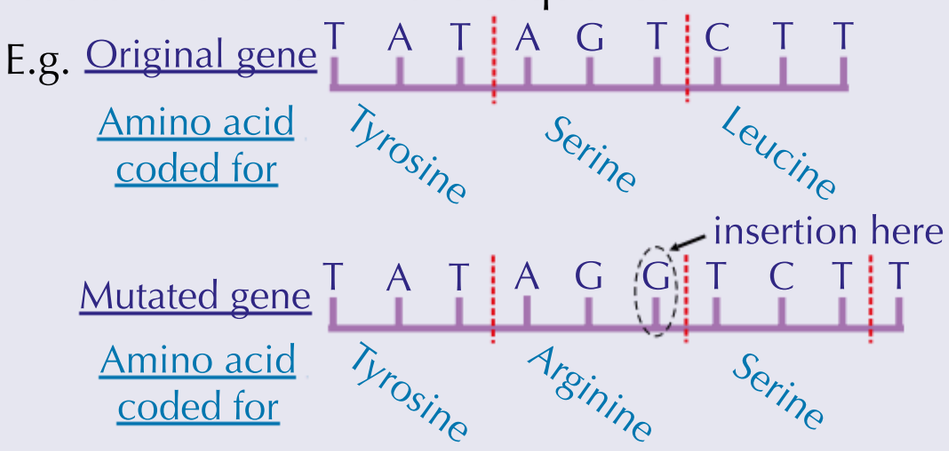
Enzymes are proteins — see page 27.

4) If there's a mutation in the non-coding DNA (see previous page), it can alter how genes are expressed.

There Are **Different Types** of **Mutation**, For Example...

Insertions

- Insertions are where a new base is inserted into the DNA base sequence where it shouldn't be.
- You should remember from page 113 that every three bases in a DNA base sequence codes for a particular amino acid.
- An insertion changes the way the groups of three bases are 'read', which can change the amino acids that they code for.
- Insertions can change more than one amino acid as they have a knock-on effect on the bases further on in the sequence.

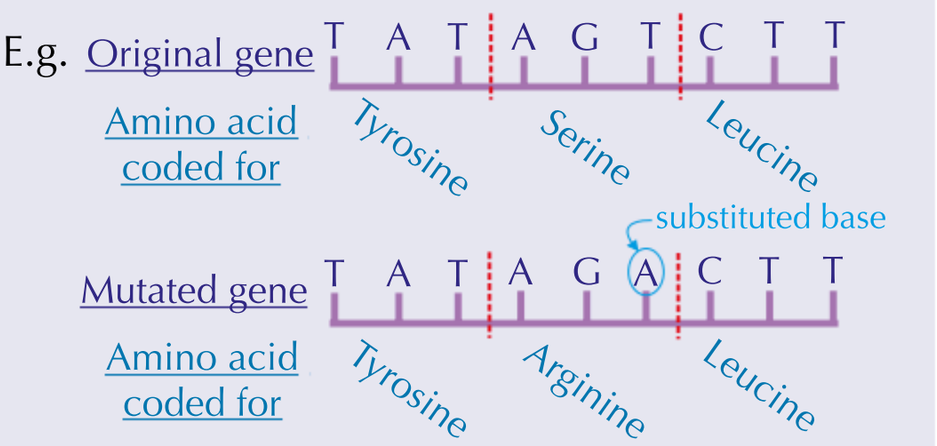


Deletions

- Deletions are when a random base is deleted from the DNA base sequence.
- Like insertions, they change the way that the base sequence is 'read' and have knock-on effects further down the sequence.

Substitutions

Substitution mutations are when a random base in the DNA base sequence is changed to a different base.



Mutations are very common, but are rarely noticeable

Mutations do sound pretty scary (especially if you're thinking about all those mutant creatures in sci-fi movies). But in reality, mutations occur all the time and hardly ever affect the structure or function of a protein. So we can usually carry on happily with our lives, without noticing anything about how our DNA is mutating.

Warm-Up & Exam Questions

Take a deep breath and go through these Warm-Up Questions one by one. Then onto the Exam Questions. Don't panic if you get something wrong — as they say, practice makes perfect...

Warm-Up Questions

1) Name the structures that contain DNA in the nucleus of a cell.
2) What is a gene?
3) Give three reasons why understanding the human genome is important for science and medicine.
4) Which DNA bases pair up according to complementary base pairing?
5) What is a mutation?
6) Explain why the function of a protein may be affected by a gene mutation.

Exam Questions

1 An organism's genetic material is made up of a chemical called DNA. GRADE 4-6 GRADE

1.1 Which of the following describes the structure of DNA?
Tick **one** box.

- ☐ A protein made up of two strands.
- ☐ A polymer made up of two strands.
- ☐ A protein made up of four strands.
- ☐ A polymer made up of four strands.

[1 mark]

1.2 Which of the following contains the largest amount of an organism's DNA?
Tick **one** box.

- ☐ A gene
- ☐ Its genome
- ☐ A chromosome

[1 mark]

1.3 Explain the relationship between DNA and the proteins produced by an organism.

[3 marks]

2 Krabbe disease is a genetic disorder caused by a mutation in the GALC gene. The GALC gene codes for an enzyme that is needed for a healthy nervous system. GRADE 7-9 GRADE

2.1 One of the mutations that can cause Krabbe disease involves the addition of nucleotides to the GALC gene. Describe the structure of a nucleotide.

[1 mark]

2.2 Outline how the enzyme coded for by the GALC gene would normally be synthesised within a cell.

[4 marks]

2.3 Explain how a mutation in the GALC gene could alter the activity of the enzyme it codes for.

[4 marks]

2.4 Mutations happen continuously, although genetic disorders such as Krabbe disease are rare. Explain why this is the case.

[1 mark]

Quick Quiz

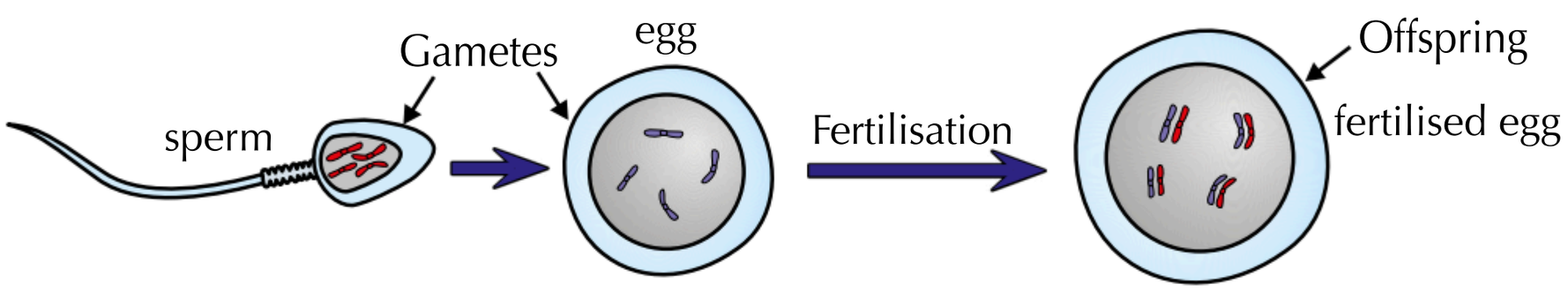
Reproduction

*Reproduction is important for all species. It can happen in **two** different ways...*

Sexual Reproduction Produces **Genetically Different Cells**

1) Sexual reproduction is where genetic information from two organisms (a male and a female) is combined to produce offspring which are genetically different to either parent.
2) In sexual reproduction, the female and male produce gametes by meiosis (see next page) — e.g. egg and sperm cells in animals.
3) In humans, each gamete contains 23 chromosomes — half the number of chromosomes in a normal cell. (Instead of having two of each chromosome, a gamete has just one of each.)
4) The egg (from the female) and the sperm cell (from the male) then fuse together (fertilisation) to form a cell with the full number of chromosomes (half from the male, half from the female).

SEXUAL REPRODUCTION involves the fusion of male and female gametes.
Because there are TWO parents, the offspring contain a mixture of their parents' genes.



5) This is why the offspring inherits features from both parents — it's received a mixture of chromosomes (and it's the chromosomes that decide how you turn out).
6) This mixture of genetic information produces variation in the offspring.
7) Flowering plants can reproduce in this way too. They also have egg cells, but their version of sperm is known as pollen.

Asexual Reproduction Produces **Genetically Identical Cells**

1) In asexual reproduction there's only one parent so the offspring are genetically identical to that parent.
2) Asexual reproduction happens by mitosis — an ordinary cell makes a new cell by dividing in two (see page 10).
3) The new cell has exactly the same genetic information (i.e. genes) as the parent cell — it's called a clone.

In ASEXUAL REPRODUCTION there's only ONE parent. There's no fusion of gametes, no mixing of chromosomes and no genetic variation between parent and offspring.
The offspring are genetically identical to the parent — they're clones.

4) Bacteria, some plants and some animals reproduce asexually.

You might need to reproduce these facts in the exam...

The main messages on this page are that: 1) sexual reproduction needs two parents and forms cells that are genetically different to the parents, so there's lots of genetic variation. And 2) asexual reproduction needs just one parent to make genetically identical cells, so there's no genetic variation in the offspring. Got it? Right, time to dig a bit deeper...

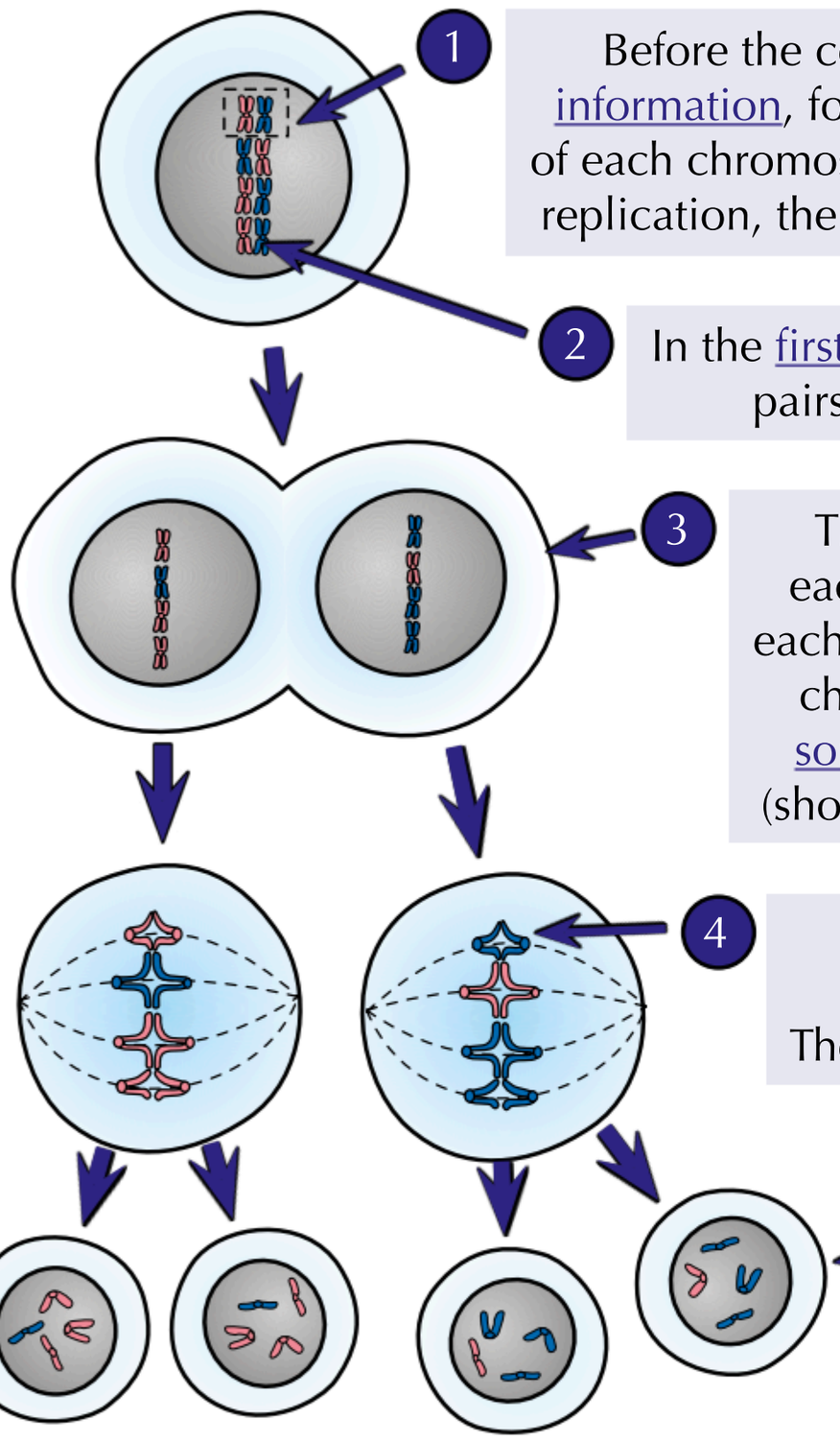
Meiosis

Quick Quiz

*If you're wondering how gametes end up with **half** the number of **chromosomes** of a normal cell, read on...*

Gametes Are Produced by Meiosis

1) As you know from the previous page, gametes only have one copy of each chromosome, so that when gamete fusion takes place, you get the right amount of chromosomes again (two copies of each).
2) To make gametes which only have half the original number of chromosomes, cells divide by meiosis. This process involves two cell divisions. In humans, it only happens in the reproductive organs (the ovaries in females and testes in males).
3) Here's how it works:



1 Before the cell starts to divide, it duplicates its genetic information, forming two-armed chromosomes — one arm of each chromosome is an exact copy of the other arm. After replication, the chromosomes arrange themselves into pairs.

Video

2 In the first division in meiosis, the chromosome pairs line up in the centre of the cell.

The genetic information is stored in DNA — see p.112.

3 The pairs are then pulled apart so each new cell only has one copy of each chromosome. Some of the male's chromosomes (shown in blue) and some of the female's chromosomes (shown in pink) go into each new cell.

Remember, each chromosome pair is made up of one chromosome inherited from each parent — see page 116.

4 In the second division, the chromosomes line up again in the centre of the cell. The arms of the chromosomes are pulled apart.

You get four gametes, each with only a single set of chromosomes in it. Each of the gametes is genetically different from the others because the chromosomes all get shuffled up during meiosis and each gamete only gets half of them, at random.

The Cell Produced by Gamete Fusion Replicates Itself

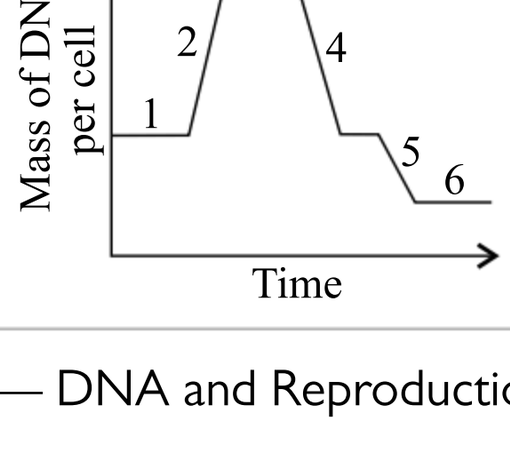
1) After two gametes have fused during fertilisation, the resulting new cell divides by mitosis to make a copy of itself.
2) Mitosis repeats many times to produce lots of new cells in an embryo.
3) As the embryo develops, these cells then start to differentiate (see page 8) into the different types of specialised cell that make up a whole organism.

There's loads on mitosis on page 10.

Now that I have your undivided attention...

In humans, meiosis only occurs in reproductive organs, for making gametes.

Q1 Human body cells contain 46 chromosomes each. The graph on the right shows how the mass of DNA per cell changed as some cells divided by meiosis in a human ovary. How many chromosomes were present in each cell when they reached stage 6? [1 mark]



Q1 Video Solution

Quick Quiz

More on Reproduction

*So is **sexual** or **asexual reproduction better**? It's a hot debate. Some organisms can't decide and do **both**.*

Variation is the Main **Advantage** of **Sexual Reproduction**

1) Offspring from sexual reproduction have a mixture of two sets of chromosomes. The organism inherits genes (and therefore features) from both parents, which produces variation in the offspring (see p.128).
2) Variation increases the chance of a species surviving a change in the environment. While a change in the environment could kill some individuals, it's likely that variation will have led to some of the offspring being able to survive in the new environment. They have a survival advantage.
3) Because individuals with characteristics that make them better adapted to the environment have a better chance of survival, they are more likely to breed successfully and pass the genes for the characteristics on. This is known as natural selection (see p.129).

We can use selective breeding to speed up natural selection. Selective breeding is where individuals with a desirable characteristic are bred to produce offspring that have the desirable characteristic too (see p.132). This means that we can increase food production, e.g. by breeding animals that produce a lot of meat.

Only Needing **One Parent** is a Big **Advantage** in **Asexual Reproduction**

The big disadvantage of asexual reproduction is that it doesn't produce variation. However, it does have some advantages over sexual reproduction...

- There only needs to be one parent. This means that asexual reproduction uses less energy than sexual reproduction, because organisms don't have to find a mate.
- This also means that asexual reproduction is faster than sexual reproduction, so if conditions are favourable, many identical offspring can be produced.

Some Organisms Can Reproduce Sexually **and** Asexually

You need to know these examples:

Malarial Parasites

Malaria is caused by a parasite that's spread by mosquitoes — see page 57. When a mosquito carrying the parasite bites a human, the parasite can be transferred to the human. The parasite reproduces sexually when it's in the mosquito and asexually when it's in the human host.

Fungi

Many species of fungus release spores, which can become new fungi when they land in a suitable place. Spores can be produced sexually and asexually. Asexually-produced spores form fungi that are genetically identical to the parent fungus. Sexually-produced spores introduce variation and are often produced in response to an unfavourable change in the environment, increasing the chance that the population will survive the change.

Plants

Loads of species of plant produce seeds sexually, but can also reproduce asexually. Asexual reproduction can take place in different ways. For example...

- Strawberry plants produce 'runners'. These are stems that grow horizontally on the surface of the soil away from a plant. At various points along the runner, a new strawberry plant forms that is identical to the original plant.
- In plants that grow from bulbs (e.g. daffodils), new bulbs can form from the main bulb and divide off. Each new bulb can grow into a new identical plant.

Each type of reproduction has its advantages...

It's generally a good thing if organisms can reproduce by both sexual and asexual reproduction — it increases their chances of being able to produce offspring, which keeps the species in existence. And nobody likes extinction...

X and Y Chromosomes

Quick Quiz

*Now for a couple of **very** important little chromosomes...*

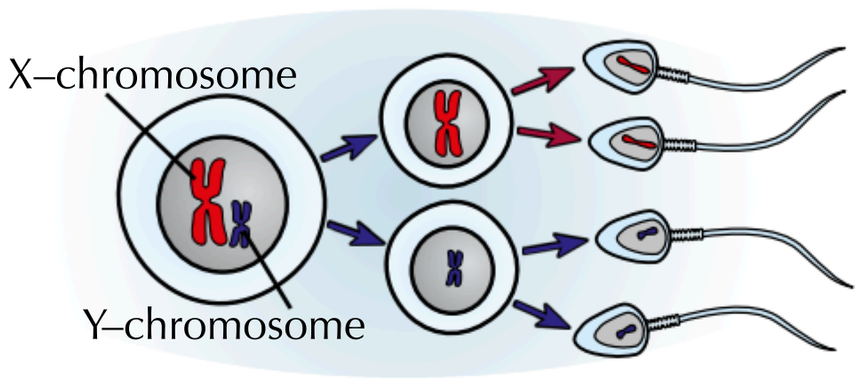
Your **Sex Chromosomes** Control Whether You're **Male** or **Female**

There are 23 pairs of chromosomes in every human body cell.
Of these, 22 are matched pairs of chromosomes that just control characteristics.
The 23rd pair are labelled XY or XX. They're the two chromosomes that decide your sex:

See p.10 for more on chromosomes.

Males have an X and a Y chromosome: XY
The Y chromosome causes male characteristics.

Females have two X chromosomes: XX
The XX combination allows female characteristics to develop.



When making sperm, the X and Y chromosomes are drawn apart in the first division in meiosis (see page 117).
There's a 50% chance each sperm cell gets an X-chromosome and a 50% chance it gets a Y-chromosome.

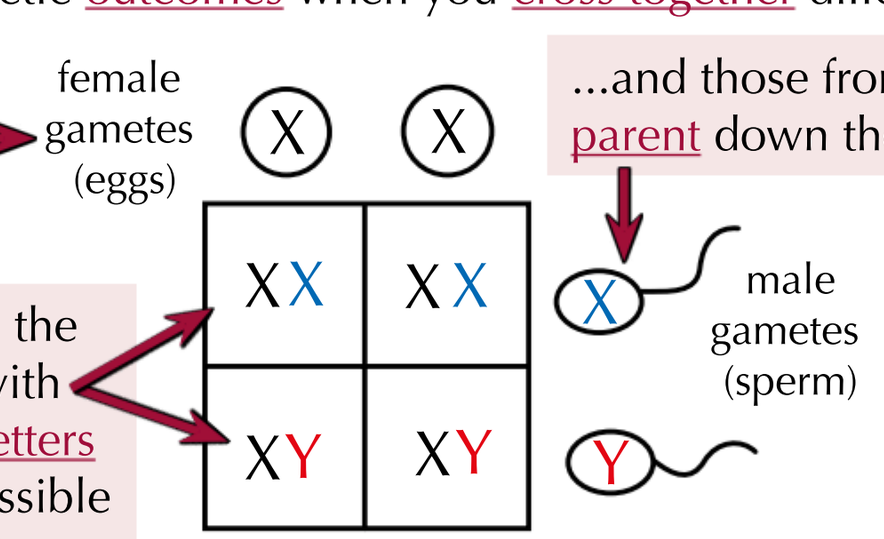
A similar thing happens when making eggs.
But the original cell has two X-chromosomes, so all the eggs have one X-chromosome.

Genetic Diagrams Show the **Possible Gamete Combinations**

To find the probability of getting a boy or a girl, you can draw a genetic diagram. These are models that are used to show all the possible genetic outcomes when you cross together different genes or chromosomes.

Put the possible gametes from one parent along the top...

...and those from the other parent down the side.

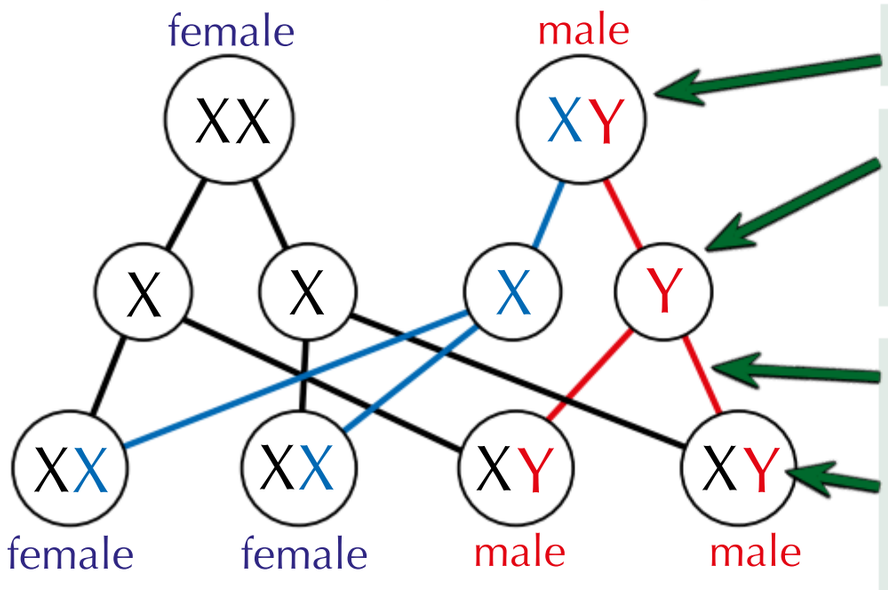


Then fill in the letters from the top and side that line up with that square. The pairs of letters in the middle show the possible combinations of the gametes.

There are two XX results and two XY results, so there's the same probability of getting a boy or a girl.

This type of genetic diagram is called a Punnett square.

The other type of genetic diagram looks a bit more complicated, but it shows exactly the same thing.



At the top are the parents.

The middle circles show the possible gametes that are formed. One gamete from the female combines with one gamete from the male (during fertilisation).

The criss-cross lines show all the possible ways the X and Y chromosomes could combine.
The possible combinations of the offspring are shown in the bottom circles. Remember, only one of these possibilities would actually happen for any one offspring.

Don't forget that this 50:50 ratio is only a probability at each pregnancy.

It's the sperm that determines the sex of the offspring

It's possible for people to have three X chromosomes, or even one X and two Y chromosomes, in their cells — but this is fairly rare. You just need to remember that XX gives female characteristics and XY gives male characteristics.

Warm-Up & Exam Questions

It's time to see how much you picked up about meiosis, reproduction and sex chromosomes...

Warm-Up Questions

1) Suggest why there is variation in the offspring of sexual reproduction.
2) How many cell divisions take place in meiosis?
3) Which type of reproduction, sexual or asexual, produces offspring most quickly?
4) What combination of sex chromosomes do human females have?

Exam Questions

1 Some species of worm can produce offspring through a process called fragmentation. In this process, fragments of the parent's body break off and undergo cell division by mitosis to develop into mature, complete organisms.

GRADE 4-6

1.1 What term is used to describe this form of reproduction?

[1 mark]

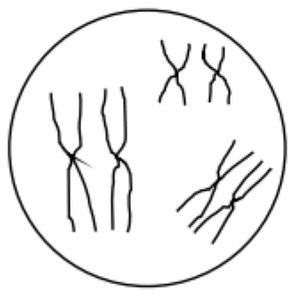
1.2 Suggest how the chromosomes in the offspring will compare to those of the parent worm.

[1 mark]

2 Mosquitoes have three pairs of chromosomes in their body cells.
Figure 1 shows a mosquito cell which is about to divide by meiosis.

GRADE 6-7

Figure 1



2.1 The cell in **Figure 1** undergoes meiosis.
State how many chromosomes will be present in each new cell produced.

[1 mark]

2.2 How many cells will be produced in total when the cell in **Figure 1** undergoes meiosis?

[1 mark]

2.3 Explain how the processes of meiosis and fertilisation lead to genetic variation in the mosquito's offspring.

[3 marks]

When a mosquito infected with malarial parasites bites a human, parasites present in the mosquito's saliva pass into the human's bloodstream.

2.4 Malarial parasites reproduce asexually whilst in a human host.
Give **three** advantages of this method of reproduction compared to sexual reproduction.

[3 marks]

2.5 Genetic variation within the malarial parasite population increases whilst in the mosquito.
Explain why.

[1 mark]

Genetic Diagrams

Quick Quiz

*Genetic diagrams aren't as scary as they look — you just need to **practise** them...*

Different Genes Control Different **Characteristics**

1) Some characteristics are controlled by a single gene. E.g. red-green colour blindness in humans and mouse fur colour. However, most characteristics are controlled by several genes interacting.
2) All genes exist in different versions called alleles (which are represented by letters in genetic diagrams).
3) You have two versions (alleles) of every gene in your body — one on each chromosome in a pair.

Homozygous — this means an organism has two alleles for a gene that are the same.

Heterozygous — this means an organism's two alleles for a particular gene are different.

4) If the two alleles are different, only one can determine what characteristic is present. The allele for the characteristic that's shown is called the dominant allele. The other one is called recessive.

Use a capital letter for dominant alleles, e.g. 'C', and show recessive alleles with small letters, e.g. 'c'.

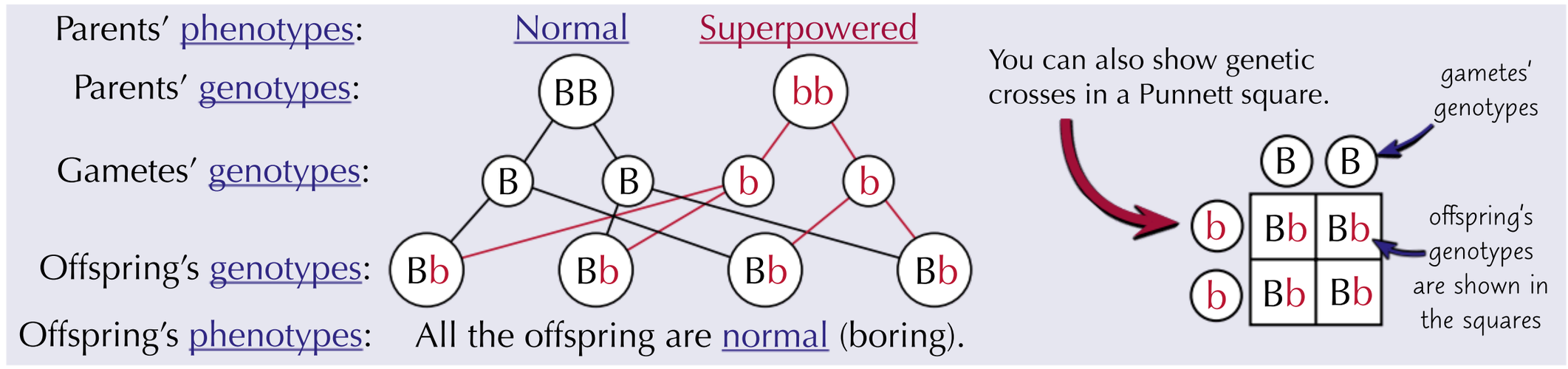
- To display a recessive characteristic, both an organism's alleles must be recessive (e.g. cc).
- To display a dominant characteristic, the organism can be either CC or Cc, because the dominant allele overrules the recessive one if the organism is heterozygous.

5) Your genotype is the combination of alleles you have. Your alleles work at a molecular level to determine what characteristics you have — your phenotype.

Genetic Diagrams Show the **Possible Alleles** of **Offspring**

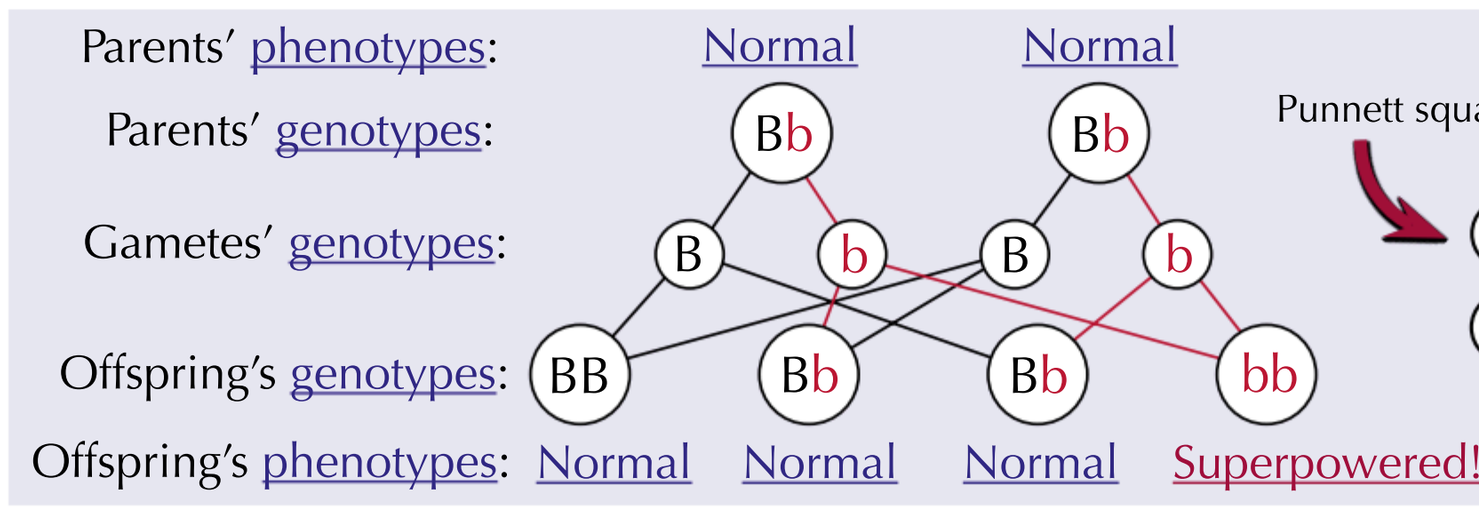
Suppose you start breeding hamsters with superpowers. The allele which causes hamsters to have superpowers is recessive ("b"), whilst normal (boring) behaviour is due to a dominant allele ("B").

1) A superpowered hamster must have the genotype bb. But a normal hamster could be BB or Bb.
2) Here's what happens if you breed from two homozygous hamsters:

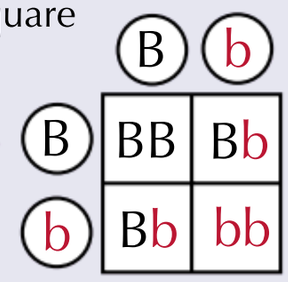


You can also show genetic crosses in a Punnett square.

3) If two of these offspring now breed, you'll get the next generation:



Punnett square



But remember — genetic diagrams only tell you probabilities. They don't say definitely what'll happen.

4) That's a 3:1 ratio of normal to superpowered offspring in this generation (a 1 in 4 or 25% probability of superpowers).

Two types of genetic diagram to learn about — what a treat...

You should know how to produce and interpret both of these types of genetic diagram before exam day. No excuses.

Quick Quiz

More Genetic Diagrams

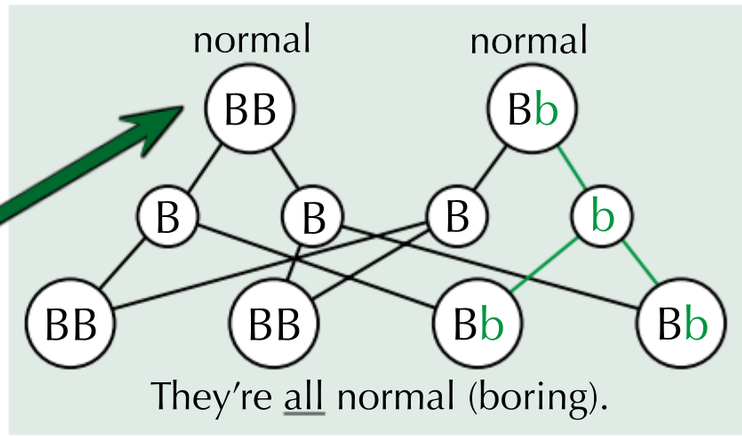
*You've got to be able to **predict** and **explain** the outcomes of crosses between individuals for each **possible combination** of **dominant** and **recessive alleles** of a gene. You should be able to draw a **genetic diagram** and **work it out** — but it'll be easier if you've seen them all before...*

BB × bb or BB × Bb — **All** the Offspring show the **Dominant** Characteristic

Let's take another look at the superpowered hamster example from page 121.

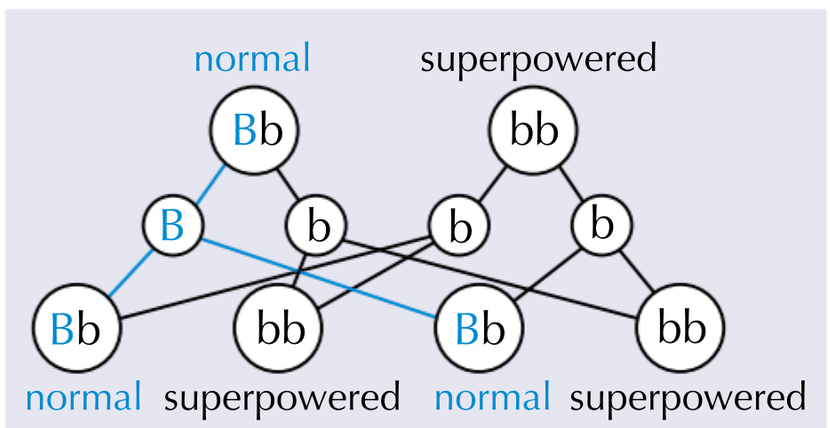
In this cross, we saw that when a homozygous dominant hamster (BB) is crossed with a homozygous recessive hamster (bb), all the offspring are normal (boring).

If you crossed a homozygous dominant hamster (BB) with a heterozygous hamster (Bb), you would also get all normal (boring) offspring.



For a reminder on the terms homozygous and heterozygous, head to page 121.

Bb × bb — Gives a **1:1 Ratio** in the Offspring



And here's another cross you could get with those superpowered hamsters...

If you crossed a heterozygous hamster (Bb) with a homozygous recessive hamster (bb), you'd expect half the offspring to be normal and half to be superpowered.

This is a 1:1 ratio.

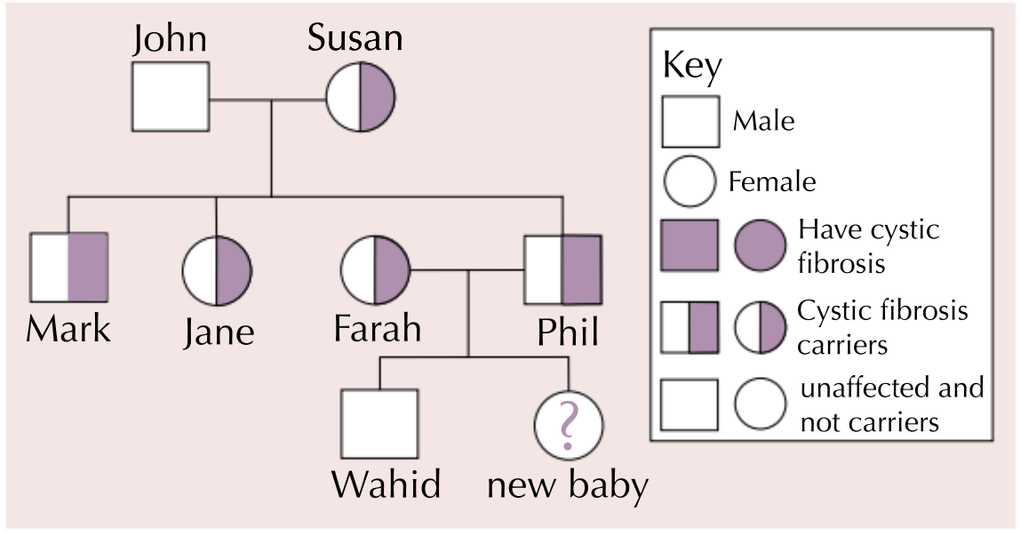
You Also Need to be Able to **Interpret Family Trees**

Knowing how inheritance works can help you to interpret a family tree — this is one for cystic fibrosis (p.123).

Many of the people shown in the family tree are carriers — this means that they have one copy of the allele for cystic fibrosis, but because the disorder is recessive, they don't show the characteristics of it.

From the family tree...

- You can tell that the allele for cystic fibrosis isn't dominant because plenty of the family carry the allele but don't have the disorder.



- You can see there is a 25% chance that the new baby will have the disorder and a 50% chance that it will be a carrier, as both of its parents are carriers but are unaffected. The case of the new baby is just the same as in the genetic diagram on the next page — the parents are both heterozygous (Ff), so the baby could be unaffected (FF), a carrier (Ff) or have cystic fibrosis (ff).

Not all inherited disorders have carriers

In the exam, you might get a family tree showing the inheritance of a dominant allele — in this case, there won't be any carriers shown. Now, here's a fascinating practice question about peas...

Q1 Round peas are caused by the dominant allele, R. The allele for wrinkly peas, r, is recessive. Using a Punnett square, predict the ratio of plants with round peas to plants with wrinkly peas for a cross between a heterozygous pea plant and a pea plant that is homozygous recessive. [3 marks]

Q1 Video Solution

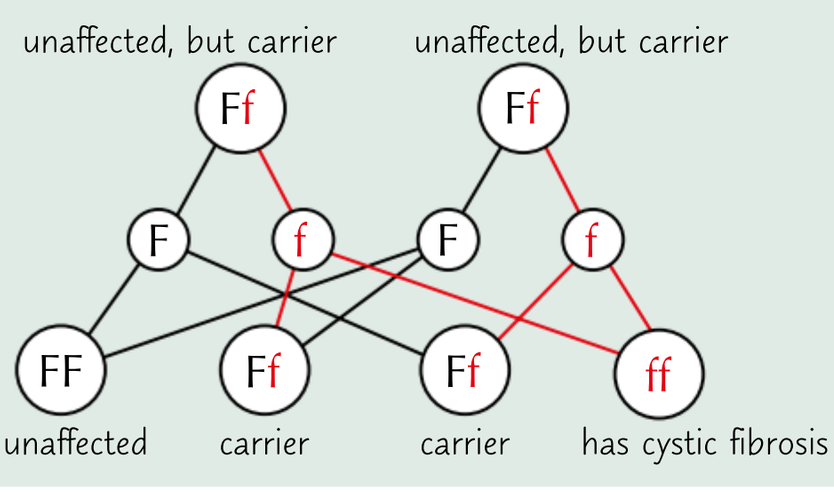
Inherited Disorders

Quick Quiz

*Some disorders can be **inherited** from your parents. Many of these can be **screened** for in embryos.*

Cystic Fibrosis is Caused by a Recessive Allele

Cystic fibrosis is a genetic disorder of the cell membranes. It results in the body producing a lot of thick sticky mucus in the air passages and in the pancreas.

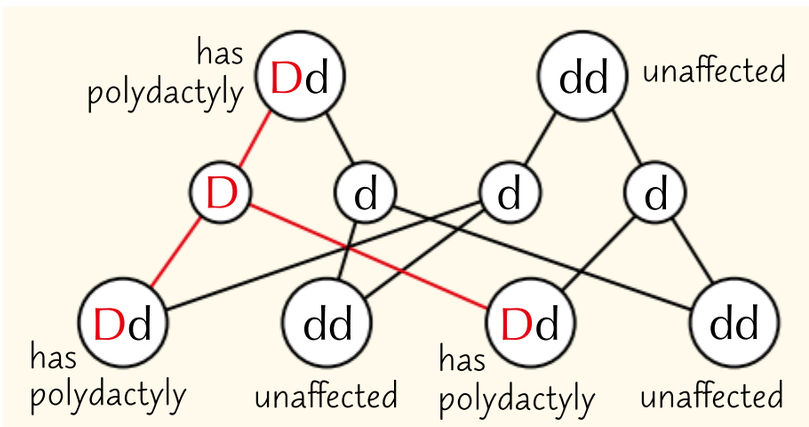


1) The allele which causes cystic fibrosis is a recessive allele, 'f'.
2) Because it's recessive, people with only one copy of the allele won't have the disorder — they're carriers.
3) For a child to have the disorder, both parents must be either carriers or have the disorder themselves.
4) As the diagram shows, there's a 1 in 4 chance of a child having the disorder if both parents are carriers.

Polydactyly is Caused by a Dominant Allele

Polydactyly is a genetic disorder where a baby is born with extra fingers or toes. It doesn't usually cause any other problems so isn't life-threatening.

1) The disorder is caused by a dominant allele, 'D', and so can be inherited if just one parent carries the defective allele.
2) The parent that has the defective allele will have the condition too since the allele is dominant.
3) As the genetic diagram shows, there's a 50% chance of a child having the disorder if one parent has one D allele.



Embryos Can Be Screened for Genetic Disorders

- During IVF, embryos are fertilised in a laboratory, and then transferred into the woman's uterus. Before being transferred, it's possible to remove a cell from each embryo and analyse its genes.
- It's also possible to get DNA from an embryo in the uterus and test that for disorders.
- Many genetic disorders can be detected in this way, such as cystic fibrosis.
- For embryos produced by IVF — after screening, any with alleles linked to disorders would be destroyed.
- For embryos in the uterus — screening could lead to the decision to terminate the pregnancy.
- Embryonic screening is quite controversial because of the decisions it can lead to — there are lots of ethical, social and economic concerns. Here are some more arguments for and against screening:

Against Embryonic Screening	For Embryonic Screening
It implies that people with genetic problems are 'undesirable' — this could increase prejudice.	It will help to stop people suffering.
There may come a point where everyone wants to screen their embryos so they can pick particular physical or personality traits.	There are laws to stop it going too far. At the moment, parents cannot even select the sex of their baby (unless it's for health reasons).
Screening is expensive.	Treating disorders costs the Government (and the taxpayers) a lot of money.

There's more on the issues raised by scientific developments on p.170.

Embryo screening — it's a tricky one...

It's great to think that we might be able to stop people from having inherited disorders that cause suffering, but there are many concerns to think about too. Try writing a balanced argument for and against embryo screening.

The Work of Mendel

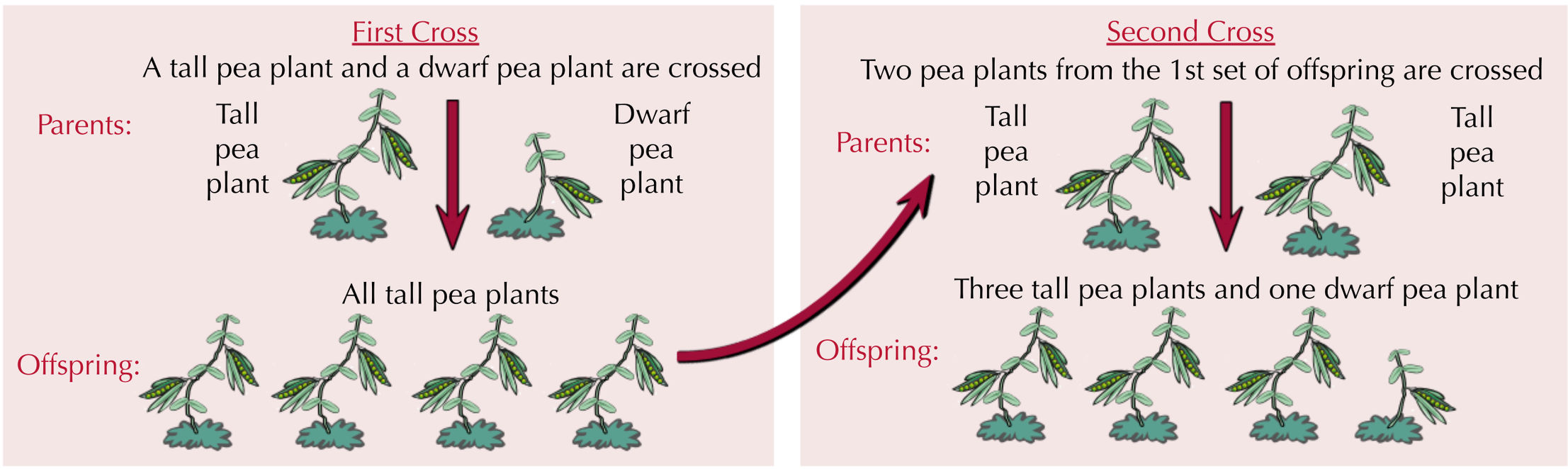
Quick Quiz

*People **didn't understand** the importance of what Mendel had discovered until after he'd died...*

Mendel Did Genetic Experiments with Pea Plants

Gregor Mendel was an Austrian monk who trained in mathematics and natural history. In his garden plot at the monastery in the mid 19th century, Mendel noted how characteristics in plants were passed on from one generation to the next. The results of his research were published in 1866 and eventually became the foundation of modern genetics.

The diagrams show two crosses for height in pea plants that Mendel carried out...



Mendel had shown that the height characteristic in pea plants was determined by separately inherited "hereditary units" passed on from each parent. The ratios of tall and dwarf plants in the offspring showed that the unit for tall plants, T, was dominant over the unit for dwarf plants, t.

Mendel Reached Three Important Conclusions

Mendel reached these three important conclusions about heredity in plants:

1) Characteristics in plants are determined by "hereditary units".
2) Hereditary units are passed on to offspring unchanged from both parents, one unit from each parent.
3) Hereditary units can be dominant or recessive — if an individual has both the dominant and the recessive unit for a characteristic, the dominant characteristic will be expressed.

It Took a While For People to Understand His Work

1) Mendel's work was new to the scientists of the day. They didn't have the background knowledge to properly understand his findings — they had no idea about genes, DNA and chromosomes.
2) It wasn't until after his death that people realised how significant his work was.
3) Using Mendel's work as a starting point, the observations of many different scientists have contributed to the understanding of genes that we have today. For example:

- In the late 1800s, scientists became familiar with chromosomes. They were able to observe how they behaved during cell division.
- Then in the early 20th century, scientists realised that there were striking similarities in the way that chromosomes and Mendel's "units" acted. Based on this, it was proposed that the "units" were found on the chromosomes. We now know these "units" as genes.
- In 1953, the structure of DNA was determined (see page 113). This allowed scientists to go on and find out exactly how genes work.

See p.169 for more on how theories develop.

Mendel looked at more than just the height of the pea plants...

For example, he also looked at the colour of the pea plant flowers and the shape of the peas. The results of his experiments contributed to what we understand about genetics and inheritance today. Three cheers for Mendel.

Warm-Up & Exam Questions

There's no better preparation for exam questions than doing... err... practice exam questions.
Hang on, what's this I see...

Warm-Up Questions

1) What are alleles?
2) Why won't someone heterozygous for the cystic fibrosis allele have the disorder?
3) What is polydactyly?
4) Outline what embryo screening involves.
5) Suggest why the importance of Mendel's work wasn't realised straight away.

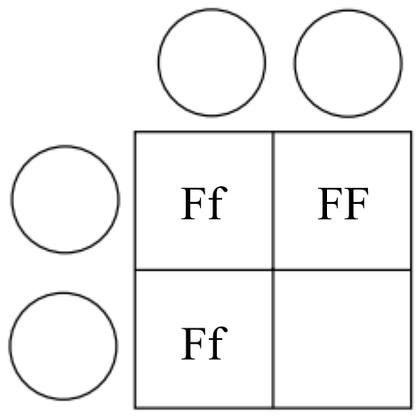
Exam Questions

1 Cystic fibrosis is a genetic disorder caused by recessive alleles. (GRADE 4-6 GRADE)

F = the normal allele **f** = the faulty allele that leads to cystic fibrosis

Figure 1 is an incomplete Punnett square showing the possible inheritance of cystic fibrosis from one couple.

Figure 1



1.1 Complete the Punnett square to show the missing offspring's genotype and the genotypes of the gametes.

[2 marks]

1.2 What proportion of the possible offspring are homozygous?

[1 mark]

1.3 State the phenotypes of the parents.

[2 marks]

2 Fruit flies usually have red eyes. However, there are a small number of white-eyed fruit flies. Having white eyes is a recessive characteristic. (GRADE 6-7 GRADE)

Two fruit flies with red eyes have the heterozygous genotype for this characteristic. They are crossed to produce offspring.

2.1 Draw a genetic diagram to show the possible phenotypes of the offspring.
Use **R** to represent the dominant allele and **r** to represent the recessive allele.

[3 marks]

2.2 State the probability that one of the fruit flies' offspring will have white eyes.

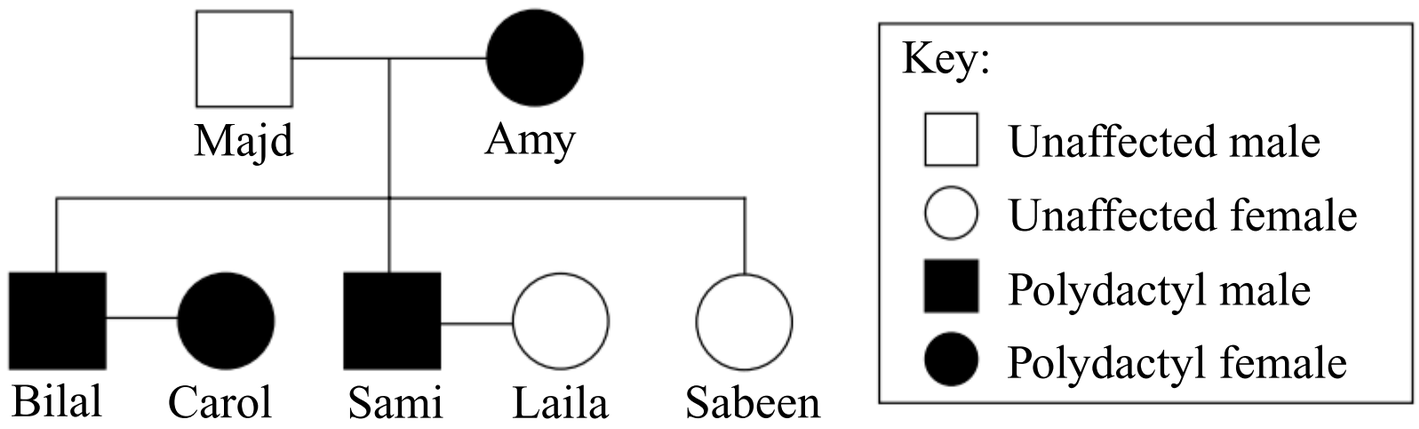
[1 mark]

Exam Questions

3 Polydactyly is a genetic disorder transmitted by the dominant allele **D**.
The corresponding recessive allele is **d**.
Figure 2 shows the family pedigree of a family with a history of polydactyly.

GRADE 6-7 GRADE

Figure 2



Using the information given above, state what Amy's genotype must be.
Explain your answer.

[2 marks]

4 Colour blindness in humans is caused by a recessive allele located on the X chromosome.
It is more common in men because men carry only one X chromosome.

GRADE 6-7 GRADE

A man who is colour blind has a child with a woman who does not have the recessive allele.

4.1 Draw a genetic diagram to show the possible sex of the offspring.

[3 marks]

4.2 The child is a boy. Explain why the boy will not be colour blind.

[1 mark]

4.3 State the probability that a daughter of this couple would be colour blind.

[1 mark]

5 Albinism is a condition characterised by the lack of pigment in the hair and skin.
It is caused by the recessive allele **a**. The dominant allele **A** results in normal pigmentation.

GRADE 7-9 GRADE

5.1 State the possible genotypes of a rabbit that shows no symptoms of albinism.

[1 mark]

A rabbit with albinism mated with a rabbit that showed no symptoms of the condition.
56% of the offspring had albinism.

5.2 Deduce the genotypes of the parent rabbits and the possible genotypes of their offspring.
Use a genetic diagram to explain your answer.

[3 marks]

5.3 From your genetic diagram, what percentage of offspring are likely to have albinism?

[1 mark]

5.4 Explain why the percentage of offspring which were born with albinism was not the same as that suggested by your genetic diagram.

[1 mark]

Revision Summary Test for Topic 6a

Hopefully you feel able to reproduce that lot. To make sure you've got it in hand, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

DNA, Genes and Mutations (p.112-114) ☑

1) Where in the cell is DNA found? What does it look like?
2) Where are genes located? What do they do?
3) What is a genome? Describe three ways that understanding the human genome is useful in science and medicine.
4) What are the three components of a nucleotide?
   Which bits form the 'backbone' of the DNA strands?
5) Which base pairs with which on the opposite strand of a DNA molecule? What's this called?
6) How many bases code for an amino acid?
7) Where are proteins made? Describe how the DNA code is used to make a protein.
8) Give an example of what non-coding DNA can do.
9) How does a newly assembled chain of amino acids become a protein with a specific function?
10) Give three examples of types of proteins with different functions, and say what their functions are.
11) Explain what a mutation is, and describe the possible severity of effects of mutations on proteins.
12) Name three different types of mutation. Bonus points for explaining what happens in each.

Reproduction and Meiosis (p.116-118) ☑

13) a) What is sexual reproduction? b) In sexual reproduction, what type of cell do the female and male produce and how are they different to normal body cells? c) Why is this difference important?
14) What is asexual reproduction? How is the outcome different from sexual reproduction?
15) How many cell divisions are there in meiosis? List the main steps in meiosis.
16) Give two reasons why each gamete is genetically different from the others.
17) What happens to the cell produced by gamete fusion?
18) Explain the advantages of sexual reproduction over asexual reproduction. And vice versa.
19) Give three examples of organisms that can reproduce by both methods and say how it works in each.

Genetic Diagrams, Inheritance and Mendel (p.119-124) ☑

20) How many pairs of chromosomes are there in ordinary human body cells?
21) What are the sex chromosomes in human females? How about males?
22) Draw a genetic diagram to show why there's the same probability of a baby being a boy or a girl.
23) Name two characteristics that are controlled by single genes.
24) What does it mean if you are homozygous for a trait? How about heterozygous?
25) When would an organism show a recessive characteristic? And a dominant characteristic?
26) What is meant by genotype? And phenotype?
27) List the possible allele combinations from a heterozygous 'Dd' crossed with another 'Dd'.
    What would the ratio of phenotypes in the offspring be?
28) Describe two inherited disorders and outline their inheritance.
29) What are the arguments against embryonic screening? How about the arguments for it?
30) What did Mendel conclude from his observations of pea plants?
31) Give three developments in genetics that showed the importance of Mendel's discoveries.

Quick Quiz

Variation

*You'll probably have noticed that not all people are **identical**. There are reasons for this.*

Organisms of the **Same Species** Have **Differences**

1) Different species look... well... different — my dog definitely doesn't look like a daisy.
2) But even organisms of the same species will usually look at least slightly different — e.g. in a room full of people you'll see different colour hair, individually shaped noses, etc.
3) These differences are called the variation within a species, and there are two types of variation — genetic variation and environmental variation.

Genetic Variation

- All plants and animals have characteristics that are similar to their parents' because an organism's characteristics are determined by the genes inherited from their parents (see p.112).
- These genes are passed on in sex cells (gametes), from which the offspring develop (see page 116).
- Most animals (and quite a few plants) get some genes from one parent and some from the other.
- This combining of genes from two parents causes genetic variation — no two of the species are genetically identical (other than identical twins).
- Some characteristics are determined only by genes (e.g. violet flower colour in plants). In animals these include eye colour, blood group and inherited disorders (e.g. haemophilia or cystic fibrosis).

Environmental Variation

- The environment, including the conditions that organisms live and grow in, also causes differences between members of the same species — this is called environmental variation.
- For example, a plant grown in plenty of sunlight would be luscious and green, but the same plant grown in darkness would grow tall and spindly and have yellow leaves. These are environmental variations.
- Environmental variation covers a wide range of differences — from losing your toes in a piranha attack, to speaking with a certain accent, to having yellow leaves, etc.

Genetic AND Environmental Variation

- Most characteristics (e.g. body weight, height, skin colour, condition of teeth, academic or athletic prowess, etc.) are determined by a mixture of genetic and environmental factors.
- For example, the maximum height that an animal or plant could grow to is determined by its genes. But whether it actually grows that tall depends on its environment (e.g. how much food it gets).

Mutations Introduce **Variation**

Mutations are covered in loads more detail on page 114.

1) Mutations are changes to the sequence of bases in DNA. These mutations happen continuously. As you saw on page 114, mutations can lead to changes in the protein that a gene codes for.

Most mutations have no effect on the protein the gene codes for, so most mutations have no effect on the organism's phenotype. Some have a small influence on phenotype and so only alter characteristics slightly. However, although it's very rare, mutations can result in a new phenotype being seen in a species.

2) If the environment changes, and the new phenotype makes an individual more suited to the new environment, it can become common throughout the species relatively quickly by natural selection — see the next page.

You can't blame all of your faults on your parents...

Although the genes that you inherit from your parents are really important at determining what characteristics you have, the conditions in which you live usually play a role too. However, this includes how your parents raised you...

Evolution

Quick Quiz

THEORY OF EVOLUTION: All of today's species have evolved from simple life forms that first started to develop over three billion years ago.

Only the **Fittest Survive**

Charles Darwin used the observations he made on a huge round-the-world trip, along with experiments, discussions and new knowledge of fossils and geology, to suggest the theory of evolution by natural selection.

Charles Darwin

1) Darwin knew that organisms in a species show wide variation in their characteristics (phenotypic variation). He also knew that organisms have to compete for limited resources in an ecosystem.
2) Darwin concluded that the organisms with the most suitable characteristics for the environment would be more successful competitors and would be more likely to survive. This idea is called the 'survival of the fittest'.
3) The successful organisms that survive are more likely to reproduce and pass on the genes for the characteristics that made them successful to their offspring.
4) The organisms that are less well adapted would be less likely to survive and reproduce, so they are less likely to pass on their genes to the next generation.
5) Over time, beneficial characteristics become more common in the population and the species changes — it evolves.

See page 169 for more on how theories develop.

- Darwin's theory wasn't perfect. Because the relevant scientific knowledge wasn't available at the time, he couldn't give a good explanation for why new characteristics appeared or exactly how individual organisms passed on beneficial adaptations to their offspring.
- We now know that phenotype is controlled by genes. New phenotypic variations arise because of genetic variants produced by mutations (changes in DNA — see page 114). Beneficial variations are passed on to future generations in the genes that parents contribute to their offspring.

The Development of a **New Species** is Called **Speciation**

1) Over a long period of time, the phenotype of organisms can change so much because of natural selection that a completely new species is formed. This is called speciation.
2) Speciation happens when populations of the same species change enough to become reproductively isolated — this means that they can't interbreed to produce fertile offspring.

There's more on speciation on p.137.

Extinction is When **No Individuals** of a Species **Remain**

The fossil record contains many species that don't exist any more. Species may become extinct because:

- The environment changes too quickly (e.g. destruction of habitat).
- A new predator kills them all (e.g. humans hunting them).
- A new disease kills them all.
- They can't compete with another (new) species for food.
- A catastrophic event happens that kills them all (e.g. a volcanic eruption or a collision with an asteroid).

Dodos are now extinct. Humans not only hunted them, but introduced other animals which ate all their eggs, and we destroyed the forest where they lived.

Natural selection — the fittest pass on their genes...

... and over time, the characteristics that improve their survival become more common in the population.

Q1 The sugary nectar in some orchid flowers is found at the end of a long tube behind the flower. There are moth species with long tongues that can reach the nectar. Explain how natural selection could have led to the moths developing long tongues. [4 marks]

Q1 Video Solution

Quick Quiz

More About Evolution

*There's a lot of **evidence** for the theory of evolution by natural selection.*
*But back in the day, poor Charles Darwin **didn't** have half as much evidence to convince people.*

Not Everyone Agreed with Darwin...

When Darwin proposed his theory in his book "On the Origin of Species" in 1859, his idea was very controversial for various reasons...

- It went against common religious beliefs about how life on Earth developed — it was the first plausible explanation for the existence of life on Earth without the need for a "Creator" (God).
- Darwin couldn't explain why these new, useful characteristics appeared or how they were passed on from individual organisms to their offspring. But then he didn't know anything about genes or mutations — they weren't discovered 'til 50 years after his theory was published.
- There wasn't enough evidence to convince many scientists, because not many other studies had been done into how organisms change over time.

...and Lamarck had Different Ideas

There were different scientific hypotheses about evolution around at the same time.

Jean-Baptiste Lamarck (1744-1829) argued that changes that an organism acquires during its lifetime will be passed on to its offspring — e.g. he thought that if a characteristic was used a lot by an organism, then it would become more developed during its lifetime, and the organism's offspring would inherit the acquired characteristic.

For example, using this theory, if a rabbit used its legs to run a lot (to escape predators), then its legs would get longer. The offspring of that rabbit would then be born with longer legs.

Scientists May Develop Different Hypotheses from Similar Observations

1) Scientists might develop different hypotheses because they have different beliefs (e.g. religious) or they have been influenced by different people (e.g. other scientists and their way of thinking)... or they just darn well think differently.

There's more about making hypotheses on p.169.

2) The only way to find out whose hypothesis is right is to find evidence to support or disprove each one.
3) For example, Lamarck and Darwin had different hypotheses to explain how evolution happens. In the end...

LAMARCK'S hypothesis was eventually REJECTED because...	DARWIN'S hypothesis is now ACCEPTED because...
...experiments didn't support his hypothesis. You can see it for yourself, e.g. if you dye a hamster's fur bright pink (not recommended), its offspring will still be born with the normal fur colour because the new characteristic won't have been passed on.	...the discovery of genetics supported his idea — it provided an explanation of how organisms born with beneficial characteristics can pass them on (i.e. via their genes).
	...other evidence was found by looking at fossils of different ages (the fossil record) — this allows you to see how changes in organisms developed slowly over time.
	...the relatively recent discovery of how bacteria are able to evolve to become resistant to antibiotics supports evolution by natural selection.

Here's to crazy new ideas...

Science is fuelled by new ideas — without them, we'd be pretty stuck. These new ideas need to be well tested though before they can become an accepted theory. If there's not enough evidence to support them, they'll be rejected.

Warm-Up & Exam Questions

You need to test your knowledge with a few Warm-Up Questions, followed by some Exam Questions...

Warm-Up Questions

1) Explain what is meant by environmental variation.
2) According to the theory of evolution, what have all of today's species evolved from?
3) Give three factors that can lead to a species becoming extinct.
4) Explain why Darwin's theory of evolution was considered controversial when it was first proposed.

Exam Questions

1 Helen and Stephanie are identical twins. This means they have identical DNA. GRADE 4-6 GRADE

1.1 Helen weighs 7 kg more than Stephanie.
Explain whether this is due to genes, environmental factors or both.

[2 marks]

1.2 Stephanie has a birthmark on her shoulder. Helen doesn't.
State whether birthmarks are caused by genes and explain your answer.

[1 mark]

2 Genetic variation in a population arises partly due to mutations. GRADE 6-7 GRADE

2.1 Give **one** cause of genetic variation in a population, other than mutations.

[1 mark]

2.2 Explain how mutations can increase variation in a species.

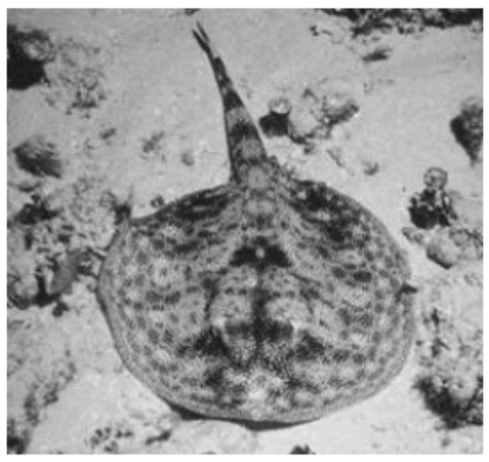
[3 marks]

2.3 Some mutations are neutral, having no effect on an organism, but some do have an impact.
Suggest how some mutations may be beneficial.

[2 marks]

3* **Figure 1** shows a type of stingray. The stingray's appearance mimics a flat rock. It spends most of its time on a rocky sea bed. GRADE 7-9 GRADE

Figure 1



Describe and explain how the stingray might have evolved to look like this.

[4 marks]

Quick Quiz

Selective Breeding

*'**Selective breeding**' sounds like it has the potential to be a tricky topic, but it's actually dead simple. You take the **best** plants or animals and breed them together to get the best possible **offspring**. That's it.*

Selective Breeding — Breed the Best to Get More of the Best

Selective breeding is when humans artificially select the plants or animals that are going to breed so that the genes for particular characteristics remain in the population. Organisms are selectively bred to develop features that are useful or attractive, for example:

- Animals that produce more meat or milk.
- Crops with disease resistance.
- Dogs with a good, gentle temperament.
- Decorative plants with big or unusual flowers.

This is the basic process involved in selective breeding:

1) From your existing stock, select the ones which have the characteristics you're after.
2) Breed them with each other.
3) Select the best of the offspring, and breed them together.
4) Continue this process over several generations, and the desirable trait gets stronger and stronger. Eventually, all the offspring will have the characteristic.

Selective breeding is also known as 'artificial selection'.

In agriculture (farming), selective breeding can be used to improve yields. E.g. to improve meat yields, a farmer could breed together the cows and bulls with the best characteristics for producing meat, e.g. large size. After doing this for several generations the farmer would get cows with a very high meat yield.

Selective breeding is nothing new — people have been doing it for thousands of years. It's how we ended up with edible crops from wild plants and how we got domesticated animals like cows and dogs.

The Main Drawback is a Reduction in the Gene Pool

1) The main problem with selective breeding is that it reduces the gene pool — the number of different alleles (forms of a gene) in a population. This is because the farmer keeps breeding from the "best" animals or plants — which are all closely related. This is known as inbreeding.

   Inbreeding can cause health problems because there's more chance of the organisms inheriting harmful genetic defects when the gene pool is limited. Some dog breeds are particularly susceptible to certain defects because of inbreeding — e.g. pugs often have breathing problems.

2) There can also be serious problems if a new disease appears, because there's not much variation in the population. All the stock are closely related to each other, so if one of them is going to be killed by a new disease, the others are also likely to succumb to it.

Selective breeding → Reduction in the number of different alleles (forms of a gene) → Less chance of any resistant alleles being present in the population

Best male + best female = best offspring

Different breeds of dog came from selective breeding. For example, somebody thought 'I really like this small, yappy wolf — I'll breed it with this other one'. After thousands of generations, we got poodles.

Q1 Explain how you could selectively breed for floppy ears in rabbits. [4 marks]

Q1 Video Solution

Genetic Engineering

Quick Quiz

*Genetic engineering is an area of science with **exciting possibilities**, but there might be **dangers** too...*

Genetic Engineering Transfers Genes Between Organisms

The basic idea of genetic engineering is to transfer a gene responsible for a desirable characteristic from one organism's genome into another organism, so that it also has the desired characteristic.

1) A useful gene is isolated (cut) from one organism's genome using enzymes, then inserted into a vector.
2) The vector is usually a virus or a bacterial plasmid (a fancy piece of circular DNA found in bacterial cells), depending on the type of organism that the gene is being transferred to.
3) When the vector is introduced to the target organism, the useful gene is inserted into its cell(s).

- Bacteria have been genetically modified to produce human insulin that can be used to treat diabetes.
- Genetically modified (GM) crops have had their genes modified, e.g. to improve the size and quality of their fruit, increase the crop yield or make them resistant to disease, insects and herbicides (chemicals used to kill weeds).
- Sheep have been genetically engineered to produce substances, like drugs, in their milk that can be used to treat human diseases.
- Scientists can use genetic modification treatments for inherited diseases caused by faulty genes, e.g. by inserting working genes into people with the disease. This is called gene therapy.

In some cases, the transfer of the gene is carried out when the organism receiving the gene is at an early stage of development (e.g. egg or embryo). This means that the organism develops with the characteristic coded for by the gene.

Genetic Engineering is a **Controversial Topic**

See p. 170 for more on ethical dilemmas.

1) Genetic engineering is an exciting area of science, which has the potential for solving many problems (e.g. treating diseases, more efficient food production etc.), but not everyone thinks it's a great idea.
2) There are worries about the long-term effects of genetic engineering — that changing an organism's genes might accidentally create unplanned problems, which could get passed on to future generations.

Pros of GM crops	Cons of GM crops
The characteristics chosen for GM crops can increase the yield, making more food.	Some people say that growing GM crops will affect the number of wild flowers (and so the population of insects) that live in and around the crops — reducing farmland biodiversity.
People living in developing nations often lack nutrients in their diets. GM crops could be engineered to contain the nutrient that's missing. For example, 'golden rice' is a GM rice crop that contains beta-carotene — lack of this substance causes blindness.	Not everyone is convinced that GM crops are safe and some people are concerned that we might not fully understand the effects of eating them on human health. E.g. people are worried they may develop allergies to the food — although there's probably no more risk for this than for eating usual foods.
GM crops are already being grown in some places, often without any problems.	A big concern is that transplanted genes may get out into the natural environment. For example, the herbicide resistance gene may be picked up by weeds, creating a new 'superweed' variety.

If only there was a gene to make revision easier...

At the end of the day, it's down to the Government to weigh up all the evidence before making a decision on how scientific knowledge is used. All that the scientists can do is make sure the Government has all the information it needs.

Quick Quiz

Cloning

__Plants__ and __animals__ can be __cloned__, and there are several ways of doing it. Read on...

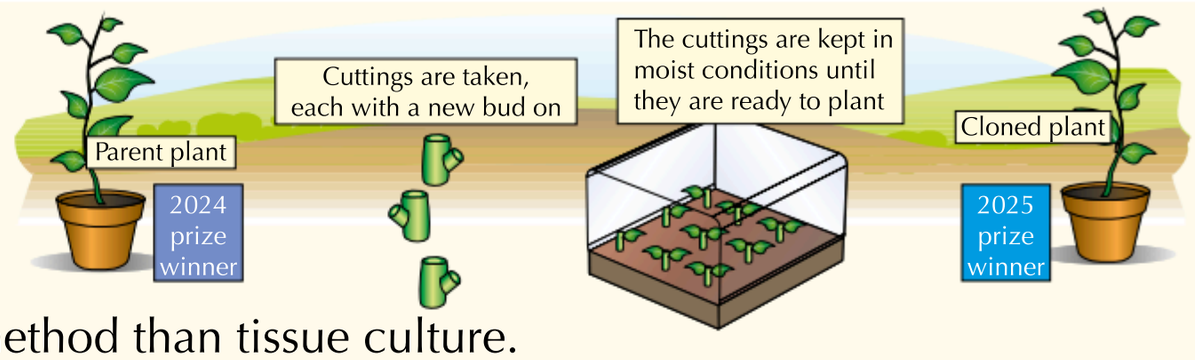
__Plants__ Can Be __Cloned__ by __Tissue Culture__ and from __Cuttings__

TISSUE CULTURE

This is where a few plant cells are put in a growth medium with hormones, and they grow into new plants — clones of the parent plant. These plants can be made very quickly, in very little space, and be grown all year. Tissue culture is used by scientists to preserve rare plants that are hard to reproduce naturally and by plant nurseries to produce lots of stock quickly.

CUTTINGS

- Gardeners can take cuttings from good parent plants, and then plant them to produce genetically identical copies (clones) of the parent plant.
- These plants can be produced quickly and cheaply. This is an older, simpler method than tissue culture.



You Can Make __Animal Clones__ Using __Embryo Transplants__ or __Adult Cells__

EMBRYO TRANSPLANTS

Farmers can produce cloned offspring from their best bull and cow — using embryo transplants.

1) Sperm cells are taken from a prize bull and egg cells are taken from a prize cow. The sperm are then used to artificially fertilise an egg cell. The embryo that develops is then split many times (to form clones) before any cells become specialised.
2) These cloned embryos can then be implanted into lots of other cows where they grow into baby calves (which will all be genetically identical to each other).
3) Hundreds of "ideal" offspring can be produced every year from the best bull and cow.

ADULT CELL CLONING

1) This involves taking an unfertilised egg cell and removing its nucleus. The nucleus is then removed from an adult body cell (e.g. skin cell) and is inserted into the 'empty' egg cell.
2) The egg cell is then stimulated by an electric shock — this makes it divide, just like a normal embryo.
3) When the embryo is a ball of cells, it's implanted into the uterus of an adult female. It grows into a genetically identical copy (clone) of the original adult body cell as it has the same genetic information.
4) This technique was used to create Dolly — the famous cloned sheep.

There are __Many Issues__ Surrounding __Cloning__

Benefits	Concerns
The study of animal clones could lead to greater understanding of the development of the embryo, and of ageing and age-related disorders.	You can get a "reduced gene pool" (see p.132). If a population are all closely related and a new disease appears, they could all be wiped out — there may be no allele in the population giving resistance to it.
Cloning quickly gets you lots of "ideal" offspring.	It's possible that cloned animals might not be as healthy as normal ones, e.g. Dolly the sheep had arthritis, which tends to occur in older sheep (but the jury's still out on if this was due to cloning).
Cloning could be used to help preserve endangered species.	Some people worry that humans might be cloned in the future. If it was allowed, any success may follow many unsuccessful attempts, e.g. children born severely disabled.

Cloning can be a pretty controversial topic...

...especially when it's to do with cloning animals (and especially humans). Is it healthy scientific progress, or are we trying to 'play God'? It's one of those questions science alone can't answer — ethics must be considered too (see p.170).

Warm-Up & Exam Questions

By doing these Warm-Up Questions, you'll soon find out if you've got the basic facts straight.

Warm-Up Questions

1) Why might a plant nursery use selective breeding?
2) What potential issues can selective breeding cause?
3) Name one useful product that humans have genetically modified bacteria to produce.
4) Briefly outline how plant clones can be produced using tissue culture.

Exam Questions

1 Organisms can be genetically modified.
This means an organism's genes can be altered to alter its characteristics.

GRADE 4-6 GRADE

1.1 Give **one** function of the enzymes used in genetic engineering.

[1 mark]

1.2 Suggest **one** useful way that plants can be genetically modified.

[1 mark]

1.3 Some people think that it is wrong to genetically modify crop plants.
Give **two** different objections that people might have.

[2 marks]

2 The characteristics of two varieties of wheat plants are shown in **Table 1**.

GRADE 6-7 GRADE

Table 1

Variety	Grain yield	Resistance to bad weather
Tall stems	High	Low
Dwarf stems	Low	High

Describe how selective breeding could be used to create a wheat plant with a high grain yield and high resistance to bad weather.

[3 marks]

3 In 1997 scientists at the Roslin Institute announced the birth of Dolly the sheep, the first mammal to be cloned using adult cell cloning.

GRADE 6-7 GRADE

3.1 Explain how a sheep can be cloned using adult cell cloning.

[4 marks]

3.2 Dolly was a female sheep. What would have been the gender of the sheep that Dolly was created from? Explain your answer.

[1 mark]

3.3 Give **one** potential problem with adult cell cloning.

[1 mark]

3.4 Give **one** reason why cloning may be beneficial in the field of medicine.

[1 mark]

Quick Quiz

Fossils

*Fossils are great. If they're **well-preserved**, you can see what oldy-worldy creatures **looked** like.*
*They also show how living things have **evolved**. Although we're not sure how life started in the first place...*

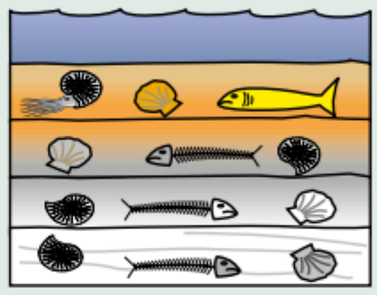
Fossils are the **Remains** of Plants and Animals

Fossils are the remains of organisms from millions of years ago, which are found in rocks.
They provide the evidence that organisms lived ages ago. Fossils can tell us a lot about how much or how little organisms have changed (evolved) over time. Fossils form in rocks in one of three ways:

From GRADUAL REPLACEMENT by minerals

- Things like teeth, shells, bones etc., which don't decay easily, can last a long time when buried.
- They're eventually replaced by minerals as they decay, forming a rock-like substance shaped like the original hard part.
- The surrounding sediments also turn to rock, but the fossil stays distinct inside the rock and eventually someone digs it up.

Most fossils are formed this way.



From CASTS and IMPRESSIONS

- Sometimes, fossils are formed when an organism is buried in a soft material like clay. The clay later hardens around it and the organism decays, leaving a cast of itself. An animal's burrow or a plant's roots (rootlet traces) can be preserved as casts.
- Things like footprints can also be pressed into these materials when soft, leaving an impression when it hardens.

From PRESERVATION in places where no decay happens

- In amber (a clear yellow 'stone' made from fossilised resin) and tar pits there's no oxygen or moisture so decay microbes can't survive.
- In glaciers it's too cold for the decay microbes to work.
- Peat bogs are too acidic for decay microbes.
  (A fully preserved man they named 'Pete Marsh' was found in a bog.)

But **No One Knows** How Life Began

Fossils show how much or how little different organisms have changed (evolved) as life has developed on Earth over millions of years. But where did the first living thing come from...

1) There are various hypotheses suggesting how life first came into being, but no one really knows.
2) Maybe the first life forms came into existence in a primordial swamp (or under the sea) here on Earth. Maybe simple organic molecules were brought to Earth on comets — these could have then become more complex organic molecules, and eventually very simple life forms.
3) These hypotheses can't be supported or disproved because there's a lack of good, valid evidence:

- Many early forms of life were soft-bodied, and soft tissue tends to decay away completely — so the fossil record is incomplete.
- Fossils that did form millions of years ago may have been destroyed by geological activity, e.g. the movement of tectonic plates may have crushed fossils already formed in the rock.

Don't get bogged down by all this information...

It's a bit mind-boggling really how fossils can still exist even millions of years after the organism died. It's thought that only a fraction of the species that have existed have been discovered as fossils — but we've learnt a lot from them.

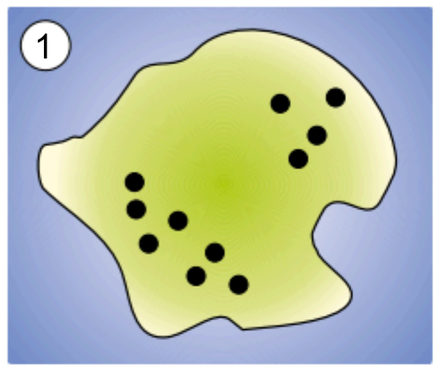
Speciation

Quick Quiz

Evolution *(see page 129) leads to the development of lots of* ***different species****.*

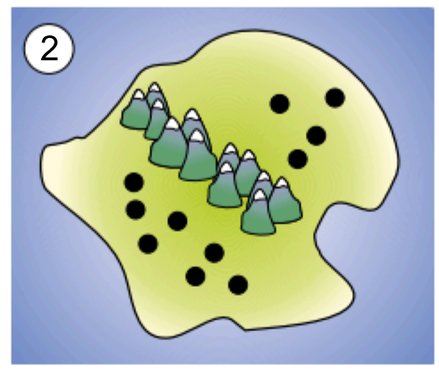
Speciation is the Development of a **New Species**

- A species is a group of similar organisms that can reproduce to give fertile offspring.
- Speciation occurs when populations of the same species become so different that they can no longer successfully interbreed to produce fertile offspring.
- Isolation (where populations of a species are separated) and natural selection lead to speciation.
  Here's how...

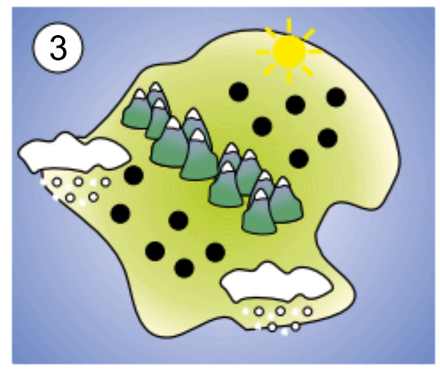


● = individual organism

Two populations of the same species.

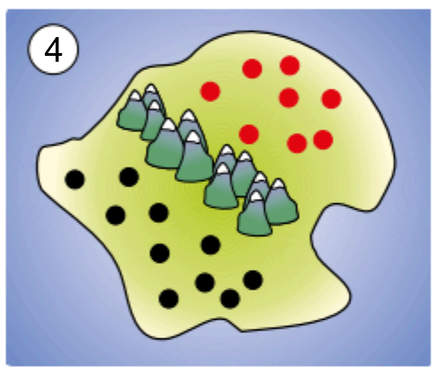


A physical barrier, e.g. mountain range, flood or earthquake, geographically isolates the two populations.



Conditions on each side of the barrier will be slightly different, e.g. different climates.

Video



Because the environment is different on each side, different characteristics will become more common in each population due to natural selection operating differently on the populations.

1) Each population shows genetic variation — they have a wide range of alleles.
2) In each population, individuals with characteristics that make them better adapted to their environment have a better chance of survival and so are more likely to breed successfully.
3) So the alleles that control the beneficial characteristics are more likely to be passed on to the next generation.

Eventually, individuals from the different populations will have changed so much that they won't be able to breed with one another to produce fertile offspring. The two groups will have become separate species.

Wallace Was a **Pioneer** of the **Theory of Speciation**

Alfred Russel Wallace was a scientist working at the same time as Charles Darwin.
He was one of the early scientists working on the idea of speciation.
His observations greatly contributed to how we understand speciation today.
Our current understanding developed as more evidence became available over time.

There's more on Darwin and natural selection on page 129.

Wallace...

1) ...independently came up with the idea of natural selection.
2) ...made lots of observations as he travelled the world, which provided evidence for the theory.
3) ...did lots of work on warning colours, which are used by some species (e.g. butterflies) to deter predators from eating them (an example of a beneficial characteristic that evolved by natural selection).

Wallace published work on natural selection together with Darwin in 1858. This then prompted Darwin to publish 'On the Origin of Species' in 1859.

New species can develop when populations become isolated

Remember — speciation happens if two or more populations of the same species change so much that they can no longer breed together to produce fertile offspring. It can occur when populations are separated from each other.

Antibiotic-Resistant Bacteria

Quick Quiz

*The discovery of **antibiotics**, like **penicillin**, was a huge benefit to medicine — suddenly bacterial infections that had often been fatal could be **cured**. But unfortunately they might **not** be a **permanent solution**.*

Bacteria can Evolve and Become Antibiotic-Resistant

1) Like all organisms, bacteria sometimes develop random mutations (see p.114) in their DNA. These can lead to changes in the bacteria's characteristics, e.g. being less affected by a particular antibiotic. This can lead to antibiotic-resistant strains forming as the gene for antibiotic resistance becomes more common in the population.
2) To make matters worse, because bacteria are so rapid at reproducing, they can evolve quite quickly.
3) For the bacterium, the ability to resist antibiotics is a big advantage. It's better able to survive, even in a host who's being treated to get rid of the infection, and so it lives for longer and reproduces many more times. This increases the population size of the antibiotic-resistant strain.

The gene for antibiotic resistance becomes more common in the population because of natural selection — see p.129.

- Antibiotic-resistant strains are a problem for people who become infected with these bacteria because they aren't immune to the new strain and there's no effective treatment. This means that the infection easily spreads between people. Sometimes drug companies can come up with a new antibiotic that's effective, but 'superbugs' that are resistant to most known antibiotics are becoming more common.
- MRSA is a relatively common 'superbug' that's really hard to get rid of. It often affects people in hospitals and can be fatal if it enters their bloodstream.

Antibiotic Resistance is Becoming More Common

1) For the last few decades, we've been able to deal with bacterial infections pretty easily using antibiotics. The death rate from infectious bacterial diseases (e.g. pneumonia) has fallen dramatically.
2) But the problem of antibiotic resistance is getting worse.
3) The more often antibiotics are used, the bigger the problem of antibiotic resistance becomes, so...
   - ...it's important that doctors only prescribe antibiotics when they really need to — not for non-serious conditions or infections caused by viruses.
   - ...it's important that you take all the antibiotics prescribed:

     Taking the full course makes sure that all the bacteria are destroyed, which means that there are none left to mutate and develop into antibiotic-resistant strains.
   - ...it's important that the use of antibiotics in farming is restricted.

     Antibiotics are given to animals to prevent them becoming ill and make them grow faster. This can lead to the development of antibiotic-resistant bacteria in the animals, which can spread to humans, e.g. during meat preparation and consumption. There is increasing concern about their overuse.

It's not that antibiotics actually cause resistance — they just create a situation where naturally resistant bacteria have an advantage and so increase in numbers.

The increase in antibiotic resistance has encouraged drug companies to work on developing new antibiotics that are effective against the resistant strains. Unfortunately, the rate of development is slow, which means we're unlikely to be able to keep up with the demand for new drugs as more antibiotic-resistant strains develop and spread. It's also a very costly process.

Superbugs — more horror movie than comic character...

Microorganisms that are resistant to all our drugs are pretty worrying. To combat them, we're going to have to change the ways we use antibiotics, as well as developing new ones. We'll need to get better at preventing infections too.

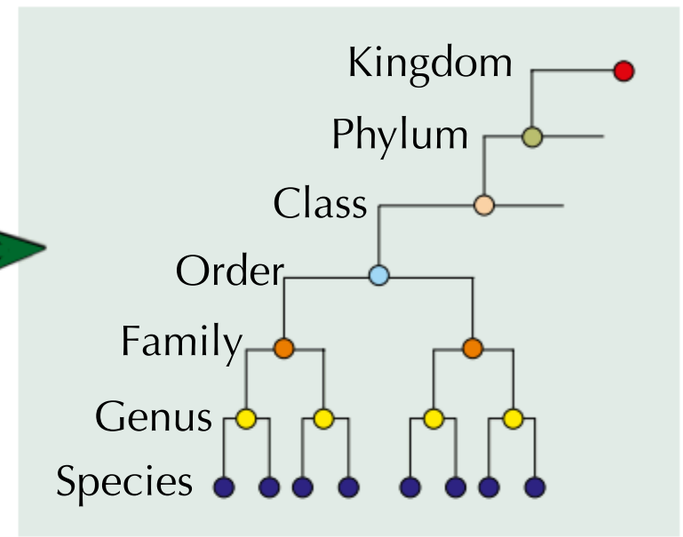
Classification

Quick Quiz

*It seems to be a basic human urge to want to **classify** things — that's the case in **biology** anyway...*

Classification is Organising **Living Organisms** into Groups

1) Traditionally, organisms have been classified according to a system first proposed in the 1700's by Carl Linnaeus, which groups living things according to their characteristics and the structures that make them up.
2) In this system (known as the Linnaean system), living things are first divided into kingdoms (e.g. the plant kingdom).
3) The kingdoms are then subdivided into smaller and smaller groups — phylum, class, order, family, genus, species.
4) As knowledge of the biochemical processes taking place inside organisms developed and microscopes improved (which allowed us to find out more about the internal structures of organisms), scientists put forward new models of classification.



5) In 1990, Carl Woese proposed the three-domain system. Using evidence gathered from new chemical analysis techniques (such as RNA sequence analysis) he found that in some cases, species thought to be closely related in traditional classification systems are in fact not as closely related as first thought.
6) In the three-domain system, organisms are first of all split into three large groups called domains:

ARCHAEA	Once thought to be primitive bacteria, they're actually a different type of prokaryotic cell. They were first found in extreme places such as hot springs and salt lakes.
BACTERIA	Contains true bacteria like *E. coli* and *Staphylococcus*. Although they often look similar to Archaea, there are lots of biochemical differences between them.
EUKARYOTA	Includes a broad range of organisms including fungi (page 54), plants, animals and protists (page 54).

7) These are then subdivided into smaller groups — kingdom, phylum, class, order, family, genus, species.

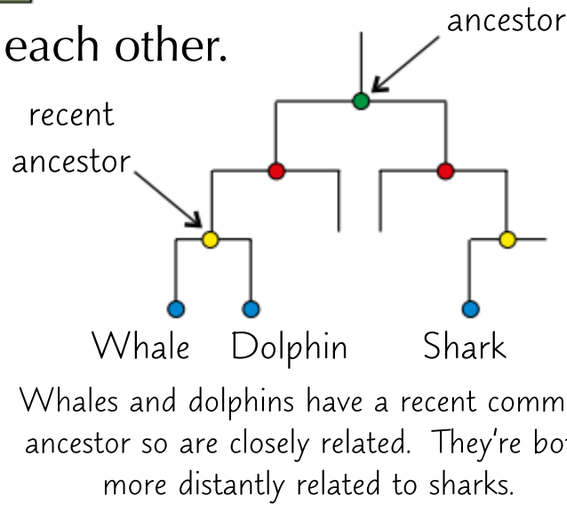
Organisms Are **Named** According to the **Binomial System**

- In the binomial system, every organism is given its own two-part Latin name.
- The first part refers to the genus that the organism belongs to. This gives you information on the organism's ancestry. The second part refers to the species.
- The binomial system is used worldwide and means that scientists in different countries or who speak different languages all refer to a particular species by the same name — avoiding potential confusion.

E.g. humans are known as *Homo sapiens*. '*Homo*' is the genus and '*sapiens*' is the species.

Evolutionary Trees Show **Evolutionary Relationships**

- Evolutionary trees show how scientists think different species are related to each other.
- They show common ancestors and relationships between species. The more recent the common ancestor, the more closely related the two species — and the more characteristics they're likely to share.
- Scientists analyse lots of different types of data to work out evolutionary relationships. For living organisms, they use the current classification data (e.g. DNA analysis and structural similarities). For extinct species, they use information from the fossil record (see page 136).



Whales and dolphins have a recent common ancestor so are closely related. They're both more distantly related to sharks.

Advances in technology have changed how we classify things...

...although some organisms have kept their misleading names, e.g. the honey badger is not a badger.

Q1 The evolutionary tree on the right shows the relationship between four species, A-D. Which two species shown in the tree are the most closely related? [1 mark]

Q1 Video Solution

Warm-Up & Exam Questions

The end of the topic is in sight now — just a few more questions to check you've been paying attention.

Warm-Up Questions

1) Suggest what makes low-oxygen environments suitable for the formation of fossils.
2) Why might natural selection work differently on two isolated populations of a species?
3) Suggest a situation where antibiotics could be prescribed inappropriately.
4) Explain why it's important that people take the full course of antibiotics they are prescribed.
5) What species does the Eurasian beaver, *Castor fiber*, belong to?

Exam Questions

1 **Figure 1** shows a section of an evolutionary tree. GRADE (4-6) GRADE

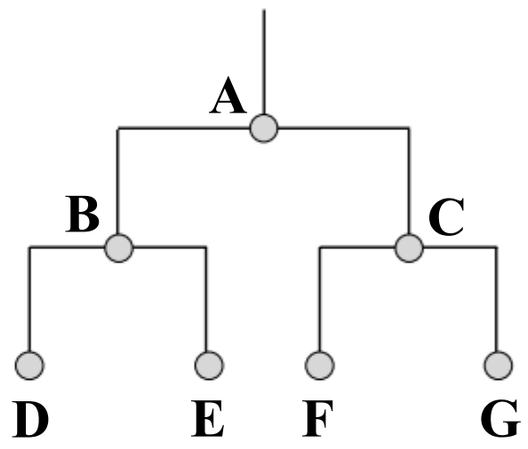
1.1 Which species is the most recent common ancestor of Species **F** and Species **G**?

[1 mark]

1.2 Would you expect Species **D** to look similar to Species **E**? Give a reason for your answer.

[1 mark]

Figure 1



2 *Staphylococcus aureus* (*S. aureus*) is a common bacterium that is found on the skin and mucous membranes. It can cause serious illness in people with weakened immune systems. Some strains of the bacterium have developed resistance to the antibiotic meticillin, and are known as MRSA. GRADE (6-7) GRADE

2.1 **Table 1** shows the different stages that led to *S. aureus* becoming resistant to meticillin. The stages are shown in the wrong order. Write out the letters **A-D** to show the stages in the correct order.

Table 1

	Stage
A	The gene for meticillin resistance became more common in the population over time, eventually giving a large proportion of the population resistance.
B	Individual bacteria with the mutated genes were more likely to survive and reproduce in a host being treated with meticillin.
C	Random mutations in the DNA of *S. aureus* led to it being less affected by meticillin.
D	The gene for meticillin resistance was passed on to lots of offspring, which also survived and reproduced.

[2 marks]

2.2 Suggest why MRSA is a more serious problem in hospitals than in wider society.

[2 marks]

2.3 Explain why antibiotic resistance in bacteria is a concern for humans.

[1 mark]

2.4 Suggest why the National Health Service (NHS) is trying to reduce the use of antibiotics.

[3 marks]

Revision Summary Test for Topic 6b

Time to see how your knowledge of Topic 6b has evolved. To check there are no gaps, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Variation and Evolution (p.128-130)

1) What is genetic variation? Why do species have genetic variation?
2) How does the environment cause variation? Include an example in your answer.
3) What is most variation a result of? Use an example to explain your answer.
4) Describe how mutations lead to variation, including the frequency of changes to phenotype.
5) What is the theory of evolution? Describe Darwin's theory of evolution by natural selection.
6) What is extinction? List five possible reasons for extinction.
7) Give three reasons why Darwin's theory wasn't immediately accepted.
8) What was Lamarck's theory of evolution? Why was it eventually rejected?
9) Give three pieces of evidence, identified since Darwin's time, that meant his theory was accepted*.

Selective Breeding, Genetic Engineering and Cloning (p.132-134)

10) Give four examples of characteristics that organisms might be selectively bred for.
11) How do you go about selectively breeding for a particular characteristic?
12) What's the main problem with selective breeding? Give two reasons why this is a problem.
13) What is genetic engineering? Give a brief outline of the process.
14) Why do we do this whole genetic engineering malarky? Four examples, please.
15) Give some advantages of GM crops. Why are some people against GM crops?
16) Give two methods of cloning plants. Describe how they're done, and what they're used for.
17) Describe two different ways of cloning animals.
18) Give three issues raised by cloning. Why is cloning still actively researched?

Fossils, Speciation and Antibiotic-Resistant Bacteria (p.136-138)

19) What are fossils? Briefly describe three ways fossils can form.
20) Why can scientists not say for certain how life on Earth began?
21) How is a species defined? What is meant by speciation?
22) Give three possible causes of geographical isolation.
    Why does geographical isolation lead to speciation?
23) Who was Alfred Russell Wallace? What is he famous for?
24) How can bacteria become antibiotic-resistant?
25) Why is antibiotic-resistance a huge deal? And why is it still a problem despite further research?
26) How can the development of antibiotic-resistance be minimised?

Classification (p.139)

27) What is the order of subdivisions in the Linnaean system, starting from kingdom?
28) What has resulted in new models of classification being put forward?
29) What are the three domains in the three-domain system of classification? Briefly describe each one.
30) What is the binomial system of naming organisms? Why is it used?
31) What is an evolutionary tree? How do scientists work them out?

*eventually

Quick Quiz

Competition

Ecology *is all about* ***organisms*** *and the* ***environment*** *they live in, and how the two* ***interact****. Simple.*

First **Learn** Some Words to Help You **Understand Ecology**...

Habitat	The place where an organism lives.
Population	All the organisms of one species living in a habitat.
Community	The populations of different species living in a habitat.
Abiotic factors	Non-living factors of the environment, e.g. temperature.
Biotic factors	Living factors of the environment, e.g. food.
Ecosystem	The interaction of a community of living organisms (biotic) with the non-living (abiotic) parts of their environment.

There's more about abiotic and biotic factors on the next page.

Organisms **Compete** with Each Other and **Depend** on Each Other

Organisms need things from their environment and from other organisms in order to survive and reproduce:

- Plants need light and space, as well as water and mineral ions (nutrients) from the soil.
- Animals need space (territory), food, water and mates.

Organisms compete with other species (and members of their own species) for the same resources.

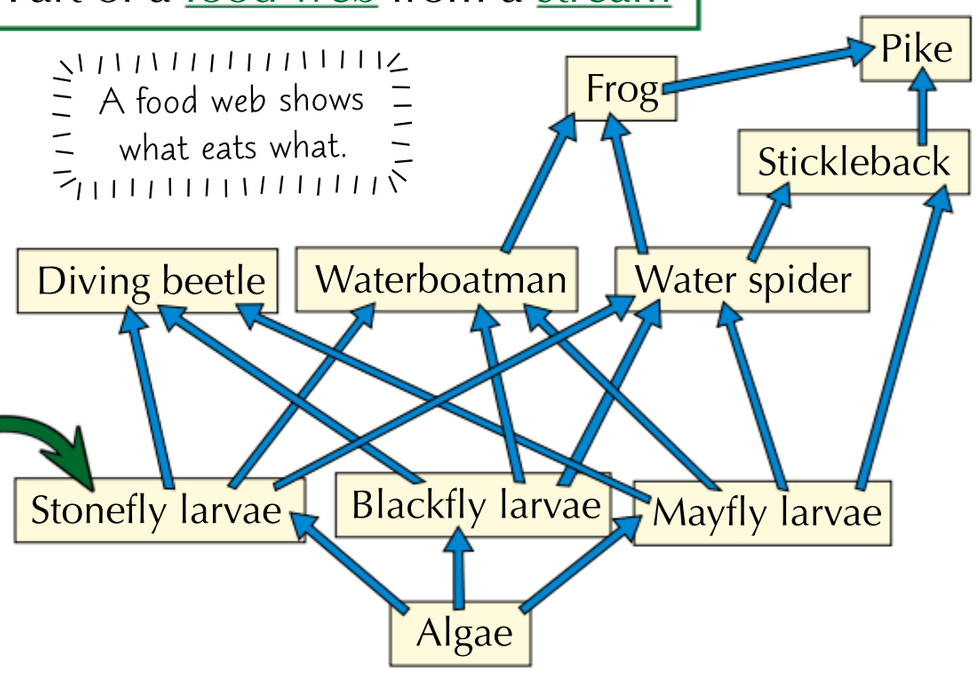
In a community, each species depends on other species for things such as food, shelter, pollination and seed dispersal — this is called interdependence.

The interdependence of all the living things in an ecosystem means that any major change in the ecosystem (e.g. one species being removed) can have far-reaching effects. For example:

Stonefly larvae are particularly sensitive to pollution. Suppose pollution killed them in this stream. Here are some effects this might have on other organisms in the food web:

Part of a food web from a stream

A food web shows what eats what.



Organism	Effect of loss of stonefly larvae	Effect on population
Blackfly/ mayfly larvae	Less competition for algae	Increase
	More likely to be eaten by predators	Decrease
Water spider	Less food	Decrease
Stickleback	Less food (if water spider or mayfly larvae numbers decrease)	Decrease

Food webs are very complex and these effects are difficult to predict accurately.

In some communities, all the species and environmental factors are in balance so that the population sizes are roughly constant (they may go up and down in cycles — see p.145). These are called stable communities. Stable communities include tropical rainforests and ancient oak woodlands.

Survival — the prize for being a winner

Make sure you know what things organisms compete for in an ecosystem. Then try these...

Q1 Give three things plants compete for in an ecosystem. [3 marks]

Q2 Using the food web above, suggest what might happen to the frog population if the stickleback population decreased. [2 marks]

Q2 Video Solution

Abiotic and Biotic Factors

Quick Quiz

*The environment in which organisms live **changes** all the time. The things that change are either **abiotic** (non-living) or **biotic** (living) factors. These can have a big **effect** on a community...*

Abiotic Factors Can **Vary** in an **Ecosystem**...

Light intensity can vary within an ecosystem because of shading caused by, e.g. tree cover.

Abiotic factors are non-living factors. For example:

- Moisture level
- Light intensity
- Temperature
- Carbon dioxide level (for plants)
- Wind intensity and direction
- Oxygen level (for aquatic animals)
- Soil pH and mineral content

A change in the environment could be an increase or decrease in an abiotic factor, e.g. an increase in temperature. These changes can affect the size of populations in a community. This means they can also affect the population sizes of other organisms that depend on them (see previous page).

For example:

A decrease in light intensity, temperature or level of carbon dioxide could decrease the rate of photosynthesis in a plant species (see p.72). This could affect plant growth and cause a decrease in the population size.

A decrease in the mineral content of the soil (e.g. a lack of nitrates) could cause nutrient deficiencies (see p.68). This could also affect plant growth and cause a decrease in the population size.

Animals depend on plants for food, so a decrease in a plant population could affect the animal species in a community.

...and So Can **Biotic Factors**

Biotic factors are living factors. For example:

- New predators arriving
- Competition — one species may outcompete another so that numbers are too low to breed
- New pathogens
- Availability of food

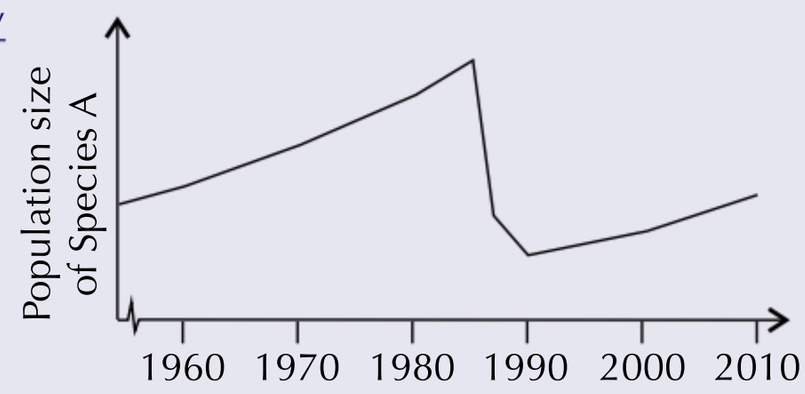
Changes in biotic factors can affect the size of populations in a community, which can have knock-on effects because of interdependence (see previous page).

For example:

A new predator could cause a decrease in the prey population. There's more about predator-prey populations on p.145.

Red and grey squirrels live in the same habitat and eat the same food. Grey squirrels outcompete the red squirrels — so the population of red squirrels is decreasing.

This graph shows the effect of a new pathogen on Species A. The population size of species A was increasing up until 1985, when it decreased rapidly until 1990 — suggesting that 1985 was the year that the new pathogen arrived. The population started to rise again after 1990.



A = not, biotic = living, so abiotic means non-living

Some human activities can affect the abiotic factors of ecosystems — see pages 157 to 159 for more. As you can see from this page, this can affect some organisms directly and other organisms indirectly. It's all a delicate balance...

Quick Quiz

Adaptations

*Life exists in so many **different environments** because the **organisms** that live in them have **adapted** to them.*

Adaptations Allow Organisms to Survive

Organisms, including microorganisms, are adapted to live in different environmental conditions. The features or characteristics that allow them to do this are called adaptations. Adaptations can be:

1) Structural

These are features of an organism's body structure — such as shape or colour. For example:

Arctic animals like the Arctic fox have white fur so they're camouflaged against the snow. This helps them avoid predators and sneak up on prey.

Animals that live in cold places (like whales) have a thick layer of blubber (fat) and a low surface area to volume ratio to help them retain heat.

Animals that live in hot places (like camels) have a thin layer of fat and a large surface area to volume ratio to help them lose heat.

2) Behavioural

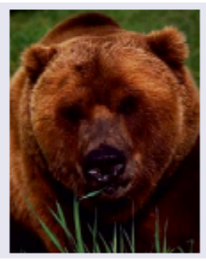
These are ways that organisms behave. Many species (e.g. swallows) migrate to warmer climates during the winter to avoid the problems of living in cold conditions.

3) Functional

These are things that go on inside an organism's body that can be related to processes like reproduction and metabolism (all the chemical reactions happening in the body). For example:

Desert animals conserve water by producing very little sweat and small amounts of concentrated urine.

Brown bears hibernate over winter. They lower their metabolism which conserves energy, so they don't have to hunt when there's not much food about.



Extremophiles Have a Huge Variety of Adaptations...

...so that they can live in a wide range of environments:

Some organisms are known as extremophiles — they're adapted to live in very extreme conditions. For example...

- some can live at high temperatures (e.g. in super hot volcanic vents),
- others can live at high salt concentration (e.g. very salty lakes),
- others can live at high pressure (e.g. bacteria in deep sea vents).

Organisms can adapt to the most extreme environments

Q1 Video Solution

An exam question might ask how an organism is adapted to its environment. Look at its characteristics (e.g. colour/shape) and the conditions it has to cope with (e.g. temperature) and you'll be sorted.

Q1 The diagram on the right shows a penguin. Penguins live in the cold, icy environment of the Antarctic. They swim in the sea to hunt for fish to eat. Some penguins also huddle together in large groups to keep warm.

a) What type of adaptation is being described when penguins 'huddle together'? [1 mark]

b) Explain one structural adaptation a penguin has to its environment. [2 marks]

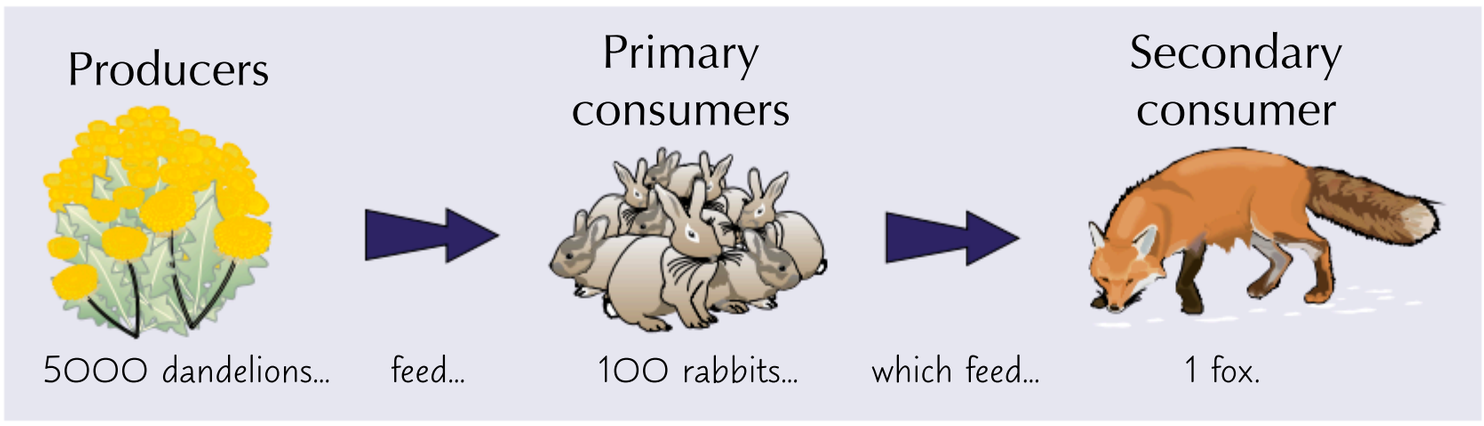
Food Chains

Quick Quiz

*You might remember **food webs** from page 142. Well, you'll be pleased to hear that **food chains** are a similar idea — except that you only really show one part of a food web in a food chain. Read on to find out more...*

Food Chains Show **What's Eaten by What** in an Ecosystem

Here's an example of a food chain:



Consumers are organisms that eat other organisms. 'Primary' means 'first', so primary consumers are the first consumers in a food chain. Secondary consumers are second and tertiary consumers are third.

1) Food chains always start with a producer.

Producers make (produce) their own food using energy from the Sun. They're usually green plants or algae — they make glucose by photosynthesis (see page 71). Some of this glucose is used to make other biological molecules in the plant. These biological molecules are the plant's biomass — the mass of living material. Biomass can be thought of as energy stored in a plant.

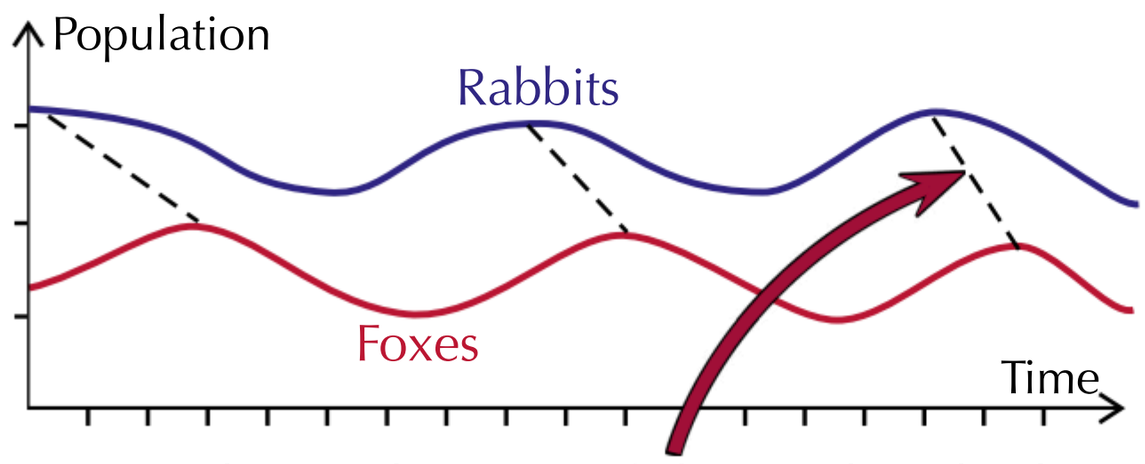
2) Energy is transferred through living organisms in an ecosystem when organisms eat other organisms.
3) Producers are eaten by primary consumers. Primary consumers are then eaten by secondary consumers and secondary consumers are eaten by tertiary consumers.

Populations of **Prey** and **Predators** Go in **Cycles**

Consumers that hunt and kill other animals are called predators, and their prey are what they eat. In a stable community (see page 142):

1) The population of any species is usually limited by the amount of food available.
2) If the population of the prey increases, then so will the population of the predators.
3) However, as the population of predators increases, the number of prey will decrease.

E.g.:



More grass means more rabbits.
More rabbits means more foxes.
But more foxes means fewer rabbits.
Fewer rabbits will mean fewer foxes again.
This up and down pattern continues...

Predator-prey cycles are always out of phase with each other. This is because it takes a while for one population to respond to changes in the other population. E.g. when the number of rabbits goes up, the number of foxes doesn't increase immediately because it takes time for them to reproduce.

A food chain shows part of a food web

Q1 Video Solution

There's usually a plant at the start of a food chain, but sometimes organisms like algae can be there too.

Q1 Look at the following food chain for a particular area: grass → grasshopper → rat → snake

a) Name the producer in the food chain. [1 mark]
b) How many consumers are there in the food chain? [1 mark]
c) Name the primary consumer in the food chain. [1 mark]
d) All the rats in the area are killed.
Explain two effects that this could have on the food chain. [4 marks]

Warm-Up & Exam Questions

This ecology topic's a busy one — so make sure you've really got these first few pages stuck in your head before moving on and learning the rest. These questions should help you out.

Warm-Up Questions

1) What is the correct scientific term for all the different species in a habitat?
2) What is an ecosystem?
3) Give four examples of abiotic factors that could affect a land-based plant species.
4) What is meant by a structural adaptation?
5) What is meant by a producer in a food chain?

Exam Questions

1 **Figure 1** shows a food chain for a particular area:

GRADE 4-6

Figure 1

algae → shrimp → sea turtle → tiger shark

1.1 What term is used to describe the tiger shark in **Figure 1**? Tick **one** box.

☐ producer ☐ primary consumer ☐ secondary consumer ☐ tertiary consumer

[1 mark]

1.2 Explain the importance of the algae in **Figure 1**.

[2 marks]

2 The Harris's antelope squirrel lives in hot deserts in parts of the USA and Mexico. It has grey fur, small ears and does not sweat. It has sharp claws which enable it to dig burrows, in which it lives. Above ground, during the hottest parts of the day it often holds its large tail over its head, or lies in the shade with its limbs spread out wide.

GRADE 6-7

2.1 Give **one** functional adaptation the Harris's antelope squirrel has to its environment.

[1 mark]

2.2 Explain **one** behavioural adaptation the Harris's antelope squirrel has to its environment.

[2 marks]

3 Cutthroat trout are present in lakes in Yellowstone National Park. In the last few decades, lake trout have been introduced to the lakes. However, lake trout have emerged as predators of the cutthroat trout.

GRADE 6-7

3.1 Explain how the introduction of the lake trout might cause the population sizes of both species of fish to fluctuate over time.

[5 marks]

3.2 Give **two** other biotic factors that could affect the size of the cutthroat trout population.

[2 marks]

PRACTICAL

Using Quadrats

Quick Quiz

*Studying **ecology** gives you the chance to get your hands **dirty** and look at some real organisms in the **wild**.*

Organisms Live in Different Places Because The Environment Varies

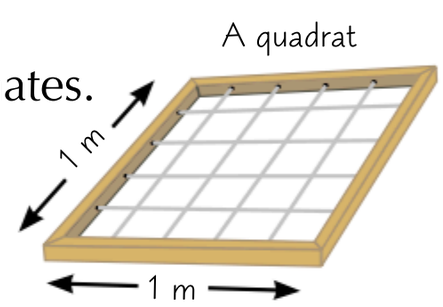
1) The distribution of an organism is where it's found within a habitat, e.g. within in a playing field.
2) Distribution is affected by environmental factors (see p.143). An organism might be more common in one area than another due to differences in environmental factors between the two areas. E.g. in the playing field, daisies may be more common in the open where there's more light than under trees.
3) There are a couple of ways to study the distribution of an organism. You can:
   - measure how common an organism is in two sample areas (e.g. with quadrats) and compare them.
   - study how distribution changes across an area, e.g. by placing quadrats along a transect (p.148).

Both of these methods give quantitative data (numbers) about the distribution.

Use Quadrats to Study The Distribution of Small Organisms

A quadrat is a square frame enclosing a known area, e.g. 1 m^2. Here's a way to compare how common an organism is in two sample areas (e.g. shady and sunny spots in that playing field) using a 1 m^2 quadrat:

1) Place a 1 m^2 quadrat on the ground at a random point within the first sample area.
   E.g. divide the area into a grid and use a random number generator to pick coordinates.
2) Count all the organisms within the quadrat.
3) Repeat steps 1 and 2 as many times as you can.
4) Work out the mean number of organisms per quadrat within the first sample area.



EXAMPLE

Navina counted the number of daisies in 7 quadrats within her first sample area and recorded the following results: 18, 20, 22, 23, 23, 23, 25

MATHS SKILLS

Here the mean is: $\frac{\text{total number of organisms}}{\text{number of quadrats}} = \frac{154}{7}$ = 22 daisies per quadrat

5) Repeat steps 1 to 4 in the second sample area, then compare the two means.
   E.g. you might find 2 daisies per m^2 in the shade, and 22 daisies per m^2 (lots more) in the open field.

You Can Also Estimate the Population Size of an Organism in One Area

EXAMPLE

Students used quadrats, each with an area of 0.5 m^2, to randomly sample daisies on an open field. The students found a mean of 10.5 daisies per quadrat. The field had an area of 800 m^2. Estimate the population of daisies on the field.

MATHS SKILLS

The population size of an organism is sometimes called its abundance.

1) Work out the mean number of daisies per m^2 by dividing 1 m^2 by the area of the quadrat and then multiplying by the number of daisies in one quadrat. — 1 ÷ 0.5 = 2; 2 × 10.5 = 21 daisies per m^2
2) Then multiply the mean by the total area (in m^2) of the habitat. — 800 × 21 = 16 800 daisies on the open field

If your quadrat has an area of 1 m^2, the mean number of organisms per m^2 is just the same as the mean number per quadrat.

Don't try to study elephants using a quadrat

Make sure you put your quadrat down in a random place before you start counting. Just don't lose it.

Q1 A 1200 m^2 field was randomly sampled for buttercups using a quadrat with an area of 0.25 m^2. A mean of 0.75 buttercups were found per quadrat. Estimate the total population of buttercups. [2 marks]

Q1 Video Solution

Using Transects

PRACTICAL

Quick Quiz

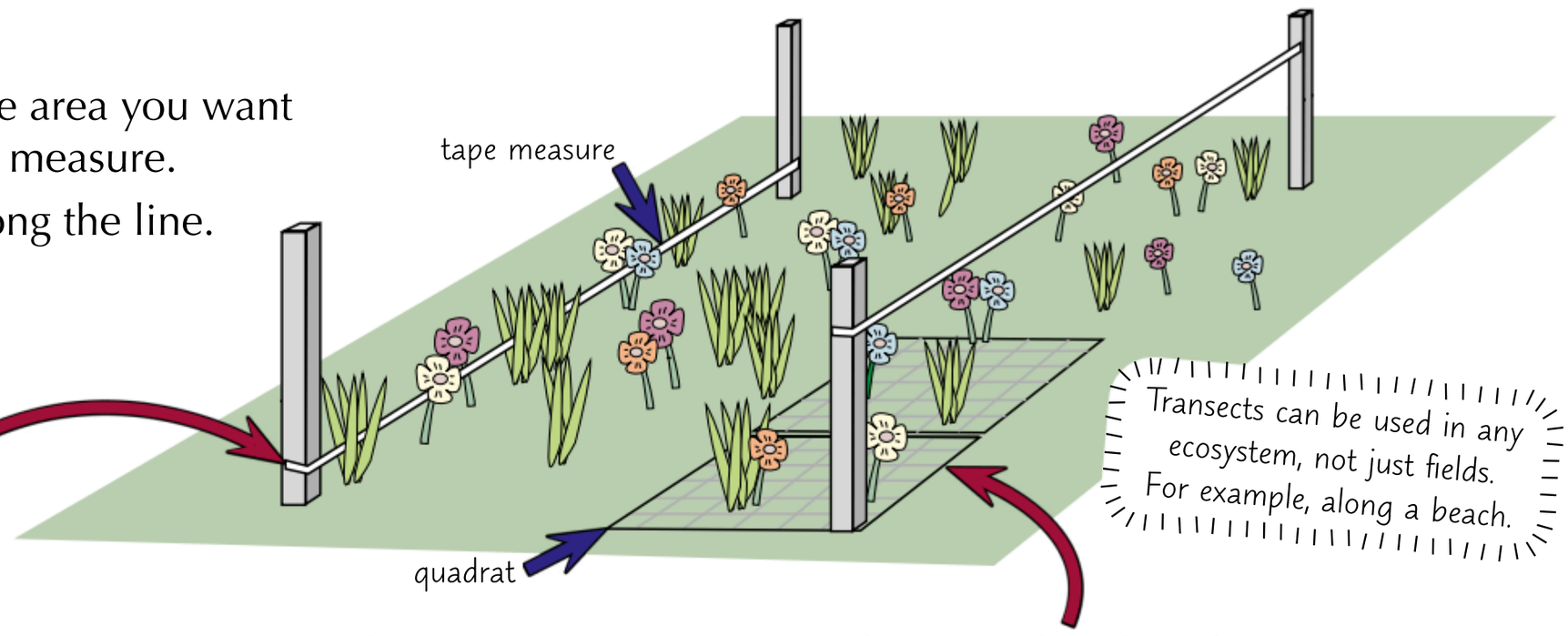
*So, now you think you've learnt all about **distribution**. Well **hold on** — there's more **ecology fun** to be had.*

Use **Transects** to **Study** The **Distribution** of Organisms **Along a Line**

You can use lines called transects to help find out how organisms (like plants) are distributed across an area — e.g. if an organism becomes more or less common as you move from a hedge into the middle of a field. Here's what to do:

1) Mark out a line in the area you want to study using a tape measure.
2) Then collect data along the line. You can do this by...

... counting all the organisms you're interested in that touch the line, or...



... using quadrats (see previous page). These can be placed next to each other along the line or at intervals, for example, every 2 m.

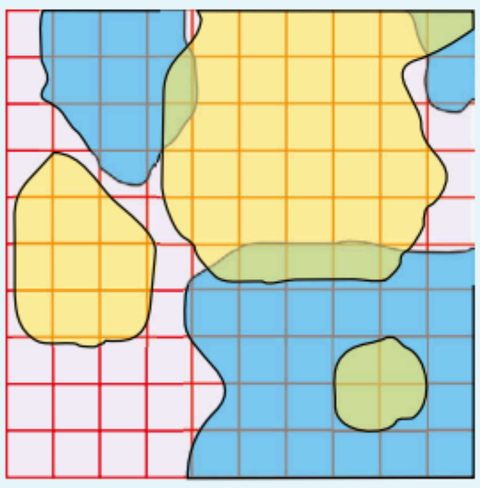
You Can **Estimate** the **Percentage Cover** of a **Quadrat**

It might be difficult to count all the individual organisms in a quadrat (e.g. if they're grass). Instead, you can calculate the percentage cover. This means estimating the percentage area of the quadrat covered by a particular type of organism, e.g. by counting the number of little squares covered by the organisms.

EXAMPLE

Some students were measuring the distribution of organisms from one corner of a school playing field to another, using quadrats placed at regular intervals along a transect. The picture below shows one of the quadrats. Calculate the percentage cover of each organism, A and B.

MATHS SKILLS



You count a square if it's more than half covered.

1) Count the number of squares covered by organism A.
2) Make this into a percentage — divide the number of squares covered by the organism by the total number of squares in the quadrat (100), then multiply the result by 100.
3) Do the same for organism B.

Type A = 42 squares

(42/100) × 100
= 0.42 × 100 = 42%

Type B = 47 squares

(47/100) × 100
= 0.47 × 100 = 47%

You don't need fancy kit to study the distribution of organisms

So if you want to measure the distribution of an organism across an area, you could use a transect. You can use them alone or along with quadrats. Using percentage cover instead of number of organisms is a good way of studying the distribution of plants, as there may be too many to count, or it may be hard to see individuals.

Environmental Change & The Water Cycle

Quick Quiz

*Time to learn why an organism's distribution may **change** over time. Then it's on to the good old **water cycle**.*

Environmental Changes Affect The **Distribution** of **Organisms**

A change in an organisms's distribution means a change in where an organism lives. Environmental changes that can affect distribution include:

1) A change in the AVAILABILITY of WATER. For example:

The distribution of some animal and plant species in the tropics changes between the wet and the dry seasons — i.e. the times of year where there is more or less rainfall, and so more or less water available. E.g. each year in Africa, large numbers of giant wildebeest migrate, moving north and then back south as the rainfall patterns change.

2) A change in TEMPERATURE. For example:

The distribution of bird species in Germany is changing because of a rise in average temperature. E.g. the European bee-eater bird is a Mediterranean species but it's now present in parts of Germany.

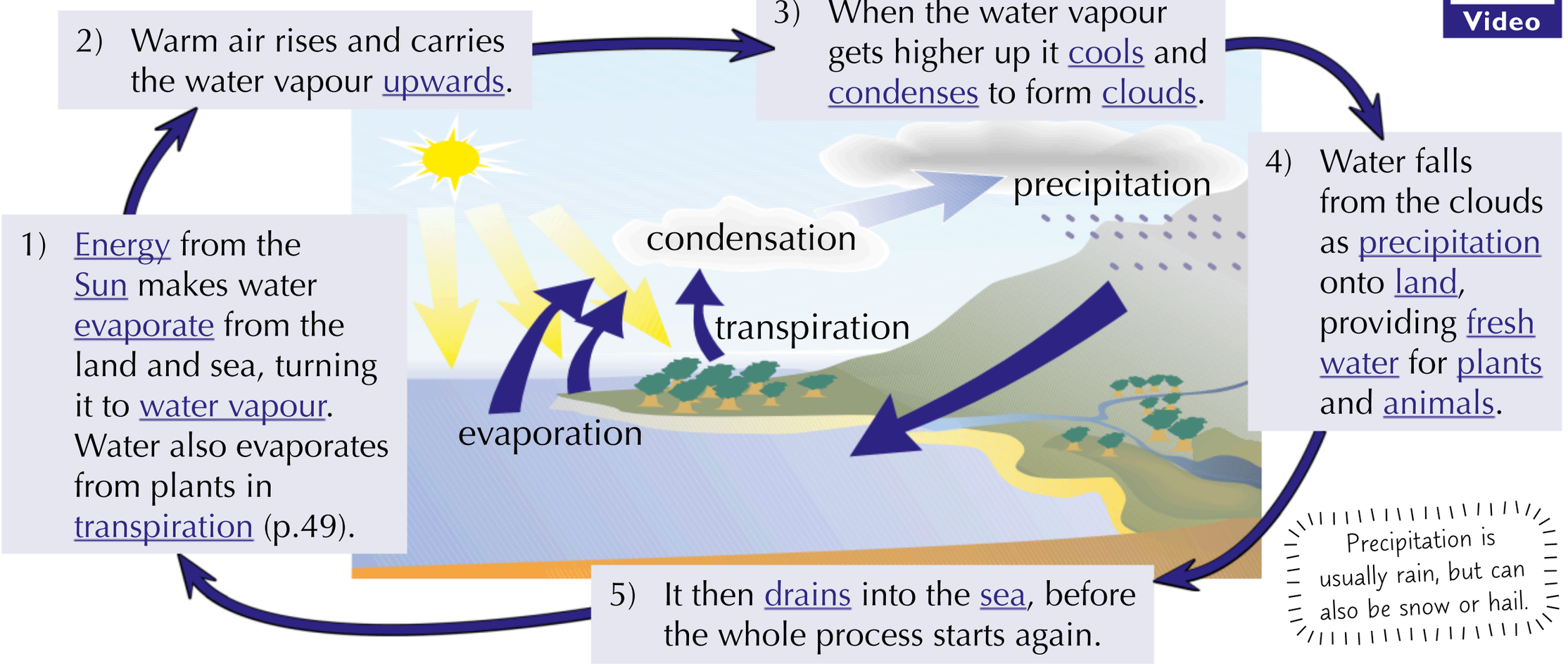
3) A change in the composition of ATMOSPHERIC GASES. For example:

The distribution of some species changes in areas where there is more air pollution. E.g. some species of lichen can't grow in areas where sulfur dioxide is given out by certain industrial processes.

These environmental changes can be caused by seasonal factors, geographical factors or human interaction. For example, the rise in average temperature is due to global warming, which has been caused by human activity (see pages 157-158).

The **Water Cycle** Means Water is **Endlessly Recycled**

Video



Evaporation, transpiration, condensation, precipitation

The water cycle is pretty straightforward, so there's absolutely no excuse not to learn it inside out. The most important thing to remember is that it's a cycle — a continuous process with no beginning or end. Water that falls to the ground as rain (or any other kind of precipitation for that matter) will eventually end up back in the clouds again and again.

The Carbon Cycle

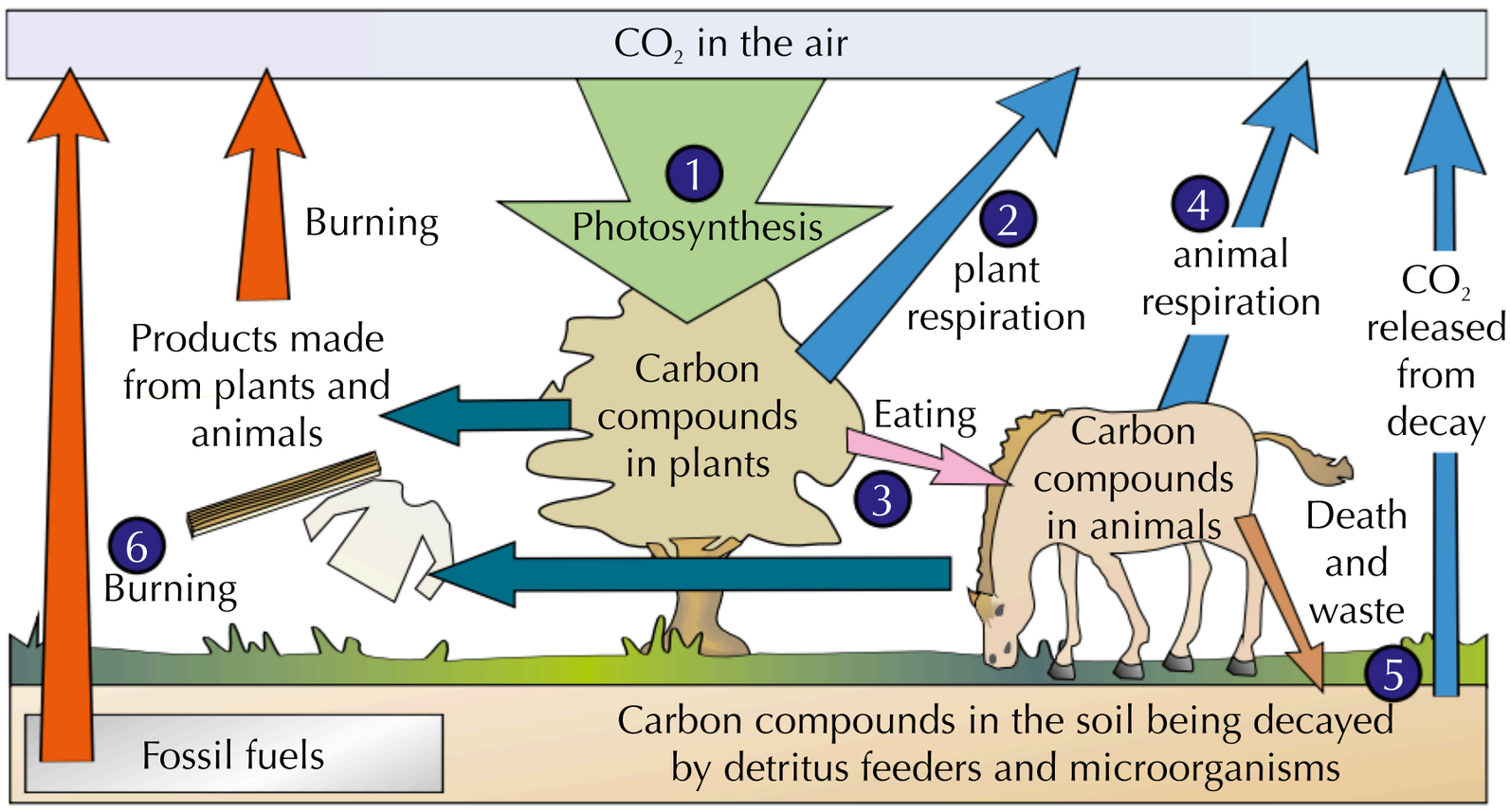
Quick Quiz

Recycling *— a buzzword for us but old school for nature.* ***Nutrients*** *in ecosystems are being* ***recycled*** *non-stop.*

Elements are **Cycled** Back to the Start of the Food Chain by **Decay**

1) Living things are made of materials they take from the world around them. E.g. plants turn elements like carbon, oxygen, hydrogen and nitrogen from the soil and the air into the complex compounds (carbohydrates, proteins and fats) that make up living organisms. These get passed up the food chain.
2) These materials are returned to the environment in waste products, or when the organisms die and decay.
3) Materials decay because they're broken down (digested) by microorganisms. Decay puts the stuff that plants need to grow (e.g. mineral ions) back into the soil.
4) In a stable community, the materials that are taken out of the soil and used by plants etc. are balanced by those that are put back in. There's a constant cycle happening.

The **Constant Cycling** of **Carbon** is called the **Carbon Cycle**



Starting at the top with the big green arrow...

1. CO_2 is removed from the atmosphere by green plants and algae during photosynthesis. The carbon is used to make glucose, which can be turned into carbohydrates, fats and proteins that are used in the growth of the plants and algae.
2. When plants and algae respire, some carbon is returned to the atmosphere as CO_2.
3. When plants and algae are eaten by animals, some carbon becomes part of the fats and proteins in their bodies. The carbon then moves through the food chain.
4. When animals respire, some carbon is returned to the atmosphere as CO_2.
5. When plants, algae and animals die, other animals (called detritus feeders) and microorganisms feed on their remains. When these organisms respire, CO_2 is returned to the atmosphere. Animals also produce waste that is broken down by detritus feeders and microorganisms.
6. Fossil fuels are made of decayed plant and animal matter. The combustion (burning) of wood and fossil fuels also releases CO_2 back into the air.

So the carbon is constantly being cycled — from the air, through food chains (via plants, algae and animals, and detritus feeders and microorganisms) and eventually back out into the air again.

What goes around comes around...

Carbon is very important for living things — it's the basis for all the organic molecules in our bodies, as well as the molecules that make up this book... and your exam paper. That's right — you've got carbon to thank for revision.

Warm-Up & Exam Questions

You can't just stare at these pages and expect all of the information to go in. Especially the practical pages with the maths examples. Do these questions to see how well you really know the stuff.

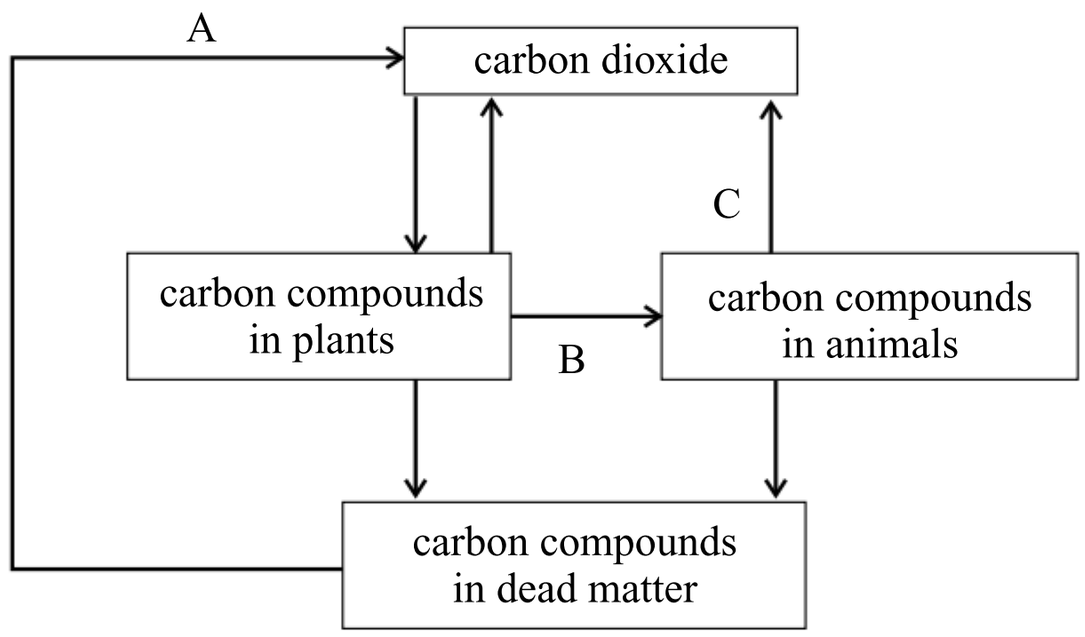
Warm-Up Questions

1) A student counted the number of daisies in five quadrats and recorded the following results: 14, 11, 11, 12, 12. What is the mean number of daises per quadrat?
2) What is a transect?
3) Give three reasons why environmental factors in an area may change.
4) What is the role of microorganisms in the carbon cycle?
5) How is the carbon in fossil fuels returned to the atmosphere?

Exam Questions

1 **Figure 1** shows a simplified version of the carbon cycle. GRADE 4-6

Figure 1



1.1 Name the process that is occurring at stage **B**.

[1 mark]

1.2 Name the process that is occurring at stage **C**.

[1 mark]

1.3 Explain how carbon is released from dead matter in the soil (stage **A**).

[2 marks]

1.4 Name the only process in the carbon cycle that removes carbon dioxide from the air.

[1 mark]

2 The water cycle describes the constant movement of water molecules on the Earth. GRADE 6-7

2.1 In the water cycle, how does water move from plants into the air?

[1 mark]

2.2 Describe the shortest route a water molecule could take through the water cycle to get from an ocean to a garden pond.

[4 marks]

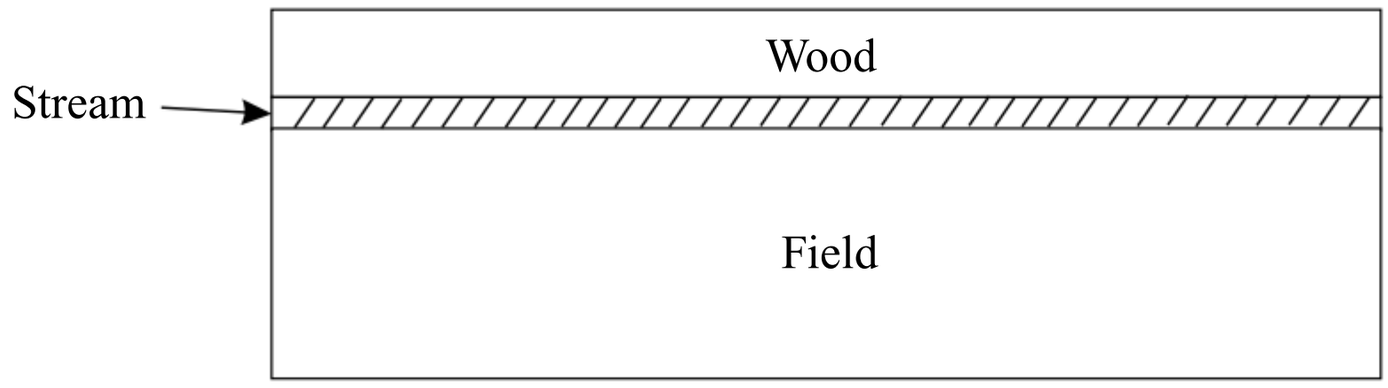
Exam Questions

PRACTICAL

3 Some students investigated the distribution of poppies across a field next to a wood. A sketch of the area is shown in **Figure 2**.

GRADE 6-7 GRADE

Figure 2



The students' results are shown in **Table 1**.

Table 1

Number of poppies per m^2	5	9	14	19	26
Distance from wood (m)	2	4	6	8	10

3.1 Describe how the students could have used quadrats to obtain the results in **Table 1**.

[3 marks]

3.2 Describe the trend in the results in **Table 1**.

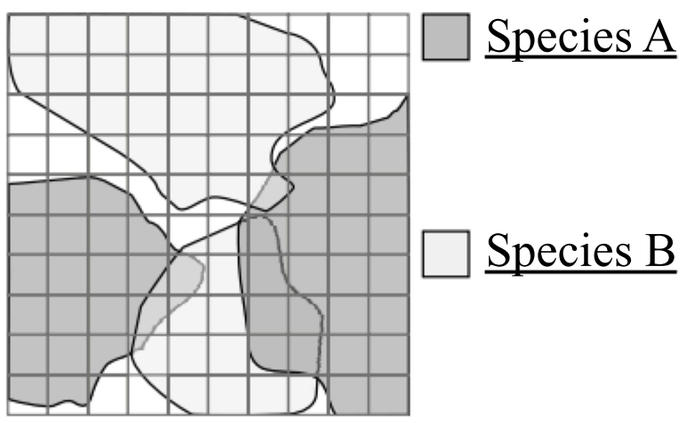
[1 mark]

3.3 The students suggest that a change in light intensity across the field may affect the distribution of the poppies. Give **two** additional factors which may also affect the distribution of the poppies.

[2 marks]

The students decided to investigate the distribution of two grass species in the field as well. **Figure 3** shows a sketch that the students made of the two grass species in one of their quadrats.

Figure 3



3.4 Use **Figure 3** to estimate the percentage cover of species A and species B in the quadrat.

[2 marks]

3.5 Give **one** reason why the students may have decided to estimate the percentage cover of the two grasses.

[1 mark]

4* An area of woodland is cleared to build a house. The tree trunks are taken away to be dried and used in furniture making. The smaller tree branches are used for firewood. The green plants are piled up at the edge of the building site.

GRADE 7-9 GRADE

Describe how carbon stored in the vegetation that has been cleared could be returned to the atmosphere.

[6 marks]

Decay

Quick Quiz

*Microorganisms **break down** plant and animal material and waste to get **energy**. This process of **decay** allows the material contained within the organisms to be released, so that it can cycle through the ecosystem.*

The **Rate of Decay** is Affected by **Several Factors**

Microorganisms such as bacteria and fungi, as well as detritus feeders (p.150) are the critters responsible for decay. They're a bit like Goldilocks — things have to be just right for them to work at their best.

1) TEMPERATURE — Warmer temperatures make things decompose (decay) quicker because they increase the rate that the enzymes (see p.27) involved in decomposition work at. If it's too hot though, decomposition slows down or stops because the enzymes are destroyed and the organisms die. Really cold temperatures slow the rate of decomposition too.

As well as these three factors, the more microorganisms and detritus feeders there are, the faster decomposition happens.

2) OXYGEN AVAILABILITY — Many organisms need oxygen to respire, so can't survive without it. The microorganisms involved in anaerobic decay (see below) don't need oxygen though.

3) WATER AVAILABILITY — Decay is faster in moist environments because the organisms involved in decay need water to carry out biological processes.

Compost is decomposed organic matter (e.g. food waste) that is used as a natural fertiliser for crops and garden plants. Farmers and gardeners use the three factors above to try to provide the ideal conditions for quick decay to make compost.

Biogas is Made by **Anaerobic Decay** of **Waste Material**

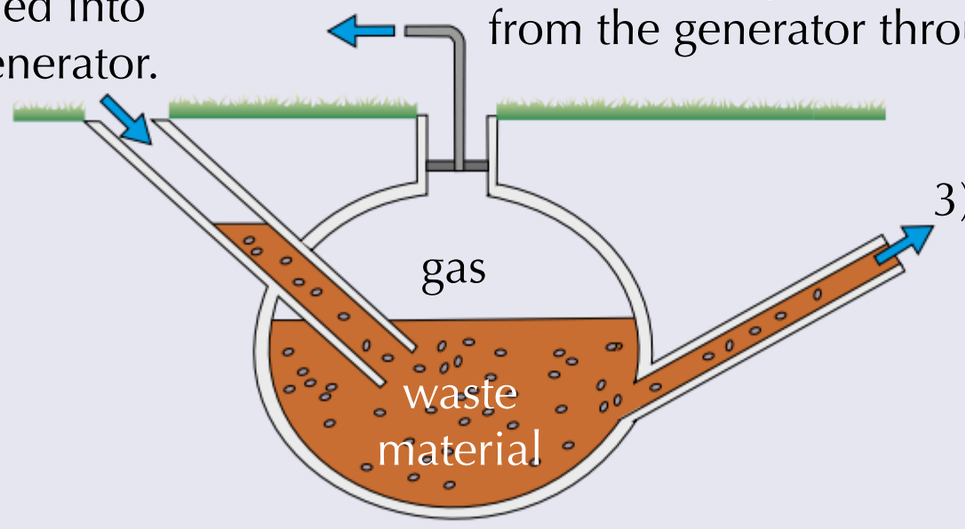
Biogas is mainly made up of methane, which can be burned as a fuel.

Lots of different microorganisms are used to produce biogas. They decay plant and animal waste anaerobically (without oxygen). This type of decay produces methane gas.

Biogas is made in a simple fermenter called a digester or generator.
These need to be kept at a constant temperature to keep the microorganisms respiring.

1) Waste material is added into the generator.

2) Biogas is produced as the waste material decays. The biogas is removed from the generator through an outlet.



3) The digested material is removed (and can be used as fertiliser).

Biogas can't be stored as a liquid (it needs too high a pressure), so it has to be used straight away — for heating, cooking, lighting, or to power a turbine to generate electricity.

It might not be pretty, but it is pretty useful...

Biogas generators are quite neat really. The decomposition of waste material is a process that naturally happens anyway — by carrying out the decomposition in generators we can harness the biogas produced and use it to produce energy for different processes. We can also use the process to clean our sewage and other wastes to avoid pollution. Win-win.

Quick Quiz

Investigating Decay

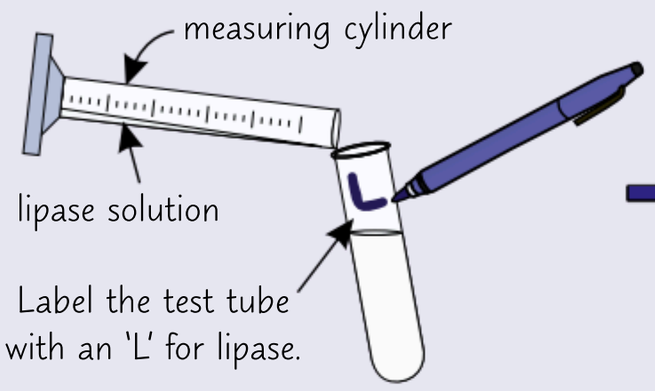
PRACTICAL

*It's time for a **practical**. I hope you're not looking forward to a **milky brew**, because this might just put you off.*

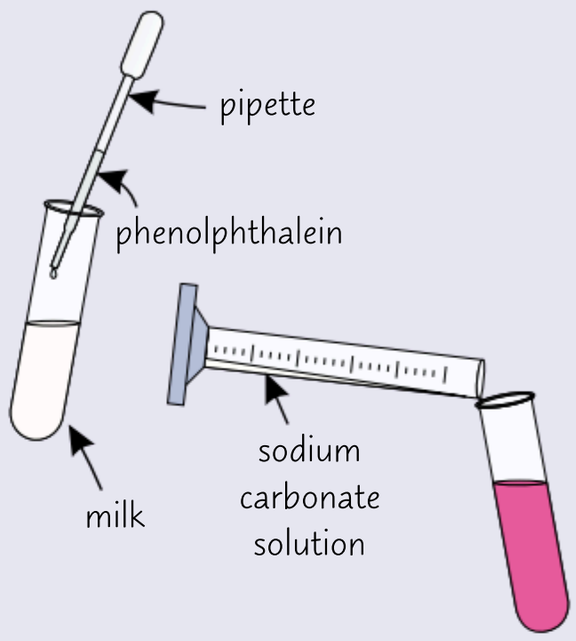
You Can **Investigate** the **Effect** of **Temperature** on the **Rate of Decay**

One way to investigate decay is by observing the action of the enzyme lipase on a sample of milk that has been made alkaline. When the lipase breaks the milk down, the pH of the milk decreases. You can detect this pH change using an indicator dye called phenolphthalein — it has a pink colour when the pH is around 10, but becomes colourless when the pH falls below 8.3.

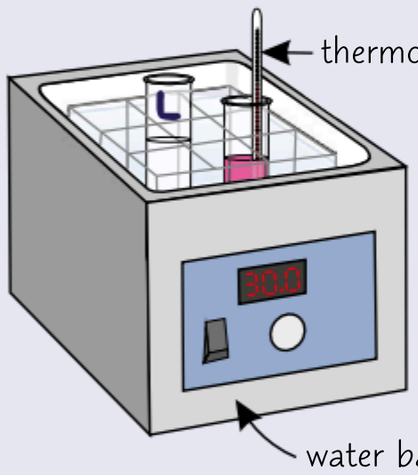
1 Measure out 5 cm^3 of lipase solution into a test tube.



2 Measure out 5 cm^3 of milk into a different test tube. Add 5 drops of phenolphthalein indicator. Then measure out 7 cm^3 of sodium carbonate solution and add it to the tube. This makes the solution in the tube alkaline, so it should turn pink.

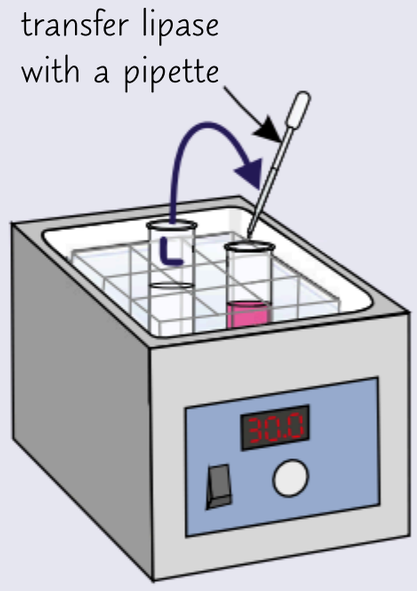


3 Put both tubes into a water bath set to 30 °C and leave them to reach that temperature.

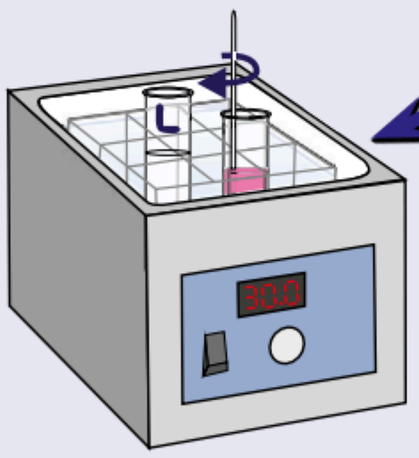


You could stick a thermometer into the milk tube to check this.

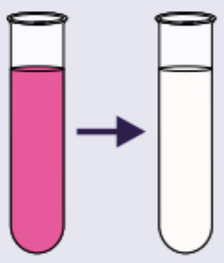
4 Once the tubes have reached 30 °C, use a calibrated dropping pipette (a dropping pipette with a scale) to put 1 cm^3 of the lipase solution into the milk tube and start a stopwatch straight away.



5 Stir the contents of the tube with a glass rod. The enzyme will start to decompose the milk.



6 As soon as the solution loses its pink colour, stop the stopwatch and record how long the colour change took in a table.



Repeat the experiment at a range of different temperatures (e.g. 10 °C, 20 °C, 40 °C, 50 °C). Carry out the experiment three times at each temperature, then calculate the mean time taken for the colour change to occur at each temperature. (You can make a water bath capable of temperatures below room temperature by adding ice cubes to a beaker of water and measuring the temperature with a thermometer.)

You can use your results to calculate the rate of decay with this formula. If you've measured time in seconds, the units will be s^{-1}, since rate is given per unit of time.

$$\text{Rate} = \frac{1000}{\text{time}}$$

You've met this formula before, on page 28.

It's important to start and stop the stopwatch quickly

Q1 Video Solution

You might want to grab a friend for this one — one person can stir whilst the other keeps the time.

Q1 A student was investigating the effect of temperature on the rate of decay of milk by measuring the time taken for a change in the pH. The results are shown in the table.

a) Calculate the rate of decay at 20 °C in s^{-1}. Give your answer to 3 significant figures. [1 mark]

b) Describe the trend in rate of decay shown by the results. [1 mark]

Temperature (°C)	10	20	30	40
Time taken (s)	600	480	350	240

Warm-Up & Exam Questions

This material isn't very glamorous, but we need processes like decay so that nutrients and other useful materials don't get locked up in dead organisms. Here are some questions to test your knowledge.

Warm-Up Questions

1) Give two types of organism that are responsible for the decomposition of materials.
2) What is compost?
3) What colour is phenolphthalein indicator in a solution with a pH of 10?
4) Suggest what you could add to milk to turn it alkaline.

Exam Questions

1 Biogas can be made in a generator through the decomposition of food waste. GRADE 4-6 GRADE

1.1 Name the main gas found in biogas.

[1 mark]

1.2 Explain why it is important that little or no oxygen is present in a biogas generator.

[2 marks]

1.3 Suggest **one** possible use of biogas produced in this way.

[1 mark]

PRACTICAL

2 Some students were investigating the effect of temperature on the decay of milk by lipase. They added phenolphthalein to alkaline milk, turning it pink, followed by the lipase. They then started a stopwatch and waited to see a colour change. A water bath was used to control the temperature of the reactants. At the first temperature investigated, it took 280 seconds for the solution to lose its pink colour. GRADE 6-7 GRADE

2.1 Calculate the rate of decay. Give your answer in s^{-1}.

[1 mark]

2.2 Give **one** variable the students should have controlled in this experiment and describe how they should have controlled it.

[2 marks]

2.3 Give **one** possible source of error in this experiment and suggest how it could be reduced.

[2 marks]

3* A gardener wants to increase the rate of decomposition in his compost bin. GRADE 7-9 GRADE

Describe the factors he should consider changing to speed up the rate of decomposition and explain how each factor affects the rate of decomposition.

[6 marks]

Revision Summary Test for Topic 7a

Phew — plenty to take on board there with that lot. To make sure it doesn't decay away, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Summary Test

Competition, Abiotic and Biotic Factors, and Adaptations (p.142-144) ☑

1) What is meant by the terms 'habitat', 'population', 'community' and 'ecosystem'?
2) Give four resources that plants compete for to survive and reproduce. Do the same for animals.
3) What does a food web show?
4) What is meant by interdependence?
5) What is a stable community?
6) What are abiotic and biotic factors?
7) Give two abiotic factors and give examples of how they might affect communities in ecosystems. Do the same for two biotic factors.
8) What is meant by the following types of adaptation? Give an example for each.
   a) structural adaptations, b) behavioural adaptations, c) functional adaptations.
9) What are extremophiles? Give three examples of where they are found.

Food Chains (p.145) ☑

10) What's the term for the type of organism that food chains start with? What are they eaten by?
11) How do plants make glucose? How does this glucose become biomass?
12) Explain why the populations of predators and prey often change in cycles.

Quadrats and Transects (p.147-148) ☑

13) What is meant by an organism's distribution?
14) Describe how you would use a quadrat to compare the distribution of clover plants in two areas.
15) If you know the mean number of organisms per m^2 in an area, how could you estimate the population of organisms in that area?
16) What are transects used for? How would you use one?
17) How would you go about estimating the percentage cover of a quadrat of a certain species?

Environmental Change, and The Water and Carbon Cycles (p.149-150) ☑

18) Give three environmental changes that can affect the distribution of organisms. What can cause these environmental changes?
19) Describe the main processes in the water cycle.
20) Name four elements that living things take from the environment.
21) How does carbon in the atmosphere enter food chains?
22) Give some ways that carbon compounds in plants and animals end up back in the air.

Decay (p.153-154) ☑

23) Give three factors that affect the rate of decomposition and explain how they affect it.
24) Is biogas produced by aerobic or anaerobic decay?
25) Which type of gas does biogas mainly contain? What does this make biogas useful for?
26) Describe a practical you could carry out to investigate the effect of temperature on the rate of decay of milk.

Biodiversity and Waste Management

Quick Quiz

*Unfortunately, human activity can **negatively affect** the **planet** and its **variety of life**. Read on for bad news...*

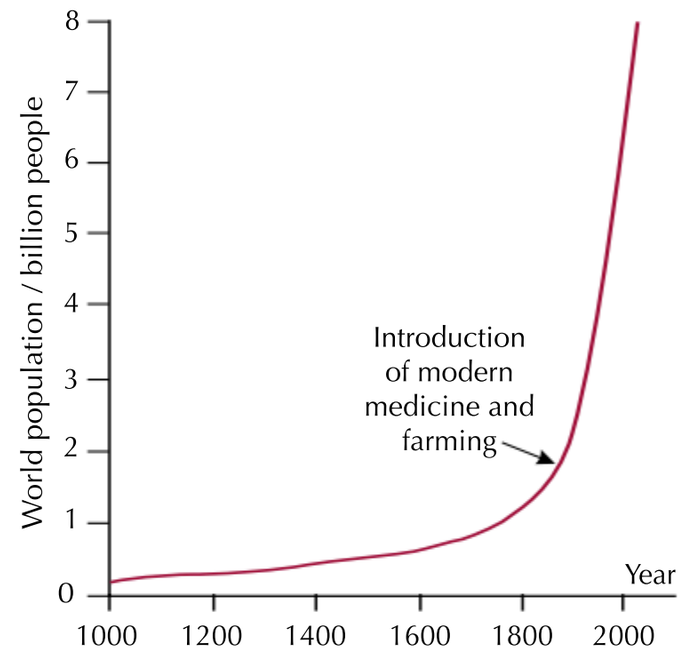
Earth's **Biodiversity** is Important

Biodiversity is the variety of different species of organisms on Earth, or within an ecosystem.

- High biodiversity is important. It makes sure that ecosystems (p.142) are stable because it reduces how dependent different species need to be on each other for things like shelter and food. Different species can also help to maintain the right physical environment for each other (e.g. the acidity of the soil).
- For the human species to survive, it's important that a good level of biodiversity is maintained.
- Lots of human actions, including waste production (see below) and deforestation (see p.159), as well as global warming (see next page) are reducing biodiversity. However, it's only recently that we've started taking measures to stop this from continuing.

There are Over **Eight Billion People** in the World

1) The population of the world is currently rising very quickly, and it's not slowing down — look at the graph...
2) This is mostly due to modern medicine and farming methods, which have reduced the number of people dying from disease and hunger.
3) This is great for all of us humans, but it means our actions are now having a bigger effect on the environment we live in:



Our increasing population puts pressure on the environment, as we take the resources we need to survive.

People around the world are also demanding a higher standard of living (and so demand luxuries to make life more comfortable — cars, computers, etc.). So we use more raw materials (e.g. oil to make plastics), but we also use more energy for the manufacturing processes. This all means we're taking more and more resources from the environment more and more quickly.

Unfortunately, many raw materials are being used up quicker than they're being replaced. So if we carry on like we are, one day we're going to run out.

We're Also Producing **More Waste**

As we make more and more things, we produce more and more waste, including waste chemicals. And without proper handling, this increasing amount of waste will lead to higher amounts of harmful pollution. Pollution affects water, land and air and kills plants and animals, reducing biodiversity.

Water	Land	Air
• Sewage and toxic chemicals from industry can pollute lakes, rivers and oceans, affecting the plants and animals that rely on them for survival (including humans). • Chemicals used on land (e.g. fertilisers, pesticides and herbicides) can be washed into water.	• We use toxic chemicals for farming (e.g. pesticides and herbicides). • Nuclear waste is buried underground. • We dump a lot of household waste in landfill sites.	• Smoke and acidic gases released into the atmosphere can pollute the air, e.g. sulfur dioxide can cause acid rain.

More people, more mess, less space, fewer resources...

...you might call it a recipe for disaster. As you can see, there's a lot of work for us to do in improving our waste management and finding ways to deal with a growing human population, whilst keeping living standards high.

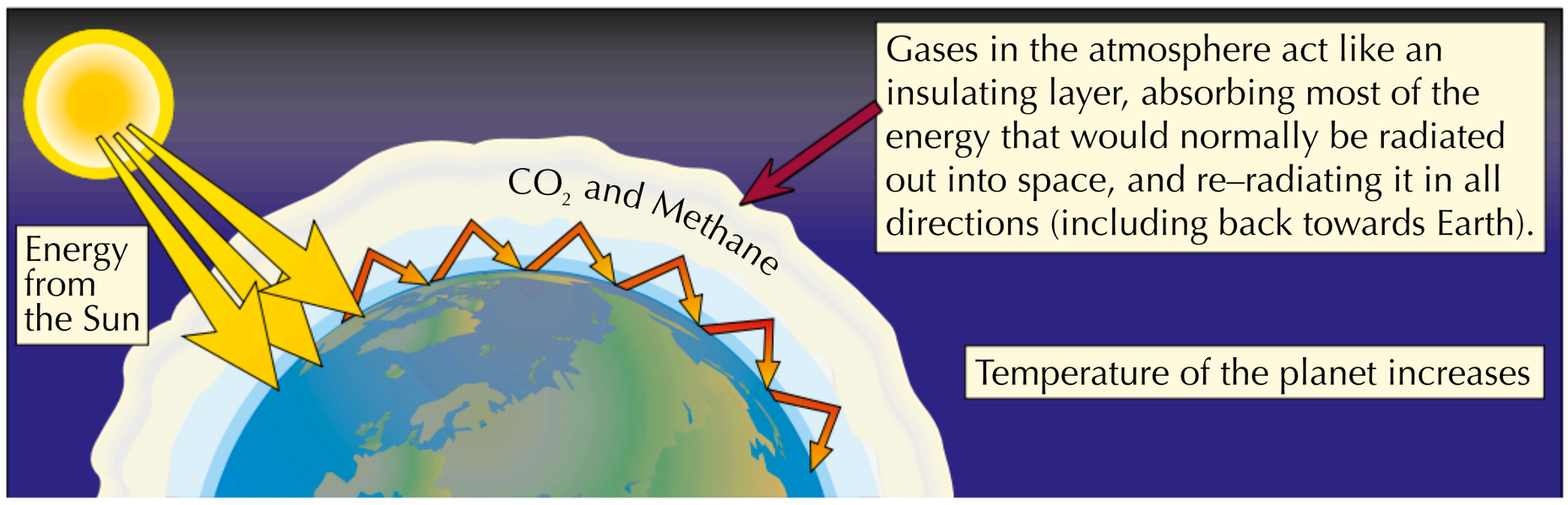
Quick Quiz

Global Warming

*The Earth is getting **warmer**. Climate scientists study how global warming is **changing** our planet, and the **effects** on people and nature — sadly, it's not as simple as everyone having nicer summers.*

Carbon Dioxide and Methane Trap Energy from the Sun

1) The temperature of the Earth is a balance between the energy it gets from the Sun and the energy it radiates back out into space.



2) Without the insulating layer of gases, there'd be nothing to keep any energy in, and at night we'd quickly get very cold. But recently we've started to worry that this effect is getting a bit out of hand...

3) The gases in the atmosphere that help keep the energy in are called "greenhouse gases". Carbon dioxide (CO_2) and methane are the main ones we worry about because the levels of these two gases are rising quite sharply (see next page).

4) The Earth is gradually heating up because of the increasing levels of greenhouse gases — this is global warming. Global warming is a type of climate change and causes other types of climate change, e.g. changing rainfall patterns.

The Consequences of Global Warming Could be Pretty Serious

There are several reasons to be worried about global warming. Here are a few:

Changes in Distribution

- Some wild animal and plant species are becoming more widely distributed as temperatures increase and the amount of rainfall changes. E.g. species that need warmer temperatures are spreading further as the conditions they thrive in exist over a wider area.
- Other species are becoming less widely distributed, e.g. species that need cooler temperatures have smaller ranges as the conditions they thrive in exist over a smaller area.

Rising Sea Levels

- Higher temperatures cause seawater to expand and ice to melt, and this has led to sea levels rising a little bit over the last 100 years. This is beginning to increase the frequency of flooding in some areas.
- If sea level keeps rising, it'll be bad news for people and animals living in low-lying places, and could result in the loss of habitats (where organisms live).

Decreasing Biodiversity

Biodiversity (see p.157) could be reduced if some species are unable to survive a change in the climate, so become extinct.

Changes in Migration Patterns

E.g. some birds may be migrating further north, as more northern areas are getting warmer.

We need to act fast, before the worst effects of climate change

Scientists accept that global warming is happening and that human activity has caused most of the recent warming. However, they don't know exactly what the effects will be — this will largely depend on how bad the warming gets.

Deforestation and Land Use

Quick Quiz

Trees *and* ***peat bogs*** *trap carbon dioxide and* ***lock it up****. The problems start when it* ***escapes****...*

Humans Use **Lots of Land** for **Lots of Purposes**

1) We use land for things like building, quarrying, farming and dumping waste.
2) This means that there's less land available for other organisms.
3) Sometimes, the way we use land has a bad effect on the environment...

Deforestation is the **Cutting Down** of **Forests**

Deforestation is done for various reasons, including:

- To clear land for farming (e.g. cattle or rice crops) to provide more food.
- To grow crops from which biofuels based on ethanol can be produced.

This causes big problems when it's done on a large scale, such as cutting down rainforests in tropical areas.

LESS CARBON DIOXIDE TAKEN IN

- Cutting down loads of trees means that the amount of carbon dioxide removed from the atmosphere during photosynthesis is reduced.
- Trees 'lock up' some of the carbon that they absorb during photosynthesis in their wood, which can remove it from the atmosphere for hundreds of years. Removing trees means that less is locked up.

More CO_2 in the atmosphere causes global warming (see previous page), which leads to climate change.

MORE CARBON DIOXIDE IN THE ATMOSPHERE

- Carbon dioxide is released when trees are burnt to clear land. (Carbon in wood doesn't contribute to atmospheric pollution until it's released by burning.)
- Microorganisms feeding on bits of dead wood release carbon dioxide as a waste product of respiration.

LESS BIODIVERSITY

- Habitats like forests can contain a huge number of different species of plants and animals, so when they are destroyed there is a danger of many species becoming extinct — biodiversity is reduced.

Destroying Peat Bogs Adds **More CO_2** to the Atmosphere

1) Bogs are areas of land that are acidic and waterlogged. Plants that live here don't fully decay when they die, because there's not enough oxygen, and so they gradually build up to form peat.
2) So the carbon in the plants is stored in the peat instead of being released into the atmosphere.
3) However, peat bogs are often drained so that the area can be used as farmland, or the peat is cut up and dried to use as fuel. It's also sold to gardeners as compost. Peat is being used faster than it forms.

- When peat is drained, it comes into more contact with air and some microorganisms start to decompose it. When these microorganisms respire, they use oxygen and release carbon dioxide, contributing to global warming (see the previous page).
- Carbon dioxide is also released when peat is burned as a fuel.
- Destroying the bogs also destroys (or reduces the area of) the habitats of some of the animals, plants and microorganisms that live there, so reduces biodiversity.

We need to use land, but we also need a healthy environment

We can't really avoid using land — we've got to if we want to do things like grow enough food or build enough houses for people. The human population is increasing so it's likely that we'll use even more land in the future — we'll have to find a way to manage land use to reduce the negative effects on the environment. It's a tricky balancing act.

Maintaining Ecosystems and Biodiversity

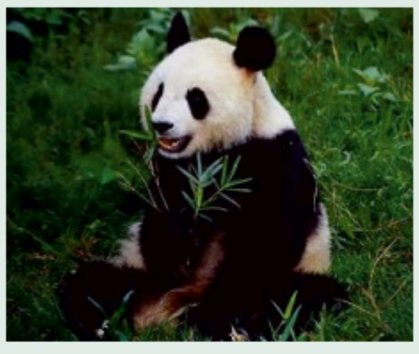
Quick Quiz

*It's really important that biodiversity is **maintained** as damage to ecosystems or populations of species can be **hard to undo**. This page is about some of the different **methods** that can be used to maintain biodiversity.*

Programmes Can be Set Up to **Protect Ecosystems** and **Biodiversity**

Biodiversity needs maintaining at a high enough level to make sure that ecosystems are stable (see p.142). In some areas, programmes have been set up by concerned citizens and scientists to minimise damage by human activities (see p.157) to ecosystems and biodiversity. Here are a few examples:

- Breeding programmes help prevent endangered species from becoming extinct. These are where animals are bred in captivity to make sure the species survives if it dies out in the wild. Individuals can sometimes be released into the wild to boost or re-establish a population.



- Programmes to protect and regenerate rare habitats like mangroves, heathland and coral reefs help to protect the species that live there — preserving the ecosystem and biodiversity in the area.
- Hedgerows and field margins (areas of land around the edges of fields where wild flowers and grasses are left to grow) can be reintroduced around fields on farms where only a single type of crop is grown. These provide a habitat for a wider variety of organisms than could survive in a single crop habitat.
- Some governments have introduced regulations and programmes to reduce the level of deforestation taking place and the amount of carbon dioxide being released into the atmosphere by businesses. This could reduce the increase of global warming (see page 158).
- People are encouraged to recycle to reduce the amount of waste that gets dumped in landfill sites. This could reduce the amount of land taken over for landfill, leaving ecosystems in place.

Conflicting Pressures Can Affect **How Biodiversity is Maintained**

Sadly for noble biodiversity warriors, maintaining biodiversity isn't as simple as you would hope. There are lots of conflicting pressures that have to be taken into account. For example:

- Protecting biodiversity costs money. For example, governments sometimes pay farmers a subsidy to reintroduce hedgerows and field margins to their land. It can also cost money to keep a watch on whether the programmes and regulations designed to maintain biodiversity are being followed. There can be conflict between protecting biodiversity and saving money — money may be prioritised for other things.
- Protecting biodiversity may come at a cost to local people's livelihoods. For example, reducing deforestation is great for biodiversity, but may leave people from the tree-felling industry unemployed. This could affect the local economy if people move away with their family to find work.
- There can be conflict between protecting biodiversity and protecting food security. Certain organisms can be seen as pests by farmers (e.g. locusts and foxes) and are killed to protect crops and livestock so that more food can be produced. However, this can affect the food chain and biodiversity.
- Many people want to protect biodiversity in the face of development, but sometimes land is in such high demand that previously untouched land with high biodiversity has to be used for development, e.g. for housing developments on the edge of towns, or for new agricultural land in developing countries.

So we should do what we can — within some limits

Like many situations in ecology, maintaining biodiversity isn't black and white. Before the people in charge can make decisions on the best way to go forward, they need to consider lots of factors to work out what needs prioritising.

Warm-Up & Exam Questions

I hope you've got all of that important information in your head. There's a lot to remember here, so have a flick back when you're doing these questions in case you've forgotten any little details.

Warm-Up Questions

1) What is meant by the term 'biodiversity'?
2) Give two greenhouse gases.
3) Give two reasons why an area of forest may be cut down.
4) How are populations of endangered species preserved by breeding programmes?
5) Explain how biodiversity can be increased in areas that farm single crops.

Exam Questions

1 Humans are producing increasing amounts of waste. This has negative consequences for biodiversity and the environment. GRADE 4-6 GRADE

1.1 Give **two** types of waste that pollute the air.

[2 marks]

1.2 Describe **one** way in which waste produced by humans pollutes water.

[1 mark]

1.3 Give **two** reasons why humans are producing increasingly more waste.

[2 marks]

2 Global warming may lead to a decline in biodiversity in the future. GRADE 6-7 GRADE

2.1 Apart from a decline in biodiversity, suggest **two** possible biological consequences of global warming.

[2 marks]

2.2 Describe **two** measures being used by some governments in order to reduce global warming.

[2 marks]

3 Peat bogs can be drained, so that the peat can be cut up and used as a source of fuel. GRADE 6-7 GRADE

3.1 Explain why draining peat bogs and using the peat as a fuel can contribute to global warming.

[4 marks]

3.2 Apart from contributing to global warming, explain **one** other negative consequence of destroying peat bogs.

[2 marks]

Quick Quiz

Trophic Levels

*And now for something slightly different... The word 'trophic' comes from the Greek word **trophe** meaning '**nourishment**'. This page must contain something about food then. Probably worth reading on I reckon...*

Food Chains Can be Divided into Trophic Levels

1) Trophic levels are the different stages of a food chain. They consist of one or more organisms that perform a specific role in the food chain.
2) They are named after their location in the food chain using numbers. The first level is called trophic level 1. Each level after that is numbered in order based on how far along the food chain the organisms in the trophic level are. For example, the organisms second in line in the food chain belong to trophic level 2, organisms third in line are in trophic level 3, and so on.

Trophic Level 1

Producers are the organisms at the starting point of a food chain, e.g. plants and algae. They're called producers because they make their own food by photosynthesis using energy from the Sun.

Trophic Level 2

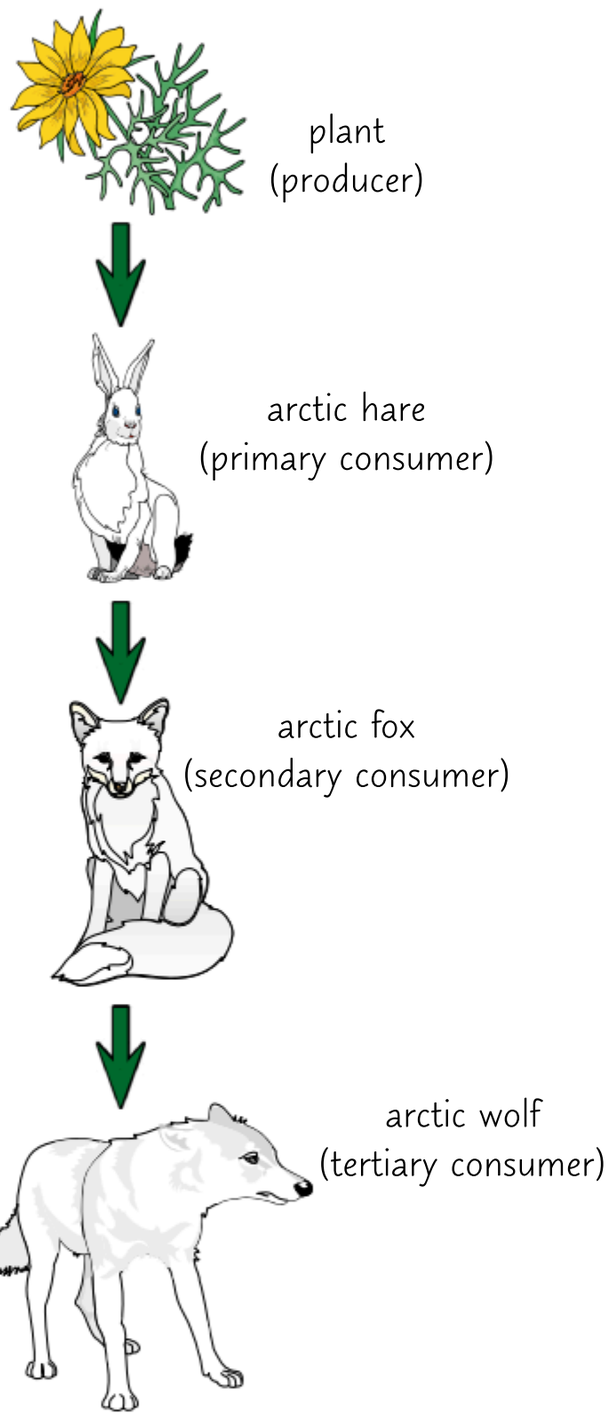
Herbivores that eat the plants and algae are primary consumers.

Trophic Level 3

Carnivores that eat the primary consumers are secondary consumers. They are meat eaters.

Trophic Level 4

Carnivores that eat other carnivores (the secondary consumers) are tertiary consumers. Carnivores that have no predators are at the top of the food chain, so they're always in the highest trophic level. They're known as apex predators.



There can be more than four trophic levels in a food chain, but there are only usually four or five because so much energy is lost from the food chain at each trophic level (see p.164).

Decomposers Break Down Uneaten Remains and Waste

- Decomposers such as bacteria and fungi play an important role in ecosystems.
- They decompose any dead plant or animal material left in an environment.
- They can do this by secreting (releasing) enzymes that break the dead stuff down into small soluble food molecules. These then diffuse into the microorganisms.

This process also releases nutrients into the environment, which the producers need in order to grow.

Plants produce, animals consume and decomposers have what's left...

So, trophic level is just the fancy term for a stage in a food chain. Each trophic level always contains organisms that perform the same role. For example, the organisms in trophic level 3 are always the secondary consumers. And don't forget about those environmental champions, the decomposers. They clean up after everyone else.

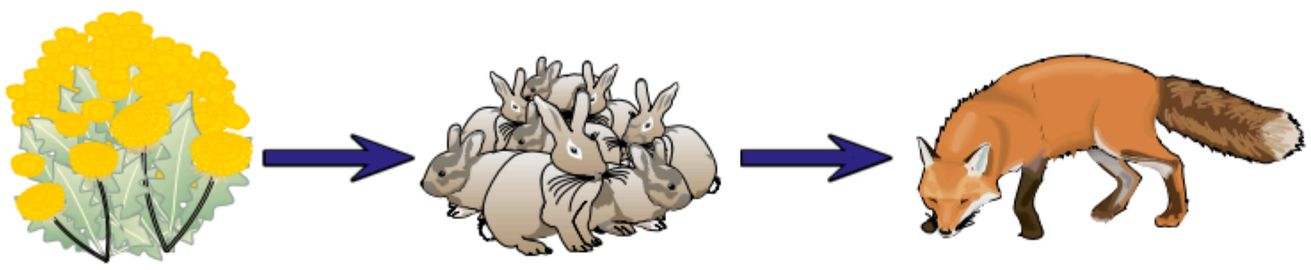
Pyramids of Biomass

Quick Quiz

Pyramids of biomass *are a handy way of illustrating the amount of* ***biomass*** *at* ***each level*** *in a food chain.*

Pyramids of Biomass Show the Relative Mass of Each Trophic Level

There's less energy and less biomass every time you move up a stage (trophic level) in a food chain. There are usually fewer organisms every time you move up a level too:



100 dandelions... feed... 10 rabbits... which feed... one fox.

Biomass is just the mass of living material.

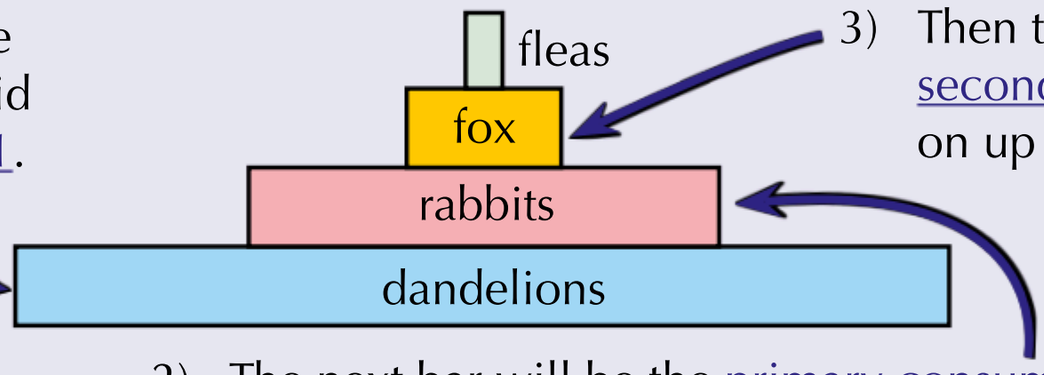
This isn't always true though. If 500 fleas are feeding on the fox, the number of organisms has increased as you move up to that stage in the food chain. So it's often better to think about biomass instead of number of organisms when looking at the food chain.

Constructing a Pyramid of Biomass

You can use information about biomass to construct a pyramid of biomass to represent a food chain:

- Each bar on a pyramid of biomass shows the relative mass of living material at a trophic level — basically how much all the organisms at each level would "weigh" if you put them all together.
- So the one fox above would have a big biomass and the hundreds of fleas would have a very small biomass. Biomass pyramids are practically always pyramid-shaped:

1) The big bar along the bottom of the pyramid shows trophic level 1. It always represents the producer.



3) Then the next bar is the secondary consumer and so on up the food chain. Easy.

2) The next bar will be the primary consumer.

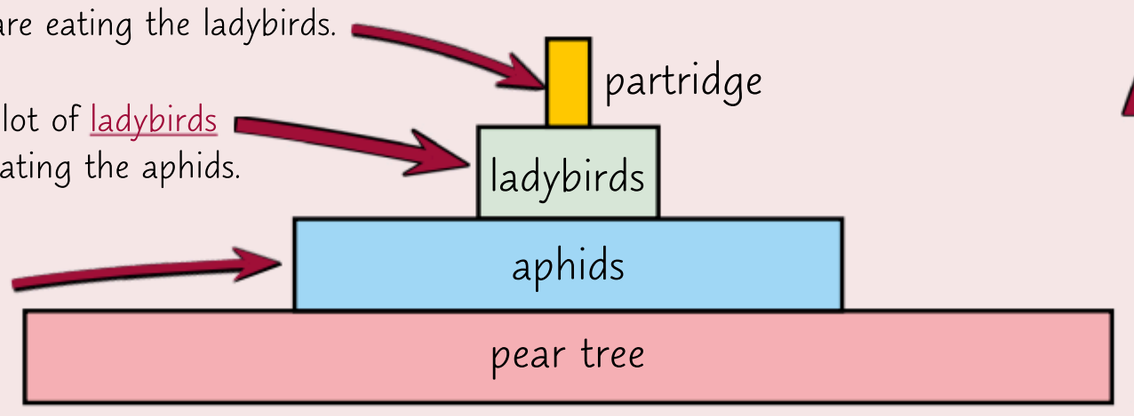
Pyramids of Biomass Give You Information About Food Chains

The biomass at each stage of a food chain on a pyramid of biomass should be drawn to scale. If you're given actual numbers, you can use them to draw bars of the correct scale. Don't forget that the order of organisms in the pyramid must follow the order of the food chain, and each bar must be labelled.

A few partridges are eating the ladybirds.

Quite a lot of ladybirds are then eating the aphids.

Lots (probably thousands) of aphids are feeding on a great big tree.



Biomass and energy are still decreasing as you go up the levels — it's just that one tree can have a very big biomass, and can fix a lot of the Sun's energy using all those leaves.

Use a sharp pencil and a ruler to draw pyramids of biomass

Q1 Video Solution

If you need to draw a pyramid of biomass to scale in the exam, you may be given a grid or some graph paper to draw it on. You'll need to work out a sensible scale to use — think about how easy your diagram will be to draw and interpret and make sure that your pyramid takes up at least half of the space available.

Q1 Draw an accurate pyramid of biomass for this food chain. Include the scale you have used. [4 marks]

	Clover	→	Snails	→	Thrushes	→	Buzzard
Biomass:	96 kg		42 kg		8 kg		1 kg

Quick Quiz

Biomass Transfer

*So biomass moves between trophic levels in a food chain, but **not all of it** can be transferred.*
*This page shows you how you can make a **rough calculation** to figure out **how much** gets transferred.*

Biomass is **Lost** Between Each **Trophic Level**

1) Producers, such as green plants and algae, use energy transferred by light from the Sun to make food (glucose) during photosynthesis. Of the energy that hits these producers, only about 1% is transferred for photosynthesis.

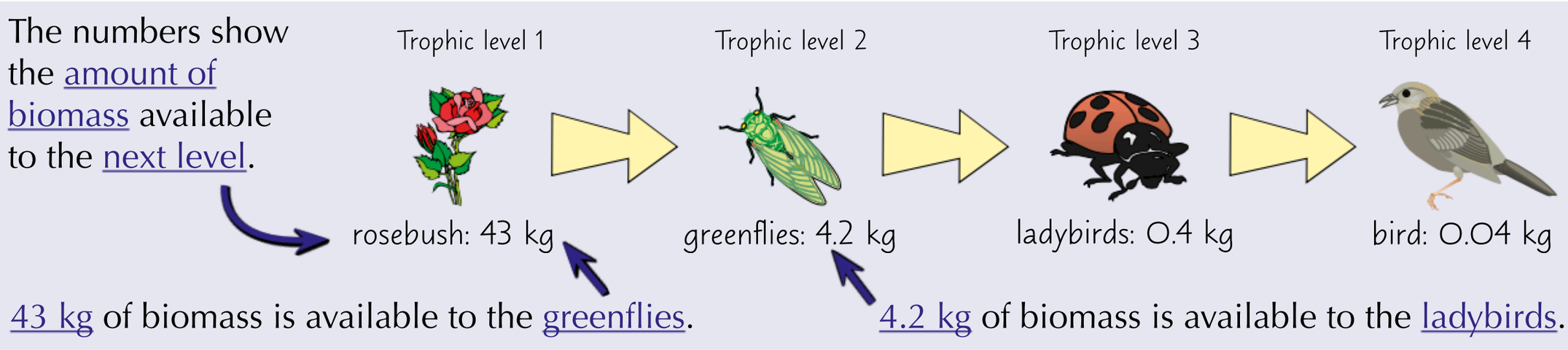
Energy from the Sun is the source of energy for nearly all life on Earth.

2) Some of the glucose is used by the plants and algae to make biological molecules. These molecules make up the plant's biomass — the mass of living material. Biomass stores energy.
3) Biomass is transferred through a food chain in an ecosystem when organisms eat other organisms. However, only about 10% of the biomass is passed from one trophic level to the next.
4) Biomass is lost between the levels for a few different reasons:

- Organisms don't always eat every single part of the organism they're consuming, for example, if some of it is inedible (e.g. bone). So not all the biomass can be passed to the next trophic level.
- Organisms don't absorb all of the stuff in the food they ingest (take in). The stuff that they don't absorb is egested (released) as faeces.
- Some of the biomass taken in is converted into other substances that are lost as waste. For example, glucose (obtained from the biomass) is used in respiration to provide energy for movement and keeping warm, etc., rather than to make more biomass. Respiration produces carbon dioxide and water as by-products. Urea is another waste substance, which is released in urine with water when the proteins in the biomass are broken down.

This biomass loss is why the number of organisms at each level usually gets smaller.

You Can **Calculate** the **Efficiency** of **Biomass Transfer**



1) You can work out how much biomass has been lost at each level:
biomass lost at a trophic level = biomass available at the previous level – biomass available at that level
So, the biomass lost at the 2nd trophic level above = 43 kg – 4.2 kg = 38.8 kg.
2) You can also calculate the efficiency of biomass transfer between trophic levels:

$$\text{efficiency} = \frac{\text{biomass transferred to the next level}}{\text{biomass available at the previous level}} \times 100$$

So, the efficiency of biomass transfer from the 1st trophic level = 4.2 kg ÷ 43 kg × 100
= 9.76744... = 9.8%.

Life on Earth is fuelled by sunlight

Remember, hardly any of the biomass from one trophic level is passed on to the next level — most is lost.

Q2 Video Solution

Q1 Give two ways that biomass is lost between trophic levels. [2 marks]

Q2 A shark with 110 kg of available biomass eats some large fish with 995 kg of available biomass.
a) Calculate the amount of biomass lost between the fish and the shark. [1 mark]
b) Determine the percentage efficiency of biomass transfer between the fish and the shark. Give your answer to three significant figures. [1 mark]

Food Security and Farming

Quick Quiz

*With more than **8 billion** people on our planet who all need **feeding**, maintaining our **food supply** is essential.*

Food Security is Having Enough Food to Feed a Population

There's a wide range of things that can threaten food security. For example:

- The world population keeps increasing, with a high birth rate in many developing countries.
- As diets in developed countries change, the demand for certain foods to be imported from developing countries can increase. This means that already scarce food resources can become more scarce.
- New pests and pathogens (e.g. bacteria and viruses) or changes in the environmental conditions (e.g. a lack of rain) can result in the loss of crops and livestock, and can lead to widespread famine.
- The high input costs of farming (e.g. the price of seeds, machinery and livestock) can make it too expensive for people in some countries to start or maintain food production, meaning that there sometimes aren't enough people producing food in these areas to feed the people.
- In some parts of the world, there are conflicts that affect the availability of food and water.

Overfishing is Decreasing Fish Stocks

1) Fish stocks are declining in the oceans because we're fishing so much.
2) This means there's less fish for us to eat, the ocean's food chains are affected and some species of fish may disappear altogether in some areas — e.g. cod are at risk of disappearing from the north west Atlantic. This isn't sustainable.
3) To tackle this problem, we need to maintain fish stocks at a level where the fish continue to breed. This is sustainable food production.
4) Fish stocks can be maintained (conserved) in these ways:

Sustainable methods of food production are needed so that enough food can be made to feed everyone now and in the future, without using resources faster than they renew.

Fishing Quotas

Limiting the number and size of fish that can be caught in certain areas. This prevents certain species from being overfished.

Net Size

There are different limits of the mesh size of the fish net, depending on what's being fished. This is to reduce the number of 'unwanted' fish that are accidentally caught, e.g. shrimp caught along with cod. Using a bigger mesh size will let the 'unwanted' species escape. It also means that younger fish will slip through the net, allowing them to reach breeding age.

Food Production Can be Made More Efficient

Limiting the movement of livestock and keeping them in a temperature-controlled environment reduces the transfer of energy from livestock to the environment as the animals use less energy moving around and controlling their own body temperature. This means that more energy is available for growth, so more food can be produced from the same input of resources — the farming is more efficient.

Livestock

Livestock, like calves and chickens, can be factory farmed where they're raised in small pens.

Fish

Fish can be factory farmed in cages where their movement is restricted.

Animal Feed

Some animals are fed high-protein food to further increase their growth.

Some factory farming methods are controversial. Disease can spread easily in animals kept so close together and some people think that making animals live in unnatural and uncomfortable conditions is cruel.

Factory farmed chickens are also called battery chickens.

Fishing practices have become 'too' effective

We can actually reduce food security by gathering too much food — this is why there's so much hype about sustainability these days. We need to find methods to produce as much food as we need without damaging future supplies — turn over for a few ideas, some of which are more ethically controversial than others...

Quick Quiz

Biotechnology

*Biotechnology is where **living things** and **biological processes** are manipulated to produce a **useful product**.*

Mycoprotein — Food from **Fungi**

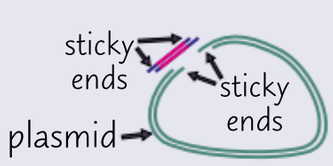
- Using modern biotechnology techniques, large amounts of microorganisms can be cultured (grown) industrially under controlled conditions in large vats for use as a food source.
- Mycoprotein is used to make high-protein meat substitutes for vegetarian meals, e.g. Quorn™.
- It's made from the fungus *Fusarium*, which is grown in aerobic conditions on glucose syrup, which it uses as food.
- The fungal biomass is harvested and purified to produce the mycoprotein.

Aerobic conditions are where oxygen is present.

Bacteria can be **Engineered** to Produce **Human Insulin**

Genetic engineering is transferring a useful gene from one organism to another (see page 133). For example:

1) The insulin gene is cut out of a human chromosome using a restriction enzyme, leaving sticky ends.



2) A plasmid (a loop of DNA) removed from a bacterium is cut open using the same restriction enzyme — leaving the same sticky ends.

Restriction enzymes recognise specific sequences of DNA and cut the DNA at these points. This leaves one of the DNA strands with unpaired bases called a 'sticky end'.

3) The plasmid and insulin gene are mixed together with ligase (an enzyme) to join the sticky ends and form recombinant DNA (two different bits of DNA stuck together).

4) The recombinant DNA is inserted into a bacterium.

5) The modified bacterium is grown in a vat under controlled conditions. You end up with millions of bacteria that produce insulin.

6) The insulin can be harvested and purified to treat people with diabetes.

Crops Can be **Genetically Modified**

Many people in the world today don't have enough food to eat (or the diet they have isn't varied). This mostly happens in developing countries — like those in Africa and parts of Asia.

Biotechnology could help...

- Genetically modified (GM) crops can be produced that are resistant to pests — improving crop yields.
- They can be genetically modified to grow better in drought conditions — again improving crop yields.
- And some crops can be modified to provide more nutritional value, e.g. 'Golden Rice' has been genetically engineered to produce a chemical that's converted in the body to vitamin A.

...But not everyone agrees

- Many argue that people go hungry because they can't afford to buy food, not because there isn't any food about. So they say that you need to tackle poverty first.
- There are fears that countries may become dependent on companies selling GM seeds.
- Sometimes it's poor soil that causes crops to fail, and so even GM crops won't survive.

It looks hard, but it's just like a fancy cut and paste...

Genetic modification is a powerful biotechnological tool that could help solve a wide range of problems.

Q1 Video Solution

Q1 Rennin is an enzyme used to make cheese. It is naturally produced by stomach cells in cows, which contain the rennin gene. Suggest how a bacterial cell could be genetically engineered to produce rennin. [3 marks]

Warm-Up & Exam Questions

By doing these questions, you'll soon find out if you've got the basic facts straight.

Warm-Up Questions

1) What is the name for the different stages of a food chain?
2) What is biomass?
3) How do fishing quotas limit overfishing?
4) The movement of factory farmed animals is often limited. Explain why.

Exam Questions

1 In an ecosystem, a single robin has a mass of 18 g and eats caterpillars. There are 75 caterpillars available to the robin that each have a mass of 1.5 g. The caterpillars feed on 40 stinging nettles that together have a mass of 1000 g.

GRADE 4-6 GRADE

1.1 Calculate the total biomass of caterpillars available to the robin.

[1 mark]

1.2 Calculate the efficiency of biomass transfer between the robin and the caterpillars. Use the equation:

efficiency = (biomass transferred to the next level ÷ biomass available at the previous level) × 100

[1 mark]

2 Current methods of fishing and farming may have to be changed in order to be more sustainable, so that there is greater human food security in the future.

GRADE 6-7 GRADE

2.1 Explain what is meant by the term 'food security'.

[1 mark]

2.2 State **two** threats to global food security and explain why each one is a threat.

[4 marks]

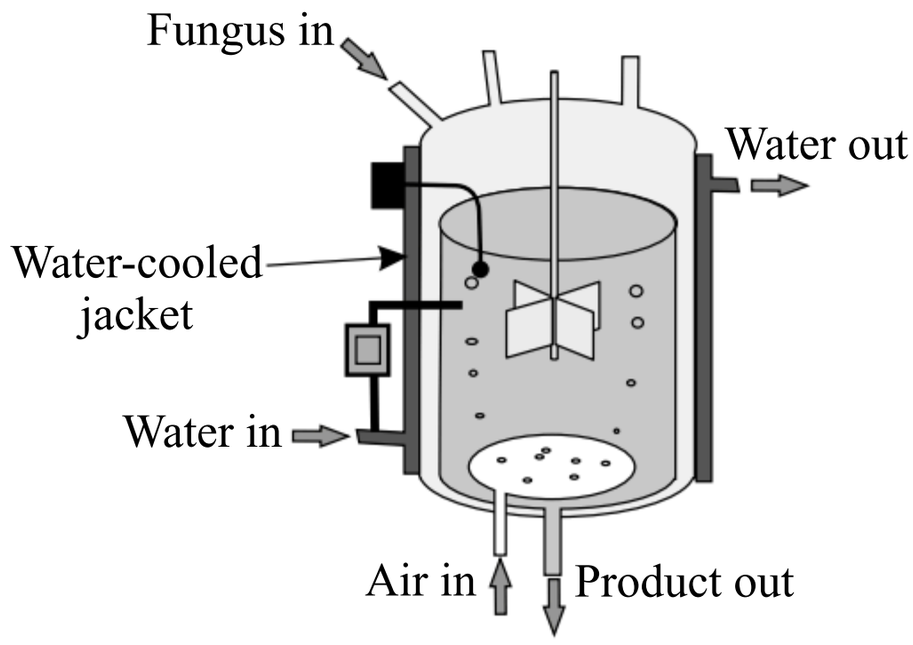
2.3 Explain **one** reason why regulations on the net size used by fishing boats may help to lessen the impact of overfishing on certain fish species.

[2 marks]

3 **Figure 1** shows a fermenter that could be used to produce mycoprotein.

GRADE 6-7 GRADE

Figure 1



3.1 Explain the purpose of the air supply in the fermenter.

[2 marks]

3.2 Suggest and explain why a water-cooled jacket is used around the fermenter.

[2 marks]

3.3 Give **one** substance not labelled in **Figure 1** that would also need to be added to the fermenter in order to produce mycoprotein. Explain your choice.

[2 marks]

Revision Summary Test for Topic 7b

Summary Test

Lots going on in that section. To make sure you've eaten it all up, you can:

- Tackle the revision summary test below, or scan the QR code to do it online.
  The questions are hard, but they'll show you how well you really know your stuff.
- Track your progress online and see which areas need more work.
- Compare your answers with sample answers for the test here: cgpbooks.co.uk/Leaf

Human Impacts on the Planet (p.157-160) ☐

1) What is biodiversity? Why is high biodiversity in an ecosystem important? ☐
2) Describe how an increasing population is having a bigger effect on the environment. ☐
3) Describe how water, land and air are each being affected by an increased production of waste. ☐
4) Name two greenhouse gases. How are the levels of these linked to global warming? ☐
5) Outline four consequences of global warming. ☐
6) Give four human uses of land that mean less land is available for other organisms. ☐
7) Why might humans carry out deforestation? Why is deforestation on a large scale bad for the environment? ☐
8) Why does the destruction of peat bogs add more carbon dioxide to the atmosphere? ☐
9) Describe some programmes that have been put in place to minimise the damage caused by human activities to ecosystems and biodiversity. ☐
10) Give four conflicting pressures when trying to maintain biodiversity. ☐

Trophic Levels and Biomass (p.162-164) ☐

11) What is a trophic level? ☐
12) In which trophic level are the following found?
    a) tertiary consumers, b) producers, c) secondary consumers, d) primary consumers ☐
13) What tends to be the maximum number of trophic levels you'd see in a food chain? Why? ☐
14) What role do decomposers play in ecosystems? ☐
15) What does a pyramid of biomass show? ☐
16) Does the bottom bar on a pyramid of biomass represent the tertiary consumer or the producer? ☐
17) How does biomass change as you move up the bars in a pyramid of biomass? ☐
18) Approximately what percentage of biomass is passed from one trophic level to the next? ☐
19) Explain why not much biomass gets transferred from one trophic level to the next. ☐
20) Give the formula for calculating the efficiency of biomass transfer. ☐

Food Security and Biotechnology (p.165-166) ☐

21) Describe five factors that affect the level of food security. ☐
22) How can fish stocks be maintained at a sustainable level? ☐
23) How can the efficiency of food production in livestock farming be improved? ☐
24) What is mycoprotein? How is it made? ☐
25) How can bacteria be used to produce human insulin? ☐
26) Give two examples of how crops could be genetically modified to increase yields. ☐
27) What is 'Golden Rice'? ☐

The Scientific Method

*This section **isn't** about how to 'do' science — but it does show you the way **most scientists** work.*

Scientists Come Up With **Hypotheses** — Then **Test** Them

1) Scientists try to explain things. They start by observing something they don't understand.
2) They then come up with a hypothesis — a possible explanation for what they've observed.
3) Next they make a prediction based on the hypothesis. They test the prediction by gathering evidence (i.e. data) from investigations. If the evidence backs up their prediction, then they're a step closer to figuring out if the hypothesis is true.
4) They can share their findings at conferences, or in peer-reviewed journals.
5) Once other scientists know about a hypothesis, they'll start basing their own predictions on it and carry out their own experiments. They'll also try to reproduce the original experiments to check the results — and if all the experiments in the world back up the hypothesis, then scientists start to think it is true.
6) However, if results from lots of other experiments don't fit with the hypothesis, then it may need to be modified or scrapped altogether.

Peer-review is where other scientists check that methods, results and scientific explanations seem sensible before they're published. It helps to detect errors and false claims, but it doesn't mean that findings are correct — just that they're not wrong in any obvious way.

Hundreds of years ago, we thought demons caused illness.

Then we thought it was caused by 'bad blood' (and treated it with leeches).

Now we've collected more evidence, we know that illnesses that can be spread between people are due to microorganisms.

If **All** the **Evidence** Supports a Hypothesis, It's **Accepted** — For Now

Accepted hypotheses often become the basis of theories. Our currently accepted theories are the ones that have survived many rounds of testing over the years. However, theories never become totally indisputable fact. If new evidence comes along that can't be explained using the existing theory, then the hypothesising and testing is likely to start all over again.

Theories Can Involve **Different Types** of **Models**

Scientists use models to explain observations and make predictions, for example:

1) Representational models — simplified descriptions or pictures of what's going on in real life. E.g. the lock and key model of enzyme action is a simplified way of showing how enzymes work (see p.27). It can be used to explain why enzymes only catalyse particular reactions.
2) Computational models — computers are used to make simulations of complex real-life processes (e.g. climate change). Computational models are used when there's lots of different variables to consider, and because you can easily change their design to take into account new data.

Scientists test models by carrying out experiments to check that the predictions made by the model happen as expected.

All models have limitations on what they can explain or predict. E.g. in climate change models, it's hard to take into account all the biological and chemical processes that influence climate. It can also be difficult to include regional variations in climate.

Scientific models are constantly being refined...

The scientific method has developed over time. Aristotle (a Greek philosopher) was the first to realise that theories need to be based on observations. Muslim scholars then introduced the ideas of creating and testing hypotheses.

Communication & Issues Created by Science

*Science has helped us to **make progress** in loads of areas, from medicine to space travel. But it **can't answer everything**, and it can raise some **issues** too. Scientific discoveries also need to be **communicated**...*

It's Important to **Communicate** Scientific Discoveries to the **Public**

Some scientific discoveries show that people should change their habits, or they might provide ideas that could be developed into new technology. So scientists need to tell the world about their discoveries.

E.g. gene technologies are used in genetic engineering to produce genetically modified (GM) crops. Information about these crops needs to be communicated to farmers who might benefit from growing them and to the public, so they can make informed decisions about the food they buy and eat.

Scientific **Evidence** can be **Presented** in a **Biased Way**

1) Media reports of scientific discoveries aren't peer-reviewed. So, even though news stories are often based on data that has been peer-reviewed, the data might be presented in a way that is over-simplified or inaccurate, making it open to misinterpretation.
2) People who want to make a point can sometimes present data in a biased way (sometimes without meaning to). For example, a scientist might overemphasise a relationship in some of their data, or a newspaper article might report data supporting an idea without giving any evidence against it.

Scientific Developments are **Great**, but they can **Raise Issues**

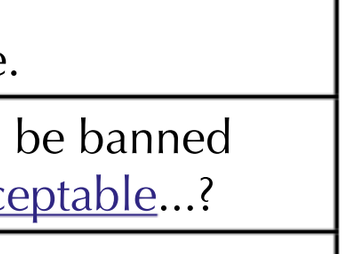
Economic issues	Society can't always afford to do what scientists recommend (e.g. investing in alternative energy sources) without cutting back elsewhere.
Social issues	Decisions based on scientific evidence affect people — e.g. should alcohol be banned (to prevent health problems)? Would the effect on people's lifestyles be acceptable...?
Personal issues	Some decisions will affect individuals. For example, someone might support renewable energy, but object if a wind farm is built next to their house.
Environmental issues	Human activity often affects the natural environment — e.g. GM crops may help to produce more food, but some people think they could cause environmental problems (see p.133).

Science **Can't Answer Every Question** — Especially **Ethical** Ones

- To answer scientific questions, scientists need data to provide evidence for their hypotheses.
- Some questions can't be answered yet because there's not enough data to support a theory, or because the data can't currently be collected. As more evidence is collected, more questions will be answered (like what the impact of global warming on sea levels will be).

But there will always be questions that experiments can't help us to answer...

Question: Should new drugs that boost 'brain power' be available to the public?



PRO
- Could improve your concentration or memory.

CONS
- Could give people an unfair advantage in exams.
- People might be pressured into taking them so that they could work more effectively, and for longer hours.

There are often issues with new scientific developments...

The trouble is, there's often no clear answer where these issues are concerned. Different people have different views, depending on their priorities — this is why understanding ethical arguments is important in science, especially biology.

Risk

Scientific discoveries *are often great, but they can prove* ***risky****. With dangers all around, you've got to be aware of hazards — this includes* ***how likely*** *they are to* ***cause harm*** *and* ***how serious*** *the effects may be.*

Every **Hazard** has a **Risk** Attached to it

HAZARD — something that could potentially cause harm.

RISK — the chance that a particular hazard will cause harm.

There are risks that seem pretty obvious, or we've known about for a while (e.g. the risk of having a car accident while in a moving car) and new risks that scientists are unsure of that arise from scientific advances, (e.g. whether nanoparticles used in cosmetics and sun cream might be harming the cells in our bodies).

You can estimate the size of a risk based on how many times something happens in a big sample over a given period. For example, you could assess the risk of a driver crashing by recording how many people in a group of 100 000 drivers crashed their cars over a year.

People Make Their **Own Decisions** About Risk

Something's generally considered high risk if there is a high chance of it happening and if its consequences are serious. But there are lots of factors that can affect how people feel about taking different risks:

Consequences

Most people are happier to accept a higher probability of an accident if the consequences are short-lived and fairly minor. E.g. you're more likely to cut your finger when using a knife than you are to die when scuba-diving — cutting your finger is fairly minor compared to death so people are more willing to take that risk.

Familiarity

How risky people think things are isn't always accurate. They tend to view familiar activities as low-risk and unfamiliar activities as high-risk — even if that's not the case (e.g. cycling on roads is often high-risk, but some people are happy to do it because it's a familiar activity).

Visibility

People may underestimate the risk of things with long-term or invisible effects, e.g. using tanning beds.

Choice

People tend to be more willing to accept a risk if they choose to do something, compared to having the risk imposed on them (e.g. having a nuclear power station built next door).

Investigations Can be **Hazardous**

Hazards from science experiments might include:

- Microorganisms, e.g. some bacteria can make you ill.
- Chemicals, e.g. sulfuric acid can burn your skin and alcohols catch fire easily.
- Fire, e.g. an unattended Bunsen burner is a fire hazard.
- Electricity, e.g. faulty electrical equipment could give you a shock.

Part of planning an investigation is making sure that it's safe. You should always make sure that you identify all the hazards that you might encounter. Then you should think of ways of reducing the risks from these hazards. For example:

- If you're working with sulfuric acid, always wear gloves and safety goggles. This will reduce the risk of the acid coming into contact with your skin and eyes.
- If you're using a Bunsen burner, stand it on a heat proof mat. This will reduce the risk of starting a fire.

You can find out about potential hazards by looking in textbooks, doing some internet research, or asking your teacher.

The pros and cons of new technology must be weighed up...

The world's a dangerous place and it's impossible to rule out the chance of an accident altogether. But if you can recognise hazards and take steps to reduce the risks, you're more likely to stay safe and avoid a serious mishap.

Designing Investigations

*Dig out your lab coat and dust down your badly-scratched safety goggles... it's **investigation time**.*

Investigations **Produce Evidence** to **Support** or **Disprove** a **Hypothesis**

1) Like all great scientists, you'll need to be able to come up with a hypothesis to test an observation (see p.169).
2) To determine whether a hypothesis is right, you need to do an investigation to gather evidence. To do this, you'll need to use your hypothesis to make a prediction — something you think will happen that you can test.
3) Investigations are used to see if there are patterns or relationships between two variables, e.g. to see if there's a relationship between the variables 'number of spots' and 'size of feet'.

1. Observation: People have big feet and spots.
2. Hypothesis: Having big feet causes spots.
3. Prediction: People who have bigger feet will have more spots.

Evidence Needs to be **Repeatable, Reproducible** and **Valid**

REPEATABLE — if the same person does the same experiment again, they'll get similar results.

REPRODUCIBLE — if someone else does the experiment, or a different method or piece of equipment is used, the results will still be similar.

VALID RESULTS — results that are both repeatable and reproducible AND they answer the original question. They come from experiments that were designed to be a fair test.

If data is repeatable and reproducible, it's reliable and scientists are more likely to have confidence in it.

To Make an Investigation a **Fair Test** You Have to **Control the Variables**

1) In a lab experiment you usually change one variable and measure how it affects another.
2) To make it a fair test, everything else that could affect the results should stay the same — otherwise you can't tell if the thing you're changing is causing the results or not.
3) Because you can't always control all the variables, you often need to use a control experiment — an experiment that's kept under the same conditions as the rest of the investigation, but doesn't have anything done to it. This is so that you can see what happens when you don't change anything at all.

The different kinds of variables are:

INDEPENDENT	The variable that you change
DEPENDENT	The variable that you measure
CONTROL	The variables you keep the same

E.g. you could find how temperature affects the rate of an enzyme-controlled reaction. The independent variable is the temperature. The dependent variable is the rate of reaction. The control variables include the concentration and amounts of reactants, pH, the time period you measure, etc.

The **Bigger** the **Sample Size** the **Better**

- A sample should represent the whole population — a small sample can't do that as well. It's also harder to spot anomalies if your sample size is too small.
- Scientists have to be realistic when choosing sample sizes, though. For example, if you were studying the effects of air pollution on the health of people living in a city, it'd be great to study everyone in the city (a huge sample), but it'd take ages and cost a tonne. It'd be more realistic to study a thousand people, with a range of ages, genders, and races to represent the population as a whole.

Designing an investigation is an involved process...

For your exam, you don't just need to be able to plan your own investigations — you should also be able to look at someone else's plan and be able to evaluate it. That also involves being able to suggest any improvements or changes.

Collecting Data

*You've designed the perfect investigation — now it's time to get stuck in and **collect some data**.*

Your Data Should be **Repeatable**, **Reproducible**, **Accurate** and **Precise**

- To check repeatability you need to repeat the readings and check that the results are similar. You need to repeat each reading at least three times.
- To make sure your results are reproducible you can cross check them by taking a second set of readings with another instrument (or a different observer).
- Your data also needs to be precise and accurate:

PRECISE RESULTS — where the data is all really close to the mean (average) of your repeated results (i.e. not spread out).

ACCURATE RESULTS — results that are really close to the true answer.

Repeat	Data set 1	Data set 2
1	12	11
2	14	17
3	13	14
Mean	13	14

Data set 1 is more precise than data set 2.

The accuracy of your results usually depends on your method.
E.g. estimating the amount of gas released in a reaction by counting bubbles isn't very accurate because you might miss some of the bubbles and they might have different volumes. It's more accurate to measure the volume of gas released using a gas syringe.

Your **Equipment** has to be **Right for the Job**

Sensitivity

The measuring equipment you use has to be sensitive enough to measure the changes you're looking for. E.g. if you need to measure changes of 1 cm^3 you need to use a measuring cylinder or burette that can measure in steps of 1 cm^3.

Resolution

Resolution is the smallest change a measuring instrument can detect.
E.g. some mass balances have a resolution of 1 g, some have a resolution of 0.1 g, and some are even more sensitive.

Calibration

Equipment also needs to be calibrated by measuring a known value. If there's a difference between the measured and known value, you can use this to correct the inaccuracy of the equipment.

Look out for **Errors** and **Anomalous Results**

Type of error	Definition	Example	How to correct
RANDOM	Unpredictable differences in results caused by things like human errors in measuring.	When taking a reading from a ruler, you have to estimate or round a reading when it's between two marks.	By taking repeat readings and finding the mean.
SYSTEMATIC	When measurements are wrong by a consistent amount every time.	When taking a reading from a ruler, you read from the end rather than from the 0 cm mark every time.	By accounting for them, e.g. by adjusting your measurements.

A systematic error caused by using equipment that isn't zeroed properly is called a ZERO ERROR. E.g. if a mass balance always reads 1 gram with nothing on it, all your measurements will be 1 gram too heavy. You can correct this error by subtracting 1 gram from all your results for mass.

If there's no systematic error, then doing repeats and calculating a mean could make your results more accurate.

An ANOMALOUS RESULT is one that doesn't fit in with the rest of your data at all. You should investigate it and try to work out what happened. If you can work out what happened (e.g. you measured something totally wrong) you can ignore it when processing your results.

Collecting data is what investigations are all about

Part of designing a good investigation is making sure that any data you collect is accurate, precise, repeatable and reproducible. Using properly calibrated equipment will help make sure that the data that you collect is accurate.

Processing Data

*Processing your data means doing some **calculations** with it to make it **more useful**.*

Data Needs to be **Organised**

- Tables are dead useful for organising data.
- When you draw a table, use a ruler and make sure each column has a heading (including the units).
- Here's a table about the kinds of biscuits made by Cookies and Goods Production Ltd.:

The burgundy row is the header row

Categories of biscuit go down the left side

Biscuit	Production (bpd)				
	Day 1	Day 2	Day 3	Day 4	Day 5
Bourbon	234	221	182	270	261
Custard cream	351	342	295	309	342
Pink wafer	121	137	97	82	101

Units — here it's biscuits per day

Repeats go in the header row under a main header

You input your results in these rows underneath the header

You Might Have to **Process Your Data**

There are different ways you can calculate the average of a set of results — they each tell you something different about your data:

- MEAN — calculated by adding all the values together and dividing by the total number of values used. When people say to calculate 'the average', they are usually talking about the mean.
- MODE — the value that occurs the most in a set of results.
- MEDIAN — the middle number when you write your results in numerical order.

EXAMPLE

The results of an experiment to find the volume of gas produced in an enzyme-controlled reaction are shown below. Calculate the mean volume, the mode and the median.

MATHS SKILLS

Repeat (cm^3)					Mean (cm^3)	Mode (cm^3)	Median (cm^3)
1	2	3	4	5			
11	14	20	15	20	(11 + 14 + 15 + 20 + 20) ÷ 5 = 16	20	15

Round to the **Lowest Number** of **Significant Figures**

The first significant figure of a number is the first digit that's not zero. The second and third significant figures come straight after (even if they're zeros). You should be aware of significant figures in calculations.

1) In any calculation, you should round to the lowest number of significant figures (s.f.) given.
2) Remember to write down how many significant figures you've rounded to after your answer.
3) If your calculation has multiple steps, only round the final answer, or it won't be as accurate.

EXAMPLE

A plant produces 10.2 cm^3 of oxygen in 6.5 minutes whilst photosynthesising. Calculate the rate of photosynthesis.

MATHS SKILLS

rate = 10.2 cm^3 ÷ 6.5 min = 1.5692... = 1.6 cm^3/min (2 s.f.)

10.2: 3 s.f. — 6.5: 2 s.f. — Final answer should be rounded to 2 s.f.

Don't forget your calculator...

In the exam you could be given some data and be expected to process it in some way. Make sure you keep an eye on significant figures in your answers, always remember to write down your working, and then check it when you're done.

Presenting Data

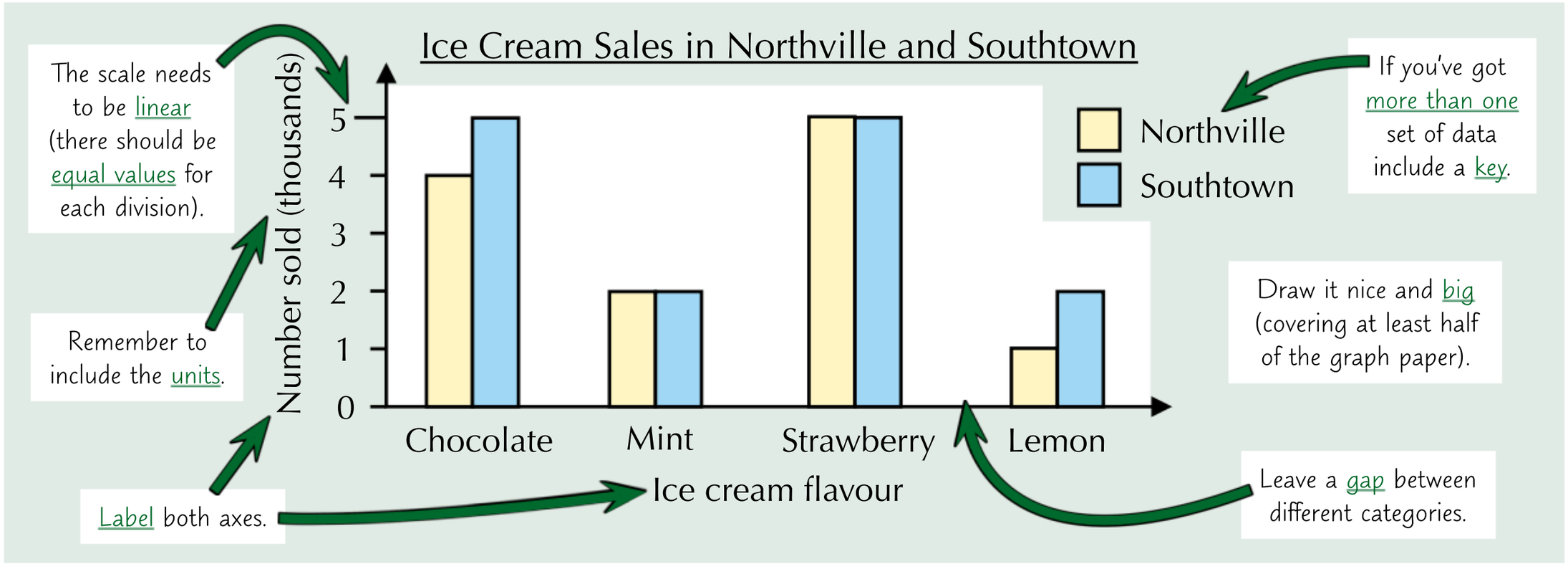
*Once you've processed your data, you can present your results in a nice **chart** or **graph**. This will help you to **spot any patterns** in your data. The **type** of data you record will affect how you should **present** it.*

If Your Data Comes in **Categories**, Present it in a **Bar Chart**

You should represent your data with a bar chart if your independent variable is:

- CATEGORIC — data that comes in distinct categories (e.g. flower colour, blood group).
- DISCRETE — data that can be counted in chunks, where there's no in-between value (e.g. number of bacteria — you can't have half a bacterium).

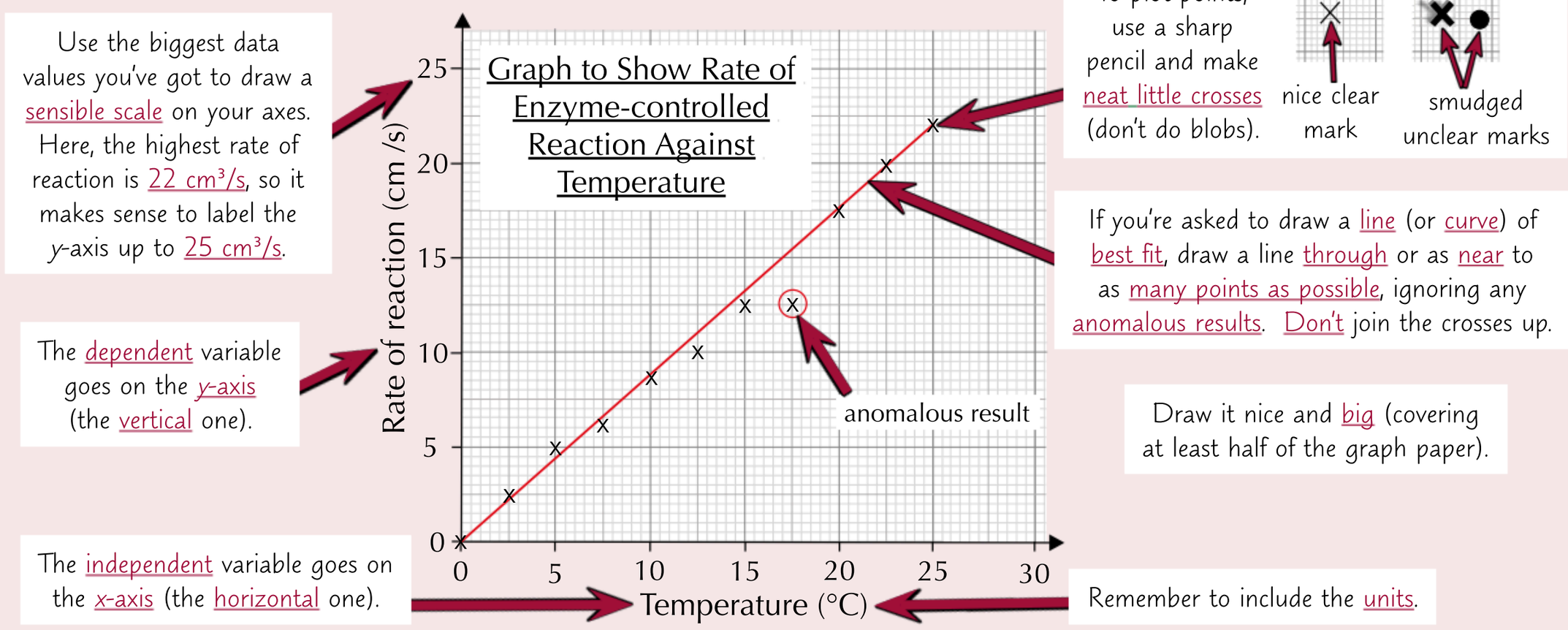
There are some golden rules you need to follow for drawing bar charts:



If Your Data is **Continuous**, Plot a **Graph**

If both variables are continuous (numerical data that can have any value within a range, e.g. length, volume, temperature) you should use a graph to display the data.

Here are the rules for plotting points on a graph:



Make sure you present your data correctly...

If your data is categoric you should present it on a bar chart. If your data is continuous, present it on a graph. Plotting data, on either a chart or a graph, is really useful for uncovering trends in your results. Turn over for more on that.

Interpreting Graphs

*Graph's aren't just fun to plot, they're also really useful for showing **trends** in your data.*

Graphs Can Give You a Lot of **Information** About Your **Data**

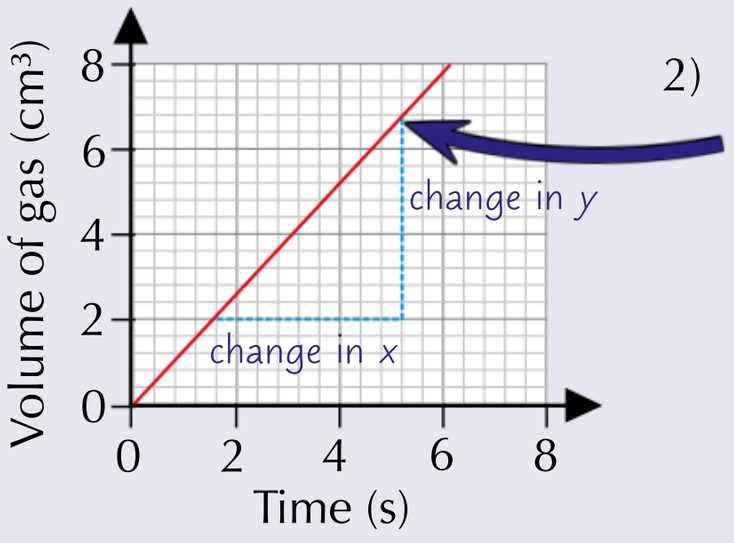
The gradient (slope) of a graph tells you how quickly the dependent variable changes if you change the independent variable.

$$\text{gradient} = \frac{\text{change in } y}{\text{change in } x}$$

This graph shows the volume of gas produced in a reaction against time. The graph is linear (it's a straight line graph), so you can simply calculate the gradient of the line to find out the rate of reaction.

1) To calculate the gradient, pick two points on the line that are easy to read and a good distance apart.

You can use this method to calculate other rates from a graph, so long as you always make *x* time.



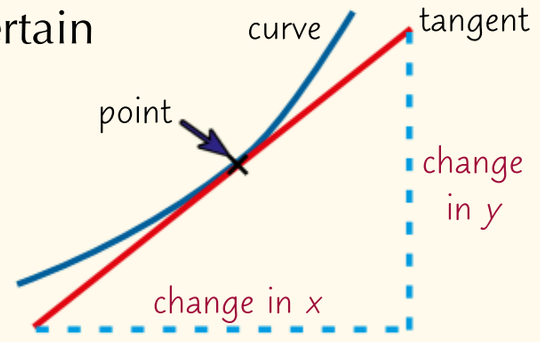
2) Draw a line down from the top point and a line across from the bottom point to make a triangle. The line drawn down the side of the triangle is the change in *y* and the line across the bottom is the change in *x*.

$$\text{Rate} = \text{gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{6.8 - 2}{5.2 - 1.6} = \frac{4.8\ \text{cm}^3}{3.6\ \text{s}} = 1.3\ \text{cm}^3/\text{s}$$

The units of the gradient are (units of *y*) ÷ (units of *x*).

The intercept of a graph is where the line of best fit crosses one of the axes. The *x*-intercept is where the line of best fit crosses the *x*-axis and the *y*-intercept is where it crosses the *y*-axis.

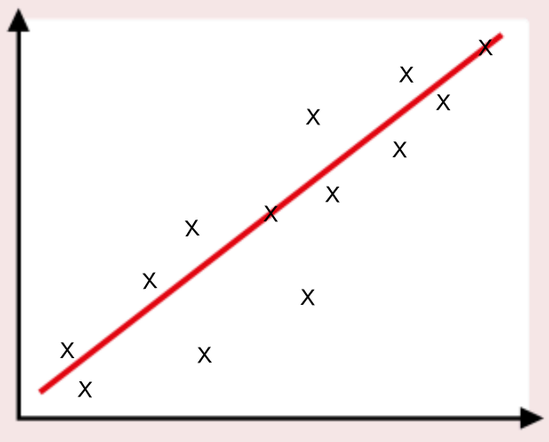
To find the gradient of a curve at a certain point, draw a tangent to the curve at that point. This is a straight line that touches the curve at that point only. Then just find the gradient of the tangent in the same way as above.



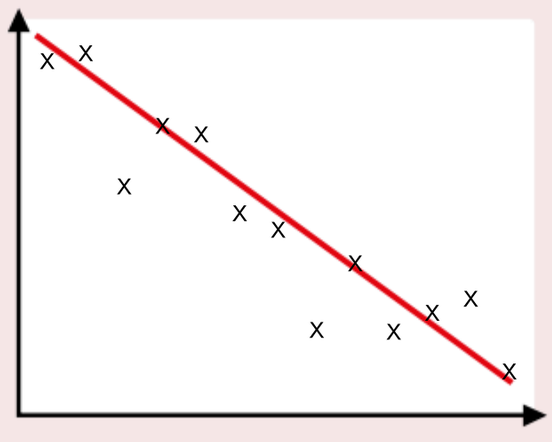
Graphs Show the **Relationship** Between Two Variables

You can get three types of correlation (relationship) between variables:

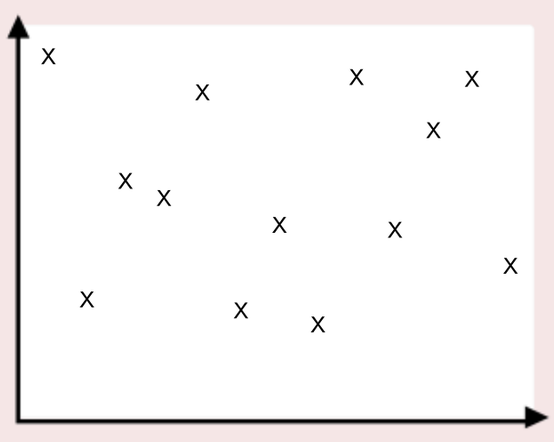
POSITIVE CORRELATION:
as one variable increases the other increases.



INVERSE (NEGATIVE) CORRELATION:
as one variable increases the other decreases.



NO CORRELATION:
no relationship between the two variables.



Just because there's correlation, it doesn't mean the change in one variable is causing the change in the other — there might be other factors involved (see page 179).

Science is all about finding relationships between things

The line of best fit is pretty useful — it shows if there is a correlation between the two variables in the investigation. Positive gradient = uphill line = positive correlation. Negative gradient = downhill line = negative correlation.

Units

*Graphs and maths skills are all very well, but the numbers don't mean much if you don't get the **units** right. Handily, you can **convert units** using **scaling prefixes**. This can save you from having to write a lot of 0's...*

S.I. Units Are Used All Round the World

It wouldn't be all that useful if I defined volume in terms of bath tubs, you defined it in terms of egg-cups and my pal Sarwat defined it in terms of balloons — it'd make it really hard to compare our data.

To stop this happening, scientists came up with a set of standard units, called S.I. units, that all scientists use to measure their data. Here are some S.I. units you'll see in biology:

Quantity	SI Base Unit
mass	kilogram, kg
length	metre, m
time	second, s

Scaling Prefixes Can Be Used for Large and Small Quantities

1) Quantities come in a huge range of sizes. For example, the volume of a swimming pool might be around 2 000 000 000 cm^3, while the volume of a cup is around 250 cm^3.

2) To make the size of numbers more manageable, larger or smaller units are used. These are the SI base unit (e.g. metres) with a prefix in front:

prefix	tera (T)	giga (G)	mega (M)	kilo (k)	deci (d)	centi (c)	milli (m)	micro (μ)	nano (n)
multiple of unit	10^{12}	10^{9}	1 000 000 (10^{6})	1000	0.1	0.01	0.001	0.000001 (10^{-6})	10^{-9}

3) These prefixes tell you how much bigger or smaller a unit is than the base unit. So one kilometre is one thousand metres.

4) To convert between units, you need to know the conversion factor:

The conversion factor is the number of times the smaller unit goes into the larger unit.

- To go from a bigger unit (like m) to a smaller unit (like cm), you multiply by the conversion factor.
- To go from a smaller unit (like g) to a bigger unit (like kg), you divide by the conversion factor.

5) Here are some conversions that'll be useful for GCSE biology:

Mass can have units of kg and g.

kg ⇄ g (× 1000 / ÷ 1000)

Length can have lots of units, including mm, μm and nm.

mm ⇄ μm ⇄ nm (× 1000 / ÷ 1000)

Time can have units of min and s.

min ⇄ s (× 60 / ÷ 60)

Volume can have units of m^3, dm^3 and cm^3.

m^3 ⇄ dm^3 ⇄ cm^3 (× 1000 / ÷ 1000)

Always Check The Values Used in Equations Have the Right Units

Always make sure the values you put into an equation or formula have the right units. In some cases you'll have to convert a value into different units before you can put it into the equation.

To make sure your units are correct, it can help to write them down on each line of your calculation.

EXAMPLE

A cell is viewed under a microscope with × 25 magnification. The image of the cell is 1.6 mm wide. Calculate the real width of the cell in μm.

MATHS SKILLS

1) Convert the image measurement from mm into μm by multiplying by the correct conversion factor.

1.6 mm → (× 1000) → 1600 μm

2) Find the real width using:
real size = image size ÷ magnification.

real size = image size ÷ magnification
= 1600 ÷ 25 = 64 μm

S.I. units help scientists to compare data...

It's easy to get in a muddle when converting between units, but there's a handy way to check you've done it right. If you're moving from a smaller unit to a larger unit (e.g. g to kg) the number should get smaller, and vice versa.

Maths Skills

*No, you've not opened up the wrong book — this is indeed a page on **maths**. Maths comes up a fair bit in biology, so you'd better be **comfortable** using it. Read on for some of the **key** maths skills you need...*

Standard Form Is Often Used For **Really Big** or **Really Small** Numbers

You need to know how to deal with numbers in standard form. They always look like this:

A is a number between 1 and 10, but not including 10.

$$A \times 10^{n}$$

n is the number of places the decimal point would move if you wrote the number out fully. It's negative for numbers less than 1, and positive for numbers greater than 1.

So 200 000 kg would be written as 2×10^{5} kg in standard form. And 0.0003 mm is 3×10^{-4} mm.

Do the **Same Thing** to **Both Sides** When **Rearranging Formulas**

Often you need to rearrange a formula to be able to solve it for the variable you want. You just need to do the same thing to each side until you get that variable on its own.

EXAMPLE

Rearrange magnification = $\frac{\text{image size}}{\text{real size}}$ to solve for 'real size'.

1) You're trying to get real size on its own.
2) Multiply both sides by real size to get rid of the fraction.
3) Simplify to tidy things up.
4) Divide both sides by magnification.
5) Simplify again.

$$\text{magnification} = \frac{\text{image size}}{\text{real size}}$$

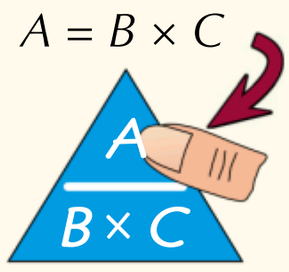
$$\text{magnification} \times \text{real size} = \frac{\text{image size}}{\text{real size}} \times \text{real size}$$

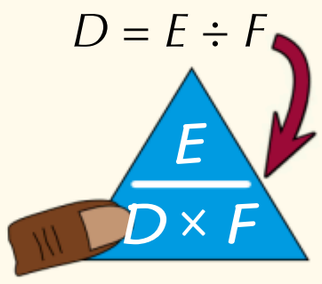
$$\text{magnification} \times \text{real size} = \text{image size}$$

$$\frac{\text{magnification} \times \text{real size}}{\text{magnification}} = \frac{\text{image size}}{\text{magnification}}$$

$$\text{real size} = \frac{\text{image size}}{\text{magnification}}$$

You can put simple formulas into formula triangles if you find them helpful:





To use them, cover up what you want to find and write down what's left showing.

Make Sure You Know How to Use Your **Calculator** Correctly

Here are a few things to remember to make sure you don't lose marks for silly calculation mistakes:

- You need to make sure that what you put in your calculator is the same calculation as is on the page, even if it looks a bit different.
- For example, to put $\frac{11.5+6.8}{3}$ into any standard calculator, you type (11.5 + 6.8) ÷ 3. Don't forget the brackets.
  You could also use the fraction button on a scientific calculator.

Your calculator will follow the order of operations for what you've typed. If you forget the brackets, it won't do the calculation you want and you'll get 13.766... instead of 6.1.

Use exact values from previous calculations in your calculator rather than rounding. Make sure you know how to find these values on your calculator — e.g. by using the 'Ans' button or scrolling up.

Rearranging formulas is a really important skill...

Some people love formula triangles, some people not so much. You can use them as a tool if you want to, but you need to know how to rearrange formulas properly, not least because complicated formulas don't work in triangles.

Drawing Conclusions

Once you've carried out an experiment and processed your data, it's time to work out ***what your data shows****.*

You Can **Only Conclude** What the Data Shows and **NO MORE**

1) Drawing conclusions might seem pretty straightforward — you just look at your data and say what pattern or relationship you see between the dependent and independent variables...

The table on the right shows the heights of pea plant seedlings grown for three weeks with different fertilisers.

Fertiliser	Mean growth / mm
A	13.5
B	19.5
None	5.5

CONCLUSION:
Fertiliser B makes pea plant seedlings grow taller over a three week period than fertiliser A.

2) ...but you've got to be really careful that your conclusion matches the data you've got and doesn't go any further.

You can't conclude that fertiliser B makes any other type of plant grow taller than fertiliser A — the results could be totally different.

3) You also need to be able to use your results to justify your conclusion (i.e. back up your conclusion with some specific data).

Over the three week period, fertiliser B made the pea plants grow 6 mm more on average than fertiliser A.

4) When writing a conclusion you need to refer back to the original hypothesis and say whether the data supports it or not:

If the hypothesis for this experiment was that adding fertiliser would increase the growth of plants and that different types of fertiliser would affect growth by different amounts, then the data supports the hypothesis.

Correlation DOES NOT Mean **Cause**

If two things are correlated (i.e. there's a relationship between them) it doesn't necessarily mean a change in one variable is causing the change in the other — this is really important, so don't forget it. There are three possible reasons for a correlation:

1. CHANCE: It might seem strange, but two things can show a correlation purely due to chance:

   E.g. a study might find a correlation between people's hair colour and their frisbee skills. But, other scientists who investigate it don't get a correlation — the results of the first study are just a fluke.

2. LINKED BY A 3RD VARIABLE: A lot of the time it may look as if a change in one variable is causing a change in the other, but it isn't — a third variable links the two things:

   E.g. there's a correlation between a higher water temperature and the amount of shark attacks. This isn't because warmer water makes sharks go crazy. The attacks are linked by a third variable — the number of people swimming (i.e. more people are swimming when the water's hotter, meaning more chance of a shark attack).

3. CAUSE: Sometimes a change in one variable does cause a change in the other. You can only make this conclusion when you've controlled all variables that could be affecting the result:

   E.g. there is a correlation between smoking and lung cancer. This is because chemicals in tobacco smoke cause lung cancer. This conclusion was only made once other variables (such as age and exposure to other things that cause cancer) had been controlled. Once they were controlled, the effect of smoking was isolated and shown to increase the risk of someone getting lung cancer.

Data can support or disprove a hypothesis — but it can't prove it

At the end of an investigation, refer back to the hypothesis you made at the start of the investigation to see whether your data supports or disproves it. And remember, you can't take your conclusion beyond what your data says.

Uncertainties and Evaluations

Uncertainty *is how sure you can really be about your data. There's a little bit of* ***maths*** *to do, and also a formula to learn, and then after that it's on to* ***evaluations****. Best get cracking with this page then...*

Uncertainty is the Amount of Error Your Measurements Might Have

1) The measurements you make will have some uncertainty in them, for example, due to limits in the resolution of the equipment you use (see page 173).
2) When you repeat a measurement, you often get a slightly different figure each time you do it due to random error. So random errors are another source of uncertainty — they can cause uncertainty in the mean of a set of results.
3) You can calculate the uncertainty of a mean result using the equation:
4) The larger the range, the less precise your results are and the more uncertainty there will be in your results. Uncertainties are shown using the '±' symbol.

The range is the largest value minus the smallest value.

$$\text{uncertainty} = \frac{\text{range}}{2}$$

EXAMPLE

The table below shows the results of a respiration experiment to determine the volume of carbon dioxide produced. Calculate the uncertainty of the mean.

MATHS SKILLS

Repeat	1	2	3	mean
Volume of CO_2 (cm^3)	20.2	19.8	20.0	20.0

1) First work out the range:
Range = largest value − smallest value
= 20.2 − 19.8 = 0.4 cm^3
2) Use the range to find the uncertainty: Uncertainty = range ÷ 2 = 0.4 ÷ 2 = 0.2 cm^3
So the uncertainty of the mean = 20.0 ± 0.2 cm^3

5) Measuring a greater amount of something helps to reduce uncertainty. E.g. measuring the amount of product over a longer period will reduce the percentage uncertainty in rate of reaction experiments.

Evaluations — Describe How it Could be Improved

1) An evaluation is a critical analysis of the whole investigation:

- You should comment on the method — was it valid? Did you control all the other variables to make it a fair test?
- Comment on the quality of the results — was there enough evidence to reach a valid conclusion? Were the results repeatable, reproducible, accurate and precise?
- Were there anomalous results? If there were, try to explain them — were they caused by errors in measurement? Were there other variables that could have affected the results? You should comment on the level of uncertainty in your results too.

2) Answering these questions will allow you to say how confident you are that your conclusion is right.
3) Then you can suggest any changes to the method that would improve the quality of the results and give you more confidence in your conclusion. E.g. you might increase the number of measurements you took. Taking more measurements at narrower intervals could give you a more accurate result:

Enzymes have an optimum temperature (a temperature they work best at). Say you do an experiment to find an enzyme's optimum temperature, with measurements at 10 °C, 20 °C, 30 °C, 40 °C and 50 °C. The results tell you the optimum is 40 °C. You could then repeat the experiment, taking more measurements around 40 °C to get a more accurate value for the optimum.

4) You could also make more predictions based on your conclusion, then further experiments could be carried out to test them.

When suggesting improvements to the investigation, always make sure that you say why you think this would make the results better.

The smaller the uncertainty, the more precise your results...

Remember the equation for uncertainty. You never know when you might need it — you could be expected to use it in the exams. You need to make sure all the data is in the same units though. E.g. if you had some measurements in metres and some in centimetres, you'd need to convert them all to the same units before you start calculating uncertainty.

Measuring Substances

Get your lab coats on, it's time to find out about the skills you'll need in ***experiments****...*

Use the **Right Apparatus** to Take **Readings**

1) Mass

Start by putting the container you are going to weigh your substance into on a balance. Set the balance to exactly zero and then weigh out the correct amount of your substance.

2) Temperature

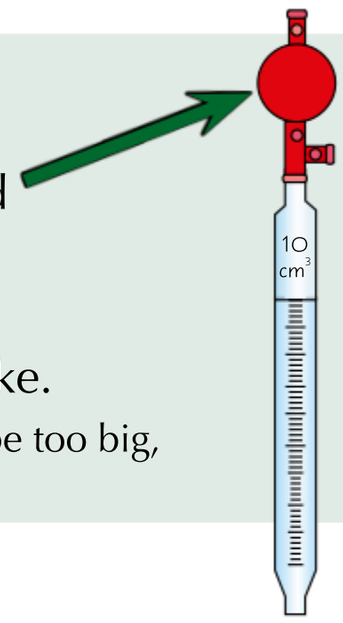
You can use a thermometer to measure the temperature of a solution. Make sure that the bulb of the thermometer is completely submerged in the solution and that you wait for the temperature to stabilise before you take your initial reading. Read off the scale on the thermometer at eye level to make sure it's correct.

When you're reading off a scale, write down the value of the graduation that the amount is closest to.

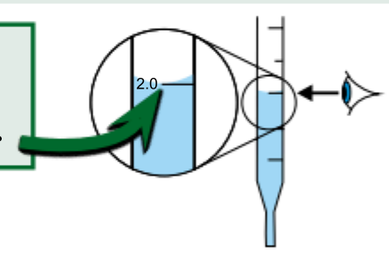
3) Volume of a Liquid

- Using a pipette — Pipettes are used to suck up and transfer volumes of liquid between containers. Dropping pipettes are used to transfer drops of liquid. Graduated pipettes are used to transfer accurate volumes. A pipette filler is attached to the end of a graduated pipette, to control the amount of liquid being drawn up.
- Using a measuring cylinder — Measuring cylinders come in all different sizes. Make sure you choose one that's the right size for the measurement you want to make.

  E.g. it's no good using a huge 1 dm^3 cylinder to measure out 2 cm^3 of a liquid — the graduations will be too big, and you'll end up with massive errors. It'd be much better to use one that measures up to 10 cm^3.

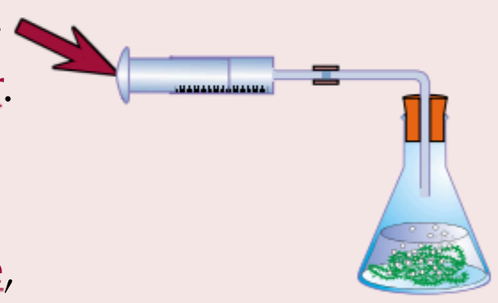


Always read the volume from the bottom of the meniscus (the curved upper surface of the liquid) when it's at eye level.



4) Volume of a Gas

- Using a gas syringe is the most accurate way to measure the volume of gas.
- Alternatively, you can use an upturned measuring cylinder filled with water. The gas will displace the water so you can read the volume off the scale.
- Or you can count the bubbles produced or measure the length of a gas bubble drawn along a tube (see p.74). These methods are less accurate, but will give you relative amounts of gas to compare results.



When you're measuring a gas, you need to make sure that the equipment is set up so that none of the gas can escape, otherwise your results won't be accurate.

5) pH

- Indicators are dyes that change colour depending on whether they're in an acid or an alkali — you just need to add a couple of drops of the indicator to the solution you're interested in.

  E.g. universal indicator is a mixture of indicators that changes colour gradually as pH changes. It's useful for estimating the pH of a solution based on its colour.
- Indicator paper is useful if you don't want to colour the entire solution that you're testing. It changes colour depending on the pH of the solution it touches. You can also hold a piece of damp indicator paper in a gas sample to test its pH.
- pH meters have a digital display that gives an accurate value for the pH of a solution.

Blue litmus paper turns red in acidic conditions and red litmus paper turns blue in alkaline conditions.

Read off the scale carefully when taking readings

If you're reading off a thermometer, a pipette or a measuring cylinder, make sure you take the readings at eye level.

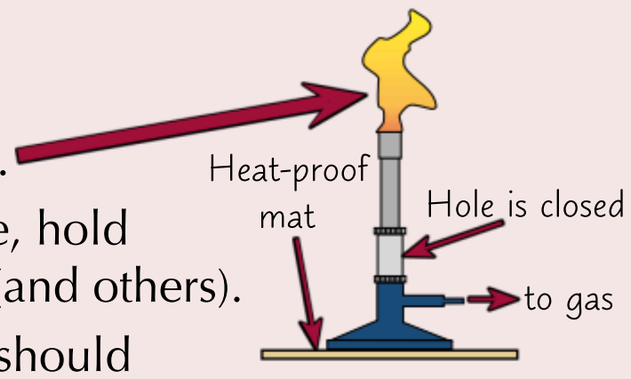
Heating, Safety and Ethics

*Experiments can be pretty **dangerous**, so you need to be aware of the **hazards** and take **sensible precautions**.*

Bunsen Burners Have a **Naked Flame**

Bunsen burners are good for heating things quickly. But you need to use them safely:

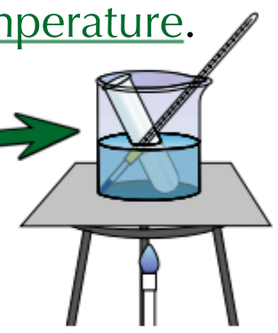
- Always use a Bunsen burner on a heat-proof mat.
- If your Bunsen burner is alight but not heating anything, make sure you close the hole so that the flame becomes yellow and clearly visible.
- Use the blue flame to heat things. If you're heating a vessel in the flame, hold it at the top (e.g. with tongs) and point the opening away from yourself (and others).
- If you're heating something over the flame (e.g. a beaker of water), you should put a tripod and gauze over the Bunsen burner before you light it, and place the vessel on this.



The **Temperature** of **Electric Water Baths** and **Heaters** Can Be **Set**

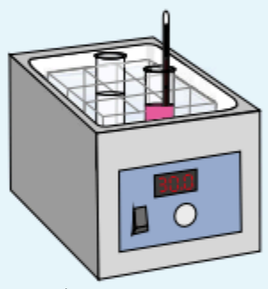
A water bath is a container filled with water that can be heated to a specific temperature.

- A simple water bath can be made by heating a beaker of water over a Bunsen burner and monitoring the temperature with a thermometer. However, it's difficult to keep the temperature of the water constant.
- An electric water bath will monitor and adjust the temperature for you.



Method

1) Set the temperature on the water bath, and allow the water to heat up.
2) Place the vessel containing your substance in the water bath using test tube holders or tongs. The level of water outside the vessel should be just above the level of the substance in the vessel.
3) The substance will then be warmed to the same temperature as the water. As the substance in the vessel is surrounded by water, the heating is very even. Use a thermometer to check its temperature.



Electric heaters are often made up of a metal plate that can be heated to a specified temperature. You place the vessel containing the substance you want to heat on top of the hot plate. The vessel is only heated from below, so you'll usually have to stir the substance inside to make sure it's heated evenly.

Think about **Safety Precautions Before** your Experiment

Both methods and chemicals carry risks, e.g. the chemicals you're using may be hazardous — they might be flammable (catch fire easily), or they might irritate or burn your skin. It's important to...

- wear sensible clothing when you're in the lab (e.g. open shoes won't protect your feet from spillages).
- wear a lab coat to protect your skin and clothing when you're doing an experiment. You may need to also wear safety goggles and gloves.
- be aware of general safety in the lab, e.g. keep anything flammable away from lit Bunsen burners, don't directly touch any hot equipment, handle glassware carefully so it doesn't break, etc.

You Need to Think About **Ethical Issues** In Your Experiments

Any organisms involved in your investigations need to be treated safely and ethically.

1) Animals need to be treated humanely:
   - they should be handled carefully.
   - wild animals captured (e.g. for investigating distribution) should be returned to their original habitat.
   - animals kept in the lab should be well cared for, e.g. not kept in overcrowded conditions.
2) If you are carrying out an experiment involving other students (e.g. investigating the effect of caffeine on reaction time), they should not be forced to participate against their will or feel pressured to take part.

Potometers

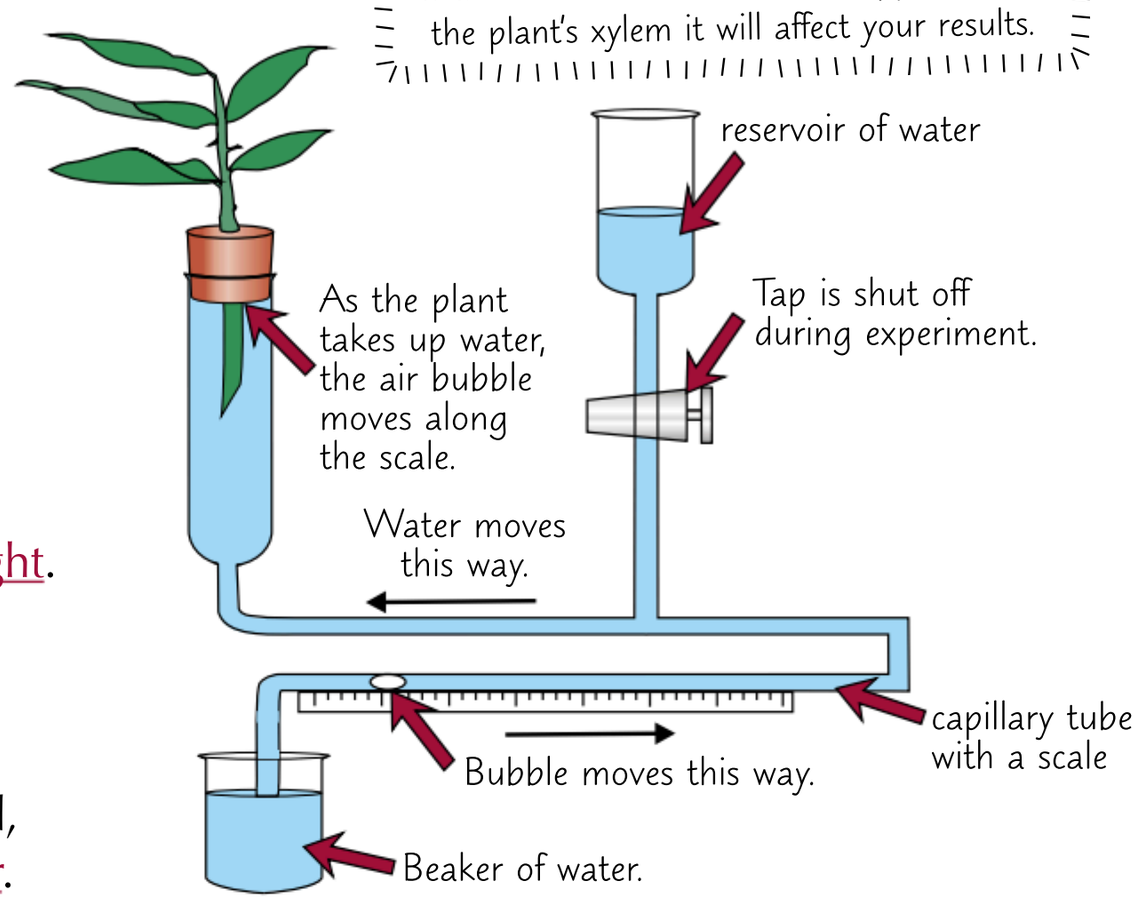
*Next up, **potometers**, then on to how to use a **microscope** and a **ruler** to work out the size of a single cell...*

Potometers Should Be Set Up **Underwater**

A potometer is a special piece of apparatus used to measure the water uptake by a plant, e.g. to estimate the transpiration rate of a plant (see page 50).
Here's how to set one up:

1) Cut a shoot underwater to prevent air from entering the xylem. Cut it at a slant to increase the surface area available for water uptake.
2) Assemble the potometer in water and insert the shoot under water, so no air can enter.
3) Remove the apparatus from the water but keep the end of the capillary tube submerged in a beaker of water.
4) Check that the apparatus is watertight and airtight.
5) Dry the leaves, allow time for the shoot to acclimatise and then shut the tap.
6) Remove the end of the capillary tube from the beaker of water until one air bubble has formed, then put the end of the tube back into the water.

If there are air bubbles in the apparatus or the plant's xylem it will affect your results.



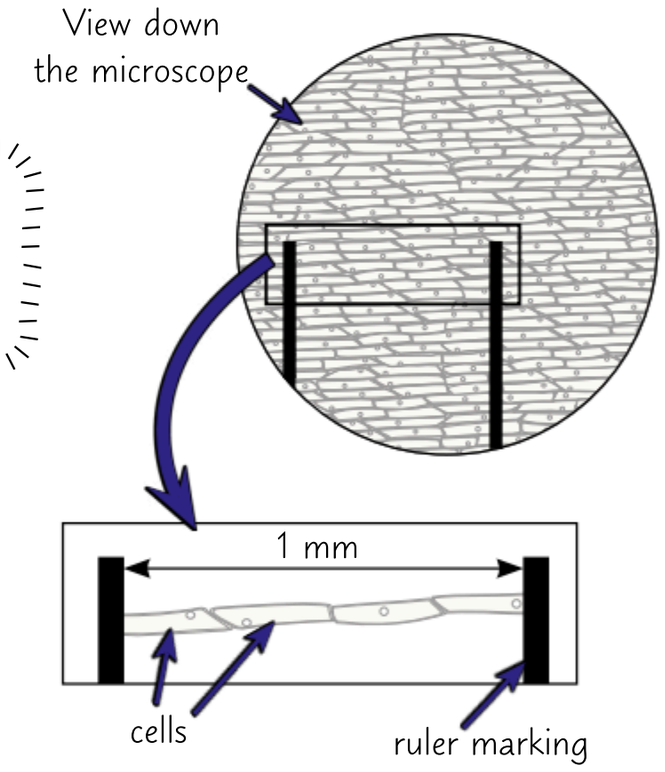
You Can **Measure** the **Size** of a **Single Cell**

When viewing cells under a microscope, you might need to work out their size.

To work out the size of a single cell:

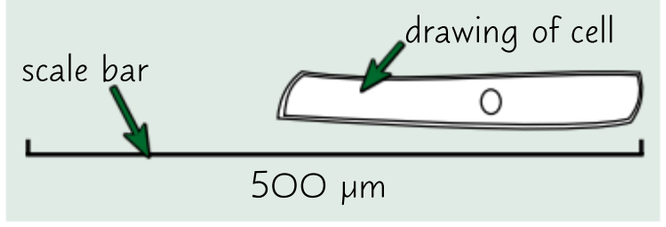
1) Place a clear ruler on top of your microscope slide. Clip the ruler and slide onto the stage.
2) Select the objective lens that gives an overall magnification of x 100.
3) Adjust the focus to get a clear image of the cells.
4) Move the ruler so that the cells are lined up along 1 mm. Then count the number of cells along this 1 mm sample.
5) 1 mm = 1000 µm. So to calculate the length of a single cell in µm, you just need to divide 1000 µm by the number of cells in the sample.
E.g. if you counted 4 cells in 1 mm, the length of a single cell would be: 1000 ÷ 4 = 250 µm

You can read all about using a microscope on page 6.



Use the **Cell Size** to Work out the **Length** of a **Scale Bar**

- If you draw a diagram of a cell you've observed under a microscope, you might want to include a scale bar.
- Once you know the size of one cell, you can use it to calculate how long your scale bar should be.
- To draw a 500 µm scale bar, just use this formula:



$$\text{scale bar length (µm)} = \frac{\text{drawn length of cell (µm)} \times 500}{\text{actual length of cell (µm)}}$$

You can draw an accurate diagram of a cell when you know its size

These techniques might seem a bit tricky at first, but if you practice them they'll become second nature. Honest.

Sampling

*You need to be able to carry out **sampling** that'll give you **non-biased results**. First up **why**, then **how**...*

You **Can't** Usually Study **Every Single Organism**

When you're investigating a population, it's generally not possible to study every single organism in the population. This means that you need to take samples of the population you're interested in.

The sample data will be used to draw conclusions about the whole population, so it's important that it accurately represents the whole population.
To make sure a sample represents the population, it should be random.

If a sample doesn't represent the population as a whole, it's said to be biased.

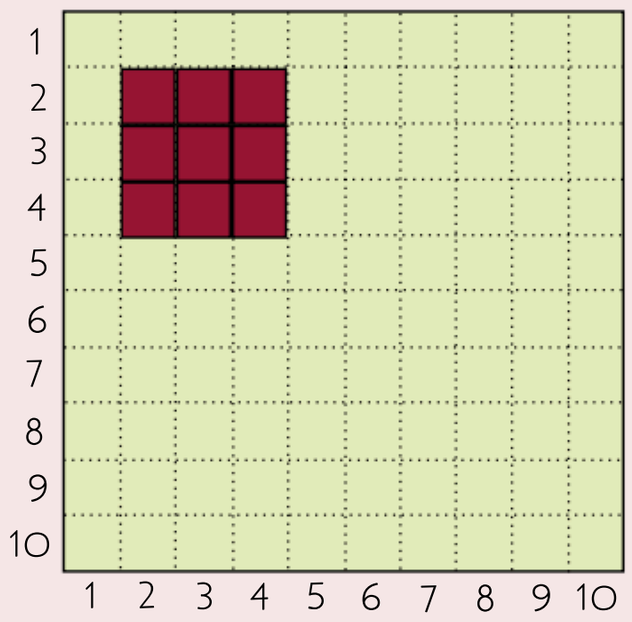
Organisms Should Be Sampled At **Random Sites** in an Area

- If you're interested in the distribution of an organism in an area, or its population size, you can take population samples in the area you're interested in using quadrats or transects (see pages 147-148).
- If you only take samples from one part of the area, your results will be biased — they may not give an accurate representation of the whole area.
- To make sure that your sampling isn't biased, you need to use a method of choosing sampling sites in which every site has an equal chance of being chosen. For example:

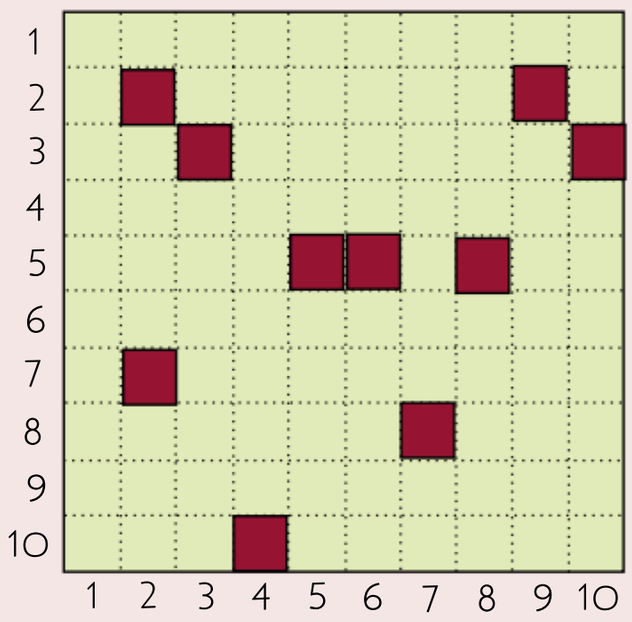
If you're looking at plant species in a field...

1) Divide the field into a grid.
2) Label the grid along the bottom and up the side with numbers.
3) Use a random number generator (on a computer or calculator) to select coordinates, e.g. (2,6).
4) Take your samples at these coordinates.

Non-random sampling
Only looks at a small part of the field.



Random sampling
Randomly selects squares from all over the field.



Health Data Should be Taken from **Randomly Selected People**

As mentioned above, it's not practical (or even possible) to study an entire human population. You need to use random sampling to choose members of the population you're interested in. E.g.

A health professional is investigating how many people diagnosed with Type 2 diabetes in a particular country also have heart disease:

1) All the people who have been diagnosed with Type 2 diabetes in the country of interest are identified by hospital records. In total, there are 270 196 people. These people are assigned a number between 1 and 270 196.
2) A random number generator is used to choose the sample group (e.g. it selects the individuals #72 063, #11 822, #193 123, etc.)
3) The proportion of people in the sample that have heart disease can be used to estimate the total number of people with Type 2 diabetes that also have heart disease.

Random number generators stop bias creeping in

Sampling is an important part of an investigation. It needs to be done randomly, or the data won't be worth much.

Comparing Results

*After collecting all your data, a few **calculations** might be needed to work out what it actually shows.*

Percentage Change Allows you to Compare Results

When investigating the change in a variable, you may want to compare results that didn't have the same initial value. For example, you may want to compare the change in mass of potato cylinders left in different concentrations of sugar solution that had different initial masses (see page 18).

One way to do this is to calculate the percentage change. You work it out like this:

$$\text{percentage (\%) change} = \frac{\text{final value} - \text{original value}}{\text{original value}} \times 100$$

EXAMPLE

A student is investigating the effect of the concentration of sugar solution on potato cells. She records the mass of potato cylinders before and after placing them in sugar solutions of different concentrations. The table below shows some of her results.

MATHS SKILLS

Potato cylinder	Concentration (mol/dm^3)	Mass at start (g)	Mass at end (g)
1	0.0	7.5	8.7
2	1.0	8.0	6.8

Which potato cylinder had the largest percentage change?

1) Stick each set of results into the equation:

$$\%\ \text{change} = \frac{\text{final value} - \text{original value}}{\text{original value}} \times 100$$

1. $\frac{8.7 - 7.5}{7.5} \times 100 = 16\%$

The mass at the start is the original value. The mass at the end is the final value.

2. $\frac{6.8 - 8.0}{8.0} \times 100 = -15\%$

Here, the percentage change is negative, which means the mass has decreased.

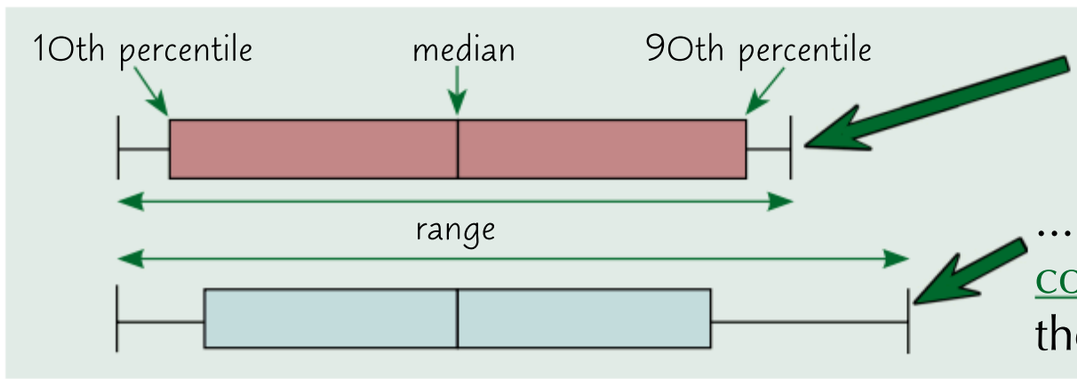
2) Compare the results. **16% is greater than 15%, so the potato cylinder in the 0.0 mol/dm^3 sugar solution had the largest percentage change.**

Percentiles Tell you Where in Your Data Set a Data Point Lies

1) Percentiles are useful if you want to compare the value of one data point to the rest of your data.
2) To find a percentile, you rank your data from smallest to largest, then divide it into one hundred equal chunks. Each chunk is one percentile.
3) This means that each percentile represents one percent of the data, and so the value of a percentile tells you what percentage of the data has a value lower than the data points in that percentile.

E.g. Mike is in the 90th percentile for height in his group of friends. This means that 90% of the group are shorter than Mike.

4) Percentiles can be used to give a more realistic idea of the spread of data than the range (see p.180) — by finding the range between the 10th and 90th percentiles in a data set (the middle 80% of the data), you can look at the spread of the data while ignoring any outlying results.



An outlier is a value that's much larger or smaller than the rest of the values in a data set.

The median is the middle value (see p.174). It's also the 50th percentile.

Good practical skills are needed when you're doing an investigation...

...but you also need to know about them for your exams. You're guaranteed to be tested on your practical knowledge, so if you've merrily skipped through this section in a bit of a rush, you'd better go back and read it through again.

Practice Exams

Once you've been through all the questions in this book, you should feel pretty confident about the exams. As final preparation, here is a set of **practice exams** to really get you set for the real thing. The time allowed for each paper is 1 hour 45 minutes. These papers are designed to give you the best possible preparation for your exams.

CGP Practice Exam Paper GCSE Biology

GCSE Biology

Paper 1

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
- Do all rough work on the paper.
- Cross out any work you do not want to be marked.

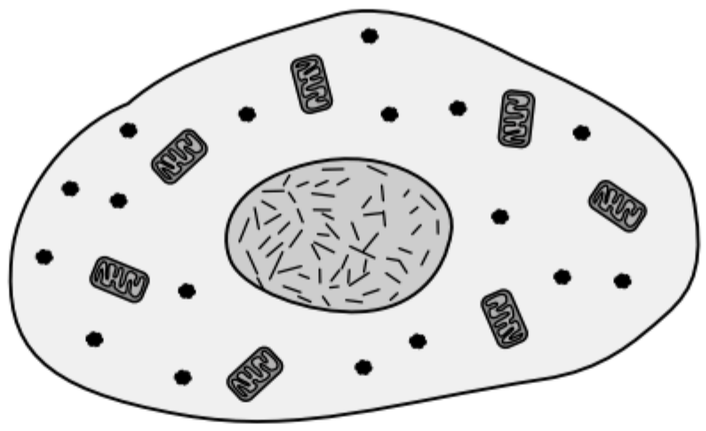
Information for candidates
- The marks available are given in brackets at the end of each question.
- There are 100 marks available for this paper.
- You are allowed to use a calculator.
- You should use good English and present your answers in a clear and organised way.

Advice to candidates
- In calculations show clearly how you worked out your answers.

For examiner's use							
Q	Attempt Nº			Q	Attempt Nº		
	1	2	3		1	2	3
1				6			
2				7			
3				8			
4				9			
5				10			
				Total			

1 **Figure 1** shows an animal cell.

Figure 1



1.1 Give **two** ways in which the cell in **Figure 1** is different from a bacterial cell.

.....

.....

.....

[2 marks]

1.2 Not all animal cells have the same structure as the cell in **Figure 1**.
Explain why.

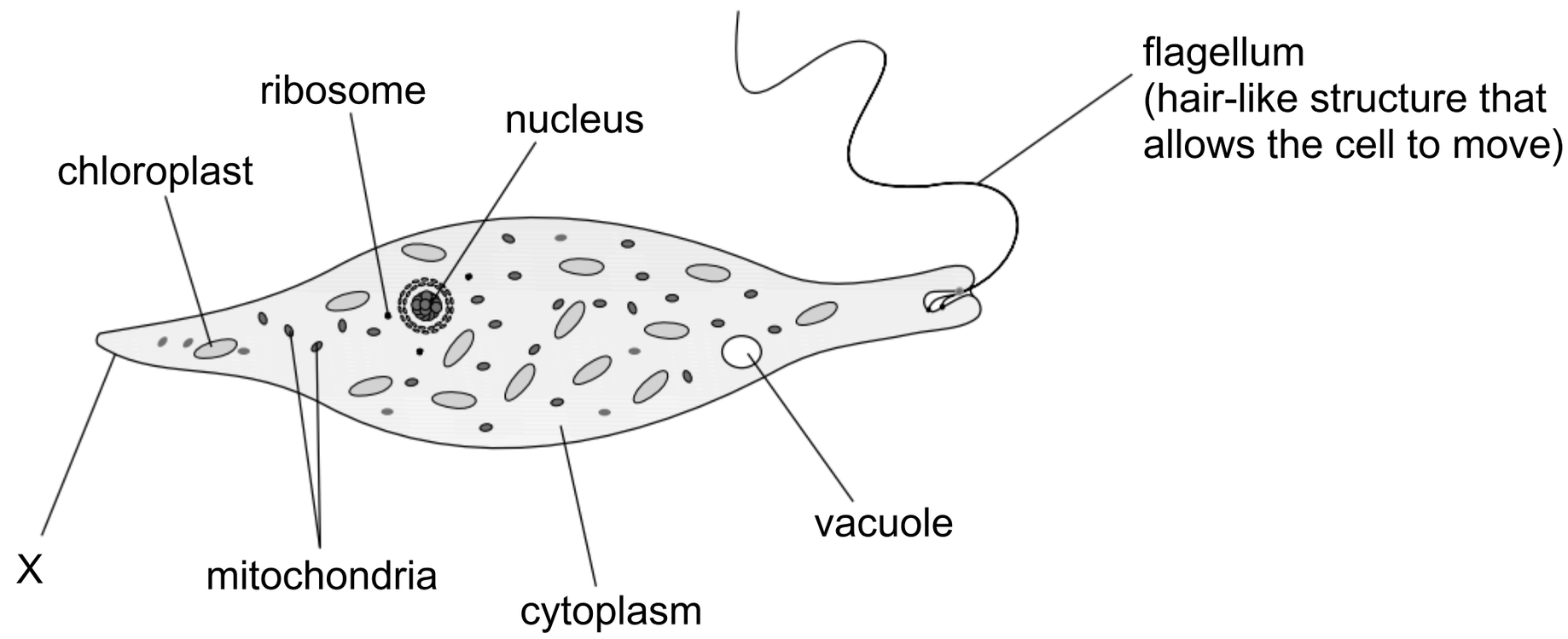
.....

.....

[1 mark]

Figure 2 shows a single-celled organism called *Euglena*, found in pond water.

Figure 2



1.3 Name part **X**.

.....

[1 mark]

Question 1 continues on the next page

1.4 *Euglena* is a eukaryote. Which of the following is **not** a eukaryote?
Tick **one** box.

☐ sperm cell ☐ muscle cell ☐ fruit fly ☐ *E. coli* bacteria

[1 mark]

1.5 A scientist viewed an individual *Euglena* under a microscope with × 150 magnification.
The real length of the *Euglena* was calculated to be 0.054 mm.
Calculate the length of the image of the Euglena. Use the formula:

$$\text{magnification} = \frac{\text{image size}}{\text{real size}}$$

.....

.....

..... mm

[2 marks]

1.6 When *Euglena* was first discovered, scientists disagreed over whether it was a plant or an animal.
Compare the features of plant and animal cells.
Include details of their features in your answer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6 marks]

2 Antibiotics can be used to kill bacteria and help a patient fight disease.

Scientists developing antibiotics were comparing the effectiveness of two different antibiotics, X and Y, against a bacterium. They grew the bacteria on agar jelly in 20 Petri dishes — forming an even covering of bacteria on the top of the agar.

They took three identical discs of paper and soaked each of them in one of three different liquids, shown in **Table 1**.

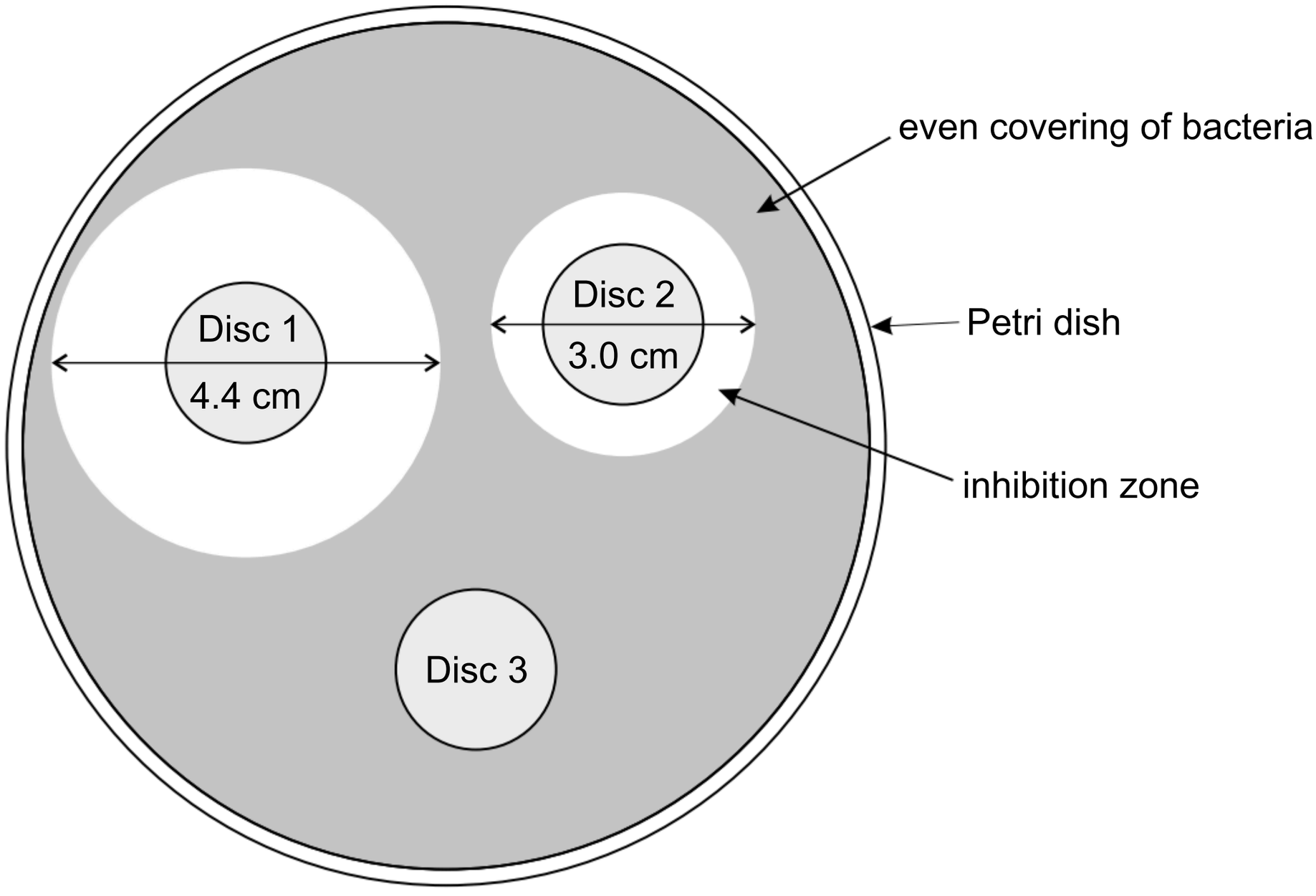
Table 1

Disc	Liquid soaked in
1	Antibiotic X
2	Antibiotic Y
3	Distilled water

The discs were then placed onto the agar in each dish. The dishes were kept in an incubator at 35 °C for two days, and then they were examined.

Figure 3 shows a typical dish.

Figure 3



Question 2 continues on the next page

2.1 Explain the purpose of Disc **3**.

.....

.....

[2 marks]

2.2 Give **two** things that the scientists should have done to avoid contaminating the bacterial cultures.

.....

.....

[2 marks]

2.3 In school labs, the maximum incubation temperature is 25 °C.
Explain why.

.....

[1 mark]

2.4 The diameter of each inhibition zone is shown on **Figure 3**.
Calculate the areas of the inhibition zones for antibiotics **X** and **Y**.
Use the equation area = πr^2.
Use the areas to compare the effectiveness of the two antibiotics.

.....

.....

.....

.....

.....

[3 marks]

The most effective antibiotic was manufactured and prescribed to patients.
Two years later, scientists repeated the experiment with a fresh sample of the bacteria and found that no inhibition zone was produced around the disc soaked in this antibiotic.

2.5 Suggest an explanation for this observation.

.....

[1 mark]

3 Measles, mumps and rubella are all examples of communicable diseases.

3.1 How is measles spread between people?
Tick **one** box.

- ☐ By droplets from an infected person's sneeze or cough.
- ☐ By sexual contact.
- ☐ By eating contaminated food.
- ☐ By a vector.

[1 mark]

3.2 The MMR vaccine helps to protect against measles, mumps and rubella.
Explain how vaccination helps to protect the body against a disease.

.....

.....

.....

.....

.....

.....

[4 marks]

Zika virus disease is another example of a communicable disease.
Symptoms of infection include fever, skin rashes, muscle and joint pain, and headaches. The virus that causes the disease is spread by a mosquito vector.

3.3 Suggest **two** ways that the spread of the Zika virus disease could be reduced.

.....

.....

.....

[2 marks]

Question 3 continues on the next page

3.4 Which of the following communicable diseases is also spread by a vector?
Tick **one** box.

- ☐ influenza
- ☐ malaria
- ☐ HIV
- ☐ athlete's foot

[1 mark]

3.5 Tuberculosis (TB) is a communicable disease caused by *M. tuberculosis* bacteria.
A strain of this bacteria was observed to have a mean division time of 18 hours.
How many cells would be produced from one bacterium of this strain in 3 days?

.....

.....

..... cells

[2 marks]

4 Different chemical reagents can be used to test for the presence of certain molecules in samples of food.

4.1 A student prepared a food sample in order to test whether the sample contained protein.
What reagent should be used for this test?
Tick **one** box.

- ☐ Benedict's solution
- ☐ iodine solution
- ☐ biuret solution
- ☐ Sudan III stain solution

[1 mark]

The student tested four different food samples for reducing sugars.
The results are shown in **Table 2**.

Table 2

Sample	Colour of sample
A	blue
B	brick-red
C	yellow
D	green

4.2 Name the reagent that the student would have used to test for reducing sugars.

.....

[1 mark]

4.3 Which of the samples in **Table 2** didn't contain reducing sugars?
Tick **one** box.

- ☐ A
- ☐ B
- ☐ C
- ☐ D

[1 mark]

Question 4 continues on the next page

Lactose is a reducing sugar commonly found in dairy products, such as milk and cheese. An enzyme called lactase breaks down lactose during digestion. The resulting products are the sugars glucose and galactose. These are absorbed into the blood from the small intestine.

4.4 Describe how the small intestine is adapted to absorb molecules such as glucose.

.....

.....

.....

.....

[3 marks]

4.5 Lactose intolerance is a digestive problem caused by insufficient production of lactase.

To test for lactose intolerance, the patient is given a drink of lactose solution.
A blood sample is then taken every 30 minutes for two hours.
The blood is tested to see how much sugar it contains.
Suggest what will happen to the blood sugar level of a person who is lactose intolerant during the test. Explain your answer.

.....

.....

.....

.....

[2 marks]

5 The lung is a specialised gas exchange organ.

5.1 What is meant by the term 'organ'?

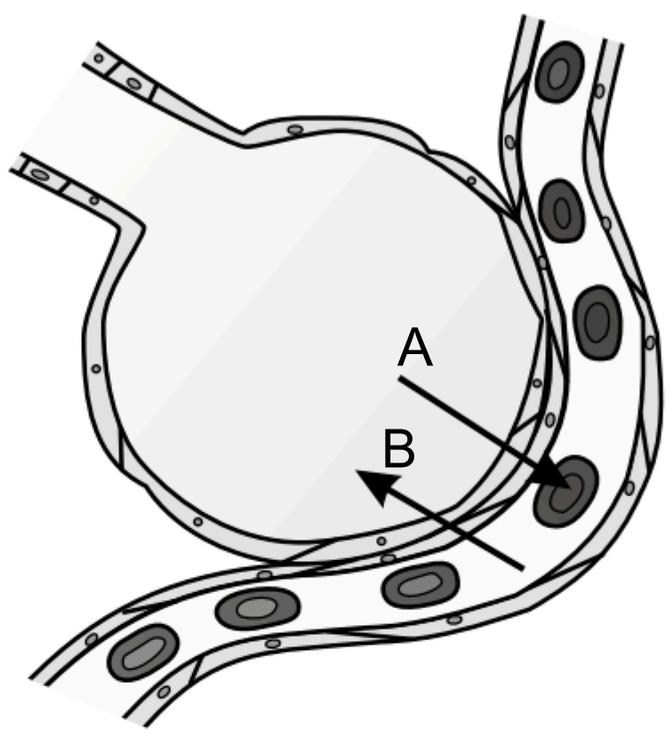
.....

.....

[1 mark]

The lung contains millions of air sacs called alveoli.
Figure 4 shows an alveolus and a blood capillary.

Figure 4



5.2 The arrows in **Figure 4** show the net movement of two gases, **A** and **B**.
Name gases **A** and **B**.

A .....

B .....

[1 mark]

5.3 Alveoli are surrounded by a network of capillaries.
Explain why.

.....

.....

.....

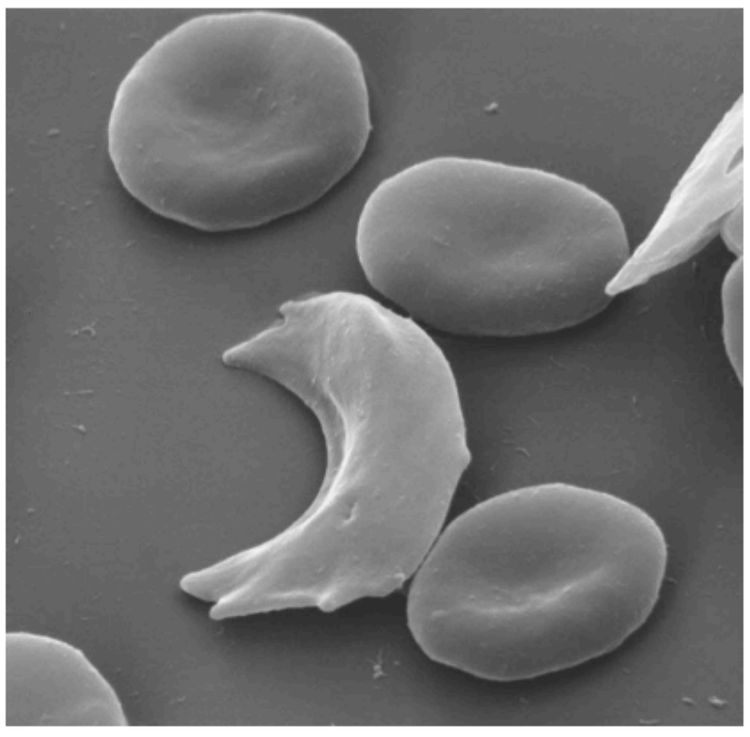
[2 marks]

Question 5 continues on the next page

5.4 **Figure 4** shows red blood cells in a capillary. Red blood cells are flexible, which allows them to fit through capillaries.

Sickle cell anaemia is a genetic disorder of the blood where the red blood cells become rigid and sickle-shaped. **Figure 5** shows a sickle-shaped red blood cell and some normal red blood cells.

Figure 5



EYE OF SCIENCE/SCIENCE PHOTO LIBRARY

Suggest why a person with sickle cell anaemia may experience breathlessness.

.....

.....

.....

.....

.....

.....

[3 marks]

6 Blood flows around the body in arteries, veins and capillaries.

6.1 **Table 3** shows some of the features of these three different types of blood vessel.

Table 3

	Type of blood vessel		
	A	**B**	**C**
Walls	Thick, muscular	Very thin	Thin
Presence of valves	No	No	Yes
Pressure of blood in vessels	High	Low	Low

Identify blood vessels **A**, **B** and **C**.

A .....

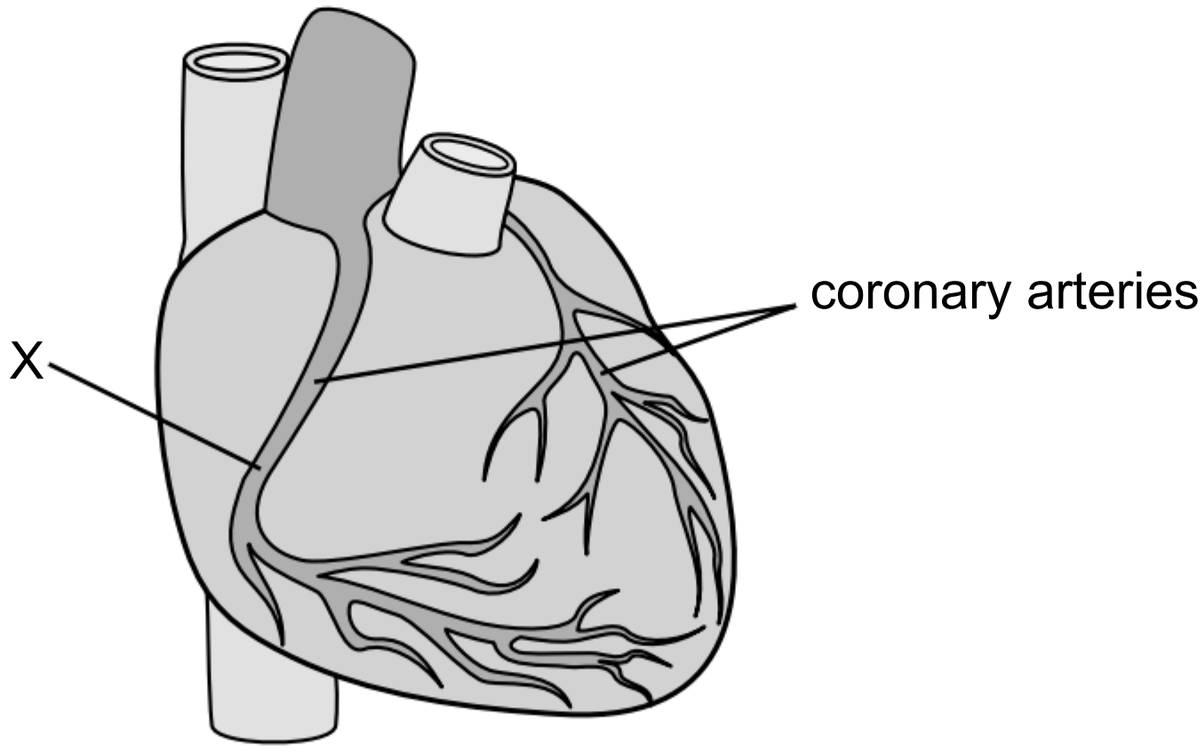
B .....

C .....

[2 marks]

Figure 6 shows a human heart.

Figure 6



6.2 Explain what would happen if the coronary artery was blocked at the point labelled **X**.

.....

.....

[2 marks]

Question 6 continues on the next page

There are fatty deposits on the wall of one of a patient's coronary arteries.
The patient's doctor recommends that the patient is treated using a stent.

6.3 Explain how having a stent fitted could help the patient.

.....

.....

[1 mark]

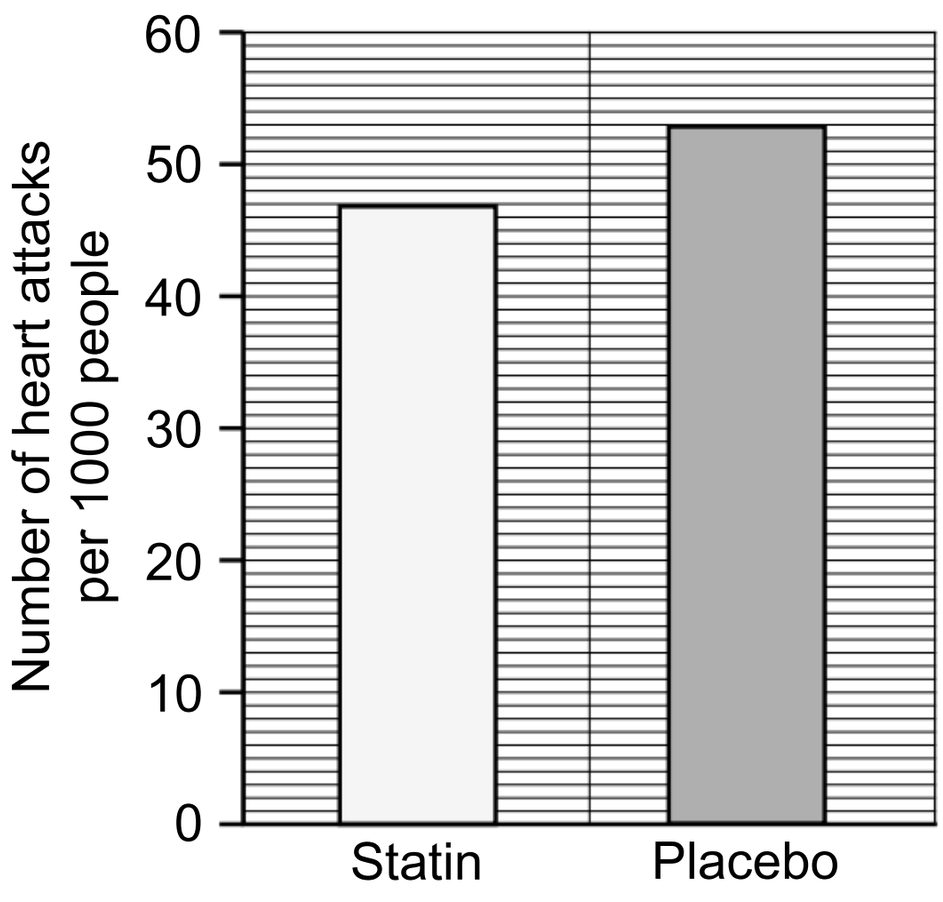
Statins are a type of drug used to reduce the risk of coronary heart disease.

A new statin has been developed and was initially put through preclinical testing.
Following this, the drug was given to thirty healthy volunteers in a clinical trial.

A trial was then carried out on 2000 patients who had previously had a heart attack — half were given the drug and half were given a placebo.

Figure 7 shows the results after five years of treatment.

Figure 7



6.4 In preclinical testing, new drugs are tested on human cells and tissues and on live animals.
Give **two** reasons why preclinical tests are carried out.

.....

.....

[2 marks]

6.5 During the clinical trial, what was the main reason for giving the drug to healthy human volunteers?

.....

[1 mark]

6.6 Describe the results of the trial.

.....

.....

[1 mark]

6.7 Give **one** reason why the results of the trial could be considered valid.

.....

.....

[1 mark]

Turn over for the next question

7 A student did an experiment to investigate the effect of temperature on the action of the enzyme amylase. The method used is shown below.

1. Add a set volume of starch solution to a test tube and the same volume of amylase solution to another.
2. Place the test tubes in a water bath at 10 °C.
3. Allow the starch and amylase solutions to reach the temperature of the water bath, then mix them together and return the mixture to the water bath.
4. Take a small sample of the mixture every ten seconds and test for starch.
5. Stop the experiment when starch is no longer present in the sample.
6. Repeat the experiment at different temperatures.

7.1 What happens to the starch solution during the experiment?

.....

[1 mark]

7.2 Give **one** variable that the student should have controlled during the experiment.

.....

[1 mark]

7.3 Suggest how the student could have tested for the presence of starch in **Step 4**.

.....

.....

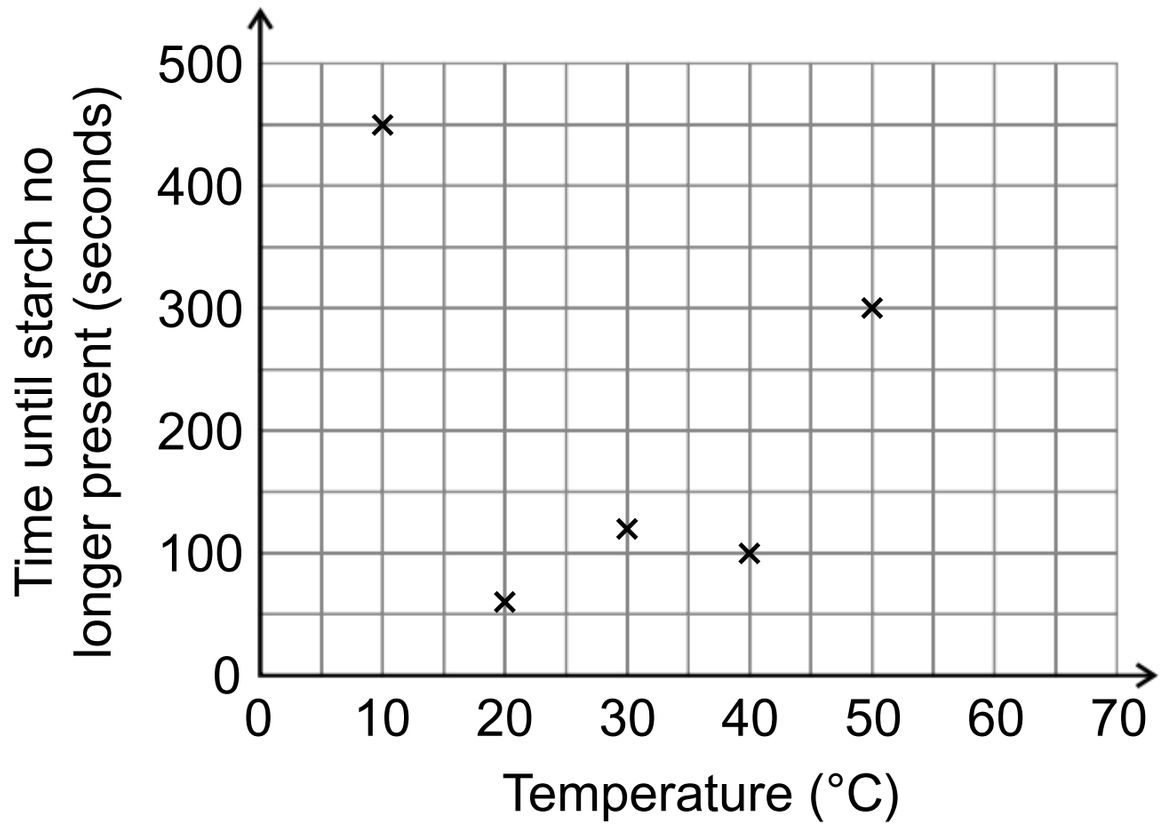
.....

.....

[3 marks]

Figure 8 shows a graph of the student's results.

Figure 8



7.4 Calculate the rate of the reaction at 40 °C.
Use the formula:

$$\text{Rate} = \frac{1000}{\text{time}}$$

.....

.....

..... s^{-1}

[2 marks]

7.5 At 60 °C, the amount of starch in the mixture did not decrease.
Explain why.

.....

.....

.....

[3 marks]

Question 7 continues on the next page

7.6 The student thinks that one of the results shown in **Figure 8** is likely to be anomalous. Explain which is the anomalous result and suggest what the student might have done to cause this anomalous result.

.....

.....

.....

[3 marks]

7.7 What conclusion can be drawn from the results shown in **Figure 8**?

.....

.....

[1 mark]

The student plans to alter the experiment in order to investigate the effect of pH on the action of amylase.

7.8 Describe how the pH could be altered in the experiment.

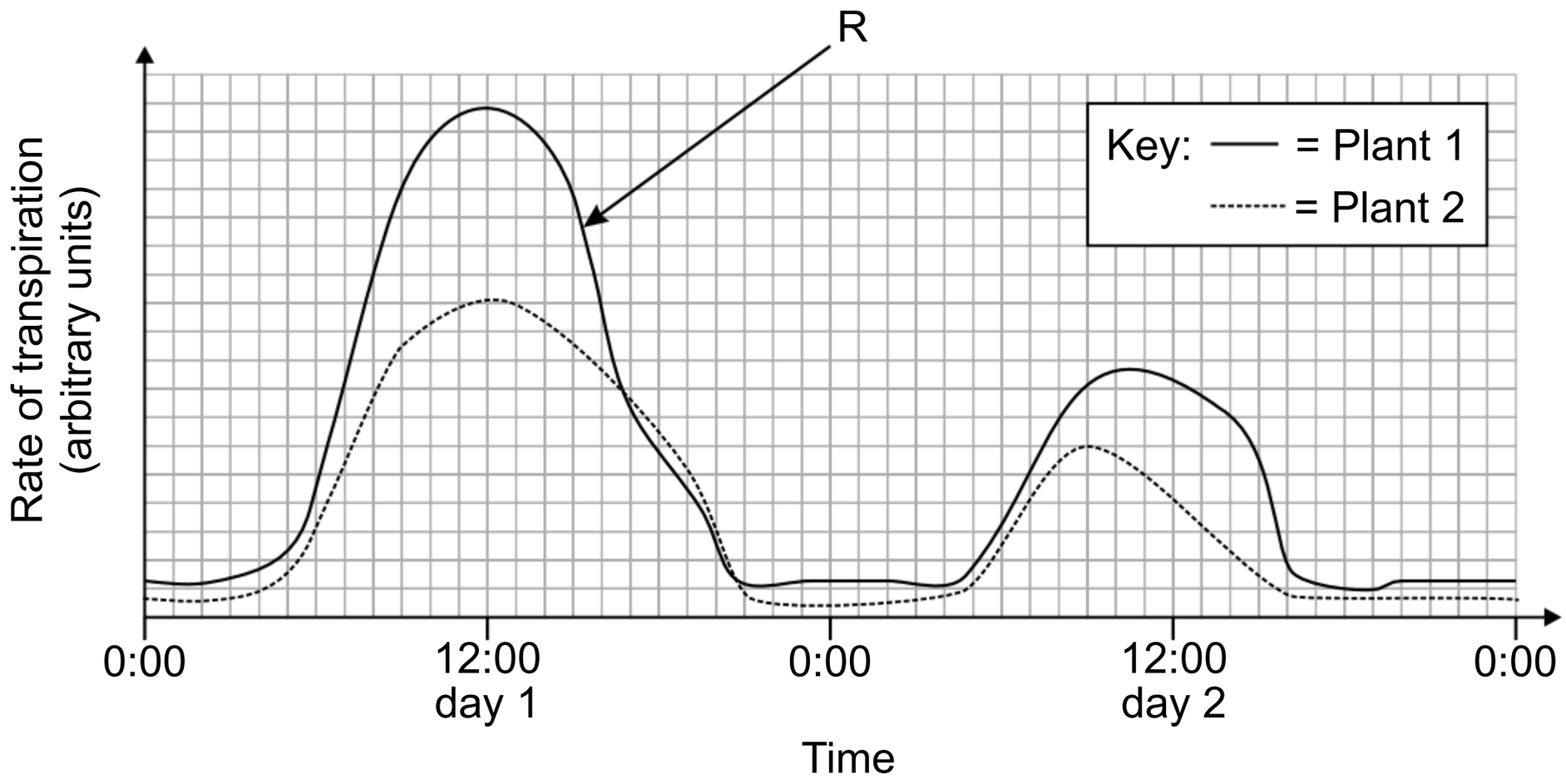
.....

.....

[1 mark]

8 A scientist measured the rate of transpiration in two plants over 48 hours. The results are shown in **Figure 9**.

Figure 9



8.1 At what time on **day 2** was the rate of transpiration highest for **plant 2**?

.....

[1 mark]

8.2 The rate of transpiration for both plants was slower on **day 2** than on **day 1**. Suggest **one** explanation for this.

.....

.....

.....

[2 marks]

8.3 At time **R** on the graph, **plant 1** was wilting.
Suggest **one** explanation for this.

.....

.....

[1 mark]

Question 8 continues on the next page

8.4 Describe how a transpiration stream moves water through a plant.

.....

.....

.....

.....

.....

.....

[3 marks]

9 Herbicides are chemicals that are used to kill unwanted plants.
Some herbicides work by disrupting the cell cycle within plants.

Describe the stages of the cell cycle and suggest why disrupting the cell cycle by using herbicides can kill plants.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6 marks]

Turn over for the next question

10 An investigation into how a student's heart rate changes during and after exercise is conducted. A portable heart rate monitor is used to measure the student's heart rate. The student's resting heart rate is taken before running around the school track for two minutes. The student then rests again.
The results are shown in **Table 4**.

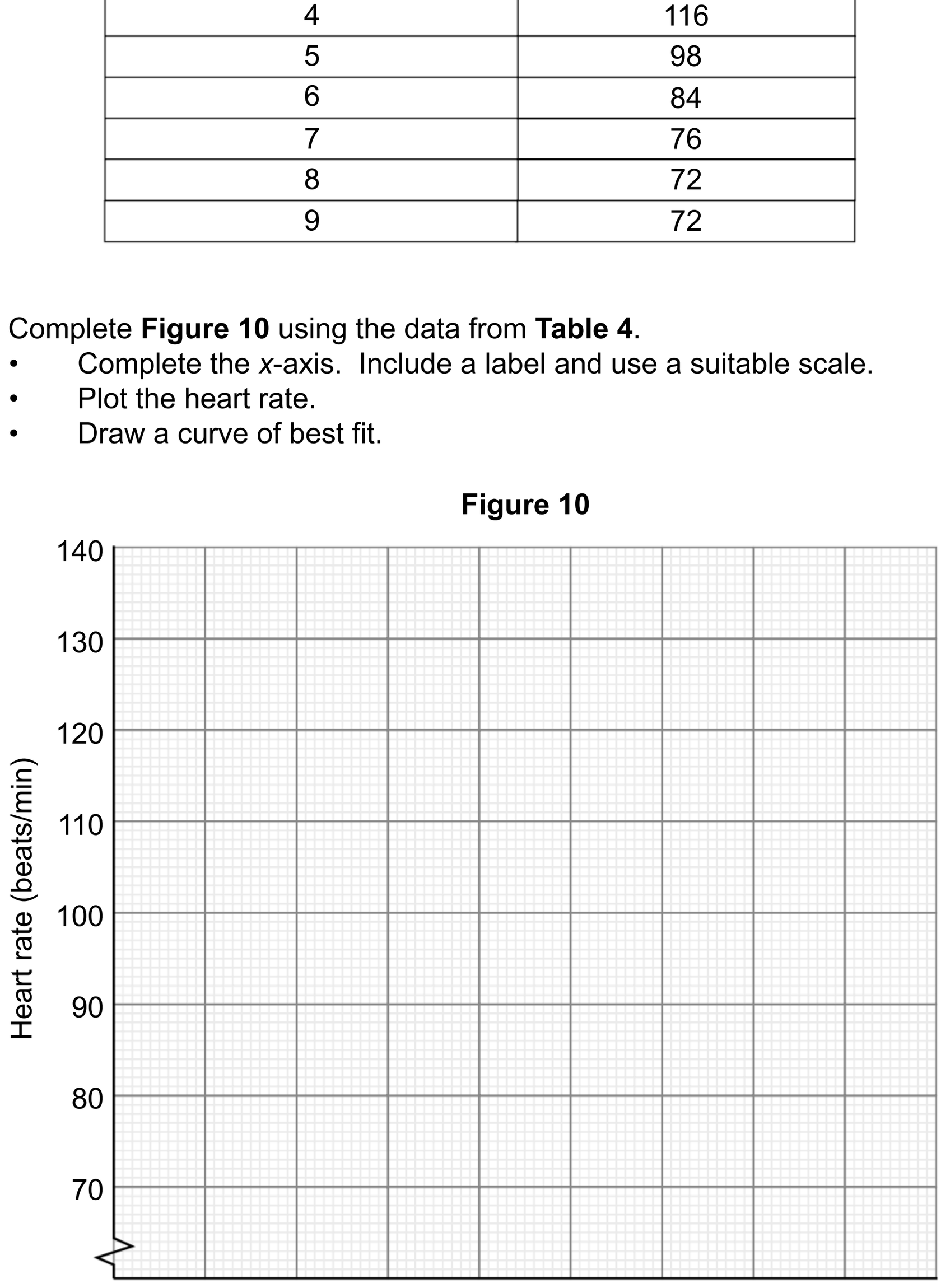
Table 4

Time after first heart rate measurement taken (min)	First run — Heart rate (beats/min)
0	72
1	118
2	132
3	129
4	116
5	98
6	84
7	76
8	72
9	72

10.1 Complete **Figure 10** using the data from **Table 4**.

- Complete the *x*-axis. Include a label and use a suitable scale.
- Plot the heart rate.
- Draw a curve of best fit.

Figure 10



[4 marks]

10.2 Use your graph in **Figure 10** to estimate the student's heart rate after thirty seconds.

heart rate = ... (beats/min)

[1 mark]

10.3 Give the dependent variable in this experiment.

.....

[1 mark]

10.4 Describe how the student's heart rate changes with exercise.
Explain why this change occurs.

.....

.....

.....

.....

.....

[4 marks]

At the beginning of the run, the student respires aerobically.

10.5 Complete the word equation for aerobic respiration.

..... + oxygen → carbon dioxide + ...

[2 marks]

10.6 As the run continues, the student's muscles start to respire anaerobically.
Suggest why this causes the student's heart rate to be higher than normal for a while after the exercise is finished.

.....

.....

.....

[3 marks]

END OF QUESTIONS

CGP Practice Exam Paper GCSE Biology

GCSE Biology

Paper 2

Higher Tier

In addition to this paper you should have:
- A ruler.
- A calculator.

Centre name					
Centre number					
Candidate number					

Time allowed:
- 1 hour 45 minutes

Surname
Other names
Candidate signature

Instructions to candidates
- Write your name and other details in the spaces provided above.
- Answer **all** questions in the spaces provided.
- Do all rough work on the paper.
- Cross out any work you do not want to be marked.

Information for candidates
- The marks available are given in brackets at the end of each question.
- There are 100 marks available for this paper.
- You are allowed to use a calculator.
- You should use good English and present your answers in a clear and organised way.

Advice to candidates
- In calculations show clearly how you worked out your answers.

For examiner's use							
Q	Attempt Nº			Q	Attempt Nº		
	1	2	3		1	2	3
1				7			
2				8			
3				9			
4				10			
5				11			
6							
Total							

1 **Figure 1** shows the amount of biomass contained within an area of plants.
It shows how much biomass from the plants is transferred to each trophic level in a food chain.

Figure 1

plants	→	grasshoppers	→	mice	→	snakes
120 kg		12 kg		1.4 kg		0.15 kg

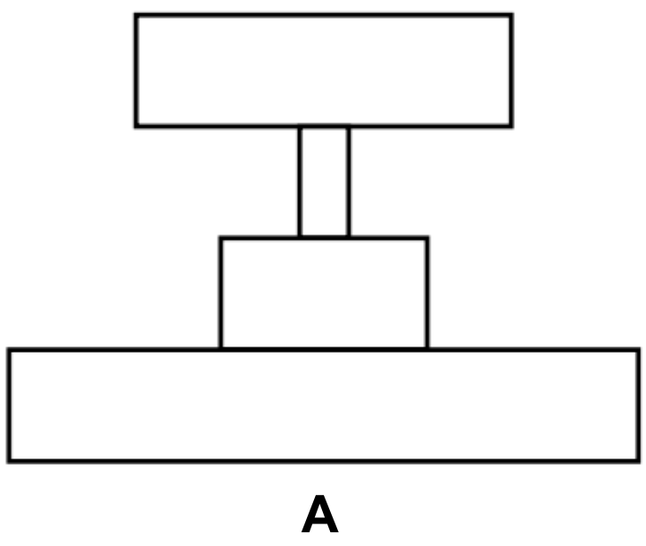
1.1 Plants are the producers in this food chain.
What is meant by the term 'producer'?

.....

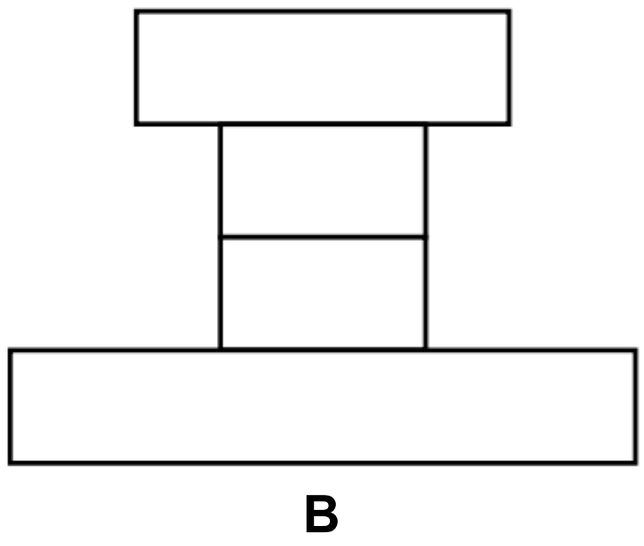
[1 mark]

Figure 2 shows four different pyramids of biomass.

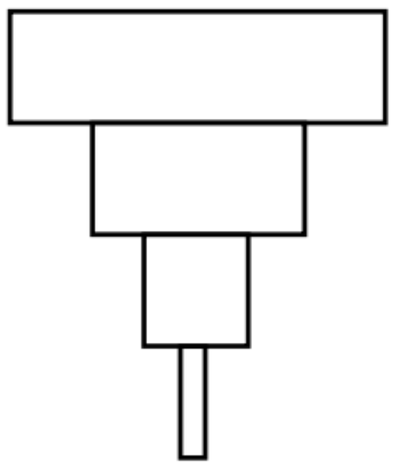
Figure 2



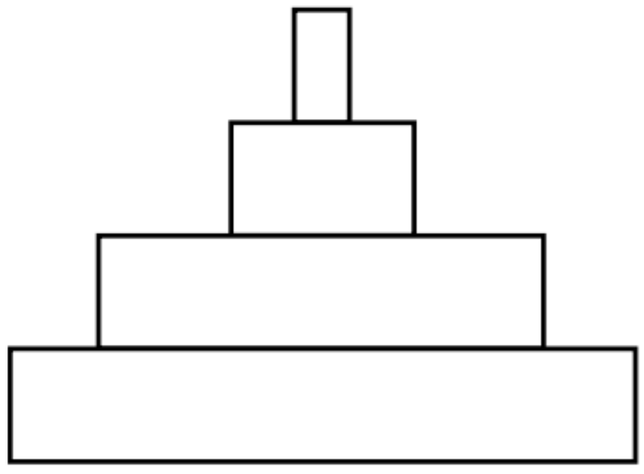
A



B



C



D

1.2 Which pyramid of biomass (**A**, **B**, **C** or **D**) represents this food chain?
Give a reason for your answer.

Pyramid .....

Reason .....

.....

[2 marks]

Question 1 continues on the next page

1.3 Calculate the efficiency of biomass transfer between the grasshoppers and mice.

Use the equation:

$$\text{efficiency} = \frac{\text{biomass transferred to the next level}}{\text{biomass available at the previous level}} \times 100$$

.....

.....

.....

Efficiency = ... %

[1 mark]

1.4 The food chain in **Figure 1** has four trophic levels.
Most food chains have no more than five trophic levels.
Suggest why the length of food chains is limited in this way.

.....

.....

.....

[2 marks]

2 A student was investigating the distribution of buttercups in an area around a school.
The student counted the number of buttercups in 10 quadrats in five different fields.
The quadrat measured 1 m^2. The results are shown in **Table 1**.

Table 1

Field	Mean number of buttercups per quadrat
A	10
B	35
C	21
D	37
E	21

2.1 What is the median of the data in **Table 1**?

.....

[1 mark]

2.2 A week later, the student repeated the experiment in a sixth field, Field **F**.
The results for each quadrat are shown below:

6 15 9 14 20 5 3 11 10 7

Using this data, calculate the mean number of buttercups per m^2 in Field **F**.

.....

.....

Mean = ... buttercups per m^2

[2 marks]

2.3 Field **F** measures 90 m by 120 m.
Estimate the population of buttercups in Field **F**.

.....

.....

Estimated population = ... buttercups

[2 marks]

Question 2 continues on the next page

2.4 In some areas of the fields, the buttercups were closely packed together.
How could the student have estimated the area covered by the buttercups in a quadrat, if they were difficult to count?

.....

.....

[1 mark]

The student observed that the distribution of buttercups changed across Field **A**.
Buttercups grow well in damp soil, so the student thinks that the change in the distribution of buttercups is due to variability in the moisture level of the soil across the field. The student wants to investigate this.

2.5 Suggest a hypothesis about the distribution of buttercups in Field **A**, based on the student's observations.

.....

.....

[1 mark]

2.6 Describe how the student could investigate this hypothesis.

.....

.....

.....

.....

.....

.....

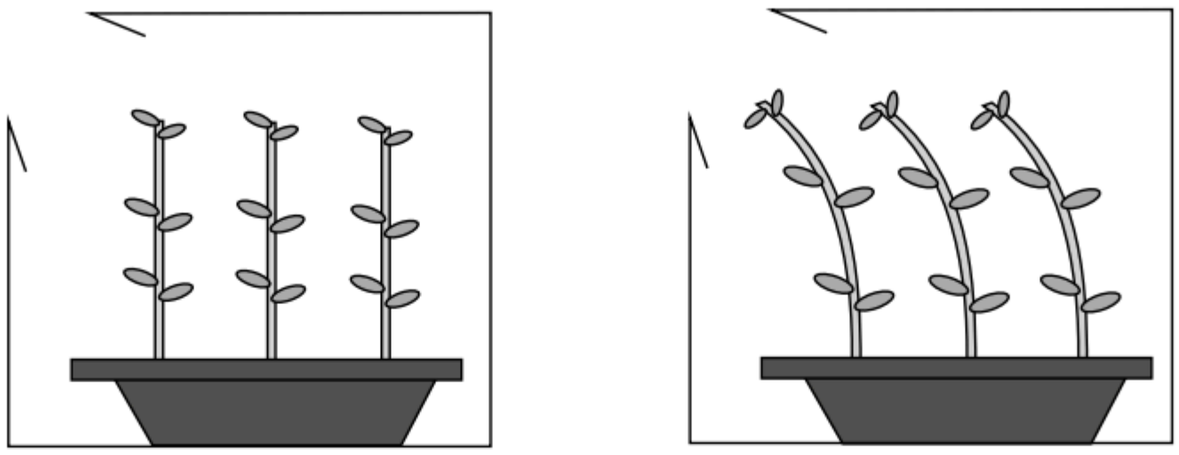
.....

[4 marks]

3 A student grew three plants in a windowsill tray.
The plants were then put in a cardboard box with a cut-out hole.

Figure 3 shows the same plants before and after three days in the cardboard box.

Figure 3



Start of experiment

After three days

3.1 Explain what caused the plants' response.

.....

.....

.....

.....

[3 marks]

3.2 Name the response shown by the plants in this experiment.

.....

[1 mark]

3.3 The student removed one of the plants from the tray.
It was observed that the roots of the plant were growing downwards.
Name the stimulus that causes the roots of a plant to grow downwards.

.....

[1 mark]

Turn over for the next question

4 **Figure 4** is an extract from a report by a lifeboat crew member.

Figure 4

> "We were very concerned when we received news of a man lost overboard tonight because the sea is extremely cold at this time of year. Fortunately, we found him quickly and were able to rescue him before he suffered any serious ill effects. His skin was very cold when we picked him up, but his core body temperature was normal."

4.1 Describe how the brain obtains information about the body's core temperature and skin temperature.

.....

.....

.....

.....

[2 marks]

4.2 Explain **one** way in which the man's body may have helped prevent his core temperature from falling whilst he was in the sea.

.....

.....

.....

.....

[2 marks]

5 ADH is a hormone involved in the regulation of the water content of the body.

5.1 From which gland is ADH released? Tick **one** box.

☐ pancreas ☐ thyroid ☐ adrenal gland ☐ pituitary gland

[1 mark]

5.2 Describe what happens if the brain detects that the water content of the blood is too **low**.

.....

.....

[2 marks]

5.3 The drug ecstasy stimulates the release of ADH. Taking ecstasy can result in too much water entering the body cells, which can cause cell damage.
Suggest why taking ecstasy can lead to too much water entering cells from the blood.

.....

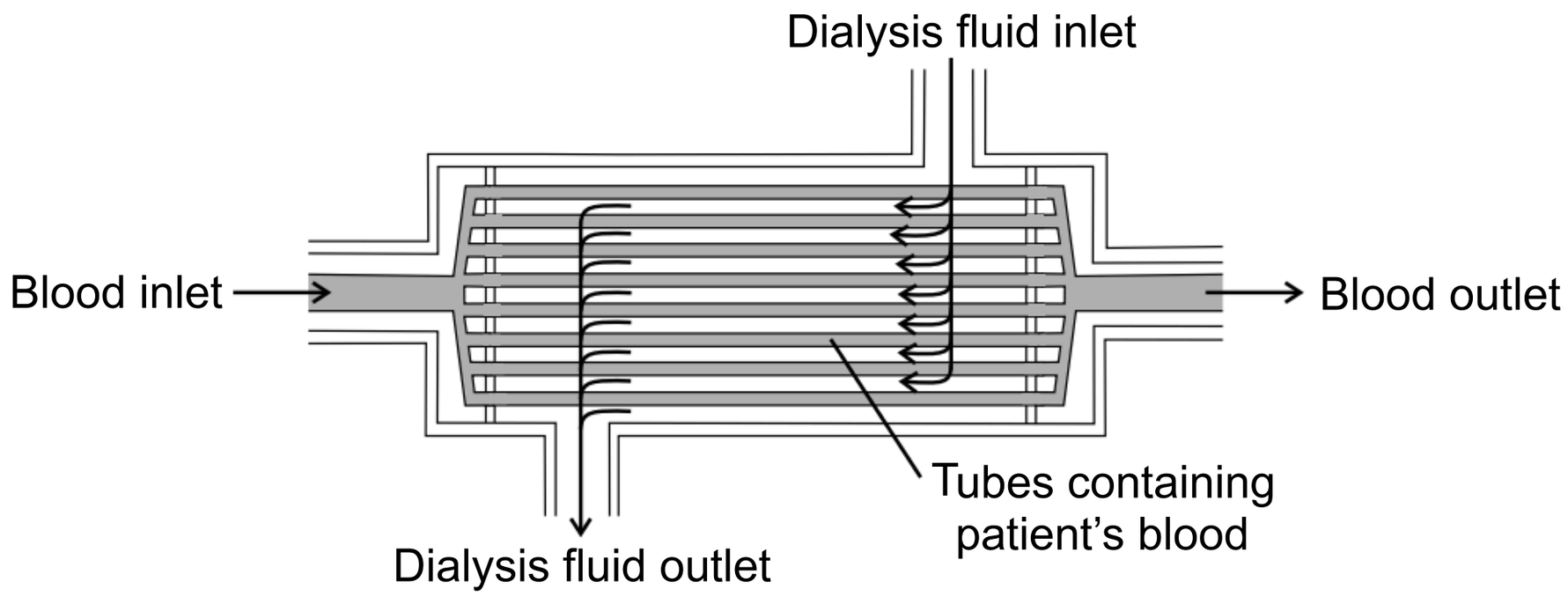
.....

.....

[2 marks]

People suffering from kidney failure may be treated using a kidney dialysis machine.
Figure 5 shows a type of dialysis machine.

Figure 5



5.4 Explain how a dialysis machine works.

.....

.....

.....

.....

[3 marks]

Turn over for the next question

6 The peppered moth is an insect that lives on the trunks of trees in Britain.
The moths are prey for birds such as thrushes.

The peppered moth exists in two varieties:

- A light-coloured variety that is better camouflaged on tree trunks in unpolluted areas.
- A dark-coloured variety that is better camouflaged on sooty tree trunks in badly polluted areas.

Figures 6 and **7** show these two varieties of moths on different tree trunks.

Figure 6

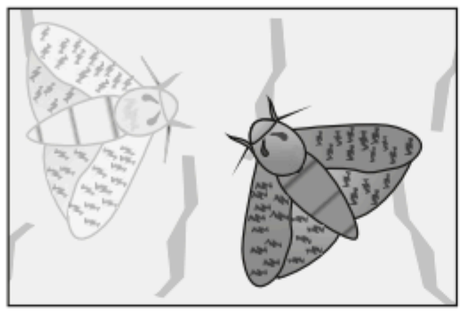
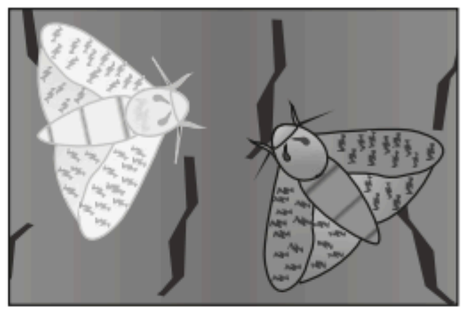


Figure 7



The dark variety of the moth was first recorded in the North of England in 1848.

It became increasingly common in polluted areas until the 1960s, when the number of soot covered trees declined because of the introduction of new laws.

6.1 The binomial name of the peppered moth is *Biston betularia*.
What is the moth's genus?

.....

[1 mark]

6.2 Which variety of moth has a better chance of survival in a soot polluted area?

.....

[1 mark]

6.3 Using the idea of natural selection, explain why the variety of moth given in **6.2** became more common in soot polluted areas.

.....

.....

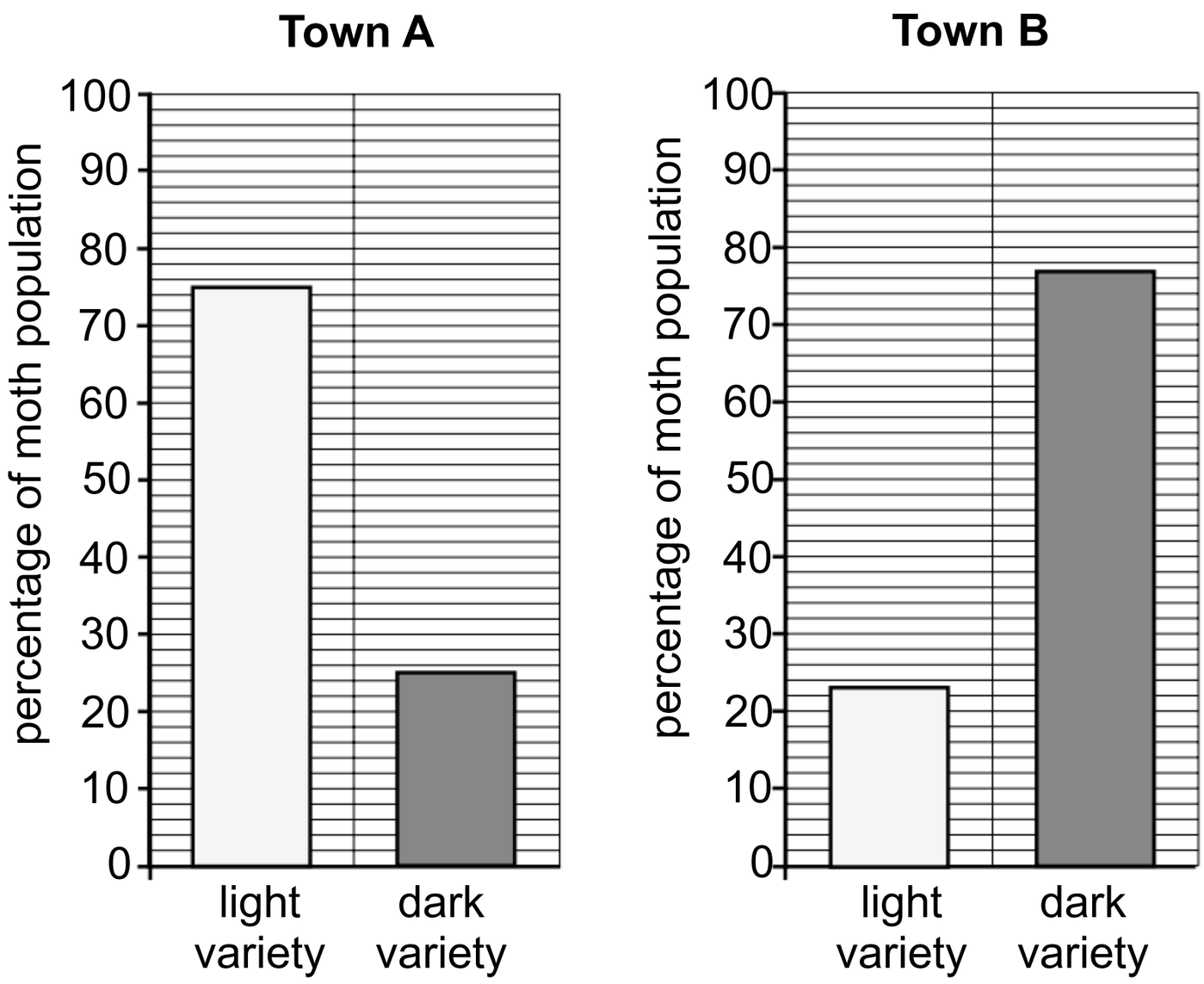
.....

.....

[3 marks]

Figure 8 shows the percentages of dark- and light-coloured peppered moths in two different towns.

Figure 8



6.4 Which town is the most polluted, **A** or **B**?
Explain your answer.

.....

.....

.....

[1 mark]

Turn over for the next question

7 A scientist was investigating the reflex actions of males and females.

The scientist made the following hypothesis:

'Males have faster reaction times than females.'

The reaction times of eight participants were tested in the investigation.
Each participant was tapped just below the knee with a small rubber hammer.
When the leg was tapped it automatically kicked outwards at the knee.

The scientist recorded how long it took each participant to respond to the stimulus of the tap on the leg. Each participant did the test 20 times, and a mean reaction time was calculated (to 2 decimal places).

The results are shown in **Table 2**.

Table 2

Sex	Participant	Age (years)	Mean reaction time (s)
Female	1	29	0.05
	2	26	0.06
	3	24	0.06
	4	27	0.04
Male	5	19	0.05
	6	22	0.04
	7	25	0.04
	8	20	0.05

7.1 How can you tell that the participants' response was a reflex?
Give **two** reasons.

.....

.....

.....

.....

[2 marks]

7.2 What was the dependent variable in this experiment?
Tick **one** box.

☐ age ☐ stimulus ☐ reaction time ☐ sex

[1 mark]

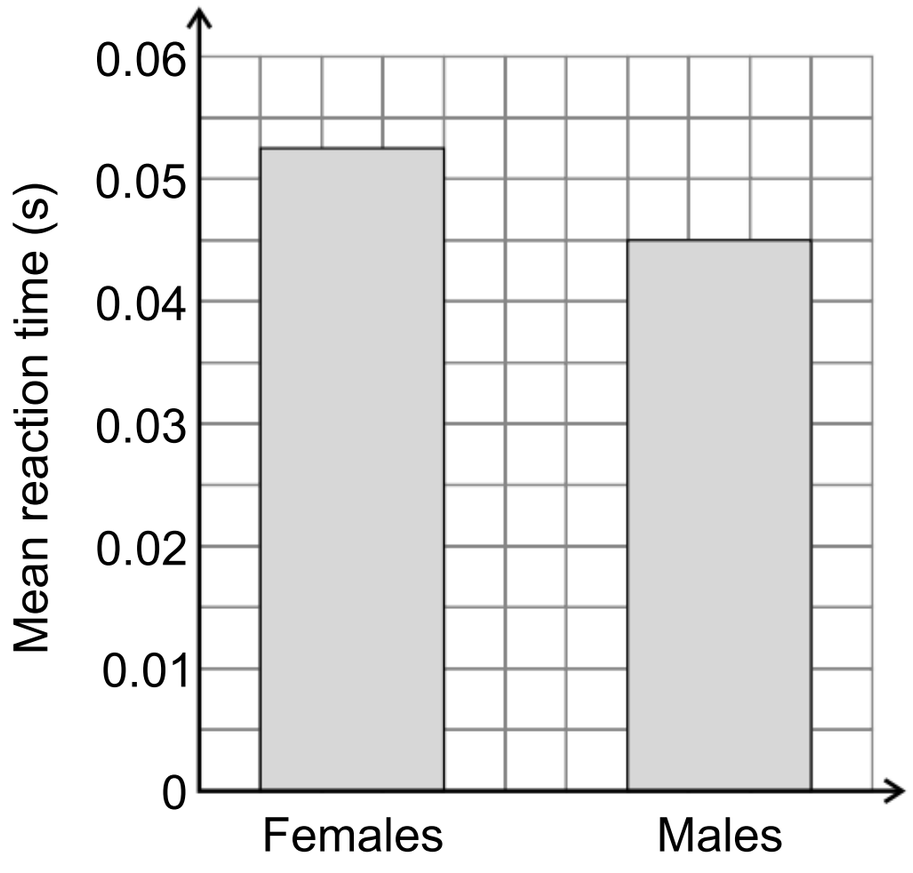
7.3 What was the independent variable in this experiment?
Tick **one** box.

☐ age ☐ stimulus ☐ reaction time ☐ sex

[1 mark]

7.4 **Figure 9** shows the mean reaction times for males and females in the investigation.

Figure 9



What can be concluded from the data in **Figure 9**?
Refer back to the scientist's hypothesis in your answer.

.....

.....

[2 marks]

7.5 Suggest **two** variables that the scientist should have controlled in this experiment.

.....

.....

.....

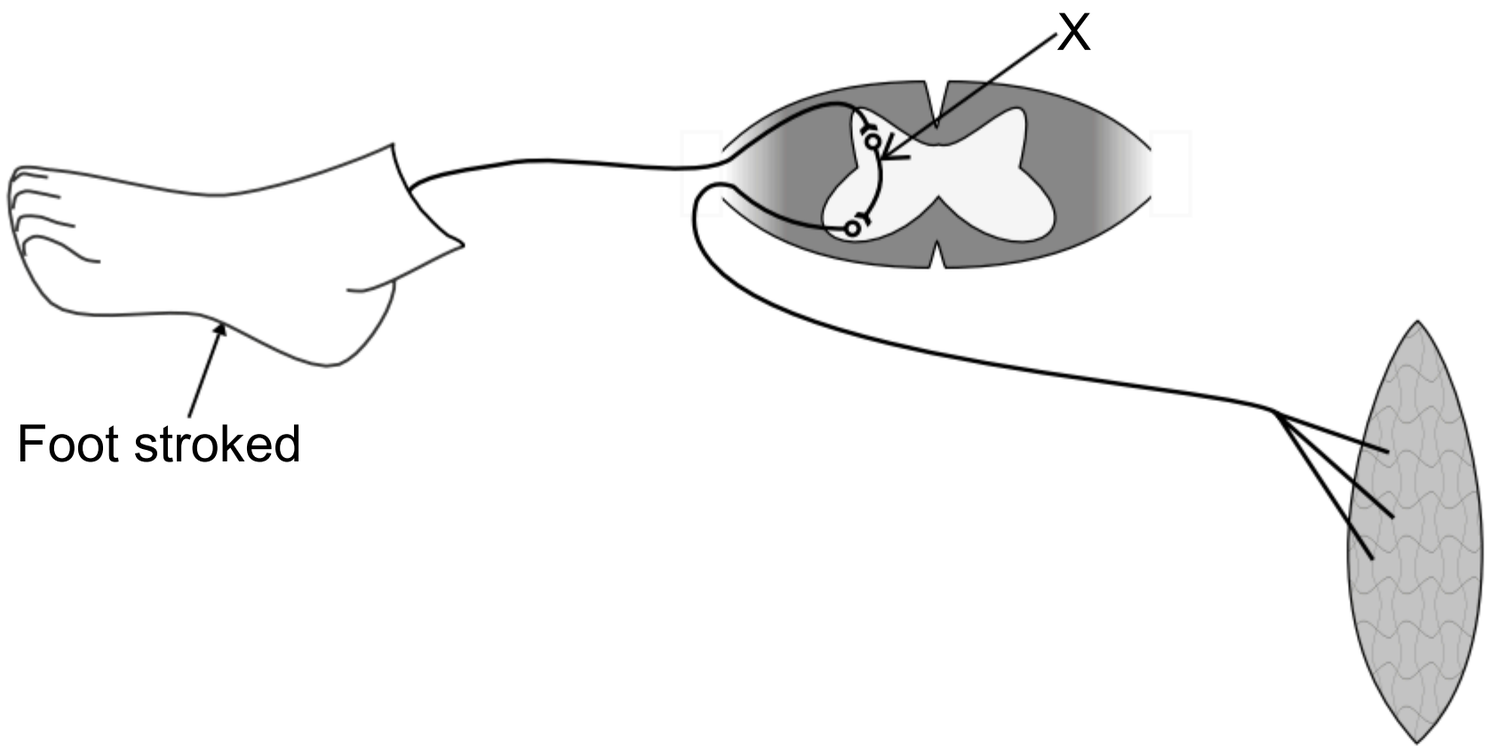
.....

[2 marks]

Question 7 continues on the next page

Another example of a reflex is the Babinski reflex, in which a baby curls its big toe upwards when the sole of its foot is stroked. **Figure 10** shows the parts of the nervous system involved in this reflex.

Figure 10



7.6 Name part **X** shown in **Figure 10**.

.....

[1 mark]

7.7 The touch stimulus is detected by receptors in the skin and causes a reflex response. Describe the path taken by a nervous impulse in this reflex, beginning at the receptors.

.....

.....

.....

.....

.....

.....

.....

.....

.....

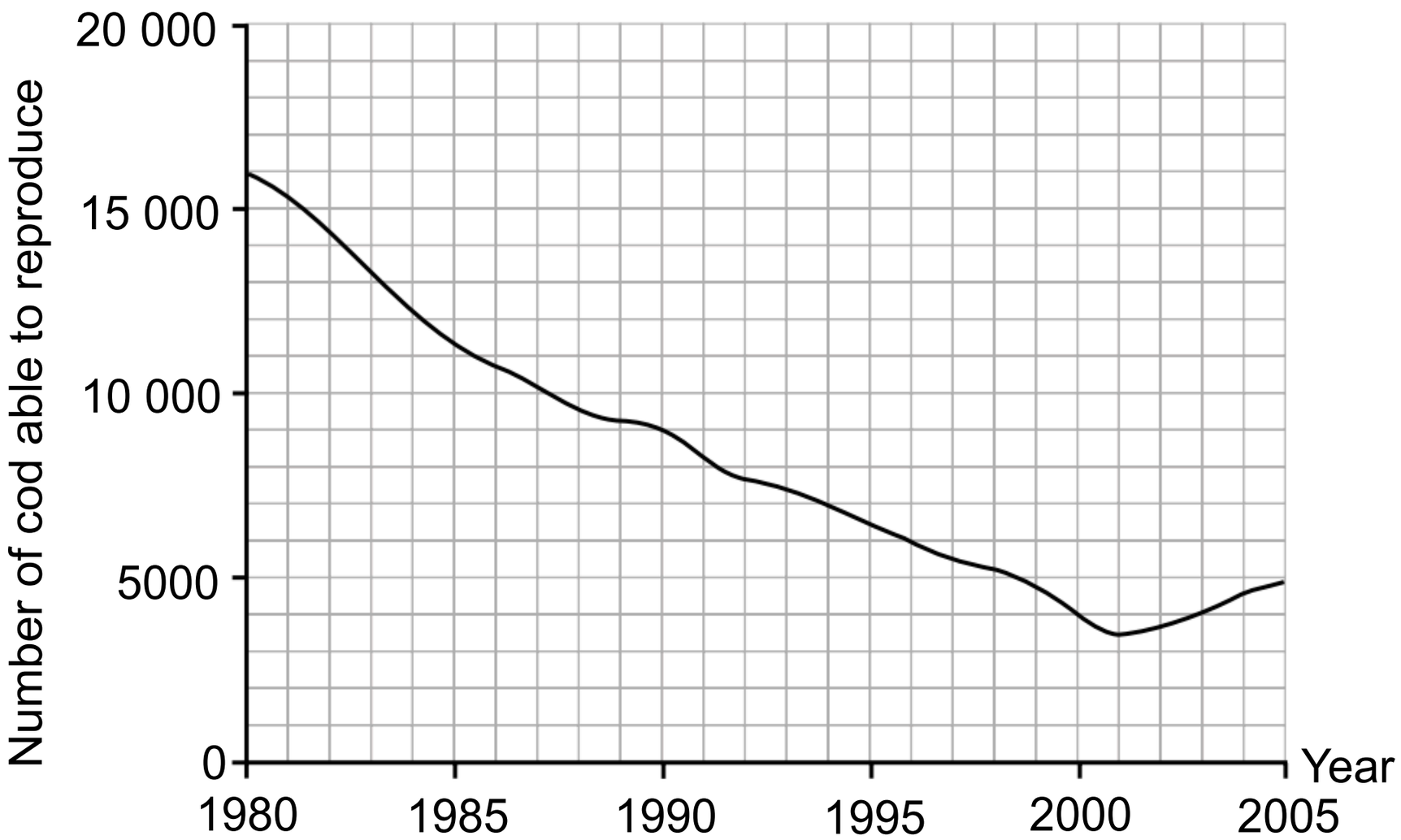
.....

[6 marks]

8 Many people fear that cod supplies from the North Sea may run out in the near future. To prevent this, cod stocks need to be fished sustainably.

Figure 11 shows how the population of cod has changed between 1980 and 2005 in the North Sea, just off the coast of Norway.

Figure 11



8.1 Explain what is meant by producing food sustainably.

.....

.....

[1 mark]

8.2 Calculate the percentage decrease in North Sea cod numbers between 1980 and 2000.

.....

.....

Percentage decrease = ... %

[2 marks]

Question 8 continues on the next page

8.3 Calculate the rate at which the cod population was decreasing between 1980 and 2000.

.....

.....

.....

Rate = ... cod/year

[2 marks]

8.4 Describe the trend shown by **Figure 11**.
Suggest reasons for the trend you have described.

.....

.....

.....

.....

.....

[3 marks]

An increasing demand for food has led scientists to look for new sustainable sources of food. An example of a sustainable food source is mycoprotein.

8.5 Name the fungus that is the main source of mycoprotein.

.....

[1 mark]

8.6 Describe how mycoprotein is produced.

.....

.....

.....

.....

.....

[3 marks]

9 Various research programmes aim to improve human health through research into the human genome.

9.1 What is meant by the term ‘genome’?

.....

[1 mark]

Cystic fibrosis is an inherited disorder caused by a recessive allele.

9.2 How could research into the human genome have the potential to help people with cystic fibrosis?

.....

.....

[1 mark]

A couple have a baby boy. The doctor tells them that the baby has inherited cystic fibrosis. Neither parent shows signs of the disorder.

9.3 Construct a Punnett square to show how the baby inherited cystic fibrosis. Use the diagram to work out the probability of any future child of the couple inheriting cystic fibrosis.

Use **F** to represent the dominant allele and **f** to represent the recessive allele.

Probability = ...

[4 marks]

Question 9 continues on the next page

9.4 Is the baby homozygous or heterozygous for cystic fibrosis?
Explain your answer.

.....

.....

[1 mark]

The doctor tells the couple that if they have another child, the fetus can be tested by embryonic screening to see if it will have cystic fibrosis.

Embryo screening can also be done during IVF treatment, so that doctors can select a healthy embryo to implant in the mother. The other embryos are discarded.

At the moment, UK regulations only allow embryo screening when there is "a significant risk of a serious genetic condition being present in the embryo." This means that it is only allowed when a child of the person carrying the faulty allele would be likely to suffer from the disorder, and the disorder is serious.

9.5 Could embryos be screened for polydactyly under the current regulations?
Explain your answer.

.....

.....

[1 mark]

9.6 Suggest **one** reason why a person might be opposed to screening embryos for inherited conditions.

.....

.....

[1 mark]

10 Reproduction can be sexual or asexual.

10.1 Give **three** differences between sexual and asexual reproduction.

.....

.....

.....

.....

.....

[3 marks]

In humans, sexual reproduction involves the reproductive system.

10.2 The male reproductive system includes the testes.
What hormone is produced by the testes?

.....

[1 mark]

10.3 The female reproductive system includes the ovaries.

The ovaries produce the hormone oestrogen.
Describe the role of oestrogen in the menstrual cycle.

.....

.....

.....

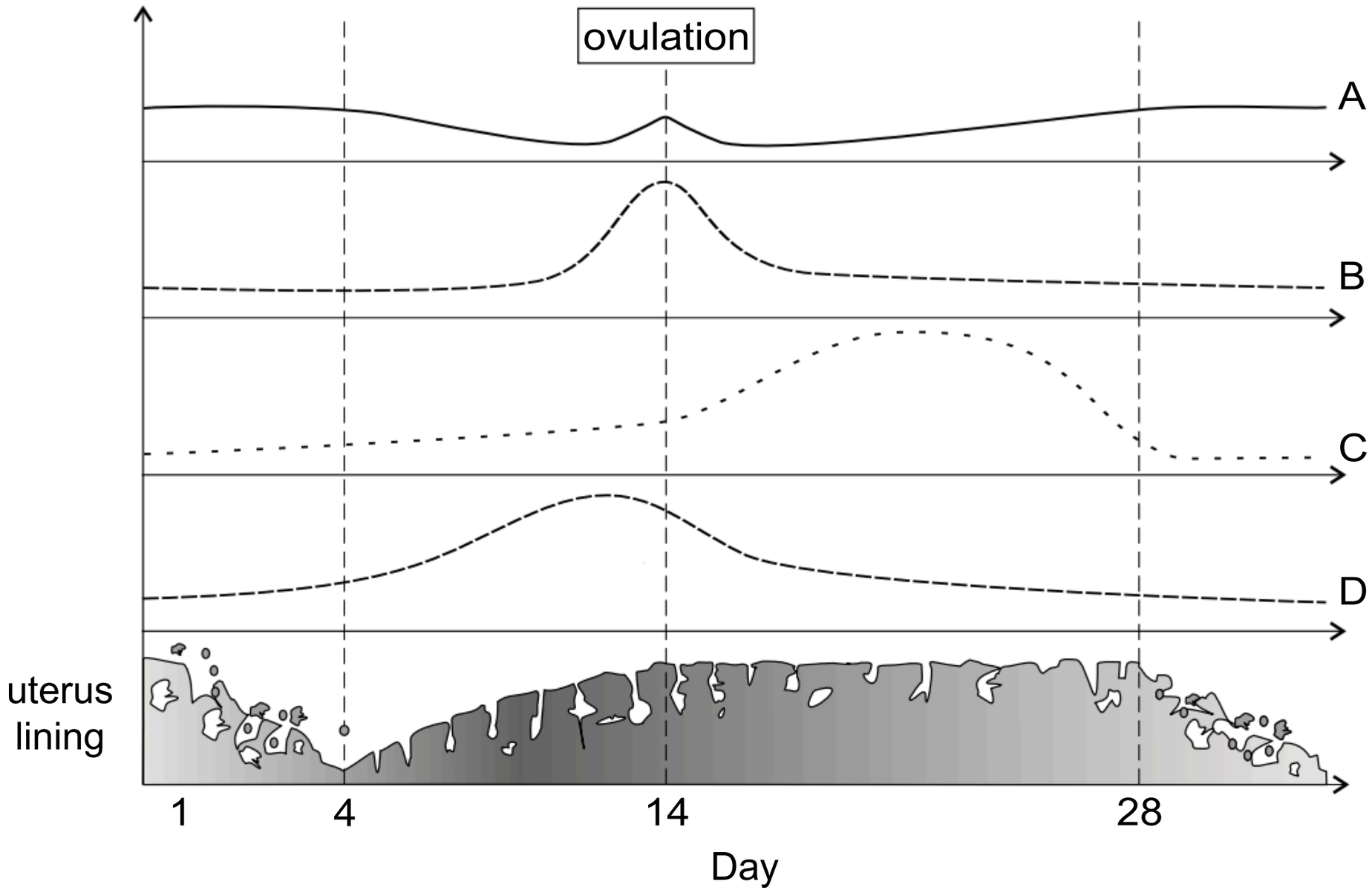
.....

[2 marks]

Question 10 continues on the next page

Figure 12 shows the fluctuations in the levels of four different hormones during one 28 day menstrual cycle.

Figure 12



10.4 Which line in **Figure 12** represents progesterone?
Explain your answer.

Line .....

Explanation .....

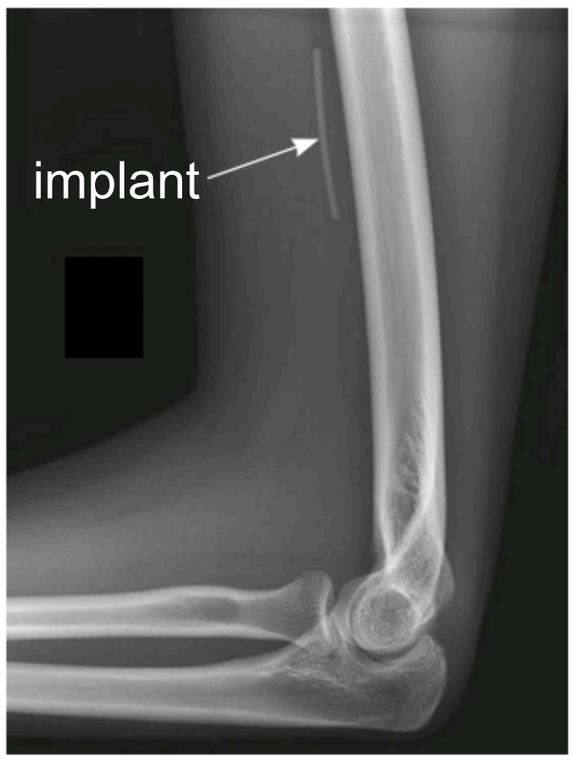
.....

.....

[2 marks]

Some methods of contraception use reproductive hormones to control fertility. One such method is the contraceptive implant, shown in **Figure 13**.

Figure 13



The contraceptive implant:

- is a small, plastic rod that is inserted by a doctor or nurse
- releases progesterone, which reduces fertility
- is effective for three years
- is over 99% effective at preventing pregnancy
- can cause side effects such as headaches and nausea
- can be made less effective by certain medications

Figure 14 shows a condom. The condom is a barrier method of contraception.

Figure 14

The condom:

- is worn over the penis during intercourse
- prevents sperm from entering the vagina
- can only be used once
- is 98% effective at preventing pregnancy when used correctly
- protects against sexually transmitted disease (STDs)

10.5 Use the information to evaluate the implant and condom as methods of contraception. Give a conclusion of which you think is the better method of contraception. Justify your conclusion.

.....

.....

.....

.....

.....

.....

.....

.....

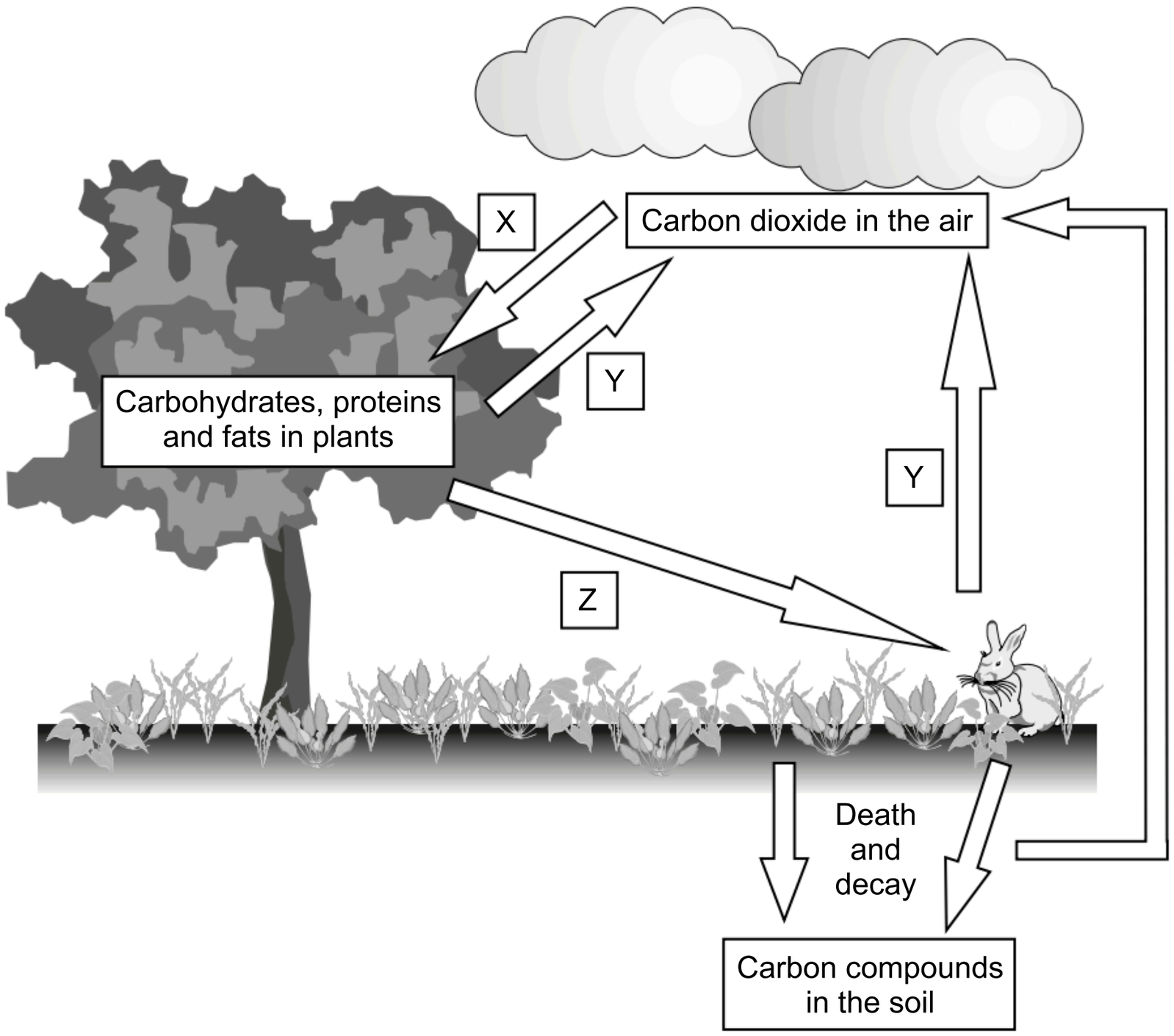
.....

[4 marks]

Turn over for the next question

11 **Figure 15** shows the carbon cycle.

Figure 15



11.1 Describe what is occurring at points **X**, **Y** and **Z** in the cycle.

.....

.....

.....

.....

.....

[3 marks]

11.2 How are microorganisms involved in the carbon cycle?

.....

.....

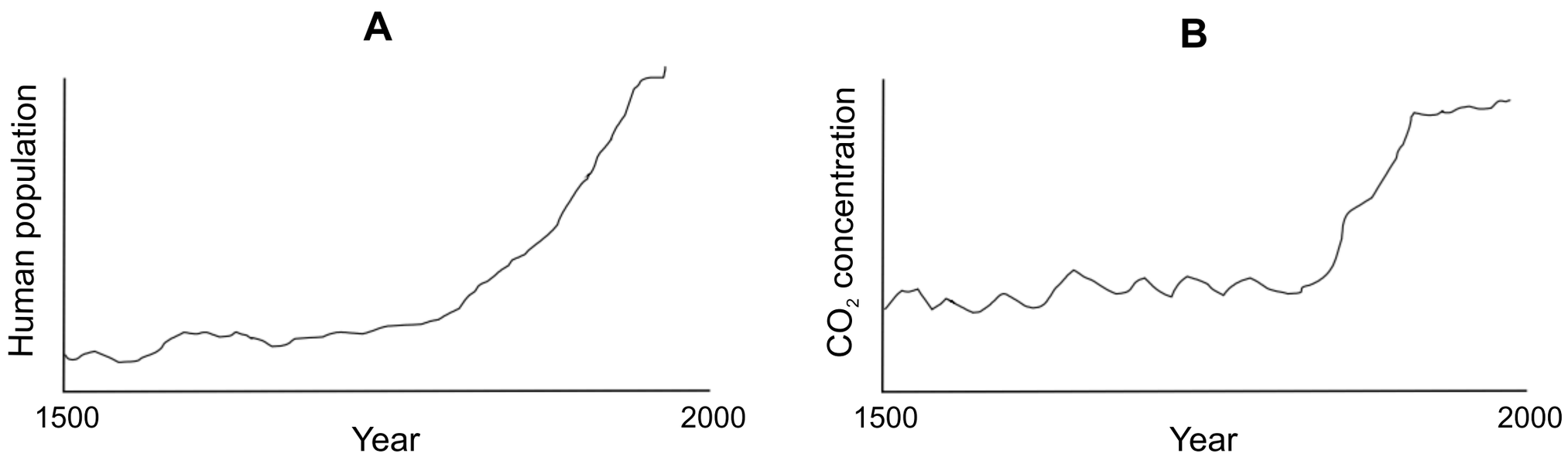
.....

[2 marks]

A scientist was examining some data to see if there is a link between the global human population and the carbon dioxide concentration in the atmosphere.

Figure 16 shows the two graphs that the scientist examined.

Figure 16



11.3 Describe the relationship between the two graphs.

.....

.....

[1 mark]

11.4 On their own, the graphs in **Figure 16** do not prove that the increased human population caused the increased carbon dioxide concentration. Give **two** reasons why not.

.....

.....

.....

.....

[2 marks]

As the global human population increases, deforestation tends to increase. Deforestation is a big problem affecting rainforests in tropical areas.

11.5 Give **two** reasons why deforestation occurs in tropical areas.

.....

.....

[2 marks]

Question 11 continues on the next page

11.6 Deforestation is one factor that can cause the amount of carbon dioxide in the atmosphere to increase. The rising concentration of carbon dioxide in our atmosphere has been linked to global warming.
Describe **two** possible biological consequences of global warming.

.....

.....

.....

.....

.....

.....

[2 marks]

END OF QUESTIONS

Answers

Topic 1a — Cells and Microscopy

Page 5
Microscopy

Q1 real size = image size ÷ magnification
= 2.4 mm ÷ 40
= 0.06 mm ***[1 mark]***
0.06 × 1000 = **60 µm** ***[1 mark]***

Page 7
Warm-Up Questions

1) mitochondria

2) Plant cells have a rigid cell wall, they have a permanent vacuole and they contain chloroplasts.

3) Any two from: e.g. prokaryotic cells are smaller than eukaryotic cells. / Prokaryotic cells don't have mitochondria but eukaryotic cells do. / Prokaryotic cells don't have a true nucleus but eukaryotic cells do. / Prokaryotic cells have circular DNA but eukaryotic cells don't.

4) electron microscope

5) 4.5×10^{-4} µm

Exam Questions

1 nucleus ***[1 mark]***

Remember, DNA in prokaryotic cells floats freely in the cytoplasm — it's not stored in a nucleus.

2.1 C ***[1 mark]***

2.2 They absorb light needed for photosynthesis to make food for the plant ***[1 mark]***.

2.3 They're where proteins are made in the cell ***[1 mark]***.

3.1 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***
Level 1: There is a brief explanation of how to prepare a slide or how to use a light microscope. ***[1 to 2 marks]***
Level 2: There is some explanation of how to prepare a slide and use a light microscope. ***[3 to 4 marks]***
Level 3: There is a clear and detailed explanation of how to prepare a slide and use a light microscope. ***[5 to 6 marks]***

Here are some points your answer may include:

To prepare a slide:
Add a drop of water to the middle of a clean slide.
Cut up an onion and separate it out into layers.
Use tweezers to peel off some epidermal tissue from the bottom of one of the layers.
Use tweezers to place the epidermal tissue into the water on the slide.
Add a drop of iodine solution/stain.
Place a cover slip on top by standing it upright on the slide, next to the water droplet, then carefully tilting and lowering it so it covers the onion tissue without trapping any air bubbles.

To use a light microscope:
Clip the slide onto the stage.
Select the lowest-powered objective lens.
Use the coarse adjustment knob to move the stage up to just below the objective lens (without looking down the eyepiece).
Look down the eyepiece and use the coarse adjustment knob to move the stage downwards until the image is roughly in focus.
Adjust the focus with the fine adjustment knob, until a clear image of the cells is visible.
To see the cells with greater magnification, swap to a higher-powered objective lens and refocus.

3.2 E.g. real size = 75 µm ÷ 1000 = 0.075 mm
magnification = image size ÷ real size
= 7.5 mm ÷ 0.075 mm = **× 100** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

You could have converted the image size to µm here instead — the important thing is that both values have the same units before you stick them in the formula.

Page 10
Chromosomes and Mitosis

Q1 a) 11 ÷ (62 + 11) = 0.150...
0.150... × 100 = **15%** ***[1 mark]***

b) E.g. she could see the X-shaped chromosomes in the middle of the cells. / She could see the arms of the chromosomes being pulled apart ***[1 mark]***.

Page 11
Binary Fission

Q1 Lots of nutrients ***[1 mark]*** and a warm environment ***[1 mark]***.

Q2 2 hours = 60 × 2 = 120 minutes
120 minutes + 40 minutes = 160 minutes
160 ÷ 20 = 8 divisions ***[1 mark]***
$2^8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$
= **256 cells** ***[1 mark]***

Page 12
Warm-Up Questions

1) The cell has a hair-like shape, which gives it a large surface area to absorb water and minerals from the soil.

2) It's the process by which a cell changes to become specialised for its job.

3) Copies of the plant can be made by taking stem cells from the meristem and growing them into new, genetically identical plants (clones).

4) in the nucleus

5) prokaryotic

Exam Questions

1.1 The amount of DNA is doubling ***[1 mark]*** so that there is one copy for each new cell ***[1 mark]***.

1.2 The two new cells separate ***[1 mark]***.

1.3 two ***[1 mark]***

2.1 They could be grown into a particular type of cell, which can then be used to replace faulty cells ***[1 mark]***.

2.2 Embryonic stem cells have the potential to develop into any kind of cell, whereas adult stem cells can only develop into certain types of cell ***[1 mark]***.

2.3 E.g. bone marrow ***[1 mark]***.

3 3 hours = 60 × 3 = 180 minutes
180 ÷ 30 = 6 divisions
$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$ **64 cells** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

Page 14
Culturing Microorganisms

Q1 a) A ***[1 mark]***

b) diameter = 13 mm
radius = 13 ÷ 2 = 6.5 mm ***[1 mark]***
$\pi r^2 = \pi \times 6.5^2 = 132.7...$
$=$ **133 mm²** ***[1 mark]***

c) E.g. a paper disc soaked in sterile water ***[1 mark]***.

Page 15
Warm-Up Questions

1) an inoculating loop / a sterile dropping pipette and spreader

2) To prevent contamination of a culture with unwanted microorganisms (that could affect results).

Exam Questions

1.1 B ***[1 mark]***

1.2 To show that any difference in the growth of the bacteria is only due to the effect of the antiseptic ***[1 mark]***.

2.1 radius = 15 ÷ 2 = 7.5 mm
$\pi r^2 = \pi \times 56.25 = 176.7... =$ **177 mm²** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

2.2 The strain of bacteria used in the experiment was resistant to antibiotic Z ***[1 mark]***.

Topic 1b — Exchanging Substances

Page 17
Diffusion

Q1 a) The ink will diffuse / spread out through the water ***[1 mark]***. This is because the ink particles will move from where there is a higher concentration of them (the drop of ink) to where there is a lower concentration of them (the surrounding water) ***[1 mark]***.

b) The ink particles will diffuse / spread out faster ***[1 mark]***.

Page 20
Exchange Surfaces

Q1 Surface area:
$(2 \times 2) \times 2 = 8$
$(2 \times 1) \times 4 = 8$
$8 + 8 = 16\ \mu m^2$ ***[1 mark]***
Volume:
$2 \times 2 \times 1 = 4\ \mu m^3$ ***[1 mark]***
So the surface area to volume ratio is 16 : 4, or **4 : 1** ***[1 mark]***.

Page 21
Exchanging Substances

Q1 E.g. they have a large surface area. / They have a moist lining for dissolving gases. / They have very thin walls. / They have a good blood supply ***[1 mark]***.

Q2 The damage to the villi is likely to reduce the surface area for absorption ***[1 mark]***. Therefore, less iron can be absorbed from the digested food in the small intestine into the blood ***[1 mark]***.

Pages 23-24
Warm-Up Questions

1) The higher the temperature, the faster the rate of diffusion because the particles have more energy and so move around faster.

2) Osmosis and active transport.

3) They're thin, they have a large surface area, they have lots of blood vessels and they're often ventilated.

4) E.g. alveoli (in the lung) and villi (in the small intestine).

5) Oxygen and water vapour.

6) A good blood supply helps to maintain the concentration gradient. This increases the diffusion rate of oxygen from the water into the blood, and the diffusion rate of carbon dioxide from the blood into the water.

Exam Questions

1 The time taken will decrease ***[1 mark]***. There will be more ammonia gas particles at the end of the tube where they are injected, meaning a greater concentration gradient between that end and the end where the litmus paper is ***[1 mark]***. This means an increased rate of diffusion, so the gas particles will reach the litmus paper more quickly ***[1 mark]***.

2.1 The potato cylinder in tube D, because this tube contains the most concentrated sugar solution so this cylinder will have lost the most water ***[1 mark]*** by osmosis ***[1 mark]***.

2.2 Tube A contained distilled water, so some of the water moved by osmosis into the potato cylinder ***[1 mark]*** from an area of high water concentration to an area of low water concentration ***[1 mark]***.

3.1 By diffusion down their concentration gradients into the blood ***[1 mark]***, and by active transport, against their concentration gradient ***[1 mark]***.

3.2 E.g. they increase the surface area to maximise absorption ***[1 mark]***. They have a thin wall/a single layer of surface cells to reduce the distance across which diffusion occurs ***[1 mark]***. They have a good blood supply which maintains the concentration gradient for the uptake of substances ***[1 mark]***.

4.1 diffusion ***[1 mark]***

The acid must by moving by diffusion here as it is moving from a region of higher concentration (outside the cubes) to a region of lower concentration (inside the cubes, which don't contain any acid to start with).

4.2 (835 + 825 + 842 + 838) ÷ 4 = **835 s**
[2 marks for correct answer, otherwise 1 mark for correct working.]

4.3 Surface area: (10 × 10) × 6 = 600 mm^2
Volume: 10 × 10 × 10 = 1000 mm^3
So the surface area to volume ratio is 600 : 1000, which simplifies to **3 : 5**
[3 marks for correct answer, otherwise 1 mark for correct surface area and 1 mark for correct volume.]

600 and 1000 are both divisible by 200. If you don't spot this straight away, you could simplify the ratio by first dividing both sides by 100, then by 2.

4.4 As the size of the gelatine cube increases, the time taken for the cube to become yellow increases ***[1 mark]***. This is because there is a greater distance for the acid to travel in the bigger cubes / the bigger cubes have a smaller surface area to volume ratio ***[1 mark]***, which decreases the rate of diffusion ***[1 mark]***.

Topic 2a — Organisation, Enzymes and Circulation

Page 28
Investigating Enzymatic Reactions

Q1 2 minutes = 2 × 60 = 120 seconds
36 ÷ 120 = **0.3 cm^3/s** ***[1 mark]***

Pages 32-33
Warm-Up Questions

1) A group of organs working together to perform a particular function.

2) The pH at which the enzyme works best.

3) a) amylase
b) protease
c) lipase

Carbohydrases (e.g. amylase) break down carbohydrates, such as starch. Proteases break down proteins, and lipases break down lipids (fats).

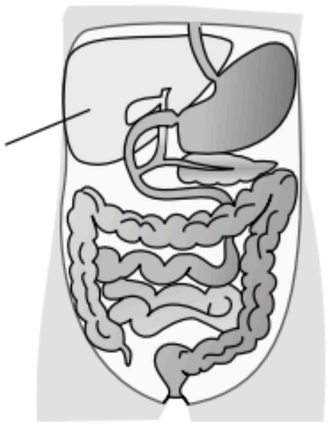
4) a) (simple) sugars
b) amino acids
c) glycerol and fatty acids

5) stomach, pancreas, small intestine

6) E.g. break up the food using a pestle and mortar. Then transfer the ground up food to a beaker and add some distilled water. Next, stir the mixture with a glass rod, and finally filter the solution using a funnel lined with filter paper.

Exam Questions

1.1 Place where bile is produced



[1 mark]

1.2 It neutralises the hydrochloric acid from the stomach ***[1 mark]***.

1.3 It emulsifies fats ***[1 mark]*** to give a much bigger surface area of fat for the enzyme lipase to work on ***[1 mark]***.

2.1 The enzyme has a specific shape which will only fit with one type of substrate ***[1 mark]***.

2.2 It would break the bonds in the enzyme and change the shape of the enzyme's active site/denature the enzyme ***[1 mark]***. This would mean the substrate would no longer fit into it so the enzyme wouldn't work anymore ***[1 mark]***.

A similar thing happens when the pH or the temperature is too high — the bonds are disrupted and the shape of the active site may change.

3.1 Accept answers between 38 °C and 40 °C ***[1 mark]***.

3.2 Enzyme B, because it has an unusually high optimum temperature which it would need to work in the hot vent ***[1 mark]***.

4 The food sample contains starch ***[1 mark]*** and lipids ***[1 mark]***, but no proteins ***[1 mark]***.

5.1 To prevent the starch coming into contact with amylase in the syringe, which would have started the reaction before the student started the stop clock ***[1 mark]***.

5.2 Rate = 1000 ÷ 60 = 16.666... = **17 s^{-1} (2 s.f.)** ***[1 mark]***

5.3 Repeat the experiment using buffers with a range of different pH values and compare the results ***[1 mark]***.

Page 36
Circulatory System — Blood Vessels

Q1 2.175 l × 1000 = 2175 ml ***[1 mark]***
2175 ÷ 8.7 = **250 ml/min** ***[1 mark]***

Page 37
Circulatory System — Blood

Q1 They help the blood to clot at a wound ***[1 mark]***.

Q2 To carry oxygen from the lungs to all the cells in the body ***[1 mark]***.

Pages 38-39
Warm-Up Questions

1) bronchi

2) right ventricle

3) A device that is implanted under the skin and has a wire going to the heart. It produces an electric current to keep the heart beating regularly.

4) They carry blood back to the heart.

5) platelets

Exam Questions

1.1 495 ÷ 12 = 41.25 = **41 breaths per minute** ***[1 mark]***

You might be wondering why the answer has been rounded to 41 when the calculator said 41.25. That's because you can't get a quarter of a breath.

1.2 Resting heart rate is controlled by a group of cells in the right atrium wall that act as a pacemaker ***[1 mark]***.

2.1 It is biconcave, which gives it a large surface area for absorbing oxygen ***[1 mark]***.

2.2 E.g. it contains haemoglobin ***[1 mark]*** which binds with oxygen so it can be carried to the body tissues ***[1 mark]***. / It has no nucleus ***[1 mark]*** so there is space to carry oxygen ***[1 mark]***.

2.3 White blood cells ***[1 mark]***. They defend the body against infection ***[1 mark]***.

2.4 They have permeable walls ***[1 mark]***, so substances can diffuse in and out ***[1 mark]***. Their walls are usually only one cell thick ***[1 mark]***, which increases the rate of diffusion by decreasing the distance over which it occurs ***[1 mark]***.

2.5 150 s ÷ 60 = 2.5 min
1155 ÷ 2.5 = **462 ml/min** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

Make sure you read the question carefully — if you gave your answer in ml/second you wouldn't have got all the marks available.

3.1 A — aorta, B — vena cava, C — left atrium ***[1 mark]***

3.2 It pumps blood around the body ***[1 mark]***.

3.3 To prevent the backflow of blood ***[1 mark]***.

3.4 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There are some relevant points describing how deoxygenated blood passes through the heart to reach the lungs but the answer is missing some detail. ***[1 to 2 marks]***

Level 2: There is a clear, detailed description of how deoxygenated blood passes through the heart to reach the lungs. ***[3 to 4 marks]***

Here are some points your answer may include:
Deoxygenated blood enters the right atrium through the vena cava.
The right atrium contracts.
The blood passes through a valve into the right ventricle.
The right ventricle contracts, forcing the blood through the pulmonary artery and out of the heart towards the lungs.

4.1 E.g. the vein has a bigger lumen/thinner wall/valves ***[1 mark]***. / The artery has a smaller lumen/thicker wall/ no valves ***[1 mark]***.

To answer this question, you needed to think about how the structure of veins and arteries differ — and how you could tell them apart by just looking at them.

4.2 The vein, because it has a thinner wall/less muscle in its wall/isn't as strong ***[1 mark]***.

Topic 2b — Disease and Plant Organisation

Page 43
Warm-Up Questions

1) the coronary arteries

2) a) a stent

b) E.g. complications during the operation, infection from surgery and developing a blood clot/thrombosis.

3) A mechanical device that is put into a person to pump blood if their own heart fails.

Exam Questions

1.1 fatty material ***[1 mark]***

1.2 The fatty material causes the coronary arteries to become narrow, so blood flow to the heart muscle is restricted ***[1 mark]***. This reduces (or stops) the delivery of oxygen to the heart muscle ***[1 mark]***.

2.1 It could mean that blood flows in both directions rather than just forward ***[1 mark]***, so the blood wouldn't circulate as effectively as normal ***[1 mark]***.

2.2 Biological valves ***[1 mark]*** — these are replacement valves that can be taken from humans or other mammals ***[1 mark]***.
Mechanical valves ***[1 mark]*** — these are man-made replacement valves ***[1 mark]***.

2.3 E.g. it requires surgery, which could lead to bleeding/ infection. / There could be problems with blood clots ***[1 mark]***.

3.1 Statins are drugs that can reduce the amount of 'bad' cholesterol present in the bloodstream ***[1 mark]***. This slows down the rate of fatty deposits forming, which reduces the likelihood of coronary heart disease developing ***[1 mark]***.

3.2 E.g. the patient has to remember to take them every day. / Statins can sometimes cause negative side effects. / The effect of taking statins isn't instant ***[1 mark]***.

Page 47
Warm-Up Questions

1) The state of physical and mental wellbeing.

2) It can be spread from person to person or between people and animals.

3) The presence of certain substances in the body. / The presence of certain substances in the environment.

4) E.g. money is needed to research/treat the diseases. / People with non-communicable diseases may be unable to work, which may reduce the country's economy.

Exam Questions

1.1 A risk factor is something that is linked to an increase in the likelihood that a person will develop a certain disease during their lifetime ***[1 mark]***.

1.2 E.g. smoking ***[1 mark]***.

1.3 E.g. ionising radiation ***[1 mark]***.

2.1 Uncontrolled cell division ***[1 mark]***.

2.2 malignant ***[1 mark]***

2.3 Tumour cells can break off and spread to other parts of the body by travelling in the bloodstream ***[1 mark]***. The malignant cells then invade healthy tissues elsewhere in the body and form secondary tumours ***[1 mark]***.

3.1 A disease that cannot spread between people or between animals and people ***[1 mark]***.

3.2 Because of the problem with their immune system, their body is less likely to be able to defend itself against the pathogen that causes influenza ***[1 mark]***.

Page 50

Transpiration and Stomata

Q1 Aloe vera, because the transpiration rate will be higher in the hot, dry area ***[1 mark]***, so the aloe vera will have fewer stomata to help conserve water ***[1 mark]***.

Pages 51-52

Warm-Up Questions

1) palisade mesophyll tissue

2) Meristem tissue is found at the growing tips of roots and shoots and is able to differentiate into lots of different types of plant cell.

3) False

Exam Questions

1 spongy mesophyll tissue ***[1 mark]***

2.1 xylem vessels ***[1 mark]***

2.2 They are made of dead cells joined end to end ***[1 mark]*** with no end walls between them and therefore a hole down the middle ***[1 mark]***. They're strengthened with a material called lignin ***[1 mark]***.

2.3 transpiration ***[1 mark]***

3.1 phloem ***[1 mark]***

3.2 They are made of columns of elongated living cells with small pores in the end walls ***[1 mark]*** to allow cell sap to flow through ***[1 mark]***.

3.3 translocation ***[1 mark]***

4.1 The water level in the vase would decrease more slowly ***[1 mark]***, as the rate of transpiration would be lower ***[1 mark]***. This is because water molecules have less energy at lower temperatures, meaning they don't diffuse/evaporate as quickly from the leaves ***[1 mark]***.

4.2 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: The correct diagram is identified and the answer links the shape of the guard cell to the closing of the stomata at night. ***[1 to 2 marks]***

Level 2: The correct diagram is identified and the answer clearly explains how the shape of the guard cell changes at night, allowing the stomata to close to reduce water loss when gas exchange is not required. ***[3 to 4 marks]***

Here are some points your answer may include:
A, because the guard cell is most flaccid.
Flaccid guard cells mean that the stomata would be closed.
The stomata close at night as they don't need to be open for gas exchange as photosynthesis doesn't happen in the dark.
The closing of the stomata at night prevents water being lost through transpiration.

5.1 $(10 + 11 + 9) \div 3 =$ **10%** ***[2 marks for correct answer, otherwise 1 mark for correct working]***

5.2 The movement of air from the fan sweeps away water vapour, maintaining a low concentration of water outside the leaf ***[1 mark]*** and increasing the rate at which water is lost through diffusion ***[1 mark]***. This means that the plants next to the fan would lose more water (and therefore more mass) than the plants in a still room in the same amount of time ***[1 mark]***.

5.3 The rate of transpiration would be slower ***[1 mark]*** since most water loss occurs through the stomata, which are on the underside of the leaves ***[1 mark]***.

5.4 E.g. you could put a new group of 3 basil plants in a separate room ***[1 mark]*** and increase the humidity in the room by misting/spraying the air with water ***[1 mark]***.

If you've thought of another sensible way to increase or decrease the humidity around the plants, you'd still get the mark in the exam.

Topic 3 — Infection and Response

Page 58

Warm-Up Questions

1) A communicable disease is a disease that is easily spread.

2) Vaccinations can stop people from developing the disease, so that they are unlikely pass it on to someone else.

3) E.g. measles, HIV and tobacco mosaic virus.

4) A red skin rash.

5) It causes purple or black spots to develop on the leaves, which can then turn yellow and drop off.

6) By sexual contact.

Exam Questions

1.1 Initial symptoms are flu-like ***[1 mark]***.

1.2 Pain when the infected person urinates ***[1 mark]***. Thick yellow or green discharge from the vagina or penis ***[1 mark]***.

1.3 Any two from: e.g. fever / stomach cramps / vomiting / diarrhoea ***[2 marks]***.

2.1 A microorganism that can cause disease ***[1 mark]***.

2.2 The tobacco mosaic virus ***[1 mark]***. It can reduce the growth of the plant ***[1 mark]***.

2.3 By sexual contact ***[1 mark]***, or by exchanging bodily fluids such as blood ***[1 mark]***.

3.1 To prevent the contamination of food by disease-causing pathogens that may be on the chefs' hands ***[1 mark]***.

3.2 Malaria is spread by vectors ***[1 mark]*** and not through contaminated food/surfaces / skin to skin contact ***[1 mark]***.

3.3 E.g. mosquitoes are the vectors of malaria, so destroying mosquitoes will prevent malaria being spread between people ***[1 mark]***.

Page 60
Fighting Disease — Vaccination

Q1 Basia's white blood cells recognise the antigens on the flu virus and rapidly produce antibodies, which kill the pathogen ***[1 mark]***. Cassian's white blood cells don't recognise the antigens, so it takes a while for them to produce antibodies and he becomes ill in the meantime ***[1 mark]***.

Pages 63-64
Warm-Up Questions

1) The skin acts as a barrier to pathogens and secretes antimicrobial substances that kill them.

2) To destroy pathogens that enter the body.

3) foxgloves

4) Whether the drug works and produces the effect you're looking for.

5) A placebo is a substance that's like the real drug but doesn't do anything.

6) Using a placebo allows the doctor to see the actual difference the drug makes (to a patient's illness/ symptoms).

Exam Questions

1.1 They trap particles that could contain pathogens ***[1 mark]***.

1.2 They waft the mucus up to the back of the throat where it can be swallowed ***[1 mark]***.

1.3 (hydrochloric) acid ***[1 mark]***

2.1 human volunteers ***[1 mark]***

Drugs are tested on human cells, live animals and human tissue in pre-clinical trials. If a drug makes it through the pre-clinical trials, it's then tested on human volunteers in a clinical trial.

2.2 In a double blind trial neither the patient nor the doctor ***[1 mark]*** knows who is receiving the drug and who is receiving the placebo until all the results have been gathered ***[1 mark]***.

2.3 dosage ***[1 mark]***, efficacy/how well the drug works ***[1 mark]***

2.4 Peer review is where other scientists check the work to make sure it is valid/has been carried out rigorously ***[1 mark]***. It's done help to prevent false claims being made about the results ***[1 mark]***.

3.1 Engulfing foreign cells and digesting them / phagocytosis ***[1 mark]***.

3.2 A particular antibody will only lock onto a specific type of antigen from one type of pathogen ***[1 mark]***.

4.1 *Penicillium* mould / *Penicillium notatum* ***[1 mark]***

4.2 Overuse of antibiotics may increase the rate of development of resistant strains of bacteria ***[1 mark]***. This will make bacterial infections harder to treat in the future ***[1 mark]***.

4.3 Antibiotics cannot kill viruses ***[1 mark]***.

5.1 E.g. a painkiller/aspirin ***[1 mark]***.

5.2 Rubella is a virus and viruses reproduce using your body cells ***[1 mark]***, which makes it very difficult to develop drugs that destroy just the virus without killing the body's cells ***[1 mark]***.

5.3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of how vaccination against rubella can prevent a person catching the disease or how having a large proportion of vaccinated individuals in a population reduces the risk of rubella for people who are not vaccinated. ***[1 to 2 marks]***

Level 2: There is some explanation of how vaccination against rubella can prevent a person catching the disease and how having a large proportion of vaccinated individuals in a population reduces the risk of rubella for people who are not vaccinated. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of how vaccination against rubella can prevent a person catching the disease and how having a large proportion of vaccinated individuals in a population reduces the risk of rubella for people who are not vaccinated. ***[5 to 6 marks]***

Here are some points your answer may include:
When a person is vaccinated against rubella, they are injected with dead or inactive rubella viruses.
The dead or inactive viruses carry antigens, which cause the body to produce antibodies to attack them.
If live rubella viruses infect the body after this, white blood cells can rapidly mass produce antibodies to defeat the virus.
If a large proportion of the population is vaccinated against rubella, then there are fewer people who are likely to pass the disease on.
This means that even someone who hasn't been vaccinated is less likely to catch the disease.

5.4 E.g. they may be worried that they will have a bad reaction to a vaccine. / Vaccines don't always work and so the person might not be given immunity ***[1 mark]***.

Page 67
More Uses of Monoclonal Antibodies

Q1 The fluorescent antibodies will bind to the antigens on the bacteria ***[1 mark]***, so any bacteria that are present in the sample will be detected / will glow ***[1 mark]***.

Page 69
Warm-Up Questions

1) a (B-)lymphocyte

2) A (B-)lymphocyte and a tumour cell.

3) stunted growth

4) It acts as a physical defence, preventing pathogens from entering the plant.

Exam Questions

1 chlorosis ***[1 mark]***

2.1 A waxy cuticle on leaves ***[1 mark]***, cellulose cell walls ***[1 mark]***, layers of dead cells around stems, e.g. bark ***[1 mark]***.

2.2 Magnesium ions are needed for making chlorophyll, which is needed for photosynthesis ***[1 mark]***.

2.3 Any two from: e.g. stunted growth / spots on the leaves / patches of decay / abnormal growths / malformed leaves or stems. ***[2 marks — 1 mark for each correct answer.]***

3.1 A radioactive substance ***[1 mark]***, a toxic drug ***[1 mark]***, a chemical which stops cells growing and dividing ***[1 mark]***.

3.2 The side effects of an antibody-based drug are lower than for standard treatment methods ***[1 mark]***, because monoclonal antibodies target specific cells ***[1 mark]***.

4 A will not turn blue if the woman is not pregnant ***[1 mark]***, so B is included to show that the test has worked/the testing stick is not faulty if the woman is not pregnant ***[1 mark]***. If B turns blue, it shows that the urine has travelled all the way up the stick/past A on the stick ***[1 mark]***.

Topic 4 — Bioenergetics

Page 75
Light Intensity and Farming

Q1 Using a greenhouse means that the farmer can keep conditions just right for photosynthesis ***[1 mark]***. The plants will grow faster, and a good crop can be harvested more often, which can then be sold to make a profit ***[1 mark]***.

Q2 $\text{light intensity} \propto \frac{1}{\text{distance}^2}$

$1 \div 15^2 = 0.00444...$ a.u. ***[1 mark]***
$1 \div 5^2 = 0.04$ a.u. ***[1 mark]***
$0.04 \div 0.00444... = 9$ ***[1 mark]***

Pages 76-77
Warm-Up Questions

1) glucose

2) A limiting factor is something that stops photosynthesis from happening any faster.

3) If the temperature's too high, the plant's enzymes will be denatured, so the rate of photosynthesis rapidly decreases.

Remember, if the temperature is too high for an enzyme, the bonds holding it together may break. This can cause the shape of the enzyme's active site to change, denaturing the enzyme.

4) E.g. oxygen production.

5) $\text{light intensity} \propto \frac{1}{\text{distance}^2}$

6) To increase the plants' growth so that more/bigger tomatoes are produced and a bigger profit can be made from selling them.

Exam Questions

1.1 carbon dioxide + water ***[1 mark]*** $\xrightarrow{\text{light}}$ glucose + oxygen ***[1 mark]***

1.2 cellulose ***[1 mark]***

1.3 Any two from: e.g. for respiration. / For storage as lipids/ oils or fats. / For storage as insoluble starch. / For making amino acids, which are then made into proteins. ***[2 marks]***

2.1 By counting the number of bubbles produced / by measuring the volume of gas produced, in a given time/ at regular intervals ***[1 mark]***.

2.2 Dependent variable — rate of photosynthesis/number of bubbles in a given time/volume of gas in a given time ***[1 mark]***.
Independent variable — light intensity ***[1 mark]***.

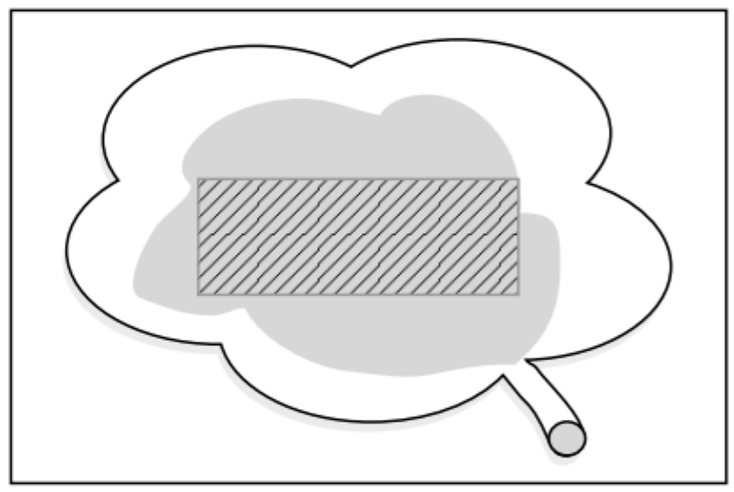
2.3 E.g. carbon dioxide concentration in the water/ temperature/the plant being used ***[1 mark]***.

3.1 At low light intensities, increasing the CO_2 concentration has no effect ***[1 mark]***, but at higher light intensities, increasing the concentration of CO_2 increases the maximum rate of photosynthesis ***[1 mark]***.

3.2 The rate of photosynthesis does not continue to increase because temperature or the level of carbon dioxide becomes the limiting factor ***[1 mark]***.

You don't know if the temperature was kept constant or not, so either the level of carbon dioxide or temperature could have been the limiting factor here — there's no way of knowing.

4.1



[1 mark]

4.2 Plants need both chlorophyll and light to photosynthesise and produce starch — there is only chlorophyll in the green area of the plant ***[1 mark]***, and light can only reach parts of the leaf not covered by black paper ***[1 mark]***.

5 $\text{light intensity} = \frac{1}{d^2}$

$= \frac{1}{7.5^2}$

$= 0.01777... =$ **0.018 a.u. (2 s.f.)**

[2 marks for correct answer, otherwise 1 mark for correct working.]

Page 80
Exercise

Q1 Running ***[1 mark]***. It raises the pulse rate the most, so it is the most vigorous type of exercise ***[1 mark]***. The more vigorous the exercise, the more anaerobic respiration will be taking place in the muscles ***[1 mark]***. Anaerobic respiration produces lactic acid, so running will lead to the greatest build up of lactic acid in the blood ***[1 mark]***.

Page 81

Warm-Up Questions

1) True

2) Glucose and oxygen.

3) Breathing rate increases, breath volume increases and heart rate increases.

Exam Questions

1.1 To release energy ***[1 mark]***.

1.2 Any two from: e.g. aerobic respiration uses oxygen, anaerobic respiration does not. / Glucose is broken down fully during aerobic respiration but is only partially broken down during anaerobic respiration. / Aerobic respiration doesn't produce lactic acid, anaerobic respiration does. / Aerobic respiration releases more energy than anaerobic respiration. ***[2 marks]***

1.3 glucose + oxygen ***[1 mark]*** → carbon dioxide + water ***[1 mark]***

2.1 glucose → lactic acid ***[1 mark]***

2.2 During vigorous exercise, the body can't supply enough oxygen to the muscles ***[1 mark]***. It uses anaerobic respiration to provide energy without using oxygen, which keeps the muscles going for longer ***[1 mark]***.

3.1 Because during exercise the muscles need more energy from respiration ***[1 mark]***, and this respiration requires oxygen ***[1 mark]***.

3.2 Because there is an oxygen debt / oxygen is needed to react with the lactic acid that has built up ***[1 mark]***.

Topic 5a — The Nervous System

Page 85

Synapses and Reflexes

Q1 A rapid, automatic response to a stimulus that doesn't involve the conscious part of the brain ***[1 mark]***.

Q2 a) muscle ***[1 mark]***

b) The heat stimulus is detected by receptors in the hand ***[1 mark]***, which send impulses along a sensory neurone to the CNS ***[1 mark]***. The impulses are transferred to a relay neurone ***[1 mark]***. They are then transferred to a motor neurone and travel along it to the effector/muscle ***[1 mark]***.

Page 86

Investigating Reaction Time

Q1 a) 242 + 256 + 253 + 249 + 235 = 1235 ***[1 mark]***
1235 ÷ 5 = **247 ms** ***[1 mark]***

b) Any two from: e.g. the hand each person used to click the mouse / the computer equipment/programme used / the amount of energy drink they consumed / the type of energy drink used / the time between consuming the energy drink and taking the test ***[2 marks]***.

Page 87

Warm-Up Questions

1) The maintenance of a stable internal environment in response to changes in both internal and external conditions.

2) E.g. body temperature, blood glucose level, water content.

3) synapse

4) sensory neurone, relay neurone, motor neurone

5) E.g. age / gender / drugs / caffeine.

Exam Questions

1.1 Stimulus: appearance of red triangle ***[1 mark]***
Receptors: cells in the eye / light receptor cells ***[1 mark]***
Effectors: muscles (in hand controlling mouse) ***[1 mark]***

1.2 343 × 3 = 1029
1029 – 328 – 346 = **355 ms** ***[2 marks for the correct answer, otherwise 1 mark for the correct calculation]***

2.1 motor neurone ***[1 mark]***

2.2 Muscle ***[1 mark]***, which contracts to move the baby's finger ***[1 mark]***.

2.3 When the electrical impulse reaches the end of the neurone, it stimulates the release of a chemical ***[1 mark]***. The chemical diffuses across the synapse to activate an electrical impulse in the next neurone ***[1 mark]***.

2.4 In a baby older than 6 months, the pathway will involve conscious parts of the brain, whereas in a newborn baby it won't. / In a baby older than 6 months, the response will not be produced as rapidly as in the newborn baby ***[1 mark]***.

The response in the baby older than 6 months is not a reflex — it chooses whether it wants to grasp an object. Remember, reflexes are automatic — they don't involve conscious parts of the brain, which makes the response much faster.

Pages 93-94

Warm-Up Questions

1) cerebellum

2) cornea

3) The lenses refract the light rays so they focus on the retina.

4) It changes the shape of the cornea using lasers, so the amount that light is refracted into the eye is changed.

5) thermoregulatory centre

Exam Questions

1.1 A: cornea ***[1 mark]***
B: pupil ***[1 mark]***

1.2 To control the diameter of the pupil/the amount of light entering the eye ***[1 mark]***.

1.3 Information is sent using impulses ***[1 mark]***, via the optic nerve ***[1 mark]***.

2.1 cerebral cortex ***[1 mark]***

2.2 E.g. the doctor may be concerned that removing the tumour will cause physical damage to the brain ***[1 mark]***. The doctor may also be concerned that removing the tumour will cause increased problems with brain function ***[1 mark]***.

3.1 Vasoconstriction ***[1 mark]***. Less blood flows through the blood vessels close to the surface of the skin ***[1 mark]***. This reduces the transfer of energy from the skin to the environment, so helps to keep the body warm ***[1 mark]***.

3.2 Sweat glands respond by stopping sweat production ***[1 mark]***, because sweat transfers energy from the body to the environment when it evaporates ***[1 mark]***.

3.3 Skin temperature will increase, so the sweat glands will respond by producing lots of sweat ***[1 mark]***. Drinking water will help to restore the water balance in the body ***[1 mark]***.

3.4 The blood vessels would widen/dilate / vasodilation would occur ***[1 mark]***. This allows more blood to flow near to the surface of the skin ***[1 mark]*** so more energy can be transferred to the surroundings, cooling the body ***[1 mark]***.

4.1 A, because the pupil is dilated/wider in this eye ***[1 mark]***, which allows more light to enter the eye/reach the light receptors in the eye ***[1 mark]***.

4.2 radial muscles ***[1 mark]***

4.3 Reflex responses happen very quickly/are automatic ***[1 mark]*** so the eye can adjust quickly to prevent the retina being damaged by bright light ***[1 mark]***.

5.1 The ciliary muscles relax ***[1 mark]*** allowing the suspensory ligaments to pull tight ***[1 mark]***, which results in the lens becoming thinner/less rounded ***[1 mark]***.

5.2 If the lens cannot form a rounded shape, light from nearby objects won't be refracted/bent enough to be focused on the retina ***[1 mark]***. This means that people with presbyopia will be unable to focus on nearby objects ***[1 mark]***.

Topic 5b — Hormones

Page 97

Controlling Blood Glucose

Q1 Curve 1, because the secretion rate is high when the blood glucose level is low / the secretion rate decreases as the blood glucose level rises ***[1 mark]***. Glucagon increases the blood glucose level, so it is secreted when the blood glucose level becomes too low ***[1 mark]***.

Page 98

Warm-Up Questions

1) In the blood.
2) E.g. pituitary gland, thyroid gland, adrenal gland, pancreas, testes.
3) A hormonal response.
4) Removes glucose from the blood. / Makes the liver turn glucose into glycogen for storage.
5) A condition in which the body becomes resistant to its own insulin.

Exam Questions

1.1 Organ A is the pancreas ***[1 mark]***.
Organ B is the liver ***[1 mark]***.

1.2 Eating carbohydrates ***[1 mark]***.

1.3 The pancreas/organ A does not produce enough insulin, or any insulin at all ***[1 mark]***. This means that the liver/organ B is unable to remove glucose from the blood for storage ***[1 mark]***. So the blood glucose level is able to rise to a dangerously high level ***[1 mark]***.

1.4 It makes the liver/organ B turn glycogen into glucose for release into the blood ***[1 mark]***.

2 The pituitary gland releases hormones that act on other glands, directing them to release hormones ***[1 mark]***. If a pituitary hormone that acts on the thyroid gland is not released ***[1 mark]***, then the thyroid gland may also stop releasing hormones, resulting in the symptoms described ***[1 mark]***.

Page 99

The Kidneys

Q1 The kidneys will reabsorb less water ***[1 mark]***, so too much water may be lost from the body in urine ***[1 mark]***.

Page 101

Warm-Up Questions

1) The deamination of amino acids in the liver produces ammonia as a waste product. This is then converted into urea in the liver.
2) (excess) ions, (excess) water
3) ADH/anti-diuretic hormone
4) pituitary gland
5) It allows ions and waste substances to diffuse out of the blood into the dialysis fluid, but not large molecules like proteins (so they stay in the blood).

Exam Questions

1.1

Substances	Blood	Filtrate
glucose	✗	✓
urea	✗	✓
ions	✗	✓

[2 marks — 1 mark for each correct row of the table]

Water is also filtered out of the blood into the filtrate. However, it's not all filtered out (otherwise the blood wouldn't be a liquid anymore) so the missing word in the table can't be water.

1.2 selective reabsorption ***[1 mark]***

2.1 The urea diffuses out through the dialysis membrane / into the dialysis fluid ***[1 mark]***, because the concentration of urea in the blood is higher than in the dialysis fluid / because the dialysis fluid contains no urea ***[1 mark]***.

2.2 The dialysis fluid contains the same concentration of glucose as healthy blood ***[1 mark]***, so there is no concentration gradient and no net movement of glucose ***[1 mark]***.

2.3 E.g. he will not have to have dialysis sessions three times every week for hours each time / A kidney transplant is cheaper than dialysis in the long term ***[1 mark]***.

2.4 E.g. there are long waiting lists for kidneys. / There's a risk that the donor kidney will be rejected by the patient's immune system ***[1 mark]***.

3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief explanation of why the runner's urine was a darker colour than normal. ***[1 to 2 marks]***

Level 2: There is some explanation of why the runner's urine was a darker colour than normal. ***[3 to 4 marks]***

Level 3: There is a clear and detailed explanation of why the runner's urine was a darker colour than normal. ***[5 to 6 marks]***

Here are some points your answer may include:
As the runner ran, she sweated, resulting in water loss and less water in her blood.
Receptors in her brain detected the decreased water content of her blood.
The coordination centre in the brain received this information and coordinated a response by the pituitary gland.
The pituitary gland released ADH (anti-diuretic hormone) into the blood.
The ADH caused the kidneys to reabsorb more water, resulting in less water being released in the runner's urine.
So the urine was more concentrated and appeared darker in colour.

Page 106

Warm-Up Questions

1) FSH/follicle-stimulating hormone

2) Any three from: e.g. progesterone-only pill / contraceptive implant / contraceptive injection / intrauterine systems.

3) E.g. it doesn't always work. / It can be expensive. / Too many eggs can be stimulated, resulting in multiple pregnancies.

4) They stimulate several eggs to mature.

5) thyroid

Exam Questions

1.1 It inhibits it ***[1 mark]***.

1.2 progesterone ***[1 mark]***

1.3 day 14 ***[1 mark]***

2.1 Keeping oestrogen levels permanently high inhibits production of FSH ***[1 mark]***, so no eggs mature / so egg development and production stop ***[1 mark]***.

2.2 E.g. by stimulating the production of thick cervical mucus ***[1 mark]***, which prevents any sperm getting through and reaching the egg ***[1 mark]***.

2.3 E.g. the contraceptive implant ***[1 mark]***, as this is effective for three years once inserted and so does not have to be thought about on a daily basis ***[1 mark]***. / The contraceptive injection ***[1 mark]***, as this is effective for 2 to 3 months and so does not have to be thought about on a daily basis ***[1 mark]***. / The contraceptive patch ***[1 mark]***, as each patch lasts one week, so does not have to be thought about on a daily basis ***[1 mark]***.

2.4 E.g. male/female sterilisation ***[1 mark]***.

3 The cat's heart rate would increase ***[1 mark]*** because adrenaline is released ***[1 mark]***. The increased heart rate increases the rate of delivery of glucose and oxygen ***[1 mark]*** to the brain and muscle cells ***[1 mark]***, which prepares the cat to escape danger/the dog by fight or flight ***[1 mark]***.

Page 110

Warm-Up Questions

1) auxin

2) E.g. to stimulate seed germination / to control dormancy / to stimulate flowering / to grow larger fruit.

3) To control the time at which fruit ripens. / To speed up the ripening of fruits.

Exam Questions

1.1 auxin ***[1 mark]***

1.2 She could use a rooting powder containing auxin ***[1 mark]*** to help the cutting produce roots rapidly and start growing as a new plant ***[1 mark]***.

1.3 gibberellin ***[1 mark]***

2.1 The shoot has grown upwards, away from gravity ***[1 mark]***.

2.2 Auxin accumulated on the lower side of the shoot ***[1 mark]***. This caused the lower side of the shoot to elongate faster, so the shoot bends upwards ***[1 mark]***.

2.3 Any two from: e.g. the number of beans on the soil ***[1 mark]*** / the type of beans used ***[1 mark]*** / the amount of water in the soil ***[1 mark]*** / the depth of the soil ***[1 mark]*** / the temperature the beans are kept at ***[1 mark]*** / the length of time the beans are left for ***[1 mark]***.

Topic 6a — DNA and Reproduction

Page 113

The Structure of DNA and Protein Synthesis

Q1 a) Four ***[1 mark]***.

b) GAGGATGCACTA ***[1 mark]***.

Page 115

Warm-Up Questions

1) chromosomes

2) A small section of DNA found on a chromosome that codes for a particular sequence of amino acids that are put together to make a specific protein.

3) E.g. it allows scientists to identify genes in the genome that are linked to different types of disease. Knowing which genes are linked to inherited diseases could help us to understand them better and could help us to develop effective treatments for them. Scientists can look at genomes to trace the migration of certain populations of people around the world.

4) A and T. C and G.

5) A mutation is a random change in an organism's DNA.

6) Mutations change the DNA base sequence. This can alter the amino acids that the bases code for. This can change the shape of the protein the amino acids make up, which can affect its function.

Exam Questions

1.1 A polymer made up of two strands. ***[1 mark]***

1.2 Its genome. ***[1 mark]***

Remember, an organism's genome is its entire set of genetic material. A gene is a short section of DNA and a chromosome is a really long structure, which contains genes.

1.3 DNA contains genes ***[1 mark]***. Each gene codes for a particular sequence of amino acids ***[1 mark]***, which are put together to make a specific protein ***[1 mark]***.

2.1 A nucleotide consists of a sugar, a phosphate group and one 'base' ***[1 mark]***.

2.2 The code from the GALC gene would be copied into mRNA. / A template would be made using the code in the GALC gene ***[1 mark]***. The code would then be carried to a ribosome ***[1 mark]***. Amino acids would be brought to the ribosome in the correct order by carrier molecules ***[1 mark]***. The amino acids would then be joined together to make the correct enzyme ***[1 mark]***.

2.3 The mutation in the GALC gene changes its base sequence ***[1 mark]***. This alters the amino acids that the bases code for ***[1 mark]***, which changes the shape of the enzyme and its active site ***[1 mark]***. A change in the shape of the active site will affect the ability of the substrate to bind to it, altering enzyme activity ***[1 mark]***.

2.4 Most mutations have very little or no effect on the production of a specific protein / change a specific protein to such a small extent that its function or appearance is unaffected ***[1 mark]***.

Page 117
Meiosis

Q1 23 ***[1 mark]***

Page 120
Warm-Up Questions

1) Because there are two parents, the offspring contain a mixture of their parents' genes. This mixture of genetic information produces variation.

2) two

3) asexual

4) XX

Exam Questions

1.1 asexual ***[1 mark]***

1.2 They will be genetically identical ***[1 mark]***.

Remember, asexual reproduction produces clones — offspring are exactly the same as the parent.

2.1 three ***[1 mark]***

When a cell undergoes meiosis, each new cell ends up with half the number of chromosomes as in the original cell.

2.2 four ***[1 mark]***

Remember, when a cell undergoes meiosis, four gametes are produced — it doesn't matter whether you're talking about human cells or mosquito cells.

2.3 Meiosis produces gametes that are genetically different to each other ***[1 mark]***. A male gamete and a female gamete then combine at fertilisation ***[1 mark]***, so the offspring inherits a mixture of chromosomes from both parents ***[1 mark]***.

2.4 Any three from: e.g. only one parent is needed ***[1 mark]***. / It uses less energy ***[1 mark]***. / It is faster ***[1 mark]***. / It's possible to produce many identical offspring ***[1 mark]***.

2.5 The malarial parasite reproduces sexually whilst in the mosquito, which leads to genetic variation ***[1 mark]***.

Page 122
More Genetic Diagrams

Q1

	R	r
r	Rr	rr
r	Rr	rr

round peas : wrinkly peas

1 : 1

[1 mark for correct gametes, 1 mark for correct offspring genotypes and 1 mark for correct ratio.]

Pages 125-126
Warm-Up Questions

1) Different versions of the same gene.

2) Because the allele which causes cystic fibrosis is recessive, so you have to have two recessive alleles to have the disorder. Heterozygous people have one dominant and one recessive allele.

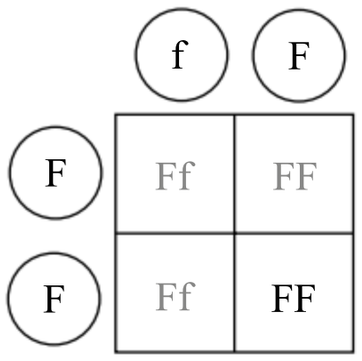
3) It's a genetic disorder where a baby is born with extra fingers or toes.

4) Taking a cell from an embryo and analysing its genes in order to detect genetic disorders.

5) Scientists of the time didn't have the background knowledge to properly understand Mendel's findings because they didn't know about genes, DNA and chromosomes.

Exam Questions

1.1

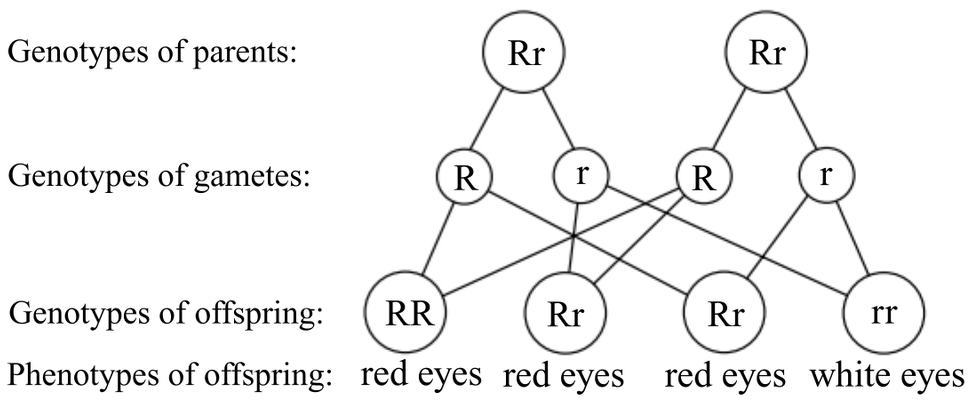


[1 mark for correct genotype of offspring, 1 mark for correct genotypes of gametes]

1.2 1 in 2 / 50% ***[1 mark]***

1.3 One is unaffected ***[1 mark]*** and the other is a carrier of cystic fibrosis ***[1 mark]***.

2.1 E.g.



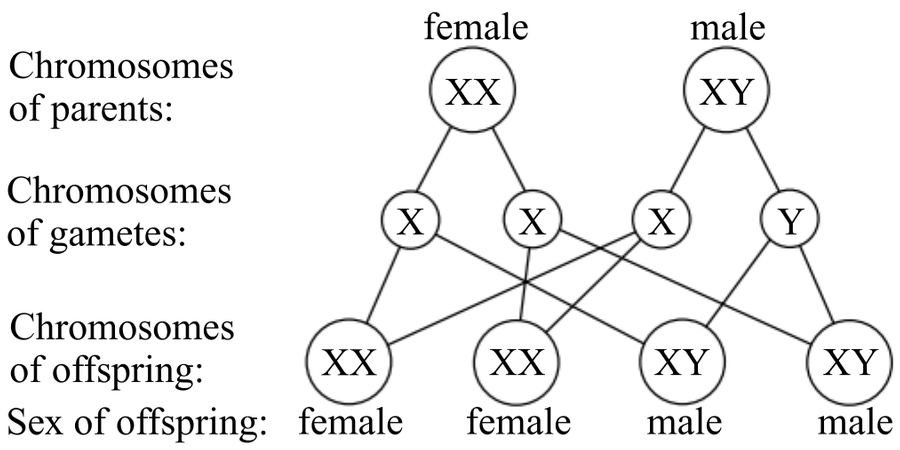
[1 mark for correct genotypes of the parents, 1 mark for correct genotypes of offspring, 1 mark for correct phenotypes of offspring]

You could have drawn a Punnett square instead here.

2.2 1 in 4 / 25% ***[1 mark]***

3 Dd ***[1 mark[***. Polydactyly is a dominant disorder, so if she was DD all of her children would be affected ***[1 mark]***.

4.1



[1 mark for correct chromosomes in parents, 1 mark for correct chromosomes in offspring, 1 mark for correct sex of offspring]

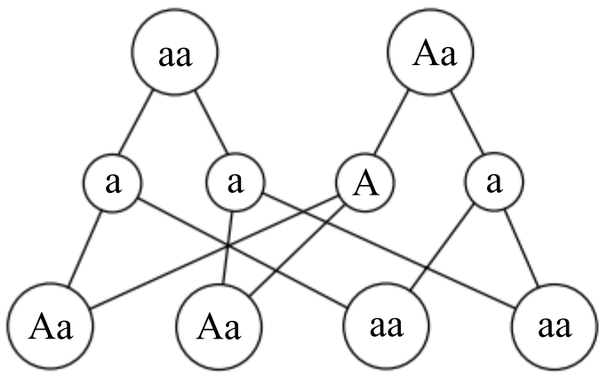
4.2 Male children will not inherit the colour blindness allele because they don't inherit an X chromosome from their father ***[1 mark]***.

4.3 0 / 0% ***[1 mark]***

A daughter of this couple would inherit the recessive colour blindness allele from her father, but also a dominant allele from her mother, so she would not be colour blind.

5.1 AA, Aa ***[1 mark]***

5.2 E.g.



[1 mark for parent with genotype aa, 1 mark for parent with genotype Aa, 1 mark for correct genotypes of offspring]

5.3 50% ***[1 mark]***

Offspring with the genotype aa will have albinism.

5.4 Fertilisation is random/the genetic diagram only shows the probability of the outcome, so the numbers of offspring produced will not always be exactly in those proportions ***[1 mark]***.

Topic 6b — Variation, Evolution and Classification

Page 129 Evolution

Q1 There was a variety of tongue lengths in the moth population ***[1 mark]***. Moths with longer tongues got more food/nectar and were more likely to survive ***[1 mark]***. These moths were more likely to reproduce and pass on the genes responsible for their long tongues ***[1 mark]***. So, over time, longer tongues became more common in the moth population ***[1 mark]***.

Page 131 Warm-Up Questions

1) Differences between members of the same species that have been caused by the environment/conditions something lives in.

2) Simple life forms that first started to develop over three billion years ago.

3) Any three from: e.g. the environment changes too quickly. / A new predator kills all the individuals. / A new disease kills all the individuals. / They can't compete with another species for food. / A catastrophic event (e.g. a volcanic eruption or collision with an astroid) kills all the individuals.

4) It contradicted the common religious beliefs of the time that all life on earth was created by God. Darwin lacked the evidence that he needed to convince many other scientists. He also didn't have any explanation for why new beneficial characteristics appeared, or how they were passed on to offspring.

Exam Questions

1.1 The difference in weight must be caused by the environment ***[1 mark]***, because the twins have exactly the same genes ***[1 mark]***.

In this case, the environment can mean the amount of food each twin eats or the amount of exercise they each do.

1.2 No, because if they were caused by genes both twins should have the birthmark ***[1 mark]***.

2.1 sexual reproduction/the combining of genes from two parents ***[1 mark]***

2.2 Mutations change the sequence of DNA bases ***[1 mark]***, which can change the protein produced by a gene ***[1 mark]*** and lead to new characteristics, increasing variation ***[1 mark]***.

2.3 E.g. if the environment changes, some mutations may give individuals a phenotype that makes them more suited to the new environment ***[1 mark]***. This would make them more likely to survive (than individuals without the phenotype) ***[1 mark]***.

3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There are some relevant points describing how the stingray has evolved but the answer is missing some detail. ***[1 to 2 marks]***

Level 2: There is a clear, detailed explanation of how the stingray has evolved through natural selection. ***[3 to 4 marks]***

Here are some points your answer may include:
Ancestors of this stingray showed variation in their appearance.
The stingrays that looked more like flat rocks were better camouflaged and so less likely to be seen and eaten by predators/more likely to survive.
This means they were more likely to reproduce.
As a result, the genes that caused the stingrays to look more like flat rocks were more likely to be passed on to the next generation.
Over time, the flat rock appearance became more common in the population and the stingray evolved.

Page 132
Selective Breeding

Q1 Select rabbits with floppy ears ***[1 mark]*** and breed them together to produce offspring ***[1 mark]***. Select offspring with floppy ears and breed them together ***[1 mark]***. Repeat this over many generations until all of the offspring have floppy ears ***[1 mark]***.

Page 135
Warm-Up Questions

1) E.g. to produce plants with large or unusual flowers.

2) Selective breeding reduces the gene pool. This causes an increased chance of organisms inheriting harmful genetic defects. There is also an increased chance that a population could be wiped out by a new disease.

3) E.g. insulin.

4) A few plant cells are put in a growth medium with hormones, and they then grow into new plants that are clones of the parent plant.

Exam Questions

1.1 E.g. to isolate/cut the gene out of the organism's DNA ***[1 mark]***.

1.2 E.g. to improve the size/quality of their fruit. / To make them make them resistant to disease/insects/herbicides ***[1 mark]***.

1.3 Any two from: e.g. some people are not convinced that GM crops are safe and are concerned that we might not fully understand the effects of eating them on human health. / Some people say that growing GM crops will negatively affect the number of wild flowers and insects that live in and around the crops. / There is concern that the transplanted genes could get out into the natural environment, which could lead to the creation of 'superweeds'. ***[2 marks]***

2 The tall and dwarf wheat plants could be bred together ***[1 mark]***. The best of the offspring/the offspring with the highest grain yield and highest bad weather resistance could then be bred together [1 mark], and this process repeated over several generations [1 mark].

3.1 Remove the nucleus from the unfertilised egg cell of a sheep ***[1 mark]***. Insert a nucleus from an adult sheep's body cell into the empty egg cell ***[1 mark]***. Stimulate the egg cell with an electric shock to make it divide ***[1 mark]***. When the embryo is a ball of cells, implant it into the uterus of a female sheep to develop ***[1 mark]***.

3.2 Female. Dolly is a clone, so she must have exactly the same genetic material as her parent ***[1 mark]***.

3.3 E.g. cloning can lead to a "reduced gene pool". / It's possible that cloned animals might not be as healthy as normal ones. / Some people worry that it may lead to humans being cloned in the future. ***[1 mark]***

3.4 E.g. the study of animal clones could lead to greater understanding of the development of the embryo/ageing/ age-related disorders ***[1 mark]***.

Page 139
Classification

Q1 B and C ***[1 mark]***.

Page 140
Warm-Up Questions

1) The microbes that cause decay can't survive in low oxygen conditions, so the dead organisms are preserved rather than decayed.

2) The two populations may be in different environments and so different characteristics will be beneficial. Natural selection will work differently in each population so that individuals with the beneficial characteristics are more likely to survive and reproduce.

3) Any one from: e.g. when the illness is only minor. / When the infection is being caused by a virus.

4) Taking the complete course makes sure that all the bacteria are destroyed. This means that there are none left to mutate and develop into antibiotic-resistant strains.

5) *fiber*

Exam Questions

1.1 Species C ***[1 mark]***

1.2 Yes, you would expect Species D to look similar to Species E because they share a recent common ancestor, so they are closely related/have similar genes ***[1 mark]***.

2.1 C, B, D, A ***[2 marks for correct answer, otherwise 1 mark for three stages in the correct order]***

2.2 E.g. people in hospital are more likely to have weakened immune systems than people in the rest of society ***[1 mark]***, and so are more likely to develop serious illness as a result of MRSA infection ***[1 mark]***.

2.3 E.g. it can lead to infections becoming more widespread/difficult to control ***[1 mark]***.

2.4 When antibiotics are used, resistant bacteria have an advantage over non-resistant bacteria ***[1 mark]***, so they will increase in number meaning the resistance spreads ***[1 mark]***. Reducing the use of antibiotics will slow/ reduce the spread of antibiotic resistance ***[1 mark]***.

Topic 7a — Ecology and Cycles

Page 142
Competition

Q1 Any three from: light / space / water / mineral ions ***[1 mark for each correct answer, up to 3 marks]***.

Q2 E.g. the frog population might increase as there might be more water spiders available for them to eat ***[1 mark]*** because there will be fewer sticklebacks to eat the water spiders / less competition for food ***[1 mark]***. / The frog population might decrease as they are more likely to be eaten by pike ***[1 mark]*** because there will be fewer sticklebacks for the pikes to eat ***[1 mark]***.

Page 144
Adaptations

Q1 a) A behavioural adaptation ***[1 mark]***.

b) E.g. it has flippers ***[1 mark]*** so it can swim for food ***[1 mark]***. / A thick layer of fat ***[1 mark]*** so it retains heat ***[1 mark]***. / A low surface area to volume ratio ***[1 mark]*** so it retains heat ***[1 mark]***.

Page 145
Food Chains

Q1 a) grass ***[1 mark]***

b) three ***[1 mark]***

c) grasshopper ***[1 mark]***

d) The population of grasshoppers could increase ***[1 mark]*** as there's nothing to eat them ***[1 mark]***. The population of snakes could decrease ***[1 mark]*** as there's nothing for them to eat ***[1 mark]***.

Page 146
Warm-Up Questions

1) a community

2) The interaction of a community of living organisms (biotic) with the non-living (abiotic) parts of their environment.

3) Any four from: moisture level / light intensity / temperature / carbon dioxide level / wind intensity / wind direction / soil pH / mineral content of soil.

4) Structural adaptations are features of an organism's body structure that allow an organism to live in an environment.

5) A producer is an organism that makes its own food using energy from the Sun.

Exam Questions

1.1 tertiary consumer ***[1 mark]***

1.2 The algae are producers ***[1 mark]***. They are the source of biomass /energy for the food chain ***[1 mark]***.

2.1 It doesn't sweat ***[1 mark]***.

2.2 E.g. it lives in burrows ***[1 mark]*** so it can avoid the heat above ground ***[1 mark]***. / It holds its tail over its head ***[1 mark]*** so it can shelter from the sun ***[1 mark]***. / It lies in the shade ***[1 mark]*** so it can shelter from the sun ***[1 mark]***. / It lies with its limbs spread out wide ***[1 mark]*** so it can lose more heat (as its surface area to volume ratio is increased) ***[1 mark]***.

3.1 At first the population size of the cutthroat trout would decrease ***[1 mark]***, as the lake trout would eat the cutthroat trout ***[1 mark]***. This would lead to a decline in the population of the lake trout (as they'd have less to eat) ***[1 mark]***, allowing an increase in the population of the cutthroat trout ***[1 mark]***. The population size of the lake trout would then increase (as they'd have more to eat) and the cycle would start again ***[1 mark]***.

3.2 Any two from: availability of food / competition for resources / new pathogens ***[1 mark for each correct answer, up to 2 marks]***.

Page 147
Using Quadrats

Q1 $1 \div 0.25 = 4$ quadrats per m^2
$0.75 \times 4 = 3$ buttercups per m^2 ***[1 mark]***.
$3 \times 1200 =$ **3600 buttercups in total** ***[1 mark]***.

Pages 151-152
Warm-Up Questions

1) $(14 + 11 + 11 + 12 + 12) \div 5 = 12$

2) A line used to help find out how organisms are distributed across an area.

3) E.g. due to seasonal factors / geographic factors / human interaction.

4) To break down/decay dead matter and animal waste.

5) Through burning.

Exam Questions

1.1 eating ***[1 mark]***

1.2 respiration ***[1 mark]***

1.3 Detritus feeders and microorganisms break down/ decay the dead matter ***[1 mark]*** and return carbon to the air as carbon dioxide through respiration ***[1 mark]***.

1.4 photosynthesis ***[1 mark]***

2.1 By the process of transpiration / water evaporates from the plants ***[1 mark]***.

2.2 Energy from the Sun evaporates water from the ocean ***[1 mark]***, so the water molecule enters the atmosphere as water vapour ***[1 mark]***. The warm water vapour rises, then cools and condenses into clouds ***[1 mark]***. The water molecule then falls from the clouds as precipitation/rain/ snow/hail into a garden pond ***[1 mark]***.

3.1 E.g. they could have placed quadrats at regular intervals ***[1 mark]*** along a transect/in a straight line from the wood to the opposite side of the field ***[1 mark]***, and counted the number of poppies in each quadrat ***[1 mark]***.

3.2 E.g. the number of poppies increases with increasing distance from the wood ***[1 mark]***.

3.3 Any two from: e.g. moisture level / soil pH / soil mineral content / wind intensity / wind direction.
[2 marks — 1 mark for each correct answer.]

3.4 Species A = 47 squares out of 100
= (47 ÷ 100) × 100 = 47% ***[1 mark]***
Species B = 48 squares out of 100
= (48 ÷ 100) × 100 = 48% ***[1 mark]***

Remember, to calculate percentage cover of an organism in a quadrat you count the number of squares which are more than half covered by the organism.

3.5 E.g. because there may have been too many blades of grass to count each one individually / it's hard to count individual blades of grass ***[1 mark]***.

4 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***
Level 1: There is a brief description of one or two ways in which carbon stored in the vegetation could be returned to the atmosphere. ***[1 to 2 marks]***
Level 2: There is some description of more than two of the ways carbon stored in the vegetation could be returned to the atmosphere. ***[3 to 4 marks]***
Level 3: There is a clear and detailed description of all the ways carbon stored in the vegetation could be returned to the atmosphere. ***[5 to 6 marks]***

Here are some points your answer may include:
Carbon stored in the small branches will be returned to the atmosphere as carbon dioxide when the branches are burnt.
The green plants could be eaten by animals, which will release some carbon as carbon dioxide during respiration.
The green plants could also be broken down by microorganisms/detritivores which will release carbon as carbon dioxide during respiration.
The wood that is taken away to be made into furniture will eventually return the carbon to the atmosphere through decay/burning when its lifespan as furniture is over.

Page 154

Investigating Decay

Q1 a) $\text{Rate} = \frac{1000}{\text{time}}$

$\text{Rate} = \frac{1000}{480} = 2.083333... =$ **2.08 s^{-1} (3 s.f.)** ***[1 mark]***

b) As the temperature increases, the rate of decay increases ***[1 mark]***.

Page 155

Warm-Up Questions

1) Any two from: e.g. bacteria / fungi / detritus feeders.

2) Compost is decomposed organic matter that is used as a natural fertiliser.

3) pink

4) E.g. sodium carbonate.

Exam Questions

1.1 methane gas ***[1 mark]***

1.2 The type of decay which produces biogas involves anaerobic respiration ***[1 mark]***, which occurs when oxygen is not present ***[1 mark]***.

1.3 E.g. heating / cooking / lighting / powering a turbine to generate electricity ***[1 mark]***.

2.1 1000 ÷ 280 = 3.5714... = **3.57 s^{-1} (3 s.f)** ***[1 mark]***

2.2 E.g. the amount of milk used for each repeat ***[1 mark]***. This could be controlled by using a measuring cylinder/ pipette to accurately measure the volume of milk used ***[1 mark]***.

You could have also said the amount of phenolphathlein or lipase used.

2.3 E.g. judging when the solution has lost its pink colour ***[1 mark]***. This effect of the error could be reduced by getting the same person to judge the colour change each time ***[1 mark]***.

3 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***
Level 1: There is a brief description of factors that could be changed to increase the rate of decomposition. ***[1 to 2 marks]***
Level 2: There is a description of factors that could be changed to increase the rate of decomposition and an explanation of how some of these factors affect the rate of decomposition. ***[3 to 4 marks]***
Level 3: There is a clear and detailed description of the factors that could be changed to increase the rate of decomposition and a full explanation of how each factor affects the rate of decomposition. ***[5 to 6 marks]***

Here are some points your answer may include:
The gardener may need to change the temperature of his compost bin. A warm temperature will increase the rate of decomposition because the enzymes of the decay organisms will be able to work at a higher rate. However, if the temperature is too high, it will decrease the rate of decomposition. This is because the enzymes of the decay organisms will be destroyed and the decay organisms will die.

The gardener may need to increase the water content of his compost bin. A moist environment allows decay to happen faster because decay organisms need water to carry out their biological processes.
He may also need to increase the air flow through the compost bin. The more air flow, the greater the oxygen availability. This will increase the rate of decomposition by those microorganisms involved in decay that respire aerobically.
He could also consider adding additional decay microorganisms and detritus feeders to his compost bin, as the more of these organisms there are, the faster decomposition will take place.

Topic 7b — Ecosystems, Biodiversity & Food Production

Page 161
Warm-Up Questions

1) The variety of different species of organisms on Earth, or within an ecosystem.
2) E.g. carbon dioxide, methane.
3) E.g. to clear land for farming or to grow crops for biofuels.
4) Breeding programmes breed endangered animals in captivity to make sure the species survives if they die out in the wild. Individuals can sometimes be released into the wild to boost or re-establish a population.
5) Hedgerows and field margins can be reintroduced around single-crop fields. These provide a habitat for organisms that would otherwise be unable to live in the area.

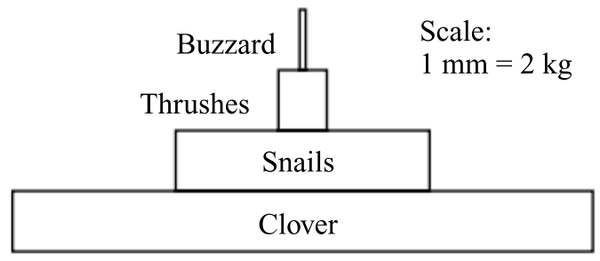
Hedgerows and field margins mean that more species of wild plants can grow. Remember, plants are often the producers in food chains — so the more variety there is in the plants that are growing, the more organisms that will be able to survive in the area.

Exam Questions

1.1 E.g. smoke ***[1 mark]***, acidic gases ***[1 mark]***.
1.2 E.g. sewage/toxic chemicals from industry can pollute lakes/rivers/oceans. / Fertilisers/pesticides/herbicides/ toxic chemicals used on land can be washed into water ***[1 mark]***.
1.3 E.g. the human population size is increasing ***[1 mark]***, people around the world are demanding a higher standard of living ***[1 mark]***.
2.1 E.g. changes in species distributions ***[1 mark]***, changes in migration patterns ***[1 mark]***.
2.2 Governments have introduced regulations and programmes to reduce deforestation ***[1 mark]***. They have also introduced regulations to limit the amount of carbon dioxide released by businesses ***[1 mark]***.
3.1 When peat is drained, it comes into more contact with air and some microorganisms start to decompose it ***[1 mark]***. This releases carbon dioxide ***[1 mark]***. Carbon dioxide is also released when peat is burned as a fuel ***[1 mark]***. Carbon dioxide is a greenhouse gas, which contributes to global warming ***[1 mark]***.
3.2 E.g. destroying peat bogs reduces biodiversity ***[1 mark]***, as it destroys the habitats of the organisms that live in the bog ***[1 mark]***.

Page 163
Pyramids of Biomass

Q1 E.g.



[2 marks for drawing all bars correctly to scale, otherwise 1 mark for any three bars drawn correctly to scale. 1 mark for all bars correctly labelled, 1 mark for a correct scale.]

Page 164
Biomass Transfer

Q1 Any two from: e.g. organisms don't always eat every single part of the organism they're consuming. / Organisms don't absorb all of the material in the food they ingest. / Some of the biomass is converted into other substances and released as waste. ***[1 mark for each correct answer, up to 2 marks]***

Q2 a) 995 – 110 = **885 kg** ***[1 mark]***
b) (110 ÷ 995) × 100 = 11.055... = **11.1% (3 s.f.)** ***[1 mark]***

Page 166
Biotechnology

Q1 A plasmid could be removed from the bacterial cell and cut open with an enzyme ***[1 mark]***. The rennin gene could be cut out of a cow chromosome using an enzyme ***[1 mark]***. The plasmid and rennin gene could then be joined together and the plasmid re-inserted into the bacterial cell ***[1 mark]***.

Page 167
Warm-Up Questions

1) trophic levels
2) Biomass is the mass of living material.
3) By reducing the number of fish that are caught in certain areas and preventing fish that are too small from being caught.
4) To reduce the transfer of energy from the livestock to the environment, which means more energy is available for growth. This increases the efficiency of farming / means more food can be produced from the same input of resources.

Exam Questions

1.1 $75 \times 1.5 =$ **112.5 g** ***[1 mark]***

1.2 $(18 \div 112.5) \times 100 =$ **16%** ***[1 mark]***

2.1 Food security is having enough food to feed a population ***[1 mark]***.

2.2 Any two from: e.g. the increasing human population ***[1 mark]***, because more people will require more food ***[1 mark]***. / Increased demand for certain foods ***[1 mark]*** because this means that these food resources will become more scarce ***[1 mark]***. / New pests/pathogens ***[1 mark]*** because these may result in the loss of crops and livestock ***[1 mark]***. / Environmental conditions ***[1 mark]*** because these may result in the loss of crops and livestock ***[1 mark]***. / High input costs of farming ***[1 mark]*** because these can make it too expensive for people in some countries to farm ***[1 mark]***. / Conflicts in some parts of the world ***[1 mark]*** because they can affect the availability of food and water ***[1 mark]***.

2.3 A larger mesh size will let unwanted species of fish escape ***[1 mark]*** reducing the number of fish that are caught accidentally and discarded ***[1 mark]***. / A larger mesh size will allow smaller and younger fish to escape ***[1 mark]***, allowing them to continue to breed and maintain the fish population ***[1 mark]***.

3.1 To supply oxygen ***[1 mark]*** for fungus to respire aerobically ***[1 mark]***.

3.2 E.g. to control the temperature inside the fermenter ***[1 mark]*** because if the temperature gets too high it will kill the fungus/denature the fungus's enzymes ***[1 mark]***.

3.3 E.g. glucose syrup ***[1 mark]*** as this is a food source for the fungus ***[1 mark]***.

Practice Paper 1

1.1 Any two from: e.g. it has a 'true' nucleus / it contains mitochondria / it doesn't have a cell wall / it doesn't contain plasmids ***[2 marks]***.

1.2 E.g. most cells have a structure that is specialised for their function, so they will contain different subcellular structures ***[1 mark]***.

1.3 cell membrane ***[1 mark]***

1.4 *E. coli* bacteria ***[1 mark]***

1.5 image size = magnification × real size
$= 150 \times 0.054$
$=$ **8.1 mm** ***[2 marks for correct answer, otherwise 1 mark for correct working]***

1.6 How to grade your answer:

Level 0: No relevant information is given. ***[No marks]***

Level 1: There is a brief comparison of plant and animal cells, including at least one similarity and one difference. ***[1 to 2 marks]***

Level 2: There is a comparison of plant and animal cells, including at least two similarities and two differences. Some descriptions of subcellular structures are included. ***[3 to 4 marks]***

Level 3: There is a detailed comparison of plant and animal cells, including at least three similarities and three differences. Detailed descriptions of subcellular structures are included. ***[5 to 6 marks]***

Here are some points your answer may include:
Similarities:
Both plant and animal cells have a nucleus, which controls the cell's activities.
Both plant and animal cells contain cytoplasm, which is where most of the cell's chemical reactions take place.
Plant cells and animal cells both have a cell membrane, which controls what goes in and out of the cell.
Mitochondria are found in both plant cells and animal cells — these are where most of the reactions for aerobic respiration take place.
Both plant cells and animal cells have ribosomes, which are where proteins are made.
Differences:
Chloroplasts, the site of photosynthesis, are present in plant cells, but not in animal cells.
Plant cells have a cell wall, which supports and strengthens the cell, but animal cells do not.
Plant cells contain a permanent vacuole, containing cell sap, but animal cells do not.

2.1 It is a control ***[1 mark]***. It shows that any difference between the growth of the bacteria is due to the effect of the antibiotics alone ***[1 mark]***.

2.2 Any two from: e.g. used sterilised Petri dishes / used sterilised culture mediums / sterilised the inoculating loop used to transfer the bacteria (by passing it through a hot flame) / used a sterile dropping pipette and spreader / lightly taped on the lids of the Petri dishes after transferring the bacteria / stored the Petri dishes upside down ***[2 marks]***.

2.3 Because harmful pathogens are more likely to grow above this temperature ***[1 mark]***.

2.4 Antibiotic X:
4.4 cm ÷ 2 = 2.2 cm
$\pi \times 2.2^2 = 15.205... =$ **15 cm² (2 s.f.)** ***[1 mark]***

Antibiotic Y:
3.0 cm ÷ 2 = 1.5 cm
$\pi \times 1.5^2 = 7.068... =$ **7.1 cm² (2 s.f.)** ***[1 mark]***

The inhibition zone of antibiotic X is greater than the inhibition zone of antibiotic Y, so antibiotic X is more effective against the bacteria than antibiotic Y ***[1 mark]***.

2.5 The bacteria could have developed resistance to the antibiotic ***[1 mark]***.

3.1 By droplets from an infected person's sneeze or cough ***[1 mark]***.

3.2 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There are some relevant points explaining how vaccination helps to protect the body against disease but the answer is missing some detail. ***[1 to 2 marks]***

Level 2: There is a clear, detailed explanation of how vaccination helps to protect the body against disease. ***[3 to 4 marks]***

Here are some points your answer may include:
The body is injected with small amounts of dead or inactive pathogens (which are harmless).
The pathogens carry antigens, which cause the white blood cells in the body to produce antibodies.
Antibodies attack/kill the pathogens.
If live pathogens of the same type appear again, the white blood cells can rapidly mass-produce antibodies to kill the pathogens so the person is less likely to get ill.

3.3 Any two from: e.g. stop the mosquitoes from breeding / protect people from mosquito bites using mosquito nets / protect people from mosquito bites by using insecticides (to kill the mosquitoes) ***[2 marks]***.

3.4 malaria ***[1 mark]***

3.5 3 days = 3 × 24 = 72 hours
72 ÷ 18 = 4 divisions
$2^4 = 2 \times 2 \times 2 \times 2 =$ **16 cells** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

4.1 biuret solution ***[1 mark]***

4.2 Benedict's solution ***[1 mark]***

4.3 A ***[1 mark]***

4.4 It is covered in many villi that provide a large surface area for diffusion (and active transport) to occur across ***[1 mark]***. The villi have a thin wall/single layer of surface cells, which decreases the distance for diffusion to occur across ***[1 mark]***. They also have a good blood supply to assist quick absorption ***[1 mark]***.

4.5 Their blood sugar level will not rise very much or not rise at all ***[1 mark]***. This is because a person with lactose intolerance has little or no lactase to break down the lactose in the drink, so there will be little or no sugar to be absorbed from the small intestine ***[1 mark]***.

5.1 A group of different tissues that work together to perform a certain function ***[1 mark]***.

5.2 A = oxygen, B = carbon dioxide ***[1 mark]***

The job of the lungs is to transfer oxygen to the blood and remove waste carbon dioxide from it.

5.3 So that they have a good blood supply ***[1 mark]*** to maximise the rate of diffusion of the gases into and out of the alveoli ***[1 mark]***.

A good blood supply makes sure that the concentration gradient between the blood and the air in the alveoli is maintained, which leads to a higher rate of diffusion.

5.4 E.g. the function of red blood cells is to transport oxygen to respiring cells ***[1 mark]***. The sickle-shaped cells are more rigid than normal red blood cells and they are the wrong shape, which could result in them being unable to fit through the capillaries ***[1 mark]***. This could reduce/block the blood supply to respiring cells, meaning that not enough oxygen is transported to the cells, causing breathlessness ***[1 mark]***.

If you don't have enough oxygen for respiration, your breathing rate increases to get more oxygen into your blood. In a person with sickle cell anaemia, the lack of oxygen may cause their breathing rate to increase so much that they feel breathless.

6.1 A — artery, B — capillary, C — vein ***[2 marks for all three correct, otherwise 1 mark for two correct.]***

6.2 The blood supply to the area below the blockage would be cut off/reduced ***[1 mark]***. Not enough oxygen would reach this part of the heart muscle, resulting in cells being unable to respire/damage/death of the muscle tissue/a heart attack ***[1 mark]***.

6.3 A stent will keep the coronary artery open, making sure that enough blood can reach the heart muscle ***[1 mark]***.

6.4 Any two from: to test drug efficacy/to see whether the drug works and produces the desired effect. / To find out about the drug's toxicity/how harmful the drug is. / To find the best dosage of drug to use/the correct concentration and how often it should be taken ***[2 marks]***.

6.5 To make sure it did not cause any harmful side effects when the body is working normally ***[1 mark]***.

6.6 The statin decreased the risk of having a heart attack for people who had already had one heart attack ***[1 mark]***.

6.7 E.g. a control/placebo was used ***[1 mark]***.

7.1 It is broken down into sugars/maltose ***[1 mark]***.

7.2 E.g. the concentration of starch / the concentration of amylase / the amount of mixing / the volume of the sample removed ***[1 mark]***.

7.3 E.g. they could have added the sample to a drop of iodine solution (e.g. in a well of a spotting tile) ***[1 mark]***. If starch was present the iodine solution would have changed to blue-black ***[1 mark]***. If starch was no longer present, the iodine solution would have remained browny-orange ***[1 mark]***.

7.4 Time of reaction at 40 °C = 100 s.
Rate = 1000 ÷ 100 = **10 s^{-1}** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

7.5 The temperature was too high ***[1 mark]***, causing the enzyme to denature / the shape of the enzyme's active site to change ***[1 mark]***. This means the starch could no longer fit in the active site of the enzyme, so it could not be broken down ***[1 mark]***.

7.6 The result for 20 °C is anomalous ***[1 mark]*** because the time taken until starch is no longer present is quicker than expected/quicker than at 30 or 40 °C ***[1 mark]***. To cause this result, the student may not have used the correct volume of starch/amylase solution. / The student may have started timing the experiment too late. / The student may have stopped the experiment too early ***[1 mark]***.

7.7 As temperature increases, the time until the starch is no longer present increases ***[1 mark]***.

7.8 By using buffer solutions with a range of different pH values ***[1 mark]***.

8.1 9:00 a.m. ***[1 mark]***

8.2 E.g. day 2 was less bright, so the stomata weren't fully open and so less water could move out of the leaves. / Day 2 was colder, so the water evaporated/diffused more slowly. / Day 2 was less windy, so the water vapour was carried away more slowly, meaning that diffusion couldn't happen as quickly. / Day 2 was wetter/more humid, so there was a smaller diffusion gradient between the inside and outside of the leaf so diffusion couldn't happen as quickly ***[1 mark for reason, 1 mark for explanation]***.

The rate of transpiration varies throughout the day due to the changing light intensity, but it can also be affected by the temperature, the air flow and the humidity around the leaves.

8.3 The plant has lost too much water/has lost water faster than it could be replaced through the roots ***[1 mark]***.

8.4 Transpiration creates a slight shortage of water in the leaf ***[1 mark]***. More water is drawn up from the rest of the plant through the xylem vessels to replace it ***[1 mark]***. This in turn means that more water is drawn up from the roots, and so there's a constant transpiration stream of water through the plant ***[1 mark]***.

9 How to grade your answer:

Level 0: No relevant information. ***[No marks]***

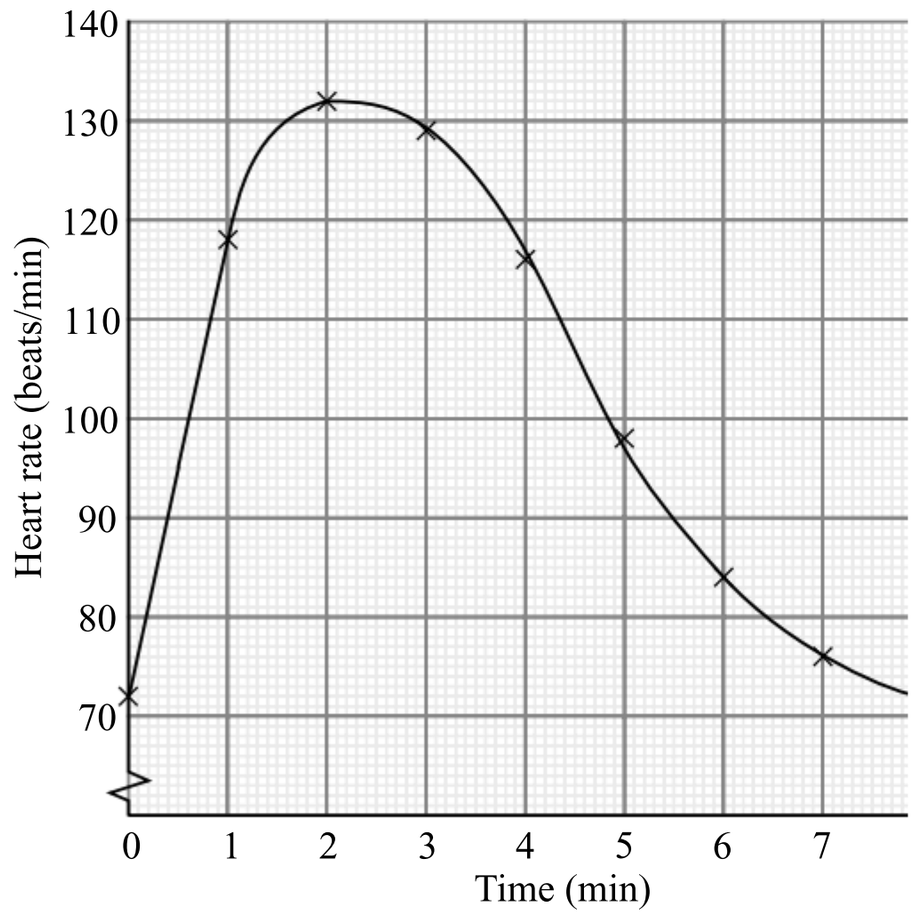
Level 1: There is a basic description of at least two of the stages of the cell cycle or a suggestion of why disrupting the cell cycle may kill plants is stated. ***[1 to 2 marks]***

Level 2: There is a clear description of at least two stages of the cell cycle and a suggestion of why disrupting the cell cycle may kill plants is stated. ***[3 to 4 marks]***

Level 3: There is a detailed description of each stage of the cell cycle and a suggestion of why disrupting the cell cycle may kill plants is stated. ***[5 to 6 marks]***

Here are some points your answer may include:
The cell cycle involves the division of cells to form two identical cells.
Before division takes place the cell grows and increases the amount of subcellular structures it has, such as mitochondria and ribosomes.
It then duplicates its DNA so there's one copy for each new cell.
Next, mitosis takes place. The chromosomes line up at the centre of the cell and cell fibres pull them apart.
The two arms of each chromosome go to opposite ends of the cell.
Membranes form around each of the sets of chromosomes to become the nuclei of the two new cells. Finally the cytoplasm and cell membrane divide so two identical daughter cells are formed.
Disrupting the cell cycle can kill plants as plants are multicellular organisms which use mitosis to grow and to replace damaged cells. If they can no longer do this, they may die.

10.1



[1 mark for label and suitable scale for the x-axis, 2 marks for all points correctly plotted (or 1 mark for 8 points plotted correctly), 1 mark for a smooth curve of best fit.]

10.2 95 beats/min ***[1 mark. Allow any value correctly read from 30 seconds on your own graph.]***

10.3 heart rate ***[1 mark]***

10.4 The student's heart rate increases during exercise ***[1 mark]***. This is because more energy is needed when exercising, so the student respires more ***[1 mark]***. Increased respiration means that the student's cells need more oxygen ***[1 mark]***, so the student's heart rate increases so that the blood flows more quickly to the cells to deliver oxygen ***[1 mark]***.

10.5 glucose ***[1 mark]*** + oxygen → carbon dioxide + water ***[1 mark]***

10.6 E.g. anaerobic respiration causes a build up of lactic acid in the muscles ***[1 mark]***. After exercise, extra oxygen is needed to react with lactic acid / the lactic acid is transported in the blood from the muscles to the liver to be converted back to glucose ***[1 mark]***, so the heart rate remains high to transport oxygen to the cells faster / to transport lactic acid away from the muscles to the liver faster ***[1 mark]***.

Practice Paper 2

1.1 A producer is a photosynthetic organism that makes its own food using energy from the Sun ***[1 mark]***.

1.2 D ***[1 mark]***, because the biomass of the organisms decreases at each trophic level and the bars on this pyramid get smaller at each trophic level ***[1 mark]***.

1.3 efficiency = (1.4 kg ÷ 12 kg) × 100
= 0.1166... × 100 = **12% (2 s.f.)** ***[1 mark]***

1.4 Energy is lost at each level of a food chain ***[1 mark]***. After about five levels, the amount of energy being passed on is not sufficient to support another level of organisms ***[1 mark]***.

2.1 21 ***[1 mark]***

To find the median, you write out all the results from lowest to highest, e.g. 10, 21, 21, 35, 37. 21 is in the middle of the list, so it is the median.

2.2 (6 + 15 + 9 + 14 + 20 + 5 + 3 + 11 + 10 + 7) ÷ 10
= 100 ÷ 10
= **10 buttercups per m^2** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

2.3 90 × 120 = 10 800 m^2 ***[1 mark]***
10 × 10 800 = **108 000 buttercups** ***[1 mark]***
Allow incorrect value from 2.2.

2.4 The student could have calculated the percentage cover ***[1 mark]***.

2.5 E.g. more buttercups grow where there is a higher moisture level in the soil ***[1 mark]***.

2.6 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***

Level 1: There is a brief description of how the student could investigate whether the change in distribution of buttercups is due to variability in the moisture level of the soil. ***[1 to 2 marks]***

Level 2: There is a detailed description of how the student could investigate whether the change in distribution of buttercups is due to variability in the moisture level of the soil. ***[3 to 4 marks]***

Here are some points your answer may include:
The student could use a transect across Field A.
To do this, a line should be marked out across the field.
Then the student should record the number of buttercups in quadrats placed next to each other/at intervals along the line.
The student should also measure the moisture level of the soil at each sampling point (e.g. with a probe).

3.1 Light coming through the hole in the box caused more auxin to accumulate on the shaded sides of the shoots ***[1 mark]***. This made the cells on the shaded sides of the plants grow faster ***[1 mark]***, so the shoots bent towards the light ***[1 mark]***.

3.2 phototropism ***[1 mark]***

3.3 gravity ***[1 mark]***

4.1 The thermoregulatory centre in the brain contains receptors that monitor the temperature of blood flowing through the brain ***[1 mark]***. It also receives impulses from temperature receptors in the skin, giving information about skin temperature ***[1 mark]***.

4.2 E.g. blood vessels supplying skin capillaries would have constricted ***[1 mark]*** to reduce the amount of energy transferred from his skin to the environment ***[1 mark]***. / He would have started to shiver ***[1 mark]***. This needs respiration, which transfers some energy to warm the body ***[1 mark]***.

5.1 pituitary gland ***[1 mark]***

5.2 It sends a message to the pituitary gland to release more ADH ***[1 mark]***, so more water is reabsorbed from the kidney tubules ***[1 mark]***.

5.3 The increased level of ADH due to taking ecstasy causes the kidneys to reabsorb too much water into the blood ***[1 mark]***. If the water content of the blood is higher than the body cells, water may enter the cells due to osmosis ***[1 mark]***.

5.4 The person's blood flows between partially permeable membranes ***[1 mark]***. The dialysis fluid contains the same concentration of glucose and dissolved ions as healthy blood, which ensures that glucose and other useful ions are not lost ***[1 mark]***. Waste products/urea and excess ions and water are able to diffuse out of the blood into the dialysis fluid ***[1 mark]***.

6.1 *Biston* ***[1 mark]***

6.2 the dark variety ***[1 mark]***

6.3 The dark variety is less likely to be eaten by predators in soot polluted areas (because they are better camouflaged) ***[1 mark]*** so they are more likely to survive to reproduce ***[1 mark]***, meaning that the genes for the characteristics that made them successful / genes for dark colouring are more likely to be passed on to the next generation and become more common in the population ***[1 mark]***.

6.4 Town B, because it contains a higher proportion of dark moths ***[1 mark]***.

7.1 Their reaction time was very fast ***[1 mark]***.
Their response was involuntary/automatic ***[1 mark]***.

If you have to think about what response to give then it's not a reflex action.

7.2 reaction time ***[1 mark]***

7.3 sex ***[1 mark]***

7.4 The males in this experiment had a faster mean reaction time than the females ***[1 mark]***, so the data supports the scientist's hypothesis ***[1 mark]***.

7.5 Any two from: e.g. the age of the participants. / The strength of the tap on the knee. / Caffeine consumption of the participants prior to the investigation ***[2 marks]***.

7.6 relay neurone ***[1 mark]***

7.7 How to grade your answer:

Level 0: There is no relevant information. ***[No marks]***
Level 1: There is a brief description of some parts of the path taken by a nervous impulse in the reflex. ***[1 to 2 marks]***
Level 2: There is some description of the path taken by a nervous impulse in the reflex, but some detail is missing. ***[3 to 4 marks]***
Level 3: There is a clear and detailed description of the path taken by a nervous impulse in the reflex. ***[5 to 6 marks]***

Here are some points your answer may include:
The impulse travels along a sensory neurone to the central nervous system/spinal cord.
When the impulse reaches a synapse between the sensory neurone and a relay neurone, it triggers chemicals to be released.
These chemicals cause impulses to be sent along the relay neurone.
When the impulse reaches a synapse between the relay neurone and a motor neurone, chemicals are released again which cause impulses to be sent along the motor neurone.
The impulse then reaches the muscle, which contracts to curl the big toe upwards.

8.1 Producing enough food without using resources faster than they renew ***[1 mark]***.

8.2 4000 – 16 000 = –12 000
(–12 000 ÷ 16 000) × 100 = –75%, so the percentage decrease is 75% ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

A negative value for percentage change indicates a decrease.

8.3 16 000 – 4000 = 12 000 cod
2000 – 1980 = 20 years
12 000 ÷ 20 = **600 cod/year** ***[2 marks for correct answer, otherwise 1 mark for correct working.]***

8.4 E.g. the number of cod able to reproduce declined from 16 000 to about 3500 between 1980 and 2001, then rose to about 5000 by 2005 ***[1 mark]***. Cod numbers may have declined because levels of fishing increased / larger/better fishing nets were used so more cod were caught ***[1 mark]***. They may have then risen because of the introduction of fishing quotas / restrictions on the type/size of nets that could be used ***[1 mark]***.

8.5 *Fusarium* ***[1 mark]***

8.6 E.g. the fungus is grown on glucose syrup ***[1 mark]*** in aerobic conditions ***[1 mark]***. The fungal biomass is then harvested and purified to make mycoprotein ***[1 mark]***.

9.1 The entire set of genetic material in an organism ***[1 mark]***.

9.2 Knowing more about the cystic fibrosis gene could help us to understand the disorder better / develop effective treatments for it ***[1 mark]***.

9.3 E.g.

mother's alleles

	F	f
F	FF	Ff
f	Ff	ff

father's alleles

ff offspring has cystic fibrosis.
Probability = 0.25 / 25% / 1:3 / 1 in 4 / ¼
[1 mark for correctly identifying parents' genotypes, 1 mark for correctly identifying possible genotypes of the offspring, 1 mark for correctly identifying cystic fibrosis genotype, 1 mark for identifying the correct probability of a child having cystic fibrosis.]

The parents must both be carriers, since neither of them show any sign of the disorder. This means they both have one copy of the recessive allele for cystic fibrosis — so they're both Ff.

9.4 Homozygous, because he has two alleles the same / both of his alleles are recessive ***[1 mark]***.

9.5 No, because polydactyly isn't a significant health issue / embryos are only screened for serious genetic disorders ***[1 mark]***.

People with polydactyly have extra fingers or toes, but it doesn't cause any other problems, so it isn't life threatening.

9.6 E.g. there may come a point where everyone wants to screen their embryos so they can pick the most 'desirable' one. / The rejected embryos are destroyed — they could have developed into humans ***[1 mark]***.

10.1 Any three from: e.g. sexual reproduction involves the fusion of male and female gametes whereas asexual reproduction doesn't involve gametes. / Sexual reproduction involves two parents whereas asexual reproduction only involves one parent. / Sexual reproduction produces offspring that are genetically different to their parents whereas asexual reproduction produces offspring that are genetically identical to their parents (clones). / Sexual reproduction produces offspring that are genetically different to each other whereas asexual reproduction produces offspring that are genetically identical. ***[3 marks]***

10.2 testosterone ***[1 mark]***

10.3 It causes the lining of the uterus to grow ***[1 mark]***. It also stimulates the release of luteinising hormone (LH) and inhibits the release of follicle stimulating hormone (FSH) ***[1 mark]***.

10.4 C ***[1 mark]*** — e.g. because the level corresponds to the thickness of the uterus lining, which is maintained by progesterone ***[1 mark]***.

There are a few different ways that you can tell that this line represents progesterone. As long as you're able to justify your answer, you'll get the marks.

10.5 Advantages of the implant: e.g. it works for three years / protection is always available / you do not have to remember to take a pill every day / it is highly effective.
Disadvantages of the implant: e.g. it has to be inserted by a doctor or nurse which may be painful/inconvenient / if certain medications are taken, a different contraceptive method would have to be used / you could suffer from side effects / it does not protect against STDs.
Advantages of the condom: e.g. it protects against STDs / there are no side effects / it is highly effective if used correctly / it is not affected by medication.
Disadvantages of the condom: e.g. it needs to be available at the time of intercourse / if it isn't used correctly it may not be effective / it can only be used once.
[3 marks for at least one advantage and one disadvantage of each method, otherwise 2 marks for at least two advantages or disadvantages of either method or 1 mark for one advantage or disadvantage of either method. 1 mark for giving a justified conclusion that refers to at least one advantage and one disadvantage of the chosen method.]

11.1 At point X, CO_2 is being removed from the atmosphere by photosynthesis ***[1 mark]***. At point Y, CO_2 is being released into the atmosphere by plants and animals respiring ***[1 mark]***. At point Z, carbon compounds in the plants are being transferred to animals as they eat the plants ***[1 mark]***.

11.2 Microorganisms break down waste products and dead organisms ***[1 mark]*** and release carbon dioxide into the atmosphere as they respire ***[1 mark]***.

11.3 The two sets of data show the same pattern/increase at the same time ***[1 mark]***.

11.4 Any two from: e.g. the two things may follow similar patterns by chance. / Some other factor may have caused both increases. / The concentration of carbon dioxide varied even when human population was low/fairly constant. ***[2 marks]***

11.5 Any two from: e.g. to increase the amount of land available for farming cattle. / To increase the amount of land available for rice crops. / To increase the amount of land available for growing crops from which biofuels are produced. ***[2 marks]***

11.6 Any two from: e.g. habitats in low-lying areas will be lost due to rising sea levels ***[1 mark]***. / The distribution of organisms could change due to changes in temperature and rainfall ***[1 mark]***. / Biodiversity could be reduced ***[1 mark]***. / There could be changes in migration patterns (e.g. some birds may migrate further north) ***[1 mark]***.

Glossary

A

Abiotic factor A non-living factor of the environment.

Accommodation The ability to focus on near or distant objects by changing the shape of the lens in the eye.

Accurate result A result that is very close to the true answer.

Active transport The movement of particles against a concentration gradient (i.e. from an area of lower concentration to an area of higher concentration) using energy transferred during respiration.

Adaptation A feature or characteristic that helps an organism to survive in its natural environment.

Aerobic respiration Respiration taking place in the presence of oxygen.

Allele An alternative version of a gene.

Alveolus A tiny air sac in the lungs, where gas exchange occurs.

Amino acid A small molecule that is a building block of proteins.

Anaerobic respiration Respiration taking place in the absence of oxygen.

Anomalous result A result that doesn't seem to fit with the rest of the data.

Antibiotic A drug used to kill or prevent the growth of bacteria.

Antibiotic resistance When bacteria aren't killed by an antibiotic.

Antibody A protein produced by white blood cells in response to the presence of an antigen.

Antigen A molecule on the surface of a cell or a pathogen. Foreign antigens trigger white blood cells to produce antibodies.

Antitoxin A protein produced by white blood cells that counteracts toxins made by invading bacteria.

Artery A blood vessel that carries blood away from the heart.

Asexual reproduction Where organisms reproduce by mitosis to produce genetically identical offspring.

Auxin A plant hormone that controls the growth of a plant in response to different stimuli.

B

Behavioural adaptation A way in which an organism behaves that helps it to survive in its environment.

Bias Prejudice towards or against something.

Binary fission A type of simple cell division carried out by prokaryotic cells, where the cell makes copies of its genetic material before splitting into two daughter cells.

Binomial system The system used in classification for naming organisms using a two-part Latin name.

Biodiversity The variety of different species of organisms on Earth, or within an ecosystem.

Biomass The mass of living material in an organism or a group of organisms.

Biotic factor A living factor of the environment.

C

Capillary A type of blood vessel involved in the exchange of materials at tissues.

Carbohydrase A type of digestive enzyme that catalyses the breakdown of a carbohydrate into sugars.

Cardiovascular disease Disease of the heart or blood vessels.

Catalyst A substance that increases the speed of a reaction, without being changed or used up.

Categoric data Data that comes in distinct categories (e.g. flower colour or blood group).

Cell membrane A membrane surrounding a cell, which holds it together and controls what goes in and out.

Cell wall A structure surrounding some cell types, which gives strength and support.

Cellulose A molecule which strengthens cell walls in plants and algae.

Central nervous system (CNS) The brain and spinal cord. It's where reflexes and actions are coordinated.

Chlorophyll A green substance found in chloroplasts which absorbs light for photosynthesis.

Chloroplast A structure found in plant cells and algae. It is the site of photosynthesis.

Chromosome A long molecule of DNA found in the nucleus. Each chromosome carries many genes.

Clinical trial A set of drug tests on human volunteers.

Clone An organism that is genetically identical to another organism.

Communicable disease A disease that can spread between individuals.

Community The populations of different species living in a habitat.

Continuous data Numerical data that can have any value within a range (e.g. length, volume or temperature).

Contraceptive A method of preventing pregnancy, which can be hormonal or non-hormonal.

Control experiment An experiment that's kept under the same conditions as the rest of an investigation, but doesn't have anything done to it.

Control variable A variable in an experiment that is kept the same.

Coordination centre An organ (e.g. the brain, spinal cord or pancreas) that processes information from receptors and organises a response from the effectors.

Coronary artery A blood vessel which supplies blood to the heart muscle.

Coronary heart disease A disease in which the coronary arteries are narrowed by the build up of fatty deposits.

Correlation A relationship between two variables.

Cutting (plants) A small piece of a plant (usually with a new bud on it) that can be grown into a new plant.

Cystic fibrosis An inherited disorder of the cell membranes caused by a recessive allele.

Cytoplasm A gel-like substance in a cell where most of the chemical reactions take place.

Glossary

D

Deforestation The cutting down of forests (large areas of trees).

Dependent variable The variable in an experiment that is measured.

Diabetes A condition that affects the body's ability to control its blood glucose level.

Dialysis A way of artificially filtering the blood to remove waste products and keep the concentration of dissolved ions in the blood at normal levels. It is used to treat patients with kidney failure.

Differentiation The process by which a cell becomes specialised for its job.

Diffusion The spreading out of particles from an area of higher concentration to an area of lower concentration.

Discrete data Numerical data that can only take certain values (e.g. number of people).

Distribution Where organisms are found in a particular area.

DNA Deoxyribonucleic acid. The molecule in cells that stores genetic information.

Dominant allele The allele for the characteristic that's shown by an organism if two different alleles are present for that characteristic.

Double-blind trial A clinical trial where neither the doctors nor the patients know who has received the drug and who has received the placebo until all the results have been gathered.

E

Ecosystem The interaction of a community of living organisms with the abiotic parts of their environment.

Effector Either a muscle or gland which responds to nervous impulses.

Efficacy Whether something, e.g. a drug, works or not.

Endothermic reaction A reaction where energy is transferred from the environment.

Enzyme A protein that acts as a biological catalyst.

Eukaryotic cell A complex cell, such as a plant or animal cell.

Evolution The changing of the inherited characteristics of a population over time.

Excretion The removal of waste products from the body.

Exothermic reaction A reaction that transfers energy to the environment.

Extinction When no living individuals of a species remain.

Extremophile An organism that's adapted to live in extreme conditions.

F

Family tree A diagram that shows how a characteristic is inherited in a group of related people.

Fermentation The process of anaerobic respiration in yeast cells.

Fertilisation The fusion of male and female gametes during sexual reproduction.

Fertility The ability to conceive a child.

Fishing quota A limit on the number and size of fish that can be caught in a certain area.

Food security Having enough food to feed the population.

Fossil The remains of an organism from many years ago, which is found in rock.

Fossil record The history of life on Earth preserved as fossils.

Functional adaptation Something that goes on inside an organism's body that helps it to survive in its environment.

G

Gamete A sex cell, e.g. an egg cell or a sperm cell in animals.

Gene A short section of DNA, found on a chromosome, which contains the instructions needed to make a protein (and so controls the development of a characteristic).

Genetic engineering The process of cutting out a useful gene from one organism's genome and inserting it into another organism's cell(s).

Genetically modified (GM) crop A crop which has had its genes modified through genetic engineering.

Genome All of the genetic material in an organism.

Genotype What alleles you have, e.g. Tt.

Geotropism See gravitropism.

Gibberellin A type of plant hormone that stimulates seed germination, stem growth and flowering.

Gland An organ that hormones are produced and secreted from.

Global warming The rise in the average global temperature.

Glucagon A hormone produced and secreted by the pancreas when blood glucose level is too low.

Glycogen A molecule that acts as a store of glucose in liver and muscle cells.

Gravitropism The growth of a plant in response to gravity. Also known as geotropism.

Guard cell A type of cell found on either side of a stoma. A pair of these cells control the stoma's size.

H

Habitat The place where an organism lives.

Haemoglobin A red pigment found in red blood cells that carries oxygen.

Hazard Something that has the potential to cause harm (e.g. fire, electricity, etc.).

Heterozygous Where an organism has two alleles for a particular gene that are different.

Homeostasis The regulation of conditions inside your body (and cells) to maintain a stable internal environment, in response to changes in both internal and external conditions.

Homozygous Where an organism has two alleles for a particular gene that are the same.

Hormone A chemical messenger which travels in the blood to activate target cells.

Hybridoma A cell made by fusing a mouse lymphocyte with a tumour cell, which is used to produce lots of monoclonal antibodies.

Hyperopia Long-sightedness.

Hypothesis A possible explanation for a scientific observation.

Glossary

I

Inbreeding When closely related animals or plants are bred together.

Independent variable The variable in an experiment that is changed.

Inherited disorder A disorder caused by a faulty allele, which can be passed on to an individual's offspring.

Insulin A hormone produced and secreted by the pancreas when blood glucose level is too high.

Interdependence Where, in a community, each species depends on other species for things such as food, shelter, pollination and seed dispersal.

IVF *In vitro* fertilisation. The artificial fertilisation of eggs in the lab.

L

Limiting factor A factor which prevents a reaction from going any faster.

Lipase A type of digestive enzyme that catalyses the breakdown of lipids into fatty acids and glycerol.

M

Mean (average) A type of average found by adding up all the data and dividing by the number of values.

Median (average) The middle value in a set of data when the values are put in order of size.

Meiosis A type of cell division where a cell divides twice to produce four genetically different gametes. It occurs in the reproductive organs.

Menstrual cycle A monthly sequence of events during which the body prepares the lining of the uterus (womb) in case it receives a fertilised egg, and releases an egg from an ovary. The uterus lining then breaks down if the egg has not been fertilised.

Meristem tissue Tissue found at the growing tips of plant shoots and roots that is able to differentiate.

Metabolism All the chemical reactions that happen in a cell or the body.

Mitochondria Structures in a cell which are the site of most of the reactions for aerobic respiration.

Mitosis A type of cell division where a cell reproduces itself by splitting to form two identical offspring.

Mode (average) The most common value in a set of data.

Monoclonal antibodies Antibodies produced from lots of clones of a single white blood cell. This means that all the antibodies are identical and will only target one specific antigen.

Motor neurone A nerve cell that carries electrical impulses from the CNS to effectors.

MRSA A strain of antibiotic-resistant bacteria.

Mutation A random change in an organism's DNA.

Mycoprotein Protein produced by fungi, in particular *Fusarium*.

Myopia Short-sightedness.

N

Natural selection The process by which species evolve.

Negative feedback A mechanism that restores a level back to optimum in a system.

Nervous system The organ system in animals that allows them to respond to changes in their environment.

Neurone A nerve cell. Neurones transmit information around the body, including to and from the CNS.

Non-communicable disease A disease that cannot spread between individuals.

Nucleotide A repeating unit in DNA and RNA that consists of a sugar, a phosphate group and a base.

Nucleus (of a cell) A structure found in animal and plant cells which contains the genetic material.

O

Obesity A condition where a person has an excessive amount of body fat, to the point where it poses a risk to their health.

Optimum dose The dose of a drug that is most effective and has few side effects.

Optimum level (in the body) A level of something (e.g. water, ions or glucose) that enables the body to work at its best.

Organ A group of different tissues that work together to perform a certain function.

Organ system A group of organs working together to perform a particular function.

Osmosis The movement of water molecules across a partially permeable membrane from a more dilute solution to a more concentrated solution.

Oxygen debt The amount of extra oxygen your body needs after exercise to react with the build up of lactic acid and remove it from cells.

P

Partially permeable membrane A membrane with tiny holes in it, which lets some molecules through it but not others.

Pathogen A microorganism that causes disease, e.g. a bacterium, virus, protist or fungus.

Peer review The process where scientists check each other's results and explanations to make sure they're scientific.

Permanent vacuole A structure in plant cells that contains cell sap.

Phagocytosis The process by which white blood cells engulf foreign cells and digest them.

Phenotype The characteristics you have, e.g. brown eyes.

Phloem A type of plant tissue which transports dissolved sugars around the plant.

Photosynthesis The process by which plants use energy to convert carbon dioxide and water into glucose and oxygen.

Phototropism The growth of a plant in response to light.

Placebo A substance that is like a drug being tested, but which doesn't do anything.

Plasma The liquid component of blood, which carries blood cells and other substances around the body.

Glossary

Platelet A small fragment of a cell found in the blood, which helps blood to clot at a wound.

Polydactyly An inherited disorder, caused by a dominant allele, where a person has extra fingers or toes.

Precise result A result that is close to the mean.

Predator An animal that hunts and kills other animals for food.

Prey An animal that is hunted and killed by another animal for food.

Primary consumer An organism in a food chain that feeds on a producer.

Producer An organism at the start of a food chain that makes its own food using energy from the Sun.

Prokaryotic cell A small, simple cell, e.g. a bacterium.

Protease A type of digestive enzyme that catalyses the breakdown of proteins into amino acids.

Protein A large biological molecule made up of long chains of amino acids.

Protist A type of pathogen. Protists are often transferred to other organisms by a vector.

Punnett square A type of genetic diagram.

Pyramid of biomass A diagram to represent the biomass at each stage of a food chain.

Q

Quadrat A square frame enclosing a known area. It is used to study the distribution of organisms.

R

Random error A difference in the results of an experiment caused by things like human error in measuring.

Range The difference between the smallest and largest values in a set of data.

Receptor A group of cells that are sensitive to a stimulus (e.g. receptor cells in the eye detect light).

Recessive allele An allele whose characteristic only appears in an organism if there are two copies present.

Reflex A fast, automatic response to a stimulus.

Relay neurone A nerve cell that carries electrical impulses from sensory neurones to motor neurones.

Repeatable result A result that will come out the same if the experiment is repeated by the same person using the same method and equipment.

Reproducible result A result that will come out the same if someone different does the experiment, or a slightly different method or piece of equipment is used.

Respiration The process of breaking down glucose to transfer energy, which occurs in every cell.

Ribosome A structure in a cell, where proteins are made.

Risk The chance that a hazard will cause harm.

Risk factor Something that is linked to an increased likelihood that a person will develop a certain disease.

S

Secondary consumer An organism in a food chain that eats a primary consumer.

Selective breeding (artificial selection) When humans artificially select the plants or animals that are going to breed, so that the genes for particular characteristics remain in the population.

Sensory neurone A nerve cell that carries electrical impulses from a receptor in a sense organ to the CNS.

Sex chromosome (humans) One of the 23rd pair of chromosomes, X or Y. Together they determine whether an individual is male or female.

Sexual reproduction Where two gametes combine at fertilisation to produce a genetically different new individual.

Speciation The development of a new species.

Species A group of similar organisms that can reproduce to give fertile offspring.

Stable community A community where all the species and environmental factors are in balance, so that the population sizes are roughly constant.

Statins A group of medicinal drugs that are used to decrease the risk of heart and circulatory disease.

Stem cell An undifferentiated cell which has the ability to become one of many different types of cell, or to produce more stem cells.

Stent A wire mesh tube that's inserted inside an artery to help keep it open.

Stimulus A change in the environment.

Stoma A tiny hole in the surface of a leaf.

Structural adaptation A feature of an organism's body structure that helps it to survive in its environment.

Synapse The connection between two neurones.

Systematic error An error that is consistently made every time throughout an experiment.

T

Tertiary consumer An organism in a food chain that eats a secondary consumer.

Theory A hypothesis which has been accepted by the scientific community because there is good evidence to back it up.

Thermoregulatory centre An area of the brain which controls and monitors body temperature.

Tissue A group of similar cells that work together to carry out a particular function.

Tissue culture (plants) A method of cloning plants in which a few plant cells are put on a growth medium containing hormones and allowed to grow into new plants.

Toxicity How harmful something is, e.g. a drug.

Toxin A poison. Toxins are often produced by bacteria.

Transect A line which can be used to study the distribution of organisms across an area.

Translocation The movement of dissolved sugars around a plant.

Transpiration stream The movement of water from a plant's roots, through the xylem and out of the leaves.

Trophic level A stage in a food chain.

Tumour A growth of abnormal cells.

Glossary

U

Uncertainty The amount of error your results might have.

Urea A waste product produced from the breakdown of amino acids in the liver.

V

Vaccination The injection of dead or inactive microorganisms, in order to produce an immune response that will help to protect you against a particular pathogen in the future.

Valid result A result that is repeatable, reproducible and answers the original question.

Valve A structure within the heart or a vein which prevents blood flowing in the wrong direction.

Variable A factor in an investigation that can change or be changed (e.g. temperature or concentration).

Variation The differences that exist between individuals.

Vasoconstriction When blood vessels supplying the skin capillaries constrict (get narrower).

Vasodilation When blood vessels supplying the skin capillaries dilate (get wider).

Vector (in disease) An organism that transfers a disease from one animal or plant to another, which doesn't get the disease itself.

Vector (in genetic engineering) Something used to transfer DNA into a cell, e.g. a virus or a bacterial plasmid.

Vein A blood vessel that carries blood to the heart.

Virus A tiny pathogen that can only replicate within host body cells.

W

White blood cell A blood cell that is part of the immune system, defending the body against disease.

X

Xylem A type of plant tissue which transports water and mineral ions around the plant.

Z

Zero error A type of systematic error caused by using a piece of equipment that isn't zeroed properly.

Index

A

abiotic factors 142, 143
abstinence 103
accommodation 89
accuracy (of results) 173
active sites 27
active transport 19
adaptations 144
adrenal glands 96, 105
adrenaline 105
adult cell cloning 134
aerobic respiration 79
alleles 121
alveoli 21, 34
ammonia 99
amylase 28-30
anaerobic respiration 79
ancestors 139
animal cells 4
anomalous results 173
antibiotic resistance 61, 138
antibiotics 61, 138
  investigating the effects of 13
antibodies 37, 59
anti-diuretic hormone (ADH) 99
antimicrobials 59
antiretroviral drugs 56
antitoxins 37, 59
aorta 35
apparatus 181
Archaea 139
arteries 36
artificial hearts 42
asexual reproduction 116, 118
aspirin 61
atria 35
auxins 107, 109

B

bacteria 4, 11, 13, 54, 57, 139
balances 181
bar charts 175
barrier methods
  (of contraception) 103
basal metabolic rate 105
behavioural adaptations 144
Benedict's reagent 31
benign tumours 46
bias 170
bile 29
binary fission 11
binomial system 139
biodiversity 157-160
  maintenance of 160
biogas 153
biological heart valves 42
biomass 164
biomass transfer 164
biotechnology 166
biotic factors 142, 143
Biuret test 31
blind studies 62
body temperature 92
brain 88
breathing rate 34
bronchi 34
bronchioles 34
Bunsen burners 182

C

caffeine (effect on reaction time) 86
calculators 178
cancer 67
capillaries 36
carbohydrases 29
carbon cycle 150
catalysts 27
categoric data 174
cell
  cycle 10
  division 10, 11
  elongation 107
  membranes 4, 17
  walls 4
cells 4
  drawing 6
  specialised 8
central nervous system (CNS) 84, 85
cerebellum 88
cerebral cortex 88
chlorophyll 4, 71
chloroplasts 4, 71
cholesterol 41
chromosomes 10, 112
  X and Y 119
cilia 59
ciliary muscles 89
class (classification) 139
classification 139
climate change 158
clinical trials 62
cloning 134
communicable diseases 44, 54, 55
communities 142
complementary base pairing 113
computational models 169
concave lenses 90
concentration gradients 17
conclusions 179
condensation 149
condoms 57, 103
contact lenses 91
contraception 103
control variables 172
convex lenses 90
corneas 89
coronary arteries 35
coronary heart disease 41
correlation 176, 179
cover slip (slides) 6
cuttings 109, 134
cystic fibrosis 123
cytoplasm 4

D

Darwin, Charles 129, 130
decay 150, 153, 154
  investigating 154
  rate of 153
decomposers 162
defences (of plants) 68
deforestation 159
deletions (mutations) 114
dependent variables 172
detritus feeders 150
diabetes 97
dialysis 100
diaphragm (thorax) 34
diaphragms (contraceptive) 103
differentiation 8
diffusion 17, 18, 20-22
digestive enzymes 29
digestive system 26, 30
digitalis 61
discrete data 174
distribution of organisms 147-149
DNA 10, 112, 113
  nucleotides 113
dormancy (in seeds) 109
dosage 62
double-blind studies 62
double circulatory systems 35
drugs 61, 62
  development of 62

E

ecosystems 142
  maintenance of 160
effectors 83, 84
efficacy (of drugs) 62
electric heaters 182
electron microscopes 5
embryo screening 123
embryo transplants 134
endocrine system 96
endothermic reactions 71
environmental changes 149
environmental variation 128
enzymes 27-30
epidermal tissue 48
errors 173
ethene (in plants) 109
ethics (in investigations) 182

Index

Eukaryota 139
eukaryotes 4, 54
evaluations 180
evaporation 149
evolutionary trees 139
evolution by natural selection 129, 130, 137
exchange surfaces 20-22
exercise 80
exothermic reactions 78
extinction 129
eyes 89-91

F

fair tests 172
family (classification) 139
family trees 122
farming 165
fatty acids 29
fermentation 79
fertility 103, 104
  treatments 104
fight or flight response 105
filtration (kidneys) 99
Fleming, Alexander 61
flowering 109
focusing 89
follicle-stimulating hormone (FSH) 102, 104
food chains 145, 162, 163
food poisoning 57
food production 165
food security 165
formula triangles 178
fossils 136
functional adaptations 144
fungi 54, 56

G

gall bladder 30
gamete fusion 117
gametes 116, 117
gas exchange 21, 22, 34
gas syringes 181
gene pools 132, 134
genes 10, 112
genetic
  diagrams 119, 121-123
  disorders 123
  engineering 133, 166
  variants 114
  variation 128
genetically modified crops 133
genomes 112
genotypes 121
genus 139
geotropism 107
germination 109
gibberellin 109
gills 22
glands 96
global warming 158
glucagon 97
glucose
  in the blood 97
  test for 31
  uses in plants 71
glycerol 29
glycogen 97
gonorrhoea 57
gradients 176
graphs 175, 176
gravitropism 107
greenhouse gases 158
greenhouses 75
guard cells 50
gullet 30

H

habitats 142
haemoglobin 37
hazards 171
heart 35
  transplants 42
  valves 42
hereditary units 124
heterozygous organisms 121
HIV 56
homeostasis 83
homozygous organisms 121
hormones 96
hybridoma cells 65
hyperopia 90
hypotheses 169, 172

I

immune system 59
independent variables 172
indicators 181
infertility 104
inhibition zones 13, 14
insertions (mutations) 114
insulin 97
intercepts (on a graph) 176
interdependence 142
intrauterine devices (IUDs) 103
inverse
  correlations 176
  proportion 75
  square law 75
in vitro fertilisation (IVF) 104
iodine test (for starch) 28
irises 89, 91
iris reflex 91
IVF 104

K

kidney failure 100
kidneys 99, 100
kingdom (classification) 139

L

lactic acid 79, 80
Lamarck, Jean-Baptiste 130
lamellae 22
large intestine 30
laser eye surgery 91
leaves
  gas exchange 22
  tissues in 22
lenses 89-91
light microscopes 5, 6
limiting factors (of photosynthesis) 71-73
linear graphs 175, 176
lines of best fit 175
Linnaean system 139
Linnaeus, Carl 139
lipases 29
lipids (test for) 31
liver 30, 97
'lock and key' model 27
long-sightedness 90
lungs 34
luteinising hormone (LH) 102, 104
lymphocytes 65

M

magnification 5
malaria 57
malignant tumours 46
mean (average) 174
mean division time 11
measles 56
measuring cylinders 181
mechanical heart valves 42
median 174
medulla (brain) 88
meiosis 117
Mendel, Gregor 124
menstrual cycle 102
meristem tissue 48
metabolism 78
microorganisms 54
  culturing 13, 14
microscopes 5, 6, 183
migration patterns 158
  human 112
mitochondria 4
mitosis 10, 117
mode 174
models (scientific) 169

Index

monoclonal antibodies 65-67
motor neurones 84, 85
MRI scanning 88
mRNA 113
MRSA 61, 138
mucus 59
muscle cells 8
muscle fatigue 80
mutations 114, 128
mycoprotein 166
myopia 90

N

natural selection 129, 137
negative correlations 176
negative feedback systems 83
  blood glucose 97
  body temperature 92
  thyroxine 105
  water content 99
nerve cells 8
nervous system 84, 85
neurones 84, 85
neuroscientists 88
non-coding DNA 113, 114
non-communicable diseases 44
nucleotides 113
nucleus 4

O

oestrogen 102, 103
optic nerve 89
order (classification) 139
organ donors 100
organ systems 26
organs 26
osmosis 18
ovaries 96, 102
overfishing 165
ovulation 102
oxygen debt 80

P

pacemakers 35
painkillers 61
palisade mesophyll tissue 48
pancreas 30, 96, 97
parasites 54
partially permeable membranes 17
pathogens 54, 55
pea plants 124
peat bogs 159
peer review 62, 169
pepsin 29, 30
percentage
  change 185
  cover 148
percentiles 185
Petri dishes 13
phagocytosis 37, 59
phenolphthalein 154
phenotypes 121
phenotype variation 129
phloem cells 8
phloem tubes 48, 49
pH meters 181
photosynthesis 71-74
  rate of 72, 73
phototropism 107
phylum (classification) 139
pipettes 181
pituitary gland 96, 102, 105
placebo effect 62
plant
  cells 4
  diseases 68
  growth 107-109
  hormones 107, 109
plasma 37
plasmids 4, 133
platelets 37
pollution 157
polydactyly 123
population (human) 157
positive correlations 176
potatoes 18
potometers 50, 183
precipitation 149
precision (of results) 173
preclinical testing 62
predator-prey cycles 145
predators 145
pregnancy tests 66
prey 145
primary consumers 145
producers 145
progesterone 102, 103
prokaryotes 4, 11
proteases 29
proteins 113
  function of 113
  test for 31
protists 54, 57
puberty 102
pulmonary artery 35
pulmonary vein 35
Punnett squares 119, 121
pupils 89, 91
pyramids of biomass 163

Q

quadrats 147, 148

R

random
  errors 173
  sampling 184
range (of data) 180
rates of reaction 28
reaction times 86
receptors 83, 84
rectum 30
red blood cells 37
reflex arcs 85
relay neurones 85
repeatability (of data) 172, 173
replacement lens surgery 91
representational models 169
reproducibility (of data) 172, 173
resolution (microscopes) 5
respiration 78, 79
retinas 89
ribcage 34
ribosomes 4, 113
ripening (of fruit) 109
risk 171
risk factors (for disease) 45, 46
root hair cells 8, 19
rooting powder 109
roots 107
rose black spot 56
rounding numbers 174

S

safety 171, 182
salivary glands 30
Salmonella 57
sample size 172
sampling 184
sclera 89
secondary consumers 145
secondary sexual
    characteristics 102
selective breeding 132
selective reabsorption 99
sensory neurones 84, 85
sex hormones 102-104
sexually transmitted diseases 57
sexual reproduction 116, 118
shivering 92
shoots 107
short-sightedness 90
SI units 177, 178
significant figures 174
slides 6
small intestine 30

Index

specialised cells 8
speciation 129, 137
species 137, 139
  in classification 137
sperm cells 8, 102
spermicide 103
spongy mesophyll tissue 48
stains (microscope) 6
standard form 5, 178
starch 28
  test for 31
statins 41
stem cells 9
stents 41
sterilisation (contraception) 103
stomach 30
stomata 22, 50
structural adaptations 144
subcellular structures 4, 8
substitutions (mutations) 114
Sudan III 31
sugars (test for) 31
superbugs 138
surface area to volume ratios 20
survival of the fittest 129
suspensory ligaments 89
sweat 92, 99
synapses 85
systematic errors 173

T

tables (of data) 174
tangents 176
target organs 96
temperature receptors 92
tertiary consumers 145
testes 96, 102
testosterone 102
theories 169
thermometers 181
thermoregulatory centre 92
thorax 34
three-domain system of classification 139
thyroid gland 96, 105
thyroid stimulating hormone (TSH) 105
thyroxine 105
tissue culture 109, 134
tissues 26
tobacco mosaic virus 56
toxicity (of drugs) 62
trachea 34
transects 148
transpiration 49, 50, 149
  rate 50
transplants (organ) 100
trophic levels 162-164
tumour markers 67
tumours 46
Type 1 diabetes 97
Type 2 diabetes 97

U

uncertainty 180
units 177, 178
urea 99
urine 99

V

vaccination 57, 60
vacuoles 4
valid results 172
valves (heart) 35, 42
variables 172
variation 128
vasoconstriction 92
vasodilation 92
vectors 54, 57, 133
veins 36
vena cava 35
ventricles 35
villi 21
viruses 54, 56
vision defects 90, 91
volumes (measuring) 181

W

Wallace, Alfred Russel 137
waste management 157
water baths 182
water cycle 149
weed killers 109
white blood cells 37, 59
Woese, Carl 139

X

X and Y chromosomes 119
xylem cells 8
xylem tubes 48, 49

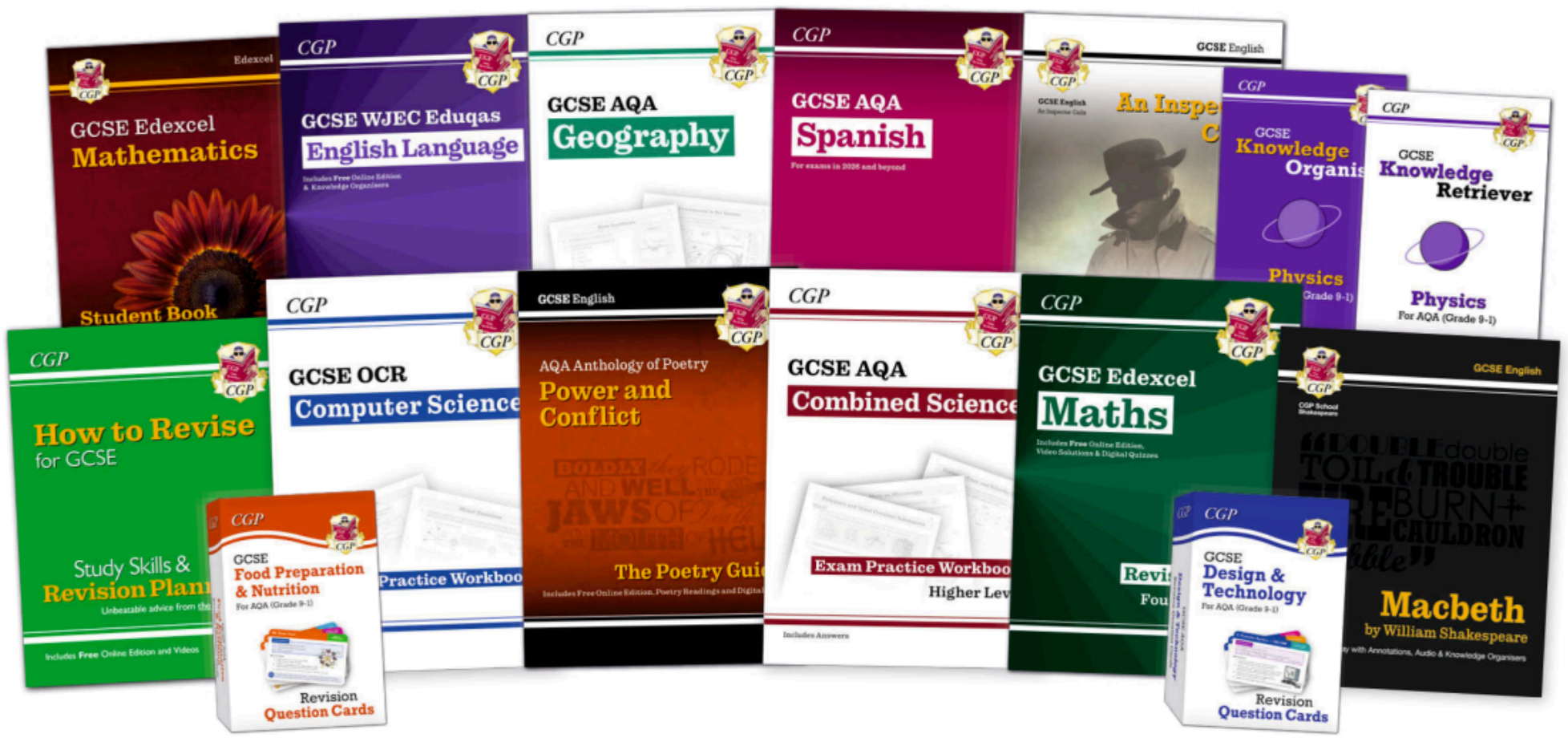
Y

yeast cells 79

Z

zero errors 173

CGP books — they might just save your life...



OK, maybe not your *life*. But when it comes to exam revision, CGP are the undisputed champions.
You can order any of our books and cards (with next-day delivery!) from us, online or by phone:

cgpbooks.co.uk • 0800 1712 712

CGP Books — The Choice of Champions!

0325 - 30593

CGP

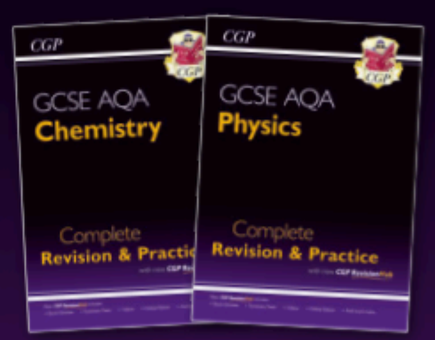
This CGP book has everything you need for GCSE Biology!

That's right — **millions of students** have aced their exams with CGP...

- **The best revision notes by miles...**
  Yep, for every single topic on the course
- **Lots of exam-style questions...**
  Oh, yes — and all the answers too
- **Plus full practice papers...**
  They're just like the real thing
- **With the new *CGP RevisionHub*...**
  Online revision that's actually useful!

You *could* choose someone else's dreary book...
...but really — why, *why* would you do that? ☺

P.S. CGP also has revision books for GCSE **Chemistry** and **Physics**!



This book contains substances known to man

www.cgpbooks.co.uk